

Prepladder Version X Qbank

Subject: Physiology

Contents

Lesson	Page
1.Synapse and Synaptic Inhibition	5
2.Somatosensory Receptors, Nociceptors, Proprioceptors, and Reflexes	34
3.Ascending and Descending Tracts of Spinal Cords	78
4.Basal Ganglia and Cerebellum	101
5.Special Senses	127
6.EEG & Sleep Learning and Memory	151
7.General Physiology	168
8.Previous Year Questions	205
9.Neuron and nerve, Action Potential, RMP, Nerve fibre classification	251
10.Functional Anatomy of Skeletal Muscle	301
11.Motor unit & Smooth muscle	325
12.Previous Year Questions	354
13.Cardiac Conducting System and Action Potentials	389
14.ECG	415
15.Hemodynamics, BP Regulation, and Regional Circulation	432
16.Cardiac Cycle and Cardiac Output	453
17.Previous Year Questions	480
18.Blood Physiology	520
19.Previous Year Questions	539
20.Renal Physiology	556
21.Tubular Transport	589
22.Acids Secretion and Counter-Current Mechanism of the Kidney	619
23.Previous Year Questions	637
24.Functional Anatomy, Mechanics, Ventilation-Perfusion, Compliance of RS	688
25.Pulmonary Function Tests	723

26.Diffusion of Gases, O ₂ and CO ₂ Transport	742
27.Regulation of respiration	771
28.Hypoxia, Altitude, and Deep sea pressures	792
29.Previous Year Questions	821
30.GIT Hormones, Secretions, and Absorption Mechanisms	863

Contents (continued)

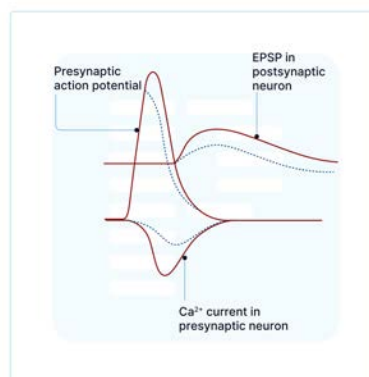
Lesson	Page
31.GIT Motility	893
32.Previous Year Questions	909
33.Previous Year Questions	935
34.Hormone Mechanism & Pituitary hormone	989
35.Adrenal Physiology and Calcium Regulation	1022
36.Thyroid Hormones & Endocrine Pancreas	1043
37.Previous Year Questions	1078
38.Reproductive Physiology	1142

Synapse and Synaptic Inhibition

1. Which of the following best describes the mechanism of action of glutamate at its receptors in the central nervous system?

- A. It inhibits metabotropic receptors, activating a G-protein, thus increasing cAMP levels.
- B. It stimulates NMDA receptors, resulting in an influx of calcium ions.
- C. It inhibits AMPA receptors, preventing sodium influx.
- D. It stimulates kainate receptors, thus decreasing chloride ion permeability.

2. As illustrated below, in examining the effects of presynaptic inhibition and facilitation on action potentials, Ca^{2+} currents, and EPSPs (excitatory postsynaptic potential), which statement regarding the dashed line recordings is false? (solid lines for controls; dashed lines for inhibition/facilitation)



- A. It can be produced by opening of voltage gated K^{+} channel
- B. Produced due to enhanced Ca^{2+} entry in the presynaptic neuron.
- C. The effect is produced by GABA
- D. Produced due to increases Cl^{-} conductance

3. Feedforward Inhibition is seen in

- A. Renshaw cells
 - B. Cerebellum
 - C. Basal Ganglia
 - D. Hippocampus
-

4. In presynaptic inhibition, which neurotransmitter is most commonly responsible for inhibiting synaptic transmission at the presynaptic terminal?

- A. Glutamate
 - B. Serotonin
 - C. GABA
 - D. Glycine
-

5. A 28-year-old woman has fluctuating muscle weakness that worsens throughout the day. Tests show a decremental response to repetitive nerve stimulation, improvement with edrophonium, and antibodies against neuromuscular junction proteins. What is the primary pathophysiological mechanism in this condition?

- A. Decreased acetylcholine release from motor nerve terminals
 - B. Reduced functional acetylcholine receptors on the postsynaptic membrane
 - C. Antibodies blocking voltage-gated calcium channels on presynaptic terminals
 - D. It is a presynaptic syndrome
-

6. Which of the following is the correct sequence of events at the neuromuscular junction?

- A. Ca^{2+} influx → ACh release → End-plate potential → Muscle contraction
 - B. ACh release → Ca^{2+} influx → End-plate potential → Muscle contraction
 - C. End-plate potential → Ca^{2+} influx → ACh release → Muscle contraction
 - D. ACh release → End-plate potential → Ca^{2+} influx → Muscle contraction
-

7. Slow Type inhibitory postsynaptic potential (IPSP) is generated by?

- A. Opening of K^{+} channels
 - B. Closing of K^{+} channels
 - C. Opening of Cl^{-} channels
 - D. Opening of Na^{+} channels
-

8. What is the primary effect of an inhibitory postsynaptic potential (IPSP) on a neuron's excitability?

- A. Increases the rate of Neurotransmitter release
 - B. Decreases membrane potential, closer to firing threshold
 - C. Increases membrane potential, farther from firing threshold
 - D. No effect on membrane potential
-

9. A 65-year-old male patient presents with neurological symptoms following a Cerebrovascular accident. Imaging confirms ischemic damage in the brain. Elevated levels of glutamate are found in the extracellular of the brain. Which of the following statements about glutamate-induced cytotoxicity is true?

- A. Cleared primarily by K⁺-dependent systems in neurons
 - B. Causes neuronal death by excessive Na⁺ influx
 - C. Cytotoxicity due to excessive Ca²⁺ influx
 - D. Elevated Na⁺ stimulates astrocytes to remove glutamate
-

10. During the generation of excitatory postsynaptic potential (EPSP), which of the following movements of ions occur at the postsynaptic membrane?

- A. Efflux of sodium ions
 - B. Influx of calcium ions
 - C. Influx of potassium ions
 - D. Influx of chloride ions
-

11. Which subtype of GABA receptors are associated with mediating fast inhibitory postsynaptic potentials through the entry of Cl⁻ into neurons? GABA-A GABA-B GABA-C

- A. Receptor 1 and 2
 - B. Receptor 2 and 3
 - C. Receptor 1 and 3
 - D. All of the above
-

12. A 63-year-old male presents with a neurodegenerative condition involving impaired glutamate regulation. Which type of glial cell is primarily involved in glutamate regulation?

- A. Schwann cells

- B. Astrocytes
 - C. Ependymal cells
 - D. Oligodendrocytes
-

13. Match the following neurotransmitters with their site of production in CNS. A. Acetylcholine (ACh) 1. Raphe nuclei B. Norepinephrine (NE) 2. Nucleus basalis of Meynert C. Dopamine 3. Locus ceruleus D. Serotonin 4. Substantia nigra

A. Acetylcholine (ACh)	1. Raphe nuclei
B. Norepinephrine (NE)	2. Nucleus basalis of Meynert
C. Dopamine	3. Locus ceruleus
D. Serotonin	4. Substantia nigra

- A. A-4, B-3, C-2, D-1
 - B. A-3, B-2, C-4, D-1
 - C. A-2, B-3, C-4, D-1
 - D. A-4, B-2, C-3, D-1
-

14. Which neurotransmitter is primarily responsible for inhibiting motor neuron activity at the level of the spinal cord to regulate muscle tone and reflexes?

- A. GABA
 - B. Glutamate
 - C. Glycine
 - D. Acetylcholine
-

15. 25-year-old patient with generalized anxiety disorder presents with persistent insomnia. Which neurotransmitter is targeted to promote relaxation and reduce excitability?

- A. Glutamate and aspartate
 - B. Gamma-aminobutyric acid (GABA)
 - C. Dopamine and serotonin
 - D. Endorphins and substance P
-

16. A 40-year-old patient presents with severe muscle stiffness and spasms, including trismus and rigidity, after stepping on a rusty nail two weeks ago. Which toxin is responsible for these symptoms by interfering with neurotransmitter release at the neuromuscular junction?

- A. Botulinum toxin
 - B. Tetanus toxin
 - C. Strychnine
 - D. Curare
-

17. Which protein, essential for regulating neurotransmitter release at the neuromuscular junction, prevents the premature fusion of synaptic vesicles?

- A. Synaptobrevin
 - B. SNAP-25
 - C. Syntaxin
 - D. Complexin
-

18. Which of the following types of vesicle at the synaptic terminal contain glutamate neurotransmitter ?

- A. Small clear vesicles
 - B. Large dense-core vesicles
 - C. Small dense-core vesicles
 - D. Endosomes
-

19. A researcher studying the precision of synaptic connections between neurons focuses on how specific neurons target each other for communication. Which of the following concepts refers to the precise targeting of synaptic connections to ensure accurate neuronal communication?

- A. Synaptic specificity
 - B. Active zones
 - C. Synaptobrevin
 - D. Postsynaptic density
-

20. A researcher studying synaptic transmission observes that communication at a particular synapse is both rapid and bidirectional. What is the most likely synapse?

- A. Chemical synapse
 - B. Axo Axonal synapse
 - C. Axodendritic synapse
 - D. Electrical synapses
-

21. A 52-year-old male patient complains of visual disturbances and difficulty walking. On detailed examination, the patient is diagnosed with multiple sclerosis

that occurs due to demyelination of neurons. Which of the following cells is involved in this condition?

- A. Microglia
 - B. Oligodendrocytes
 - C. Schwann cells
 - D. Astrocytes
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	3
Question 5	2
Question 6	1
Question 7	1
Question 8	3
Question 9	3
Question 10	2
Question 11	3
Question 12	2
Question 13	3
Question 14	3
Question 15	2
Question 16	2
Question 17	4
Question 18	1
Question 19	1
Question 20	4
Question 21	2

Solution for Question 1:

Correct Answer: B) It stimulates NMDA receptors, resulting in an influx of calcium ions.

Explanation:

Glutamate is an amino acid that acts as a neuroexcitatory molecule by stimulating NMDA receptors in the CNS, which results in an influx of calcium ions.

Type	Subtype	Features	Functions
Ionotropic receptors	AMPA (Alpha-amino-3-hydroxy-5-methylisoxazole-4-propionate) (Option C)	Influx of Na ⁺ and efflux of K ⁺ . It has a low permeability for Ca ²⁺ .	Produces fast excitatory postsynaptic potential.
Kainate receptor (Option D)	Influx of Na ⁺ and Ca ²⁺	Aids in excitatory neurotransmission	
NMDA receptor (N-methyl-D-aspartate) (Option B)	Allows large influx of Ca ²⁺ and Na ⁺ Unique Features: Glycine Binding: Essential for receptor response to glutamate. Mg ²⁺ Block: At normal membrane potentials, the channel is blocked by extracellular Mg ²⁺ . This block is removed during depolarisation by adjacent AMPA and kainate receptors. EPSP: Slower than AMPA/kainate receptors-induced EPSP.	Involved in synaptic plasticity, memory, and learning and can cause excitotoxicity in excess.	
Metabotropic receptors (Option A)	Presynaptic (mGluR2-4,6-8): Involved in limiting glutamate release.	Activation leads to changes in second messengers: IP3 and DAG: Increased intracellular levels. cAMP: Decreased intracellular levels.	Modulate synaptic plasticity and neurotransmission indirectly through second messengers.

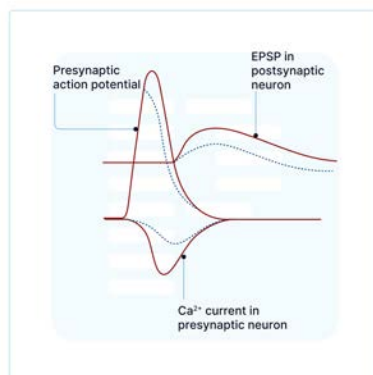
Postsynaptic (mGluR 1,5): Involved in synaptic plasticity.			
--	--	--	--

Solution for Question 2:

Correct answer: B) Produced due to enhanced Ca^{2+} entry in the presynaptic neuron.

Explanation:

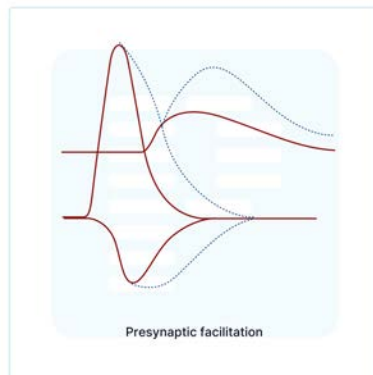
Dashed lines the records obtained during inhibition.



Presynaptic Inhibition occurs when a neuron reduces neurotransmitter release at the axon terminal. Key mechanisms include:

- When K^+ channels open, K^+ exits the neuron, causing hyperpolarization. This makes it harder for the neuron to reach the threshold for action potentials, reducing overall activity.
- During presynaptic inhibition, mediated by GABA (Option C), hyperpolarization leads to decreased opening of voltage-gated Ca^{2+} channels during action potentials. This results in less Ca^{2+} entering the presynaptic neuron.
- GABA binding opens Cl^- channels, allowing Cl^- to enter and further hyperpolarize the neuron, reinforcing the inhibitory effect.

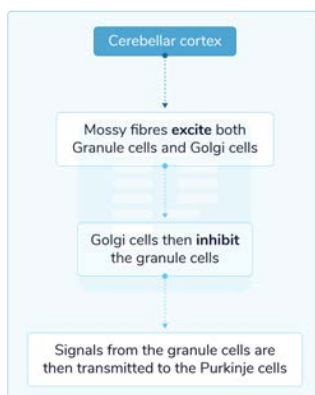
Where as Presynaptic Facilitation effectively boosts synaptic transmission by increasing Ca^{2+} levels in the presynaptic neuron, resulting in enhanced neurotransmitter release and a stronger response in the postsynaptic neuron.



Solution for Question 3:

Correct Answer: B) Cerebellum

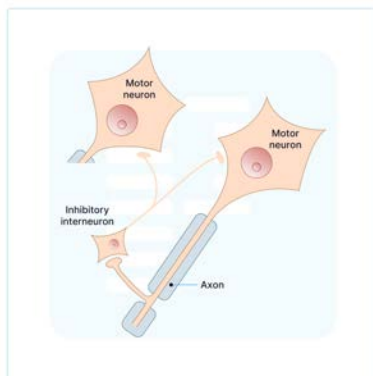
Explanation: Feedforward inhibition is a mechanism where an excitatory neuron activates an inhibitory neuron, which then inhibits a downstream neuron to prevent excessive excitation. This is commonly observed in the cerebellum, which helps regulate motor control and coordination.



Feature	Feedback Inhibition	Feedforward Inhibition
Definition	Inhibition where the output of a process inhibits its own production.	Inhibition where an excitatory neuron directly inhibits another neuron in response to input.

Mechanism	The end product of a metabolic pathway inhibits an enzyme involved in the pathway.	An excitatory neuron activates an inhibitory neuron and then inhibits a downstream neuron.
Function	It helps maintain homeostasis by regulating the production of substances.	Anticipates and prevents excessive excitation in neural circuits.
Example	The inhibition of an enzyme in a metabolic pathway by its end product (e.g., ATP inhibiting phosphofructokinase).	In the cerebellum, excitatory inputs inhibit output neurons to prevent overexcitation.

Renshaw cells (Option A) are involved in feedback inhibition rather than feedforward inhibition.



Basal ganglia (Option C) is involved in complex inhibitory circuits, but they are more commonly associated with feedback inhibition and disinhibition mechanisms rather than feedforward inhibition.

Hippocampus (Option D) is more associated with memory and learning processes rather than direct feedforward inhibition

Solution for Question 4:

Correct Answer: C) GABA

Explanation:

GABA mediates presynaptic inhibition/Indirect Inhibition via a G-protein that causes an increase in K⁺ conductance.

GABA is released onto the outer surfaces of presynaptic nerve terminals.

Mechanism of action :

- GABA opens anion channels, allowing chloride ions to diffuse into the terminal fibril.
- The negative charges of chloride ions cancel out much of the excitatory effect of incoming sodium ions.

Functions of presynaptic inhibition :

- It occurs before the signal reaches the synapse
- It's common in sensory pathways and helps minimise sideways spread and mixing of signals

Features	Indirect Inhibition	Direct Inhibition
Site of action	Presynaptic terminal	Postsynaptic neuron
Primary mechanism	Reduces neurotransmitter release.	Hyperpolarises or shunts postsynaptic neurons..
Main neurotransmitter	GABA (primarily)	GABA, Glycine (Option D)
Receptor types involved	Mainly GABA-B receptors	GABA-A, Glycine receptors
Spatial effect	It can affect multiple synapses from the same axon	Localized to specific synapses
Onset and duration	Can be longer-lasting	Usually, rapid onset and offset

Glutamate (Option A): It is typically an excitatory neurotransmitter that facilitates synaptic transmission rather than inhibiting it.

Serotonin (Option B): It can modulate neurotransmitter release and has complex effects on synaptic transmission but is not the primary neurotransmitter for presynaptic inhibition.

Glycine (Option D): It acts as an inhibitory neurotransmitter in the spinal cord rather than functioning through presynaptic inhibition in other areas of the central nervous system.

Solution for Question 5:

Correct Answer: B) Reduced functional acetylcholine receptors on the postsynaptic membrane

Explanation:

This scenario describes the classic presentation and diagnostic findings of myasthenia gravis in which functional acetylcholine receptors on the postsynaptic membrane are reduced.

Disorders related to Neuromuscular junction:

Feature	Myasthenia Gravis	Lambert-Eaton Myasthenic Syndrome (LEMS)
Primary autoantibody target	Acetylcholine receptors	Voltage-gated calcium channels (Option C)
Pathophysiology	Reduced postsynaptic acetylcholine receptors (Option B)	Decreased presynaptic acetylcholine release (Options A and D)
Fatigue pattern	Worsens with activity	Improves with brief exercise.
Reflexes	Normal or mildly reduced	Reduced or absent may improve after exercise.
Repetitive nerve stimulation	Decremental response.	Incremental response.
Response to edrophonium	Improves symptoms.	Little to no improvement.

Solution for Question 6:

Correct Answer: A) Ca^{2+} influx → ACh release → End-plate potential → Muscle contraction

Explanation:

Mechanism of synapse at Neuromuscular junction:



Solution for Question 7:

Correct Answer: A) Opening of K⁺ channels

Explanation:

Slow inhibitory postsynaptic potential (IPSP) is mediated by the opening of K⁺ channels, which allow potassium ions to flow out of the cell, resulting in hyperpolarisation of the neuron, moving its membrane potential further from the threshold for firing an action potential.

IPSP Type	Channel	Region
Slow IPSP	Opening of K ⁺ Channel (Option A)	Autonomic ganglia, smooth muscles, cardiac muscles.
Fast IPSP	Entry of Cl ⁻ Channel (Option C)	All parts of the CNS.

Closing of K⁺ channels (Option B) would typically cause depolarisation, not hyperpolarisation.

Opening of Na⁺ channels (Option D) generates late Slow Excitatory postsynaptic potential

Solution for Question 8:

Correct Answer: C) Increases membrane potential, farther from firing threshold

Explanation:

IPSPs (Inhibitory Postsynaptic Potentials) are hyperpolarizations that occur when certain neurotransmitters activate Cl^- channels on the postsynaptic membrane, allowing Cl^- to enter the cell.

- This causes the neuron to hyperpolarize, peaking 1–1.5 ms after the stimulus and then declining exponentially.
- This hyperpolarisation shifts the membrane potential further from the firing threshold, reducing the neuron's excitability and necessitating greater excitatory input to reach the threshold (Option C) .
- IPSPs can also occur through mechanisms such as: Opening of K^+ channels, allowing K^+ to exit the cell Closure of Na^+ channels Closure of Ca^{2+} channels
- Opening of K^+ channels, allowing K^+ to exit the cell
- Closure of Na^+ channels
- Closure of Ca^{2+} channels
- Opening of K^+ channels, allowing K^+ to exit the cell
- Closure of Na^+ channels
- Closure of Ca^{2+} channels

Increases the rate of neurotransmitter release (Option A): IPSPs do not influence neurotransmitter release; they affect the neuron's membrane potential.

Decreases membrane potential, closer to firing threshold (Option B): IPSPs cause hyperpolarisation, which makes the membrane potential more negative and farther from the firing threshold.

No effect on membrane potential (Option D): IPSPs affect the membrane potential by making it more negative, moving it further from the threshold.

Solution for Question 9:

Correct Answer: C) Cytotoxicity due to excessive Ca^{2+} influx

Explanation:

Excessive glutamate acts on neuronal cell bodies, causing a massive influx of Ca^{2+} , leading to neuronal cell death (Option B) .

- Na⁺-dependent uptake systems in neurons and glia clear glutamate from the brain's extracellular fluid (Option A) .
- Excessive levels of glutamate can occur in response to ischemia, anoxia, hypoglycemia, or trauma.
- During a stroke Occlusion of a cerebral artery leads to cell death in the ischemic area. Surrounding partially ischemic cells may survive but lose their ability to maintain the transmembrane Na⁺ gradient. Elevated intracellular Na⁺ levels impair astrocytes' ability to remove glutamate from the extracellular fluid (Option D) . Glutamate accumulates, leading to excitotoxic damage and cell death in the penumbra, the region around the infarcted area.
- Occlusion of a cerebral artery leads to cell death in the ischemic area.
- Surrounding partially ischemic cells may survive but lose their ability to maintain the transmembrane Na⁺ gradient.
- Elevated intracellular Na⁺ levels impair astrocytes' ability to remove glutamate from the extracellular fluid (Option D) .
- Glutamate accumulates, leading to excitotoxic damage and cell death in the penumbra, the region around the infarcted area.
- Occlusion of a cerebral artery leads to cell death in the ischemic area.
- Surrounding partially ischemic cells may survive but lose their ability to maintain the transmembrane Na⁺ gradient.
- Elevated intracellular Na⁺ levels impair astrocytes' ability to remove glutamate from the extracellular fluid (Option D) .
- Glutamate accumulates, leading to excitotoxic damage and cell death in the penumbra, the region around the infarcted area.

Solution for Question 10:

Correct Answer: B) Influx of calcium ions

Explanation:

During an EPSP, calcium ions (Ca²⁺) influx occurs when excitatory neurotransmitters bind to postsynaptic receptors, causing depolarisation and increased neuron excitability.

Excitatory Postsynaptic Potential (EPSP):

- Mechanism: Excitatory neurotransmitters bind to receptors on the postsynaptic membrane, causing the opening of Na^+ or Ca^{2+} channels. This results in the influx of Na^+ or Ca^{2+} into the postsynaptic neuron, leading to depolarization.
- Effect: A single stimulus typically does not produce a propagated action potential in the postsynaptic neuron but results in a transient partial depolarisation called an EPSP.
- Neuron Excitability: During an EPSP, the neuron's excitability to subsequent stimuli increases.
- Propagation: The area of current flow is localised, meaning the EPSP only affects a small region of the membrane and does not depolarise the entire membrane.
- Summation: Multiple EPSPs from various synaptic knobs can summate, leading to significant depolarisation and increased neuronal excitability.

Efflux of sodium ions (Option A): During EPSP, sodium ions flow into the neuron, not out of it. This influx contributes to the depolarisation of the postsynaptic membrane.

Influx of potassium ions (Option C): Potassium ions typically move out of the cell, not into the cell, during an excitatory postsynaptic potential.

Influx of chloride ions (Option D): The influx of chloride ions typically results in hyperpolarisation, producing an inhibitory postsynaptic potential (IPSP), not an excitatory postsynaptic potential (EPSP).

Solution for Question 11:

Correct Answer: C) Receptor 1 and 3

Explanation:

GABA-A and GABA-C are the two subtypes of GABA receptors associated with mediating fast inhibitory postsynaptic potential through the entry of Cl^- ions into the neurons.

Type	Receptor subtype	Subunit	Distribution	Mechanism
Inotropic receptor	GABA- A	Pentamers of various combinations: 6 α , 4 β , 4 γ , 1 δ , 1 ϵ	CNS	Activation allows Cl^- entry into neurons, mediating fast inhibitory

				postsynaptic potentials (IPSP)
GABA- C	Pentamers of three ρ subunits in various combinations	Almost exclusively in the retina.		
G-protein coupled receptor	GABA- B	No specific subunit composition	CNS	Linked via G-proteins : G_i inhibits adenylyl cyclase to open the K^+ channel, G_o inhibits or delays Ca^{2+} influx.

Solution for Question 12:

Correct Answer: B) Astrocytes

Explanation:

Astrocytes are glial cells that play an important role in regulating glutamate levels. They maintain glutamate levels by absorbing excess glutamate and preventing excitotoxicity.

Types of Glial Cells:

Oligodendrocytes (Option D)	Form myelin sheath in CNS neurons.
Astrocytes (Option B)	Support and protect neurons, provide nutrition, and control levels of glutamate. It is of 2 types: Fibrous astrocytes (found in white matter) Protoplasmic astrocytes (found in grey matter)
Schwann cells (Option A)	Form myelin sheath in PNS neurons.

Ependymal cells (Option C): These cells line the ventricles and the central canal of the Spinal cord and play a key role in the production and circulation of CSF.

Solution for Question 13:

Correct Answer: C) A-2, B-3, C-4, D-1

Explanation:

The sites of production of neurotransmitters are as follows:

Neurotransmitters	Source	Description
Acetylcholine (ACh)	Nucleus basalis of Meynert	Associated with Muscle contraction, Memory and REM sleep NT* for Sweat glands
Norepinephrine (NE)	Locus ceruleus	Regulates wakefulness and alertness.
Dopamine	Substantia nigra	Destruction causes Parkinson's disease. NT* of reward and addiction.
Serotonin	Raphe nuclei	Regulation of mood, sleep, and appetite. The highest concentration is in Platelets, enterochromaffin cells, and myenteric plexus.

*NT: Neurotransmitter

Solution for Question 14:

Correct Answer: C) Glycine

Explanation:

At the spinal cord level:

- Glycine acts as a primary inhibitory neurotransmitter.
- It plays a crucial role in regulating motor neuron activity by hyperpolarising postsynaptic neurons, thereby reducing their excitability.
- Dysfunction or deficiency in glycine can lead to increased motor neuron excitability, resulting in symptoms such as muscle spasms and exaggerated reflexes.

In CNS:

- Glycine has both inhibitory and excitatory action in the central nervous system.
- A small amount of extracellular glycine is essential for the proper functioning of the NMDA channel.
- Although glycine is an inhibitory neurotransmitter, it is necessary to activate the NMDA receptor, which plays a role in excitatory neurotransmission.

GABA (Option A) is an inhibitory neurotransmitter primarily found in the brain and less involved in spinal motor neuron regulation than glycine.

Glutamate (Option B) is an excitatory neurotransmitter in the brain that facilitates neuronal signalling.

Acetylcholine (Option D) is involved in neuromuscular transmission at the neuromuscular junction.

Solution for Question 15:

Correct Answer: B) Gamma-aminobutyric acid (GABA)

Explanation:

Gamma-aminobutyric acid (GABA), an inhibitory neurotransmitter, plays a pivotal role in reducing neuronal excitability and promoting relaxation.

Classification of Neurotransmitters based on Functions:

Table rendering failed

Both Excitatory and Inhibitory:

- Acetylcholine (ACh) has excitatory effects in certain areas of the nervous system (e.g., neuromuscular junction) and inhibitory effects (e.g., cardiac autonomic neurons).
- Glycine has Inhibitory effects in the spinal cord and facilitates NMDA receptors in the CNS, indirectly causing Excitatory effects.

Classification Based on Chemical Structure:

Class 1	Choline & Acetyl CoA: Acetylcholine (ACh)
Class 2	Biogenic Amines: Norepinephrine, Dopamine, Serotonin, Histamine

Class 3	Amino Acid Neurotransmitters: Glutamate, GABA, Aspartate, Glycine
Class 4	Purines and Gases: ATP, NO, CO
Peptide Neurotransmitters: Substance P, Neuropeptide Y	

Solution for Question 16:

Correct Answer: B) Tetanus toxin

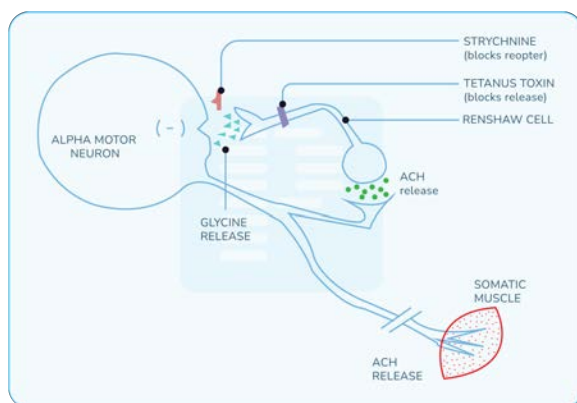
Explanation:

The clinical presentation of severe muscle stiffness, trismus, and muscle spasms following an injury with a rusty nail suggests tetanus, caused by the tetanus toxin (tetanospasmin). Tetanus toxin is produced by *Clostridium tetani* bacteria in soil and enters the body through wounds.

Mechanism of Tetanus Toxin at the synapse:

- Tetanus neurotoxin (TeNT) binds to the presynaptic membrane of the neuromuscular junction, is internalised, and then transported retroaxonally to the spinal cord.
- It induces Spastic paralysis due to the blockade of neurotransmitter releases such as GABA and glycine from presynaptic terminals of spinal inhibitory interneurons, leading to unopposed excitation of motor neurons and muscle rigidity.

The toxin breaks down synaptobrevin protein through its endopeptidase activity.



Botulinum toxins (Option A):

- These are potent neurotoxins produced by *Clostridium botulinum* bacteria.
- There are several types, with types A, B, and E being toxic to humans.
- Mechanism : Botulinum toxins A and E cleave SNAP-25 (synaptosome-associated protein-25), and B cleave Synaptobrevin, which are presynaptic membrane proteins essential for releasing acetylcholine at the neuromuscular junction leading to Flaccid paralysis.

Strychnine (Option C):

- Strychnine is a naturally occurring alkaloid found in the seeds of plant *Strychnos nux-vomica*.
- It acts by blocking glycine receptors in the spinal cord, resulting in exaggerated muscle contractions (spasms) and potentially severe muscle rigidity.

Curare (Option D):

- Curare is a neuromuscular blocking agent that competitively blocks acetylcholine receptors at the neuromuscular junction, leading to muscle paralysis.

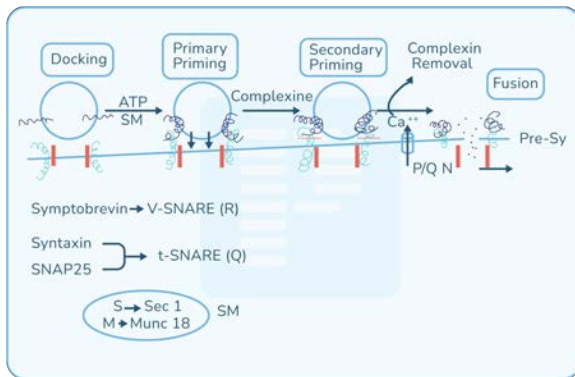
Solution for Question 17:

Correct Answer: D) Complexin

Explanation:

In the presynaptic membrane mechanism, complexin binds to the SNARE complex, preventing premature fusion of synaptic vesicles with the presynaptic membrane until calcium influx triggers its dissociation. This regulatory function ensures that neurotransmitter release occurs at the appropriate time and in response to neuronal activity.

Pre synaptic membrane Mechanism:



- Docking stage : Synaptobrevin (V-SNARE) on synaptic vesicles binds to syntaxin and SNAP-25 (t-SNARE) on the presynaptic membrane. (Option A, B, C)
- ATP and SM protein : ATP hydrolysis provides energy for protein coiling. SM proteins (Sec1/MUNC 18) regulate SNARE protein interactions.
- Priming stages : Primary priming : Protein coils bring vesicles closer to the presynaptic membrane. Secondary priming : Complexin stabilises the SNARE complex, preventing premature fusion.
- Primary priming : Protein coils bring vesicles closer to the presynaptic membrane.
- Secondary priming : Complexin stabilises the SNARE complex, preventing premature fusion.
- Primary priming : Protein coils bring vesicles closer to the presynaptic membrane.
- Secondary priming : Complexin stabilises the SNARE complex, preventing premature fusion.
- Calcium regulation : P/Q and N-type calcium channels open upon stimulation, allowing calcium influx, which removes complexin.
- Fusion stage : Synaptobrevin and syntaxin further coil with SNAP-25, facilitating the fusion of vesicular and presynaptic membranes.
- Release : Fusion leads to neurotransmitter release into the synaptic cleft.

Solution for Question 18:

Correct Answer: A) Small clear vesicles

Explanation:

Small clear vesicles contain glutamate along with other neurotransmitters.

- The vesicles and proteins within them are mostly synthesised in the neuronal cell body and transported by fast axoplasmic transport.
- Small vesicles are located near the synaptic cleft and discharge their contents into active zones.

Types of vesicles	Neurotransmitters
Small clear vesicles	Acetylcholine, GABA, glutamate
Small dense core vesicles (Option C)	Catecholamines(eg:dopamine,epinephrine and norepinephrine)
Large dense core vesicles (Option B)	Neuropeptides

Endosomes (Option D): It plays an important role in endocytosis but has No role in neurotransmitter release.

Solution for Question 19:

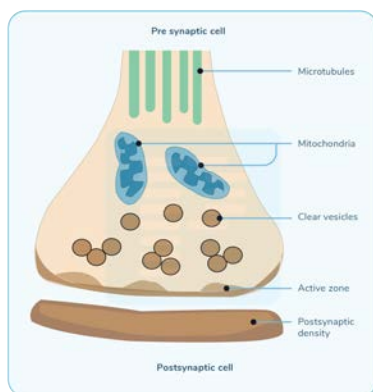
Correct Answer: D) Synaptic specificity

Explanation:

Synaptic specificity refers to the precise targeting of neurons to form connections with the appropriate partner neurons.

- It involves mechanisms that ensure accurate and specific communication between neurons, which is crucial for proper neural network formation and function.
- It is maintained by neurexins and neuroligins interaction, which stabilise synapses.

Neurexins are presynaptic cell adhesion molecules, while neuroligins are located on the postsynaptic membrane.



Active zones (Option B): These are specialised regions in the presynaptic terminal where neurotransmitter release occurs. They are crucial for the release process but do not directly involve the specific targeting or connection of neurons.

Synaptobrevin (Option C): This protein facilitates the release of neurotransmitters from vesicles in the presynaptic terminal by aiding in vesicle fusion. It does not contribute to the specific targeting or connection of neurons.

Postsynaptic density (Option D): This region on the postsynaptic membrane contains receptors and signalling proteins crucial for organising postsynaptic responses. It does not address neuronal targeting or synaptic connection precision.

Solution for Question 20:

Correct Answer: D) Electrical synapses

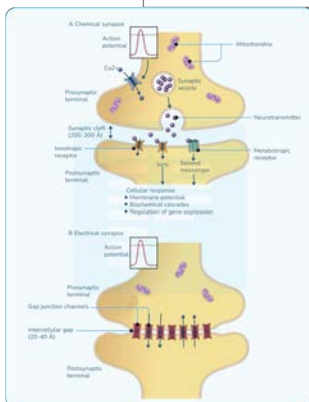
Explanation:

Electrical synapses are rapid and bidirectional due to 'Gap junctions'.

Types of Synapses:

Aspects	Electrical synapse	Chemical synapse (Option A)
Occurrence	Rare	Most common
Width of synaptic cleft	Narrow	Wide
Neurotransmitters	Absent	Present

Gap junctions	Present	Absent
Synaptic transmission	No time delay	Time delay is present
Transmission direction	Bidirectional	Unidirectional
Component	Pre-synaptic and Post-synaptic membranes (connected by gap junctions)	Pre-synaptic cleft and post-synaptic cleft
Types	-	3 types: Axodendritic (Option C) Axoaxonal (Option B) Axosomatic



Solution for Question 21:

Correct Answer: B) Oligodendrocytes

Explanation:

- Multiple sclerosis (MS) is a central nervous system (CNS) disorder characterised by the degeneration of oligodendrocytes, which are responsible for forming myelin sheaths around nerve fibres.
- Loss of myelin leads to leakage of K^+ through voltage-gated channels, which in turn leads to hyperpolarisation or abnormal depolarisation, impaired nerve signal conduction, and symptoms such as paraparesis, paresthesia, optic neuritis, dysarthria, and dysphagia.

Myelination of CNS and PNS

Characteristics	Peripheral nervous system	Central nervous system
Cells involved in Myelination	One Schwann cell for one neuron.	One Oligodendrocyte for multiple neurons.
Demyelinating disease	Localised disease (e.g., Guillain-Barré Syndrome)	Diffuse disease (e.g., Multiple Sclerosis)
Covering of unmyelinated neuron	Surrounded by the Schwann cell process.	Bare (no myelin sheath)

Microglia (Option A) are non-neural cells that act like scavenger cells and remove debris formed by infection, injury, and disease; they have no role in myelination.

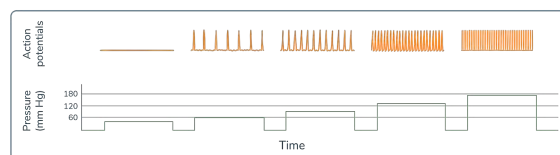
Astrocytes (Option D) are supporting cells that provide nutrition and maintain glutamate levels but don't have a role in myelin sheath formation.

Somatosensory Receptors, Nociceptors, Proprioceptors, and Reflexes

1. A 30-year-old patient reports that rubbing the injured area provides him temporary relief from pain. According to the gate-control theory of pain, what is the primary mechanism behind this relief?

- A. Increased activation of nociceptive C fibers to reduce pain perception
- B. Simultaneous activation of innocuous cutaneous mechanoreceptors
- C. $A\alpha$ and $A\beta$ fibers inhibit pain signals from C fibers in the ventral horn of spinal cord
- D. Inhibition of $A\alpha$ and $A\beta$ fibers reduces the transmission of pain signals to the brain

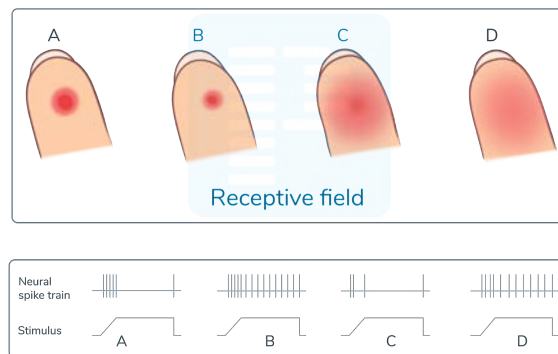
2. Action potentials in an afferent fiber from a mechanoreceptor of a single sensory unit is recorded in response to pressure stimulus. Which of the following principle is shown here:



- A. Labelled line principle
- B. Weber-Fechner Law
- C. Frequency coding principle
- D. Concept of specific nerve energy

3. The receptive fields (shown as red areas on fingertips) and firing rate of four different types of mechanoreceptors supplied by FA1, FA2, SA1 and SA2 afferent

are shown here. The figure marked with B is:



- A. Pacinian corpuscle
- B. Meissner's corpuscle
- C. Merkel's disc
- D. Ruffini's ending

4. A patient with paraplegia uses thigh stimulation to manage bladder and bowel control, leading to involuntary evacuation. Which reflex is being utilized?

- A. Righting reflex
- B. Mass reflex
- C. Hopping reflex
- D. Placing reflex

5. A 20 years old man is brought to the emergency room after stepping on a nail while walking barefoot. Upon examination, the following observations are made, Which of the following best describes the reflex pathways involved in this scenario? Right foot: Immediately withdraws upon being touched. Left foot: Planted firmly on the ground.

- A. Monosynaptic stretch reflex Polysynaptic crossed extensor reflex

- B. Polysynaptic flexor reflex Monosynaptic stretch reflex
 - C. Polysynaptic flexor reflex Polysynaptic crossed extensor reflex
 - D. Polysynaptic flexor reflex Monosynaptic stretch reflex
-

6. A 30 years old woman presents with a history of multiple sclerosis. Upon examination, she exhibits ankle clonus when the examiner briskly dorsiflexes her foot. What is the most likely cause of clonus in this patient?

- A. Hypoactive stretch reflexes
 - B. Increased GABAergic activity
 - C. Hyperactive muscle spindles
 - D. Enhanced Golgi tendon organ activation
-

7. A 35 year old weightlifter is lifting a heavier weight than usual and feels a sudden, intense muscle contraction in his biceps, the muscle suddenly relaxes, preventing a potential muscle tear. Which reflex is responsible for this protective muscle relaxation?

- A. Inverse stretch reflex
 - B. Withdrawal reflex
 - C. Stretch reflex
 - D. Crossed extensor reflex
-

8. A 45 year old weightlifter is attempting a deadlift. As he lifts the weight, he feels a sudden, intense pain in his hamstring muscle and immediately drops the weight. What is the role of the Golgi tendon organ in this scenario?

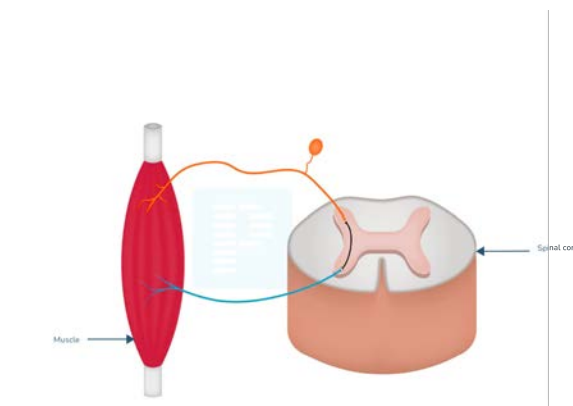
- A. To directly cause muscle contraction

- B. To prevent muscle damage by causing reflex inhibition of the contracting muscle.
 - C. To increase the sensitivity of muscle spindles
 - D. To send information about muscle length to the brain
-

9. A 28 year old man presents with experience of prolonged episodes of tingling and numbness in his right arm. A reflex hammer test on the patient's patellar tendon noted a weaker than normal knee-jerk reflex and decision to use the Jendrassik maneuver to enhance the reflex response was taken. Which of the following is directly activated by the Jendrassik maneuver?

- A. α -motor neuron
 - B. γ -motor neuron
 - C. Golgi tendon organ
 - D. Extrafusal muscle fiber
-

10. Which of the following best describes the sequence of events in the stretch reflex pathway?



- A. Spinal cord → Muscle spindle → α -motor neuron → Muscle contraction
- B. Muscle spindle → Spinal cord → α -motor neuron → Muscle contraction

- C. Spinal cord → α -motor neuron → Muscle spindle → Muscle contraction
 - D. Muscle spindle → α -motor neuron → Spinal cord → Muscle contraction
-

11. A 45 year old man presents with complaints of muscle weakness and difficulty maintaining balance. Upon examination, the physician notices that the patient has diminished reflexes. Which of the following nerve fibers is responsible for transmitting information about the velocity and length of muscle stretch from the muscle spindle to the spinal cord?

- A. Type Ib fibers
 - B. Type II fibers
 - C. $A\gamma$ fibers
 - D. Type Ia fibers
-

12. A 32 years old man came for an evaluation of muscle spasms and difficulty with fine motor tasks. Upon examination, the patient has hyperactive deep tendon reflexes and experiences a persistent tremor during voluntary movements. Which of the following types of intrafusal muscle fibers, would most likely cause this hyperactive deep tendon reflex?

- A. Dynamic nuclear bag fibers
 - B. Static nuclear bag fibers
 - C. Nuclear chain fibers
 - D. Extrafusal fibers
-

13. The muscle spindle is primarily involved in detecting changes in?

- A. Muscle tension

- B. Muscle length
 - C. Skin stretch
 - D. Joint pressure
-

14. A 35 y/o presented after a car accident and complained of severe pain. They administered morphine, knowing it acts on the descending pain modulation system. Which of the following brain regions and neurotransmitters are correctly paired and activated by morphine?

- A. Periaqueductal grey (PAG) neurons - endorphins in the dorsal horn of the spinal cord.
 - B. Periventricular nucleus neurons - norepinephrine in the dorsal horn of the spinal cord.
 - C. Periaqueductal grey (PAG) neurons - directly release serotonin in the nucleus raphe magnus.
 - D. Periventricular nucleus neurons - enkephalins in the rostral ventromedial medulla.
-

15. A researcher is studying the sensory perception abilities of different body parts in humans. She observes that some areas of the body can detect finer details and small differences in stimuli better than others. Which of the following statements best explains the reason behind the high sensitivity and discriminatory abilities in certain body areas?

- A. Small receptive fields and low receptor density.
 - B. Large receptive fields and high receptor density.
 - C. Small receptive fields and high receptor density.
 - D. Large receptive fields and low receptor density.
-

16. A 55-year-old patient develops a persistent burning pain in the left arm following a thalamic stroke. Which of the following statements about thalamic pain syndrome is false?

- A. Due to damage to the spinothalamic tract
 - B. Can result from both ischemic and hemorrhagic strokes
 - C. Typically short-lived and resolves within a few months
 - D. Antidepressants and anticonvulsants are commonly used for treatment
-

17. Which statement correctly explains the localization capability of the paleospinothalamic pathway for pain?

- A. The pathway allows precise localization of pain to specific points on the body
 - B. The pathway localizes pain only to major body parts, such as an arm or leg, due to its diffuse connectivity.
 - C. The pathway is responsible for localizing temperature changes in the body.
 - D. The pathway can localize pain to exact muscle groups involved in movement
-

18. Identify the correct statement about the neospinothalamic tract for pain:

- A. It primarily carries slow pain sensations
 - B. The first-order neurons terminate mainly in lamina II of the dorsal horn
 - C. The neurotransmitter released by the A δ fiber is glutamate
 - D. All fibers terminate exclusively in the reticular formation
-

19. A patient reports severe pain when lightly touched with a cotton swab. This symptom is most consistent with?

- A. Allodynia
 - B. Hyperalgesia
 - C. Analgesia
 - D. Hypoalgesia
-

20. A 40 years old woman comes to the clinic complaining of severe, burning pain in her right hand following a chemical spill causing tissue damage at her workplace. Which of the following substances is responsible for her pain?

- A. Dopamine
 - B. Acetylcholine
 - C. Serotonin
 - D. Bradykinin
-

21. The TRPA1 receptors are activated by all of the following nociceptive stimuli except?

- A. Mechanical stimuli
 - B. Heat
 - C. Cold
 - D. Chemical stimuli
-

22. Nociceptors have their cell bodies located in which of the following structures?

- A. Dorsal root ganglion
- B. Ventral horn of the spinal cord
- C. Lateral geniculate nucleus

D. Sympathetic ganglion

23. Which neurotransmitter is primarily involved in fast pain?

- A. Substance P
 - B. Glutamate
 - C. GABA
 - D. Serotonin
-

24. Which type of receptor is known for its rapid adaptation to stimuli, making it particularly effective at detecting vibrations and rapid changes in pressure?

- A. Merkel cells
 - B. Meissner's corpuscles
 - C. Ruffini endings
 - D. Pacinian corpuscles
-

25. A researcher is studying various sensory receptors and their responses to different stimuli. During an experiment with pacinian corpuscle, the researcher notes that the intensity of the initial electrical response in the receptor is proportionate with the mechanical pressure applied. What is the initial graded electrical response observed in sensory receptors upon stimulus application?

- A. Action potential
- B. Receptor potential
- C. Synaptic potential
- D. Resting potential

26. Which type of mechanoreceptor is primarily responsible for two-point discrimination and the ability to read Braille by detecting fine touch and texture details on the fingertips?

- A. Free Nerve Ending
- B. Hair End Organ
- C. Iggo Dome Receptor
- D. Ruffini Corpuscle

27. Which type of receptor is most likely responsible for detecting the rapid changes in pressure and vibrations in the patient's fingertips?

- A. Meissner Corpuscles
- B. Pacinian Corpuscles
- C. Hair End Organ
- D. Warmth Receptors

28. A patient reports an unusual sensitivity to temperature changes, and you suspect a disorder related to thermoreceptor function. Which of the following statements accurately describes the nerve fibers involved in transmitting signals from thermoreceptors?

- A. Cold receptors: myelinated type C fibers
- B. Warmth receptors: myelinated type A δ fibers
- C. Cold receptors: myelinated type A δ fibers
- D. Warmth receptors: myelinated type A β fibers

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	3
Question 4	2
Question 5	3
Question 6	3
Question 7	1
Question 8	2
Question 9	2
Question 10	2
Question 11	4
Question 12	1
Question 13	2
Question 14	1
Question 15	3
Question 16	3
Question 17	2
Question 18	3
Question 19	1
Question 20	4
Question 21	2
Question 22	1

Question 23	2
Question 24	4
Question 25	2
Question 26	3
Question 27	2
Question 28	3

Solution for Question 1:

Correct Answer: B) Simultaneous activation of innocuous cutaneous mechanoreceptors

Explanation:

Rubbing or shaking an injured area decreases the pain due to the injury. This is due to simultaneous activation of innocuous cutaneous mechanoreceptors thereby reducing pain by affecting dorsal horn neuron responsiveness. This is called the gate-control mechanism of pain modulation.

Gate control theory: Developed By Patrick David Wall and Ronald Melzack

- It suggests that pain perception is influenced by the interaction of different nerve fibers in the spinal cord.
- Large-diameter $A\alpha$ and $A\beta$ fibers inhibit pain signals from C fibers in the dorsal horn of spinal cord by activating inhibitory interneurons, effectively "closing the gate". (Option C, D ruled out)
- When small-diameter fibers are more active, they "open the gate," allowing pain signals to reach the brain.

Influencing Factors:

- Psychological factors, including cognitive and emotional states affect gate control.
- Depression can keep the gate open more often, increasing pain perception
- Unhealthy lifestyles can also result in open gates.

Application:

- According to the gate-control theory, electrical stimulation of large nerve fibers should inhibit pain induced by C fibers alone.

- Transcutaneous Electrical Nerve Stimulation (TENS) uses electrodes to activate Aα and Aβ fibers near the injury, thereby using the gate-control mechanism for pain relief.
- Acupuncture.

Analgesic systems:

Table rendering failed

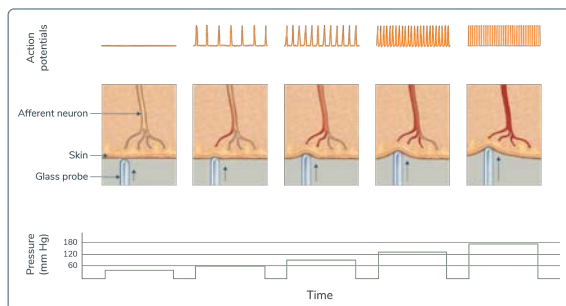
Increased activation of nociceptive C fibres to reduce pain perception (Option A) is not correct as increased activation of nociceptive C fibres would actually enhance pain perception, not reduce it.

Solution for Question 2:

Correct answer: C) Frequency coding principle

Explanation:

The principle shown here is the Frequency Coding Principle.



- The Frequency Coding Principle states that the intensity of a stimulus is encoded by the frequency of action potentials generated in the afferent fibers.
- As pressure increases, the firing rate of action potentials in the mechanoreceptor increases, conveying information about the strength of the stimulus
- Action potentials in an afferent fiber from a mechanoreceptor of a single sensory unit increase in frequency as branches of the afferent neuron are stimulated by pressure of increasing magnitude.

Labelled Line Principle (Option A) refers to the idea that specific pathways in the nervous system convey specific types of sensory information.

Weber-Fechner Law (Option B) relates to the perception of changes in stimulus intensity and the relationship between the actual change and perceived change.

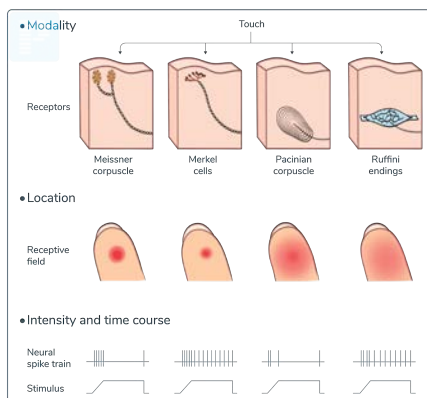
Concept of Specific Nerve Energy (Option D) suggests that each type of sensory nerve fiber is dedicated to a specific type of sensory information, regardless of how it is activated.

Solution for Question 3:

Correct answer: C) Merkel's disc

Explanation:

The figure marked with B represents Merkel's disc



The human hand contains four types of mechanoreceptors that work together to sense contact with objects:

- Merkel cells and Ruffini endings: Activate for steady pressure.
- Meissner and Pacinian corpuscles: Activate for tingling and vibratory sensations.

Location of a stimulus is encoded by spatial distribution of the population of receptors activated. A receptor fires only when the skin close to its sensory terminals is touched. The receptive fields of mechanoreceptors (shown as red areas on fingertips) differ in size and response to touch.

- Merkel cells and Meissner corpuscles provide the most precise localization as they have the smallest receptive fields and are most sensitive to pressure applied by a small probe.

Stimulus intensity is signaled by firing rates of individual receptors; duration of stimulus is signaled by time course of firing. The spike trains indicate action potentials elicited by pressure

from a small probe at the center of each receptive field.

- Meissner and Pacinian corpuscles adapt rapidly, the others adapt slowly

Accommodation: Refers to a gradual increase in threshold caused by prolonged slowly rising sub-threshold nerve depolarization, resulting from the inactivation of sodium channels.

Adaptation: If a supra-threshold stimulus persists unchanged for several minutes without a change in position or amplitude, the neural response diminishes and sensation is lost, a condition called receptor adaptation.

Solution for Question 4:

Correct Answer: B) Mass reflex

Explanation:

Mass reflex is due to excessive activity in the spinal cord causing massive discharge in large portions of the cord. Mass reflex can sometimes be used to give patients with paraplegia a degree of bladder and bowel control. They can be trained to initiate urination and defecation by stroking or pinching their thighs, thus producing an intentional mass reflex.

Reflex	Description	Key Points
Mass Reflex	Excessive activity in the spinal cord causing massive discharge in large portions of the cord. (Option B)	Results from activation of large numbers of reverberating circuits in the cord at once. Can last for minutes Stimulus: strong pain stimulus or excessive filling of a viscus (e.g., bladder or gut). Effects: Strong flexor spasm in skeletal muscles, evacuation of colon and bladder, increased arterial pressure, profuse sweating.

Righting Reflex	Reflex involving uncoordinated movements to raise the body to a standing position when laid on the side. (Option A ruled out)	Demonstrates complex reflexes associated with posture integrated in the spinal cord. Animals with a healed transected thoracic cord can right itself from lying position and even walk using both forelimbs and hindlimbs. Consists of: Labyrinthine Righting Reflex, Neck Righting Reflex, Body Righting Reflex, Limb Righting Reflex, Optical Righting Reflex.
Hopping Reflex	Short steps taken by an individual to balance the body when pushed laterally. (Option C ruled out)	Helps maintain balance and stability after lateral push. Impaired following decortication. Indicates that motor cortex is the center for integration of hopping reactions.
Placing Reflex	Placing feet firmly on the ground to attain and maintain a new stable position after lateral push. (Option D ruled out)	Ensures stability after the initial balance provided by the hopping reaction. Impaired following decortication. Indicates that motor cortex is the center for integration of placing reactions.

Solution for Question 5:

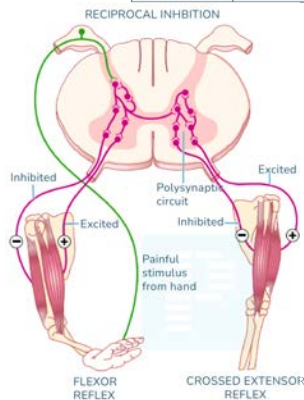
Correct Answer: C) Right - Polysynaptic flexor reflex; Left - Polysynaptic crossed extensor reflex.

Explanation:

- Right foot withdrawal: Stepping on a nail triggers the withdrawal reflex, also known as the flexor reflex, which is polysynaptic.
- Left foot response: The withdrawal reflex also involves a crossed extensor reflex which is polysynaptic reflex.

Reflex	Withdrawal Reflex	Flexor Reflex	Crossed Extensor Reflex
Description	A protective reflex that causes withdrawal of a limb or body part from a stimulus.	A withdrawal reflex in which the primary response is flexion of a limb.	A component of the withdrawal reflex that occurs in the opposite limb.

Receptor	Nociceptors (e.g., pain receptors)	Nociceptors	Nociceptors in the contralateral limb are indirectly activated.
Afferent Fiber	Group III and IV (pain) and Group II (pressure)	Group III and IV (pain) and Group II (pressure)	Signals are relayed from the stimulated side.
Synapses	Polysynaptic (Options A, D ruled out)	Polysynaptic	Polysynaptic (Option B ruled out)
Efferent Fiber	α -motor neurons to flexors	α -motor neurons to flexors	α -motor neurons to extensors
Response	Flexion of ipsilateral limb or body part to move it away from the stimulus.	Contraction of flexor muscles and relaxation of extensor muscles in the stimulated limb.	Extension of the contralateral limb.
Clinical Significance	It is a protective mechanism, absence or weakness of this can indicate spinal cord injury or dysfunction.	Part of the withdrawal reflex and essential for rapid response to painful stimuli.	Provides support and balance during withdrawal of the opposite limb.



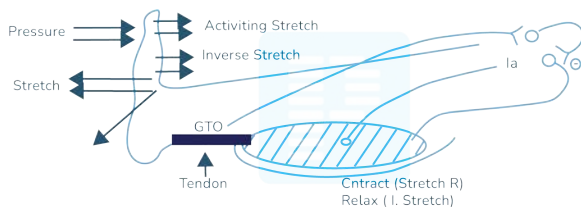
Solution for Question 6:

Correct Answer: C) Hyperactive muscle spindles

Explanation:

- The patient's history of multiple sclerosis and the presence of ankle clonus strongly suggest damage to the descending motor pathways.
- This damage leads to hyperactive muscle spindles and, consequently, clonus.
- Clonus is characterized by rhythmic muscle contractions in response to sustained stretch.

Table rendering failed



Increased GABAergic activity (Option B) in the spinal cord would actually reduce clonus as GABA is an inhibitory neurotransmitter.

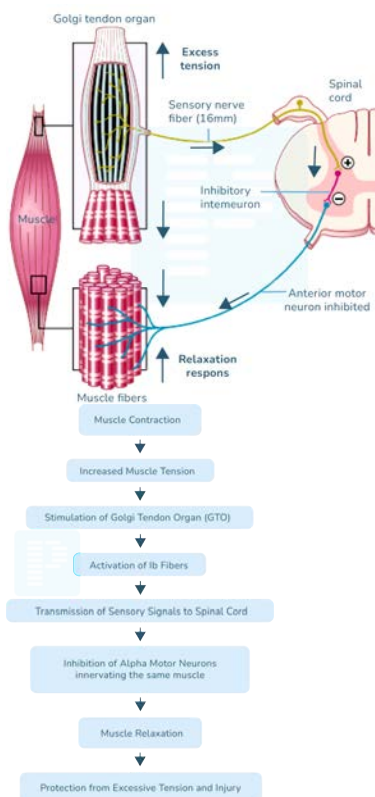
Enhanced Golgi tendon organ activation (Option D): Golgi tendon organs (GTOs) are involved in the inverse stretch reflex, which causes muscle relaxation in response to increased tension.

Solution for Question 7:

Correct Answer: A) Inverse stretch reflex

Explanation:

- Inverse Stretch Reflex is activated when there is excessive tension in a muscle, as detected by the Golgi tendon organ (GTO).
- When the tension becomes too high, the GTO sends signals to the spinal cord, which in turn inhibits the alpha motor neurons innervating the same muscle, leading to muscle relaxation.



Feature	Stretch Reflex (Option C ruled out)	Inverse Stretch Reflex
Receptor	Muscle Spindle	Golgi Tendon Organ
Stimulus	Muscle Stretch	Muscle Tension
Response	Muscle Contraction	Muscle Relaxation
Function	Maintain Muscle Length, Posture	Protect from Excessive Force
Type of Reflex	Monosynaptic	Polysynaptic

Withdrawal Reflex (Option B) is a protective mechanism that causes a body part to automatically move away from a source of pain or discomfort.

Crossed extensor reflex (Option D) causes the opposite limb to extend when a withdrawal reflex is initiated in one limb.

Solution for Question 8:

Correct Answer: B) To prevent muscle damage by causing reflex inhibition of the contracting muscle.

Explanation:

The Golgi tendon organ, when activated by excessive muscle tension, triggers a protective reflex that inhibits the contracting muscle, preventing tendon rupture or muscle tearing.

Golgi tendon organ	
Location and Structure	Muscle tendons. Encapsulated sensory receptors. Around 10-15 muscle fibers are connected.
Stimulus	Tension in the muscle, as reflected by tension in the tendon. Sudden, strong muscle contractions increase tension in the tendon
Function	Control and prevent excessive muscle tension. Operates as a negative feedback system.
Reflex Pathway	When stimulated, signals are transmitted to the spinal cord, triggering the inverse stretch reflex (also called autogenic inhibition). The signal activates an inhibitory interneuron in the spinal cord. The interneuron inhibits the anterior motor neuron of the respective muscle, causing relaxation, this is a bisynaptic pathway. This protects the muscle and tendon from excessive tension that might cause tearing.
Afferent Fiber	Golgi tendon organ to the spinal cord via large, fast-conducting type Ib nerve fibers.
Vs Muscle spindle	Muscle spindle - muscle length. (Option D ruled out) Golgi tendon organ - muscle tension.

To directly cause muscle contraction (Option A) : Alpha motor neurons are responsible for muscle contraction

To increase the sensitivity of muscle spindles (Option C): Gamma motor neurons are responsible for regulating the sensitivity of muscle spindles

Solution for Question 9:

Correct Answer: B) γ -motor neuron

Explanation:

- The Jendrassik maneuver, which can involve actions such as clenching the teeth or pulling on the hands, activates γ -motor neurons.
- γ -motor neurons, also called gamma efferent fibers, innervate intrafusal muscle fibers.

Muscle contraction		
Feature	Direct	Indirect
Definition	Occurs when α -motor neurons are directly stimulated.	Occurs when the muscle spindle is activated, leading to the activation of α -motor neurons via the stretch reflex pathway.
Pathway	Monosynaptic	Monosynaptic
Purpose	Voluntary movements	Involuntary, provides damping to smooth muscle contractions.
Example	Picking up an object.	Maintaining posture, reacting to unexpected changes in load on a muscle (e.g., catching a ball).

Jendrassik Maneuver:

- Clinical technique used to enhance the sensitivity of the stretch reflex (and thus, the response of the muscle spindle).
- The maneuver involves the patient performing an isometric contraction in a muscle group distant from the one being tested. Clenching the teeth Locking the fingers and pulling the hands apart Squeezing the knees together.
- Clenching the teeth
- Locking the fingers and pulling the hands apart
- Squeezing the knees together.
- The maneuver is thought to increase γ -motor neuron activity, making the muscle spindles more sensitive to stretch.
- Clenching the teeth
- Locking the fingers and pulling the hands apart
- Squeezing the knees together.

Alpha-Gamma Coactivation:

- On voluntary muscle contraction, the brain often signals both α - and γ -motor neurons simultaneously. This is known as alpha-gamma coactivation.
- Activating γ -motor neurons causes intrafusal fibers to contract, maintaining tension in the spindle ensuring that spindle remains sensitive to stretch even during voluntary contractions.
- Example: When catching a ball, alpha-gamma coactivation keeps the muscle spindles responsive and allows for rapid adjustments to maintain grip and arm position as the weight of the ball impacts the hand.

α -motor neuron (Option A): While the knee-jerk reflex does involve the activation of α -motor neurons to cause the quadriceps muscle to contract, the Jendrassik maneuver does not directly activate these neurons.

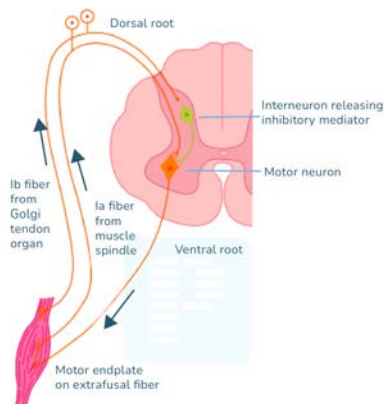
Golgi tendon organ (Option C): the sensitivity of the Golgi tendon organ is not directly affected by the Jendrassik maneuver.

Extrafusal muscle fibers (Option D) are not directly activated by the Jendrassik maneuver.

Solution for Question 10:

Correct Answer: B) Muscle spindle $\rightarrow$ Spinal cord $\rightarrow$ α -motor neuron $\rightarrow$ Muscle contraction

Explanation:



Stretch reflex pathway: Regulates muscle length and contributes to postural control. It is initiated by the stretching of muscle, which activates specialised sensory receptors.

- Muscle Spindles: Consists of intrafusal muscle fibers (nuclear bag and nuclear chain fibers), Sensory nerve endings (group Ia and II afferent fibers), and Gamma motor neuron

endings

- Consists of intrafusal muscle fibers (nuclear bag and nuclear chain fibers),
- Sensory nerve endings (group Ia and II afferent fibers), and
- Gamma motor neuron endings
- Afferent Limb: When a muscle is stretched, the muscle spindles are also stretched, stimulating the sensory endings. This stretch-induced stimulation generates receptor potentials that trigger action potentials in the afferent fibers.
- When a muscle is stretched, the muscle spindles are also stretched, stimulating the sensory endings.
- This stretch-induced stimulation generates receptor potentials that trigger action potentials in the afferent fibers.
- Integration Center: The Ia afferent fibers, have a large diameter and, conduct action potentials, transmit signals to the spinal cord through the dorsal roots. Within the spinal cord, these fibers form direct (monosynaptic) connections with alpha motor neurons located in the anterior horn.
- The Ia afferent fibers, have a large diameter and, conduct action potentials, transmit signals to the spinal cord through the dorsal roots.
- Within the spinal cord, these fibers form direct (monosynaptic) connections with alpha motor neurons located in the anterior horn.
- Efferent Limb: The activated alpha motor neurons send signals through axons, which exit the spinal cord and travel to the extrafusal muscle fibers.
- The activated alpha motor neurons send signals through axons, which exit the spinal cord and travel to the extrafusal muscle fibers.
- Effector: The arrival of action potentials at the neuromuscular junctions causes the release of acetylcholine, initiating muscle fiber contraction. This counteracts the initial stretch by shortening the muscle, thus restoring its original length.
- The arrival of action potentials at the neuromuscular junctions causes the release of acetylcholine, initiating muscle fiber contraction.
- This counteracts the initial stretch by shortening the muscle, thus restoring its original length.
- Consists of intrafusal muscle fibers (nuclear bag and nuclear chain fibers),
- Sensory nerve endings (group Ia and II afferent fibers), and
- Gamma motor neuron endings
- When a muscle is stretched, the muscle spindles are also stretched, stimulating the sensory endings.

- This stretch-induced stimulation generates receptor potentials that trigger action potentials in the afferent fibers.
- The Ia afferent fibers, have a large diameter and, conduct action potentials, transmit signals to the spinal cord through the dorsal roots.
- Within the spinal cord, these fibers form direct (monosynaptic) connections with alpha motor neurons located in the anterior horn.
- The activated alpha motor neurons send signals through axons, which exit the spinal cord and travel to the extrafusal muscle fibers.
- The arrival of action potentials at the neuromuscular junctions causes the release of acetylcholine, initiating muscle fiber contraction.
- This counteracts the initial stretch by shortening the muscle, thus restoring its original length.

Solution for Question 11:

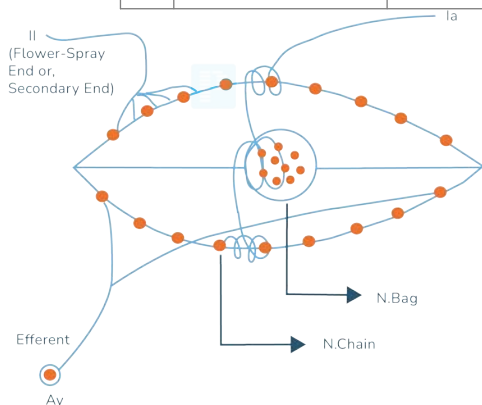
Correct Answer: D) Type Ia fibers

Explanation:

- Type Ia fibers are large, rapidly conducting sensory fibers that originate from the central portion of both nuclear bag and nuclear chain fibers within the muscle spindle.
- These fibers are responsible for transmitting information about the velocity and length of muscle stretch to the spinal cord.

	Afferent (Sensory)	Efferent (Motor)			
Fiber Type	Type Ia fibers	Type II fibers	A γ fibers	γ -dynamic (γ -d) fibers	γ -static (γ -s) fibers
Description	Large, rapidly conducting fibers	Smaller sensory fibers that innervate nuclear chain fibers.	Smaller motor fibers that innervate the intrafusal fibers.	Excite nuclear bag intrafusal fibers.	Excite nuclear chain intrafusal fibers.

ti o n					
F u n c t i o n	Transmit information about velocity and length of muscle stretch.	Transmit information about static length of the muscle. (Option B ruled out)	Adjusts sensitivity of the muscle spindle. (Option C ruled out)	Enhance the dynamic response (sensitivity to the velocity of muscle stretch).	Enhance the static response (sensitivity to steady-state muscle length).
O r i g i n	Annulospiral (primary) endings wrapped around the center of dynamic and static nuclear bag fibers and nuclear chain fibers.	Secondary endings located adjacent to the centers of static nuclear bag and nuclear chain fibers.	Gamma motor neurons in the anterior horns of the spinal cord.		



Afferent and Efferent Fibres" data-author="NA" data-hash="7851" data-license="NA" data-source="NA" data-tags="April2025" height="367" src="https://image.prepladder.com/notes/fSsUUWJu7ODggEJfqSwc1744010595.png" width="500" />

Afferent Connections of the Muscle Spindle Efferent Connections of the Muscle Spindle

Type Ib fibers (Option A) are large and rapidly conducting fibers, that originate from the Golgi tendon organs and transmit information about muscle tension, not muscle length.

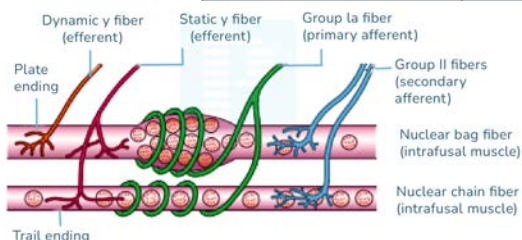
Solution for Question 12:

Correct Answer: A) Dynamic nuclear bag fibers

Explanation:

- Dynamic nuclear bag fibers are highly sensitive to the velocity of muscle stretch.
- When a muscle is stretched rapidly, these fibers fire strongly, leading to a robust activation of the stretch reflex, which causes exaggerated stretch reflexes.

Types of intrafusal fibers	
Nuclear Bag Fiber	Respond to changes in muscle length (stretch). Two subtypes: dynamic and static.
Dynamic Nuclear Bag Fiber	Detects real-time shortening/lengthening of a muscle (i.e., the velocity).
Static Nuclear Bag Fiber	Detects sustained/maintained length of a muscle but plays no role in detecting velocity. (Option B ruled out)
Nuclear Chain Fiber	Thinner and shorter than nuclear bag fibers and lack a definite bag. Static response of the muscle spindle, providing information about the steady-state muscle length. (Option C ruled out)



Extrafusal fibers (Option D) are regular contractile fibers of skeletal muscle responsible for generating force and movement. They are not a part of the muscle spindle and do not directly contribute to the stretch reflex.

Solution for Question 13:

Correct Answer: B) Muscle length

Explanation:

- Muscle spindles are sensory receptors found within skeletal muscle.
- The function of a muscle spindle is to detect changes in muscle length, also called stretch.

Muscle spindle receptor:

Muscle Spindle Receptor	
Location	Encapsulated sensory receptors in skeletal muscle, attached to tendons.
Components	Each spindle contains 3 to 12 intrafusal muscle fibers.
Sensory Endings	Stimulated by stretching.
Types of Sensory Endings	Primary Afferent Endings (Annulospiral) : Sensitive to velocity of muscle length change (dynamic response). Secondary Afferent Endings (Flower-Spray) : Sensitive to steady-state muscle length (static response).
Types of Gamma Motor Neurons	Gamma-Dynamic (γ -d) : Excites nuclear bag intrafusal fibers. Gamma-Static (γ -s) : Excites nuclear chain intrafusal fibers.
Functions	Regulating Muscle Length : Provides feedback to regulate muscle length. Damping : Smoothens muscle contractions and prevents jerkiness. Stabilizing Body Position : Maintains posture and balance by informing the spinal cord and motor centers about muscle length. Higher Motor Control : Information is transmitted to the cerebellum and cerebral cortex.

Muscle Spindle Receptor" data-author="NA" data-hash="7857" data-license="NA" data-source="NA" data-tags="April2025" height="307" src="https://image.prepladder.com/notes/llwELzoimQh1yDiUZ2681744015307.png" width="500" />

Muscle Tension (Option A): Golgi tendon organs are the receptors responsible for detecting muscle tension.

Skin stretch (Option C) is detected by cutaneous mechanoreceptors, such as Ruffini Corpuscles.

Joint Pressure (Option D): Joint capsule receptors are responsible for detecting joint pressure.

Solution for Question 14:

Correct Answer: A) Periaqueductal grey (PAG) neurons - endorphins in the dorsal horn of the spinal cord.

Explanation:

- The periaqueductal grey (PAG) matter region in the midbrain plays a crucial role in the descending pain modulation pathway.
- When stimulated by opioids like morphine, PAG neurons activate neurons in the brainstem, the nucleus raphe magnus (which releases serotonin) and the rostral ventromedial medulla (which releases norepinephrine).
- These neurons project to the dorsal horn, where they release their respective neurotransmitters to inhibit the activity of pain-transmitting neurons.

Brain Region	Neurotransmitter Released	Site	Effect
Periaqueductal grey (PAG)	Endorphins	Dorsal horn of the spinal cord	Inhibits pain transmission by activating enkephalin-containing interneurons.
Nucleus raphe magnus	Activates neurons that release serotonin.		
Rostral ventromedial medulla	Activates neurons that release norepinephrine.		
Periventricular region (PV)	Endorphins	Raphe Magnus nucleus	Activates neurons that release serotonin. (Option B and D ruled out)

Periventricular nucleus neurons - enkephalins in the rostral ventromedial medulla (Option C) is not true as PAG does not directly release serotonin; it activates serotonergic neurons in the nucleus raphe magnus.

Solution for Question 15:

Correct Answer: C) Small receptive fields, high receptor density.

Explanation:

Small receptive fields coupled with a high receptor density allow for precise localisation and detailed discrimination of sensory stimuli, making these areas highly sensitive.

Classification of Sensory	Function	Type	Specific Receptor
---------------------------	----------	------	-------------------

Receptors			
Mechanoreceptors	Detect mechanical compression or stretching of the receptor or adjacent tissues	Skin Tactile Sensibilities (epidermis and dermis)	Free nerve endings, Expanded tip endings (Merkel's discs) Spray endings, Ruffini's endings, Encapsulated endings (Meissner's corpuscles, Krause's corpuscles) Hair end-organs
Deep Tissue Sensibilities	Free nerve endings, Expanded tip endings, Spray endings, Ruffini's endings, Encapsulated endings (Pacinian corpuscles), Muscle endings (Muscle spindles, Golgi tendon receptors)		
Hearing	Sound receptors of cochlea		
Equilibrium	Vestibular receptors		
Arterial Pressure	Baroreceptors of carotid sinuses and aorta		
Thermoreceptors	Detect changes in temperature (cold or warmth)	Cold	Cold receptors
Warmth	Warm receptors		
Nociceptors (pain receptors)	Detect physical or chemical damage in the tissue	Pain	Free nerve endings
Electromagnetic Receptors	Detect light on the retina of the eye	Vision	Rods Cones
Chemoreceptors	Detect various chemical changes including taste, smell, blood oxygen levels, osmolality, and CO ₂ concentration	Taste	Receptors of taste buds
Smell	Receptors of olfactory epithelium		
Arterial Oxygen	Receptors of aortic and carotid bodies		

Osmolality	Neurons in or near supraoptic nuclei		
Blood CO ₂	Receptors in/on the surface of the medulla, aortic and carotid bodies		
Blood Glucose, Amino Acids, Fatty Acids	Receptors in hypothalamus		

Location	Receptive Field Size	Characteristics
Fingertips	Small	Fine two-point discrimination, high precision
Back	Large	Poorer two-point discrimination, low precision

Area	Receptor Density	Sensitivity and Discrimination
Fingertips, Lips, Tongue	High	Greater sensitivity, fine detail discrimination
Back, Forearm	Low	Less sensitivity, coarse discrimination

Solution for Question 16:

Correct Answer: C) Typically short-lived and resolves within a few months

Explanation:

Thalamic pain syndrome typically starts within the first six months post stroke, but can develop even after ten years. It is persistent and potentially life-long pain that does not resolve.

Definition	Thalamic pain syndrome or Dejerine Roussy syndrome or Central post-stroke pain is a rare central neuropathic pain syndrome that occurs after infarction of the ventro-posterolateral thalamus
Aetiology	Vascular lesions affecting the central somatosensory system, most commonly ischemic or hemorrhagic stroke (Option B ruled out)
Pathophysiology	Damage to the spinothalamic tract or thalamus disrupts sensory processing, leading to abnormal pain perception (Option A ruled out)

Epidemiology	Prevalence ranges from 8% to 46%
Onset	Typically within the first six months post-stroke, but can develop after ten years also
Symptoms	Sharp, burning and stabbing pain with hyperalgesia and allodynia; Pain is described as pricking, aching, freezing, shooting, etc.
Diagnosis	Based on history, clinical examination, and imaging: CT/MRI to confirm stroke
Treatment	Pharmacological: Antidepressants, Anticonvulsants, (Option D ruled out) Opioids, NMDA receptor antagonists Non-Pharmacological: Transcutaneous electrical nerve stimulation (TENS), Repetitive transcranial magnetic stimulation (rTMS), Deep brain stimulation (DBS) Motor cortex stimulation
Prognosis	Poor Persistent and potentially life-long pain (Option C)
Complications	Reduces quality of life Depression or anxiety Sleep disturbance Drug dependence Poor social interactions Increased suicide risk

Solution for Question 17:

Correct Answer: B) The pathway localizes pain only to major body parts, such as an arm or leg, due to its diffuse connectivity.

Explanation:

Slow-chronic pain can usually be localized only to a major part of the body, such as one arm or leg, but not to a specific point on the arm or leg. This phenomenon is due to the multisynaptic, diffuse connectivity of the Paleo-Spinothalamic pathway, which is why patients have difficulty localizing the source in some chronic types of pain.

Paleo-Spinothalamic Pathway

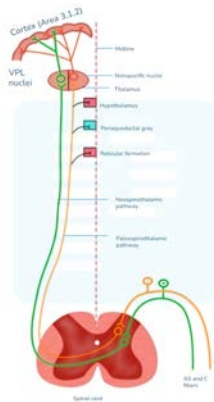


Table rendering failed

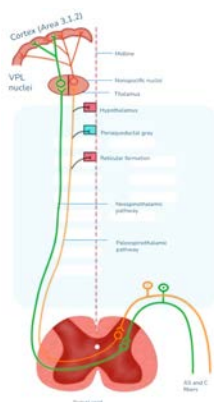
Solution for Question 18:

Correct Answer: C) The neurotransmitter released by the A δ fiber is glutamate

Explanation:

Glutamate is the primary neurotransmitter at A δ fiber endings.

Neo-Spinothalamic Pathway :



Aspect	Description
Fiber Type	A δ (A-delta) fibers

Function	Carries fast pain sensations, primarily mechanical and acute thermal pain (Option A ruled out)
First-Order Neurons	Terminate mainly in lamina I of the dorsal horn of the spinal cord (Option B ruled out)
Second-Order Neurons	Cross to the opposite side of the spinal cord through the anterior commissure. Ascend in the anterolateral pathway. The fibers terminate mainly in the ventrobasal complex, and a few fibers also terminate in the posterior nuclear complex of the thalamus. Sparse projections to midbrain reticular formation, periaqueductal grey and hypothalamus (Option D ruled out)
Third-Order Neurons	Originate from specific thalamic nuclei. The signals are then transmitted to other basal areas of the brain as well as somatosensory cortex
Localization of Pain	Fast pain is better localised than slow-chronic pain. Exact localisation requires simultaneous stimulation of tactile receptors
Neurotransmitter	Glutamate (primary neurotransmitter at Aδ fiber endings) (Option C)
Topographic Organization	In spinal cord: Lower body fibers placed laterally Upper body fibers placed medially Sacral segment fibers occupy most lateral part Cervical segment fibers occupy most medial part
Special Features	Pain perception persists despite removing somatic sensory areas of the cerebral cortex, indicating that the brain stem, thalamus, and lower centers handle this function. The cortex only helps in interpreting pain quality, not perception.

Solution for Question 19:

Correct Answer: A) Allodynia

Explanation: Allodynia refers to an excessive response to mild or normally innocuous stimuli.

Example: Painful sensation from a light touch which is never painful.

Table rendering failed

Analgesia (Option C) refers to the complete loss of pain sensation. It may not be associated with the loss of other sensations.

Hypoalgesia (Option D) refers to decreased sensitivity and raised threshold to painful stimuli.

Solution for Question 20:

Correct Answer: D) Bradykinin

Explanation:

Based on the above scenario, the patient most likely has a chemical burn. Bradykinin is involved in mediating the chemical nociceptive stimuli experienced by the patient.

Chemical Nociceptive stimuli

- Both TRPA1 and TRPV1 detects chemical stimuli
- Bradykinin (most painful chemical stimuli)- B1 & B2 receptors (Option D)
- Bradykinin is a potent inflammatory mediator that is released during tissue injury and contributes significantly to the sensation of pain
- Chemical burns can cause the release of bradykinin and other inflammatory mediators, which activate pain pathways.

Other chemical mediators:

- Serotonin- 5HT receptors
- Histamine- H3/H4 receptors
- Nerve endings also have Acid-Sensing Ion Channel (ASIC) receptors that detect H⁺ and cause acid induced pain.

Dopamine (Option A) is involved in motor pathway and not sensory

Acetylcholine (Option B) is primarily involved in neuromuscular transmission and autonomic nervous system function

Serotonin (Option C) is involved in the pain pathway, but is not the primary mediator in chemical nociceptive stimuli

Solution for Question 21:

Correct Answer: B) Heat

Explanation:

Transient Receptor Potential Ankyrin 1 (TRPA1) is activated by mechanical, cold and chemical stimuli.

Types of nociceptive stimuli and their receptors:

- Nociceptors can be activated by mechanical, chemical or thermal stimuli.

Mechanical Nociceptors:

Respond to strong pressure (eg: from sharp objects)

The receptors for mechanical stimuli at free nerve ending include: Transient Receptor Potential Ankyrin 1 (TRPA1) (also for cold and chemical stimuli) (Option A, C & D ruled out)

Mechanical stimuli can also release ATP, which acts on purinergic receptors such as P2X and P2Y and cause pain

Thermal Nociceptors:

Cold detection:

TRP Ankyrin 1 - It is sensitive when the temperature is $< 17^{\circ}\text{C}$ and

TRP Melastatin 8 - It is sensitive when the temperature is between $8-28^{\circ}\text{C}$.

Heat detection:

TRP Vanilloid 1 (also detects acid and chemicals) (Option B ruled out) - Activated when temperature is $> 43^{\circ}\text{C}$, H^{+} ions and capsaicin (the active ingredient of chili).

TRPV2/V3/V4 - It is activated when temperature is $> 52^{\circ}\text{C}$.

Chemical Nociceptors:

Both TRPA1 and TRPV1 detects chemical stimuli

Nerve endings also have Acid-Sensing Ion Channel (ASIC) receptors that detect H^{+} and cause acid induced pain

Bradykinin (most painful chemical stimuli) - B1 & B2 receptors

Serotonin - 5HT receptors

Histamine - H3/H4 receptors

Solution for Question 22:

Correct Answer: A) Dorsal root ganglion

Explanation:

- Nociceptors arise from cell bodies in the dorsal root ganglion that send one axon to the periphery and another to the spinal cord or brainstem.
- Nociceptors (pain receptors) are basically the free nerve endings that detect pain sensations.
- Location of nociceptors: Arise from cell bodies in the dorsal root ganglion that send one axon to the periphery and another to the spinal cord or brainstem (Option A) .
- Nociceptors activate Transient receptor potential (TRP) channels on nerve endings, which causes the neurons to depolarize and generate action potentials.

Ventral horn of the spinal cord (Option B) contains motor neuron cell bodies.

Lateral geniculate nucleus (Option C) is part of the thalamus and is involved in visual processing.

Sympathetic ganglion (Option D) is part of the autonomic nervous system and does not contain the cell bodies of nociceptors .

Solution for Question 23:

Correct Answer: B) Glutamate

Explanation:

Nociceptors:

- These are responsible for detection of pain.
- These receptors are basically free nerve endings.
- There are two types of pain:

	Fast pain (First Pain)	Slow pain (Second Pain)
Fiber Type	A δ fiber Myelinated Large diameter (2-5 μ m)	C fiber Unmyelinated Smaller diameter (0.4 - 1.2 μ m)
Location	Body surface	Deeper tissues
Conduction velocity	Very rapid (~12- 35 m/s)	Very slow (~ 0.5 - 2 m/s)
Neurotransmitter	Glutamate (Option B)	Substance P (Option A Ruled out)

Description	Rapid response Mediates the discriminative aspects of pain (localisation and intensity)	Delayed, dull, intense, diffuse and unpleasant feeling associated with noxious stimulus
-------------	--	---

Solution for Question 24:

Correct Answer: D) Pacinian corpuscles

Explanation: Pacinian receptors are rapidly adapting receptors sensitive to vibrations and rapid pressure changes

Adaptation of receptors:

- One of the most important characteristics of sensory receptors is that they can adapt partially or completely to any constant stimulus after a period of time.
- When a continuous stimulus is applied to a receptor, the initial response occurs at a high impulse rate, then at a progressively slower rate, which ultimately decreases to very few or none at all.

Mechanism:

- Adaptation occurs in 2 ways :

1	In Rapidly Adapting Receptors It is a faster mechanism. The receptors are viscoelastic in nature so that when a stimulus is applied to one side of the receptor, it is transmitted instantly by the viscous component to the same side of the central nerve fiber, eliciting a response Within a few seconds, the fluid in the receptors redistributes and the response is no longer elicited
2	In Slowly Adapting Receptors It is a slower mechanism called accommodation This occurs directly in the nerve fiber. If there is a continued stimulus, then the tip of the receptor becomes accommodated to the stimulus. This occurs from the progressive inactivation of sodium channels in the nerve fiber membrane.

Rapidly adapting receptors	Pacinian, Meissner's and hearing organ
Slow adapting receptors	Merkel's, Ruffini's and free nerve endings (Option A & C ruled out)

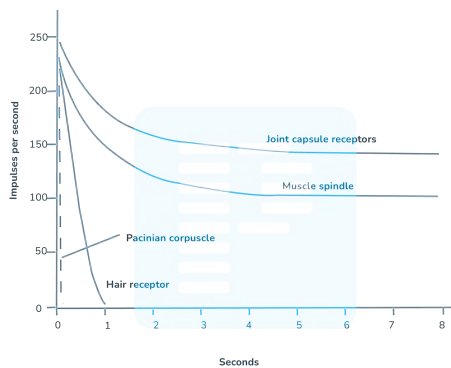


Figure 47-5 Adaptation of different types of receptors showing rapid adaptation of some receptors and slow adaptation of others.

Meissner's corpuscles (Option B) are also rapidly adapting receptors but are sensitive to light touch rather than vibrations.

Solution for Question 25:

Correct Answer: B) Receptor potential

Explanation:

Receptor potential is the graded depolarizing potential in a sensory receptor in response to a stimulus. Its magnitude varies with the intensity of the stimulus.

Receptor Potentials

- A receptor potential is a type of graded potential that occurs in sensory receptors. It is a local, non-propagated change in membrane potential that occurs in response to a stimulus.
- Receptor potential is not an action potential; instead, it causes the sensory neuron to fire action potentials.
- Once the receptor potential reaches a critical threshold, it triggers an action potential at the sensory nerve's first node of Ranvier.

Receptor Potential" data-author="NA" data-hash="7858" data-license="NA" data-source="NA" data-tags="April2025" height="430" src="https://image.prepladder.com/notes/4dckPzu4YYs0VLdSI2ED1744015597.png" width="500" />

Factors affecting receptor potential:

- Proportional Response: Amplitude of receptor potential is generally proportional to stimulus intensity.
- Nature: Non-propagated, depolarizing potential in sensory receptors.
- Threshold: When it reaches a critical level, an action potential is generated.
- Stimulus Intensity: Stronger stimuli lead to larger receptor potentials.

Mechanism of receptor potential generation in Pacinian Corpuscle

Table rendering failed

sensory nerve fiber" data-author="NA" data-hash="7859" data-license="NA" data-source="NA" data-tags="April2025" height="358" src="https://image.prepladder.com/notes/RwU5xVtZreVvDRUsznng1744015663.png" width="500" />

Action potential (Option A) is an all-or-none electrical signal that occurs after the receptor potential reaches a threshold.

Synaptic potential (Option C) occurs at synapses between neurons, not in sensory receptors.

Resting potential (Option D) is the baseline electrical state of a cell when it is not stimulated.

Solution for Question 26:

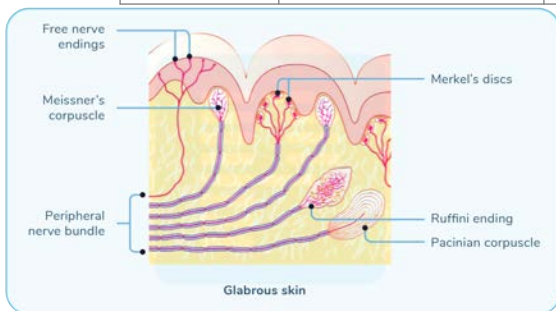
Correct Answer: C) Iggo Dome Receptor

Explanation:

- Merkel discs are often grouped together in a receptor organ called touch domes, which project upward against the underside of the epithelium of the skin.
- This upward projection causes the epithelium to protrude, forming a dome (Iggo dome receptor, named after a scientist, Iggo) and creating an extremely sensitive receptor.
- Blind individuals rely on the fine tactile acuity (two-point discrimination) of their fingertips to read Braille, which has dots spaced 2.5 mm apart.

Rapidly Adapting Tactile Receptors	Location	Function	Response

Merkel's Disc (Option C ruled out)	Underneath the epithelium of the skin	Detect continuous touch and pressure, Transmit strong initial signal and weaker sustained signal, Localise touch sensations (Two point discrimination), Determine texture	Transmit initially a strong response but partially adapting, then a continuing weaker response that adapts slowly. Extremely sensitive when grouped in a receptor organ called touch domes
Ruffini Corpuscles (Option D ruled out)	Deeper layers of the skin and deeper internal tissues, joint capsules	Important for signaling continuous states of deformation of tissue such as heavy prolonged touch and pressure Help to signal the degree of joint rotation	Very slow adapting, providing continuous feedback about deformation and prolonged pressure.
Free Nerve Ending (Option A ruled out)	Found everywhere in skin	Detect touch and pressure	Can detect even light contact



Hair End Organ (Option B) detects movement of objects on the body surface or initial contact with the body

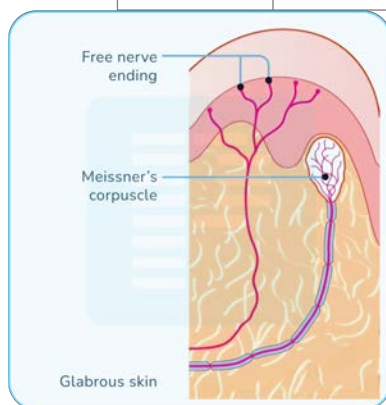
Solution for Question 27:

Correct Answer: B) Pacinian Corpuscles

Explanation:

Pacinian Corpuscles are rapidly adapting receptors that are particularly sensitive to rapid changes in pressure and vibrations of tissues.

Rapidly Adapting Tactile Receptors	Structure	Function	Location	Adaptation
Pacinian Corpuscle (Option B)	Encapsulated nerve ending	Detect tissue vibration or other rapid changes in the mechanical state of tissues Stimulated by rapid local compression of the tissue	Immediately beneath the skin and deep in the fascial tissues of the body	Adapts in a few hundredths of a second
Meissner Corpuscle (Option A ruled out)	Elongated, encapsulated nerve ending of a large myelinated sensory nerve fiber	Sensitive to movement of objects over the skin and low-frequency vibrations crucial for spatial location discernment of touch sensations	Non-hairy skin, particularly abundant in fingertips and lips.	Adapts in a fraction of a second
Hair End Organ (Option C ruled out)	Nerve fiber entwined at the base of each hair	Detects : Movement of objects on the body surface or Initial contact with the body	Hairy skin	Adapts readily



Pacinian Corpuscles" data-author="NA" data-hash="7863" data-license="NA" data-source="NA" data-tags="April2025" height="383" src="https://image.prepladder.com/notes/8UhmWxb4yJ5FYOyml55O1744016225.png" width="500" />

Solution for Question 28:

Correct Answer: C) Cold receptors: myelinated type A δ fibers

Explanation:

Cold receptors primarily use small, myelinated type A δ fibers to transmit signals, which conduct at a speed of approximately 20 m/sec.

Type	Location	Transmission Pathway	Speed
Cold Receptors (Option C)	Under the skin at spots	Type A δ fibers (myelinated), some Type C fibers (unmyelinated) (Option A ruled out)	~20 m/sec (A δ fibers)
Warmth Receptors (Option B&D; Ruled out)	Under the skin at spots	Unmyelinated Type C fibers	0.4-2 m/sec

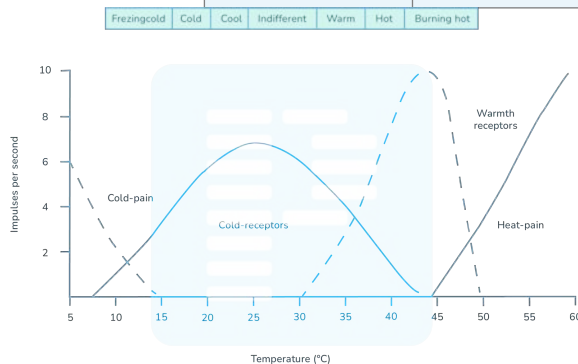


Figure 49-10. Discharge frequencies at different skin temperatures of a cold pain fiber, a cold fiber, a warmth fiber, and a heat pain fiber.

Type of Nerve Fiber	Temperature Range
Cold Receptors	10-40°C (peak ~24°C)
Warmth Receptors	>30°C (fade ~49°C)
Cold Pain Fibers	Very cold temperatures
Heat Pain Fibers	Start ~45°C

- Transmitted in parallel pathways with pain signals.
- Signals travel in the tract of Lissauer and terminate in laminae I, II, and III of dorsal horns.
- Processed signals by cord neurons cross to the opposite anterolateral sensory tract and terminate in both brain stem's reticular areas and the thalamus's ventrobasal complex.
- Removal of cortical postcentral gyrus reduces, but does not eliminate, temperature distinction ability.

Ascending and Descending Tracts of Spinal Cords

1. Which layer of the cerebral cortex is primarily responsible for receiving input sensory signals and sending them to other layers?

- A. Layer I
 - B. Layer III
 - C. Layer IV
 - D. Layer VI
-

2. A patient presents to the clinic with difficulty feeling the precise location of a stimulus on their skin and reports problems distinguishing between different textures and shapes. Additionally, they are having trouble detecting rapid changes in vibrations. Which of the following somatosensory pathways is most likely affected in this patient, given their symptoms?

- A. Dorsal Column-Medial Lemniscal System
 - B. Lateral Spinothalamic Tract
 - C. Anterior Spinothalamic Tract
 - D. Corticospinal Tract
-

3. A 45 Y/o man presents with numbness and loss of proprioception in his right leg and lower torso. Upon examination, the physician notes a loss of vibration sense and two-point discrimination in the affected areas. MRI shows a lesion in the medulla. Which of the following nucleus is most likely affected?

- A. Nucleus cuneatus
- B. Nucleus gracilis

- C. Ventral posterolateral nucleus (VPL)
 - D. Nucleus ambiguus
-

4. A patient presents with a loss of pain and temperature sensation on the left side of the body, below the level of a spinal cord lesion. Upon examination, the patient also exhibits impaired sensation of crude touch and itch on the same side but fine touch and proprioception sensations are intact . Which sensory pathway is most likely affected in this patient?

- A. Dorsal Column-Medial Lemniscal System
 - B. Anterolateral Pathway
 - C. Corticospinal Tract
 - D. Spinocerebellar Tract
-

5. Which of the following best describes the sensory and motor deficits in Brown-Séquard Syndrome?

- A. Contralateral loss of pain and temperature; ipsilateral loss of proprioception and vibration.
 - B. Ipsilateral loss of pain and temperature; contralateral loss of proprioception and vibration.
 - C. Ipsilateral loss of all sensations at the lesion; contralateral motor loss below the lesion.
 - D. Contralateral loss of vibration and fine touch ; ipsilateral loss of temperature, and crude touch.
-

6. Which of the following statements regarding removal of each somatosensory area is correct?

- A. Somatosensory Area I: complete loss of pain and temperature sensations.
 - B. Somatosensory Area II: inability of precise localization of sensations
 - C. Somatosensory Association Areas: inability to recognize two sides of an object
 - D. Somatosensory Area I: enhance ability to judge the shape of an object
-

7. Which of the following statements accurately describes the representation in motor and sensory homunculus?

- A. Hand and digits have large cortical representation in the motor homunculus
 - B. Feet represented at the bottom of both the precentral gyrus and postcentral gyrus
 - C. Trunk and back have large cortical representation in both homunculi
 - D. Cortical representation is proportional to the size of the body part in both homunculi
-

8. A 45-year-old man recently had his right leg amputated below the knee. Weeks after surgery, he feels sensations and pain in the missing leg area, sometimes triggered by touching his thigh. Which statement best explains the patient's condition and suggests a treatment?

- A. Brain reorganization of cortical brain area & Spinal cord stimulation may help.
 - B. Infection at the amputation site causes pain & Antibiotics should be prescribed.
 - C. Reorganization at spinal cord neuronal connection leading to pain.
 - D. Pain is due to rejection of the prosthetic limb & Discontinue use immediately.
-

9. Which law explains the phenomenon of phantom limb pain?

- A. Muller's Doctrine of Specific Nerve Energy
- B. Bell-Magendie Law

- C. Law of Projection
 - D. Weber-Fechner Law
-

10. A 45 Y/o right-handed woman presents with difficulty in performing coordinated movements and a lack of fluidity in her actions, particularly when performing tasks that require the integration of both hands, such as typing or playing the piano. Which of the following brain regions is most likely implicated in her condition?

- A. Primary Motor Cortex
 - B. Premotor Cortex
 - C. Supplementary Motor Cortex
 - D. Cortical Association Areas
-

11. Which of the following is the major contributor to the corticospinal tract in terms of percentage of descending fibers?

- A. Primary motor cortex
 - B. Premotor cortex
 - C. Supplementary cortex
 - D. Somatosensory area
-

12. Which of the following is not true regarding extrapyramidal tracts?

- A. They are descending tracts
- B. The tectospinal tract functions through gamma motor neurons
- C. Pontine and medullary reticulospinal tracts are excitatory and inhibitory, respectively
- D. The rubrospinal tract is involved in the lateral motor pathway

13. Which of the following extra-pyramidal tracts contributes to the lateral motor pathway?

- A. Vestibulospinal tract
 - B. Rubrospinal tract
 - C. Reticulospinal tract
 - D. Tectospinal tract
-

14. A 60 y/o man presents with complaints of weakness and difficulty walking. Upon examination, he is found to have grade 1 power in right leg muscle and an exaggerated knee-jerk reflex. Which of the following is the most likely diagnosis?

- A. Lower motor neuron lesion of the L4
 - B. Upper motor neuron lesion of the L4
 - C. Upper motor neuron lesion of the right internal capsule
 - D. Lower motor neuron lesion of the right internal capsule
-

15. A 65-year-old man complains of difficulty moving his arms and states that they feel "stiff." Upon examination, both arms show increased resistance to passive movement. The resistance is present throughout the range of motion and does not change with the speed of movement. What is the most likely cause of the patient's symptoms?

- A. Lower motor neuron lesion
- B. Cerebellar lesion
- C. Lesion of the pyramidal tract
- D. Lesion of the extrapyramidal tract

16. A 17 y/o male is brought after being struck by a car. He is unconscious and exhibiting abnormal posturing. His arms are stiffly extended, his forearms are pronated, his wrists and fingers are flexed, and his legs are extended with his toes pointed inward. Which of the following is the most likely diagnosis?

- A. Decorticate rigidity
 - B. Decerebrate rigidity
 - C. Cerebellar ataxia
 - D. Spinal shock
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	2
Question 4	2
Question 5	1
Question 6	3
Question 7	1
Question 8	1
Question 9	3
Question 10	3
Question 11	4
Question 12	2
Question 13	2
Question 14	3
Question 15	4
Question 16	2

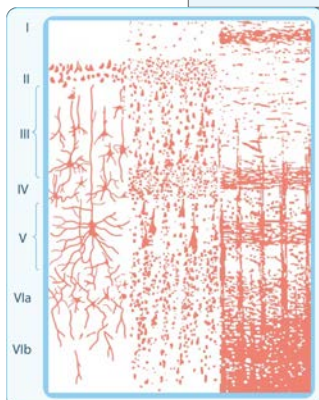
Solution for Question 1:

Correct Answer: C) Layer IV

Explanation: Layer IV is the primary input layer where sensory signals first enter the cortex. It processes these signals and then distributes them to other layers.

Cerebral Cortex Layers and Functions: Has six layers of neurons.

Layer	Function
I- Molecular layer	Receives nonspecific input signals from lower brain centers. Control the excitability of stimulated areas.
II- External granular layer	Receives nonspecific input and sends axons to the opposite cortex through the corpus callosum.
III- External pyramidal layer	Sends axons to related cortical areas on the opposite side through the corpus callosum.
IV- Internal granular layer	The primary layer for receiving input sensory signals, which then spread to the surface and deeper layers.
V- Internal pyramidal layer	Sends large axons to distant nervous system parts like basal ganglia, brain stem, and spinal cord. Controls signal transmission.
VI- Fusiform/ polymorphic layer	Sends axons to the thalamus. Regulates excitability of incoming sensory signals.



Solution for Question 2:

Correct Answer: A) Dorsal Column-Medial Lemniscal System

Explanation:

The Dorsal Column-Medial Lemniscal System handles precise touch localization, texture, vibration, and proprioception. The patient's difficulty with these sensations suggests an issue with this system.

Somatosensory pathways:

- Two primary somatosensory pathways that transmit sensory signals from the body to the brain: the dorsal column-medial lemniscal system and the anterolateral system.

Pathway	Sensory Modalities
Dorsal Column-Medial Lemniscal System (Option A)	Touch: Provides precise localization and fine gradations of intensity. Vibration: Detects rapid changes in stimulus through phasic sensations. Proprioception: Contributes to the sense of body position and movement. Pressure: Allows for the perception of fine degrees of pressure intensity.
Anterolateral System (Option B & C)	Lateral Spinothalamic Tract: Transmits sensations of pain and temperature (warmth and cold). Anterior Spinothalamic Tract: Carries crude touch and pressure for general skin contact, along with tickle, itch, and sexual sensations.

Corticospinal Tract is involved in motor control, not sensory discrimination. (Option D)

Solution for Question 3:

Correct Answer: B) Nucleus gracilis

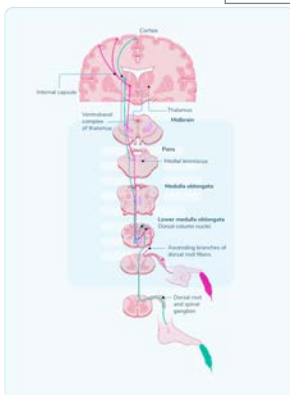
Explanation:

Based on the above scenario, the patient is most likely having a lesion in the Nucleus gracilis

Dorsal column-medial lemniscal pathway:

Pathway	Sensory receptors → Spinal cord (dorsal roots) → Brain (somatosensory cortex)
Neurons	

First-order neurons	Originate in mechanoreceptors, and enter spinal cord via dorsal roots Pseudo-unipolar neurons with cell bodies in the dorsal root ganglion Split into medial and lateral branches: Medial branch: Travels up dorsal column to brain Lateral branch: Synapses in dorsal horn, forming local reflexes, spinocerebellar tracts Fibers from lower thoracic, sacral, and lumbar regions (T7-T12, L, S) ascend to medulla without synapsing, forming the fasciculus gracilis Fibers from cervical and upper thoracic regions (C, T1-T6) ascend to medulla without synapsing, forming the fasciculus cuneatus
Second-order neurons	Synapse in dorsal medulla (cuneate nucleus for upper body, gracile nucleus for lower body) Cross to opposite brain stem side, travel to thalamus (medial lemniscus)
Third-order neurons	Synapse in thalamus (ventrobasal complex) Project to somatosensory cortex (postcentral gyrus, lateral parietal cortex)



Nucleus cuneatus : Transmits sensory information from the upper body (arms and upper torso). The patient's symptoms do not involve these areas. (Option A)

Ventral posterolateral nucleus (VPL) : Located in the thalamus, not the medulla. This option is incorrect because the lesion is in the medulla. (Option C)

Nucleus ambiguus : Involved in motor control of the pharyngeal and laryngeal muscles. It is not related to the dorsal column pathway or the sensory symptoms described. (Option D)

Solution for Question 4:

Correct Answer: B) Anterolateral Pathway

Explanation:

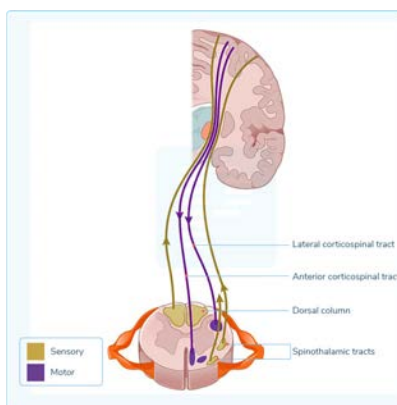
Based on the above scenario, the patient most likely has a lesion in the anterolateral pathway.

Anterolateral Pathway

- Transmits sensory signals such as pain, temperature, crude touch, itch, sexual sensations, and crude pressure from the opposite side of the body.
- Damage results in loss of pain and temperature sensation on the opposite side below the lesion.
- Characteristics: Slower transmission, less precise spatial localization, reduced ability to transmit rapidly changing signals.

Pathway:

Table rendering failed

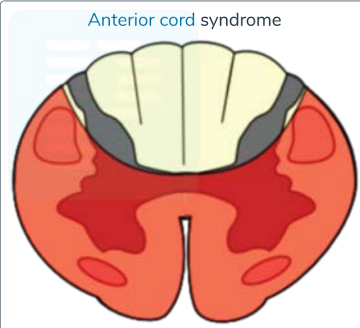
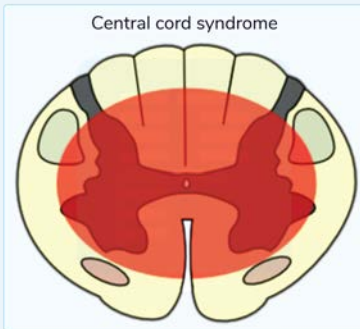
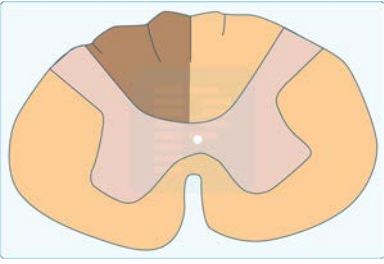


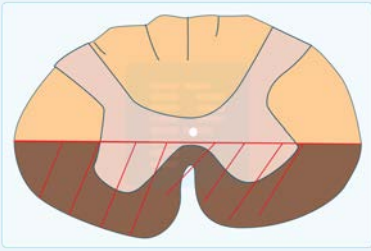
Solution for Question 5:

Correct Answer: A) Contralateral loss of pain and temperature; ipsilateral loss of proprioception and vibration.

Explanation:

Spinal cord syndromes:

Condition	Description	Clinical Findings
Brown-Séquard Syndrome	Anterior cord syndrome  Hemisection of the spinal cord.	Ipsilateral loss of all sensation at the level of the lesion; ipsilateral LMN signs (e.g., flaccid paralysis) at the level of the lesion; Ipsilateral UMN signs below the level of lesion (due to corticospinal tract damage); Ipsilateral loss of proprioception, vibration, and fine touch below the level of lesion (due to dorsal column damage); Contralateral loss of pain, temperature, and crude touch below the level of lesion (due to spinothalamic tract damage); If a lesion occurs above T1, it may present with ipsilateral Horner syndrome due to damage of the oculosympathetic pathway.
Anterior Spinal Artery Occlusion	Central cord syndrome  Spinal cord infarction sparing dorsal horns and columns. Watershed area mid-thoracic; ... (content truncated)	Presents with UMN signs below lesion (corticospinal tract), LMN signs at the level of lesion (anterior horn), and Loss of pain and temperature sensation below the lesion (spinothalamic tract).
Tabes Dorsalis	 Degeneration/demyelination of dorsal columns and roots caused by <i>Treponema pallidum</i> (tertiary syphilis).	Progressive sensory ataxia (impaired proprioception and poor coordination); Positive Romberg sign; Absent deep tendon reflexes (DTRs); Seen in the terminal stage of syphilis Associated with shooting pain, Argyll Robertson pupils, and Charcot joints.

Syringomyelia	 A fluid-filled, gliosis-lined cavity within the spinal cord. Often associated with Chiari I malformation.	"Cape-like" loss of pain and temperature sensation in bilateral upper extremities (due to damage of fibers crossing in anterior white commissure); LMN deficits if anterior horns are affected; Most lesions occur between C2 and T9.
---------------	--	--

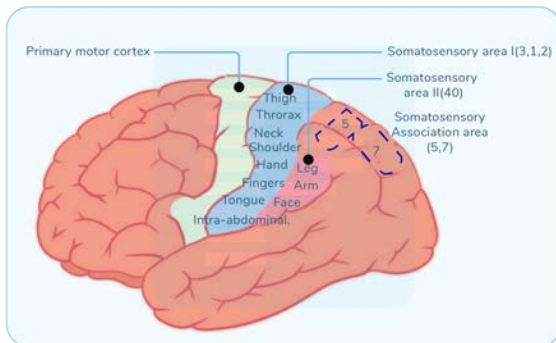
Solution for Question 6:

Correct Answer: C) Somatosensory Association Areas : inability to recognize two side of an object

Explanation:

Feature	Somatosensory Area I (SSA 1)	Somatosensory Area II (SSA 2)	Somatosensory Association Area
Location	Postcentral gyrus (Brodmann's areas 3, 1, and 2).	Superior wall of Sylvian fissure	Brodmann's areas 5 and 7, located in parietal cortex behind SSA I
Representation and localization	High localization of body parts. Nose, lips, mouth and face laterally; head, neck, shoulder and lower body part medially	Poor localization (Option B) Face is represented anteriorly, arms centrally and legs posteriorly	Combines information from the primary somatosensory area, leading to complex sensations, such as feeling an object like a knife or ball.
Function	Primary area for somatic sensation	Less known functions; Relies on input from SSA I	Decipher deeper meanings of sensory information from SSA
Sensory Information	Receive inputs from thalamic and cortical areas	Signals enter from brain stem, SSA I and from other sensory areas (visual and auditory)	Receives signals from SSA I, thalamus, visual cortex and auditory cortex

Effect of Excision	B/L excision of leads to: Loss of precise localization of sensations Inability to judge pressure against the body, weight, texture, or shape of the object (astereognosis) (Option D). Pain and temperature senses preserved but poorly localized (Option A)	Removal of parts of SSA II does not affect the response of neurons in SSA I	U/L removal leads to: Amorphosynthesis : Difficulty recognizing objects on one side. (Option C) Hemineglect: Denial of the existence of the opposite side of the body. Constructional apraxia: Difficulty in constructing a picture.
--------------------	---	--	--



Solution for Question 7:

Correct Answer: A) Hand and digits have large cortical representation in both homunculi

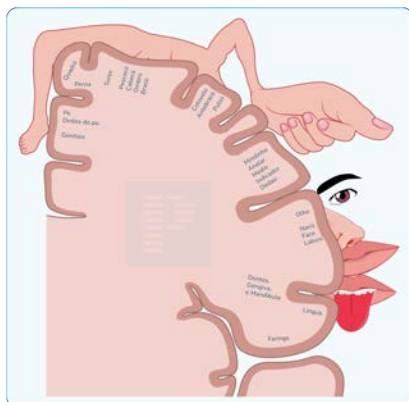
Explanation:

Homunculus represents either the motor or the sensory distribution along the cerebral cortex of the brain.

Feature	Motor Homunculus	Sensory Homunculus
Location	Precentral gyrus	Postcentral gyrus
Representation	Feet at the top of the gyrus (Option B); face at the bottom	Legs on top and head at the bottom of the gyrus (Option B)
Cortical Representation (Option D)	Proportionate to the skill involved in fine, voluntary activities	Proportional to the number of sensory receptors that sense muscle position and changes in muscle length
Large Representation Areas	Hand, digits (Option A), lips, jaw, and tongue (areas involved in	Face (especially lips), Hand (especially digits) (Option A)

(Option C)	speech and hand movements)	
Small Representation Areas	Trunk and lower limb	Trunk and lower limb
Specialization	Large area for lips, jaw, and tongue due to specialization in speech (vocalization of the cortex)	Large area for face mainly lips) due to speech Large area for hand(mainly digits) due to cutaneous sensations from areas involved in skilled activities

Sensory Homunculus:



Motor Homunculus:



Solution for Question 8:

Correct Answer: A) Brain reorganization of cortical brain area & Spinal cord stimulation may help.

Explanation:

Cortical plasticity

- Refers to the ability of neurons to modify their connections under specific circumstances.
- Cortical structures are capable of reorganization (plasticity) after loss of a body part, neighboring areas take over the cortical area of the lost part. (Option A).
- Cortical connections of sensory units become weak with disuse and strengthen with use.

Phantom Limb Pain

- Perceived due to cortical plasticity
- In the phantom limb phenomenon, the limb actually does not exist (amputated), but the patient complains that he feels pain or itch in the limb
- Based on the evidence, brain reorganization occurs when sensory input is cut off. For example: In leg amputees, the ventral posterior thalamic nucleus can perceive the signal that once received input from leg and foot now responds to stump (thigh) stimulation

Management :

Solution for Question 9:

Correct Answer: C) Law of Projection

Explanation: The law of Projection forms the basis of phantom limb

Laws of Sensory Physiology

Bell-Magendie Law :

- States that the dorsal root of the spinal cord is sensory and the ventral root of the spinal cord is motor in function. (Option B)

Muller's Doctrine of Specific Nerve Energy :

- States that sensation evoked by a stimulus that generates impulse in the specific sensory pathway depends on the precise area of the brain that is eventually activated by the stimulus. (Option A)

Law of Projection :

- States that no matter where a specific sensory pathway is stimulated along its course to the cortex, the sensation is referred to the location of the receptors.
- Forms the basis of the phantom limb. (Option C)

Weber-Fechner Law :

- States that the magnitude of sensation felt is proportionate to the logarithm of the intensity of the stimulus. (Option D)

Interpreted signal strength = $\text{Log (Stimulus) + Constant}$

- Emphasizes that the greater the background sensory intensity, the greater an additional change must be for the individual to detect the change.

Labelled Line Theory

- States that each sensory receptor is specialized to detect specific stimuli and sends signals via dedicated nerve fibers to the corresponding area of the somatosensory cortex.

Solution for Question 10:

Correct Answer: C) Supplementary Motor Cortex

Explanation:

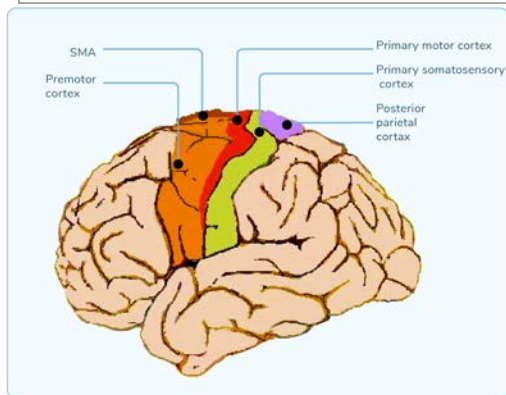
Based on the above scenario, the patient most likely has a lesion in the supplementary motor cortex, which is important for complex bimanual coordination.

Cortical Motor areas:

The motor cortex is located in the frontal lobe.
Primary Motor Cortex: Corresponds to Brodmann Area 4 It lies in the first convolution of the frontal lobes anterior to the central sulcus Responsible for executing voluntary movements and controlling fine motor skills (Option A)
Premotor Cortex: Corresponds to Brodmann area 6 Lies 1-3 cm anterior to the primary motor area Responsible for setting posture before movement (Option B)
Supplementary Motor Cortex: Lies mainly in the longitudinal fissure Responsible for Bimanual coordination (Option C)

Cortical Association Areas:

Commands for voluntary movements originate in cortical association areas (Option D)



Solution for Question 11:

Correct Answer: D) Somatosensory area

Explanation:

Corticospinal tract:

- The corticospinal tract, also known as the pyramidal tract, is the most important descending tract from the motor cortex.
- Contribution:
 - Primary motor cortex- 30% (Option A)
 - Premotor and supplementary cortex- 30% (Options B&C;)
 - Somatosensory area- 40% (Option D)
- After leaving the cortex, the tract passes through the posterior limb of the internal capsule, then through the brainstem and into the medulla.
- In the medulla, most of the fibers cross to the opposite side, forming the medullary pyramids.
- These crossed fibers descend as the lateral corticospinal tract finally terminates in the spinal cord.
- Some fibers which do not cross to the opposite side in the medulla pass ipsilaterally down to the spinal cord as the ventral corticospinal tract.



Solution for Question 12:

Correct Answer: B) The tectospinal tract functions through gamma motor neurons

Explanation:

Extrapyramidal tracts:

- These include the reticulospinal tract, vestibulospinal tract, tectospinal tract and rubrospinal tract.
- All of them originate from the brain and terminate in the spinal cord. Hence, all are descending tracts (Option A)

Tract	Reticulospinal tract	Vestibulospinal tract	Tectospinal tract	Rubrospinal tract
Origin	Reticular activating system	Medial and inferior vestibular nuclei	Superior colliculus of the midbrain	Red nucleus
Characteristics	Two types: Pontine (excitatory) and medullary (inhibitory) tracts (Option C) They are involved in maintaining posture and modulating muscle tone via gamma motor neurons (Option B)	It controls antigravity muscles, such as proximal limb extensors, to control posture and balance	Controls head and eye movements	Controls distal limb muscles The tract excites flexor motor neurons and inhibits extensor motor neurons

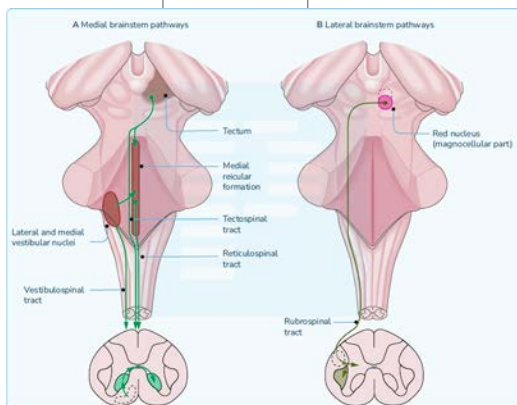
Motor pathway	Medial	Medial	Medial	Lateral (Option D)
---------------	--------	--------	--------	--------------------

Solution for Question 13:

Correct Answer: B) Rubrospinal tract

Explanation:

	Lateral motor pathway	Medial motor pathway
Comprises of:	Corticospinal tract Rubrospinal tract	Reticulospinal tract Vestibulospinal tract Tectospinal tract
Pathway and control	Terminates in the dorsolateral area of spinal gray matter and controls distal muscles	Terminates in ventromedial area of spinal gray matter and controls axial and proximal muscles
Function	Fine, skilled, voluntary motor movements	Maintaining erect posture



Solution for Question 14:

Correct Answer: C) Upper motor neuron lesion of the right internal capsule

Explanation :

- The internal capsule contains corticospinal tract fibers, and a lesion in this location will cause an upper motor neuron lesion.

Upper motor neurons (UMN):

- Neurons in the cortex and brainstem that activate lower motor neurons.
- The pyramidal tract and extrapyramidal tract are both considered UMN's because they both influence alpha motor neurons.
- All neurons above the alpha motor neuron are upper motor neurons.

Lower motor neurons (LMN) :

- Spinal and cranial motor neurons that directly innervate skeletal muscles.
- The alpha motor neuron is considered the lower motor neuron.
- The lower motor neuron is the final common pathway for both upper motor neurons and the stretch reflex.
- The corticospinal tract is excitatory to lower motor neurons while the extrapyramidal tract is inhibitory.

Feature	UMN Lesion	LMN Lesion
Atrophy	None	Severe (loss of trophic factor)
Fasciculation	None	Present
Tone	Hypertonia (spasticity & rigidity)	Hypotonia (Decreased)
Deep tendon reflex	Hyperactive	Hypoactive or absent
Muscle group affected	Regional	Segmental/distal
Babinski sign	Positive	Absent

An upper motor neuron lesion at the level of the L4 spinal nerve root would not affect the knee-jerk reflex. (Option B)

Solution for Question 15:

Correct Answer: D) Lesion of the extrapyramidal tract

Explanation :

- Rigidity is defined as an increased resistance to passive movement that is not velocity-dependent, meaning the resistance remains constant regardless of the speed of movement.
- Rigidity is a feature of extrapyramidal lesions.

Feature	Spasticity	Rigidity
Definition	Velocity-dependent resistance	Velocity-independent resistance
Mechanism	Lesion of the pyramidal tract (Option C)	Lesion of the extrapyramidal tract
Clinical Finding	Resistance increases with speed	Resistance is continuous and does not vary with the speed of movement
Example	Clasp-knife spasticity	Lead pipe rigidity, cogwheel rigidity
Associated with	Weakness	Hypertonia

Lower motor neuron lesions are characterized by hypotonia (decreased muscle tone), muscle wasting (atrophy), weakened or absent reflexes, and flaccid paralysis. (Option A)

Cerebellar lesions can cause motor dysfunction, hypotonia, not rigidity. (Option B)

Solution for Question 16:

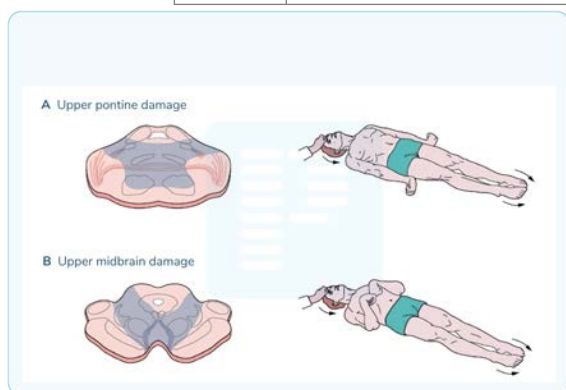
Correct Answer: B) Decerebrate rigidity

Explanation :

The patient's posturing is characteristic of decerebrate rigidity.

Feature	Decerebrate Rigidity	Decorticate Rigidity
Definition	A state of severe hypertonia characterized by extension of all four limbs.	A state in which the upper extremities are flexed and the lower extremities are extended.
Cause	Damage to the lower midbrain and upper pons	Damage to the upper midbrain

Mechanism	Loss of descending inhibition to the pontine reticular system causes overactivity of this system, which excites antigravity muscles.	Rubrospinal tract excitation of flexor muscles in the upper extremities results in flexion due to damage above the red nucleus.
Posture	Arms adducted and extended Forearms pronated Wrists and fingers flexed Legs extended Feet plantar flexed	Arms adducted and flexed Wrists and fingers flexed Legs extended Plantar flexion of the feet (Option A)
Clinical Significance	Severe brain stem damage; poor prognosis	Better prognosis than decerebrate rigidity



Cerebellar ataxia typically manifests as uncoordinated and inaccurate movements, often with intention tremor rather than the fixed, rigid posturing. (Option C)

Spinal shock can occur after spinal cord injury and presents as a flaccid paralysis with loss of reflexes below the level of the lesion. (Option D)

Basal Ganglia and Cerebellum

1. Which of the following statements about the basal ganglia and their associated neurotransmitters is **CORRECT**?

- A. The caudate nucleus primarily uses acetylcholine as its neurotransmitter.
 - B. The globus pallidus internus (GPI) exerts an excitatory influence on the thalamus.
 - C. The subthalamic nucleus (STN) uses glutamate as its principal neurotransmitter to excite the globus pallidus internus (GPI).
 - D. The substantia nigra pars reticulata (SNr) inhibits the thalamus using dopamine.
-

2. Which statement correctly describes the effect of each basal ganglia pathway?

- A. The direct pathway decreases inhibitory output from the internal segment of the globus pallidus.
 - B. The indirect pathway increases inhibitory output from the internal segment of the globus pallidus.
 - C. The direct pathway increases inhibitory output from the external segment of the globus pallidus.
 - D. The indirect pathway decreases inhibitory output from the external segment of the globus pallidus.
-

3. A 55-year-old man presents with resting tremors, bradykinesia, rigidity, and postural instability. His symptoms have progressively worsened over the past year. Based on the clinical presentation, which of the following is the most likely underlying pathology?

- A. Degeneration of dopaminergic neurons in the substantia nigra pars compacta
- B. Degeneration of cholinergic neurons in the basal forebrain

- C. Degeneration of GABAergic neurons in the striatum
 - D. Degeneration of serotonergic neurons in the raphe nuclei
-

4. Which statement correctly describes the cerebellar structure and its nuclei?

- A. Anterior and posterior lobes regulate posture and balance
 - B. Flocculonodular lobe coordinates fine motor skills.
 - C. Dentate nucleus is key in motor planning and execution.
 - D. Granule cells are the primary output neurons of the cerebellar cortex.
-

5. Which of the following accurately describes the primary function of the mossy fibers in the internal circuitry of the cerebellum?

- A. Transmit motor commands to the cerebellum
 - B. Provide excitatory input to granule cells.
 - C. Relay sensory information to the deep cerebellar nuclei.
 - D. inhibit the activity of the deep cerebellar nuclei.
-

6. Which of the following statements about the functional divisions of the cerebellum is incorrect?

- A. The vestibulocerebellum controls balance and eye movements.
 - B. Spinocerebellum coordinates axial and proximal limb movements
 - C. Cerebrocerebellum handles movement planning and programming
 - D. The flocculonodular lobe coordinates fine motor movements.
-

7. A 72-year-old man presents with uncoordinated movements and difficulty in performing rapid alternating tasks, speech difficulty, and experiences rapid, tremulous eye movements. Which of the following features is not typically associated with cerebellar disease?

- A. Hypertonia
- B. Dysarthria
- C. Ataxia
- D. Rebound Phenomenon

8. Match the parts of the Papez circuit with their correct functions: Part of Papez Circuit Function 1) Cingulate Gyrus A) Conversion of short-term memory to long-term memory 2) Parahippocampal Gyrus B) Controls heart rate, blood pressure, cognitive and emotional processing 3) Hippocampus C) Responsible for controlling recent memories of the brain 4) Mammillary Body and Anterior Nucleus of Thalamus D) Spatial memory, the memory of three-dimensional space

Part of Papez Circuit	Function
1) Cingulate Gyrus	A) Conversion of short-term memory to long-term memory
2) Parahippocampal Gyrus	B) Controls heart rate, blood pressure, cognitive and emotional processing
3) Hippocampus	C) Responsible for controlling recent memories of the brain
4) Mammillary Body and Anterior Nucleus of Thalamus	D) Spatial memory, the memory of three-dimensional space

- A. 1 - A, 2 - B, 3 - D, 4 - C
- B. 1 - B, 2 - D, 3 - A, 4 - C
- C. 1 - D, 2 - C, 3 - A, 4 - B
- D. 1 - C, 2 - A, 3 - B, 4 - D

9. Which of the following statements about the amygdala is incorrect?

- A. Controls emotions and motivations
 - B. Connected with the prefrontal cortex & hippocampus for emotional memory processing
 - C. Damage can result in a loss of sexual behaviour.
 - D. The prefrontal cortex develops working memory
-

10. A 24-year-old man is brought to the emergency room after a motor vehicle accident. He displays significant behavioral changes including placing non-food objects in his mouth, increased sexual advances towards staff, and appears to have no fear when approached by the medical team. Which of the following brain structures is MOST likely to be damaged in this patient?

- A. Cerebellum
 - B. Amygdala
 - C. Hippocampus
 - D. Parietal lobe
-

11. Which of the following nuclei has a function of thermoregulation?

- A. Medial preoptic nuclei
 - B. Lateral nuclei
 - C. Para-ventricle nuclei
 - D. Arcuate nuclei
-

12. A 45-year-old man presents with significant behavioral changes following a traumatic brain injury. He experiences frequent unprovoked outbursts of anger and aggression, triggered by minimal stimuli, along with emotional blunting and decreased pleasure. Based on his presentation, which brain region is MOST likely affected?

- A. Hippocampus
 - B. Medial forebrain bundle
 - C. Central gray area
 - D. Ventromedial hypothalamus
-

13. In a patient experiencing severe hemorrhage resulting in isotonic fluid loss, which of the following accurately describes the role of osmoreceptors and their effect on the thirst mechanism?

- A. Increased blood osmolality stimulates the supra-optic nucleus to secrete ADH
 - B. Angiotensin II directly stimulates the lateral nucleus of the hypothalamus
 - C. Osmoreceptors in the OVLT and MnPO are stimulated by decreased blood osmolality
 - D. Stimulated SFO osmoreceptors increase ANP secretion, which reduces ADH.
-

14. Which of the following statements is true about the updated theory of control of feeding?

- A. Both orexigenic and anorexigenic signals are present in the pre-ventricular nucleus
- B. Leptin, Peptide YY and cortisol are all anorexigenic signals
- C. POMC Fragmentation produces alpha MSH promotes food intake
- D. Orexin peptide produced from the lateral nucleus of the hypothalamus plays an essential role in regulating sleep

15. Which statement best describes the center-surround model of the basal ganglia?

- A. The direct pathway broadly inhibits motor actions, while the indirect pathway facilitates specific actions.
 - B. The direct pathway facilitates specific motor actions, while the indirect pathway suppresses competing actions.
 - C. The indirect pathway reduces inhibitory output from the GPi, while the direct pathway increases inhibitory output.
 - D. Both pathways uniformly facilitate all motor actions
-

16. Which of the following statements about climbing and mossy fibers is correct?

- A. Climbing and mossy fibers provide the primary input to the cerebellar cortex
 - B. The Vestibulocerebellar serves as the pathway for climbing fibers.
 - C. Mossy fibers are the terminal endings of the olivocerebellar tracts.
 - D. Mossy fibers use aspartate to provide excitatory signals to the Purkinje cells.
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	1
Question 4	3
Question 5	2
Question 6	4
Question 7	1
Question 8	2
Question 9	3
Question 10	2
Question 11	1
Question 12	4
Question 13	2
Question 14	4
Question 15	2
Question 16	1

Solution for Question 1:

Correct Answer: C) The subthalamic nucleus (STN) uses glutamate as its principal neurotransmitter to excite the globus pallidus internus (GPi).

Explanation:

The subthalamic nucleus (STN) uses glutamate to excite the GPi.

The basal ganglia are a group of subcortical nuclei in the brain involved in regulating movement and coordinating motor activity. Key parts include:

- Striatum : Consists of the caudate nucleus and putamen. It's the main input center for the basal ganglia, receiving information from the cerebral cortex.
- Globus Pallidus : Divided into the external (GPe) and internal (GPi) segments. It plays a role in the regulation of voluntary movement by processing signals from the striatum.
- Substantia Nigra : Located in the midbrain and has two parts: the pars compacta (which produces dopamine) and the pars reticulata. It is involved in movement control and has a key role in the dopamine system.
- Subthalamic Nucleus : Located below the thalamus, it contributes to motor control by influencing the output of the globus pallidus.

The basal ganglia use several key neurotransmitters to regulate motor control:

- The caudate nucleus primarily uses GABA as its neurotransmitter, not acetylcholine. (Option A)
- The globus pallidus internus (GPi) exerts an inhibitory influence on the thalamus. (Option B)
- The substantia nigra pars reticulata (SNr) inhibits the thalamus using GABA, not dopamine. (Option D)

Solution for Question 2:

Correct Answer: A) The direct pathway decreases inhibitory output from the internal segment of the globus pallidus.

Explanation:

The basal ganglia play a crucial role in regulating motor control, movement coordination, and certain aspects of cognition.

Direct Pathway:

- Promotes movement by reducing inhibitory output from the GPi to the thalamus.
- Mechanism: The striatum inhibits the GPi, leading to less inhibition of the thalamus and increased motor activity.

Indirect Pathway:

- Inhibits movement by increasing inhibitory output from the GPe to the thalamus.
- Mechanism: The striatum inhibits the GPe, resulting in less inhibition of the STN, which then excites the GPi. This increased activity of the GPi leads to more inhibition of the thalamus and decreased motor activity.

indirect pathway" data-author="NA" data-hash="555" data-license="NA" data-source="NA" data-tags="January2025" height="500" src="https://image.prepladder.com/notes/ShbNM6MM8gtJNqbjOtVH1736069514.png" width="696" />

	Direct Pathway	Indirect Pathway
Function	Facilitates motor activity	Suppresses motor activity
Input	Excitatory input from the cerebral cortex	Excitatory input from the cerebral cortex
Target	Striatum (caudate nucleus and putamen)	Striatum (caudate nucleus and putamen)
Striatum Output	Inhibits internal segment of globus pallidus (GPi)	Inhibits external segment of globus pallidus (GPe)
GPi/SNr Effect	Reduced inhibition of thalamus	Increased inhibition of thalamus
Thalamus Effect	Increased thalamic activation	Decreased thalamic activation
Motor Cortex Effect	Enhanced motor activity	Reduced motor activity
Subthalamic Nucleus (STN)	Not directly involved	Activated by reduced inhibition from GPe

Solution for Question 3:

Correct Answer: A) Degeneration of dopaminergic neurons in the substantia nigra pars compacta

Explanation:

- The patient's symptoms are characteristic of Parkinson's disease, which is primarily caused by the degeneration of dopaminergic neurons in the substantia nigra pars compacta.
- This leads to a decrease in dopamine levels in the striatum, affecting the basal ganglia circuitry and resulting in the hallmark motor symptoms of the disease.
- Disorders affecting the basal ganglia can result in a range of motor symptoms, leading to either reduced (hypokinetic) or excessive (hyperkinetic) movement.

Hypokinetic Diseases

- Parkinsonism: Cause: Reduced dopamine in the substantia nigra. Symptoms: Akinesia, bradykinesia, cogwheel rigidity, resting tremor, shuffling gait, stooped posture, reduced facial expression, small handwriting, and mood/cognitive issues.
- Cause: Reduced dopamine in the substantia nigra.
- Symptoms: Akinesia, bradykinesia, cogwheel rigidity, resting tremor, shuffling gait, stooped posture, reduced facial expression, small handwriting, and mood/cognitive issues.
- Cause: Reduced dopamine in the substantia nigra.
- Symptoms: Akinesia, bradykinesia, cogwheel rigidity, resting tremor, shuffling gait, stooped posture, reduced facial expression, small handwriting, and mood/cognitive issues.

Hyperkinetic Diseases

- Hemiballismus : Damage to the subthalamic nucleus, leading to decreased inhibition of the thalamus and excessive movement.
- Damage to the subthalamic nucleus, leading to decreased inhibition of the thalamus and excessive movement.
- Huntington's Disease : Degeneration of the indirect pathway in the striatum, causing involuntary movements similar to hemiballismus.
- Degeneration of the indirect pathway in the striatum, causing involuntary movements similar to hemiballismus.
- Dystonia : Dysfunction in the direct pathway, resulting in sustained, twisting movements.
- Dysfunction in the direct pathway, resulting in sustained, twisting movements.
- Damage to the subthalamic nucleus, leading to decreased inhibition of the thalamus and excessive movement.
- Degeneration of the indirect pathway in the striatum, causing involuntary movements similar to hemiballismus.
- Dysfunction in the direct pathway, resulting in sustained, twisting movements.

Degeneration of cholinergic neurons in the basal forebrain is associated with Alzheimer's disease, which primarily affects memory and cognitive functions rather than motor control.
(Option B)

GABAergic neurons in the striatum are involved in the motor control pathways, but their degeneration is more characteristic of Huntington's disease which involves a different pattern of motor dysfunction, including chorea, which is not consistent with the symptoms described.
(Option C)

Degeneration of serotonergic neurons in the raphe nuclei is related to mood disorders and certain types of depression, rather than the motor symptoms seen in Parkinson's disease.

(Option D)

Solution for Question 4:

Correct Answer: C) Dentate nucleus is key in motor planning and execution

Explanation:

Overview of the anatomy of the cerebellum, including its lobes, lobules, cortex, and nuclei:

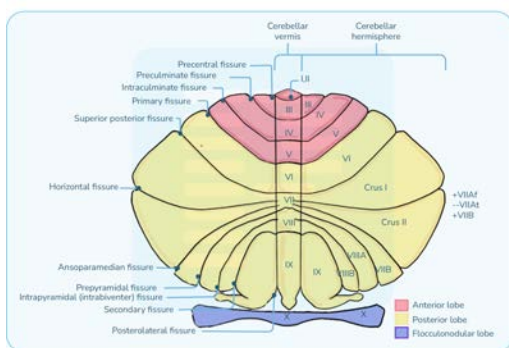
- Lobes: Anterior Lobe: Located superiorly and anteriorly, it is involved in the coordination of voluntary movements and posture. Posterior Lobe: Positioned between the primary fissure and the posterior cerebellar notch, it plays a crucial role in the planning and execution of complex movements. Flocculonodular Lobe: Located at the inferior part of the cerebellum, it is crucial for maintaining balance and coordinating eye movements.
- Anterior Lobe: Located superiorly and anteriorly, it is involved in the coordination of voluntary movements and posture.
- Posterior Lobe: Positioned between the primary fissure and the posterior cerebellar notch, it plays a crucial role in the planning and execution of complex movements.
- Flocculonodular Lobe: Located at the inferior part of the cerebellum, it is crucial for maintaining balance and coordinating eye movements.
- Lobules: The cerebellum is divided into lobules by fissures. Major lobules include: Lobule I–V: Part of the anterior lobe. Lobule VI–VII: Part of the posterior lobe. Lobule VIII–IX: Located in the posterior lobe and flocculonodular lobe.
- The cerebellum is divided into lobules by fissures. Major lobules include: Lobule I–V: Part of the anterior lobe. Lobule VI–VII: Part of the posterior lobe. Lobule VIII–IX: Located in the posterior lobe and flocculonodular lobe.
- Lobule I–V: Part of the anterior lobe.
- Lobule VI–VII: Part of the posterior lobe.
- Lobule VIII–IX: Located in the posterior lobe and flocculonodular lobe.
- Cortex: The cerebellar cortex is organized into three layers: Molecular Layer: Contains the dendritic trees of Purkinje cells and interneurons like stellate and basket cells. Purkinje Cell Layer: Contains Purkinje cells, which are the primary output neurons of the cerebellar cortex, sending inhibitory signals to the deep cerebellar nuclei. Granular Layer: Contains granule cells, which receive input from the mossy fibers and send excitatory projections to the

Purkinje cells.

- The cerebellar cortex is organized into three layers: Molecular Layer: Contains the dendritic trees of Purkinje cells and interneurons like stellate and basket cells. Purkinje Cell Layer: Contains Purkinje cells, which are the primary output neurons of the cerebellar cortex, sending inhibitory signals to the deep cerebellar nuclei. Granular Layer: Contains granule cells, which receive input from the mossy fibers and send excitatory projections to the Purkinje cells.
- Molecular Layer: Contains the dendritic trees of Purkinje cells and interneurons like stellate and basket cells.
- Purkinje Cell Layer: Contains Purkinje cells, which are the primary output neurons of the cerebellar cortex, sending inhibitory signals to the deep cerebellar nuclei.
- Granular Layer: Contains granule cells, which receive input from the mossy fibers and send excitatory projections to the Purkinje cells.
- Nuclei: The deep cerebellar nuclei are the primary output nuclei of the cerebellum and include dentate nucleus, emboliform nucleus, globose nucleus, and fastigial nucleus .
- The deep cerebellar nuclei are the primary output nuclei of the cerebellum and include dentate nucleus, emboliform nucleus, globose nucleus, and fastigial nucleus .
- Anterior Lobe: Located superiorly and anteriorly, it is involved in the coordination of voluntary movements and posture.
- Posterior Lobe: Positioned between the primary fissure and the posterior cerebellar notch, it plays a crucial role in the planning and execution of complex movements.
- Flocculonodular Lobe: Located at the inferior part of the cerebellum, it is crucial for maintaining balance and coordinating eye movements.
- The cerebellum is divided into lobules by fissures. Major lobules include: Lobule I–V: Part of the anterior lobe. Lobule VI–VII: Part of the posterior lobe. Lobule VIII–IX: Located in the posterior lobe and flocculonodular lobe.
- Lobule I–V: Part of the anterior lobe.
- Lobule VI–VII: Part of the posterior lobe.
- Lobule VIII–IX: Located in the posterior lobe and flocculonodular lobe.
- Lobule I–V: Part of the anterior lobe.
- Lobule VI–VII: Part of the posterior lobe.
- Lobule VIII–IX: Located in the posterior lobe and flocculonodular lobe.
- The cerebellar cortex is organized into three layers: Molecular Layer: Contains the dendritic trees of Purkinje cells and interneurons like stellate and basket cells. Purkinje Cell Layer: Contains Purkinje cells, which are the primary output neurons of the cerebellar cortex, sending inhibitory signals to the deep cerebellar nuclei. Granular Layer: Contains granule cells, which receive input from the mossy fibers and send excitatory projections to the

Purkinje cells.

- **Molecular Layer:** Contains the dendritic trees of Purkinje cells and interneurons like stellate and basket cells.
- **Purkinje Cell Layer:** Contains Purkinje cells, which are the primary output neurons of the cerebellar cortex, sending inhibitory signals to the deep cerebellar nuclei.
- **Granular Layer:** Contains granule cells, which receive input from the mossy fibers and send excitatory projections to the Purkinje cells.
- **Molecular Layer:** Contains the dendritic trees of Purkinje cells and interneurons like stellate and basket cells.
- **Purkinje Cell Layer:** Contains Purkinje cells, which are the primary output neurons of the cerebellar cortex, sending inhibitory signals to the deep cerebellar nuclei.
- **Granular Layer:** Contains granule cells, which receive input from the mossy fibers and send excitatory projections to the Purkinje cells.
- The deep cerebellar nuclei are the primary output nuclei of the cerebellum and include dentate nucleus, emboliform nucleus, globose nucleus, and fastigial nucleus .



The anterior and posterior lobes of the cerebellum are primarily involved in the coordination of voluntary movements and fine motor control. Posture and balance are more closely associated with the flocculonodular lobe and its connections. (Option A)

The cerebellum is actually divided into three main lobes: the anterior, posterior, and flocculonodular lobes. The flocculonodular lobe primarily contributes to balance and eye movements rather than fine motor skills. (Option B)

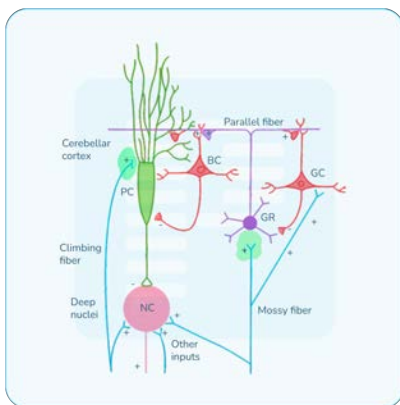
Granule cells in the cerebellar cortex are not the primary output neurons. Instead, Purkinje cells are the main output neurons, sending inhibitory signals to the deep cerebellar nuclei. Granule cells send excitatory input to Purkinje cells. (Option D)

Solution for Question 5:

Correct Answer: B) Provide excitatory input to granule cells

Explanation:

- Mossy fibers provide excitatory input to the granule cells in the cerebellar cortex. Granule cells, in turn, project their axons as parallel fibers, which make excitatory synapses on the Purkinje cells.
- The following components of the cerebellar internal circuit integrate sensory and motor information to fine-tune motor activities and support motor learning.



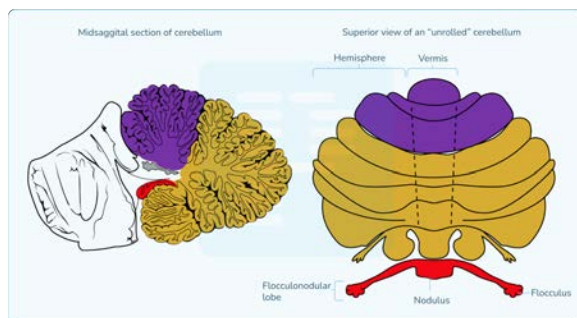
- Mossy fibers do not directly transmit motor commands from the cerebral cortex.
- They carry sensory and other inputs to the cerebellum, which then processes and integrates these inputs to help coordinate motor commands. (Option A)
- Mossy fibers do not relay sensory information directly to the deep cerebellar nuclei.
- They first synapse in the granule cell layer of the cerebellar cortex. From there, the processed information is relayed to the deep cerebellar nuclei via Purkinje cells. (Option C)
- Inhibitory control of the deep cerebellar nuclei is primarily mediated by Purkinje cells, which send inhibitory signals to the deep cerebellar nuclei.
- Mossy fibers provide excitatory input to the cerebellar cortex, which indirectly influences the activity of the deep nuclei. (Option D)

Solution for Question 6:

Correct Answer: D) Flocculonodular lobe coordinates fine motor movements

Explanation:

The flocculonodular Lobe is a part of the vestibulocerebellum , which is important for balance and eye movements , not fine motor control.



Functional Division	Function	Description	Location
Spinocerebellum	Coordination of axial and proximal limb movements (Option B)	Coordinates and smoothens voluntary movements, particularly those involving the axial body and proximal limbs.	Central portion of the cerebellum
Cerebrocerebellum	Planning and programming of movements (Option C)	Involved in motor planning and timing, especially for complex, precise movements.	Lateral parts of the cerebellar hemispheres
Vestibulocerebellum	Balance and eye movements (Option A)	Modulates motor commands to maintain posture and balance. Integrates information from the vestibular system to coordinate head and eye movements.	Flocculonodular lobe
Flocculonodular Lobe	Balance and eye movements	Part of the vestibulocerebellum, responsible for balance and equilibrium of the body.	Flocculonodular lobe

Solution for Question 7:

Correct Answer: A) Hypertonia

Explanation: Hypotonia is a characteristic of cerebellar diseases, not hypertonia. It occurs on the same side as the cerebellar lesion.

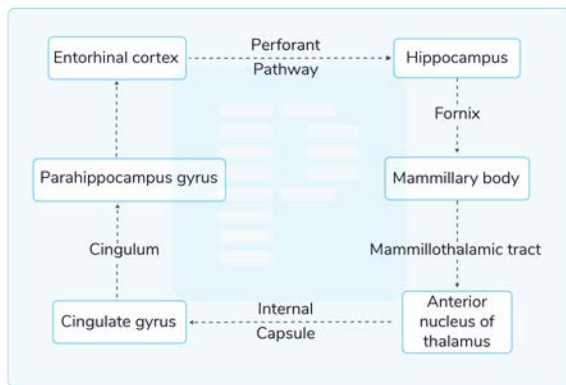
Clinical Features of Cerebellar Diseases

Hypotonia	Decreased muscle tone, often on the same side as the cerebellar lesion.
Ataxia (Option C)	Uncoordinated movements, often leading to a "drunken" gait.
Dysdiadochokinesia	Inability to perform rapid alternating movements, such as flipping the hands back and forth.
Dysmetria (Past pointing)	Movements that overshoot or undershoot the intended target.
Intention tremor	Tremors occurring when moving toward a target
Dysarthria (Option B)	Jumbled speech due to lack of coordination in the muscles involved in speech
Nystagmus	Rapid, tremulous eye movements
Rebound Phenomenon (Option D)	Limb jerking back after resistance is suddenly removed
Decomposition of movement	Can't move multiple joints simultaneously.
No sensory deficit	Not typically present in cerebellar diseases.

Solution for Question 8:

Correct Answer: B) 1 - B, 2 - D, 3 - A, 4 - C

Explanation: The Papez circuit is a neurocircuitry of the limbic system involved in learning and memory, emotion, and social behavior.



It has the following components:

Part of Papez Circuit	Function
Cingulate Gyrus	Controls heart rate, blood pressure, cognitive and emotional processing.
Parahippocampal Gyrus and Entorhinal Cortex	Spatial memory, the memory of three-dimensional space.
Hippocampus	Responsible for the conversion of short-term memory to long-term memory.
Mammillary Body and Anterior Nucleus of Thalamus	Responsible for controlling recent memories of the brain.

Solution for Question 9:

Correct Answer: C) Damage can result in a loss of sexual behaviour.

Explanation:

- Damage to the amygdala typically results in hypersexuality, not a loss of sexual behavior, as seen in Kluver-Bucy syndrome.
- Amygdala is part of limbic system central, controlling emotions and motivations. It is connected with the prefrontal cortex & hippocampus via amygdala circuitry for the acquisition, consolidation, and retrieval of memories associated with emotions. (Option A and B)

Damage to the amygdala typically results in hypersexuality, not a loss of sexual behavior, as seen in Kluver-Bucy syndrome.

Amygdala is part of limbic system central, controlling emotions and motivations. It is connected with the prefrontal cortex & hippocampus via amygdala circuitry for the acquisition, consolidation, and retrieval of memories associated with emotions. (Option A and B)

Amygdala Circuit

Part of the Amygdala Circuit	Function
Amygdala	Controls emotions and motivations.
Prefrontal Cortex	Responsible for the development of working memory (Option D)
Hippocampus	Converts short-term memory to long-term memory

Solution for Question 10:

Correct Answer: B) Amygdala

Explanation:

The case presents a patient with classic symptoms of Klüver-Bucy syndrome , including a lack of fear, oral tendencies, and hypersexuality.

- It is a rare behavioral disorder that can occur after damage to the anterior temporal lobes, including the amygdalae , which are almond-shaped groups of nuclei located deep within the temporal lobes.
- The amygdala is responsible for processing emotions, especially fear and reward.

The symptoms of Klüver-Bucy syndrome are varied but can include the following:

- Lack of Fear : Individuals may show no fear of objects or situations that would normally elicit a fear response.
 - Excessive Curiosity : They may display an intense interest in exploring their surroundings, often in a non-directed and inappropriate manner.
 - Memory Impairment : Short-term memory is often significantly affected, leading to an inability to recall recent events.
 - Oral Tendencies : A tendency to put objects in their mouth, even those that are not edible.
 - Hypersexuality : An increased and often inappropriate sexual drive.
-

Solution for Question 11:

Correct Answer: A) Medial preoptic nuclei

Explanation:

Medial Preoptic Nucleus is primarily responsible for thermoregulation in response to heat.

Anterior region of the Hypothalamus with functions:

Nucleus	Functions
Supraoptic	ADH secretion & a small amount of Oxytocin
Para-ventricular	Oxytocin, TRH, CRH, ANS activity regulations & a small amount of ADH
Preoptic • Medial • Ventrolateral (VLPO)	• GnRH, Thermoregulation (responses to Hot) • Sleep-promoting area (GABA)
Anterior	Thermoregulation (Hot), Sexual behavior
Supra-chiasmatic	Controls biological rhythm (circadian)
Lateral	Controls thirst and hunger

Solution for Question 12:

Correct Answer: D) Ventromedial hypothalamus

Explanation:

The ventromedial hypothalamus plays a crucial role in inhibiting rage responses generated by other parts of the limbic system, such as the periventricular zone and lateral hypothalamus. Damage to the ventromedial hypothalamus can lead to disinhibition of these rage centers, resulting in the aggressive behavior exhibited by the patient. Additionally, the ventromedial hypothalamus is involved in promoting feelings of satiety and tranquility, while damage to this area can lead to overeating and hyperactivity.

Centres	Location	Function/Response
Primary Reward Cent	Medial forebrain bundle	Lateral hypothalamus: Strong stimulation leads to rage; weaker

ers	Lateral and ventromedial nuclei of the hypothalamus	stimulation induces reward Plays a dual role in both aggressive and rewarding behaviours
Primary Punishment Centers	The central grey area surrounding the aqueduct of Sylvius in the mesencephalon	Elicits potent aversive responses
Primary Rage Centers	The periventricular zone of the hypothalamus Lateral hypothalamus	Intense stimulation triggers rage responses: Defensive posturing, aggression, autonomic responses like piloerection and pupil dilation
Inhibitory Mechanisms	Ventromedial nuclei of the hypothalamus Hippocampi Anterior limbic cortex: Anterior cingulate gyri and subcallosal gyri	Suppress or inhibit rage responses Maintain emotional control and regulate aggressive behaviors

Hippocampus (Option A) – Primarily involved in memory and learning; damage causes memory deficits rather than aggression.

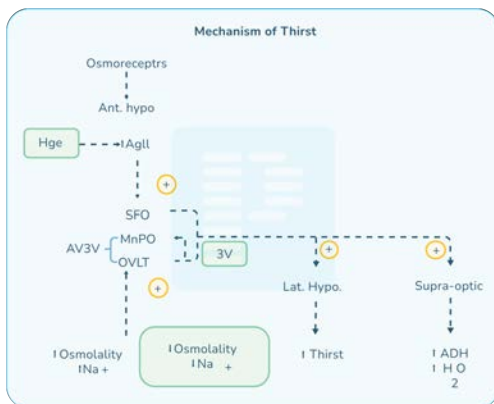
Medial forebrain bundle (Option B) – Associated with reward and motivation, not directly responsible for aggressive outbursts.

Central gray area (Option C) – Plays a role in pain modulation and defensive behaviors, but not in spontaneous, unprovoked aggression.

Solution for Question 13:

Correct Answer: B) Angiotensin II directly stimulates the lateral nucleus of the hypothalamus

Explanation:



The hypothalamus primarily regulates the thirst mechanism, and the lateral nucleus houses the thirst centre.

- The osmoreceptors in the hypothalamus These three osmoreceptors Median Preoptic Nucleus (MnPO), Organum Vasculosum Laminae Terminalis (OVLT), and Sub Fornical Nucleus (SFO) play key roles in detecting changes in blood osmolality or sodium ion concentration. An increase in blood osmolality or sodium ions stimulates the OVLT, which then activates MnPO through interconnected pathways. It does not directly stimulate the supra-optic nucleus (Option A)
- These three osmoreceptors Median Preoptic Nucleus (MnPO), Organum Vasculosum Laminae Terminalis (OVLT), and Sub Fornical Nucleus (SFO) play key roles in detecting changes in blood osmolality or sodium ion concentration.
- An increase in blood osmolality or sodium ions stimulates the OVLT, which then activates MnPO through interconnected pathways. It does not directly stimulate the supra-optic nucleus (Option A)
- These three osmoreceptors Median Preoptic Nucleus (MnPO), Organum Vasculosum Laminae Terminalis (OVLT), and Sub Fornical Nucleus (SFO) play key roles in detecting changes in blood osmolality or sodium ion concentration.
- An increase in blood osmolality or sodium ions stimulates the OVLT, which then activates MnPO through interconnected pathways. It does not directly stimulate the supra-optic nucleus (Option A)
- Response to isotonic loss of fluid (e.g. severe haemorrhage) Severe hemorrhage results in isotonic fluid loss from the blood, triggering an increase in renin secretion. Renin secretion leads to elevated levels of angiotensin 2 (Ag II) , which acts as a stimulus for the SFO. Despite the blood osmolality remaining constant during haemorrhage, the SFO can still be stimulated by the presence of angiotensin 2. (Option B)
- Severe hemorrhage results in isotonic fluid loss from the blood, triggering an increase in renin secretion.

- Renin secretion leads to elevated levels of angiotensin 2 (Ag II) , which acts as a stimulus for the SFO.
- Despite the blood osmolality remaining constant during haemorrhage, the SFO can still be stimulated by the presence of angiotensin 2. (Option B)
- Severe hemorrhage results in isotonic fluid loss from the blood, triggering an increase in renin secretion.
- Renin secretion leads to elevated levels of angiotensin 2 (Ag II) , which acts as a stimulus for the SFO.
- Despite the blood osmolality remaining constant during haemorrhage, the SFO can still be stimulated by the presence of angiotensin 2. (Option B)
- Pathways of signals Signals from stimulated osmoreceptors primarily target two brain regions: Lateral nucleus of the hypothalamus, stimulating thirst and causing the person to feel thirsty. Supra-optic nucleus of the hypothalamus, which produces ADH (antidiuretic hormone). ADH acts on the kidneys to reabsorb water, thereby reducing osmolality or sodium ion concentration in the body.
- Signals from stimulated osmoreceptors primarily target two brain regions: Lateral nucleus of the hypothalamus, stimulating thirst and causing the person to feel thirsty. Supra-optic nucleus of the hypothalamus, which produces ADH (antidiuretic hormone).
- Lateral nucleus of the hypothalamus, stimulating thirst and causing the person to feel thirsty.
- Supra-optic nucleus of the hypothalamus, which produces ADH (antidiuretic hormone).
- ADH acts on the kidneys to reabsorb water, thereby reducing osmolality or sodium ion concentration in the body.
- Signals from stimulated osmoreceptors primarily target two brain regions: Lateral nucleus of the hypothalamus, stimulating thirst and causing the person to feel thirsty. Supra-optic nucleus of the hypothalamus, which produces ADH (antidiuretic hormone).
- Lateral nucleus of the hypothalamus, stimulating thirst and causing the person to feel thirsty.
- Supra-optic nucleus of the hypothalamus, which produces ADH (antidiuretic hormone).
- ADH acts on the kidneys to reabsorb water, thereby reducing osmolality or sodium ion concentration in the body.
- Lateral nucleus of the hypothalamus, stimulating thirst and causing the person to feel thirsty.
- Supra-optic nucleus of the hypothalamus, which produces ADH (antidiuretic hormone).

Stimulated osmoreceptors in the SFO increase the secretion of ADH. (Option C)

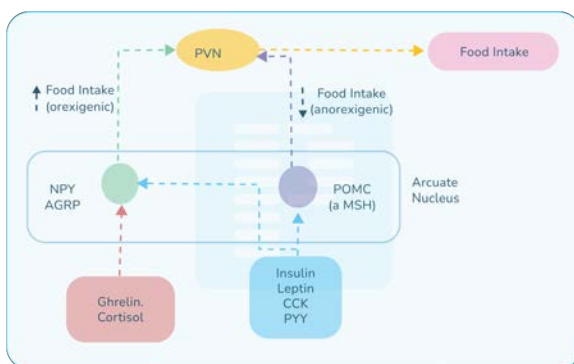
Solution for Question 14:

Correct Answer: D) Orexin peptide produced from the lateral nucleus of the hypothalamus plays an essential role in regulating sleep

Explanation:

Orexin peptide produced from the lateral nucleus of the hypothalamus plays an essential role in regulating sleep and also increases food intake.

Theory of control of feeding:



Arcuate Nucleus Overview:

- Located in the hypothalamus, it integrates orexigenic and anorexigenic signals to regulate feeding behaviour. (Option A)

Neuronal Types:

- POMC Neurons: Produce alpha-MSH, an anorexigenic signal that reduces food intake. (Option C)
- Produce alpha-MSH, an anorexigenic signal that reduces food intake. (Option C)
- NPY and AGRP Neurons: Produce orexigenic signals that stimulate the hypothalamic pre-ventricular nucleus, increasing food intake.
- Produce orexigenic signals that stimulate the hypothalamic pre-ventricular nucleus, increasing food intake.
- Produce alpha-MSH, an anorexigenic signal that reduces food intake. (Option C)
- Produce orexigenic signals that stimulate the hypothalamic pre-ventricular nucleus, increasing food intake.

Peripheral Signals:

- Anorexigenic Signals (reduce food intake): Insulin, leptin, CCK, and Peptide YY stimulate POMC neurons. (Option B)

- Insulin, leptin, CCK, and Peptide YY stimulate POMC neurons. (Option B)
- Orexigenic Signals (increase food intake): Ghrelin and cortisol stimulate NPY and AGRP neurons.
- Ghrelin and cortisol stimulate NPY and AGRP neurons.
- Insulin, leptin, CCK, and Peptide YY stimulate POMC neurons. (Option B)
- Ghrelin and cortisol stimulate NPY and AGRP neurons.

Leptin and Obesity:

- Leptin, indicating fat cell mass, acts as an anorexigenic signal in the hypothalamus.
- High leptin levels in obese individuals indicate a large fat cell mass, promoting reduced food intake.

Role of Orexin:

- Produced by the lateral nucleus of the hypothalamus, orexin regulates sleep and increases food intake.

Solution for Question 15:

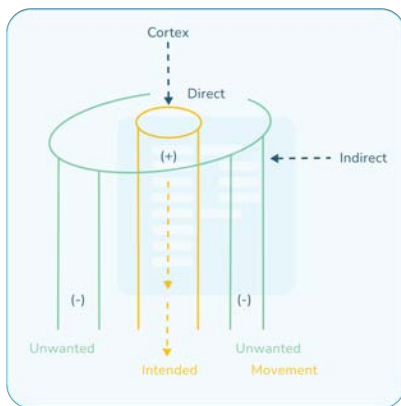
Correct Answer: B) The direct pathway facilitates specific motor actions, while the indirect pathway suppresses competing actions.

Explanation:

Center-Surround Model:

The centre-surround model describes how the basal ganglia circuitry provides focused activation (centre) while suppressing surrounding activity (surround) to enhance motor control and precision.

- Centre: The direct pathway acts as a "centre" mechanism by facilitating specific motor actions and enhancing desired movements. It reduces inhibitory signals from the GPi/SNr to the thalamus, thereby increasing thalamic activity related to the motor cortex.
- Surround: The indirect pathway acts as a "surround" mechanism by inhibiting unwanted movements and refining motor control. It increases inhibition in broader motor regions by enhancing the inhibition of the GPi/SNr, thus decreasing thalamic activation.



- The direct pathway does not broadly inhibit motor actions; it facilitates specific actions by reducing inhibition from the GPi. The indirect pathway does not facilitate actions; it suppresses competing actions by increasing inhibition. (Option A)
- The indirect pathway increases inhibitory output from the GPi, not reduces it. The direct pathway reduces inhibitory output from the GPi. (Option C)
- The pathways do not uniformly facilitate all motor actions. (Option D)

Solution for Question 16:

Correct Answer: A) Climbing and mossy fibers provide the primary input to the cerebellar cortex.

Explanation :

Climbing fibers and mossy fibers are the main afferent pathways that bring excitatory input to the cerebellar cortex.

Afferent Fibers to Cerebellar Cortex

Source/Tracts	Fiber Type	Neurotransmitter	Function
Olivocerebellar tracts	Climbing Fibers	Aspartate	Proprioceptive input from the entire body
Vestibulocerebellar	Mossy Fibers	Glutamate	Receives information from the labyrinth.
Spinocerebellar	Mossy Fibers	Glutamate	Receives information from the Muscle spindle, the Golgi Tendon Organ.

Cuneocerebellar	Mossy Fibers	Glutamate	Receives information from the Muscle spindle, Golgi Tendon Organ.
Tectocerebellar	Mossy Fibers	Glutamate	Receives information from visual & auditory.
Pontocerebellar	Mossy Fibers	Glutamate	Receives information from the motor & other parts of the cerebral cortex.

Special Senses

1. Which of the following cells in the olfactory bulb lacks an axon?

- A. Periglomerular cells
 - B. Mitral cells
 - C. Tufted cells
 - D. Granule cell
-

2. Which of the following statements accurately describes the pathway of taste perception from the tongue to the brain?

- A. Taste signals from the taste buds are first transmitted to the gustatory cortex via CN VII, IX & X
 - B. The taste receptors generate nerve impulses relayed to the nucleus of the solitary tract in the medulla.
 - C. From the nucleus of the solitary tract, taste signals travel to the ventral posterolateral nucleus of the thalamus
 - D. The gustatory cortex, responsible for taste perception, is located in the medulla oblongata,
-

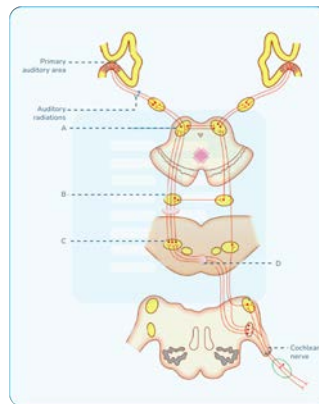
3. Which of the following regions is responsible for the perception of smell?

- A. Piriform cortex
- B. Entorhinal cortex
- C. Amygdala
- D. Anterior olfactory nucleus

4. A 38-year-old man presents to the ophthalmology clinic with complaints of sensitivity to light and occasional blurriness in his vision. During the physical examination, When light is shone into his right eye, both pupils constrict normally and When light is shone into his left eye, only the left pupil constricts. Given these observations, which of the following connections of the pathway is most likely affected?

- A. Ciliary ganglion and sphincter pupillae
 - B. Edinger-Westphal nucleus and ciliary ganglion
 - C. pretectal nucleus and Edinger-Westphal nucleus
 - D. optic tract and pretectal nucleus
-

5. Identify the labelled structures A, B, C, and D in the auditory pathway diagram.



- A. A: Inferior colliculus, B: Superior olivary nucleus, C: Dorsal cochlear nucleus, D: Medial geniculate body
- B. A: Inferior Colliculus, B: Lateral lemniscus, C: Superior olivary nucleus, D: Ventral cochlear nucleus
- C. A: Inferior Colliculus, B: Lateral Lemniscus, C: Superior olivary nucleus, D: Trapezoid Body

D. A: Primary auditory area, B: Lateral lemniscus, C: Superior olivary nucleus, D: Spiral ganglion

6. Which of the following statements about colour vision abnormalities is incorrect?

- A. Protanopia is the loss of L cones, resulting in green-blue vision only
 - B. Deuteranopia is the loss of M cones, resulting in red-green vision only.
 - C. Loss of L and M cones has X-linked inheritance, occurring in about 1% of males.
 - D. Loss of S cones is rare and has autosomal inheritance via chromosome 7.
-

7. Which of the following statements is correct regarding the structure of the cochlea?

- A. The cochlea is a part of the outer ear and contains the ossicles.
 - B. The cochlea is divided into three fluid-filled compartments: the scala vestibuli, scala media, and scala tympani.
 - C. The organ of Corti, which contains hair cells responsible for hearing, is located in the scala vestibuli.
 - D. The basilar membrane separates the scala media from the scala tympani and is responsible for transmitting sound waves directly to the auditory nerve.
-

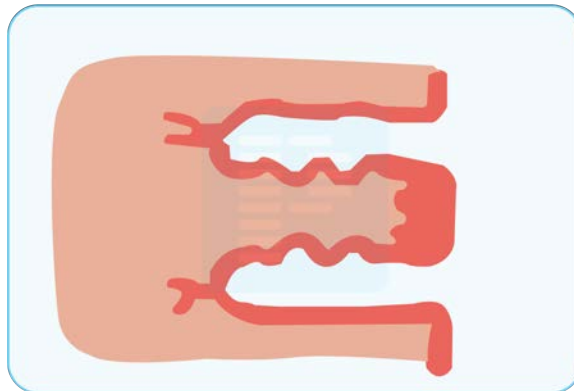
8. Match the following types of taste with their respective mechanisms. Type of Taste Mechanism 1) Salt A) Mediated by H⁺ ions. 2) Sour B) Mediated by T1R2 and T1R3 GPCRs 3) Sweet C) Mediated by T1R1 and T1R3 GPCRs 4) Bitter D) Mediated by Na⁺-selective channels. 5) Umami E) Mediated by T2R family GPCRs.

Type of Taste	Mechanism
1) Salt	A) Mediated by H ⁺ ions.

2) Sour	B) Mediated by T1R2 and T1R3 GPCRs
3) Sweet	C) Mediated by T1R1 and T1R3 GPCRs
4) Bitter	D) Mediated by Na ⁺ -selective channels.
5) Umami	E) Mediated by T2R family GPCRs.

- A. 1 - D, 2 - A, 3 - B, 4 - E, 5 - C
- B. 1 - D, 2 - A, 3 - E, 4 - B, 5 - C
- C. 1 - D, 2 - A, 3 - C, 4 - E, 5 - B
- D. 1 - D, 2 - E, 3 - B, 4 - C, 5 - A

9. What is the location and the number of taste buds approximately present in the image given below



- A. Up to 100, two lateral ends of the tongue
- B. Up to 5, base of the tongue
- C. Up to 100, tip of the tongue
- D. Up to 25, base of the tongue

10. In the visual pathway, which of the following correctly identifies the first, second, and third-order neurons and their respective locations?

- A. First-order: Bipolar cells in the retina; Second-order: Ganglion cells in the retina; Third-order: Neurons in the LGN of the thalamus.
 - B. First-order: Photoreceptor cells in the retina; Second-order: Bipolar cells in the retina; Third-order: Ganglion cells in the retina.
 - C. First-order: Bipolar cells in the retina; Second-order: Ganglion cells in the retina; Third-order: Neurons in the superior colliculus.
 - D. First-order: Photoreceptor cells in the retina; Second-order: Ganglion cells in the retina; Third-order: Neurons in the LGN of the thalamus.
-

11. What is the primary mechanism by which an odorant molecule leads to the depolarisation of an olfactory cell?

- A. close sodium ion channels
 - B. Triggers the breakdown of cyclic adenosine monophosphate
 - C. Initiating a signalling cascade that ultimately opens sodium ion channels.
 - D. The odorant molecule is transported into the olfactory cell, where it directly depolarises
-

12. Which of the following statements about bipolar sensory neurons involved in the olfactory pathway is incorrect?

- A. Only first and second-order neurons are sufficient to reach the level of the cerebral cortex
 - B. Bypasses the thalamus and directly enters the cortex
 - C. Each bipolar sensory neuron has 6-12 cilia at the top of the olfactory epithelium
 - D. Olfactory bulb is the site where first-order neurons synapse with bipolar sensory neurons
-

13. Arrange in order the mechanism of sound transmission i) Depolarization of hair cells ii) Bending of stereocilia iii) Production of neurotransmitter glutamate iv) K & Ca enter via the channel

- A. iv, iii, ii, i
 - B. iv, i, ii, iii
 - C. ii, i, iii, iv
 - D. ii, iv, i, iii
-

14. Which of the following statements accurately describes the difference between outer and inner hair cells in the human cochlea?

- A. Outer hair cells primarily send auditory signals to the brain, while inner hair cells are involved in amplifying sound.
 - B. Inner hair cells are responsible for sending the majority of auditory information to the brain, whereas outer hair cells serve to amplify and fine-tune the sound vibrations.
 - C. Both outer and inner hair cells have the same function of sending auditory signals to the brain.
 - D. Outer hair cells are fewer in number compared to inner hair cells and play a critical role in sound transmission.
-

15. A 25-year-old woman visits her ophthalmologist complaining of difficulty seeing in low-light conditions. She reports that her vision is fine during the day but deteriorates significantly at night. Based on your understanding of the visual pathway mechanisms in light versus dark conditions, which of the following best explains her symptoms?

- A. Increased activity of rod cells in bright light.
- B. Decreased neurotransmitter release from photoreceptors in the dark.

- C. Increased depolarisation of photoreceptors in the presence of light.
 - D. Decreased cGMP levels in photoreceptors during darkness.
-

16. Which of the following statements correctly differentiates between the structure of rods and cones in the human retina?

- A. Rods contain the photopigment iodopsin, whereas cones contain rhodopsin.
 - B. Rods are primarily responsible for colour vision, whereas cones are responsible for night vision.
 - C. Rods have a longer outer segment with more disk-like structures, whereas cones have a shorter outer segment with fewer disk-like structures.
 - D. Rods are concentrated in the fovea, whereas cones are distributed throughout the retina.
-

17. Which of the following best explains why rods are more sensitive to low light than cones?

- A. Individual connections to ganglion cells
 - B. High convergence onto ganglion cells
 - C. Lower density in the retina
 - D. Larger size of photoreceptors
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	1
Question 4	3
Question 5	3
Question 6	2
Question 7	2
Question 8	3
Question 9	1
Question 10	4
Question 11	3
Question 12	4
Question 13	4
Question 14	2
Question 15	2
Question 16	3
Question 17	2

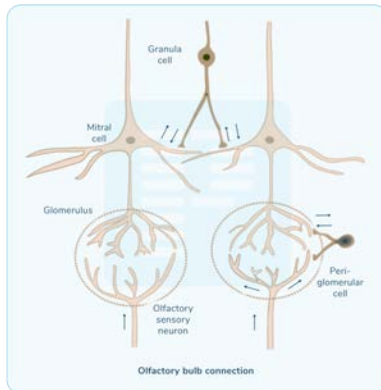
Solution for Question 1:

Answer: (D) Granule cell

Explanation:

Granule cells are inhibitory neurons present in the olfactory bulb. Their unique characteristic is the loss of axons, but have multiple branched-out dendrites.

Olfactory bulb connection:



The synapse between the bipolar neuron and the second-order neuron occurs within the olfactory bulb.

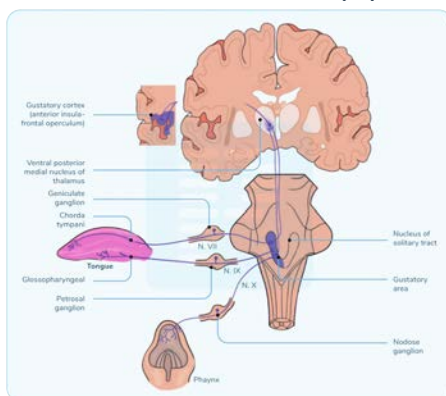
- The second-order neuron belongs to the categories of mitral and tufted cells , which serve as projection neurons.
- The specialised structure where these connections occur is termed glomeruli within the olfactory bulb.
- Between the first-order (bipolar) and second-order (mitral and tufted) neurons, two additional cell types exist: Periglomerular cells function as inhibitory neurons. Granule cells are also inhibitory neurons ; they lack axons but possess dendrites, enabling dendrodendritic synapses.
- Periglomerular cells function as inhibitory neurons.
- Granule cells are also inhibitory neurons ; they lack axons but possess dendrites, enabling dendrodendritic synapses.
- Both types of inhibitory neurons play roles in refining and concentrating olfactory signals.
- Periglomerular cells function as inhibitory neurons.
- Granule cells are also inhibitory neurons ; they lack axons but possess dendrites, enabling dendrodendritic synapses.

Solution for Question 2:

Answer: (B) The taste receptors generate nerve impulses relayed to the nucleus of the solitary tract in the medulla.

Explanation:

- **Taste Receptors on Tongue:** The journey begins with the taste receptors on the tongue, responsible for detecting the five primary tastes: sour, salty, sweet, bitter, and umami. These receptors are located in taste buds.
- **Cranial Nerves:** Taste signals are transmitted from the taste buds to the brain stem via three cranial nerves: the facial nerve (VII), the glossopharyngeal nerve (IX), and the vagus nerve (X).
- **Nucleus of the Solitary Tract in the Medulla:** These cranial nerves carry the taste information to a specific nucleus in the medulla oblongata called the nucleus of the solitary tract. (Option A)
- **Ventral Posterior Medial Nucleus of the Thalamus:** From the nucleus of the solitary tract, the taste signals are relayed to the ventral posterior medial nucleus of the thalamus. (Option C)
- **Gustatory Cortex:** Finally, the taste signals are transmitted from the thalamus to the gustatory cortex, the primary area responsible for processing taste perception in the brain. This region is located in the lower portion of the postcentral gyrus, extending into a part of the insular cortex. (Option D)



Solution for Question 3:

Answer: (A) Piriform cortex

Explanation:

Regions of the olfactory cortex are

olfactory cortex" data-author="NA" data-hash="579" data-license="NA" data-source="NA" data-tags="January2025" height="456" src="https://image.prepladder.com/notes/tNJnu5M5L1VDAPiraNdx1736095211.png" width="847" />

- Anterior olfactory nucleus (Option D)
- Olfactory tubercle.
- Piriform cortex: Responsible for the perception of a smell. (Option A)
- Amygdala: Mediated the emotional responses to olfactory stimuli. (Option C)
- Entorhinal cortex: Concerned with olfactory memories . (Option B)

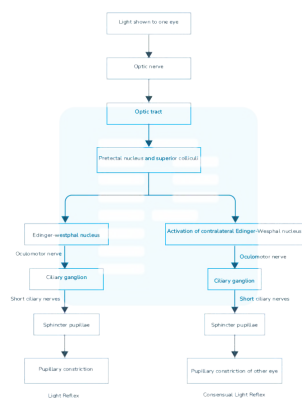
Sequence of events in Olfactory cortex:

- The second-order neuron travels to the cortex via the lateral olfactory tract.
- The lateral olfactory tract originates from the olfactory bulb, specifically from the glomeruli, and extends into different areas of the olfactory cortex.
- Some of these fibers project to the thalamus, which then relays signals to the orbitofrontal cortex.
- The orbitofrontal cortex is involved in the conscious perception and discrimination of odors.
- The pathway traverses the mediodorsal (dorsomedial) thalamic nucleus (MDT) and the orbitofrontal cortex, forming the new olfactory cortex together.

Solution for Question 4:

If there is a lesion between the pretectal nucleus and the contralateral Edinger-Westphal nucleus, the ability to coordinate the consensual light reflex between the two eyes will be impaired, resulting in failure of light reflex in the right eye when the light is shone on the left eye.

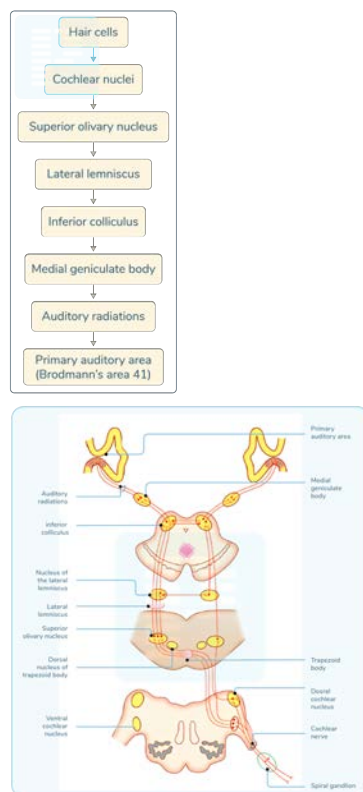
- This would result in the left pupil constricting when light is shone into the left eye (direct reflex intact) but failing to cause the right pupil to constrict (impaired consensual reflex) as supposedly the connections are impaired between the right Edinger-Westphal nucleus and the pretectal nucleus.



Solution for Question 5:

Explanation:

The marked structures correspond to the auditory pathway, shown in the image below.



Solution for Question 6:

Deuteranopia is the loss of M cones, resulting in loss of green colour perception and leading to red-blue vision only.

Color Vision Type	Cone Type affected	Perceived vision	Inheritance
Protanopia (Red defect) (Option A)	L Cones	Green-blue vision	X-linked
Deuteranopia (Green defect) (Option B)	M Cones	Red-blue vision	X-linked
Tritanopia (Blue defect) (Option D)	S Cones	Red-green vision	Autosomal (linked to chromosome 7)

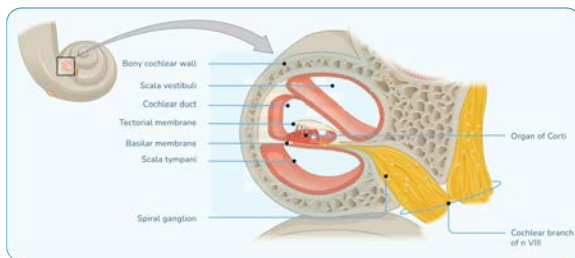
Solution for Question 7:

Answer: B. The cochlea is divided into three fluid-filled compartments: the scala vestibuli, scala media, and scala tympani.

Explanation:

- The cochlea is a spiral-shaped organ in the inner ear, essential for hearing.
- It consists of three main compartments: Scala vestibuli: The upper chamber filled with perilymph. Scala media (cochlear duct): The middle chamber filled with endolymph and contains the organ of Corti. Scala tympani: The lower chamber filled with perilymph.
- Scala vestibuli: The upper chamber filled with perilymph.
- Scala media (cochlear duct): The middle chamber filled with endolymph and contains the organ of Corti.
- Scala tympani: The lower chamber filled with perilymph.

- The organ of Corti is located in the scala media and contains hair cells that detect sound vibrations and convert them into electrical signals sent to the brain.
- Scala vestibuli: The upper chamber filled with perilymph.
- Scala media (cochlear duct): The middle chamber filled with endolymph and contains the organ of Corti.
- Scala tympani: The lower chamber filled with perilymph.



Option A. The cochlea is a part of the outer ear and contains the ossicles.

- The cochlea is part of the inner ear, not the outer ear.
- The ossicles (malleus, incus, and stapes) are located in the middle ear, not in the cochlea.

Option C. The organ of Corti, which contains hair cells responsible for hearing, is located in the scala vestibuli.

- The organ of Corti is located in the scala media, not the scala vestibuli.
- The scala vestibuli is one of the fluid-filled compartments but does not contain the organ of Corti.

Option D. The basilar membrane separates the scala media from the scala tympani and is responsible for transmitting sound waves directly to the auditory nerve.

- While the basilar membrane does separate the scala media from the scala tympani and plays a crucial role in sound wave propagation, it does not transmit sound waves directly to the auditory nerve.
- The hair cells on the basilar membrane convert mechanical vibrations into electrical signals, which are then transmitted to the auditory nerve by synaptic connections.

Solution for Question 8:

Humans have five established basic tastes: sweet, sour, bitter, salt, and umami.

- All these tastes are recognized by binding of the taste chemical to a protein receptor molecule that lies on the taste receptor cell to initiate the receptor potential.
- Each taste has a different mechanism for triggering receptor potential.

Type of Taste	Receptor Type	Mechanism
Salt	Ionotropic Receptor (ENaC)	Mediated by Na ⁺ -selective channels
Sour	Ionotropic Receptor (H ⁺)	Mediated by H ⁺ ions.
Sweet	Metabotropic Receptor (GPCR)	Mediated by T1R2 and T1R3 GPCRs.
Bitter	Metabotropic Receptor (GPCR)	Mediated by T2R family GPCRs.
Umami	Metabotropic Receptor (GPCR)	Mediated by T1R1 and T1R3 GPCRs

Solution for Question 9:

Answer: (A) Up to 100, two lateral ends of the tongue

Explanation:

The image above describes a foliate papillae present the two lateral ends of the tongue and has upto 100 taste buds.

Types of Papilla in tongue:

Papilla in tongue" data-author="NA" data-hash="574" data-license="NA" data-source="NA" data-tags="January2025" height="279" src="https://image.prepladder.com/notes/5KO5G0W9rF1TfrwvbOb1736093721.png" width="748" />

Papilla in tongue-diagramatic" data-author="NA" data-hash="575" data-license="NA" data-source="NA" data-tags="January2025" height="496" src="https://image.prepladder.com/notes/DlcUg0abl71Ju4T3Qx1w1736093760.png" width="667" />

Solution for Question 10:

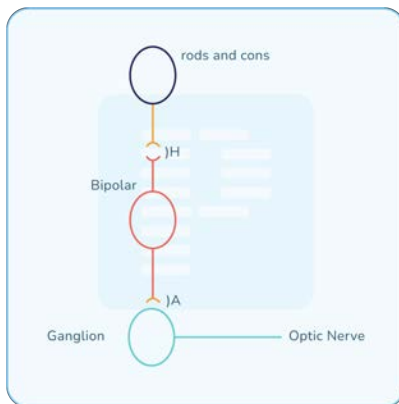
Answer: D) First-order: Photoreceptor cells in the retina; Second-order: Ganglion cells in the retina; Third-order: Neurons in the LGN of the thalamus.

Explanation:

Neurons of Visual Pathway:

- First-order neurons : Photoreceptor cells (rods and cones) in the retina.
- Second-order neurons : Ganglion cells in the retina.

Third-order neurons : Neurons in the thalamus's lateral geniculate nucleus (LGN).



Neuronal Cell Types in the Retina:

	Function
Photoreceptors (Rods and cones)	Transmit signals to the outer plexiform layer. They connect with bipolar and horizontal cells.
Horizontal Cells	Transmit signals horizontally in the outer plexiform layer. Connect rods and cones to bipolar cells.
Bipolar Cells	Transmit signals vertically from rods, cones, and horizontal cells. Connect with ganglion cells and amacrine cells in the inner plexiform layer.
Amacrine Cells	Transmit signals in two ways: directly from bipolar cells to ganglion cells or horizontally within the inner plexiform layer. Connect axons of bipolar cells to dendrites of ganglion cells or other amacrine cells.
Ganglion Cell	Transmit output signals from the retina to the brain via the optic nerve.

Interplexiform Cells	Transmit inhibitory signals retrogradely from the inner to the outer plexiform layer. Control the lateral spread of visual signals by horizontal cells, potentially enhancing visual contrast.
----------------------	---

Solution for Question 11:

Answer: (C) Initiating a signaling cascade that ultimately opens sodium ion channels.

Explanation:

Mechanism of Olfactory signaling:

Olfactory signaling" data-author="NA" data-hash="571" data-license="NA" data-source="NA" data-tags="January2025" height="444" src="https://image.prepladder.com/notes/1y2b7mCXnmpCbEZQrbxW1736092465.png" width="648" />

- Single cilia (as shown in the picture) have olfactory receptors that are G protein-coupled receptors.
- Odorants activate G protein, which increases cyclic AMP (cAMP) (Option B) as a second messenger.
- cAMP opens non-specific cation channels in the cilium, allowing calcium and sodium ions to enter, causing depolarization.
- Depolarization above a threshold (-30 mV) generates action potentials in bipolar sensory neurons.
- Action potentials are transmitted to the olfactory bulb.
- Calcium-gated chloride channels in the cilium further amplify depolarization.
- Chloride ions move out due to increased calcium, reducing negative charge and enhancing positivity inside the cell.
- This amplification process facilitates action potential generation and transmission to the olfactory bulb.

(Option A) Odorant molecules don't close sodium channels but They initiate a signaling cascade for that lead to opening of Na channel

(Option D) Odorant molecules don't need to enter the olfactory cell; binding to external receptors is sufficient to trigger the signaling cascade.

Solution for Question 12:

Answer: (D) Olfactory bulb is the site where first-order neurons synapse with bipolar sensory neurons.

Explanation: Bipolar sensory neurons are the first order neurons that synapses with the second order neuron in the olfactory bulb.

bipolar sensory neurons" data-author="NA" data-hash="570" data-license="NA" data-source="NA" data-tags="January2025" height="451" src="https://image.prepladder.com/notes/zczrmM64cJ1yo7bV2OTw1736091649.png" width="551" />

- Bipolar sensory neurons have dendrites and axons.
- Their axons extend through the cribriform plate of the ethmoid bone to the olfactory bulb.
- At the olfactory bulb, bipolar neurons synapse with second-order neurons at structures called glomeruli.
- Each bipolar neuron has 6-12 cilia containing olfactory receptors . (Option C)
- When activated by odorants, bipolar cells generate action potentials transmitted through second-order neurons to the cerebral cortex.
- Unlike other sensory pathways, the olfactory pathway requires only first and second-order neurons to reach the cerebral cortex directly, bypassing the thalamus. (Option A and B)

Other cells of olfactory epithelium include:

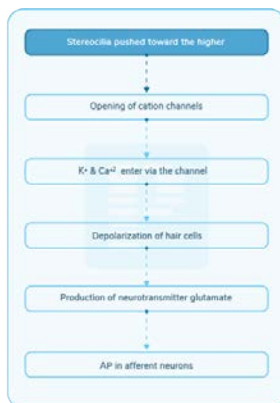
Cell type	Description
Bipolar sensory neuron	Approximately 50 million in number.
Supporting cells	Secrete mucous in which the cilia of the bipolar sensory neurons are embedded. Chemicals encountered by the nose (smell) are first dissolved in the mucous before being detected by the receptors.
Basal stem cells	Responsible for regenerating bipolar sensory neurons.

Solution for Question 13:

Answer: (D) ii, iv, i, iii

Explanation:

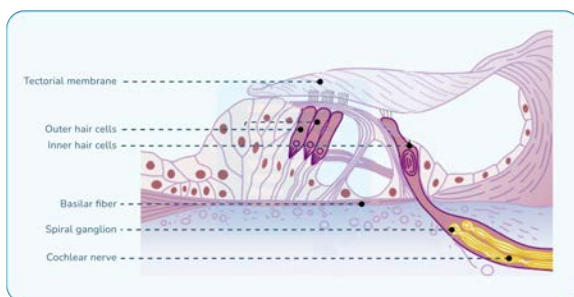
The organ of Corti, containing hair cells, are the sensory receptors for sound, required for the transmission of sound. When the basilar membrane vibrates, it causes the stereocilia (projections on the hair cells) to bend against the tectorial membrane. The bending of the stereocilia triggers a cascade of electrochemical events, ultimately generating nerve impulses in the auditory nerve fibers connected to the hair cells



Solution for Question 14:

Answer: B. Inner hair cells are responsible for sending the majority of auditory information to the brain, whereas outer hair cells serve to amplify and fine-tune sound vibrations■■■.

Explanation:



Outer hair cells (OHCs) and inner hair cells (IHCs).

Differences between outer and inner hair cells:

Feature	Outer Hair Cells (OHCs)	Inner Hair Cells (IHCs)
Rows	Three rows	Arranged in a single row
Structure	Cylindrical shape; contains prestin, a motor protein	Flask-shaped; stereocilia arranged in a linear or slightly curved pattern
Function	Amplify sound vibrations; enhance sensitivity and frequency selectivity	Convert sound vibrations into electrical signals sent to the brain
Mechanism	Change length (contract and elongate) in response to sound (electromotility)	Stereocilia deflection opens ion channels, causing depolarization and neurotransmitter release
Innervation	Predominantly efferent (motor) nerve fibers	Predominantly afferent (sensory) nerve fibers
Role in Hearing	Act as mechanical amplifiers, increasing the amplitude of low-level sounds	Primary sensory receptors for hearing; transmit auditory information to the brain
Damage Impact	Loss leads to reduced hearing sensitivity and clarity, especially for soft sounds	Damage results in significant hearing loss due to impaired signal transmission

Solution for Question 15:

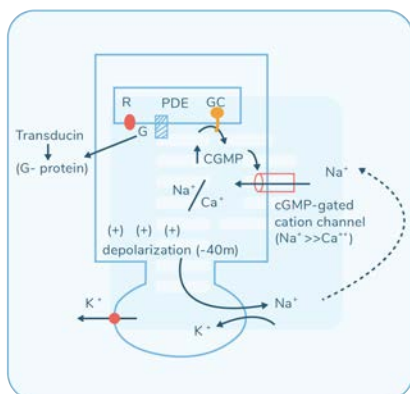
Answer: B. Decreased neurotransmitter release from photoreceptors in the dark.

Explanation:

In the dark:

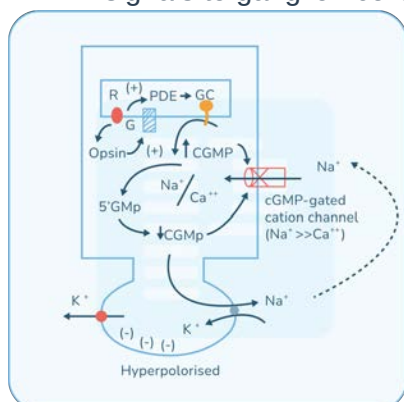
- Rods and cones are depolarised due to the high levels of cyclic guanosine monophosphate (cGMP).
- cGMP binds to and opens cGMP-gated sodium (Na^+) channels known as Funny Current Sodium Channels.
- Sodium ions (Na^+) flow into the photoreceptors, keeping them depolarised.
- Depolarised photoreceptors continuously release the neurotransmitter glutamate at their synaptic terminals.

- This glutamate release inhibits bipolar cells, preventing signal transmission to the ganglion cells and, subsequently, to the brain.



In light:

- Light photons are absorbed by photopigments (rhodopsin in rods, photopsins in cones).
- This absorption activates a G-protein called transducin.
- Transducin activates phosphodiesterase (PDE), an enzyme that breaks down cGMP.
- The decrease in cGMP levels causes cGMP-gated sodium channels to close.
- The closure of sodium channels leads to hyperpolarization of the photoreceptor membrane.
- Hyperpolarized photoreceptors reduce the release of glutamate.
- The decrease in glutamate release allows bipolar cells to become active, transmitting signals to ganglion cells and then to the brain, resulting in visual perception.



Option A. Rod cells are not primarily active in bright light; they are more sensitive to low light. In bright light, rod cells are saturated and do not function effectively.

Option C. Light exposure causes hyperpolarization of photoreceptors, not depolarization. Light reduces the intracellular cGMP levels, leading to the closure of sodium channels and hyperpolarization.

Option D. In darkness, cGMP levels are high, keeping the sodium channels open and the photoreceptors depolarized.

Solution for Question 16:

Answer: C) Rods have a longer outer segment with more disk-like structures, whereas cones have a shorter outer segment with fewer disk-like structures.

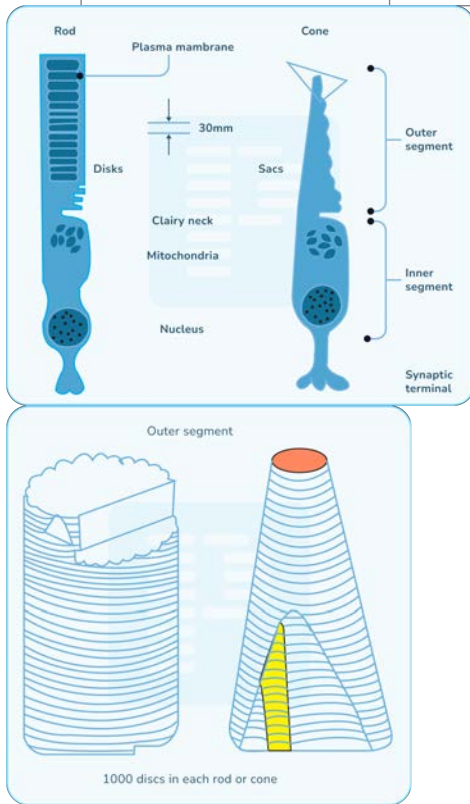
Explanation:

Structure and Function of Rods and Cones

- Rods and cones are the two main types of photoreceptor cells in the retina, each with unique structural adaptations that allow them to perform their specific roles in vision.
- Rods are optimised for night and peripheral vision, while cones enable detailed and colour vision in bright light. (Option B)
- Both rods and cones share four major structural segments, as explained: Outer segment
Inner segment Nucleus Synaptic body
- Outer segment
- Inner segment
- Nucleus
- Synaptic body

Feature	Rods	Cones
Outer Segment Shape (Option C)	Elongated and cylindrical	Shorter and tapering
Disk Structures (Option C)	Numerous separate disks	Fewer, continuous infoldings
Photopigment (Option A)	Rhodopsin	Photopsins (iodopsins)
Light Sensitivity	High (sensitive to low light)	Lower (sensitive to bright light)
Color Vision	No (monochromatic)	Yes (trichromatic)
Vision Type	Scotopic (night vision)	Photopic (daylight and color vision)
Distribution (Option D)	Throughout the retina, absent in the fovea	Concentrated in the fovea, less in the periphery

Synaptic Terminal	Spherule	Pedicle
Renewal of Disks	Continuous shedding and renewal	Continuous, slower renewal
Connection to Bipolar Cells	Many rods to one bipolar cell	One cone to one bipolar cell (in fovea)



Solution for Question 17:

Answer: B. High convergence onto ganglion cells

Explanation:

Convergence in Rods:

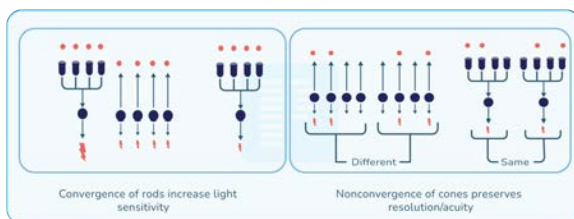
- Rods are highly convergent onto ganglion cells. This means that multiple rod cells synapse onto a single ganglion cell.

- This high level of convergence allows for the pooling of signals from many rods. Consequently, even small amounts of light can trigger a response because the signals from multiple rods are combined and sent as a single, stronger signal to the brain. This increases the sensitivity of rods to low light conditions.
- While convergence enhances sensitivity, it reduces spatial resolution. Since multiple rods converge onto a single ganglion cell, the specific location of light cannot be as accurately determined, leading to less detailed vision.

Non-Convergence in Cones:

- Cones generally have a more direct, one-to-one connection with ganglion cells, especially in the fovea.
- As a result, they are less sensitive to low light compared to rods. Each cone needs a stronger light stimulus to activate the associated ganglion cell.

The direct connection of cones to ganglion cells allows for high spatial resolution. Each cone contributes a distinct signal to the brain, enabling detailed and sharp vision.



Individual connections to ganglion cells (Option A): This is true for cones, not rods. Cones connect individually to ganglion cells, especially in the fovea, which provides high spatial resolution but lower sensitivity.

Lower density in the retina (Option C): Rods are more densely packed in the peripheral retina, which contributes to their sensitivity to low light, but the primary reason for their high sensitivity is their convergence onto ganglion cells.

Larger size of photoreceptors (Option D): The size of the photoreceptors does not significantly impact their sensitivity to light.

EEG & Sleep: Learning and Memory

1. Which of the following statements about EEG (Electroencephalogram) is incorrect?

- A. EEG measures the summated version of EPSP and IPSP generated in the brain.
- B. EEG records the dendritic potentials of pyramidal neurons.
- C. Positive dendritic potentials are known as EPSP (Excitatory Postsynaptic Potential) and negative dendritic potentials are known as IPSP (Inhibitory Postsynaptic Potential).
- D. D) Electrocorticography (ECOG) is the measurement of electrical activity specifically in the cerebral cortex by placing electrodes on the surface of scalp.

2. Match the following EEG wave types with their respective frequency ranges and associated activities. Type of Wave Frequency Range (Hz) Activity During

1) Beta A) 8-13 i) Deep sleep
2) Alpha B) 14-30 ii) Relaxation, eyes closed
3) Theta C) 2-4 iii) Alert and active thinking
4) Delta D) 5-7 iv) Light sleep, emotional stress in adults

Type of Wave	Frequency Range (Hz)	Activity During
1) Beta	A) 8-13	i) Deep sleep
2) Alpha	B) 14-30	ii) Relaxation, eyes closed
3) Theta	C) 2-4	iii) Alert and active thinking
4) Delta	D) 5-7	iv) Light sleep, emotional stress in adults

- A. 1 - B - iii, 2 - A - ii, 3 - D - iv, 4 - C - i
- B. 1 - C - i, 2 - B - iii, 3 - A - ii, 4 - D - iv
- C. 1 - B - iii, 2 - D - iv, 3 - A - ii, 4 - C - i
- D. 1 - A - ii, 2 - C - i, 3 - D - iv, 4 - B - iii

3. What type of waves are going to be seen if the patient is evaluated on EEG during deep sleep?

- A. Sleep spindles
 - B. K-Complexes
 - C. Theta waves
 - D. Delta waves
-

4. Which of the following statements about REM and NREM sleep is incorrect?

- A. During NREM sleep, there is an increase in parasympathetic outflow, leading to lower heart rate and blood pressure.
 - B. Muscle tone is completely lost during REM sleep, and the body is effectively paralyzed.
 - C. During deeper stages of NREM sleep, the subject is difficult to arouse by sensory stimuli.
 - D. During REM sleep, the subject is easily aroused by sensory stimuli.
-

5. A 65-year-old woman presents with severe insomnia, reporting that she has been unable to sleep for several days. Neurological examination reveals a lesion in the basal ganglia and adjacent anterior hypothalamus. Which of the following statements about the mechanisms of sleep is incorrect?

- A. The wake-promoting area is located in the posterior nucleus of the hypothalamus and releases histamine.
- B. The sleep-promoting area is located in the anterior nucleus of the hypothalamus and releases GABA.
- C. Decreased levels of adenosine at basal forebrain stimulate the sleep-promotion.

D. The awake state has increased NE, serotonin, and reduced levels of acetylcholine, while the opposite is true for the state of sleep

6. Match each sleep disorder with the correct stage of sleep in which it most commonly occurs

Sleep Disorder Stage of Sleep

a) Somnambulism 1) Stage 2 NREM

b) Nocturnal enuresis 2) REM Sleep

c) Narcolepsy 3) End of REM Sleep

d) Sleep Paralysis 4) Stage 3 and 4 (NREM)

Sleep Disorder	Stage of Sleep
a) Somnambulism	1) Stage 2 NREM
b) Nocturnal enuresis	2) REM Sleep
c) Narcolepsy	3) End of REM Sleep
d) Sleep Paralysis	4) Stage 3 and 4 (NREM)

- A. a-4, b-2, c-1, d-3
 - B. a-4, b-1, c-2, d-3
 - C. a-3, b-2, c-1, d-4
 - D. a-3, b-1, c-2, d-4
-

7. A 55-year-old patient presents with complaints of memory issues. He can recall general knowledge about the world, such as facts about historical events, but has difficulty remembering personal experiences, such as what he had for dinner last night or his activities from last weekend. Based on these symptoms, which of the following best describes the types of explicit memory affected in this patient?

- A. Impaired semantic memory with intact episodic memory
- B. Intact semantic memory with impaired episodic memory
- C. Impaired procedural memory with intact working memory
- D. Intact procedural memory with impaired working memory

8. A 62-year-old woman who had a stroke several months ago reports that she is having difficulty with tasks she used to perform automatically. She struggles to remember how to ride her bicycle, a skill she previously performed with ease. She also mentions difficulty in learning new motor tasks. Based on these symptoms, which type of implicit memory is most likely affected in this patient?

- A. Procedural memory
 - B. Priming
 - C. Associative learning
 - D. Procedural memory
-

9. Which type of learning involves the process by which humans and animals learn to behave in such a way as to obtain rewards and avoid punishments?

- A. Classical conditioning
 - B. Operant conditioning
 - C. Observational learning
 - D. Habituation
-

10. Which of the following parts of the brain is responsible for working memory?

- A. Prefrontal cortex
 - B. Neocortex
 - C. Ventral hippocampus
 - D. Amygdala
-

11. Which of the following statements about synaptic plasticity and memory formation is false?

- A. Long-term potentiation (LTP) primarily occurs at glutamatergic synapses in the hippocampus
 - B. Long-Term Depression (LTD) involves a significant influx of calcium into the postsynaptic neuron.
 - C. LTP leads to the phosphorylation of AMPA receptors, enhancing their activity and strengthening the synapse.
 - D. LTD is mediated by metabotropic glutamate receptors and involves dephosphorylation of AMPA receptors, leading to synaptic weakening.
-

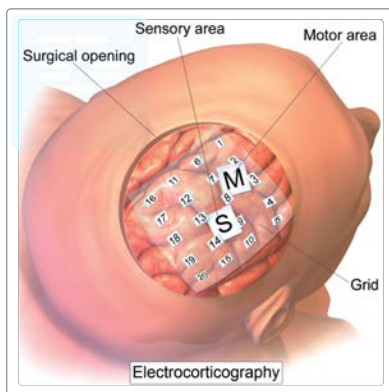
Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	4
Question 4	4
Question 5	3
Question 6	2
Question 7	2
Question 8	1
Question 9	2
Question 10	1
Question 11	2

Solution for Question 1:

Explanation:

Electrocorticography (ECOG) involves placing electrodes directly on the cerebral cortex , not over the scalp. It is an invasive procedure requiring craniotomy. It is used to evaluate epileptogenic foci in patients with refractory epilepsy.



Electroencephalogram - EEG

- Electroencephalogram (EEG) is the record of spontaneous electrical activities generated in the cerebral cortex that are picked up from the brain's surface through electrodes placed on designated sites in the scalp. (Option B ruled out)
- It detects the dendritic potentials of neurons, primarily pyramidal cells reaching the surface of the scalp.
- Positive dendritic potential is known as EPSP (Excitatory Postsynaptic Potential), and negative dendritic potential is known as IPSP (Inhibitory postsynaptic potential). (Option C ruled out)
- EEG captures the summated value of EPSP and IPSP . (Option A ruled out)
- Also, EEG is the recording of local potential from the surface of the scalp, it is not the action potential .

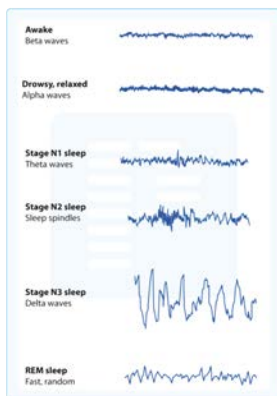
Solution for Question 2:

Explanation:

Type of Wave	Frequency Range (Hz)	Activity During
Beta	14-30	Alert and active thinking
Alpha	8-13	Relaxation, eyes closed
Theta	5-7	Light sleep, emotional stress in adults
Delta	2-4	Deep sleep

Solution for Question 3:

The patient is likely to be in NREM Stage 4 sleep, known as the deep stage of sleep, characterized by delta waves (low frequency, high amplitude).



Stage of Sleep	Characteristics
Stage N1 Sleep	Light sleep, easy to wake up, characterised by theta waves. (Option C ruled out)
Stage N2 Sleep	Deeper than stage N1, the presence of sleep spindles and K-complexes, is still relatively easy to wake up. (Option A and B ruled out)
Stage N3 Sleep	Deepest sleep, Delta waves (low frequency and high amplitude), very difficult to wake up.
REM Sleep	Rapid eye movements, high brain activity, associated with dreaming, Beta wave pattern

Solution for Question 4:

	NREM Sleep	REM Sleep
--	------------	-----------

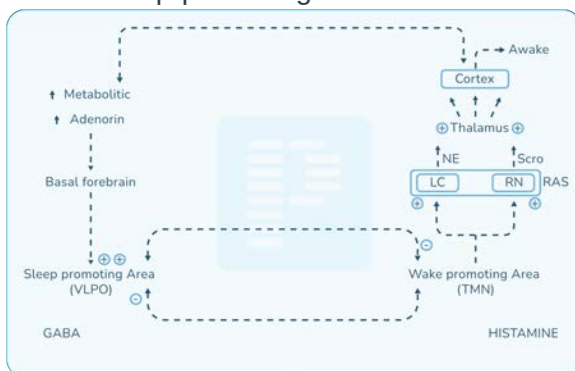
Autonomic Outflow	Increased parasympathetic outflow, leading to lower heart rate and blood pressure (Option A ruled out)	Increased sympathetic outflow
Muscle Tone and Reflexes	Intact during NREM sleep	Complete loss of muscle tone (muscle atonia) during REM sleep (Option B ruled out)
Arousability	Difficult to arouse in deeper stages (Stages 3 and 4) (Option C ruled out)	More difficult to arouse compared to NREM sleep by sensory stimuli, but may awaken spontaneously.
Eye Movement (EOG)	Absence of rapid eye movement	Presence of rapid eye movement
Brain Electrical Activity	Not associated with PGO (Ponto-Geniculate Occipital) spikes	Associated with PGO spikes
Dreaming Activity	Dreams uncertain; memorized if dreams occur	Vivid dreams; not memorized

Solution for Question 5:

There are 2 centres located in the hypothalamus, which are in constant mutual inhibition.

- Wake-Promoting Area: Located in the posterior hypothalamus- Tuberomammillary nucleus (TMN), releasing histamine. (Option A ruled out) The wake-promoting area stimulates two important nuclei of the reticular activating system, i.e., Raphe nuclei (RN) , which produce serotonin , and Locus coeruleus (LC) , which produce non-epinephrine . These act at the level of the thalamus from which all the diffuse projections will reach the cortex and the active cortex means the person will remain awake. This active cortex indefinitely produces metabolites with awakeness, primarily Adenosine . An increase in adenosine acts at the level of the basal forebrain and stimulates the sleep-promoting area.
- The wake-promoting area stimulates two important nuclei of the reticular activating system, i.e., Raphe nuclei (RN) , which produce serotonin , and Locus coeruleus (LC) , which produce non-epinephrine .
- These act at the level of the thalamus from which all the diffuse projections will reach the cortex and the active cortex means the person will remain awake.
- This active cortex indefinitely produces metabolites with awakeness, primarily Adenosine .

- An increase in adenosine acts at the level of the basal forebrain and stimulates the sleep-promoting area.
- Sleep-Promoting Area: Located in the anterior hypothalamus- Ventrolateral preoptic nucleus (VLPO), releasing GABA. (Option B ruled out)
- The wake-promoting area stimulates two important nuclei of the reticular activating system, i.e., Raphe nuclei (RN) , which produce serotonin , and Locus coeruleus (LC) , which produce non-epinephrine .
- These act at the level of the thalamus from which all the diffuse projections will reach the cortex and the active cortex means the person will remain awake.
- This active cortex indefinitely produces metabolites with awakeness, primarily Adenosine .
- An increase in adenosine acts at the level of the basal forebrain and stimulates the sleep-promoting area.



Neurotransmitter	Awake State	REM State	NREM Sleep
Norepinephrine & Serotonin	High	Low	Low
Acetylcholine	Low (Option D ruled out)	High	Low
GABA & Histamine	Low GABA, High Histamine	High GABA, Low Histamine	High GABA, Low Histamine

Solution for Question 6:

Correct Answer B) a-4, b-1, c-2, d-3

Explanation:

Stage of Sleep	Sleep disorder
----------------	----------------

Stage 3 and 4 NREM	Somnambulism (Sleepwalking): Occurs during the deeper stages of NREM sleep, specifically stages 3 and 4.
Nocturnal enuresis: Occurs in stages 3/4. Common in childhood, due to the failure of neural regulation of micturition.	
Stage 1 & 2 NREM	Sleep Talking (Somniloquy): This can occur in any stage of sleep but is most common in stages 1 and 2 of NREM sleep.
Bruxism (Teeth Grinding): Primarily occurs during stage 2 of NREM sleep.	
The transition from stage 3 to 4 NREM	Night Terrors: Typically occur during the transition from stage 3 to stage 4 of NREM sleep. Unlike nightmares, individuals usually do not remember night terrors.
REM Sleep	Narcolepsy: Involves sudden onset of REM sleep. Individuals with narcolepsy often experience excessive daytime sleepiness, sudden muscle weakness (cataplexy), sleep paralysis, and hallucinations at the onset or end of sleep. The disorder is linked to a deficiency in the neurotransmitter orexin .
Nightmares: Occur during REM sleep and involve vivid, disturbing dreams that can cause fear and anxiety. Unlike night terrors, nightmares are often remembered in detail upon waking. They can be triggered by stress, anxiety, or trauma.	

Solution for Question 7:

Correct Answer: B) Intact semantic memory with impaired episodic memory

Explanation:

The patient is able to recall general knowledge and facts about the world (intact semantic memory) but struggles to remember specific personal experiences such as what he had for dinner or activities from last weekend (impaired episodic memory). This distinction is consistent with the described symptoms and aligns with the known functions of semantic and episodic memory.

Explicit memory , also known as declarative memory, is a type of long-term memory that involves conscious recollection of information.

It is divided into two main categories:

- This refers to the storage of general knowledge about the world.
- It includes facts, concepts, and ideas that are not tied to specific personal experiences.
- For instance, knowing that the capital of France is Paris, understanding the rules of mathematics, or recognizing historical dates and events are all parts of semantic memory.
- The temporal lobes , particularly the anterior and lateral temporal cortex , are critically involved in the processing and storage of semantic memory.
- This involves the recollection of personal experiences and specific events that have occurred at particular times and places.
- It includes memories of one's own life, such as what you had for dinner last night, a vacation you took last summer, or a conversation you had with a friend.
- The hippocampus and adjacent medial temporal lobe structures are essential for the formation and retrieval of episodic memories.
- The patient's difficulty in remembering what he had for dinner or his activities from the previous weekend suggests an impairment in episodic memory. (Option A ruled out)
- Procedural memory is a type of implicit memory related to the performance of tasks, such as riding a bike or playing an instrument.
- Working memory is a short-term memory system involved in the temporary storage and manipulation of information.

The patient's symptoms do not relate to procedural memory or working memory issues but rather to long-term explicit memory. (Option C and Option D ruled out)

Solution for Question 8:

Correct Answer: A) Procedural memory

Explanation:

Implicit memory , also known as non-declarative memory, involves unconscious memory processes that influence thoughts and behaviors.

Implicit memory is divided into following subcategories:

- This type of implicit memory involves the learning and retention of motor skills and actions.
- It enables individuals to perform tasks without conscious thought, such as riding a bicycle, typing on a keyboard, or playing a musical instrument. (Option A)
- Priming refers to the unconscious influence that exposure to a stimulus has on a subsequent response to a different stimulus.
- It helps with faster recognition or processing of similar stimuli without conscious awareness.
- For example, seeing the word "yellow" might make it easier to recognize the word "banana" shortly afterward. (Option B ruled out)
- Classical Conditioning : A process where a neutral stimulus becomes associated with a significant stimulus, resulting in a learned response. For example, Pavlov's dogs learned to salivate at the sound of a bell because it was associated with food.
- A process where a neutral stimulus becomes associated with a significant stimulus, resulting in a learned response.
- For example, Pavlov's dogs learned to salivate at the sound of a bell because it was associated with food.
- Operant Conditioning : A type of learning where behavior is strengthened or weakened by rewards or punishments. For example, a rat learning to press a lever to receive food involves operant conditioning.
- A type of learning where behavior is strengthened or weakened by rewards or punishments.
- For example, a rat learning to press a lever to receive food involves operant conditioning.
- Habituation : A decrease in response to a repeated, benign stimulus. For instance, becoming accustomed to a constant background noise and eventually ignoring it.
- A decrease in response to a repeated, benign stimulus.
- For instance, becoming accustomed to a constant background noise and eventually ignoring it.
- Sensitisation : An increased response to a stimulus following a strong or noxious stimulus. For example, becoming more sensitive to a light touch after being exposed to a painful stimulus.
- An increased response to a stimulus following a strong or noxious stimulus.
- For example, becoming more sensitive to a light touch after being exposed to a painful stimulus.

Semantic memory (Option D) is a type of explicit memory that involves the storage and recall of factual information and general knowledge, such as knowing the capital of a country or understanding historical events. The patient's difficulties with motor tasks and learning new skills are not related to semantic memory.

Solution for Question 9:

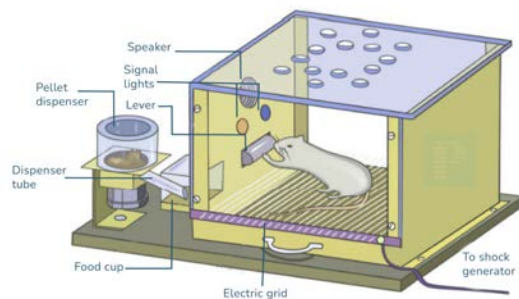
Correct Answer: B) Operant conditioning

Explanation:

Operant conditioning involves learning through the consequences of behaviors. In this type of learning, behaviors are either reinforced (strengthened) or punished (weakened) based on their consequences. B.F. Skinner's experiments with rats in Skinner boxes demonstrated this concept.

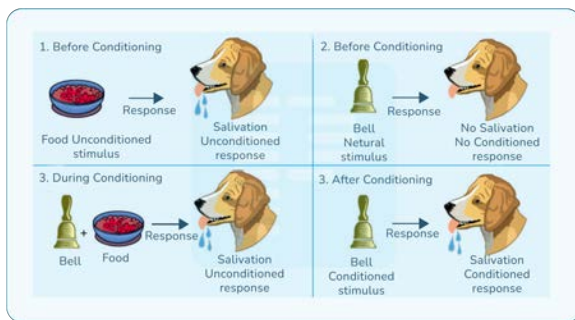
Skinner's experiment:

- In the experiment, a rat is placed in a chamber with an electrical grid on the floor and multiple switches on the wall.
- The rat learns that pressing one switch results in an electric shock, while pressing another switch or lever results in food dropping into the box.
- Over time, the rat associates pressing the food switch with receiving food, reinforcing this behavior.
- Meanwhile, the association with the electric shock stimulus diminishes in the rat's memory.



Thus, the rat associates with the stimulus to change the behaviour.

Classical conditioning (Option A) is associative learning, where an organism learns to associate two stimuli.



Observational learning (Option C): Observational learning, described by Albert Bandura, involves learning by observing others and their consequences. While observational learning can lead to behaviour modification, it focuses more on social learning and modelling rather than direct reinforcement or punishment.

Habituation (Option D) is a form of non-associative learning where an organism decreases its response to a repeated stimulus that carries no positive or negative consequences. It does not involve learning through consequences or behavioural modification.

Solution for Question 10:

Correct Answer: A) Prefrontal cortex

Explanation:

Working memory is a type of short-term memory.

- It typically lasts for about 18-20 seconds before information may fade without rehearsal.
- The maximum capacity of working memory is around 7 ± 2 'chunks' of information, such as items like fruits or numbers.
- The prefrontal cortex is the brain region primarily responsible for working memory. (Option A)

Short-term memory:

- Short-term memory retains information for durations ranging from seconds to several hours.
- The ventral hippocampus, specifically the CA1 region, is responsible for short-term memory. (Option C ruled out)
- Through rehearsal or repetition, short-term memories can be consolidated into long-term memories.

Long-term memory:

- Long-term memory retains information for years or even a lifetime.
- The neocortex serves as the centre for long-term memory storage. (Option B ruled out)

Amygdala (Option D): The amygdala is primarily involved in emotional processing and the formation of emotional memories. Although it interacts with memory systems, it does not handle working memory, which is more closely linked to the prefrontal cortex.

Solution for Question 11:

Correct Answer: B) Long-Term Depression (LTD) involves a significant influx of calcium into the postsynaptic neuron.

Explanation:

A significant influx of calcium into the postsynaptic neuron is seen in the process of Long-term potentiation strengthening the synapse.

Mechanism of memory formation:

Memory formation relies on synaptic plasticity, which refers to the strengthening or weakening of synapses based on neuronal activity.

Long-term potentiation:

- Long-term potentiation (LTP), a mechanism for memory formation, involves the strengthening of synapses, particularly in the hippocampus (CA1) and cerebellum
- It primarily occurs at glutamatergic synapses in the hippocampus. (Option A ruled out)
- Two key glutamate receptors, AMPA and NMDA, are involved in LTP.
- Glutamate binding to the AMPA receptor causes an influx of sodium ions (Na^+) into the postsynaptic neuron, leading to depolarisation.
- NMDA Receptor and Calcium Influx: The depolarization caused by AMPA receptor activation is crucial as it removes a magnesium ion (Mg^{2+}) block from the NMDA receptor channel. This removal is key because Mg^{2+} normally prevents calcium ions (Ca^{2+}) from entering the postsynaptic neuron.
- Calcium's Role: This influx of Ca^{2+} initiates a cascade of events. Firstly, Ca^{2+} binds with calmodulin, forming a complex that activates kinase enzymes.

- Kinase Activity and AMPA Receptor Phosphorylation: Kinases then phosphorylates various proteins, notably the AMPA receptor itself. Phosphorylation keeps the AMPA receptor open even after the initial stimulation ceases, leading to increased sensitivity and a strengthened synapse. (Option C ruled out)

Long-term depression:

- Long-Term Depression (LTD) is a process that weakens synapses
- LTD involves a smaller influx of calcium ions (Ca^{2+}) compared to LTP.
- Metabotropic glutamate receptors are involved in LTD, unlike the NMDA receptors in LTP. These metabotropic receptors, when activated by glutamate, lead to a low influx of Ca^{2+} into the postsynaptic neuron. This smaller influx is crucial for initiating LTD.
- Phosphatase enzymes are activated in LTD due to the low Ca^{2+} concentration. This contrasts with LTP, where a high Ca^{2+} concentration activates kinase enzymes.
- Phosphatase activity results in the dephosphorylation of AMPA receptors, leading to their closure. (Option D ruled out)

The closure of AMPA receptors inhibits the synapse, making it less responsive to stimuli. LTD, therefore, serves to weaken specific synapses, potentially filtering out irrelevant information while strengthening relevant ones through LTP.

General Physiology

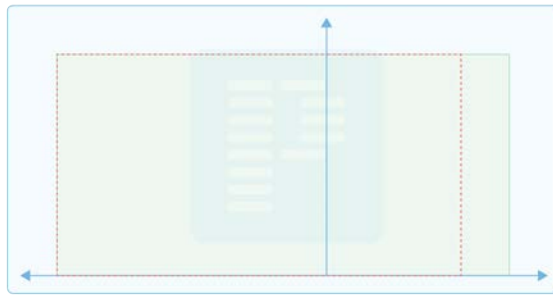
1. A patient with metastatic breast cancer is prescribed paclitaxel. What effect does paclitaxel have on microtubules?

- A. At low concentrations, paclitaxel promotes rapid polymerisation and mitotic spindle formation.
 - B. At high concentrations, paclitaxel destabilizes microtubules, causing rapid disassembly and mitotic arrest.
 - C. At high concentrations, paclitaxel stabilizes microtubules, increasing polymer mass and inducing bundle formation.
 - D. Paclitaxel shifts equilibrium towards tubulin dimers, promoting microtubule dynamics and mitosis.
-

2. Which of the following is likely to be seen if a patient is infused with 3% NaCl?

- A. Water shifts from ICF to ECF
 - B. Decreased plasma protein concentration
 - C. Increased arterial blood pressure
 - D. All of the above
-

3. Which of the following does this Darrow-Yannet diagram (Solid line represent the controlled condition and broken red-line indicates new-steady state condition) represent?



- A. Excessive NS infusion
 - B. Excessive infusion of hypertonic saline
 - C. Diarrhea
 - D. Diabetes insipidus
-

4. Oxygen moves from the alveoli into the blood across the alveolar-capillary membrane. Which of the following best describes this process?

- A. Facilitated Diffusion
 - B. Simple Diffusion
 - C. Non-ionic diffusion
 - D. Osmosis
-

5. While studying insulin secretion in pancreatic beta cells, a researcher scientist observes the fusion of the insulin-stored vesicles with the cell membrane to release insulin into the extracellular space in response to blood glucose levels. Which of the following statements about this process is correct?

- A. Exocytosis is a passive process that does not require energy.
- B. During exocytosis, vesicles fuse with the lysosome to release their contents.
- C. Exocytosis involves the transport of substances out of the cell via vesicle fusion with the cell membrane.

D. Exocytosis is primarily used to uptake nutrients and molecules from the extracellular environment.

6. Select the correct statements regarding endocytosis:

- A. Pinocytosis involves ingesting solid particles into the cell by budding small vesicles from the cell membrane.
 - B. Phagocytosis is a type of endocytosis that involves the ingestion of liquid into the cell.
 - C. Receptor-mediated endocytosis is a non-selective process where molecules are ingested into the cell without needing receptors..
 - D. All forms of endocytosis require ATP for the vesicle formation process
-

7. Which of the following statements regarding facilitated diffusion is false?

- A. It does not require ATP expenditure.
 - B. Involved transport protein is stereospecific and follows saturation kinetics.
 - C. Diffusion of ammonia in collecting ducts is a type of facilitated diffusion.
 - D. Molecules move down the chemical gradient from high to low concentration.
-

8. Oxygen moves from the alveoli into the blood across the alveolar-capillary membrane. Which of the following best describes this process?

- A. Facilitated Diffusion
 - B. Simple Diffusion
 - C. Non-ionic diffusion
 - D. Osmosis
-

9. Which of the statements below is false?

- A. An example of symport is the absorption of glucose into the small intestine.
 - B. Antiport maintains sodium-potassium balance in cells.
 - C. Secondary active transporter always depends on the primary active transporter in the symport.
 - D. The symport mechanism in myocardial cells is blocked by digitalis, which increases contraction force.
-

10. Match the following terms with their correct definitions: 1. Stereospecificity 2. Saturation Kinetics 3. Competition 4. Allosteric Regulation

a. The phenomenon where increasing the concentration of a substrate leads to a point where enzyme activity plateaus because the enzyme is fully occupied. b. The selective interaction of an enzyme or receptor with only one stereoisomer of a chiral molecule, leading to a specific product. c. The inhibition or activation of enzyme activity through the binding of a molecule at a site other than the enzyme's active site, causing a conformational change. d. The scenario where two or more molecules with similar structures vie for the same active site on an enzyme, potentially inhibiting each other's binding and activity.

- 1. Stereospecificity
- 2. Saturation Kinetics
- 3. Competition
- 4. Allosteric Regulation

- a. The phenomenon where increasing the concentration of a substrate leads to a point where enzyme activity plateaus because the enzyme is fully occupied.
- b. The selective interaction of an enzyme or receptor with only one stereoisomer of a chiral molecule, leading to a specific product.
- c. The inhibition or activation of enzyme activity through the binding of a molecule at a site other than the enzyme's active site, causing a conformational change.
- d. The scenario where two or more molecules with similar structures vie for the same active site on an enzyme, potentially inhibiting each other's binding and activity.

- A. 1-b, 2-a, 3-d, 4-c
- B. 1-a, 2-b, 3-c, 4-d

C. 1-c, 2-d, 3-a, 4-b

D. 1-b, 2-c, 3-a, 4-d

11. Identify the false statement regarding tight junctions.

- A. Tight junctions are involved in endothelial barrier function and endothelium-dependent vasodilation.
 - B. Tight junctions allow the passage of small molecules and ions between cells through the paracellular pathway.
 - C. Tight junctions are composed of proteins such as occludin, junctional adhesion molecules, and claudins.
 - D. Tight junctions are primarily found in the apical margins of cells in connective tissue.
-

12. Which of the following statements about the Na⁺-K⁺ ATPase is false?

- A. It is an electrogenic pump with a coupling ratio of 3Na⁺:2K⁺
 - B. The α subunit has intracellular Na⁺ and ATP-binding sites
 - C. The α subunit contains the phosphorylation site at Asp 376
 - D. Its activity is enhanced by ouabain and digitalis glycosides
-

13. Which of the following is considered downhill transport?

- A. Active transport
 - B. Passive transport
 - C. Primary active transport
 - D. Secondary active transport
-

14. Which of the following Connexin mutations is associated with cataracts?

- A. Connexin 26
 - B. Connexin 30
 - C. Connexin 40
 - D. Connexin 50
-

15. Which of the following correctly matches a junction type with its key features?

- A. Tight Junctions - Provide mechanical strength and link cytoskeletons of adjacent cells.
 - B. Adherens Junctions - Create a seal between adjacent cells and prevent the passage of molecules through intercellular space.
 - C. Desmosomes - Form channels between cells, allowing direct passage of small molecules and ions.
 - D. Gap Junctions - Enable rapid communication and coordination, forming channels between cells for electrical signal propagation.
-

16. Which of the following best describes the term "milieu intérieur"?

- A. Intracellular fluid
 - B. Extracellular fluid
 - C. Transcellular fluid
 - D. Intravascular fluid
-

17. Match the following cell markers with their corresponding cells: Cell marker
Cell i) Cytokeratin a) Astrocytes ii) Vimentin b) Skeletal muscle cell iii) Desmin c)
Mesenchymal cell iv) Glial fibrillary acidic protein (GFAP) d) Epithelial cell

Cell marker	Cell
i) Cytokeratin	a) Astrocytes
ii) Vimentin	b) Skeletal muscle cell
iii) Desmin	c) Mesenchymal cell
iv) Glial fibrillary acidic protein (GFAP)	d) Epithelial cell

- A. i-a ; ii-b ; iii-c ; iv-d
- B. i-d ; ii-c ; iii-b ;iv-a
- C. ii-a ; iii-d ; i-b ; iv-c
- D. ii-d ; iv-b ; iii-a ; i-c

18. Which of the following statements about microtubules is true? i) Microtubules form the cytoskeleton of organelles such as cilia and nerve axons, providing structural integrity and rigidity. ii) Microtubules function as channels allowing the passage of small molecules and ions between cells. iii) Microtubules play a dynamic role in cell division, movement, and intracellular transport by acting as conveyor belts for vesicles, granules, and organelles like mitochondria.

- A. All are correct
- B. All are correct except (iii)
- C. All are correct except (ii)
- D. Only (i) is correct

19. Which biological membrane has the highest protein-to-lipid ratio?

- A. Plasma membrane

- B. Mitochondrial inner membrane
 - C. Mitochondrial outer membrane
 - D. All have the same ratio
-

20. Which statements about the cell membrane are correct?

- A. Sphingolipids form channels for water and water-soluble substances, aiding diffusion.
 - B. Cholesterol acts as a receptor for peptide hormones, triggering intracellular processes.
 - C. Peripheral proteins penetrate the lipid bilayer and actively transport substances.
 - D. Glycocalyx contributes to negative surface charge, cell adhesion, and immune reactions.
-

21. A 45-year-old male patient is admitted to the hospital after experiencing severe dehydration due to prolonged vomiting and diarrhoea. His symptoms include dry mouth, low blood pressure, and rapid heart rate. Which of the following fluids can be used in this case?

- A. Normal saline
 - B. 5% sodium chloride
 - C. 10% dextrose
 - D. Any of the above
-

22. Match the following: Body fluids Indicators used to measure it 1. Extracellular fluid a. Inulin 2. Plasma b. Aminopyrine 3. Total body water c. Evans blue

Body fluids	Indicators used to measure it
1. Extracellular fluid	a. Inulin

2. Plasma	b. Aminopyrine
3. Total body water	c. Evans blue

- A. 1-a, 2-b, 3-c
 - B. 1-a, 2-c, 3-b
 - C. 1-c, 2-a, 3-b
 - D. 1-c, 2-b, 3-a
-

23. Which of the following is an example of transcellular fluid?

- A. Cerebrospinal fluid
 - B. Peritoneal fluid
 - C. Vitreous humor
 - D. All of the above
-

24. The plasma volume of a person is 4 liters and the hematocrit is 20%, what is the blood volume?

- A. 5 litres
 - B. 4 litres
 - C. 4.5 litres
 - D. 6 litres
-

25. Which of the following statements regarding body fluids is true?

- A. ICF makes up 40% of body weight
- B. ECF makes up 40% of body weight

- C. ICF makes up 33% of body weight
 - D. None of the above
-

26. Which of the following statements are examples of feed-forward control systems in physiology? The clotting cascade rapidly amplifies clot formation after vascular injury. Muscles adjust in advance for standing or moving. Increased heart rate before starting of physical activity. Sweating in response to increased body temperature. The smell of food triggers digestive enzyme secretion.

- A. 1, 2, 4
 - B. 2, 3, 5
 - C. 1, 3, 5
 - D. 2, 4, 5
-

27. A person moved from his hometown to Delhi to appear for an exam. The initial respiratory rate was 12 breaths per minute. After arriving at the exam center, the respiratory rate increased to 20 breaths per minute. The respiratory center corrected this to 16 breaths per minute. Calculate the regulation factor for the respiratory rate.

- A. 0.25
 - B. 2
 - C. 0.5
 - D. 6
-

28. A patient's blood glucose level in a steady-state condition before a heavy lunch was 90 mg/dL. After lunch, the blood glucose level increased to 180 mg/dL. Insulin secretion from the pancreas reduced the blood glucose level to 120 mg/dL.

Calculate the gain of the glucose regulatory mechanism.

- A. +2
 - B. -2
 - C. +3
 - D. -3
-

29. Which of the following operates based on a negative feedback mechanism?
Thyroid hormone regulation Blood glucose regulation Blood clotting cascade
Nerve Signal Propagation Temperature regulation

- A. 1, 2, and 5
 - B. 1, 3, and 4
 - C. 2, 3, and 5
 - D. 1, 2, and 4
-

30. A 3-week-old infant is being exclusively breastfed by her mother. During a recent pediatric visit, the mother expresses concern about her milk supply. The pediatrician explains that the infant's feeding patterns contribute to the regulation of milk production. Which physiological concept best describes how the mother's body responds to the infant's feeding cues to regulate milk production?

- A. Negative feedback mechanism
 - B. Positive feedback mechanism
 - C. Autonomic reflex
 - D. Intrinsic regulation mechanism
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	3
Question 4	2
Question 5	3
Question 6	4
Question 7	3
Question 8	2
Question 9	4
Question 10	1
Question 11	4
Question 12	4
Question 13	2
Question 14	4
Question 15	4
Question 16	2
Question 17	2
Question 18	3
Question 19	2
Question 20	4
Question 21	1
Question 22	2

Question 23	4
Question 24	1
Question 25	1
Question 26	2
Question 27	3
Question 28	2
Question 29	1
Question 30	2

Solution for Question 1:

Correct Answer: C) At high concentrations, paclitaxel stabilizes microtubules, increasing polymer mass and inducing bundle formation.

Explanation:

Paclitaxel primarily exerts anti-tumour effects by stabilising microtubules and altering their dynamics in a concentration-dependent manner.

Paclitaxel:

DEFINITION	An anti-tumour agent which is effective against several cancers
MECHANISM	Binds to the β -subunit of tubulin in a 1:1 stoichiometry ↓ Causes stabilisation of microtubules (Options B and D)
CONCENTRATION-DEPENDENT EFFECTS	
High Concentrations (Option C)	Low Concentrations
Microtubule polymer mass increases Induces microtubule bundle formation in interphase cells	Suppresses microtubule dynamics without altering polymer mass (Option A) Blocks mitosis by stabilising spindle microtubules
EFFECT ON CELLS	
Sensitive Cells	Resistant Cells

Mitotic Block Induces cell death	Model 1: Contain "hypo stable" microtubules with equilibrium shifted towards tubulin dimers Model 2: Equilibrium is shifted towards highly dynamic microtubules
-------------------------------------	--

Solution for Question 2:

Correct Answer: D) All of the above

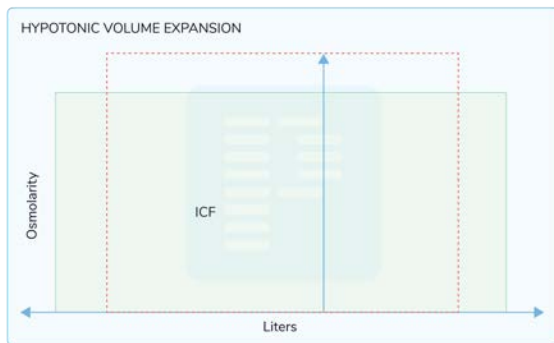
Explanation:

3% NaCl is a hypertonic solution; the changes seen will be that of hypertonic volume expansion.

Disorder	Examples	Effects
Hyperosmotic Volume Expansion	Excessive infusion of hypertonic saline.	Increase in ECF volume. Increased plasma osmolality Water shifts from ICF to ECF Decreased plasma protein concentration Increased arterial blood pressure
Hyposmotic Volume Expansion	SIADH	Increase in ECF volume. Increase in ICF volume. Decrease in plasma osmolality. No change in hematocrit. Decreased plasma protein concentration.
Isosmotic Volume Expansion	Infusion of isotonic NaCl	Increase in ECF volume. No change in ICF volume. No change in osmolality. Decreased hematocrit. Decreased plasma protein concentration.

Hypertonic Volume Expansion" data-author="NA" data-hash="7848" data-license="NA" data-source="NA" data-tags="April2025" height="262" src="https://image.prepladder.com/notes/nyCGzU1L7keWvGnol06d1744010158.png" width="500" />

Isotonic Volume Expansion" data-author="NA" data-hash="7846" data-license="NA" data-source="NA" data-tags="April2025" height="262" src="https://image.prepladder.com/notes/FukfpvgzNSUeF5TrVkMF1744009943.png" width="500" />



Solution for Question 3:

Correct Answer: C) Diarrhea

Explanation:

The Darrow-Yannet diagram shows isosmotic volume contraction; an example of that is diarrhoea.

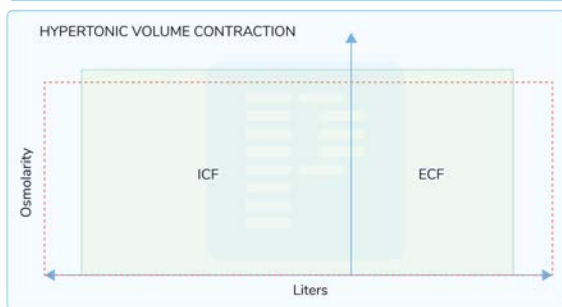
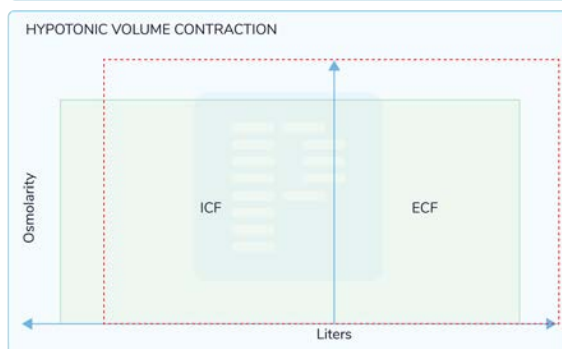
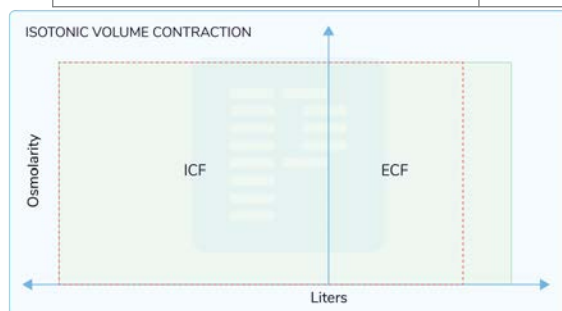
Darrow-Yannet Diagram:

- It helps to understand the plasma osmolarity and volume alterations in various diseases.
- The volume is plotted in x-axis and plasma osmolarity in y-axis.
- Broken lines indicate the flow of water between compartments driven by osmotic gradients, reflecting passive movement in response to changes in solute concentrations.
- A normal Darrow-Yannet Diagram is shown below:



Disorder	Examples	Effects
----------	----------	---------

Isosmotic volume contraction	Diarrhoea Vomiting Haemorrhage Burns	Decrease in ECF and plasma volumes Increased plasma protein concentration Increased hematocrit Decreased arterial blood pressure
Hyposmotic volume contraction	Adrenocortical insufficiency Gastric secretions loss	Decreased plasma volume Decreased ECF and plasma osmolality Water shifts from ECF to ICF Cellular swelling Decreased arterial blood pressure
Hyperosmotic volume contraction	Diabetes insipidus Excessive sweating Alcoholism Fever	Decrease in ECF and plasma volumes Increased ECF and plasma osmolality Water shifts from ICF to ECF Increased hematocrit Decreased arterial blood pressure.



Solution for Question 4:

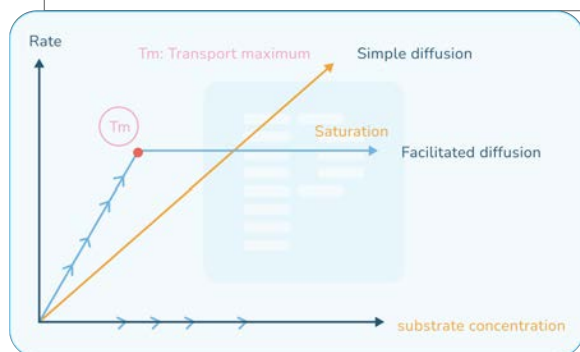
Correct Answer: B) Simple Diffusion

Explanation:

Oxygen moves from an area of higher concentration in the alveoli to an area of lower concentration in the blood through simple diffusion, which follows Fick's law of diffusion.

Types of diffusion:

Simple diffusion	Facilitated diffusion (Option A)
Move from high to low solute concentration (downhill transport) It does not require energy. Not require carrier protein (channel protein may or may not be present) No saturation: more concentration leads to a higher rate of transport. Fick's law: $\text{Rate (J)} = D \cdot A \cdot \Delta C / x$ D: Diffusion constant. A: Surface area of diffusion. X: Thickness of the membrane. ΔC: Concentration difference between membranes (most important determinant).	Movement of molecules across a membrane via carrier protein. No expenditure of energy. For example, glucose is transported by all types of GLUT. Saturation is achieved after a certain point called the plateau. The maximum transport rate occurs at the plateau or saturation point, called maximum (T_m)



Active transport requires energy to move substances against their concentration gradient, which is not the case for the passive movement of oxygen from the alveoli to the blood. (Option C)

Osmosis refers to the movement of water across a semipermeable membrane from an area of lower solute concentration to an area of higher and does not describe the movement of gases like oxygen. (Option D)

Solution for Question 5:

Correct answer: C) Exocytosis involves the transport of substances out of the cell via vesicle fusion with the cell membrane.

Explanation:

Exocytosis can be constitutive or non-constitutive.

Constitutive	Non-constitutive
Direct secretion of vesicles outside the cell.	Vesicles that do not move directly outside the cell are moved to a particular portion of the cell, where they are temporarily stored. During their temporary storage, pre-processing of these vesicles happens. As a result, more refined protein vesicles are produced. This is known as non-constitutive exocytosis.

Incorrect statements:

Exocytosis is a passive process that does not require energy.

- Exocytosis is an active process that requires energy in the form of ATP.
- This energy is needed for the vesicle fusion with the cell membrane and the subsequent release of its contents.

During exocytosis, vesicles fuse with the lysosome to release their contents.

- This statement is incorrect because they fuse directly with the cell membrane, not with lysosomes.
- Lysosomes are involved in intracellular digestion, not in the exocytotic release of substances.

Exocytosis is primarily used to uptake nutrients and molecules from the extracellular environment.

- Exocytosis expels substances from the cell, not to uptake them.
 - Endocytosis is the process used to take in nutrients and molecules from the extracellular environment.
-

Solution for Question 6:

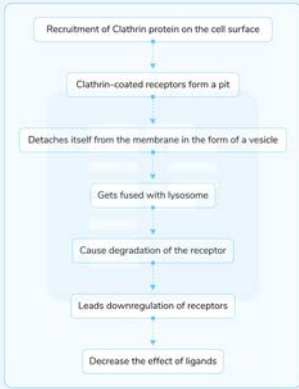
Correct Answer: D) All forms of endocytosis require ATP for the vesicle formation process.

Explanation:

All forms of endocytosis require ATP for the vesicle formation process.

Endocytosis:

- Endocytosis is an active transport process that requires energy in the form of ATP to form and move vesicles within the cell.
- Three types of endocytosis:

Pinocytosis (Option: A)	Referred to as "cell drinking". Uptake of extracellular fluids by budding small vesicles from the cell membrane.
Phagocytosis (Option: B)	Referred to as "cell eating". Engulfs large particles such as pathogens or cellular debris. into the cell.
Receptor-mediated endocytosis (Option: C)	 <pre>graph TD; A[Recruitment of Clathrin protein on the cell surface] --> B[Clathrin-coated receptors form a pit]; B --> C[Detaches itself from the membrane in the form of a vesicle]; C --> D[Gets fused with lysosome]; D --> E[Cause degradation of the receptor]; E --> F[Leads downregulation of receptors]; F --> G[Decrease the effect of ligands];</pre> ... (content truncated)

Solution for Question 7:

Correct Answer: C) Diffusion of ammonia in collecting ducts is a type of facilitated diffusion.

Explanation:

Diffusion of ammonia in collecting ducts to excrete ammonium ions into urine is a type of non-ionic diffusion. It occurs at the level of GIT and kidney for the transportation of weak acids or bases.

Facilitated diffusion does not require energy expenditure and uses transport (carrier) proteins that follow saturation kinetics, stereospecificity and competition. (Option A and B)

Molecules move from areas of high to low concentration (down their chemical gradient), and cations move to negatively charged areas whereas anions move to positively charged areas ("down" their electrical gradient). (Option D)

Solution for Question 8:

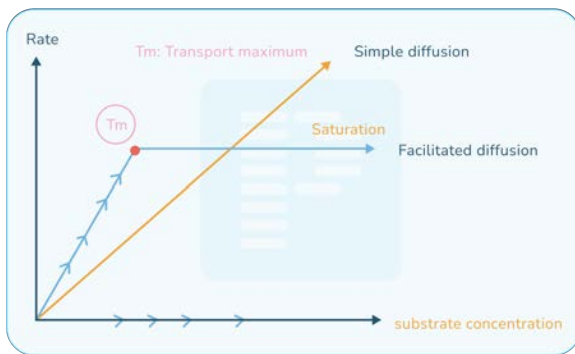
Correct Answer: B) Simple Diffusion

Explanation:

Oxygen moves from an area of higher concentration in the alveoli to an area of lower concentration in the blood through simple diffusion, which follows Fick's law of diffusion.

Types of diffusion:

Simple diffusion	Facilitated diffusion (Option A)
Move from high to low solute concentration (downhill transport) It does not require energy. Not require carrier protein (channel protein may or may not be present) No saturation: more concentration leads to a higher rate of transport. Fick's law: $\text{Rate (J)} = D \cdot A \cdot \Delta C / x$ D: Diffusion constant. A: Surface area of diffusion. X: Thickness of the membrane. ΔC: Concentration difference between membranes (most important determinant).	Movement of molecules across a membrane via carrier protein. No expenditure of energy. For example, glucose is transported by all types of GLUT. Saturation is achieved after a certain point called the plateau. The maximum transport rate occurs at the plateau or saturation point, called maximum (T_m).



Active transport requires energy to move substances against their concentration gradient, which is not the case for the passive movement of oxygen from the alveoli to the blood. (Option C)

Osmosis refers to the movement of water across a semipermeable membrane from an area of lower solute concentration to an area of higher and does not describe the movement of gases like oxygen. (Option D)

Solution for Question 9:

Correct Answer: D) The symport mechanism in myocardial cells is blocked by digitalis, which increases contraction force.

Explanation:

Digitalis blocks Antiport mechanism.

Symport	Antiport
It involves the movement of two molecules in the same direction Transport requires the binding of more than one substance to the transport protein. It is important in glucose absorption in the intestines. (Option A) Glucose and sodium move together, with sodium following the concentration gradient and glucose against the concentration gradient. The secondary active transporter is always dependent on the primary active transporter. (Option C) For example, SGLT depends on the Na-K-ATPase pump to remove Na molecules from inside the cell.	It involves the movement of two molecules in opposite directions Exchanges one substance for another It is crucial for maintaining sodium-potassium balance in cells and basal metabolism. (Option B) E.g. Sodium-Calcium exchanger Calcium moves inside and outside the cell in uphill transport (against the gradient); hence, sodium moves inside the cell along the concentration gradient. Na-K-ATPase prevents intracellular sodium accumulation. Na-K-ATPase is blocked by Ouabain or digitalis, increases calcium accumulation inside the myocardial cells and leads to increased force of contraction. (Option D)

Solution for Question 10:

Correct Answer: A) 1-b, 2-a, 3-d, 4-c

Explanation:

When any transportation occurs through carrier protein, it follows one of the three kinetics.

Stereospecificity	Saturation Kinetics	Competition
Property of enzymes or receptors to interact selectively with one stereoisomer of a chiral molecule, leading to the production of a specific enantiomerically pure product.	The phenomenon where the enzyme's activity reaches a maximum rate at a certain high substrate concentration and cannot increase further because all active sites are fully occupied, described by the Michaelis-Menten kinetics.	Multiple molecules with similar structures compete for the same active site on an enzyme, potentially blocking each other's binding and thus inhibiting each other's activity.
E.g. D-glucose and L-glucose	E.g. Glycine conjugation of salicylate	E.g. Glucose and Galactose

Allosteric Regulation - Regulation of an enzyme's activity through binding a regulatory molecule at a site other than the enzyme's active site induces a conformational change that alters the enzyme's activity.

Solution for Question 11:

Correct Answer: D) Tight junctions are primarily found in the apical margins of cells in connective tissue.

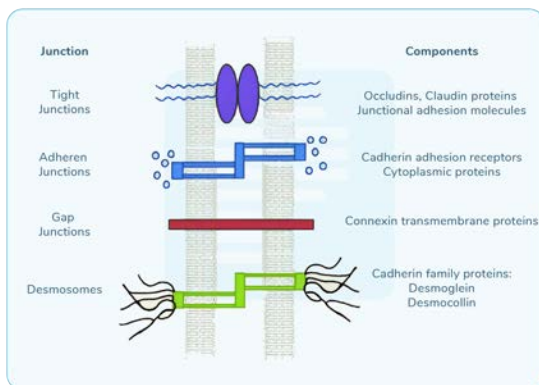
Explanation:

Tight junction: Sealing strand composed of a long row of transmembrane adhesion proteins embedded in the two interacting plasma membranes.

- The extracellular domains of these proteins join directly to one another to occlude the intercellular space.

- Tight junctions are primarily found in apical margins of epithelia cells, not connective tissue (Option D)
- Tight junctions actually limit the permeability of epithelial cells to ions and solutes, thus regulating the overall ion and solute flux rather than increasing it.
- Tight junctions are important to endothelial barrier function and endothelium-dependent vasodilation. (Option A ruled out)
- Tight junctions permit the passage of ions and small molecules between adjacent cells via the paracellular pathway. (Option B ruled out)
- Occludin, junctional adhesion molecules, and claudins are the main families of transmembrane proteins that contribute to tight junctions. (Option C ruled out)

Intercellular junctions:

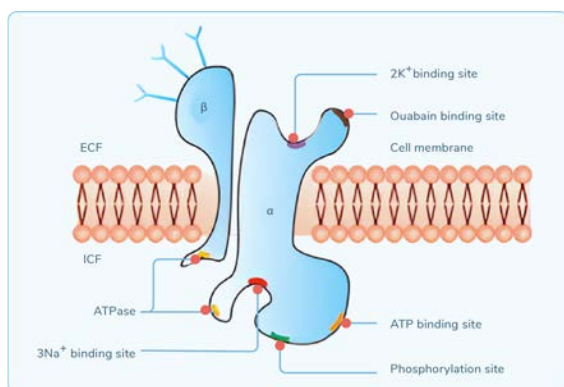


Solution for Question 12:

Correct Answer: D) Its activity is enhanced by ouabain and digitalis glycosides

Explanation :

Na, K ATPase catalyses the hydrolysis of ATP to adenosine diphosphate(ADP) and uses the energy to extrude three Na⁺ from the cell and take two K⁺ into the cell for each molecule of ATP hydrolysed.



Features:

- Its activity is inhibited by ouabain and related digitalis glycosides used to treat heart failure.
- It is an electrogenic pump with a coupling ratio of $3\text{Na}^+ : 2\text{K}^+$
- It is a heterodimer made up of α subunit of weight 1,00,000 and β subunit of weight 55,000.
- The α subunit has intracellular Na^+ and ATP-binding sites.
- The α subunit contains the phosphorylation site at Asp 376.
- Both α and β subunits are heterogeneous, with multiple isoforms ($\alpha_1, \alpha_2, \alpha_3, \beta_1, \beta_2, \beta_3$) present in different tissues.
- Thyroid hormone, aldosterone, and androgen increase the pump activity whereas dopamine in the kidney inhibits the pump activity.

Solution for Question 13:

Correct Answer: B) Passive transport

Explanation:

Passive Transport	Active Transport	Primary Active Transport	Secondary Active Transport
Moves molecules down their gradient (downhill transport) from an area of higher concentration to an area of lower concentration	Moves molecules against their gradient (uphill transport) from an area of lower concentration to an area of higher concentration.	It uses ATP directly to move molecules against their gradient from an area of lower concentration to an area of higher concentration.	It uses energy from the gradient created by primary active transport Primary active transport creates an ion gradient (e.g., Na^+ gradient by the

			Na ⁺ /K ⁺ pump) ↓ Na ⁺ /K ⁺ pump uses ATP to pump Na ⁺ out of the cell (against its gradient) ↓ A high concentration of Na ⁺ outside the cell ↓ Uses the Na ⁺ gradient to transport other molecules (e.g., glucose) into the cell
It does not require energy	Requires energy (ATP)	Requires energy (ATP)	Energy is indirectly required from the ATP used by the Na ⁺ /K ⁺ pump
Example: Diffusion of oxygen (O ₂) across a cell membrane	Example: Sodium-potassium pump (Na ⁺ /K ⁺ pump)	Example: Calcium pump (Ca ²⁺ pump)	Example: Glucose-sodium co-transport (SGLT)
Downhill transport	Not downhill transport	Not downhill transport	Not downhill transport

Solution for Question 14:

Correct Answer: D) Connexin 50

Explanation :

Connexin 46 and 50 mutations lead to cataracts.

Connexons:

- Structures that form the gap junction present in the membrane of each cell.
- 1 Connexon = 6 connexins

Connexins:

- Protein subunits, which together form connexons.
- They surround a channel.

- When lined up with the channel in the corresponding connexon in the adjacent cell, it permits substances to pass between the cells without entering the ECF.

Mutated Connexins and associated diseases:

Mutated Connexin	Disease
Connexin 30	Clouston syndrome
Connexin 30.3 and 31	Erythrokeratoderma Variabilis
Connexin 26,30 and 31	Inherited deafness
Connexin 36	Predisposition to Myoclonic Epilepsy
Connexin 37	Predisposition to Arteriosclerosis
Connexin 46 and 50	Cataract
Connexin 40	Idiopathic Atrial fibrillation
Connexin 32	Charcot-Marie-Tooth disease

Solution for Question 15:

Correct Answer: D) Gap Junctions - Enable rapid communication and coordination, forming channels between cells for electrical signal propagation.

Explanation:

Cell-to-Cell Interactions:

Mechanism	Key Components
Physical Contact	Tight junctions Adherens junctions Desmosomes Gap junctions
Chemical Messengers	Hormones Neurotransmitters growth factors
ECM Interaction	Proteins polysaccharides

Cellular Junctions:

Type	Key Features
------	--------------

Tight Junctions	Create a seal between adjacent cells. Prevent the passage of molecules through intercellular space. It is essential for maintaining tissue barriers and regulating substance movement across epithelia.
Adherens Junctions	Provide mechanical strength. Link cytoskeletons of adjacent cells.
Desmosomes	Similar to adherens junctions in function. Provide additional mechanical stability.
Gap Junctions	Form channels between cells. Allow direct passage of small molecules and ions. Enable rapid communication and coordination (e.g. - electrical signals in cardiac muscle).

Cell Adhesion Molecules (CAMs):

Type	Role
Integrins	Cell-to-cell and cell-to-ECM adhesion Involved in cell signaling, migration, and proliferation
Immunoglobulins	Cell-to-cell adhesion Immune recognition
Cadherins	Calcium-dependent Mediate cell-to-cell adhesion Crucial in tissue morphogenesis
Selectins	Transient cell-to-cell adhesion in the bloodstream Enable white blood cells to bind to endothelial cells and migrate to inflammation sites

Extracellular Matrix (ECM):

- Provides structural support, regulates cell behavior, and influences tissue development.

Chemical Messengers:

- Mechanism: Cells release and receive chemical messengers (hormones, neurotransmitters, growth factors).
- Function: Bind to specific receptors on target cells, triggering intracellular signaling pathways to regulate cellular processes.

Solution for Question 16:

Correct Answer: B) Extracellular fluid

Explanation:

"Milieu Intérieur" - The stable internal environment within an organism, maintained despite external changes.

Extracellular fluid (ECF):

Definition	Fluid outside the cells Constitutes about one-third of the total body water.
Function	It functions as the body's internal environment, or milieu intérieur since it contains the essential ions and nutrients necessary for cell survival.
Types	Interstitial Fluid: Fills the spaces between cells Lymph Tissue fluid Intravascular Fluid: The fluid within the blood vessels, mostly Plasma.

Solution for Question 17:

Correct Answer: B) i-d ; ii-c ; iii-b ; iv-a

Explanation:

Cell markers are used to identify different types of cells.

Cell marker	Cell
Cytokeratin	Epithelial cell
Vimentin	Mesenchymal cell.
Desmin	Skeletal muscle cell
Glial fibrillary acidic protein (GFAP)	Astrocytes

Solution for Question 18:

Correct Answer: C) All are correct except (ii)

Explanation:

Microtubules:

Role	Explanation	Key Points
Cellular Scaffold	Provides structural integrity as part of the cell cytoskeleton.	Constructed from polymerised tubulin Provides rigidity and shape to cell parts
Cytoskeleton of Organelles	Forms cytoskeletons of cilia, nerve axons, and mitotic spindles.	Important for cell division Maintains organisation and compartmentalisation
Dynamic Roles in Cells	Involved in cell division, movement, and intracellular transport.	Acts as conveyor belts Facilitates movement of vesicles, granules, and organelles within the cell

Solution for Question 19:

Correct Answer: B) Mitochondrial inner membrane

Explanation:

Biological membrane	Protein (%)	Lipid (%)	Ratio
Plasma membrane	~50	~50	~ 1.1
Mitochondrial inner membrane	~75	~25	~ 3.1
Myelin sheath	~15 to 30	~70 to 85	~ 0.17 to 0.4

Solution for Question 20:

Correct Answer: D) Glycocalyx contributes to negative surface charge, cell adhesion, and immune reactions.

Explanation:

Components of cell membrane:

Membrane Lipids:

Component	Description	Function
Phospholipids	Most abundant Arranged in a bilayer with hydrophilic heads outward and hydrophobic tails inward.	A barrier to water-soluble substances, allowing fat-soluble substances to pass.
Sphingolipids	It is found in smaller amounts, especially in nerve cells.	Protection Signal transmission Adhesion sites for extracellular proteins
Cholesterol	Embedded in the lipid bilayer.	Influences membrane permeability and fluidity

Membrane Proteins:

Type	Description	Function
Integral Proteins	Span the entire lipid bilayer, protruding on both sides.	Transport: Form channels for water and ions. Selective Diffusion: Facilitate specific substance diffusion Carrier Proteins: Bind and transport substances, including active transport Enzymes: Catalyze reactions Receptors: Bind ligands to trigger intracellular processes
Peripheral Proteins	Attached to the inner or outer surface, not spanning the lipid bilayer.	Function as enzymes or regulators of substance transport.

Membrane Carbohydrates - Glycocalyx:

Component	Description	Function
Glycoproteins	Proteins with carbohydrate chains	Protection Cell recognition Signaling.

Glycolipids	Lipids with carbohydrate chains	Cell recognition Communication.
Proteoglycans	Carbohydrate-rich compounds that are attached to the outer cell surface	Structural support Interaction with the extracellular matrix.

Solution for Question 21:

Correct Answer: A) Normal saline

Explanation:

Among the given options, normal Saline is the only isotonic solution. Only isotonic solutions should be used in the case of dehydration.

Isotonic solution:

Definition	Solution of equal concentration as that of red cell content & plasma. RBCs remain intact on using it.
Examples	0.9%NaCl (Normal Saline). 5% glucose. 10% mannitol. 20% urea.

Solution for Question 22:

Correct Answer: B) 1-a, 2-c, 3-b

Explanation:

Indicators used to measure body fluids:

ECF	Plasma volume	Interstitial fluid	ICF
-----	---------------	--------------------	-----

Inulin Radioactive Inulin Chlorine Bromine Mannitol Sucrose Sulphate Thiosulphate Thiocyanate Ferrocyanide	Evans Blue Dye Radio-iodinated Human Serum Albumin (RISA) Radio-iodinated Gamma Globulin Fibrinogen Radio-isotopes of Phosphorus Iron Chromium	It cannot be measured directly. It is calculated using ECF - Plasma volume.	It cannot be measured directly. It is calculated using TBW - ECF. Total body water (TBW) is measured by: Deuterium oxide (D ₂ O). Tritium oxide. Aminopyrine.
---	--	--	---

Solution for Question 23:

Correct Answer: D) All of the above

Explanation:

Transcellular Fluid:

Definition	Fluid in the lumen of structures lined by epithelium.
Examples	Fluid in the secretion of exocrine digestive glands Cerebrospinal fluid (CSF) Pleural, peritoneal, synovial and pericardial fluids Intraocular (aqueous and vitreous humor) fluids Bile Luminal fluids of the gut Fluid in the cochlea and kidney tubules
Volume	1.5% of body weight About 1L.

Solution for Question 24:

Correct Answer: A) 5 litres

Explanation:

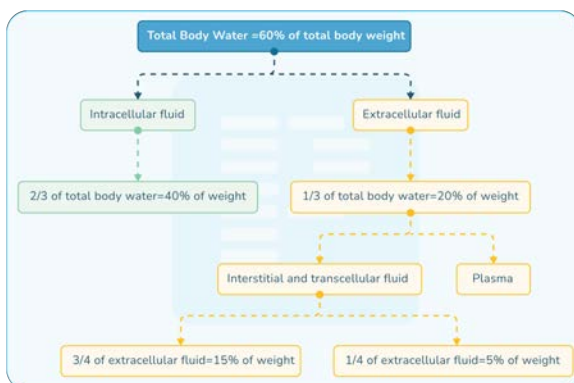
- Blood Volume = Plasma Volume/(1-Hematocrit)

$$= 4/(1-20\%)$$
$$= 4/(1-0.2)$$
$$= 4/0.8$$
$$= 5 \text{ litres}$$

Solution for Question 25:

Correct Answer: A) ICF makes up 40% of body weight

Explanation:



Solution for Question 26:

Correct Answer: B) 2, 3, 5

Explanation:

Feed-forward control system/anticipatory control system:

- Predicts changes and initiates responses to prevent deviations from the desired state before they occur
- Pre-empts disturbances, enhancing efficiency and stability by preparing in advance to maintain homeostasis

Example	Mechanism	Physiological Benefit
Anticipatory Regulation of Blood Glucose	Secretion of insulin in response to the sight or smell of food	Prevents a sharp spike in blood glucose levels after eating.
Heart Rate Adjustment during Exercise	Brain signals increase heart rate and blood flow to muscles before activity.	Ensures muscles receive adequate oxygen and nutrients from the start of exercise.
Digestive Enzyme Secretion	The cephalic phase triggers the secretion of digestive enzymes and gastric juices.	Prepares the digestive system to efficiently break down and absorb nutrients.
Thermoregulation	Anticipation of cold triggers vasoconstriction. Thermoregulation also has negative feedback component.	Maintains core body temperature, preventing a drop that could affect enzymatic activities.
Postural Control	Muscles adjust in advance for standing or moving.	Prevents falls and injuries by maintaining balance and coordination.
Feed-Forward Inhibition by the Cerebellum	Granular cells excite basket cells, which inhibit Purkinje cells, refining motor commands before they occur.	Ensures smooth, precise motor control by preventing excessive or unwanted movements.

Solution for Question 27:

Correct Answer: C) 0.5

Explanation:

The regulation factor checks the accuracy of the control system.

Regulation Factor = Change with control system / Change without control system

- Initial Respiratory Rate: 12 breaths/minute
- Post-Flight Respiratory Rate: 20 breaths/minute
- Corrected Respiratory Rate: 16 breaths/minute

Correction Achieved by the Respiratory Center = Change with control system = $(20 - 16 = 4$ breaths/minute)

Total Correction Needed = Change without control system = $(20 - 12 = 8$ breaths/minute)

- Regulation Factor = Change with control system / Change without control system = $4/8 = 0.5$

Solution for Question 28:

Correct Answer: B) -2

Explanation:

Correction achieved by the insulin secretion is: $(180 \text{ mg/dL} - 120 \text{ mg/dL} = 60 \text{ mg/dL})$

Error compared to the initial state is: $(120 \text{ mg/dL} - 90 \text{ mg/dL} = 30 \text{ mg/dL})$

Gain = Correction Achieved / Error Remaining

Substituting the values: Gain = $60 \text{ mg/dL} / 30 \text{ mg/dL} = 2$

Since the gain is negative in the context of this glucose regulatory mechanism (because an increase in blood glucose results in a decrease in gain):

Gain = -2

The degree of effectiveness with which a control system maintains constant conditions is determined by the gain of negative feedback.

Solution for Question 29:

Correct Answer: A) 1, 2, and 5

Explanation:

Thyroid hormone, blood glucose, and temperature regulation follow a negative feedback mechanism.

Negative feedback mechanism: The body responds to a change by initiating actions that counteract the change, aiming to restore equilibrium.

Examples of negative feedback mechanisms:

System	Stimulus	Response	Effect
Body Temperature	Temperature deviation	Sweating/shivering, vasodilation/constriction	Restores normal temperature
Blood Glucose	High/low blood glucose	Insulin/glucagon release	Normalises blood glucose levels
Thyroid Hormones	High/low thyroid hormones	TSH release adjustment	Maintains hormone balance
Blood Pressure	High/low blood pressure	Heart rate and vessel adjustment	Restores normal blood pressure
Calcium Homeostasis	High/low blood calcium	PTH/calcitonin release	Normalises blood calcium levels
Menstrual Cycle	Hormone level changes	FSH/LH regulation	Regulates menstrual cycle
Respiratory Rate	High CO ₂ levels	Increased breathing rate	Normalizes CO ₂ levels

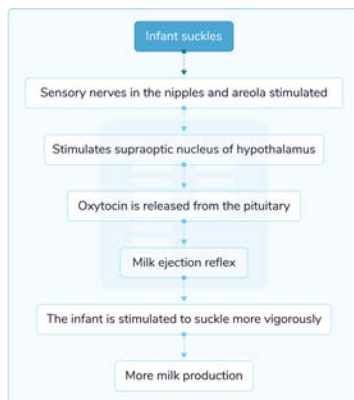
Solution for Question 30:

Correct Answer: B) Positive feedback mechanism

Explanation:

Breastfeeding is an example of a positive feedback mechanism. The initial stimulus amplifies the positive feedback response.

Breastfeeding:



Examples of positive feedback mechanisms:

Mechanism	Description	Outcome
Labour and Childbirth	Stretching of cervix stimulates oxytocin production. Oxytocin release stimulates uterine contractions, which in turn stimulate more oxytocin release.	Amplified contractions leading to childbirth.
Lactation	Baby's suckling stimulates prolactin and oxytocin release. Prolactin causes enhancement of milk production and oxytocin causes ejection of milk.	Increased milk flow during breastfeeding.
Blood Clotting	Platelets release chemicals that attract more platelets to the site of injury.	Rapid formation of a blood clot to seal the wound.
Nerve Signal Propagation	Depolarisation triggers the opening of sodium channels, causing further depolarisation and more sodium channel opening.	Rapid propagation of the action potential along the nerve fibre.

Previous Year Questions

1. Dehydration is more severe in infants compared to adults. Which of the following is a true statement about infants?

- A. Extracellular fluid (ECF) is more than intracellular fluid (ICF)
 - B. Intracellular fluid (ICF) is more than extracellular fluid (ECF)
 - C. Total body water in infants is more than in adults
 - D. Extracellular fluid equals intracellular fluid
-

2. What is the mechanism of action of botulinum toxin?

- A. Inhibits acetylcholine release from the presynaptic terminal
 - B. Inhibits acetylcholine reuptake
 - C. Blocks the postsynaptic acetylcholine receptor
 - D. Inhibits acetylcholinesterase
-

3. A patient presents with high blood pressure accompanied by a decrease in heart rate. What is the most likely physiological mechanism responsible for this response?

- A. Stimulation of baroreceptors
 - B. Inhibition of baroreceptors
 - C. Bezold-Jarisch reflex (J reflex)
 - D. Stimulation of chemoreceptors
-

4. Which electrolyte abnormality will lead to cardiac arrhythmia in patients with severe vomiting?

- A. Hypokalemia
 - B. Hyponatremia
 - C. Hyperkalemia
 - D. Hypocalcemia
-

5. Which ion is most responsible for the resting membrane potential (RMP)?

- A. Potassium (K^+)
 - B. Calcium (Ca^{2+})
 - C. Sodium (Na^+)
 - D. Chloride (Cl^-)
-

6. Which of the following best explains the role of a second messenger in cellular signaling?

- A. Act on extracellular receptors
 - B. Modulate extracellular hormone response
 - C. Modulate intracellular hormone response
 - D. Directly interact with the cell's DNA
-

7. What physiological response is most likely to occur in a person exposed to high temperatures while exercising vigorously to maintain internal body temperature

within a narrow range?

- A. Sweating and vasodilation in the skin
 - B. Shivering and vasoconstriction in the skin
 - C. Increased production of heat by the liver
 - D. Decreased heart rate (HR) and respiratory rate (RR)
-

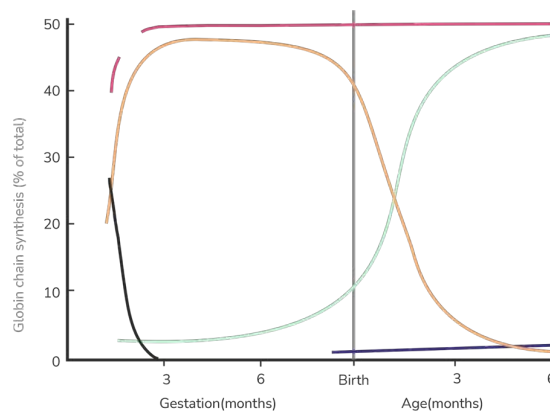
8. In negative feedback, infinite gain is seen in which of the following?

- A. Blood volume control by kidney
 - B. Temperature control in the hypothalamus
 - C. BP control by baroreceptors
 - D. Infinite gain is not possible
-

9. How do eukaryotes engage in intercellular communication?

- A. Lysosome
 - B. Peroxisome
 - C. Exosome
 - D. Genome
-

10. Which of the following curves represents gamma-globin chain production?



- A. Green curve
- B. Black curve
- C. Blue curve
- D. Orange curve

11. Which of the options below does not exemplify a feed-forward control mechanism?

- A. Temperature regulation
- B. Cephalic phase of gastric secretion
- C. Cerebellar action in motor coordination
- D. Exercise resulting in heart rate increase

12. I-125 labeled albumin is used for measurement of:

- A. Plasma volume
- B. Interstitial fluid volume
- C. ICF
- D. ECF

13. When calculating the quantity of IV fluids required for a patient, what amount of daily transient loss/insensible loss would you consider?

- A. 150-200 ml
 - B. 500-600 ml
 - C. 200-350 ml 350-400 ml
 - D. 350-400 ml
-

14. Determine the extracellular fluid (ECF) volume in a participant who received an injection of 10 grams of mannitol into their venous blood. After reaching equilibrium, the urinary excretion of mannitol was observed to be 10%, while the plasma concentration of mannitol measured 65 mg per 100 ml.

- A. 14L
 - B. 10L
 - C. 28L
 - D. 20L
-

15. Which of the following types of amino acids are present in the transmembranous part of the membrane receptor?

- A. Amphipathic
 - B. Hydrophobic
 - C. Hydrophilic
 - D. None of the above
-

16. Gibbs – Donnan equilibrium is mainly due to?

- A. Phosphate
 - B. Bicarbonate in the intracellular compartment
 - C. Calcium in the extracellular compartment
 - D. Large intracellular proteins
-

17. Lysosomal enzymes are present in?

- A. Cytosol
 - B. Mitochondria
 - C. Endoplasmic reticulum
 - D. Nucleus
-

18. Which of the following options has the lowest concentration in intracellular fluid (ICF) compared to extracellular fluid?

- A. Calcium
 - B. Magnesium
 - C. Potassium
 - D. Protein
-

19. Which of the following organelle has DNA?

- A. Mitochondria

- B. Golgi complex
 - C. SER
 - D. RER
-

20. What is the most accurate method to measure the temperature of a soldier brought to the ER from Siachen, in order to rule out hypothermia in a limited resource setting?

- A. Oral
 - B. Rectal
 - C. Infrared tympanic thermometer
 - D. Axillary
-

21. Which indicator is suitable for measuring the total amount of body water?

- A. Evan's blue
 - B. Tritium oxide
 - C. Mannitol
 - D. Sucrose
-

22. What mechanism is observed in the baroreceptor reflex among the options listed below?

- A. Feedforward
- B. Positive feedback
- C. Negative feedback
- D. Both A & B

23. Which of the following is not a dietary fiber?

- A. Cellulose
 - B. Inulin
 - C. Pectin
 - D. Gums
-

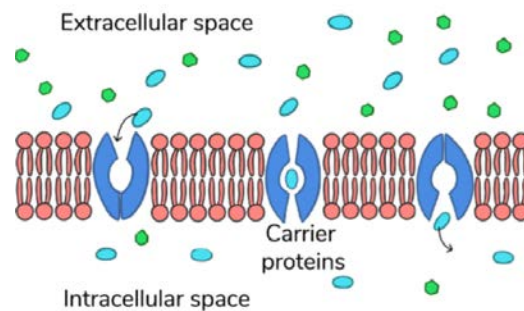
24. Which of the following is the principal constituent of serum osmolality?

- A. Potassium
 - B. Sodium
 - C. Bicarbonate
 - D. Hydrogen ions
-

25. A man was brought to the emergency after suddenly becoming unconscious while working in the field. On examination, his temperature was 105 degrees Fahrenheit, and his skin turgor was decreased. Which of the following would not be seen in the patient?

- A. Tachypnea
 - B. Hypotension
 - C. Sweating
 - D. Red hot skin
-

26. Identify the type of transport across the cell membrane given below.



- A. Simple diffusion
 - B. Facilitated diffusion
 - C. Primary active transport
 - D. Secondary active transport
-

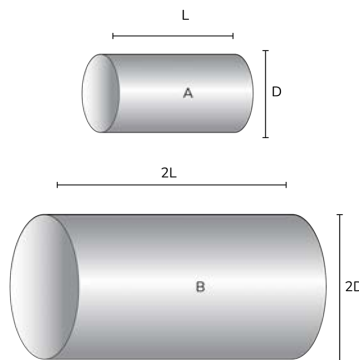
27. The signaling sequence in a type 2 membrane protein with N-terminal facing cytoplasm is located in?

- A. Inside the membrane
 - B. At the C-terminal
 - C. At the N-terminal
 - D. Both inside the membrane and at C-terminal
-

28. What is the normal insensible water loss?

- A. 100 mL/hour
- B. 150 mL/hour
- C. 50 mL/hour
- D. 200 mL/hour

29. Vessel A has a length of L and a diameter of D , and vessel B has $2L$ and $2D$. Considering the flow is laminar through both cylinders, the flow through B is how many times as much as the flow through A?



- A. 4 times
- B. 8 times
- C. 16 times
- D. 32 times

30. What is the value of normal oncotic pressure?

- A. 12 mmHg
- B. 25 mmHg
- C. 32 mmHg
- D. 54 mmHg

31. Calculate the sodium deficit in a 30-year-old man weighing 70 kg with a sodium level of 120 mEq/L.

- A. 280 mEq
 - B. 480 mEq
 - C. 840 mEq
 - D. 1400 mEq
-

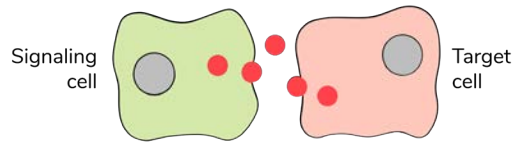
32. A body fluid sample is being studied with Na: 10 mEq/L, K: 140 mEq/L, Cl: 4 mEq/L. Identify the compartment from which the fluid has been obtained.

- A. Interstitial
 - B. Intracellular fluid
 - C. Extracellular fluid
 - D. Plasma
-

33. Which of the following cell organelles is associated with destruction of bacteria and viruses?

- A. Peroxisomes
 - B. Lysosomes
 - C. Golgi complex
 - D. Ribosomes
-

34. Which of the following types of cell-to-cell signaling is depicted in the image below?



- A. Paracrine
 - B. Autocrine
 - C. Endocrine
 - D. Merocrine
-

35. Glucose is absorbed in the intestine by?

- A. Primary active transport
 - B. Facilitated diffusion
 - C. Simple diffusion
 - D. Secondary active transport
-

36. Which molecular motor protein results in spread of the rabies virus from the site of inoculation to the central nervous system?

- A. Kinesin
 - B. Dynein
 - C. Actin
 - D. Myosin
-

37. In ORS, sodium is given along with glucose to facilitate?

- A. Co-transport
 - B. Osmosis
 - C. Facilitated diffusion
 - D. Primary active transport
-

38. 'c' wave in JVP is seen in?

- A. Iso-volumetric contraction
 - B. Slow filling at end of diastole
 - C. End of systole
 - D. Start of diastole
-

39. What is the most effective stimulus for increasing aldosterone secretion?

- A. Hyperkalemia
 - B. ACTH
 - C. Hypernatremia
 - D. Hyponatremia
-

40. What is the transporter molecule responsible for transporting Fe^{2+} into the enterocytes?

- A. Ferroportin
- B. Hephaestin
- C. Divalent metal Transporter 1
- D. Divalent Metal Transporter 2

41. Nuclear signalling pathways are all except

- A. Caveolins
 - B. NF- κ B
 - C. Importins
 - D. Rat proteins
-

42. What are the mechanisms involved in the absorption of glucose in the intestine?

- A. Primary active transport
 - B. Secondary active transport
 - C. Facilitated diffusion
 - D. Diffusion
-

43. What is the effect on blood flow when the radius of a vessel is doubled?

- A. 8 times
 - B. 16 times
 - C. 32 times
 - D. 256 times
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	1
Question 4	1
Question 5	1
Question 6	3
Question 7	1
Question 8	1
Question 9	3
Question 10	4
Question 11	4
Question 12	1
Question 13	2
Question 14	1
Question 15	2
Question 16	4
Question 17	3
Question 18	1
Question 19	1
Question 20	2
Question 21	2
Question 22	3

Question 23	4
Question 24	2
Question 25	3
Question 26	2
Question 27	1
Question 28	3
Question 29	2
Question 30	2
Question 31	3
Question 32	2
Question 33	2
Question 34	1
Question 35	4
Question 36	2
Question 37	1
Question 38	1
Question 39	1
Question 40	3
Question 41	4
Question 42	2
Question 43	2

Solution for Question 1:

Correct Answer: C) Total body water in infants is more than in adults

Explanation:

Infants have a higher total body water (TBW) percentage compared to adults, making them more susceptible to dehydration .

- Total body water (TBW) composition: Infants: ~ 75-80% of body weight Adults: ~ 55-60% of body weight
- Infants: ~ 75-80% of body weight
- Adults: ~ 55-60% of body weight
- Higher water turnover: Infants lose water more rapidly due to higher metabolic rates, increased insensible losses, and immature renal function .
- Infants: ~ 75-80% of body weight
- Adults: ~ 55-60% of body weight

Total Body Water (TBW) Composition:

- 90% of body weight in early fetal life.
- 75% of body weight at birth.
- 60% of body weight by the end of the first year, remaining consistent until puberty.
- After puberty, males have slightly more TBW compared to females.

Preterm vs. Term Neonates:

- Preterm neonates have a higher TBW compared to term neonates.

TBW Distribution:

- TBW is divided into: Extracellular Fluid (ECF) Intracellular Fluid (ICF)
- Extracellular Fluid (ECF)
- Intracellular Fluid (ICF)
- The equation is: $TBW = ECF + ICF$ $ECF = \text{plasma} + \text{interstitial fluid}$
- $ECF = \text{plasma} + \text{interstitial fluid}$
- Extracellular Fluid (ECF)
- Intracellular Fluid (ICF)
- $ECF = \text{plasma} + \text{interstitial fluid}$

ECF and ICF Changes:

- In fetuses and newborns, $ECF > ICF$ volume.
- After birth, ECF volume decreases and ICF volume increases.

Extracellular fluid (ECF) is more than intracellular fluid (ICF) (Option A) is incorrect because, while infants have a higher ECF proportion at birth (~40%) , ICF eventually exceeds ECF as they

grow.

Intracellular fluid (ICF) is more than extracellular fluid (ECF) (Option B) is incorrect at birth , but becomes true after the first year . In neonates, ECF is initially greater than ICF .

Extracellular fluid equals intracellular fluid (Option D) is incorrect because ECF and ICF are not equal at any stage ; there is always a shift, with ICF becoming dominant after infancy .

Solution for Question 2:

Correct Answer: A) Inhibits acetylcholine release from the presynaptic terminal

Explanation:

Botulinum toxin prevents acetylcholine (ACh) release at the neuromuscular junction , leading to flaccid paralysis .

- Mechanism of Action: Botulinum toxin cleaves SNARE proteins (SNAP-25, synaptobrevin, syntaxin) , which are essential for vesicular fusion with the presynaptic membrane. This prevents ACh-containing vesicles from being released into the synaptic cleft. Result: Muscle paralysis due to the lack of ACh stimulation at the nicotinic receptors .
- Botulinum toxin cleaves SNARE proteins (SNAP-25, synaptobrevin, syntaxin) , which are essential for vesicular fusion with the presynaptic membrane.
- This prevents ACh-containing vesicles from being released into the synaptic cleft.
- Result: Muscle paralysis due to the lack of ACh stimulation at the nicotinic receptors .
- Clinical Uses: Botox (cosmetic use) : Reduces wrinkles by relaxing muscles. Dystonias (e.g., cervical dystonia, blepharospasm) Spasticity (e.g., cerebral palsy, stroke-related spasticity) Hyperhidrosis (excessive sweating)
- Botox (cosmetic use) : Reduces wrinkles by relaxing muscles.
- Dystonias (e.g., cervical dystonia, blepharospasm)
- Spasticity (e.g., cerebral palsy, stroke-related spasticity)
- Hyperhidrosis (excessive sweating)
- Toxicity: Causes botulism , presenting with flaccid paralysis, ptosis, diplopia, dysphagia, and respiratory failure . Common sources: Improperly canned foods, honey (infant botulism) .
- Causes botulism , presenting with flaccid paralysis, ptosis, diplopia, dysphagia, and respiratory failure .

- Common sources: Improperly canned foods, honey (infant botulism) .
- Botulinum toxin cleaves SNARE proteins (SNAP-25, synaptobrevin, syntaxin) , which are essential for vesicular fusion with the presynaptic membrane.
- This prevents ACh-containing vesicles from being released into the synaptic cleft.
- Result: Muscle paralysis due to the lack of ACh stimulation at the nicotinic receptors .
- Botox (cosmetic use) : Reduces wrinkles by relaxing muscles.
- Dystonias (e.g., cervical dystonia, blepharospasm)
- Spasticity (e.g., cerebral palsy, stroke-related spasticity)
- Hyperhidrosis (excessive sweating)
- Causes botulism , presenting with flaccid paralysis, ptosis, diplopia, dysphagia, and respiratory failure .
- Common sources: Improperly canned foods, honey (infant botulism) .

Botulinum toxins:

- These are potent neurotoxins produced by Clostridium botulinum bacteria.
- There are several types, with types A, B, and E being toxic to humans.
- Mechanism : Botulinum toxins A and E cleave SNAP-25 (synaptosome-associated protein-25), and B cleave Synaptobrevin, which are presynaptic membrane proteins essential for releasing acetylcholine at the neuromuscular junction leading to Flaccid paralysis.

Inhibits acetylcholine reuptake (Option B) is incorrect because ACh is not reuptaken ; it is degraded by acetylcholinesterase in the synapse.

Blocks the postsynaptic acetylcholine receptor (Option C) is incorrect because this describes the action of curare and neuromuscular blockers (e.g., pancuronium, rocuronium) .

Inhibits acetylcholinesterase (Option D) is incorrect because this describes the action of acetylcholinesterase inhibitors (e.g., neostigmine, organophosphates) , which increase ACh levels rather than decrease them.

Solution for Question 3:

Answer: A) Stimulation of baroreceptors

Explanation:

The combination of high blood pressure (hypertension) and bradycardia (low heart rate) suggests activation of the baroreceptor reflex, a homeostatic mechanism that regulates blood pressure.

- Mechanism: Increased blood pressure stretches baroreceptors in the carotid sinus (CN IX) and aortic arch (CN X). Increased baroreceptor firing signals the medulla (nucleus tractus solitarius). Response: Inhibition of sympathetic outflow → ↓ heart rate (bradycardia) and ↓ vasoconstriction Activation of parasympathetic (vagal) tone → further ↓ heart rate
- Increased blood pressure stretches baroreceptors in the carotid sinus (CN IX) and aortic arch (CN X).
- Increased baroreceptor firing signals the medulla (nucleus tractus solitarius).
- Response: Inhibition of sympathetic outflow → ↓ heart rate (bradycardia) and ↓ vasoconstriction Activation of parasympathetic (vagal) tone → further ↓ heart rate
- Inhibition of sympathetic outflow → ↓ heart rate (bradycardia) and ↓ vasoconstriction
- Activation of parasympathetic (vagal) tone → further ↓ heart rate
- Increased blood pressure stretches baroreceptors in the carotid sinus (CN IX) and aortic arch (CN X).
- Increased baroreceptor firing signals the medulla (nucleus tractus solitarius).
- Response: Inhibition of sympathetic outflow → ↓ heart rate (bradycardia) and ↓ vasoconstriction Activation of parasympathetic (vagal) tone → further ↓ heart rate
- Inhibition of sympathetic outflow → ↓ heart rate (bradycardia) and ↓ vasoconstriction
- Activation of parasympathetic (vagal) tone → further ↓ heart rate
- Inhibition of sympathetic outflow → ↓ heart rate (bradycardia) and ↓ vasoconstriction
- Activation of parasympathetic (vagal) tone → further ↓ heart rate

Aspect	Details
Mechanism	Baroreceptors in the carotid sinus and the aortic arch detect BP changes Increased BP stretches receptors, sending signals to medulla via glossopharyngeal and vagus nerves Medulla inhibits vasoconstrictor center and stimulates vagal parasympathetic center
Actions	Vasodilation of blood vessels Decreased heart rate and contraction strength Reduced BP due to lower peripheral resistance and cardiac output
Importance	Maintains BP stability during activities like standing Fastest acting Tends to reset within 1-2 days for long-term regulation

This helps counteract hypertension and maintain homeostasis.

Inhibition of Baroreceptors (Option B) is incorrect because inhibition of baroreceptors occurs in hypotension, leading to tachycardia and vasoconstriction to restore blood pressure.

Bezold-Jarisch Reflex (Option C) is incorrect because this reflex causes bradycardia and hypotension, typically triggered by ventricular ischemia or excessive vagal stimulation, not hypertension.

Stimulation of Chemoreceptors (Option D) is incorrect because chemoreceptors (in the carotid bodies and medulla) primarily respond to hypoxia, hypercapnia, or acidosis, leading to increased respiratory drive and sympathetic activation (tachycardia), not bradycardia.

Solution for Question 4:

Correct Answer: A) Hypokalemia

Explanation:

Severe vomiting leads to hypokalaemia, which is a major risk factor for cardiac arrhythmias due to its effect on cardiac membrane potential and conduction.

- Mechanism: Vomiting causes loss of gastric acid (HCl) → metabolic alkalosis Alkalosis shifts potassium into cells → worsens hypokalaemia Renal compensation increases potassium excretion (due to increased bicarbonate delivery to the distal tubule) Low serum potassium disrupts cardiac repolarisation, leading to arrhythmias
- Vomiting causes loss of gastric acid (HCl) → metabolic alkalosis
- Alkalosis shifts potassium into cells → worsens hypokalaemia
- Renal compensation increases potassium excretion (due to increased bicarbonate delivery to the distal tubule)
- Low serum potassium disrupts cardiac repolarisation, leading to arrhythmias
- ECG changes in hypokalaemia: Flattened T waves U waves Prolonged QT interval → risk of ventricular arrhythmias (e.g., Torsades de Pointes, ventricular tachycardia)
- Flattened T waves
- U waves
- Prolonged QT interval → risk of ventricular arrhythmias (e.g., Torsades de Pointes, ventricular tachycardia)

- Vomiting causes loss of gastric acid (HCl) → metabolic alkalosis
- Alkalosis shifts potassium into cells → worsens hypokalaemia
- Renal compensation increases potassium excretion (due to increased bicarbonate delivery to the distal tubule)
- Low serum potassium disrupts cardiac repolarisation , leading to arrhythmias
- Flattened T waves
- U waves
- Prolonged QT interval → risk of ventricular arrhythmias (e.g., Torsades de Pointes, ventricular tachycardia)

Hyponatraemia (Option B) is incorrect because although vomiting can lead to hyponatraemia (due to volume loss and ADH secretion), it is less likely to cause direct cardiac arrhythmias .

Hyperkalaemia (Option C) is incorrect because vomiting typically leads to potassium loss , not hyperkalaemia. Hyperkalaemia is more commonly associated with renal failure, acidosis, or tissue breakdown .

Hypocalcaemia (Option D) is incorrect because hypocalcaemia causes neuromuscular symptoms (tetany, Chvostek's sign, Trousseau's sign) and QT prolongation , but it is not the primary electrolyte abnormality seen in vomiting-induced arrhythmias.

Solution for Question 5:

Correct A - Potassium (K^+) :

- Potassium ions are the primary contributors to the resting membrane potential. The RMP is mainly determined by the K^+ gradient across the cell membrane, as potassium ions move out of the cell through potassium leak channels. The loss of positively charged K^+ ions from the cell creates a negative charge inside the cell relative to the outside, establishing the RMP, which is typically around -70 mV in neurons.

Incorrect Options:

Option B - Calcium (Ca^{2+}) : Calcium ions play a crucial role in various cellular processes, including muscle contraction and neurotransmitter release. However, their influence on the RMP is relatively minimal compared to potassium. The primary role of calcium in relation to membrane potential is more related to action potentials and signal transduction rather than establishing the

RMP.

Option C - Sodium (Na^+) : Sodium ions also contribute to the membrane potential, but they are less influential than potassium in establishing the resting membrane potential. The sodium-potassium pump (Na^+/K^+ ATPase) maintains the gradient by actively transporting 3 Na^+ ions out of the cell and 2 K^+ ions into the cell, which helps in stabilizing the RMP. However, the resting membrane potential is primarily determined by the K^+ gradient.

Option D - Chloride (Cl^-) : Chloride ions contribute to the membrane potential by balancing the charge, but they are not the primary determinant of the resting membrane potential. Chloride ions typically follow the electrochemical gradients established by other ions like potassium and sodium.

Solution for Question 6:

Correct Option C - Modulate intracellular hormone response:

- Modulate intracellular hormone response: Second messengers are intracellular molecules that are generated in response to the binding of a hormone to its receptor on the cell surface. They help propagate and amplify the signal inside the cell, leading to various cellular responses, such as enzyme activation or gene expression.

Incorrect Options:

Option A - Act on extracellular receptors: Second messengers do not act directly on extracellular receptors. Instead, they work intracellularly after a receptor on the cell surface has been activated.

Option B - Modulate extracellular hormone response: Second messengers do not affect the extracellular hormone response; their function is to relay and amplify the signal inside the cell after the hormone binds to its receptor.

Option D - Directly interact with the cell's DNA: Second messengers do not directly interact with DNA. Instead, they influence cellular processes through various pathways, such as activating kinases or other signaling molecules, which then affect DNA indirectly.

Solution for Question 7:

Correct Option A - Sweating and vasodilation in the skin:

- When exercising vigorously in high temperatures, the body needs to dissipate heat to maintain a stable internal temperature. Sweating helps cool the body through evaporative cooling, while vasodilation (widening of blood vessels) in the skin increases blood flow to the surface, allowing more heat to be released into the environment.

Incorrect Options:

Option B - Shivering and vasoconstriction in the skin : Shivering and vasoconstriction are responses typically seen in cold environments. Shivering generates heat through muscle contractions, and vasoconstriction (narrowing of blood vessels) helps to retain heat. These mechanisms are not relevant in hot conditions.

Option C - Increased production of heat by the liver: The liver produces heat as a byproduct of metabolic processes, but this is not a primary response to high temperatures during vigorous exercise. The body focuses on cooling mechanisms rather than increasing heat production in such conditions.

Option D - Decreased heart rate (HR) and respiratory rate (RR): During vigorous exercise, both heart rate and respiratory rate typically increase to meet the higher oxygen demands of the muscles. A decrease in HR and RR would not be expected under these conditions; instead, they would rise to support increased physical activity.

Solution for Question 8:

Correct Option A - Blood volume control by kidney:

- The degree of effectiveness with which a control system maintains constant conditions is determined by the gain of the negative feedback control system. Kidney is the organ with infinite gain meaning the residual error is zero and there is total correction of the change from the normal value.
- The kidney is a very slow system for the regulation of blood volume. But given enough time, the kidney can maintain blood pressure, without residual error. The kidney will excrete excess water from the body and the blood pressure will return to back to normal value, which

makes the error 0.

Incorrect Options:

- Options B, C, and D are incorrect. There is residual error in these mechanisms.
-

Solution for Question 9:

Correct Option C -Exosomes:

- Intercellular communication in eukaryotes involves exchanging signaling molecules like proteins, nucleic acids, and lipids
- Exosomes, small vesicles produced within cells, play a vital role in this process. They carry bioactive molecules, including proteins and RNA, transferring them between cells to regulate gene expression, immune responses, and other physiological processes.

Incorrect Options:

Option A - Lysosome: Lysosomes are organelles involved in the degradation and recycling of cellular waste and materials.

Option B - Peroxisome: Peroxisomes are organelles involved in various metabolic processes, such as lipid metabolism and detoxification.

Option D - Genome: The genome refers to the complete set of genetic material (DNA) present in an organism or a cell.

Solution for Question 10:

Correct Option D - Orange curve:

- Orange curve, which represents gamma-globin chain production.
- In fetal development, the hemoglobin is formed by 2 gamma chains and 2 alpha chains, and gamma-globulin gradually decreases after birth as the production of other globin chains, such as alpha and beta, increases. The orange curve corresponds to the gamma chain,

which is predominant during fetal life.

Incorrect Options:

Option A: Green curve: This represents beta-globin chain production, not gamma-globin chain.

Option B: Black curve: This represents epsilon-globin chain production, not the gamma-globin chain.

Option C: Blue curve: This represents delta-globin chain production, not a gamma-globin chain.

Solution for Question 11:

Correct Option D - Exercise resulting in heart rate increase:

- Exercise-induced heart rate increase, illustrates feedback, not feedforward control. The heart rate elevates in response to immediate exercise demands, rather than anticipatorily preparing for changes.

Incorrect Options:

Option A - Temperature regulation: This is an example of a feedforward control mechanism. For instance, when the body anticipates exposure to cold temperatures, it may initiate processes like vasoconstriction and shivering to maintain or raise body temperature before it actually drops significantly.

Option B - Cephalic phase of gastric secretion: The cephalic phase of gastric secretion is indeed a feedforward control mechanism. It occurs before food reaches the stomach, and sensory stimuli such as the sight and smell of food can trigger the release of gastric juices in anticipation of digestion.

Option C - Cerebellar action in motor coordination: The cerebellum plays a significant role in motor coordination and is involved in feedforward control. It helps anticipate and make adjustments to muscle activities to coordinate movements effectively.

Solution for Question 12:

Correct Option A - Plasma volume:

- I-125 labeled albumin is commonly used to measure plasma volume. I-125 is a iodine labeled albumin that is injected into the bloodstream.
- It is impermeable to capillaries and does not dissociate into interstitial fluid and intracellular fluid. So, by measuring the dilution of the labeled albumin in the plasma, the total volume of plasma can be determined.
- Evan's blue is another substance that can be used for determining plasma volume.

Incorrect Options:

- Options B, C, and D are incorrect. I-125 labeled iodine is used to measure the plasma volume.
-

Solution for Question 13:

Correct Option B - 500-600 ml:

- Constant water evaporation occurs in our body during respiration or directly from the skin's surface, which is normal, is referred to as insensible water loss.
- Normally, 700 ml of water evaporates from our body daily. 350 ml of water is evaporated from the skin's surface and 350 ml from the respiratory tract.

Incorrect Options:

- Options A, C and D are incorrect.
-

Solution for Question 14:

Correct Option A - 14L:

- Mannitol is used to measure the extracellular fluid volume compartment because it is freely permeable to the capillary membrane and impermeable to cell membrane.

- Given in the question: Amount of mannitol injected = 10 g Fraction excreted in urine = 10% = 1 gm. Plasma concentration of mannitol = 65 mg / 100 ml
- Amount of mannitol injected = 10 g
- Fraction excreted in urine = 10% = 1 gm.
- Plasma concentration of mannitol = 65 mg / 100 ml
- Using the formula of volume distribution $(V) = (\text{Amount injected} - \text{Amount excreted}) / \text{Concentration}$
- Substituting the values into the formula: $V = (10 \text{ g} - 1 \text{ g}) / (65 \text{ mg} / 100 \text{ ml})$
- $V = (10 \text{ g} - 1 \text{ g}) / (65 \text{ mg} / 100 \text{ ml})$
- Let's convert the units to be consistent: $V = (10,000 \text{ mg} - 1,000 \text{ mg}) / (65 \text{ mg} / 100 \text{ ml})$ $V = 9,000 \text{ mg} / (65 \text{ mg} / 100 \text{ ml})$ $V = (9,000 \text{ mg} * 100 \text{ ml}) / 65 \text{ mg}$ $V = 900,000 \text{ ml} / 65 \text{ mg}$ $V \approx 13,846 \text{ ml}$
- $V = (10,000 \text{ mg} - 1,000 \text{ mg}) / (65 \text{ mg} / 100 \text{ ml})$
- $V = 9,000 \text{ mg} / (65 \text{ mg} / 100 \text{ ml})$
- $V = (9,000 \text{ mg} * 100 \text{ ml}) / 65 \text{ mg}$
- $V = 900,000 \text{ ml} / 65 \text{ mg}$
- $V \approx 13,846 \text{ ml}$
- Finally, converting milliliters to liters $V \approx 13.846 \text{ L}$.
- Therefore, the correct ECF volume is approximately 13.846 L, which is closest to option 14L.
- Amount of mannitol injected = 10 g
- Fraction excreted in urine = 10% = 1 gm.
- Plasma concentration of mannitol = 65 mg / 100 ml
- $V = (10 \text{ g} - 1 \text{ g}) / (65 \text{ mg} / 100 \text{ ml})$
- $V = (10,000 \text{ mg} - 1,000 \text{ mg}) / (65 \text{ mg} / 100 \text{ ml})$
- $V = 9,000 \text{ mg} / (65 \text{ mg} / 100 \text{ ml})$
- $V = (9,000 \text{ mg} * 100 \text{ ml}) / 65 \text{ mg}$
- $V = 900,000 \text{ ml} / 65 \text{ mg}$
- $V \approx 13,846 \text{ ml}$

Incorrect Options:

- Options B, C, and D are incorrect.

Solution for Question 15:

Correct Option B-Hydrophobic :

The transmembrane region of membrane receptors anchors the receptor within the lipid bilayer and transmits signals across the membrane.

It mainly consists of hydrophobic amino acids, which are suited for insertion into the hydrophobic interior of the lipid bilayer.

Hydrophilic and amphipathic amino acids are less common in this region.

Incorrect Options:

Option A (Amphipathic): While some membrane receptors may have amphipathic amino acids in regions that interface with the lipid bilayer and the aqueous environment, they are less commonly found in the transmembrane region itself.

Option C (Hydrophilic): Hydrophilic amino acids are more commonly found in regions of membrane receptors that are exposed to the aqueous environment rather than the transmembrane region.

Option D none of the above is incorrect

Solution for Question 16:

Correct Option D - Large intracellular proteins:

- The Gibbs-Donnan equilibrium, named after J. Willard Gibbs and Frederick G. Donnan, occurs across semipermeable membranes in the presence of large impermeable substances like proteins containing charge.
- According to Donnan and Gibbs the presence of a non-diffusible ion, the diffusible ions distribute themselves so that at equilibrium, their concentration ratios are equal.

Incorrect Options:

- Options A, B, and C are incorrect. These are diffusible ions.
-

Solution for Question 17:

Correct Option C-Endoplasmic Reticulum :

- Lysosomal enzymes are made as inactive precursors in the endoplasmic reticulum, processed in the Golgi apparatus, and activated in lysosomes at acidic pH.

Incorrect Options:

Option A: Cytosol: Lysosomal enzymes are not present within cytosol.

Option B: Mitochondria: Mitochondria are organelles involved in energy production and metabolism. They have their own distinct set of enzymes, such as those involved in oxidative phosphorylation and the Krebs cycle. Lysosomal enzymes are not typically found within mitochondria.

Option C: Endoplasmic reticulum: The endoplasmic reticulum (ER) is involved in protein synthesis, folding, and lipid metabolism. While the ER plays a role in the synthesis of lysosomal enzymes as inactive precursors, the active lysosomal enzymes are not present within the ER itself.

Option D: Nucleus: The nucleus is the central organelle that houses the cell's genetic material and is involved in gene expression and regulation. Lysosomal enzymes are not found within the nucleus. However, the nucleus contains its own set of enzymes involved in DNA replication, transcription, and other nuclear processes.

Solution for Question 18:

Correct Option A - Calcium:

- Calcium is present in the least concentration in intracellular fluid (ICF). Inside the cell, calcium ions are tightly regulated to maintain proper cellular function.
- The intracellular concentration of calcium is relatively low (0.0001 mEq/L) compared to the concentration in extracellular fluid..

Incorrect Options:

- Options B, C, and D are incorrect. Their concentration is relatively higher in intracellular fluid compared to extracellular fluid.

Solution for Question 19:

Correct option A-Mitochondria :

- Mitochondria are organelles found in eukaryotic cells that have their own DNA. They are often referred to as the "powerhouses" of the cell because they are responsible for generating most of the cell's energy through a process called cellular respiration. Mitochondria have a double membrane structure, and within their inner membrane, they house their own DNA, known as mitochondrial DNA (mtDNA).
- Mitochondrial DNA is separate from the nuclear DNA found in the cell's nucleus. It contains genes that encode for some of the proteins, RNA molecules, and other components needed for mitochondrial function and energy production.

Incorrect Options:

Option B. The Golgi complex, or Golgi apparatus, is involved in processing, modifying, and packaging proteins and lipids for transport to their specific destinations within the cell or for secretion. It does not have its own DNA.

Option C . The SER is involved in various functions, including lipid metabolism, detoxification of drugs and toxins, and calcium storage. It does not contain DNA.

Option D. The RER is responsible for the synthesis and processing of proteins. It is called "rough" because it is studded with ribosomes involved in protein synthesis. However, it does not contain its own DNA.

Solution for Question 20:

Correct Option B - Rectal:

- The rectal temperature is representative of the temperature at the core of the body and varies least with changes in environmental temperature.

Incorrect Options:

- Options A, C, and D are incorrect. Variations of temperature in these areas are usually associated with extreme environment.
-

Solution for Question 21:

Correct Option B - Tritium:

- Tritium oxide, also known as hydrogen-3, is a radioactive isotope of hydrogen. It can be used as an indicator to measure total body water because it is freely permeable to capillary and cell membrane. When tritiated water (water containing tritium) is ingested or injected into the body, it distributes uniformly throughout the total body water compartment. By measuring the concentration of tritium in body fluids, researchers can estimate the total body water content.
- Other indicators used for measuring TBW deuterium oxide and aminopyrine,

Incorrect Options:

Option A - Evans blue: Evans blue is a dye that is commonly used to measure plasma volume.

Option C and D: Mannitol and sucrose are used to measure extracellular fluid volume.

Solution for Question 22:

Correct Option C - Negative feedback:

- The baroreceptor reflex is a negative feedback mechanism that helps regulate blood pressure. It involves the baroreceptors, which are stretch receptors located in the walls of blood vessels, such as the carotid sinus and aortic arch, as well as the heart. These receptors detect changes in blood pressure and send signals to the brain, specifically the cardiovascular control center in the medulla oblongata, to initiate a response.
- An increase in BP results in activation of the nucleus tractus solitarius (NTS) which activates the CVLM (caudal ventrolateral medulla), and inhibits the RVLM (Rostral Ventrolateral Medulla) which is responsible for the inhibition of sympathetic stimulation

resulting in decreased cardiac output and contractility.

- If there is a decrease in BP it will not activate NTS and CVLM which in turn do not inhibit the RVLM and increases cardiac output by sympathetic supply.

Incorrect Options:

Option A - Feedforward: Feedforward is a mechanism in which the body anticipates changes and prepares for them before they occur. Regulation of body temperature is an example of a feedforward system.

Option B - Positive feedback: Positive feedback is a mechanism in which a change in stimuli leads to an amplification of that change. For example secretion of oxytocin leading to uterine contractions, in turn will result in further secretion of oxytocin and further uterine contractions.

Option D - Both A & B: It is an incorrect option.

Solution for Question 23:

Correct Option D-Gums :

- Gums like guar gum, locust bean gum, and xanthan gum are hydrocolloids derived from plants and used as food additives for thickening, stabilizing, and emulsifying. While they may contribute to dietary fiber content, they lack the same health benefits as natural dietary fibers.

Incorrect Options:

Option A - Cellulose: Insoluble fiber found in plant cell walls, adds bulk to stool.

Option B - Inulin: Soluble fiber found in chicory root, acts as a prebiotic.

Option C - Pectin: Soluble fiber found in fruits, used as a thickening agent, promotes satiety.

Solution for Question 24:

Correct Option B - Sodium:

- The combined presence of sodium and chloride accounts for approximately 90% of plasma osmolality. Out of a total of 295 mOsm/kg H₂O, sodium chloride contributes around 270 mOsm, while the remaining 20 mOsm is attributed to urea, glucose, protein, and other ions such as potassium and calcium.
- Whenever Na⁺ moves from one compartment to the other, Cl⁻ also moves to maintain electroneutrality, mainly contributing to osmolality.
- Therefore, it is mainly the Na⁺ that contributes to osmolality.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 25:

Correct Option C - Sweating:

- The scenario described a case of heat stroke, a severe form of heat illness that occurs when the body's heat-regulating systems fail to respond to high temperatures adequately.
- Classic signs of heat stroke include hyperthermia (body core temperature >105°F or 40.5°C), altered mental status, increased respiratory rate, and hot, dry skin.
- Sweating is often absent in heat stroke as the body's thermoregulatory mechanisms are overwhelmed.

Incorrect Options:

- Options A, B, and D are incorrect. These are seen in patients with heat stroke.

Options A, B, and D are incorrect. These are seen in patients with heat stroke.

Solution for Question 26:

Correct Options B - Facilitated diffusion :

- The image represents facilitated diffusion, a passive transport that moves substances down their concentration gradient without using cellular energy but requires the assistance of specific carrier proteins embedded in the cell membrane.
- This process is essential for transporting large or polar molecules that cannot cross the cell membrane's lipid bilayer by simple diffusion.

Incorrect Options:

Option A - Simple diffusion: This is a passive transport mechanism where substances move across the cell membrane along their concentration gradient directly through the lipid bilayer without the assistance of carrier proteins.

Option C - Primary active transport: This active transport uses cellular energy (ATP) to transport substances against their concentration gradient. This process often involves specific carrier proteins, such as the Na⁺/K⁺ ATPase pump.

Option D - Secondary active transport: This type of active transport uses the energy stored in the form of ion concentration gradients, established by primary active transport, to drive the transport of other substances against their concentration gradient.

Solution for Question 27:

Correct Option A - Inside the membrane:

- The signaling sequence in a type 2 membrane protein with N-terminal facing cytoplasm is located inside the membrane. This sequence is generally a hydrophobic alpha-helix that serves as a stop-transfer anchor, preventing the protein from being fully translocated across the membrane and causing it to become embedded in the membrane with the N-terminus facing the cytoplasm.
- The N-terminus and C-terminus of a protein refer to the amino acid residues at the beginning and end of the protein, respectively.
- Protein N-terminus: start; may hold signal for cell localization.
- Protein C-terminus: end; can drive interactions or enzyme function.

Incorrect Options:

Option B - At the C-terminal: The C-terminal of a type 2 membrane protein is located on the extracellular side of the membrane and is not involved in the signaling sequence or the process

of membrane insertion.

Option C - At the N-terminal: The N-terminal of a type 2 membrane protein is located on the cytoplasmic side of the membrane and is involved in the signaling sequence that determines the orientation of the protein in the membrane.

Option D - Both inside the membrane and at C-terminal : Only the inside of the membrane is involved in the signaling sequence of a type 2 membrane protein with N-terminal facing cytoplasm; the C-terminal is not involved in this process.

Solution for Question 28:

Correct Option C - 50 mL/hour:

- Insensible water loss is the amount of water lost from the body through evaporation, such as through the skin or respiratory tract that cannot be easily measured. Normal insensible water loss is generally considered to be around 30-50 mL/hour in adults .

Incorrect Options:

- Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 29:

Correct Option B - 8 times:

- Flow is equal to the pressure difference divided by resistance. So, flow is inversely proportional to the resistance of the vessel.
- According to the Hagen–Poiseuille equation, resistance can be directly measured by the following formula.
- $R = 8\eta L / \pi r^4$ Where, R = resistance. η = viscosity. L = length. r = radius of the vessel.
- Where,
- R = resistance.

- h = viscosity.
- L = length.
- r = radius of the vessel.
- From this equation, the resistance of a vessel is inversely proportional to the fourth power of radius and directly proportional to the length of the vessel..
- So, if the radius and length are doubled, the original and other parameters remain constant, then the resistance will be: $R = L/(r)^4$. $R = 2/(2)^4$. $R = 2/16 = 8$.
- $R = L/(r)^4$.
- $R = 2/(2)^4$.
- $R = 2/16 = 8$.
- So, the resistance will decrease by 8 times, and flow will increase by 8 times.
- Where,
- R = resistance.
- h = viscosity.
- L = length.
- r = radius of the vessel.
- $R = L/(r)^4$.
- $R = 2/(2)^4$.
- $R = 2/16 = 8$.

Incorrect Options:

- Option A, C and D are incorrect.

Solution for Question 30:

Correct Option B - 25 mmHg:

- The oncotic pressure , also known as colloid osmotic pressure , is the osmotic pressure exerted by proteins, notably albumin, in a blood vessel's plasma that usually pulls water into the circulatory system, i.e., it is a retaining force. The normal value of oncotic pressure in a healthy individual's plasma is approximately 25 mmHg.

Incorrect Options:

- Option A, C and D are incorrect.
-

Solution for Question 31:

Correct Option C-840 mEq:

- To calculate the sodium deficit, the following formula can be used:
- Sodium Deficit (mEq) = (Desired Sodium - Current Sodium) x Total body water (TBW)
- Total body water (TBW) = 60% of total body weight.
- $TBW = 0.6 \times 70 = 42$.
- Desired sodium is 135 to 145 mEq/L, so taking an average value of 140 mEq/L.
- Substituting the values in the formula: Sodium Deficit = $(140 - 120) \times 42$. Sodium Deficit = 840 mEq
- Sodium Deficit = $(140 - 120) \times 42$.
- Sodium Deficit = 840 mEq
- Sodium Deficit = $(140 - 120) \times 42$.
- Sodium Deficit = 840 mEq

Incorrect Options:

- Options A, B, and D are incorrect.
-

Solution for Question 32:

Correct Option B-Intracellular fluid :

- Given the electrolyte concentrations, the sample likely originates from intracellular fluid. The low sodium (Na^+) concentration (10 mEq/L) contrasts with typical extracellular levels (135-145 mEq/L), suggesting it's not extracellular.
- The high potassium (K^+) concentration (140 mEq/L) and low concentration of chloride (4 mEq/L) aligns with levels in the intracellular fluid compartment. Thus, the sample's

composition suggests it's intracellular fluid.

Incorrect Options :

- Options A, C, and D are incorrect. These all are extracellular fluids and the composition of the ions are not consistent with its composition.

Solution for Question 33:

Correct Option B-Lysosomes :

- Lysosomes, membrane-bound organelles, destroy bacteria and pathogens within cells. They contain enzymes that break down biomolecules, aiding in the cell's defense.
- By fusing with phagosomes, lysosomes form phagolysosomes, where enzymes degrade engulfed pathogens, killing them.

Incorrect Options:

Option A - Peroxisomes: Peroxisomes specialize in oxidative processes using molecular oxygen, producing hydrogen peroxide and utilizing catalase to break down fats..

Option C - Golgi complex: The Golgi complex, composed of membranes, prepares proteins and lipids for use both within and outside the cell.

Option D - Ribosomes: Ribosomes, consisting of RNA and protein, synthesize proteins by translating mRNA sequences. They are not involved in the destruction of bacteria or pathogens.

Solution for Question 34:

Correct Option A - Paracrine:

- In Paracrine signaling, the signaling molecule is released by a cell and acts on nearby cells to trigger a response.
- The signaling molecule can diffuse through the extracellular matrix and reach neighboring cells.

- An example of paracrine signaling is the release of growth factors by cells during tissue repair.

Incorrect Options:

Option B - Autocrine signaling:

- This mode of signaling is similar to paracrine signaling, but the signaling molecule acts on the same cell or cells of the same type that produced it. The beta-cell of the pancreas produces Insulin which inhibits the same cells for further release.

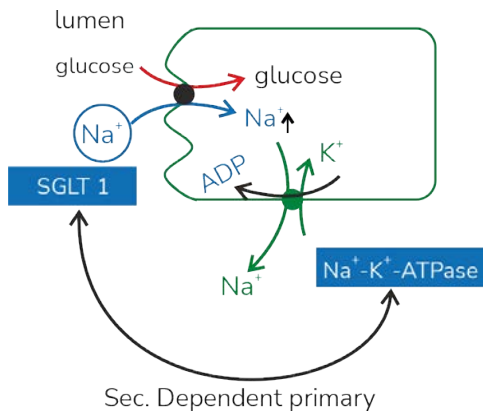
Option C - Endocrine signaling: In this mode of signaling, hormones are released into the bloodstream by endocrine glands and travel throughout the body to reach their target cells. Examples of hormones include insulin, estrogen, and testosterone.

Option D - Merocrine secretion: This mode of secretion involves the release of products by exocytosis, where the product is packaged in vesicles and released through the plasma membrane. Merocrine secretion is a mode of secretion in the salivary glands, sweat glands, and pancreas.

Solution for Question 35:

Correct Option D: Secondary active transport

- Glucose is primarily absorbed in the intestine through SGLT 1, which transports the 2 molecules of Na^+ with the gradient and 1 molecule of glucose against the gradient in the enterocyte, which is a secondary active transport.
- It is dependent on Na^+ K^+ ATPase for the removal of excess Na^+ from the enterocyte into the interstitial space when a large amount of glucose is absorbed from the cell for accommodation of the newer molecules.



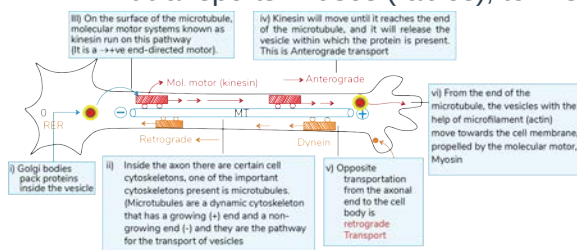
Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 36:

Correct Option B - Dynein:

- Dynein is a motor protein that facilitates retrograde transport within cells. It moves towards the minus end of microtubules, i.e., retrograde transport which is typically oriented towards the center of the cell body of the nerve.
- It transports viruses (rabies), toxins, and growth factors.



Incorrect Options:

Option A - Kinesin: Kinesin is an anterograde transport, i.e., positive end directed molecular motor protein that moves along microtubules, facilitating the transport of vesicles and mitochondria.

Options C and D: Myosin is a bi-directional molecular motor protein involved in the transportation of the substance to and fro from the positive end of the microtubules to the axon terminal with the help of actin.

Solution for Question 37:

Correct Option A - Co-transport :

- Sodium is given along with glucose to facilitate co-transport. The Na⁺/glucose cotransporter SGLT1 is an active pump that stoichiometrically transports two sodium ions along the gradient and one glucose molecule against the gradient together across the cell membrane of the enterocyte.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 38:

Correct Option A - Iso-volumetric contraction:

- The 'c' wave in the JVP is typically seen during the isovolumetric contraction phase of the cardiac cycle. It occurs during early ventricular systole when the tricuspid valve bulges back into the right atrium due to the increased pressure in the right ventricle. This causes a slight rise in the JVP.
- The isovolumetric contraction occurs immediately after the closure of the atrioventricular valves (mitral and tricuspid valves) and before the onset of ventricular ejection.

Incorrect Options:

- Options B, C, and D are incorrect.
-

Solution for Question 39:

Correct Option A - Hyperkalemia:

- Aldosterone secretion is maximally stimulated by hyperkalemia, which refers to an elevated level of potassium in the blood.
- High levels of potassium act directly on the adrenal cortex, specifically the zona glomerulosa, to increase aldosterone production and release.
- Aldosterone acts on the kidneys to promote potassium excretion and sodium reabsorption, helping to regulate electrolyte balance.

Incorrect Options:

Option B - ACTH: While ACTH (adrenocorticotropic hormone) does stimulate the secretion of aldosterone to some extent, it is not the primary factor that maximally stimulates aldosterone secretion. ACTH primarily stimulates the secretion of cortisol from the adrenal cortex.

Option C - Hyponatremia: Hyponatremia decreases the aldosterone secretion.

Option D - Hypernatremia: It also increases the secretion of aldosterone, but is not the most potent stimulus.

Solution for Question 40:

Correct Option C - Divalent Metal Transporter 1 (DMT1):

- Divalent metal transporter-1 (DMT1) is a transmembrane protein located in the apical membrane of enterocytes. It is responsible for transporting ferrous forms of iron from the intestinal lumen into the enterocyte.
- DMT1 is expressed in the duodenum, the first part of the small intestine, where iron is absorbed from the diet.

Incorrect Options:

Option A - Ferroportin: Ferroportin is a transmembrane protein located on the basolateral membrane of enterocytes. It functions as the main exporter of intracellular iron, allowing iron to be released into the bloodstream for systemic distribution.

Option B - Hephaestin: Hephaestin is a copper-dependent ferroxidase enzyme found on the basolateral membrane of enterocytes. It plays a role in oxidizing Fe^{2+} to Fe^{3+} , which enables the transfer of iron to transferrin for transport in the bloodstream.

Option D - Divalent Metal Transporter 2 (DMT2): DMT2 is a transporter primarily involved in the transport of manganese (Mn^{2+}) rather than iron.

Solution for Question 41:

Correct Option D - Rat proteins:

- These are not nuclear signalling pathways.
- Recombinant rat IFN is a bioactive protein, it is used in cell culture applications.

Incorrect Options:

Option A - Caveolins: Caveolins are a family of proteins that are involved in the formation of caveolae, which are specialised lipid rafts in the plasma membrane. They are involved in receptor independent endocytosis.

Option B - NF- κ B: NF- κ B (nuclear factor kappa-light-chain-enhancer of activated B cells) is a transcription factor that plays a central role in the regulation of immune and inflammatory responses. It is involved in various nuclear signaling pathways and is activated in response to diverse stimuli, such as cytokines, pathogens, and stress signals.

Option C - Importins: Importins are proteins that facilitate the transport of molecules into the nucleus. They are responsible for importing various molecules, including transcription factors, into the nucleus to regulate gene expression and participate in nuclear signaling pathways.

Solution for Question 42:

Correct Option B - Secondary active transport :

- Glucose is primarily absorbed in the intestine through SGLT 1, which transports the 2 molecules of Na⁺ with the gradient and 1 molecule of glucose against the gradient in the enterocyte, which is a secondary active transport.

- It is dependent on Na⁺ + K⁺ + ATPase for the removal of excess Na⁺ from the enterocyte into the interstitial space when a large amount of glucose is absorbed from the cell for accommodation of the newer molecules.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 43:

Correct Option B - 16 times:

- Flow is equal to the pressure difference divided by resistance. So, flow is inversely proportional to the resistance of the vessel.
- According to the Hagen–Poiseuille equation, resistance can be directly measured by the following formula.
- $R = \frac{8hL}{\pi r^4}$ Where, R = resistance. h = viscosity. L = length. r = radius of the vessel.
- Where,
- R = resistance.
- h = viscosity.
- L = length.
- r = radius of the vessel.
- From this equation, the resistance of a vessel is inversely proportional to the fourth power of radius.
- So, if the radius is doubled the original and other parameters remain constant, then the resistance will be: $R = \frac{1}{(r)^4}$. $R = \frac{1}{(2)^4}$. $R = \frac{1}{16} = 16$.
- $R = \frac{1}{(r)^4}$.
- $R = \frac{1}{(2)^4}$.
- $R = \frac{1}{16} = 16$.
- So, the resistance will decrease by 16 times, and flow will increase by 16 times.
- Where,
- R = resistance.
- h = viscosity.
- L = length.

- r = radius of the vessel.

- $R = 1/(r)^4$.

- $R = 1/(2)^4$.

- $R = 1/16 = 0.0625$.

Incorrect Options:

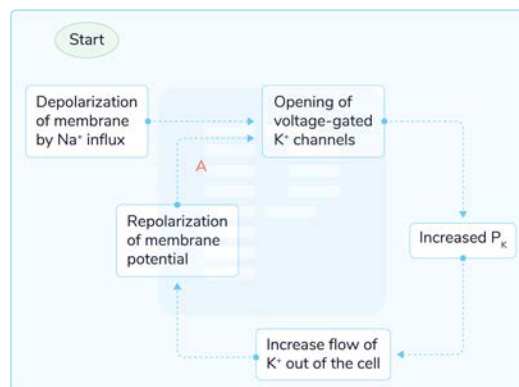
- Options A, C, and D are incorrect.

Neuron and nerve, Action Potential, RMP, Nerve fibre classification

1. Which of the following best describes the difference between temporal summation and spatial summation in neurons?

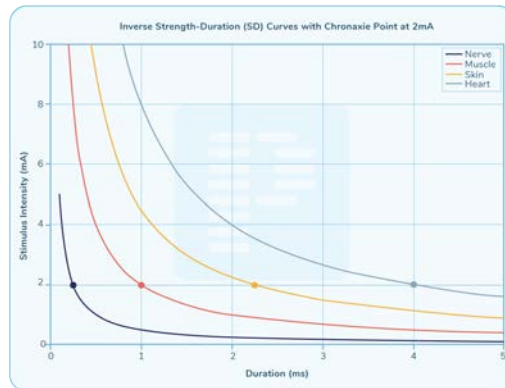
- A. There is no difference between Temporal summation and Spatial summation.
- B. Temporal summation requires signals to occur close in time, spatial summation requires simultaneous signals at different synapses.
- C. Temporal summation is not limited by the refractory period, spatial summation is.
- D. Temporal summation combines signals from different synapses, spatial summation combines signals from the same synapse.

2. Feedback control in voltage-gated ion channels in the membrane of neuron during action potential has been shown here. The phenomenon shown at point A is:



- A. Positive feedback
- B. Negative feedback
- C. Feedforward
- D. Voltage dependent opening

3. Neurologists decide to assess the excitability of different tissues by plotting the strength-duration relationship parameters on a graph. Which of the following tissues show the highest excitability?



- A. Nerve
- B. Muscle
- C. Skin
- D. Heart

4. A 35-year-old woman presents to the neurology clinic with complaints of tingling and numbness in her hands and feet. On examination, the neurologist decides to conduct nerve conduction studies to assess the function of her peripheral nerves. The results show a compound action potential (CAP) with multiple peaks and a biphasic waveform. What does the presence of multiple peaks indicate?

- A. Summation of action potentials from a single nerve fiber
- B. Different conduction velocities of various nerve fibers
- C. Biphasic nature of the waveform
- D. Single directional deflection

5. A 35-year-old man undergoes a minor surgical procedure on his hand under local anesthesia. Which of the following sensory modalities would be the FIRST to be lost?

- A. Proprioception
 - B. Discriminative Touch
 - C. Motor function
 - D. Pain sensation
-

6. Match the following nerve fibre type with its respective property. Fibre type Property 1. $A\alpha$ a. Preganglionic autonomic 2. $A\beta$ b. Touch 3. $A\delta$ c. Temperature 4. B d. Proprioception

Fibre type	Property
1. $A\alpha$	a. Preganglionic autonomic
2. $A\beta$	b. Touch
3. $A\delta$	c. Temperature
4. B	d. Proprioception

- A. 1-d, 2-b, 3-c, 4-a
 - B. 1-a, 2-b, 3-d, 4-c
 - C. 1-b, 2-a, 3-c, 4-d
 - D. 1-c, 2-d, 3-b, 4-a
-

7. Which of the following would decrease the conduction velocity of an action potential along an axon?

- A. Increasing the diameter of the axon.
 - B. Decreasing the capacitance of the membrane.
 - C. Increasing the axoplasmic resistance.
 - D. Increasing the myelination of the axon.
-

8. Which of the following statements about the conduction of action potentials is TRUE?

- A. Speed of action potential conduction is inversely proportional to the degree of myelination of the axon.
 - B. Saltatory conduction occurs in both myelinated and unmyelinated axons.
 - C. The action potential jumps from one node of Ranvier to the next in myelinated axons.
 - D. The myelin sheath increases the flow of ions across the axonal membrane.
-

9. What impact does hyperkalemia have on a neuron's resting membrane potential (RMP)?

- A. It causes depolarization, making the neuron more excitable.
 - B. It causes hyperpolarization, making the neuron less excitable.
 - C. It has no effect on the RMP.
 - D. It increases the activity of the Na⁺/K⁺ -ATPase pump.
-

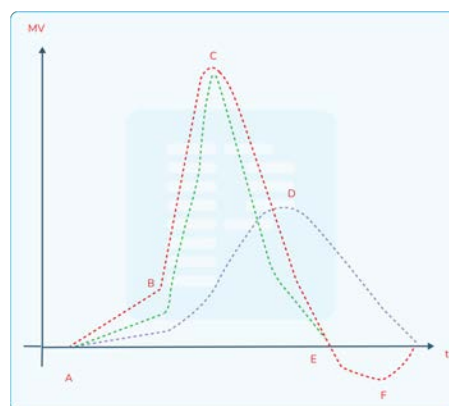
10. During an action potential, the voltage-gated sodium channel transitions through different states. Which of the following sequences accurately reflects the order of these states during a single action potential cycle starting from RMP?

- A. Closed → Open → Inactive → Open
 - B. Inactive → Closed → Open → Inactive
 - C. Closed → Inactive → Open → Closed
 - D. Closed → Open → Inactive → Closed
-

11. Which statements accurately describe the refractory period on a neuron's signaling properties?

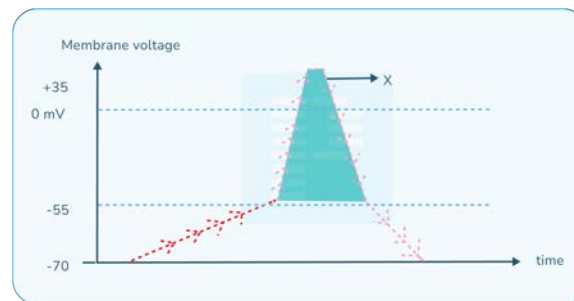
- A. Ensures action potentials move in one direction along an axon.
 - B. Restricts the length of an action potential, stopping overly long depolarizations.
 - C. Sets the intensity needed to start an action potential (threshold level).
 - D. Aids in the summation of local potentials, facilitating action potential generation.
-

12. Based on the graph showing potassium ion permeability during an action potential, at which point does the permeability of potassium ions reach its peak?



- A. Point A
- B. Point C
- C. Point D
- D. Point F

13. The graph below depicts an action potential. Which of the following statements correctly describes the state of ion channels at the point marked "X" on the graph?



- A. Na⁺ channels are closed; K⁺ channels are open.
- B. Na⁺ channels are completely inactivated; K⁺ channels are open.
- C. Na⁺ and K⁺ channels are open.
- D. Na⁺ and K⁺ channels are closed.

14. A 75-year-old bedridden patient presents with severe pressure ulcers over the sacral region and complaints of muscle weakness and numbness in the lower limb. On examination, there is evidence of compromised blood flow and significant tissue damage. Which innermost layer of the peripheral nerve is most likely affected in this patient?

- A. Perineurium
- B. Endoneurium
- C. Epineurium
- D. Mesoneurium

15. Which of the following accurately compares local potentials and action potentials?

- A. Local potentials are all-or-none; action potentials are graded and decremental.
 - B. Local potentials can summate and are decremental; action potentials are all-or-nothing and non-decremental.
 - C. Local potentials have constant amplitude; action potentials vary and diminish with distance.
 - D. Both local and action potentials propagate without diminishing and can summate.
-

16. Which of the following statements accurately describes the changes in the resting membrane potential (RMP) of neuron during depolarization and hyperpolarization?

- A. Depolarization: RMP more negative; Hyperpolarization: RMP more positive.
 - B. Depolarization: RMP more positive; Hyperpolarization: RMP more negative.
 - C. Depolarization: RMP more negative; Hyperpolarization: RMP -65 mV.
 - D. Depolarization: RMP more positive; Hyperpolarization: RMP less negative.
-

17. Match the ion with its equilibrium potential in a normal mammalian cell. Ion Equilibrium Potential (mV) 1. Sodium A. -70 2. Chloride B. +63 3. Potassium C. +132 4. Calcium D. -90

Ion	Equilibrium Potential (mV)
1. Sodium	A. -70
2. Chloride	B. +63
3. Potassium	C. +132
4. Calcium	D. -90

- A. 1-B, 2-A, 3-D, 4-C

- B. 1-D, 2-C, 3-B, 4-A
 - C. 1-C, 2-D, 3-A, 4-B
 - D. 1-A, 2-B, 3-C, 4-D
-

18. A 25-year-old woman presents with symptoms of confusion and lethargy. Her blood test shows a serum sodium level of 120 mEq/L. The intracellular concentration of sodium ($[Na^+]_i$) is 12 mEq/L. what is the equilibrium potential for sodium (in mV)?

- A. 61 mV
 - B. 72 mV
 - C. 82 mV
 - D. 92 mV
-

19. Which of the following is true regarding the leak channels?

- A. Facilitate the rapid transmission of action potentials along the axon.
 - B. Responsible for generation of synaptic potentials at the postsynaptic membrane
 - C. Help in maintaining resting membrane potential by allowing passive movement of ions
 - D. Play a crucial role in the repolarization phase of the action potential
-

20. Which of the following best explains the phenomenon where the distribution of ion species across a semipermeable membrane is influenced by the presence of a non-permeable ion, resulting in an uneven distribution of permeable ions?

- A. Nernst Equation
- B. Bohr Effect

C. Gibbs-Donnan Effect

D. Van't Hoff Law

21. Which of the following is NOT true regarding resting membrane potential?

A. It is created by diffusion of K^+ from outside to inside the cell at rest

B. The RMP of cardiac muscle is -90 mV

C. K^+ is the major determinant of RMP

D. The major contributor of RMP is through passive diffusion of ions

22. Which of the following statements accurately describes the type, direction and mediators of axonal transport? 1 Anterograde Cell body to axon terminal Dynein 2 Anterograde Axon terminal to cell body Kinesin 3 Retrograde Axon terminal to cell body Dynein 4 Retrograde Cell body to axon terminal Kinesin

1	Anterograde	Cell body to axon terminal	Dynein
2	Anterograde	Axon terminal to cell body	Kinesin
3	Retrograde	Axon terminal to cell body	Dynein
4	Retrograde	Cell body to axon terminal	Kinesin

A. 1

B. 2

C. 3

D. 4

23. Which type of neuron is primarily involved in the transmission of visual information from the retina to the brain?

- A. Unipolar neuron
 - B. Bipolar neuron
 - C. Multipolar neuron
 - D. Both multipolar and bipolar neuron
-

24. A 30-year-old man presents with progressive muscle weakness and fatigue. Electromyography (EMG) and nerve conduction studies indicate that the issue lies in the transmission of nerve impulses to his muscles, while central nervous system imaging shows no abnormalities. Which part of the neuron is most likely affected?

- A. Initial Segment
 - B. Axon
 - C. Dendrite
 - D. Axon Terminal
-

25. In which part of the neuron are neurofibrillary tangles primarily found, which are associated with neurodegenerative diseases like Alzheimer's?

- A. Dendrites
 - B. Axon
 - C. Soma
 - D. Axon terminal
-

26. A 55-year-old patient with brain tumor undergoes treatment with a novel chemotherapy drug targeting tumor cells. Despite the initial response, follow-up

imaging shows minimal response and tumor progression. Which characteristic of the blood-brain barrier is responsible for this treatment challenge?

- A. Impermeability to plasma proteins
 - B. Impermeability to lipid-soluble molecules
 - C. Presence of fenestrations in capillary walls
 - D. Lack of efflux transporters for drugs
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	2
Question 5	4
Question 6	1
Question 7	3
Question 8	3
Question 9	1
Question 10	4
Question 11	1
Question 12	3
Question 13	3
Question 14	2
Question 15	2
Question 16	2
Question 17	1
Question 18	1
Question 19	3
Question 20	3
Question 21	1
Question 22	3

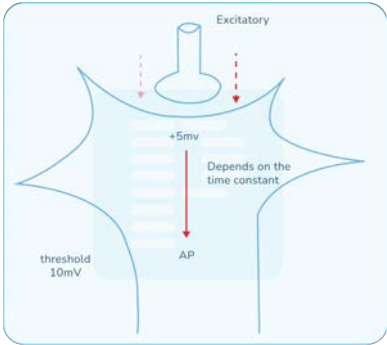
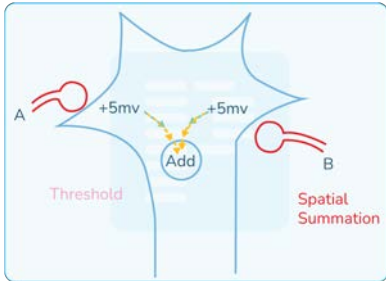
Question 23	4
Question 24	4
Question 25	3
Question 26	1

Solution for Question 1:

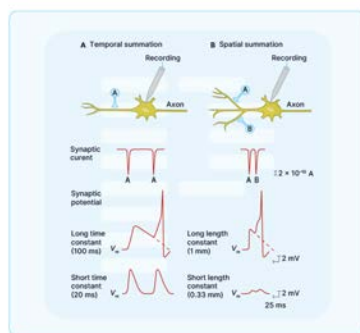
Correct Answer: B) Temporal summation requires signals to occur close in time, spatial summation requires simultaneous signals at different synapses.

Explanation:

Temporal summation occurs when multiple excitatory signals at the same synapse happen close in time to reach the threshold and trigger an action potential. Spatial summation involves excitatory signals arriving simultaneously at different synapses, summing to reach the threshold.

Feature	Temporal Summation	Spatial Summation
Definition (Option B)	 Multiple subthreshold excitatory signals at the same synapse occurring close in time.	 Excitatory signals arriving at different synapses simultaneously or within a short timeframe.
Threshold for Excitation (Option D)	Summation of signals to reach the threshold potential for triggering an action potential.	Combined effect of excitatory signals at multiple synapses to reach the threshold.
Signal Source (Option A)	Same synapse	Different synapses

Current Flow	Local depolarizing current spreads from the same synapse.	Local depolarizing currents from different synapses overlap.
Role in Action Potentials	Can trigger an action potential if signals are close enough in time to summate and reach threshold.	Can trigger an action potential if multiple signals from different locations summate to reach threshold.
Impact of Refractory Period (Option C)	Limited by the refractory period; if signals are too far apart, they won't summate.	Not directly limited by the refractory period; depends on spatial proximity of signals.
Examples	Neurons receiving frequent, rapid signals at a single synapse.	Neurons receiving simultaneous input from multiple presynaptic neurons.



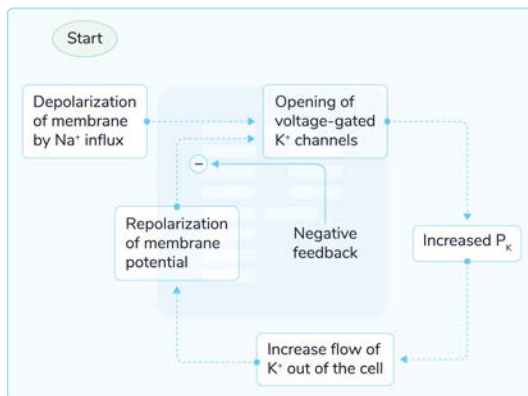
- The longer the time constant, the greater is the chance for two potentials to summate to induce an action potential (Temporal summation)
- Longer the length constant, greater the chance of Spatial summation.

Solution for Question 2:

Correct answer: B) Negative feedback

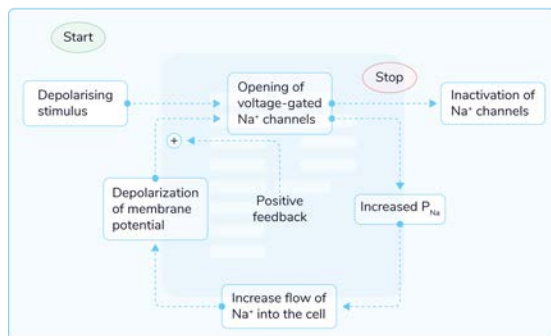
Explanation :

K⁺ channels exert Negative feedback.



- K⁺ channels provide negative feedback during an action potential by promoting repolarization.
- Their delayed opening after depolarization allows K⁺ ions to exit the cell, reducing the positive membrane potential and stabilizing the neuron's electrical activity.
- This feedback mechanism is essential for the proper functioning of neurons and for the propagation of action potentials.

Positive Feedback (Option A) : Increases the activity or response, leading to a greater output. For example, during the rapid depolarization phase of an action potential, when sodium channels open and more sodium enters, it leads to further depolarization.



Feedforward (Option C) : Anticipates changes and adjusts the response accordingly, but this is not the main focus in the context of ion channel behavior.

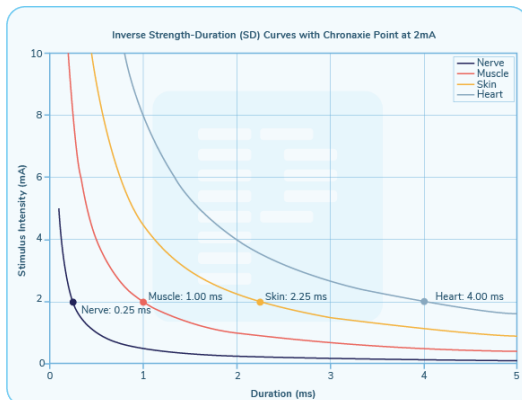
Voltage-Dependent Opening (Option D) : Refers to the way ion channels open in response to changes in membrane potential, but does not describe the feedback mechanism.

Solution for Question 3:

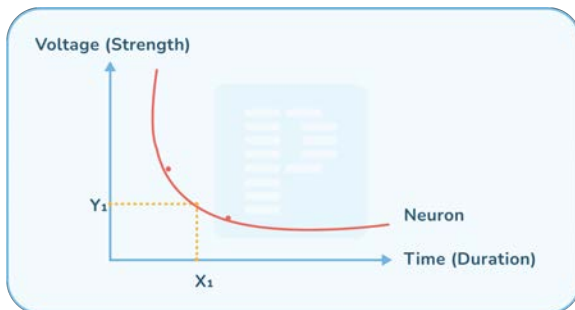
Correct Answer: A) Neuron

Explanation:

Neuron exhibits the highest chronaxie value.



Strength-Duration Relationship Parameters The strength-duration curve graphically depicts the relationship between the intensity of an electrical stimulus applied at a muscle's motor point and the duration required to produce a minimal muscle contraction.



Rheobase:

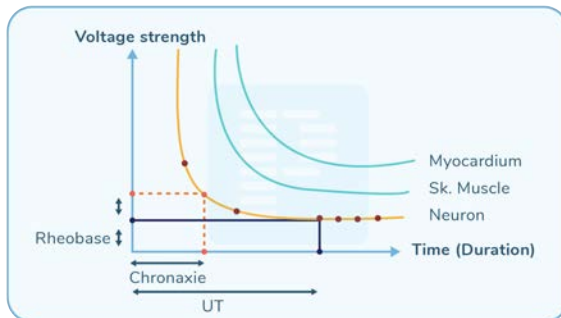
- Minimum electrical current (or voltage) needed to induce an action potential when applied indefinitely.
- Typical Values: 2 to 18 mA (or 5 to 35 volts) for healthy muscle tissue.
- Clinical Insight: Higher values in denervated muscles indicate reduced excitability.

Chronaxie:

- Minimum time required for a current that is double the rheobase intensity to stimulate the muscle.
- Normal Value: Generally below 1 ms.
- Clinical Insight: Longer chronaxie indicates lower excitability.

Utilization Time:

- Duration needed for a stimulus of rheobasic strength to excite the nerve and produce muscle contraction.
- Clinical Insight: Crucial for understanding nerve activation dynamics and stimulus efficiency.



Clinical Significance:

- Assessment: Parameters assess nerve function and excitability.
- The order of excitability among different tissues is generally: Neurons > skeletal muscles > myocardium > smooth muscles. Additionally, A fibers are the most excitable, followed by B and C fibers.
- Diagnostic Use: Evaluate nerve damage, monitor recovery, and guide rehabilitation.
- Strength-Duration Curve: Plots these parameters to reveal nerve injury extent and recovery potential.

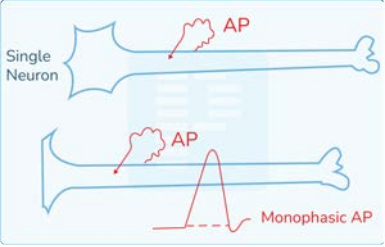
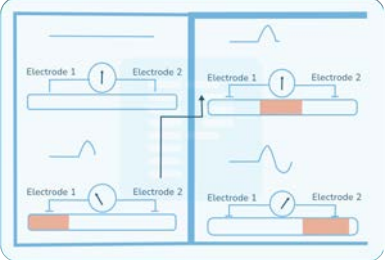
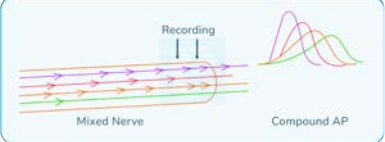
Solution for Question 4:

Correct Answer: B) Different conduction velocities of various nerve fibers.

Explanation:

The multiple peaks in a compound action potential (CAP) indicate the summation of action potentials from different types of nerve fibers (e.g., A α , A β , B, and C fibers), each with distinct conduction velocities. The different peaks reflect the sequential activation of these fibers.

Action Potential - Monophasic/ Biphasic/ Compound

 Monophasic Action Potential: Single directional deflection, recorded from a single fiber or small group.	Recording: One electrode inside the cell, the other outside.
Characteristics: Single Peak: Indicates action potential of a single fiber or small group. Amplitude: Does not significantly increase with stimulation strength Follows All-or-nothing Principle. It states that once an action potential is initiated in a nerve fiber, it will travel the entire length of the fiber if the conditions are suitable, or it will not travel at all if the conditions are not right.	
 Biphasic Action Potential: Two-phase action potential with positive and negative deflections.	Recording: Two electrodes of opposite polarity detect the wave of depolarization.
Characteristics: Waveform: Upward deflection followed by a downward deflection. Relation to CAP: Component of CAP, reflecting the activity of multiple fibers.	
 Compound Action Potential (CAP): Summation of action potentials from different nerve fibers ($A\alpha$, $A\beta$, B, C) in a mixed nerve.	Recording: Electrodes placed on the surface of the mixed nerve.

Characteristics:
 Multiple Peaks: Correspond to different conduction velocities; A α fibers first.
 Amplitude: Increases with stronger stimulation.
 Waveform: Biphasic , showing positive and negative deflections.
 Does not follow the All-or-nothing Principle.

Solution for Question 5:

Correct Answer: D) Dull, aching pain.

Explanation:

- Pain transmitted by A-d fibers that is most susceptible to blockade by local anesthetics like lidocaine.

Spike potential is defined as a rapid, transient change in membrane potential, signifying an action potential.

Spike duration is the length of time this spike potential lasts.

Table rendering failed

Susceptibility to Local Anesthesia	Pressure	Hypoxia
A γ and A δ >> A α and A β >> B >> C	A α > B > C	B > A > C

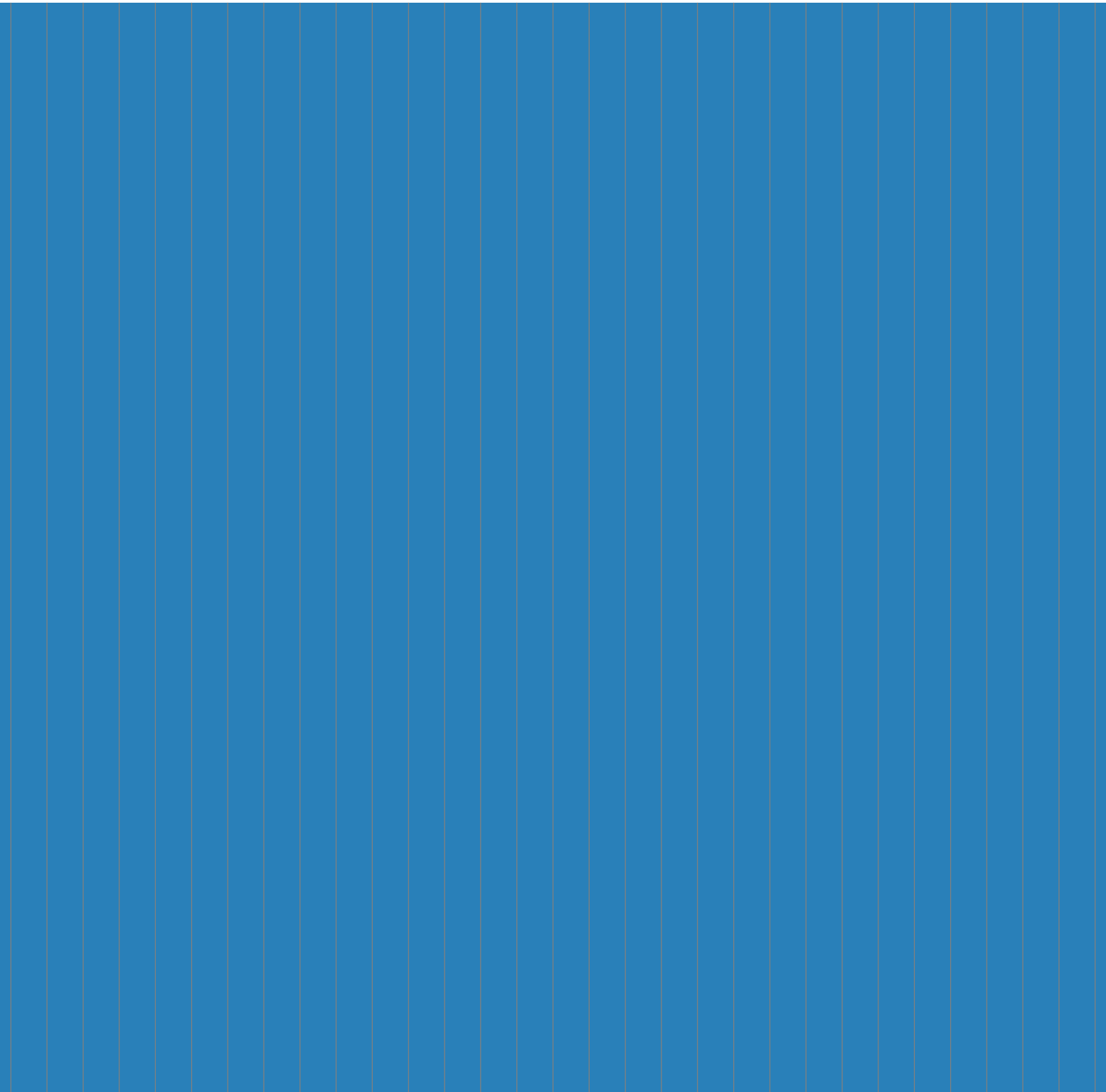
Solution for Question 6:

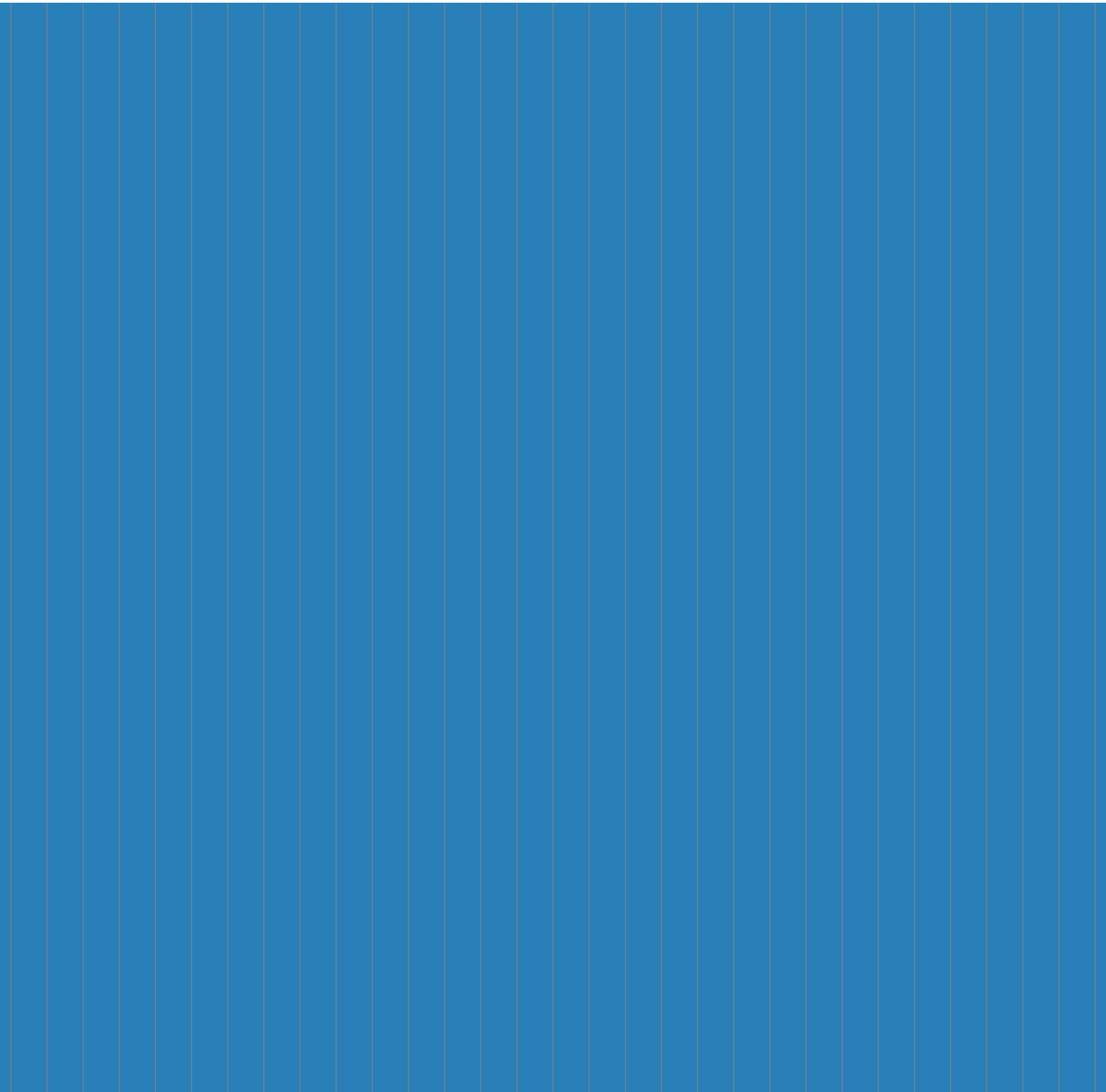
Correct Answer: A) 1-d, 2-b, 3-c, 4-a

Explanation:

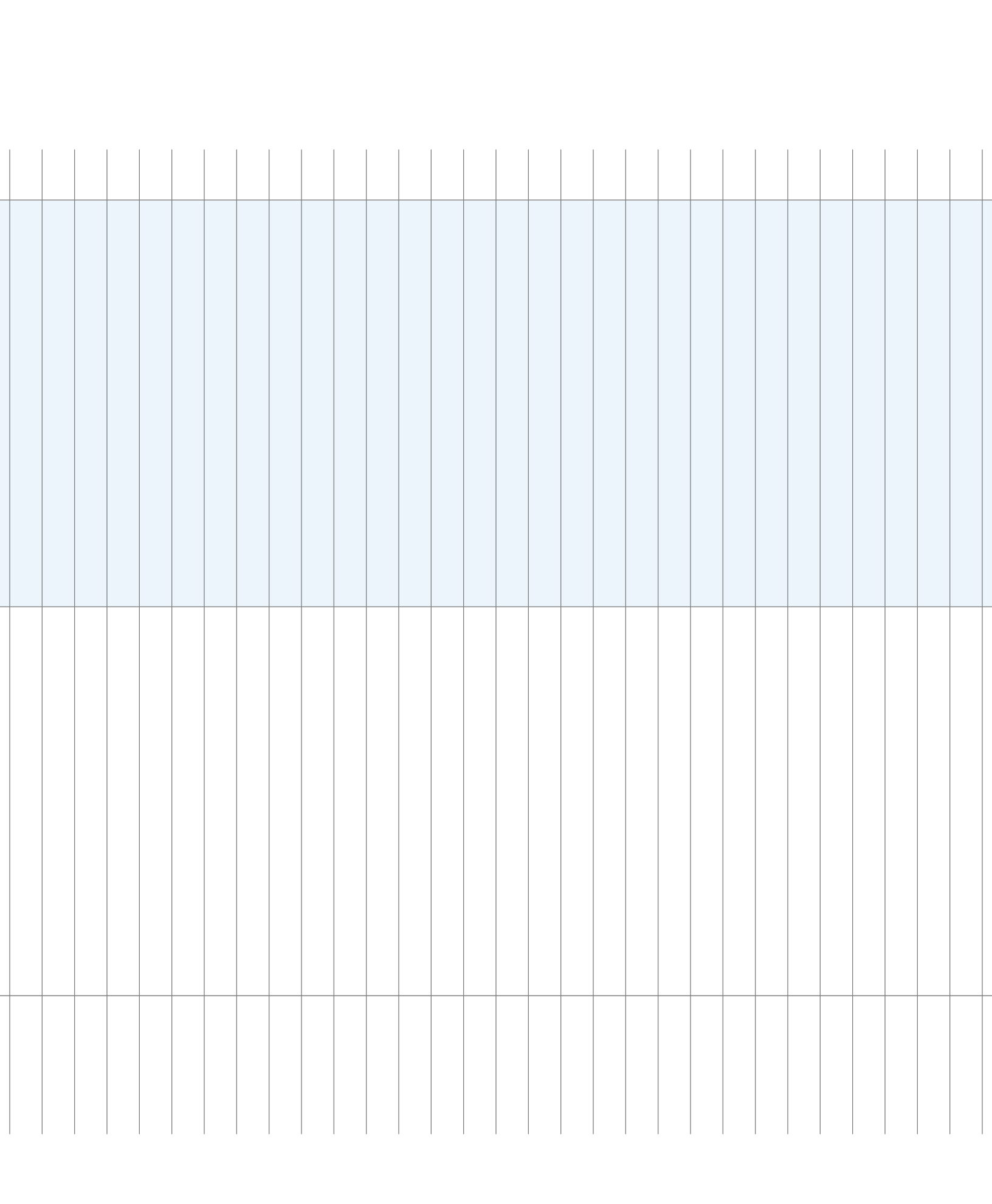
P	7	1	P	A	A	II	T	3	5	P	A	A		M	1	3	A	A	III	P	1	2	B	B		P	3	<	H
r	0	2	r	β	β		o	0	-	r	γ	γ		o	5	-	δ	δ		a	2	-			r	-	3		y
p	1	2	s				u	-	1	e				r	-	6				n	-	5			e	1		p	o
							c	7	2	s					3				(f	3					g	5			

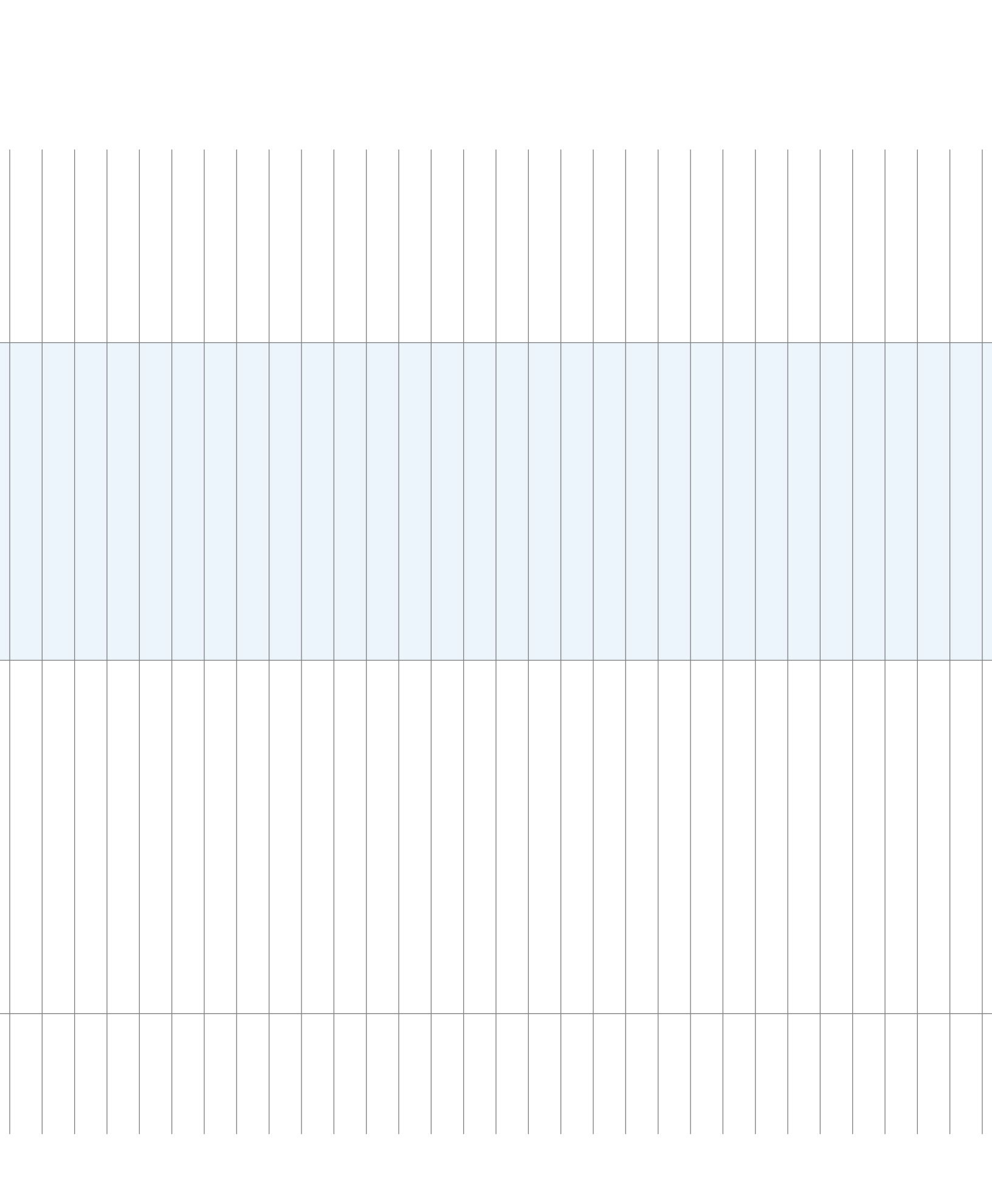
ri o c e p t i o n, s o m a t i c m o t o r	2 0	0	s u r e - M o s t s u s c e p t i b l e					h, p r e s s u r e	0		s u r e -l n t e r m e d i a t e.					t o m u s c l e s p i n d l e s	0							a s t), t e m p e r a t u r e (c o l d)	0						a n g l i o n i c a u t o n o m i c				x i a - M o s t s u s c e p t i b l e
---	--------	---	--	--	--	--	--	--	---	--	--	--	--	--	--	--	---	--	--	--	--	--	--	--	---	--	--	--	--	--	--	--	--	--	---

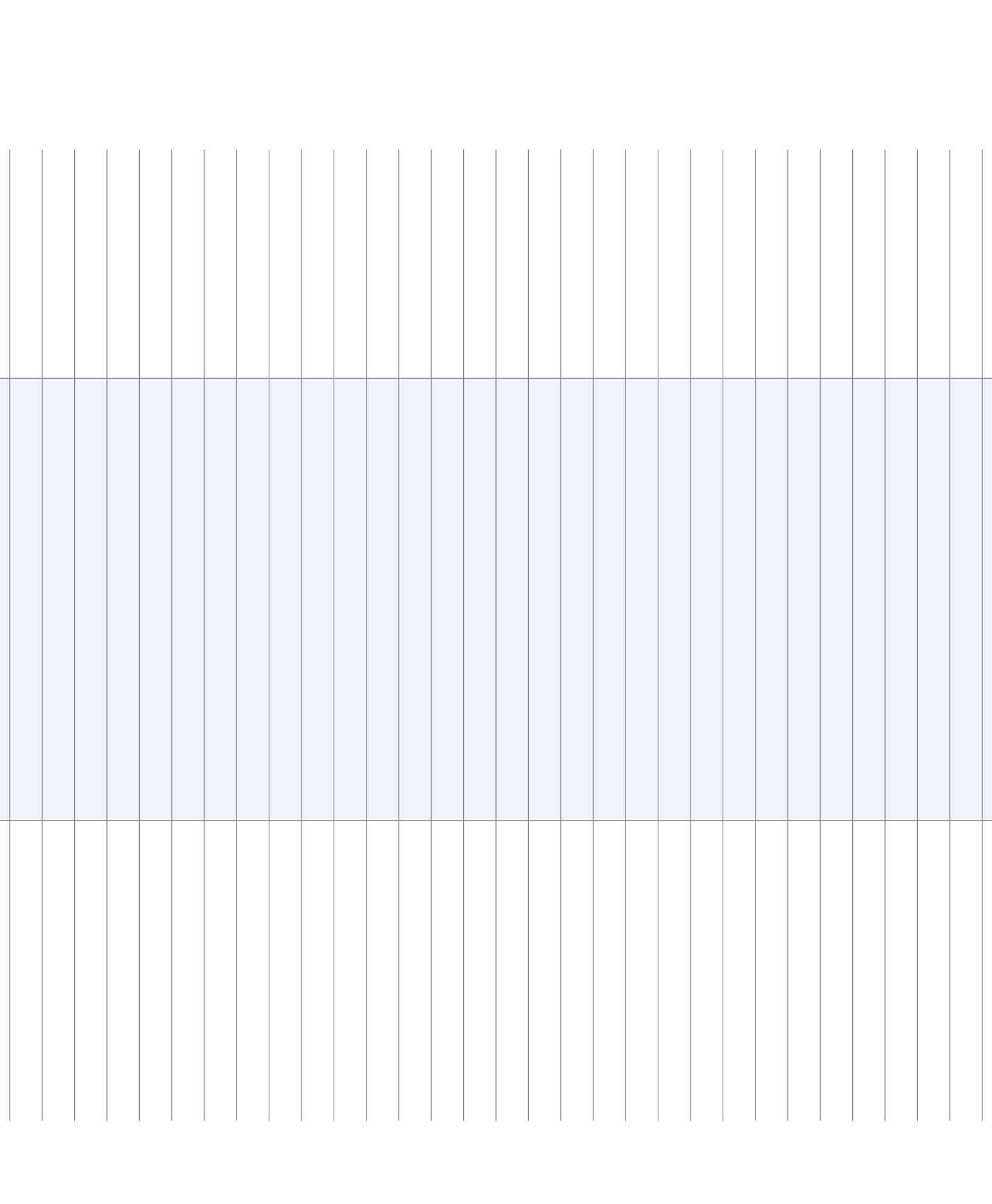




Page 273 of 1156







Classification of nerve fibres - Erlanger and Gasser and Lloyd-Hunt							
Fiber Type	Erlanger-Gasser Classification	Lloyd-Hunt Classification	Myelination	Function	Conduction Velocity (m/s)	Diameter (μm)	Sensitivity to Blockade
A α	A α	Ia, Ib	Myelinated	Proprioception, somatic motor	70-120	12-20	Pressure - Most susceptible
A β	A β	II	Touch, pressure	30-70	5-12	Pressure - Intermediate.	
A γ	A γ		Motor to muscle spindles	15-30	3-6		
A δ	A δ	III	Pain (fast), temperature (cold)	12-30	2-5		
B	B		Preganglionic autonomic	3-15	<3	Hypoxia - Most susceptible	
C (Dorsal Root)	C	IV	Unmyelinated	Pain (slow), temperature (warm)	0.5-2	0.4-1.2	Local anesthetics - Most susceptible
C (Sympathetic)	C		Postganglionic sympathetic	0.7-2.3	0.3-1.3		

Solution for Question 7:

Correct Answer: C) Increasing the axoplasmic resistance.

Explanation:

Increasing this resistance would hinder the spread of Na⁺ ions, making it more difficult to reach the threshold for action potential initiation at adjacent nodes of Ranvier.

Conduction velocity of the Action potential.	
Factors Influencing	$CV \propto \frac{1}{R.C}$ C - Capacitance of the membrane R - Axoplasmic resistance Myelination : Increases conduction velocity, enabling saltatory conduction. (Option D) Axon Diameter : Larger diameter axons exhibit faster conduction velocities. (Option A) Temperature : Higher temperatures increase ion channel opening and closing speeds affecting conduction velocity.
Classification of Nerve Fibers	Larger, myelinated fibers (like A α fibers) - Faster conduction velocity. Smaller, unmyelinated fibers (like C fibers) - Slower conduction velocity.
Clinical Significance	Local anesthetics affect unmyelinated C fibers before myelinated A fibers. A decrease in chronaxie after nerve repair suggests increased excitability and potential regeneration.
Stimulus Strength	A stronger stimulus doesn't increase the speed of an action potential in a single nerve fiber, it might activate a greater number of fibers.

A lower capacitance means that the membrane can depolarize more rapidly, leading to faster conduction. Myelination is a key factor in reducing membrane capacitance. (Option B)

Solution for Question 8:

Correct Answer: C) The action potential jumps from one node of Ranvier to the next in myelinated axons.

Explanation:

The myelin sheath prevents ion flow across the membrane, forcing the action potential to "jump" from one node of Ranvier to the next.

Conduction of action potential		
Feature	Unmyelinated	Myelinated
Conduction Speed	Slower (0.25 m/s)	Faster (up to 100 m/s)
Mechanism	Action potential flows continuously along the axon membrane. (Option B)	Action potential "jumps" between nodes of Ranvier-saltatory conduction.
Myelin Sheath	Absent	Present, decreases flow of ions, preventing ion leakage across the membrane. (Option D)
Nodes of Ranvier	Absent	Present, rapid action potential regeneration.
Energy Efficiency	Less energy efficient.	More energy efficient.
Fiber Diameter	Smaller	Larger

- Factors that increase action potential conduction velocity include increased myelination and increased axon diameter. (Option A)

Solution for Question 9:

Correct Answer: A) It causes depolarization, making the neuron more excitable.

Explanation:

- The increased extracellular potassium concentration decreases the concentration gradient for potassium to leave the cell, leading to depolarization.

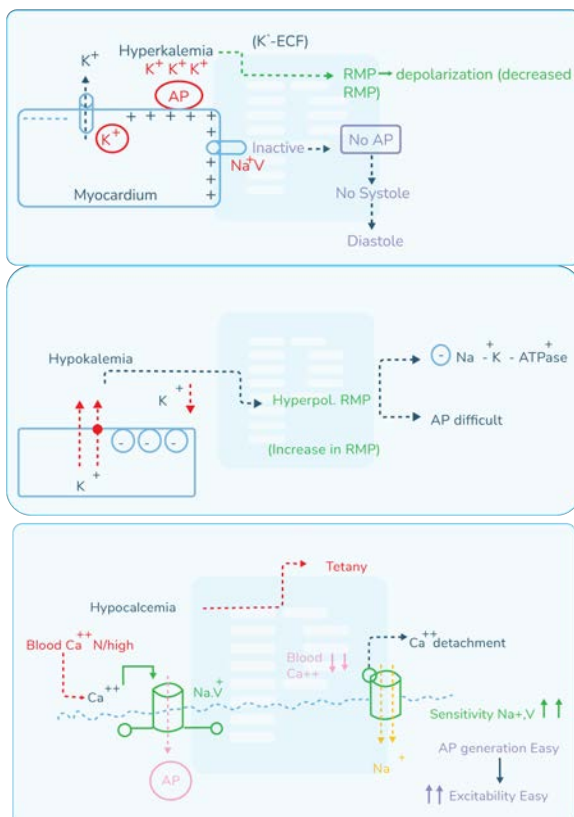
Condition	Hyperkalemia	Hypokalemia
RMP	Depolarized (Decreased RMP) (Option C)	Hyperpolarized (Increased RMP) (Option B)
Explanation	Increased extracellular potassium reduces the gradient for potassium	In hypokalemia, cells lose more potassium ions due to leaky

	to exit the cell, causing depolarization. The resting membrane potential becomes less negative than the typical -70 mV.	channels. The loss of positive charge increases the negative charge within the cell, causing hyperpolarization.
Effect on Neuron Excitability	Increased excitability can lead to spontaneous firing and even cell death. Persistent depolarization inactivates voltage-gated sodium channels, preventing action potential generation.	Decreased excitability occurs as the membrane potential is further from the threshold for action potential. Hypokalemia also inhibits the Na-K-ATPase pump.

Hypocalcemia:

- When blood calcium is normal or high, calcium remains bound to the extracellular mouth of voltage-gated Na⁺ channels, stabilizing them.
- Low blood Ca²⁺ (hypocalcemia) causes calcium to detach from the channel, leading to easier sodium entry. This detachment increases Na⁺ channel sensitivity, enhancing excitability and action potential generation.

Calcium and magnesium act as membrane stabilizers.

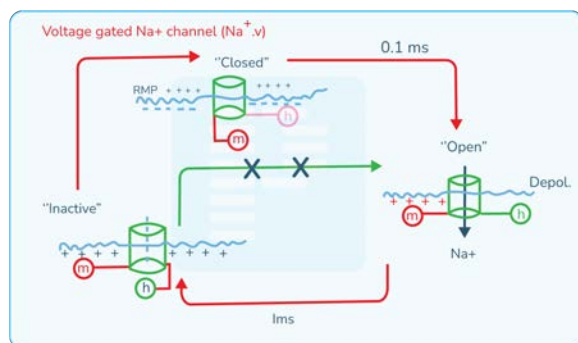


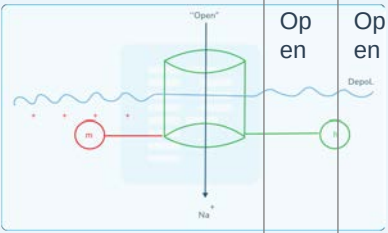
Hyperkalemia can eventually lead to increased activity of the Na⁺/K⁺-ATPase pump as a compensatory mechanism, this is not a direct or immediate effect on the RMP. (Option D)

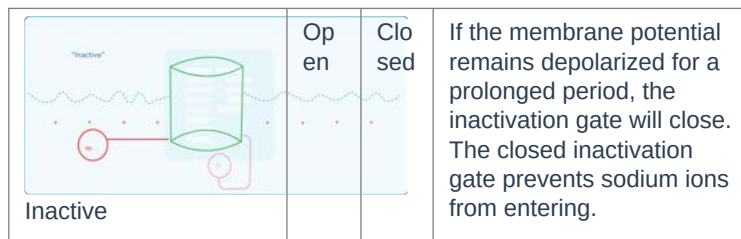
Solution for Question 10:

Correct Answer: D) Closed → Open → Inactive → Closed

Explanation:



Stage	M Gate	H Gate	Description
 Closed	Closed	Open	This is the resting state of the voltage-gated sodium channel, where the activation gate (M gate) is closed, preventing sodium ions from entering the cell.
 Open	Open	Open	When the membrane potential becomes less negative, the activation gate opens rapidly. This allows a large influx of sodium ions into the cell, leading to depolarization.



Solution for Question 11:

Correct Answer: A) Ensures action potentials move in one direction along an axon.

Explanation:

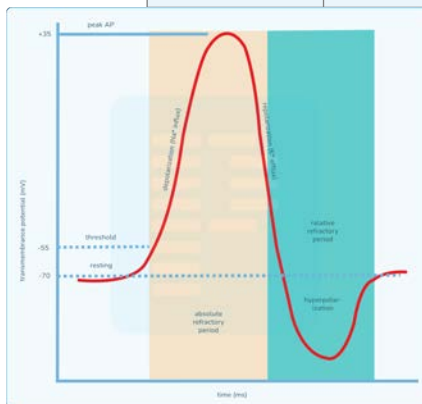
- Action potentials are self-propagating impulses that travel down an axon with constant amplitude and velocity.
- During the refractory period, the portion of the membrane that has just undergone depolarization is resistant to re-excitation.
- This resistance ensures that the action potential moves forward along the axon rather than propagating backward to already stimulated areas.

Refractory period:

- Refractory period is a period of reduced excitability in a neuron following an action potential.
- Role : Ensuring unidirectional propagation of action potentials. Modulating action potential frequency.
- Ensuring unidirectional propagation of action potentials.
- Modulating action potential frequency.
- Ensuring unidirectional propagation of action potentials.
- Modulating action potential frequency.

Characteristic	Absolute Refractory Period	Relative Refractory Period
Definition	The initial phase of the refractory period. The period during which a second action potential cannot be elicited, even with a strong stimulus.	Follows the absolute refractory period. The period during which a stronger than-normal stimulus can elicit a second action potential.
Duration	Sum of the total depolarization period and the first 1/3rd of the	Last 2/3rd of the repolarization period of the action potential.

	repolarization period of the action potential.	
Excitability	No stimulus, no matter how strong.	Stronger than normal stimuli
Voltage-Gated Na ⁺ Channels	Inactive or open state.	Transition from the inactive to the closed state.
Significance	Limits the maximum frequency of action potentials a neuron can fire.	Encoding of stimulus strength by modulating the frequency of action potentials.



The refractory period mainly governs the neuron's ability to generate a second action potential, not the duration of an individual action potential. (Option B)

While the refractory period influences a neuron's excitability, it doesn't directly determine the threshold intensity. (Option C)

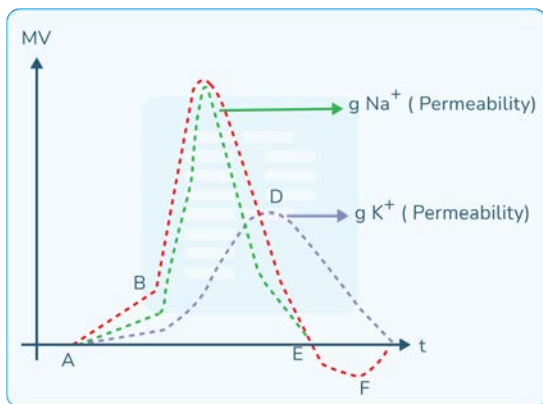
The refractory period actually makes it more difficult for local potentials to summate during its time course. (Option D)

Solution for Question 12:

Correct Answer: C) Point D

Explanation:

- Point D signifies the early phase of repolarization.
- At this stage, voltage-gated potassium channels are fully open, leading to the highest permeability to potassium ions.



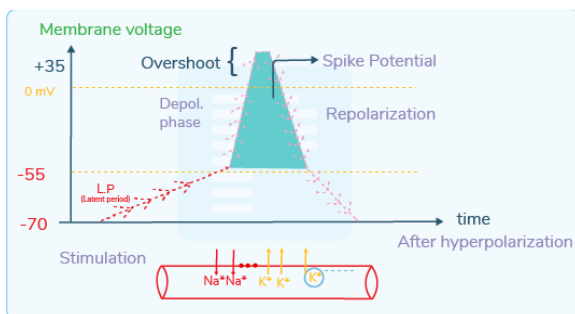
- Resting Membrane Potential (Point A on graph): At rest, the neuron's membrane potential is around -70 mV. K⁺ permeability is higher than Na⁺ permeability due to the presence of K⁺ leak channels. However, the overall permeability to both ions is relatively low.
- At rest, the neuron's membrane potential is around -70 mV.
- K⁺ permeability is higher than Na⁺ permeability due to the presence of K⁺ leak channels.
- However, the overall permeability to both ions is relatively low.
- Depolarization Phase (Between Points A and C): Voltage-gated Na⁺ channels open faster than K⁺ channels. Rapid influx of Na⁺ ions, causing a rapid rise in membrane potential, causing a steep slope.
- Voltage-gated Na⁺ channels open faster than K⁺ channels.
- Rapid influx of Na⁺ ions, causing a rapid rise in membrane potential, causing a steep slope.
- Peak of Action Potential (Point C) : Na⁺ permeability is at its maximum. K⁺ channels are also opening at this stage but have not reached their peak.
- Na⁺ permeability is at its maximum.
- K⁺ channels are also opening at this stage but have not reached their peak.
- Early Repolarization Phase (Point D) : Early phase of repolarization. Voltage-gated K⁺ channels are fully open, resulting in the highest permeability. Na⁺ channels are rapidly inactivated during this phase.
- Early phase of repolarization.
- Voltage-gated K⁺ channels are fully open, resulting in the highest permeability.
- Na⁺ channels are rapidly inactivated during this phase.
- Late Repolarization and After-hyperpolarization (Points E and F): K⁺ permeability starts to decline as the membrane potential moves towards its resting state. Slow closure of K⁺ channels results in a period of hyperpolarization, where the membrane potential falls slightly below the resting potential (Point F) .
- K⁺ permeability starts to decline as the membrane potential moves towards its resting state.

- Slow closure of K⁺ channels results in a period of hyperpolarization, where the membrane potential falls slightly below the resting potential (Point F) .
- At rest, the neuron's membrane potential is around -70 mV.
- K⁺ permeability is higher than Na⁺ permeability due to the presence of K⁺ leak channels.
- However, the overall permeability to both ions is relatively low.
- Voltage-gated Na⁺ channels open faster than K⁺ channels.
- Rapid influx of Na⁺ ions, causing a rapid rise in membrane potential, causing a steep slope.
- Na⁺ permeability is at its maximum.
- K⁺ channels are also opening at this stage but have not reached their peak.
- Early phase of repolarization.
- Voltage-gated K⁺ channels are fully open, resulting in the highest permeability.
- Na⁺ channels are rapidly inactivated during this phase.
- K⁺ permeability starts to decline as the membrane potential moves towards its resting state.
- Slow closure of K⁺ channels results in a period of hyperpolarization, where the membrane potential falls slightly below the resting potential (Point F) .

Solution for Question 13:

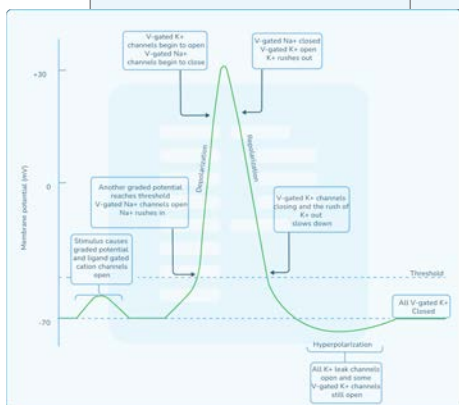
Correct Answer: B) Na⁺ channels are inactivated; K⁺ channels are open.

Explanation:



- At the peak of the action potential (point "X"), the neuron's membrane potential is briefly positive.
- The voltage-gated Na⁺ channels are inactivated, at this same point, voltage-gated K⁺ channels are open, allowing K⁺ ions to flow out of the neuron.

Phase	Membrane Potential	Na ⁺ Channels	K ⁺ Channels
Resting State	Polarized (-70 mV)	Closed	Closed
Depolarization (Rising Phase)	Less negative, towards 0 mV.	Rapidly opening, then inactivating	Slowly opens.
Overshoot (Peak)	Briefly positive (+35 mV).	Inactivated	Open
Repolarization (Falling Phase)	Returning to negative	Inactivated	Open
After-hyperpolarization	More negative.	Closed	Slowly closing
Return to Resting State (Option D)	Returning to -70 mV	Closed	Closed



At point "X", the peak of the action potential, voltage-gated Na⁺ channels are not merely closed; they are transitioning into their inactive state. (Option A)

At the peak of the action potential, the voltage-gated Na⁺ channels are transitioning into their inactive state. (Option C)

Solution for Question 14:

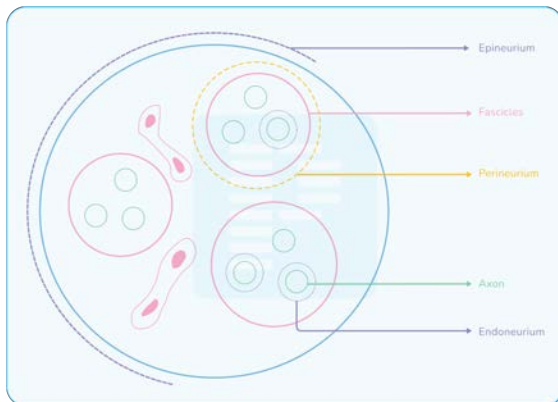
Correct Answer: C) Epineurium

Explanation:

- The fasciculi are held together by a dense layer of connective tissue called the epineurium, which surrounds the entire nerve and contains nerve fibers.

- In bedridden patients, loss of this protective fat layer can lead to pressure on nerve fibers, causing paralysis. Blood vessels travel through this connective tissue, so reduced blood supply can lead to ischemic neuritis and pain.

Connective tissue layers of Peripheral Nerve :



Endoneurium (Option B)	The endoneurium is a layer of connective tissue covering each nerve fibre within its Schwann cell and basal lamina in a peripheral nerve. Contains collagen, fibroblast, schwann cells, endothelial cells and macrophages. Endoneurium holds adjoining nerve fibres together and facilitates their aggregation to form fasciculi.
Perineurium (Option A)	A thick layer of connective tissue covering each fascicle is called Perineurium. It comprises layers of flattened cells separated by layers of collagen fibres. Controls diffusion of substances in and out of axons.
Epineurium (Option C)	The fasciculi are held together by a fairly dense layer of connective tissue that surrounds the entire nerve and is called Epineurium. Contribute to the tensile strength of the nerve. Loss of this layer in bedridden patients can lead to pressure on nerve fibres and paralysis.

Mesoneurium (Option D) is a connective tissue sheath that suspends the nerve trunk within the soft tissue and is continuous with the underlying epineurium

Solution for Question 15:

Correct Answer: B) Local potentials can summate and are decremental; action potentials are all-or-nothing and non-decremental.

Explanation:

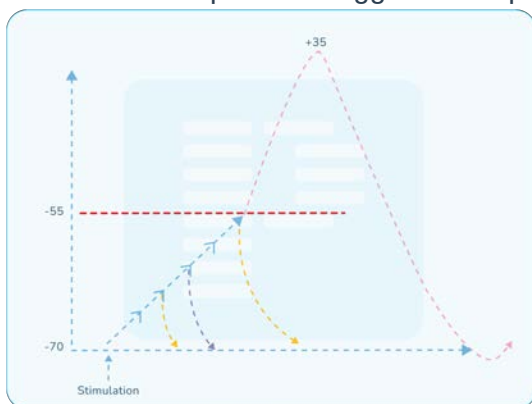
Local Potentials: Graded, decremental, and can summate.

Action Potentials: All-or-nothing, non-decremental, and do not summate.

Feature	Local Potentials	Action Potentials
Amplitude (Vs Option A&C;)	Graded; varies with stimulus strength	All-or-nothing; amplitude remains constant once threshold is reached
Propagation	Decremental; diminishes with distance	Non-decremental; maintains magnitude over long distances
Summation (Option B)	Can summate (spatial and temporal)	Cannot summate; separated by refractory periods
Function	Initiate action potentials by reaching threshold at the axon hillock	Transmit signals over long distances along nerve fibers
Ionic Basis	Involves ion movements, similar to action potentials, but often localized	Depolarization and repolarization driven by voltage-gated Na ⁺ and K ⁺ channels

Excitatory Postsynaptic Potentials (EPSPs): Local, Graded Potentials

- Depolarizing events that can trigger an action potential. Excitatory neurotransmitters bind to receptors, opening ion channels and allowing positively charged ions to enter the cell.
- Action potential triggered if depolarization reaches around -55 mV.



Action Potential Stages:

Propagation of Action Potential:

- Mechanism : Action potentials travel along the nerve fiber without diminishing. Depolarization at one site causes local current flow that triggers action potentials in adjacent regions.
- All-or-Nothing Principle : An action potential occurs only if the threshold is reached, maintaining consistent magnitude regardless of stimulus strength.

Refractory Periods:

- Absolute Refractory Period : Na^+ channels are inactivated, preventing another action potential.
- Relative Refractory Period : Requires a stronger stimulus to trigger another action potential due to partial repolarization.

Myelination and Saltatory Conduction: Myelin Sheath insulates nerve fibers, formed by Schwann cells. Action potentials "jump" between nodes of Ranvier, speeding up transmission and saving energy.

Energy Dependence: Na^+/K^+ Pump restores ionic gradients after an action potential, consuming ATP.

Factors Influencing Excitability:

- Calcium Ions (Ca^{2+}) : Low extracellular Ca^{2+} increases excitability by lowering the activation threshold for Na^+ channels; high Ca^{2+} decreases excitability.
- Local Anesthetics : Block voltage-gated Na^+ channels, reducing excitability and preventing action potentials.

Solution for Question 16:

Correct Answer: B) Depolarization: RMP more positive; Hyperpolarization: RMP more negative.

Explanation:

Depolarization: RMP becomes more positive as Na^+ enters the neuron.

Hyperpolarization: RMP becomes more negative due to excessive K^+ outflow or Cl^- influx.

Resting Membrane Potential:

- Typical Value: $\sim -65 \text{ mV}$
- Inside of the neuron is more negatively charged than the outside.

- Primarily influenced by ion concentration gradients, especially potassium (K⁺).
- Potassium Pump: Actively transports K⁺ ions into the neuron, maintaining higher internal concentration.
- Creates a tendency for K⁺ ions to diffuse out, contributing to the negative resting potential.

Depolarization:	Sudden shift from negative to positive membrane potential. Mechanism: Increased permeability to Na⁺ ions allows rapid influx of Na⁺, neutralizing the negative potential. In large nerve fibers, Na⁺ influx can cause the membrane potential to briefly become positive (overshoot). Excitatory Postsynaptic Potentials (EPSPs): Depolarizing events that can trigger an action potential. Excitatory neurotransmitters bind to receptors, opening ion channels and allowing positively charged ions to enter. Action potential is triggered if depolarization reaches around -45 mV.
Repolarization:	Return of membrane potential to resting state (-65 mV) following depolarization. Na⁺ channels close, reducing Na⁺ influx. K⁺ channels open wider, allowing K⁺ ions to exit rapidly. Restores negative membrane potential.
Hyperpolarization:	Membrane potential becomes more negative than the resting potential, below -65 mV. Inhibitory Neurotransmitters: Open channels allowing negatively charged ions (e.g., Cl⁻) to enter or positively charged (e.g., K⁺) to exit. Prolonged Potassium Channel Opening: Allows additional K⁺ outflow. Effect: Moves potential closer to the potassium Nernst potential, making the neuron less likely to fire immediately. Recovery: Gradual return to threshold potential for new action potential generation. Inhibitory Postsynaptic Potentials (IPSPs): Hyperpolarizing events that make it harder for the neuron to fire an action potential.

Solution for Question 17:

Correct Answer: A) 1-B, 2-A, 3-D, 4-C

Explanation:

Ion	Equilibrium Potential
Sodium	+63 mV
Chloride	-70 mV
Potassium	-90 mV
Calcium	+132 mV

Solution for Question 18:

Correct Answer: A) 61 mV

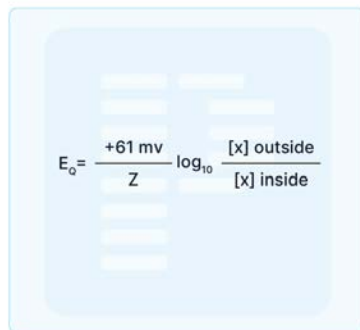
Explanation:

According to the question,

$[Na^+]_{outside} = 120 \text{ mEq/L}$

$[Na^+]_{inside} = 12 \text{ mEq/L}$

Using the Nernst equation:


$$E_{eq} = \frac{+61 \text{ mV}}{z} \log_{10} \frac{[x]_{outside}}{[x]_{inside}}$$

'z' is the electrical charge of the ion (+1 for K⁺).

Therefore,

$$E_{eq} = 61 \log (120/12)$$

$$= 61 \log (10)$$

$$= 61 \cdot (1)$$

$$= 61 \text{ mV}$$

- The Nernst equation calculates the electrical potential difference across a membrane (Nernst potential) that balances the diffusion of a specific ion, preventing further net movement despite a concentration gradient.
- The Nernst equation calculates the Nernst potential (EMF) for a single, univalent ion (an ion with a +1 or -1 charge) at normal body temperature (37°C)
- The Nernst potential magnitude depends on the ion concentration ratio across the membrane; larger differences result in higher potentials.
- Nernst equation calculates the potential for a single ion, although cell membranes are permeable to multiple ions.

Applications of the Nernst Equation:

- Explains how ion concentration differences generate electrical potential across a selectively permeable membrane.
- Basis for the Goldman-Hodgkin-Katz equation, which calculates membrane potential considering multiple ions and their permeabilities.

Limitations: It works best for systems with only one permeant ion. In living cells, this is rarely the case, making the Goldman equation a more accurate representation.

Solution for Question 19:

Correct Answer: C) Help in maintaining resting membrane potential by allowing passive movement of ions

Explanation:

Leak channels:

- Leak channels are a subtype of ion channels that passively allow ions such as Na^+/K^+ to move across the membrane.
- They are always open.
- The most important leak channel is the K^+ leak channel, which allows the passive transfer of K^+ from inside the cell to outside.

- This helps in creating the resting membrane potential of the neuron (Option C)
- The resting membrane potential is crucial for neurons in generating and propagating action potentials.

Transmission of action potentials along the axon is facilitated by voltage-gated Na⁺ channels (Option D)

Generation of synaptic potentials at the postsynaptic membrane is facilitated by ligand-gated ion channels. (Option D)

Repolarization phase of the action potential is facilitated by voltage gated K⁺ channel (Option D)

Solution for Question 20:

Correct Answer: C) Gibbs-Donnan Effect

Explanation:

Gibbs-Donnan Effect:

- It is described as the effect of an impermeable ion on the distribution of other ions to which the membrane is permeable (Option C).
- The non permeable ion influences the distribution of permeable ions in such a way that there will be an unequal distribution of these ions on either side of the membrane.
- This results in an electrochemical gradient, where the concentration of ions is balanced but not equal due to the presence of non-permeable ions.
- It helps to create and maintain an uneven distribution of ions between ECF and ICF.

Nernst Equation calculates the equilibrium potential for a particular ion across a membrane (Option A)

Bohr effect describes how pH and CO₂ concentration affect the oxygen-binding affinity of hemoglobin (Option B)

Van't Hoff law relates to the effect of temperature on the equilibrium constant for a chemical reaction (Option D)

Solution for Question 21:

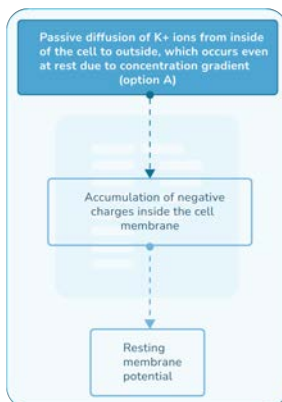
Correct Answer: A) It is created by passive diffusion of K^+ from outside to inside the cell at rest

Explanation:

Resting membrane potential (RMP):

- It is defined as the voltage difference between the exterior and interior of an excitable cell
- In any cell, K^+ is the major determinant of RMP (option C)
- The major contribution of RMP is through passive diffusion of K^+ ions followed by active diffusion through Na^+-K^+ ions (option D)

Generation of RMP:



RMP of different cells:

Cell	RMP
Neuron	-70mV
Skeletal muscle	-95mV
Smooth muscle	-60mV
Cardiac muscle	-90mV (option B)
Astrocyte	-90mV
Erythrocyte	-12mV
Photoreceptor	-40mV (dark) to -70mV (light)

Solution for Question 22:

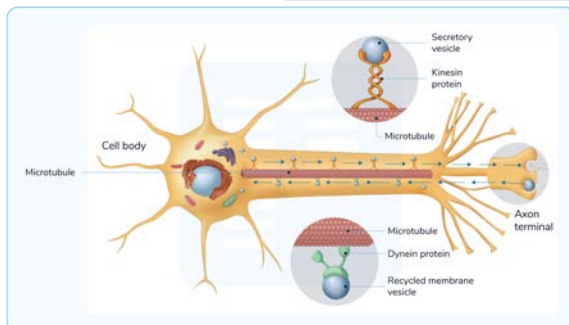
Correct Answer: C) 3

Explanation:

Axonal transport:

- It is a process in neurons which involves the transport of organelles, proteins and other cellular material along the axon
- It can be abolished by colchicine, dinitrophenol, azide, cyanide and prolonged anoxia

Type	Mediated by	Characteristics
Anterograde transport	Kinesin	It involves the transport of cellular material from soma to the axon terminal. It is responsible for the transport of proteins, synaptic vesicles necessary for synaptic transmission. Two types: Fast axoplasmic transport at a rate of 400 mm/day, mediated by kinesin. Slow axoplasmic transport at a rate of 0.5-2 mm/day, mediated by actin, neurofilaments and microtubules. Supplies material required for regeneration following nerve injury.
Retrograde transport	Dynein	It involves the transport from axon terminals to soma at a rate of about 200 mm/day. It is important for transport of viruses, toxins, nerve growth factors and reuptake of synaptic neurotransmitters.



Solution for Question 23:

Correct Answer: D) Both multipolar and bipolar neuron

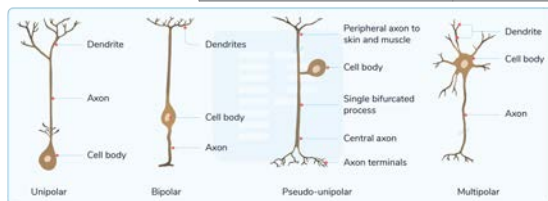
Explanation:

Transmission of visual information from the retina to the brain is done by the Ganglion cell.

- Ganglion cell is mostly multipolar but can be bipolar as well

Types of neurons:

Type	Characteristic
Unipolar neuron (Option A)	Seen in invertebrates Have one process with several receptive and releasing terminals
Bipolar neurons (Option B)	Seen in retina and olfactory epithelium Have two specialized processes Dendrite carries information to the cell Axon carries information from the cell
Pseudo-unipolar neurons	Seen in dorsal root ganglion. The developing neuron has a single process that splits into two, both of which function as axons, one reaching up to the skin and the other to the spinal cord
Multipolar neurons (Option C)	Most common Seen in ganglion cell, hippocampal pyramidal cells, cerebellar Purkinje cells, motor neurons, etc Have one axon and multiple dendrites



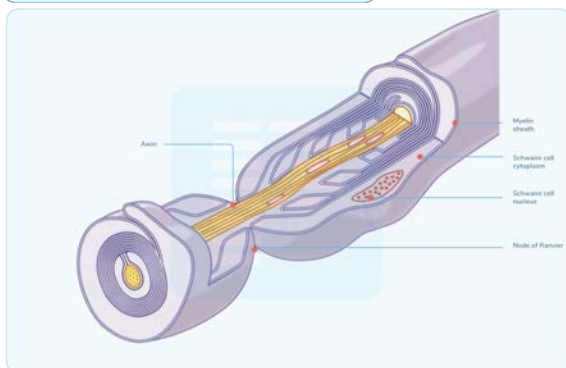
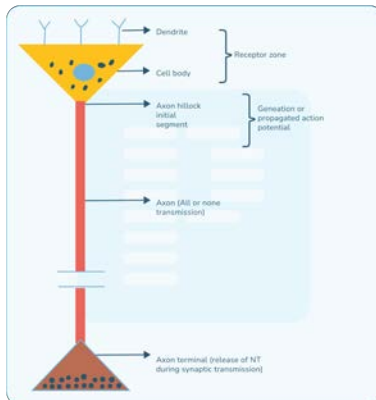
Solution for Question 24:

Correct Answer: D) Axon Terminal

Explanation:

Axon Terminal triggers the release of neurotransmitters into the synaptic cleft when action potential reaches it, facilitating communication with the next neuron or target cell.

Neurons have four important functional zones:



1. Dendritic zone (Receptor zone)	Receives information from other neurons and transmits impulses towards the cell body. Integrates local potentials from synaptic connections.
2. Axon Hillock and Initial segment (Action Potential Generation zone)	Action potentials are generated here. In motor neurons, it occurs at the axon hillock; in sensory neurons, at the initial node of Ranvier. Summates local potentials from dendrites and generates action potentials if the threshold is reached.
3. Axon (Transmission zone)	Long, slender process transmitting action potentials from the cell body to axon terminals. Ensures rapid and efficient propagation of nerve impulses over long distances.
4. Axon Terminal (Discharge zone)	Contains synaptic vesicles with neurotransmitters. When an action potential reaches the axon terminal, it triggers neurotransmitter release into the synaptic cleft for communication with other neurons or target cells.

Solution for Question 25:

Correct Answer: C) Soma

Explanation:

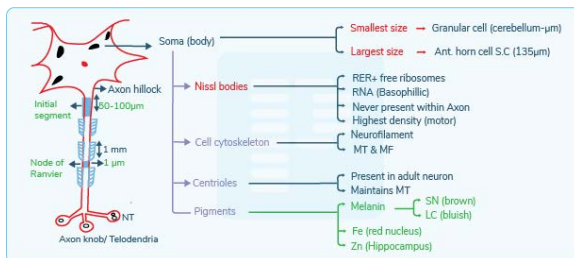
Cell body/Soma contain cytoskeletal proteins like neurofilaments, that form characteristic neurofibrillary tangles in Alzheimer's.

A neuron is the structural and functional unit of the nervous system, identified as an excitable cell due to its ability to generate action potentials when electrically stimulated.

The neuron has three parts: Cell body, Dendrite and Axon that ends at the axon terminal.

Table rendering failed

Axons often acquire a myelin sheath, a protein-lipid complex wrapped around the axon. The myelin sheath envelops the axon except at the nodes of Ranvier, periodic 1 micron gaps where the axon is unmyelinated.



Solution for Question 26:

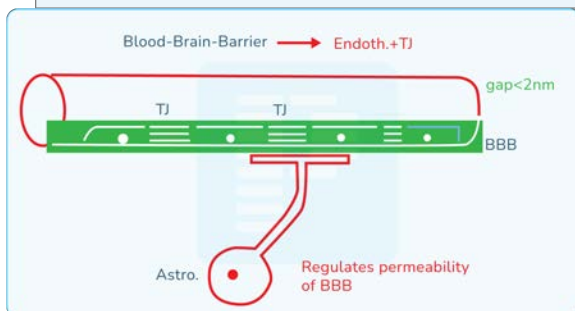
Correct Answer: A) Impermeability to plasma proteins

Explanation:

The blood-brain barrier (BBB) is impermeable to plasma proteins and most non-lipid soluble large organic molecules. This limits the effective concentration of therapeutic drugs, such as protein antibodies and non-lipid soluble drugs, in the CSF. Tight junctions between endothelial

cells are responsible for the low permeability of the BBB.

Blood-Brain Barrier (BBB)	
The BBB is the barrier between capillary endothelial cells in the brain and epithelial cells in the choroid plexus, limiting the exchange of substances into the brain.	
Anatomical Basis	Formed by two special structures: Endothelial cells in brain capillaries are joined by tight junctions. Capillaries are surrounded by astrocyte foot processes.
Permeability	Easily Penetrated: Water, CO ₂ , O ₂ , lipid-soluble substances. Not Penetrated: Proteins, polypeptides.
Circumventricular organs	The BBB is deficient in certain brain regions known as circumventricular organs, which are outside the BBB: Organum vasculosum of lamina terminalis (OVLT) Subfornical Organ Area postrema in medulla oblongata Posterior pituitary and median eminence(ventral part) of hypothalamus Pineal gland
Functions of BBB	Maintain constancy of Na ⁺ , K ⁺ , Ca ⁺⁺ , H ⁺ and Mg ⁺⁺ in brain fluids Protect neurons from harmful toxins Prevents escape of neurotransmitters from brain into circulation Disruption of BBB helps in identifying the location and extent of lesions in the brain
Clinical importance	Kernicterus: Hyperbilirubinemia damages the basal ganglia in infants due to an underdeveloped BBB. Brain Tumor : BBB is disrupted at the tumor site. Infection and Injury: BBB breakdown occurs in infected or injured regions.



Lipid-soluble substances permeate through BBB easily (Option B)

BBB does not have fenestrations. (Option C)

Efflux transporter in BBB limits the brain tissue exposure to various potential therapeutic agents.
(Option D)

Functional Anatomy of Skeletal Muscle

1. During muscle contraction, which best describes the role of the dihydropyridine receptor (DHPR) and the ryanodine receptor (RyR) in skeletal muscle?

- A. DHPR inhibits RyR to prevent calcium release.
 - B. DHPR activates RyR to release calcium from the sarcoplasmic reticulum.
 - C. RyR reuptakes calcium into the sarcoplasmic reticulum.
 - D. DHPR releases calcium directly into the muscle cytoplasm.
-

2. Match the changes in the sarcomere during muscle contraction with their corresponding descriptions: 1. A band 2. H band 3. I band 4. M line Descriptions: A. Decreases or disappears during contraction B. Length remains the same during contraction C. Becomes more prominent during contraction D. Decreases in length during contraction

- A. 1-B, 2-A, 3-C, 4-D
 - B. 1-A, 2-B, 3-C, 4-D
 - C. 1-B, 2-A, 3-D, 4-C
 - D. 1-A, 2-B, 3-D, 4-C
-

3. Which statement accurately describes the types of heat generated during muscle contraction and relaxation?

- A. Resting heat is produced during relaxation; activation heat during contraction.
- B. Activation heat is produced during relaxation; shortening heat during rest.
- C. Resting heat is produced at rest; activation, shortening, and relaxation heat during various phases of muscle activity.

D. Shortening heat is produced during rest; relaxation heat during contraction.

4. A 55-year-old male presents to the clinic with complaints of shortness of breath and fatigue during physical activity. Upon examination, the physician notes an elevated jugular venous pressure and an S3 heart sound. Echocardiography reveals dilated cardiac chambers with reduced ejection fraction. The physician explains that the heart is failing to pump efficiently, in part due to the length-tension relationship of the cardiac muscle. Which of the following best explains the difference between the length-tension relationship in skeletal muscle and cardiac muscle relevant to this patient's condition?

- A. The operating point for both skeletal and cardiac muscles is at the same point on the tension curve
 - B. Skeletal muscle generates maximal tension at near resting length, while cardiac muscle generates maximal tension at slightly shorter lengths.
 - C. Skeletal muscle maintains tension over a wider range of sarcomere lengths than cardiac muscle.
 - D. Both muscles lose tension rapidly when the sarcomere length is significantly shortened or stretched beyond optimal length.
-

5. Which statement best describes the length-tension relationship in skeletal muscle?

- A. The maximum tension is generated when the sarcomere length is 1.6 μm .
 - B. The number of cross-bridges and the tension generated decrease when the sarcomere length exceeds 2.2 μm .
 - C. Passive tension contributes to muscle tension even when the sarcomere length is optimal.
 - D. The sarcomere ruptures when the length exceeds 2.2 μm .
-

6. Rigor mortis is a postmortem change resulting in the stiffening of muscles. Which of the following best explains the biochemical basis of rigor mortis in relation to the cross-bridge cycle in skeletal muscle?

- A. Increased calcium ion concentration in the sarcoplasm leads to continuous muscle contraction.
 - B. Depletion of ATP prevents the detachment of myosin heads from actin filaments.
 - C. Accumulation of lactic acid in muscles causes a permanent contraction.
 - D. Degradation of muscle proteins leads to muscle stiffness.
-

7. Arrange the following steps in the correct sequence to describe the molecular mechanism of muscle contraction: Cross-bridge formation between myosin and actin Release of calcium ions from the sarcoplasmic reticulum Binding of calcium ions to troponin C Detachment of the myosin head from actin Hydrolysis of ATP to ADP and Pi, re-cocking the myosin head Power stroke, moving the myosin head from 90 degrees to 45 degrees

- A. 2, 3, 1, 6, 4, 5
 - B. 2, 3, 4, 1, 6, 5
 - C. 2, 3, 1, 4, 5, 6
 - D. 2, 3, 1, 6, 5, 4
-

8. Which of the following statements about the regulatory proteins of skeletal muscle is incorrect?

- A. Tropomyosin prevents actin-myosin attachment in a resting muscle.
- B. Tropomyosin is a filamentous protein, while troponin is a spherical protein consisting of 3 subunits.

C. Troponin C can bind with up to 4 calcium ions at a time, which initiates a conformational change in the troponin complex.

D. Troponin I attaches to tropomyosin, anchoring the troponin complex to the actin filament.

9. Which of the following statements is incorrect regarding the sliding filament mechanism of muscle contraction?

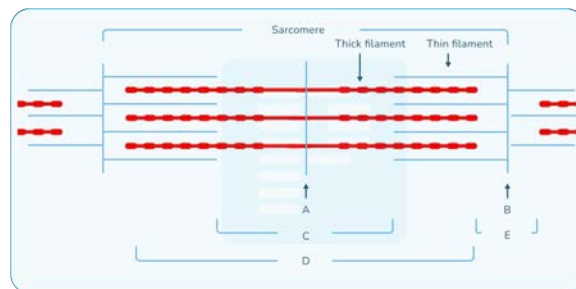
A. The sarcomere shortens as actin filaments slide over myosin filaments.

B. Contraction is triggered by action potential causing the release of calcium ions from the sarcoplasmic reticulum

C. The I-band & H-zone shortens and can disappear during full contraction.

D. Calcium ions bind to tropomyosin, allowing myosin to attach to actin.

10. Match the labeled parts of the sarcomere (A, B, C, D, E) with their correct names. 1. Z-line 2. I-band 3. M-line 4. A-band 5. H-zone



A. A-1, B-3, C-2, D-4, E-5

B. A-3, B-1, C-5, D-4, E-2

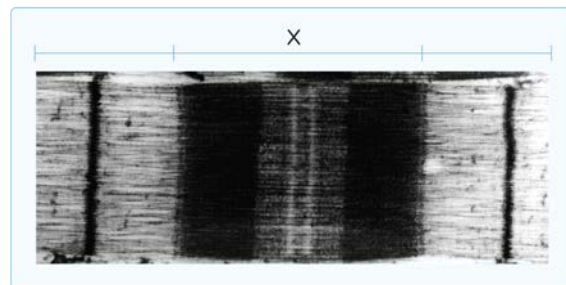
C. A-1, B-2, C-3, D-4, E-5

D. A-3, B-1, C-5, D-4, E-2

11. A 25-year-old athlete presents with muscle pain following intense exercise. A muscle biopsy is performed, and microscopic examination reveals the structural organization of muscle tissue. Which of the following correctly matches the muscle components with their descriptions?

- A. Fascicle - individual muscle cell; Fibre - bundle of myofibrils; Myofibril - contains contractile proteins; Contractile protein - forms entire muscle.
 - B. Fascicle - bundle of muscle fibres; Fibre - single muscle cell; Myofibril - contains sarcomeres; Contractile protein - actin and myosin.
 - C. Fascicle - single muscle cell; Fibre - contains sarcomeres; Myofibril - bundle of fascicles; Contractile protein - forms myofibrils
 - D. Fascicle - bundle of muscle fibres; Fibre - contains sarcomeres; Myofibril - individual muscle cell; Contractile protein - actin and myosin.
-

12. Examine the image of a sarcomere below. Which band is indicated by the X ?
{{caption_text}}



- A. A band
 - B. I band
 - C. Z disk
 - D. H zone
-

13. Which of the following statements is true regarding nerve-muscle physiology?

- A. SERCA is a protein of skeletal muscle encoded by gene ATP2A1
 - B. Secondary active transport mechanism involving the NCX protein seen in skeletal muscle.
 - C. Calsequestrin protein is present only in smooth muscle
 - D. Calreticulin protein is present in cardiac muscle
-

14. A 25-year-old man undergoing surgery for a fractured tibia experiences a sudden onset of muscle rigidity, a rapid heart rate, and a high fever shortly after receiving general anesthesia. Mutation in which of the following channels is involved in this condition?

- A. RyR1
 - B. RyR2
 - C. CaV
 - D. NaV
-

15. Which of the following statements accurately describes a unique characteristic of excitation-contraction coupling in cardiac muscle compared to skeletal muscle?

- A. T-tubules are absent, and calcium influx solely relies on diffusion through the sarcolemma.
 - B. Calcium-induced calcium release from the sarcoplasmic reticulum is mediated by ryanodine receptors alone.
 - C. Extracellular calcium entry into T-tubules contributes significantly to calcium influx during the plateau phase of the action potential.
 - D. Relaxation of cardiac muscle is primarily initiated by potassium binding to troponin.
-

16. Which of the following events directly results in the exposure of myosin-binding sites on actin filaments during skeletal muscle contraction?

- A. ATP binding to myosin heads
 - B. Depolarization of T-tubules
 - C. Binding of acetylcholine (ACh) to muscle fiber receptors
 - D. Conformational change in troponin complex
-

17. A 10-year-old boy presents with progressive muscle weakness, difficulty walking, and frequent falls. His symptoms started around the age of 5 and have gradually worsened. Physical examination reveals calf hypertrophy and a positive Gower's sign. Laboratory tests show elevated serum creatine kinase levels. Genetic testing identifies a mutation in the dystrophin gene leading to the complete absence of dystrophin protein. What is the most likely diagnosis?

- A. Duchenne Muscular Dystrophy
 - B. Becker Muscular Dystrophy
 - C. Emery-Dreifuss Muscular Dystrophy
 - D. Facioscapulohumeral Muscular Dystrophy
-

18. A 12-year-old boy presents to the OPD with complaints of difficulty climbing stairs, running, and lifting objects, and have noticed that his muscle weakness has gradually worsened. A muscle biopsy is performed, and immunohistochemical staining shows an absence of a key protein at the sarcolemma. Which of the following proteins is most likely absent, leading to this clinical presentation?

- A. Dystrophin
- B. Myosin
- C. Actin
- D. Troponin

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	3
Question 4	2
Question 5	2
Question 6	2
Question 7	1
Question 8	4
Question 9	4
Question 10	4
Question 11	2
Question 12	1
Question 13	1
Question 14	1
Question 15	3
Question 16	4
Question 17	1
Question 18	1

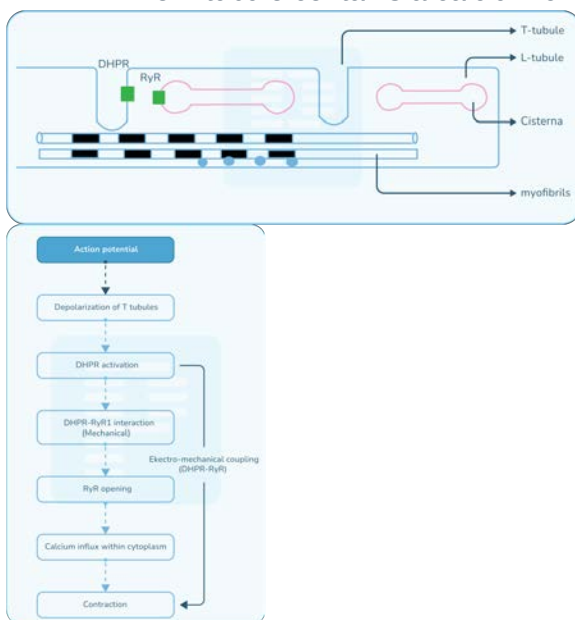
Solution for Question 1:

Correct Answer: B) DHPR activates RyR to release calcium from the sarcoplasmic reticulum.

Explanation: DHPR, located in the T-tubules, senses the action potential and undergoes a conformational change. This change directly activates RyR, the calcium release channel on the sarcoplasmic reticulum, leading to the release of calcium ions into the cytoplasm, which is crucial for muscle contraction.

Sarcotubular System:

- The Sarcotubular system is composed of the L tubule and T tubule .
- The T-tubule contains a L-type voltage gated Calcium channel, known as the DHPR (dihydropyridine receptor channel).
- The L-tubule contains a calcium channel known as RyR ryanodine receptor channel).



DHPR activates, not inhibits, RyR to facilitate calcium release during muscle contraction. (Option A)

The reuptake of calcium into the sarcoplasmic reticulum is performed by the SERCA pump, not by RyR. (Option C)

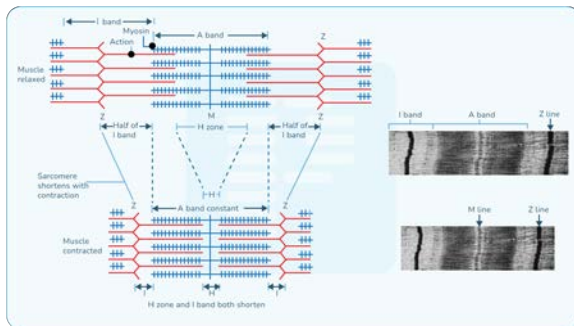
DHPR does not release calcium directly; it activates RyR to release calcium from the sarcoplasmic reticulum into the cytoplasm. (Option D)

Solution for Question 2:

Correct Answer: C) 1-B, 2-A, 3-D, 4-C

Explanation:

- A-band: The length of the A-band remains the same during muscle contraction because it corresponds to the length of the myosin filaments, which do not change.
- H-band: The H-band decreases or disappears as the actin filaments slide over the myosin filaments during contraction, reducing the region where only myosin is present.
- I-band: The length of the I-band decreases as the sarcomere shortens and the actin filaments overlap more with the myosin filaments.
- M-line: The M-line becomes more prominent as it marks the central point of the sarcomere where the thick filaments are anchored.



Solution for Question 3:

Correct Answer: C) Resting heat is produced at rest; activation, shortening, and relaxation heat during various phases of muscle activity.

Explanation:

Types of heat generated during muscle contraction and relaxation:

- Resting heat refers to the heat produced by the muscle at rest.
- It is generated by the basal metabolic processes that occur in the muscle even when it is not actively contracting.
- Activation heat is generated at the onset of muscle contraction.
- This heat results from the initial metabolic processes that prepare the muscle fibers for contraction, including the breakdown of ATP and the activation of the contractile machinery.

- Shortening heat is produced during the actual shortening of the muscle fibers.
- This occurs as the muscle generates force and performs work, converting chemical energy into mechanical energy and heat.
- Relaxation heat is generated as the muscle returns to its resting state after contraction.
- This phase involves processes such as the re-synthesis of ATP, the re-establishment of ion gradients, and the removal of metabolic byproducts.

Resting heat is not produced during relaxation. It is the heat generated by the muscle at rest, due to basal metabolic activities. Activation heat is correctly associated with the contraction phase. (Option A)

Activation heat is not produced during relaxation; it occurs at the onset of contraction. Shortening heat is not generated during rest; it is produced when muscle fibers actively shorten during contraction. (Option B)

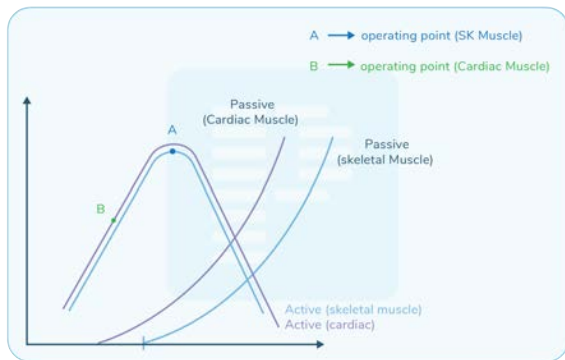
Shortening heat is not produced during rest. Relaxation heat is not generated during contraction; it occurs during the relaxation phase as the muscle returns to its resting state. (Option D)

Solution for Question 4:

Correct Answer: B) Skeletal muscle generates maximal tension at near resting length, while cardiac muscle generates maximal tension at slightly longer lengths.

Explanation:

Skeletal muscle generates maximal tension when sarcomere length is near resting length (about 2.0 to 2.2 micrometers), while cardiac muscle generates maximal tension at slightly longer lengths due to the rising part of the active tension curve, which allows for greater force production as the heart fills with more blood.



- In skeletal muscle, the operating point is at the plateau of the active tension curve, where maximal tension is achieved at resting length. In cardiac muscle, the operating point is on the rising part of the active tension curve, allowing room for additional stretching and increased force production. (Option A)
- Skeletal muscle tension decreases sharply if the sarcomere length deviates significantly from its optimal range. Cardiac muscle, due to its basket-weave fiber arrangement and early starting passive tension curve, is adapted to maintain near-maximal tension over a range of sarcomere lengths, crucial for its function in varying blood volumes. (Option C)
- Both muscle types lose tension if sarcomere lengths are significantly altered from their optimal range. However, cardiac muscle's passive tension curve starts earlier, and its fibers' basket-weave pattern allows it to recoil with significant force, enabling it to maintain function over a broader range of lengths compared to skeletal muscle. (Option D)

Solution for Question 5:

Correct Answer: B) The number of cross-bridges and tension generated decrease when the sarcomere length exceeds $2.2\ \mu\text{m}$.

Explanation:

The length-tension relationship in skeletal muscle:

- Optimal Length: The sarcomere length that produces maximum tension is between 2.0 and $2.2\ \mu\text{m}$. (Option B ruled out)
- Cross-Bridge Formation: Tension is greatest at this optimal length because cross-bridge formation between actin and myosin filaments is maximized.

- **Short Sarcomere Length:** When sarcomeres are shorter than optimal, excessive overlap of actin filaments limits cross-bridge formation, reducing tension.
- **Long Sarcomere Length:** When sarcomeres are longer than optimal, cross-bridge formation decreases as actin and myosin filaments overlap less, leading to reduced tension.
- **Maximum Stretch:** At lengths around 3.6 μm , the sarcomere can rupture due to excessive stretching. (Option D ruled out)

Passive Tension: Passive tension in muscle is the force generated by the stretching of the muscle's connective tissue structures. It arises due to the elasticity of the muscle's structural components, such as the titin protein, which resists excessive stretching.

- Passive tension increases as the muscle is stretched beyond its optimal sarcomere length, contributing to the total tension experienced by the muscle.
- While it is true that passive tension contributes to the total muscle tension at all lengths, option C is misleading because it implies that passive tension is a significant contributor even when the sarcomere length is optimal (2.0 to 2.2 μm).
- The primary focus of the optimal length is on the maximal active tension generated by cross-bridge interactions, not the passive elastic properties of the muscle.

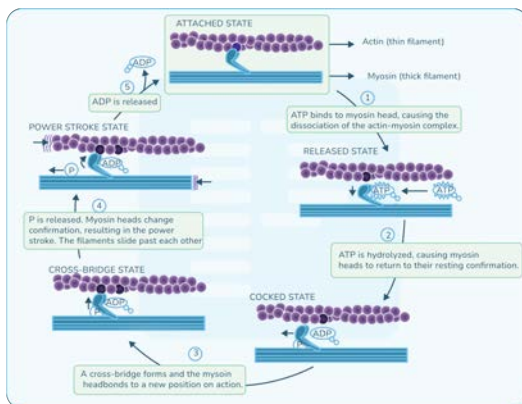
Solution for Question 6:

Correct Answer: B) Depletion of ATP prevents the detachment of myosin heads from actin filaments.

Explanation:

- In the cross-bridge cycle, ATP is essential for the detachment of myosin heads from actin filaments after a power stroke.
- In the absence of ATP, myosin heads remain bound to actin, leading to a state of continuous contraction, which is observed as muscle stiffness in rigor mortis.
- This process begins several hours after death and continues until muscle proteins begin to decompose.

Cross Bridge Formation:



- Calcium ions play a crucial role in initiating muscle contraction by binding to troponin, which allows myosin to bind to actin. However, rigor mortis occurs due to the lack of ATP, which is necessary for the detachment of myosin from actin after contraction. (Option A)
- Lactic acid accumulation in muscles is typically associated with anaerobic metabolism during vigorous exercise, leading to muscle fatigue and soreness. (Option C)
- Degradation of muscle proteins leads to the eventual resolution of rigor mortis, not its onset. As the body decomposes, proteolytic enzymes break down muscle proteins, leading to the relaxation of the muscles and the end of rigor mortis. (Option D)

Solution for Question 7:

Correct Answer: A) 2, 3, 1, 6, 4, 5

Explanation :

Sequential events during muscle contraction	
Release of calcium ions from the sarcoplasmic reticulum	An action potential triggers the release of calcium ions (Ca^{2+}) from the sarcoplasmic reticulum into the cytoplasm of the muscle cell.
Binding of calcium ions to troponin C	Calcium ions bind to troponin C on the thin filaments, causing a conformational change in the troponin complex.
Cross-bridge formation between myosin and actin	The conformational change in the troponin complex moves tropomyosin away from the myosin-binding sites on actin, allowing the myosin heads (with ADP and P_i bound) to attach to the exposed binding sites on actin.

Power stroke, moving the myosin head from 90 degrees to 45 degrees	The myosin head pivots from a 90-degree angle to a 45-degree angle, pulling the actin filament toward the center of the sarcomere. ADP and Pi are released during this movement.
Detachment of the myosin head from actin	A new ATP molecule binds to the myosin head, causing it to detach from the actin filament.
Hydrolysis of ATP to ADP and Pi, re-cocking the myosin head	The ATP is hydrolyzed to ADP and Pi, providing energy to re-cock the myosin head back to the 90-degree position, ready for another cycle of contraction.

Solution for Question 8:

Correct Answer: D) Troponin I attaches to tropomyosin, anchoring the troponin complex to the actin filament.

Explanation: Regulatory proteins play crucial roles in the contraction of skeletal muscles by controlling the interaction between actin and myosin filaments. Regulatory proteins are of 2 main types:

- Tropomyosin
- Troponin Complex (Troponin C, Troponin T & Troponin I)

Regulatory Protein	Function
Tropomyosin	Spiral-shaped filamentous protein wrapped around actin filaments. (Option B ruled out) Blocks myosin-binding sites on actin in a resting muscle, preventing muscle contraction. (Option A ruled out) Known as the gatekeeper molecule of muscle contraction.
Troponin T	Attached to tropomyosin. Anchors the troponin complex to the actin filament.
Troponin I	Acts as the inhibitory subunit of troponin . Helps position tropomyosin over myosin-binding sites on actin, preventing actin-myosin interaction in a resting muscle.
Troponin C	Binds with calcium ions, up to 4 ions at a time (Option C) The binding of calcium ions causes a conformational change in the troponin complex, moving tropomyosin away from the myosin-binding sites on actin, allowing muscle contraction.

--	--

Solution for Question 9:

Correct Answer: D) Calcium ions bind to tropomyosin, allowing myosin to attach to actin.

Explanation: The sliding filament theory describes the process by which muscles contract to produce force. It involves the following key steps:

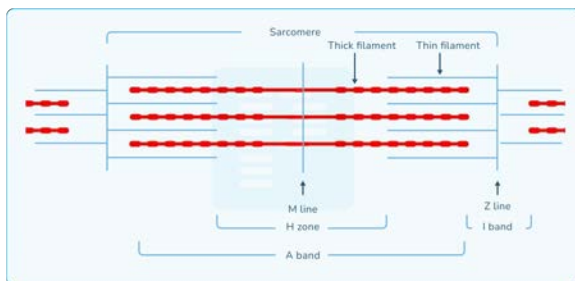
I-band shortens during contraction as the actin filaments slide over the myosin filaments. H-zone is the region within the A-band where there is only myosin and no overlapping actin. Both disappear during full contraction. (Option C ruled out)

Solution for Question 10:

Correct Answer: D) A-3, B-1, C-5, D-4, E-2

Explanation :

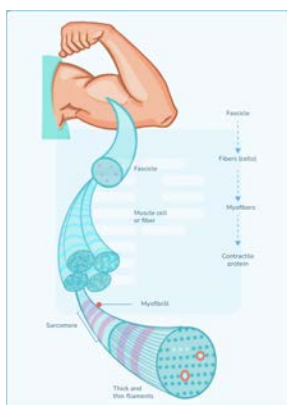
- A (M-line): The M-line is the center of the sarcomere and the H-zone, where the thick filaments are anchored.
- B (Z-line): The Z-line marks the boundary between adjacent sarcomeres and anchors the thin filaments.
- C (H-zone): The H-zone is the central region of the A-band where there are only thick filaments, with no thin filament overlap.
- D (A-band): The A-band is the length of the thick filaments, which remains constant during contraction.
- E (I-band): The I-band is the region of the sarcomere that contains only thin filaments.



Solution for Question 11:

Correct Answer: B) Fascicle - bundle of muscle fibers; Fibre - single muscle cell; Myofibril - contains sarcomeres; Contractile protein - actin and myosin

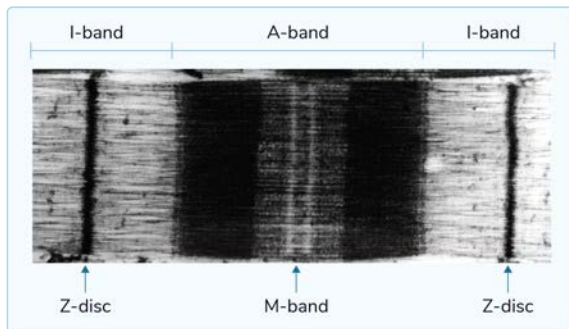
Explanation : Skeletal muscles are composed of multiple bundles called fascicles- which are made of individual muscle fibers (muscle cells)- muscle cells are made of multiple myofibrils- which are made up of contractile protein molecules- Actin and Myosin.



Solution for Question 12:

Correct Answer: B) A band

Explanation:



- A band : The dark band, anisotropic to polarized light. It contains the entire length of the myosin filaments. The ends of the actin filaments also overlap with the myosin filaments in this band.
- I band : The light band, isotropic to polarized light. It contains only actin filaments. (Option B ruled out)
- Z disk : A structure composed of filamentous proteins (alpha-actinin) that bisects the I band. Actin filaments attach to the Z disk and extend from it in both directions to interdigitate with myosin filaments. Z disks also pass from myofibril to myofibril, attaching them together across the muscle fiber. (Option C ruled out)
- H zone : Located at the center of the A band, this zone contains only myosin filaments and appears lighter than the rest of the A band. (Option D ruled out)
- M line : A structure that runs down the center of the H zone, providing structural support for the myosin filaments.

Solution for Question 13:

Correct Answer: A) SERCA is a protein molecule encoded by the gene ATP2A1

Explanation:

Sarcoplasmic reticulum Ca^{+} ATPase (SERCA) pump is a protein molecule encoded by gene ATP2A1.

- SERCA is a calcium–adenosine triphosphatase (ATPase) pump that helps transport calcium ions back into the sarcoplasmic reticulum from the muscle fiber, which leads to muscle relaxation.
- It is present in both skeletal muscle and cardiac muscle.

Sodium-Calcium Exchanger: Cardiac muscle also utilizes a sodium-calcium exchanger to remove calcium ions from the cell.

- This exchanger allows sodium to enter the cell while simultaneously transporting calcium out.
- The sodium that enters the cell during this process is then removed by the sodium-potassium ATPase pump
- Sodium-calcium exchanger (NCX) is primarily involved in calcium handling in cardiac muscle, not skeletal muscle (Option B ruled out)

Two proteins found within the sarcoplasmic reticulum include:

- Calsequestrin protein: Found in skeletal muscle, myocardium, and smooth muscle . (Option C ruled out)
- Calreticulin protein: Located specifically in smooth muscle tissue . (Option D ruled out)

Solution for Question 14:

Correct Answer: A) RyR1

Explanation: The symptoms of muscle rigidity, high fever, and rapid heart rate clinically represent Malignant Hyperthermia. A gain of mutation of the RyR1 channel causes excessive calcium release and causes muscle rigidity.

Mutation	Diseases
Gain of RyR1 Mutation	● Malignant Hyperthermia ○ Muscle rigidity, high fever, and rapid heart rate. ○ Rapid breakdown of skeletal muscle (rhabdomyolysis) and a high plasma K^+ level. ○ Tx: Dantrolene, a drug that antagonizes ryanodine receptors
Loss of RyR1 Mutation	● Central core disease ○ Hypotonia (decreased muscle tone) at birth, mild delay in child development, weakness of the facial muscles. ○ No specific treatment.
Mutation of CaV1.1	● Low potassium levels in the blood can cause a sudden onset paralysis or mild-severe form of muscle weakness which is called Hypokalemic periodic paralysis
Mutation of NaV1.4	● Due to voltage gated Na^+ channel mutation it can cause Hyperkalemic periodic paralysis
Mutation of ATP2A1	● Brody myopathies ○ muscle stiffening/cramps/spasm is seen

RyR2 (Option B) – Mutations cause catecholaminergic polymorphic ventricular tachycardia (CPVT) , leading to life-threatening arrhythmias during stress or exercise.

CaV (Option C) – Mutations in calcium voltage-gated channels cause hypokalemic periodic paralysis and certain forms of congenital myopathy .

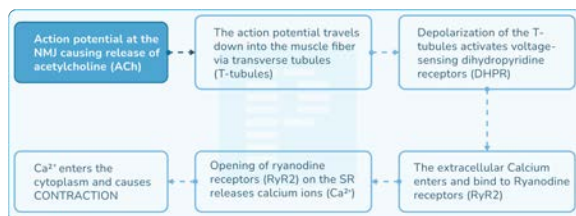
NaV (Option D) – Mutations in sodium voltage-gated channels cause paramyotonia congenita , hyperkalemic periodic paralysis , and some forms of epilepsy .

Solution for Question 15:

Correct Answer: C) Extracellular calcium entry through T-tubule membrane contributes significantly to calcium influx during the plateau phase of the action potential.

Explanation: Unlike skeletal muscle, cardiac T-tubules are open to the extracellular space, allowing extracellular calcium influx during the plateau phase of the action potential. This influx is critical for sustaining calcium levels necessary for prolonged contraction in cardiac muscle. This phenomenon is stated as Calcium-induced calcium release.

CONTRACTION IN CARDIAC MUSCLE



T-tubules are present in both cardiac and skeletal muscle , and they are essential for transmitting the action potential into the interior of the muscle fiber (Option C)

Calcium-induced calcium release in cardiac muscle involves both voltage-gated calcium channels in the T-tubules and ryanodine receptors on the SR membrane (Option C)

Relaxation of cardiac muscle is primarily initiated by the removal of calcium ions from the cytoplasm, facilitated by the sarcoplasmic reticulum calcium ATPase (SERCA) , and not by potassium binding to troponin (Option D)

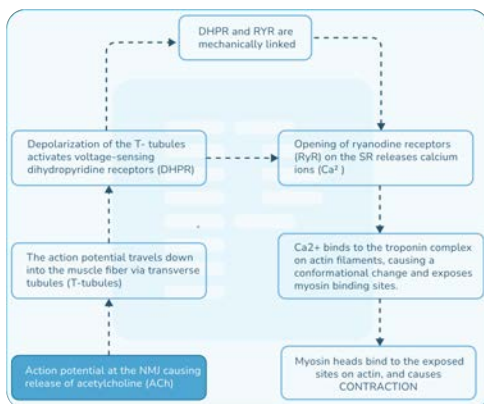
Solution for Question 16:

Correct Answer: D) Conformational change in troponin complex

Explanation:

The conformational change in the troponin complex occurs in response to elevated cytoplasmic calcium ions (Ca^{2+}), which bind to troponin. This causes tropomyosin to move away from the myosin-binding sites on actin filaments, allowing myosin heads to bind with Myosin binding sites of actin and initiate muscle contraction.

CONTRACTION OF SKELETAL MUSCLE:



ATP binding allows myosin heads to detach from actin filaments during muscle relaxation, not contraction. (Option A)

Depolarization of T-tubules allows for the propagation of the action potential into the muscle fiber but does not directly expose myosin-binding sites on actin filaments. (Option B)

Binding of acetylcholine (ACh) to muscle fiber receptors initiates muscle fiber depolarization but does not directly expose myosin-binding sites on actin filaments. (Option C)

Solution for Question 17:

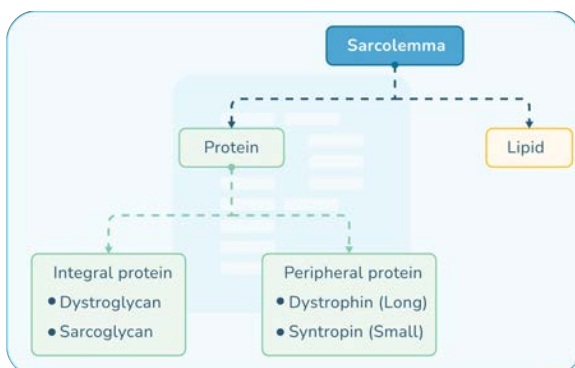
Correct Answer: A) Duchenne Muscular Dystrophy

Explanation: The progressive muscle weakness, difficulty walking, and frequent falls starting at age 5, along with calf hypertrophy, a positive Gower's sign, and elevated creatine kinase, are classic findings of Duchenne muscular dystrophy (DMD). Genetic testing confirms a dystrophin gene mutation leading to complete absence of dystrophin, which is a characteristic finding.

Becker Muscular Dystrophy (BMD) (Option B) also involves mutations in the dystrophin gene but usually presents later (adolescence or adulthood) with a slower progression and milder symptoms. Calf hypertrophy can be seen in both BMD and DMD, but the overall course and severity of BMD are milder. Emery-Dreifuss Muscular Dystrophy (EDMD) (Option C) often presents with early contractures, cardiac involvement (conduction defects, arrhythmias), and humeroperoneal muscle weakness. It does not typically involve calf hypertrophy and has a different pattern of muscle weakness compared to DMD. Facioscapulohumeral Muscular Dystrophy (FSHD) (Option D) primarily affects the muscles of the face, shoulders, and upper arms. It usually presents in adolescence or early adulthood with a distinct pattern of muscle weakness that spares the lower limbs until later stages.

Solution for Question 18:

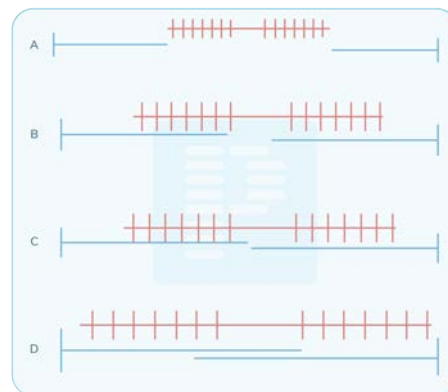
Explanation: Dystrophin is a key protein associated with the sarcolemma, the membrane surrounding muscle fibers. It plays a critical role in maintaining the structural integrity of muscle cells by linking the internal cytoskeleton to the extracellular matrix. The progressive muscle weakness, difficulty with climbing stairs and lifting objects, and absence of a key sarcolemmal protein on biopsy suggest Duchenne Muscular Dystrophy (DMD), which is caused by the absence of dystrophin.



- Myosin (Option B) and Actin (Option C) are essential contractile proteins within the muscle fibers themselves, not specifically associated with the sarcolemma.
- Troponin (Option D) is a regulatory protein complex involved in muscle contraction but is located within the muscle fiber, not at the sarcolemma.

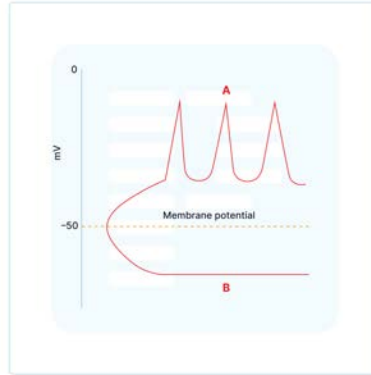
Motor unit & Smooth muscle

1. Schematic diagram of sarcomere is given below. Red coloured structure is myosin with heads. Based on the length-tension relationship of muscle fibres, between which sarcomere length does the sarcomere allow optimal muscle contraction?



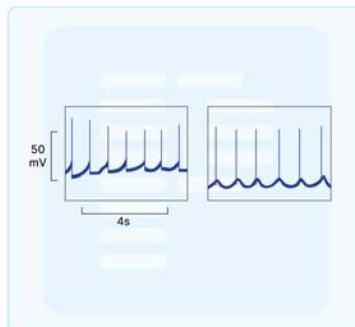
- A. A to B
- B. B to C
- C. C to D
- D. At any length

2. Effects of various agents on the membrane potential of intestinal smooth muscle is given below. Addition of which of the following substances produces the response labeled with B:



- A. Acetylcholine
- B. Cold
- C. Stretch
- D. Adrenaline

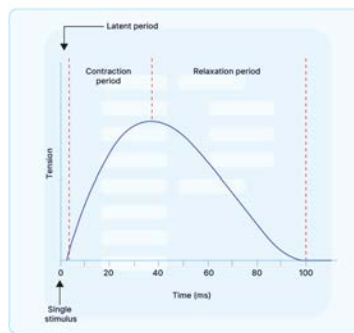
3. Electrical activity of individual muscle cells in the guinea pig is recorded. Sinusoidal fluctuation of membrane potential with firing on the rising phase of each wave is identified from the recording. Identify the muscle type:



- A. cardiac muscle
- B. skeletal muscle
- C. smooth muscle

D. none of the above

4. Graph below is showing simple muscle twitch in frog gastrocnemius muscle. Calculate tetanising frequency of this muscle.



- A. 25
 - B. 20
 - C. 10
 - D. 5
-

5. Which of the following statements about the mechanism of smooth muscle contraction is correct?

- A. Calcium ions primarily come from the sarcoplasmic reticulum
 - B. The latent period before contraction is similar to that of skeletal muscle
 - C. Calcium ions diffuse into the cell from the extracellular fluid during contraction
 - D. The concentration of calcium ions is higher inside the smooth muscle cell
-

6. Which of the following is NOT a primary source/mechanism of calcium release for smooth muscle contraction?

- A. Extracellular fluid
 - B. Sarcoplasmic reticulum
 - C. Mitochondria
 - D. Calcium-induced calcium release from intracellular stores
-

7. Which of the following is a characteristic of multi-unit smooth muscle?

- A. Fibers arranged in sheets or bundles
 - B. Fibers contract as a single unit
 - C. Discrete, separate fibers
 - D. Connected by gap junctions
-

8. A researcher investigating the electrical activity of smooth muscle cells comes across a specific group of cells in GIT similar to the cells in the S.A node of the heart that generates rhythmic activity without any stimulus. Which of the following will likely generate the above-mentioned activity in GIT cells?

- A. Spike potential
 - B. Action potential with plateau
 - C. Slow wave rhythm
 - D. Innervation of cells by ANS
-

9. Which of the following combinations of features are characteristic of smooth muscle tissue?

- A. Z-line and troponin present
 - B. Caveolae and T-Tubule present
 - C. Calmodulin and titin present
 - D. Voltage-gated calcium channels and calmodulin present
-

10. A 33-year-old competitive cyclist is preparing himself for a training program. What type of muscle fibres are important to improve his cycling performance during various stages of his race

- A. Only type I
 - B. Only type II
 - C. Both type I and type II
 - D. Smooth muscles and type II
-

11. A 23-year-old pianist comes to the clinic complaining of weakness in her fingers. She says she is unable to play a challenging piece in a song. Which of the following is the innervation ratio of the group of muscles affected in this patient?

- A. 1:1000
 - B. 1:2000
 - C. 1:600
 - D. 1:3
-

12. A patient has been experiencing occasional involuntary movement of his facial muscles for the past three days. On examination of his face, minute fasciculation of his right side facial muscles was seen. Which of the following statements regarding muscle fasciculation is true?

- A. Can occur due to late neuronal damage or irritation.
 - B. Not visible to the naked eye and can only be detected by an EMG.
 - C. Always indicative of neuromuscular disease.
 - D. It can be irregular and brief or persistent.
-

13. A 45-year-old patient presents with generalised muscle weakness. During an electromyography (EMG) study, you observe motor unit potentials (MUPs) with consistent amplitude, duration, and regular shape. What is the most likely diagnosis?

- A. Early-stage amyotrophic lateral sclerosis (ALS)
 - B. Demyelinating polyneuropathy
 - C. Normal muscle function
 - D. Myopathy with fibre degeneration
-

14. Based on the tetanisation of muscle fibres, which of the following statements is true regarding the tetanisation frequency?

- A. It is the frequency at which the muscle fibers show visible twitching
 - B. At tetanising frequency, successive contractions fuse into a smooth, continuous contraction.
 - C. If the frequency exceeds the tetanising frequency, the strength of contraction increases.
 - D. At this frequency, there are no calcium ions in the muscle sarcoplasm.
-

15. During an experimental study, a researcher stimulates a skeletal muscle with increasing frequencies of electrical impulses. The tension recordings show successive contractions that do not fully relax between stimuli. This phenomenon

is best described as?

- A. Staircase phenomenon
 - B. Incomplete tetanus
 - C. Treppe phenomenon
 - D. Complete tetanus
-

16. During a physiological study, a researcher observes incremental increases in muscle contraction height with repeated stimulations of a skeletal muscle. What is this phenomenon, also known as the Treppe phenomenon, primarily attributed to?

- A. Increased availability of ATP in the muscle fibres.
 - B. Faster pumping of calcium ions into the sarcoplasmic reticulum.
 - C. Enhanced sensitivity of the muscle fibres to acetylcholine.
 - D. Accumulation of residual calcium ions in the cytoplasm
-

17. During a clinical examination, a neurologist stimulates the motor nerve supplying a patient's slow muscle fibre. Graphical recording of muscle tension shows distinct phases. Which of the following correctly identifies the sequence of phases and their durations?

- A. Latency Period (10 ms), Contraction Period (40 ms), Relaxation Period (50 ms)
 - B. Latency Period (20 ms), Peak Contraction (60 ms), Recovery Period (30 ms)
 - C. Activation Stage (30 ms), Maximum Contraction (70 ms), Relaxation Interval (40 ms)
 - D. Initiation Phase (40 ms), Contraction Peak (50 ms), Relaxation Phase (10 ms)
-

18. During a physiology lab, students are studying the phases of a simple muscle twitch. They observe a brief delay between the application of a stimulus and the

beginning of muscle contraction. What is this initial delay called?

- A. Refractory period
 - B. Contraction period
 - C. Relaxation period
 - D. Latency period
-

19. Which of the following best describes the process of tetanisation in muscle physiology?

- A. Individual twitch contractions at low frequency.
 - B. Overlapping muscle contractions increases total strength with higher frequency.
 - C. Relaxation between action potentials.
 - D. Decrease in contraction strength with higher frequency.
-

20. Which of the following best describes the order in which the motor units are recruited?

- A. Large motor units are excited first
 - B. Small motor units are relaxed first
 - C. Large motor units are recruited first
 - D. Small motor units are excited first
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	3
Question 4	1
Question 5	3
Question 6	3
Question 7	3
Question 8	3
Question 9	4
Question 10	3
Question 11	4
Question 12	4
Question 13	3
Question 14	2
Question 15	2
Question 16	4
Question 17	1
Question 18	4
Question 19	2
Question 20	4

Solution for Question 1:

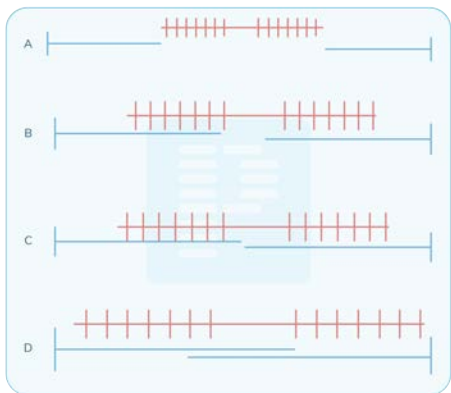
Correct Answer: B) B to C

Explanation:

When the sarcomere is at perfect length—about 2.2 micrometres—the actin and myosin overlap perfectly, allowing the muscle to contract optimally.

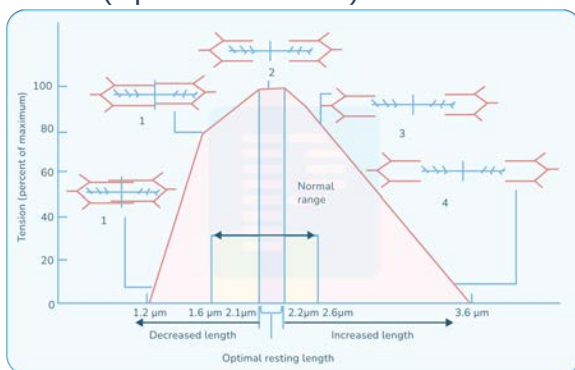
The sarcomere maintains full tension as it shortens from 2.2 micrometres to 2 micrometres, i.e., from length B to C.

- Sarcomeres are units in the muscles composed of actin and myosin filaments.
- The amount of force or tension a muscle generates depends on the sarcomere's length.



- Length A: Actin and myosin do not overlap; thus, muscle cannot generate tension. It occurs when the muscle is very stretched out.
- Length B: Actin and myosin start to overlap, thus generating tension - maximum tension occurs when the overlap is just right - 2.2 micrometre
- Length C: Sarcomere maintains full tension as it shortens from 2.2 micrometre to 2 micrometres
- Length D: When the sarcomere length is shorter than 2 micrometres, the actin filaments overlap, thus reducing tension.
- At about 1.65 micrometres, the sarcomere (Z disks) ends bump into the ends of the myosin filaments.
- Further shortening crumples the myosin filaments, and tension drops to zero.
- Thus, Length A to B: Muscle cannot generate any tension (Option A ruled out) Length B to C: Muscle starts generating tension, and the maximum is at 2.2 micrometres (Option B) Length C to D: Muscle can maintain full tension, and it allows optimum muscle contraction (Option C ruled out)

- Length A to B: Muscle cannot generate any tension (Option A ruled out)
- Length B to C: Muscle starts generating tension, and the maximum is at 2.2 micrometres (Option B)
- Length C to D: Muscle can maintain full tension, and it allows optimum muscle contraction (Option C ruled out)
- Length A to B: Muscle cannot generate any tension (Option A ruled out)
- Length B to C: Muscle starts generating tension, and the maximum is at 2.2 micrometres (Option B)
- Length C to D: Muscle can maintain full tension, and it allows optimum muscle contraction (Option C ruled out)

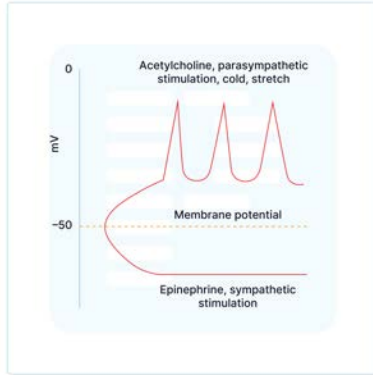


Solution for Question 2:

Correct answer: D) Adrenaline

Explanation:

The agent labeled as B in the effects on the membrane potential of intestinal smooth muscle is Adrenaline .



The addition of Adrenaline to intestinal smooth muscle results in hyperpolarization of the membrane potential, decreased frequency of action potentials, and relaxation of the muscle.

- Adrenaline (epinephrine) or norepinephrine acts as a neurotransmitter at noradrenergic nerve endings, inducing inhibitory potentials in smooth muscle.
- As a result, the muscle becomes less active, and the overall contractile force is reduced.
- In contrast, Acetylcholine (Option A ruled out) (Labeled as A) has the opposite effect. When ACh is added to intestinal smooth muscle: The membrane potential becomes less negative (depolarization). This increase in depolarization leads to more frequent spikes and greater muscle activity, increasing tonic tension and rhythmic contractions.
- The membrane potential becomes less negative (depolarization).
- This increase in depolarization leads to more frequent spikes and greater muscle activity, increasing tonic tension and rhythmic contractions.
- The membrane potential becomes less negative (depolarization).
- This increase in depolarization leads to more frequent spikes and greater muscle activity, increasing tonic tension and rhythmic contractions.

Cold (Option B): Typically reduces metabolic activity and may lead to decreased muscle contractions but does not have the specific hyperpolarizing effect of adrenaline.

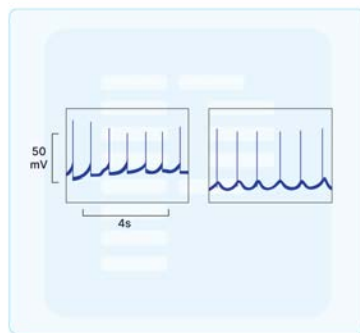
Stretch (Option C): Often activates stretch-sensitive channels, leading to depolarization and increased muscle activity, opposite to the effect of adrenaline.

Solution for Question 3:

Correct answer: C) Smooth muscle

Explanation:

The recorded electrical activity shows sinusoidal fluctuations in membrane potential with firing on the rising phase, indicating Smooth Muscle .



Left : Pacemaker-like activity with spikes firing at each peak.

Right : Sinusoidal fluctuation of membrane potential with firing on the rising phase of each wave.

Smooth Muscle Characteristics:

- Instability of Membrane Potential: Lacks a true resting value, often fluctuating between -20 to -65 mV.
- Divergent Electrical Activity: Exhibits slow sine wave-like fluctuations and spikes. Spikes may overshoot zero and last about 50 ms.
- Continuous Contraction (Tonus): Smooth muscle can contract irregularly and independently of nerve supply.

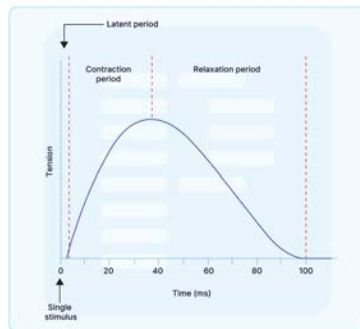
Cardiac Muscle (Option A): Exhibits rhythmic action potentials with a plateau phase, not sinusoidal fluctuations.

Skeletal Muscle (Option B): Shows rapid, all-or-nothing action potentials without any sinusoidal pattern.

Solution for Question 4:

Correct answer: A) 25Hz

Explanation:



The Tetanising frequency of this muscle is 25Hz.

- Each block corresponds to 10ms.
- Therefore contraction period = 40ms = 40/1000 seconds

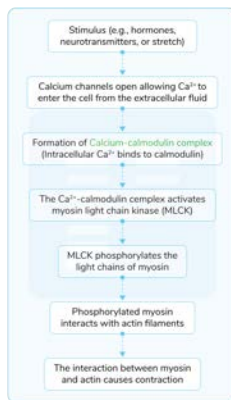
Tetanising frequency = $1 / \text{Contraction period}$

Tetanising frequency = $1000/40 = 25\text{Hz}$

Solution for Question 5:

Correct Answer: C) Calcium ions diffuse into the cell from the extracellular fluid during contraction

Explanation: When stimulated by hormones, neurotransmitters, or stretch, calcium channels in smooth muscle open, allowing calcium ions to rapidly diffuse from the extracellular fluid into the cell and initiate contraction.



- Smooth muscle can maintain contraction with less energy expenditure than skeletal muscle (Latch-bridge mechanism).
- Relaxation: Occurs when Ca^{2+} is pumped out of the cell or back into the sarcoplasmic reticulum, and myosin is dephosphorylated.

Unlike skeletal muscle, the sarcoplasmic reticulum in smooth muscle is poorly developed and is not the primary source of calcium for contraction. (Option A)

The latent period before smooth muscle contraction (200-300 milliseconds) is significantly longer than that of skeletal muscle, but it is not similar. (Option B)

The concentration of calcium ions is much higher in the extracellular fluid ($>10^{-3}$ molar) compared to inside the smooth muscle cell ($<10^{-7}$ molar). (Option D)

Solution for Question 6:

Correct Answer: C) Mitochondria

Explanation:

While mitochondria play a role in cellular calcium homeostasis, they are not considered a primary source of calcium for smooth muscle contraction. Their main function is energy production, and although they can sequester calcium, they are not a significant source of the rapid calcium increase needed for contraction.

The primary sources of calcium for smooth muscle contraction are:

- Extracellular fluid: Calcium enters the cell through voltage-gated calcium channels in the cell membrane (Option A ruled out)

- Sarcoplasmic reticulum: This intracellular organelle stores and releases calcium (Option B ruled out)
- Calcium-induced calcium release amplifies the calcium signal by triggering further release from intracellular stores (Option D ruled out)
- Voltage-gated calcium channels allow calcium influx from the extracellular space when the cell membrane is depolarised.
- Most calcium ions for smooth muscle contraction come from the extracellular fluid, not internal stores.

Solution for Question 7:

Correct Answer: C) Discrete, separate fibers

Explanation:

Multi-unit smooth muscle is characterised by discrete, separate fibres. This structure allows for more precise control, with fibres able to contract independently.

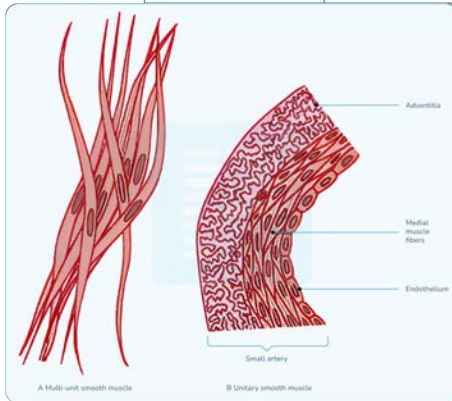
Options A, B and D are characteristics of unitary (single-unit) smooth muscle.

There are 2 types of smooth muscles:

Characteristic	Multi-Unit Smooth Muscle	Unitary (Single-Unit) Smooth Muscle
Structure	Discrete, separate fibres. (Option C)	Fibers arranged in sheets or bundles (Option A ruled out)
Innervation	Often innervated by single nerve-ending.	Less dependent on nerve signals.
Contraction	Fibers contract independently.	Fibers contract as a single unit. (Option B ruled out)
Cell Connections	Covered by a basement membrane, insulated	Connected by gap junctions (Option D ruled out)
Control	Mainly by nerve signals.	Mainly by non-nervous stimuli.
Signal Propagation	Limited between fibres.	Action potentials can spread between fibres.
Examples	Ciliary muscle, iris muscle, and piloerector muscles.	The gastrointestinal tract, bile ducts, ureters, uterus, and many blood vessels.

Also Known
As

Syncytial smooth muscle, visceral
smooth muscle.



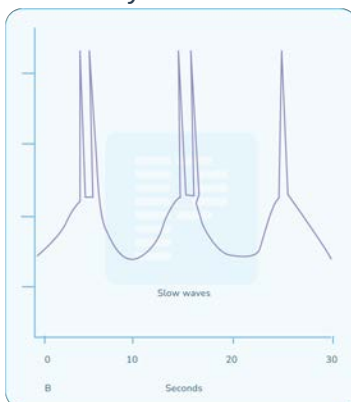
Solution for Question 8:

Correct Answer: C) Slow wave rhythm

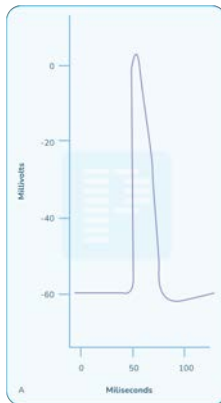
Explanation:

The rhythmic contractile activity seen in a specific group of GIT cells is a slow wave rhythm generated by pacemaker cells.

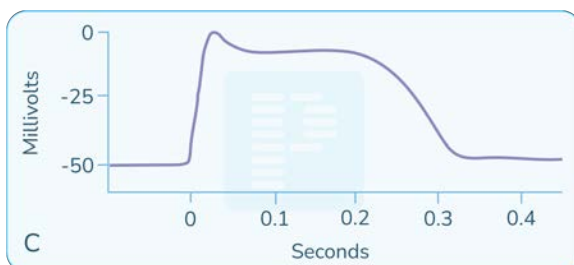
- Pacemaker cells of GIT are Interstitial cells of Cajal, similar to heart pacemaker cells.
- Slow wave rhythms are due to an excitatory process, and these are not action potentials.
- The exact cause of slow wave rhythms is not known, but two theories are suggested:
- When slow waves are strong enough, they generate action potentials, which cause rhythmical contraction, also known as pacemaker waves.



Spike Potential (Option A): Generated due to the action of hormones or neurotransmitters, these potentials are seen in the gastrointestinal tract.



Action potential with plateau (Option B): Generated in single unit smooth muscle cells such as the uterus and ureter. This potential causes prolonged contraction.



Innervation of cells by ANS (Option D): This is seen in multiunit smooth muscle cells such as those of the ciliary, iris, and piloerector muscles.

Solution for Question 9:

Correct Answer: D) Voltage-gated calcium channels and calmodulin present

Explanation: Voltage-gated calcium channels and calmodulin are characteristics of smooth muscles, whereas Z-line, troponin and titin are features of Skeletal muscle.

Characteristic features of Smooth muscles and Skeletal muscles:

Smooth muscles	Skeletal muscles
Spindle-shaped, single nucleated cells.	Multiple nucleated cells.

Neuromuscular junction is not present.	Proper neuromuscular junction is present.
Caveolae is present. (Option B ruled out)	T-tubule is present. (Option B ruled out)
Predominantly, calcium channels are present.	Predominantly, sodium channels are present.
Actin and myosin are not properly organised.	Actin and myosin are properly organised.
A dense body is present.	The Z- line is present. (Option A ruled out)
Calmodulin is present, and Titin is absent. (Option C ruled out)	Troponin (Option A ruled out) and Titin (Option C ruled out) are present.

Solution for Question 10:

Correct Answer: C) Both type I and type II

Explanation: Competitive cycling requires both endurance and high-intensity effort, so both type I and type II fibres are required.

Type I fibres provide prolonged strength of contraction over minutes to hours, and type II fibers provide maximal power for seconds to minutes.

Type of fibres	Slow fibres (Type I)	Fast fibres (Type II)
Size of fibres and their effect on contraction	Slow fibres are small, providing endurance and prolonged contraction strength for longer periods. (Option A ruled out)	Fast fibres are large and provide great strength of contraction for a very short period. (Option B, D ruled out)
Innervation	Innervated by small nerve fibres.	By large nerve fibres.
Sarcoplasmic reticulum	Less extensive sarcoplasmic reticulum, slower calcium ion release.	Extensive sarcoplasmic reticulum rapidly releases calcium ions
Blood Supply	An extensive blood vessel system supplies extra oxygen.	Less extensive blood supply.
Metabolic pathway	More number of mitochondria support a high level of oxidative metabolism.	Large amounts of glycolytic enzymes rapidly release energy through glycolysis.
Myoglobin	Present in large amounts, hence called red muscle.	Less amounts of myoglobin, hence called white muscle.

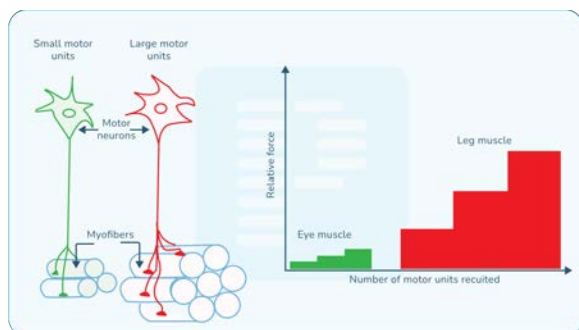
Smooth muscles and type II: Cycling requires activity in the lower leg muscles where smooth muscles are absent. (Option D ruled out)

Solution for Question 11:

Correct Answer: D) 1:3

Explanation: This patient cannot play the piano, which suggests that the small muscles of the hands are affected.

- The innervation ratio refers to the number of muscle fibres controlled by a single motor neuron.
- The innervation ratio of small-sized motor units is 1:3 to 1:6.



Difference between Small size motor unit vs Large size motor unit:

Aspects	Small-sized motor unit	Large-sized motor unit
Required for	Fine movements.	Gross movements.
Present in	Extraocular muscles (EOM) and the small muscles of our fingers.	Lower limb muscles, particularly the leg muscles, i.e., gastrocnemius and back muscles.
Innervation ratio	1:3 to 6, i.e. one single axon supplies 3-6 skeletal muscle fibres. (Option D)	1:600 to 2000, i.e., one single axon supplies 600 - 2000 skeletal muscle fibres. (Option A, B, C)

Solution for Question 12:

Correct Answer: D) It can be irregular and brief or persistent

Explanation:

Muscle fasciculation is spontaneous contractions of muscle fibres that are visible to the naked eye and can be irregular, brief, or persistent.

Features	Fasciculation	Fibrillation
Definition	Spontaneous muscle contraction of a group of muscle fibres.	Involuntary contraction of individual muscle fibres.
Visibility	Visible to the naked eye.	Not visible to the naked eye Detected on EMG. (Option B)
Cause	Irritation or spontaneous activation of motor neurons.	Damage to muscle fibers, can occur in late neuronal damage (Option A)
Significance	It is usually benign but can indicate motor neuron disorders.	Indicative of neuromuscular disease. (Option C)
Duration and pattern	Irregular and brief or persistent. (Option D)	Regular with a consistent pattern on EMG.
Sensory function	Usually intact.	Compromised if neuropathy is present.

EMG: Electromyography

Solution for Question 13:

Correct Answer: C) Normal muscle function

Explanation: EMG with consistent amplitude, duration, and regular shape MUPs is characteristic of normal muscle function despite muscle weakness, which in this case is not due to any primary muscle pathology.

Motor unit potentials are the electrical activity recorded from a motor unit during muscle contraction.

- It includes the contribution of all muscle fibres within a single motor unit.

- Electromyography (EMG) is used to record Motor Unit Potentials(MUP).
- The waveform on the EMG can help detect neuromuscular diseases. Duration of wave - time taken for the entire motor unit to fire Amplitude of wave - reflects the number and size of muscle fibres within the motor unit. Shape of wave - depends on the health of the motor unit.
- Duration of wave - time taken for the entire motor unit to fire
- Amplitude of wave - reflects the number and size of muscle fibres within the motor unit.
- Shape of wave - depends on the health of the motor unit.
- Normal wave = consistent amplitude, duration and regular shape.
- Duration of wave - time taken for the entire motor unit to fire
- Amplitude of wave - reflects the number and size of muscle fibres within the motor unit.
- Shape of wave - depends on the health of the motor unit.

In pathological conditions, such as early-stage amyotrophic lateral sclerosis (ALS) (Option A) , demyelinating polyneuropathy (Option B) , or myopathy with fibre degeneration (Option D) , EMG studies typically show abnormalities in the MUPs.

- Polyphasic pattern and longer duration: Neurogenic abnormalities.
- Polyphasic pattern and shorter duration: Myopathic abnormalities.

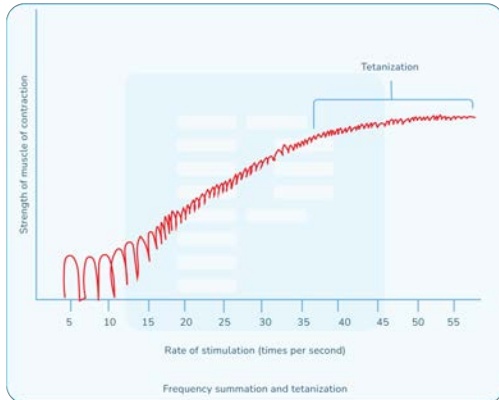
Solution for Question 14:

Correct Answer: B) At tetanising frequency, successive contractions fuse into a smooth, continuous contraction.

Explanation: At tetanising frequency, successive contractions fuse into a smooth, continuous contraction.

- At low-frequency stimulation, individual contractions occur one after another, but as the frequency increases, new contractions begin before the previous one ends; thus, the second contraction partially adds to the first.
- The total strength of contraction increases progressively.
- At a critical tetanising frequency, the contraction fuses together, and the muscle contraction appears smooth and continuous - Tetanization.
- Further, an increase in the frequency of stimulation does not increase contractile force. (Option C)

- This occurs when sufficient calcium ions are maintained in the muscle sarcoplasm even between two action potentials (AP) (Option D)
- Thus, a full contractile state is sustained without any relaxation between AP.



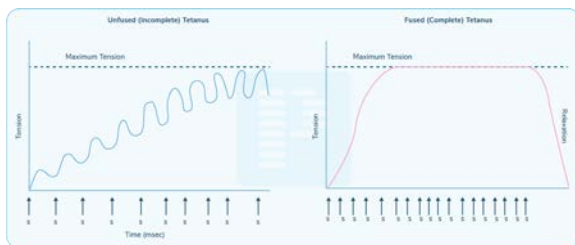
Visible twitching occurs at lower frequencies when each stimulus produces a separate contraction. This is known as unfused tetanus. (Option A)

Solution for Question 15:

Correct Answer: B) Incomplete tetanus

Explanation: Incomplete tetanus is successful stimulation at increasing frequencies, resulting in sustained but oscillating tension with incomplete relaxation between contractions.

Incomplete Tetanus	Complete Tetanus
Muscles are stimulated at an increased frequency, and each contraction begins before the previous one fully relaxes.	Muscles are stimulated at a very high frequency, so each new stimulus occurs before it can relax completely.
Sustained but oscillating tension output where the muscle shows partial relaxation between contractions.	Sustained contraction where the muscle remains in a state of maximal tension without any relaxation.
The tension curve shows successive peaks with incomplete relaxation periods in between, leading to a wave-like pattern.	The tension curve shows a sustained plateau with no dips or relaxation periods, indicating continuous maximum contraction.



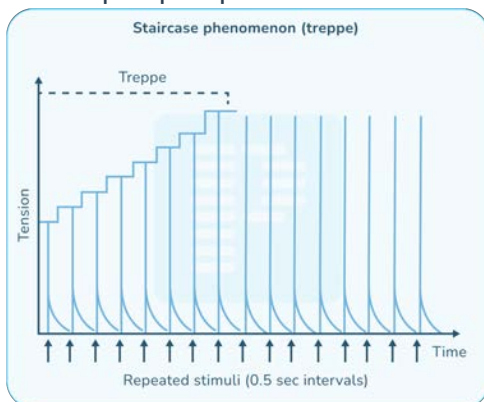
Treppe / Staircase phenomenon refers to an incremental increase in muscle contraction height with repeated stimulations due to residual calcium accumulation. (Option A and C)

Solution for Question 16:

Correct Answer: D) Accumulation of residual calcium ions in the cytoplasm

Explanation: The Treppe phenomenon, or staircase phenomenon, occurs when the muscle contraction height gradually increases with repeated stimulations after reaching baseline.

- This is primarily attributed to the accumulation of residual calcium ions in the cytoplasm that were released during the previous stimulation, as pumping Calcium ions back by the SERCA pump requires some time.



- The height of the second contraction is more than that of the first contraction, known as the Beneficial effect.
- After several contractions, there will be no further increase in height.

Increased availability of ATP in the muscle fibres (Option A):

- ATP is used in the cross-bridge cycling and relaxation processes, not in accumulating calcium ions.

Faster pumping of calcium ions into the sarcoplasmic reticulum (Option B):

- The SERCA pump pumps calcium ions back into the sarcoplasmic reticulum, depleting calcium in the cytoplasm and aiding in Relaxation.

Enhanced sensitivity of the muscle fibres to acetylcholine (Option C):

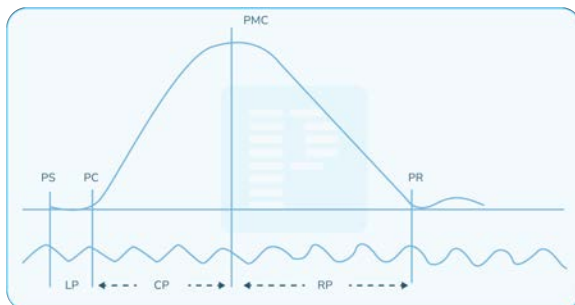
- Acetylcholine initiates muscle contraction by binding to receptors on the muscle cell membrane.
- While receptor sensitivity can affect muscle response, it does not explain the treppe phenomenon's mechanism of increase in the height of contraction..

Solution for Question 17:

Correct Answer: A) Latency Period (10 ms), Contraction Period (40 ms), Relaxation Period (50 ms)

Explanation: Slow muscle fibres exhibit a characteristic contraction pattern with a relatively longer contraction and relaxation duration than fast muscle fibres.

- A Simple Muscle Twitch occurs when a single action potential causes a brief contraction followed by relaxation.
- After nerve stimulation, the muscle fibres go through the following phases:



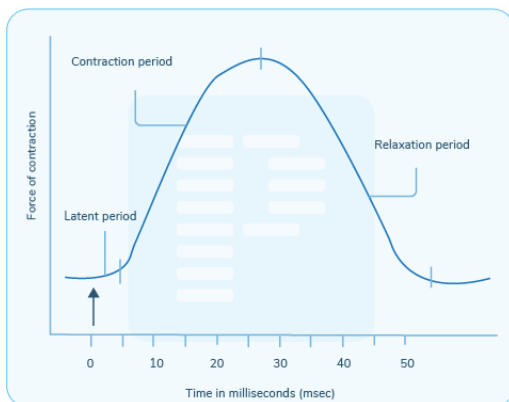
- Latent Period (LP) : Period before the onset of muscle contraction – 10 milliseconds
- Contraction Period (CP): Duration from the onset to the peak of contraction – 40 ms
- Relaxation Period (RP): Period that follows peak contraction period – 50 ms
- These phases collectively make up a total duration of around 100 milliseconds.

PS: Point of stimulus; PC: Point of onset of Contraction; PR: Point of end of Relaxation; PMC : Peak muscle tension

Solution for Question 18:

Correct Answer: D) Latency period

Explanation: The latency period is the brief delay between the application of the stimulus and the onset of muscle contraction, during which the action potential travels along the muscle fibre and calcium ions are released from the sarcoplasmic reticulum.



- Whenever the muscle contracts, it will produce some force known as increased tension generation by the muscle.
- In skeletal muscle, the Simple Muscle Twitch (SMT) is a contraction followed by a relaxation. Latency Period (LP): The interval between the initial stimulation and the beginning of muscle contraction. Contraction Period (CP) (Option B) : The time from the onset of contraction to the peak, during which force generation increases. Relaxation Period (RP) (Option C) : The phase during which the muscle relaxes and force/tension decreases.
- Latency Period (LP): The interval between the initial stimulation and the beginning of muscle contraction.
- Contraction Period (CP) (Option B) : The time from the onset of contraction to the peak, during which force generation increases.
- Relaxation Period (RP) (Option C) : The phase during which the muscle relaxes and force/tension decreases.
- Latency Period (LP): The interval between the initial stimulation and the beginning of muscle contraction.

- Contraction Period (CP) (Option B) : The time from the onset of contraction to the peak, during which force generation increases.
- Relaxation Period (RP) (Option C) : The phase during which the muscle relaxes and force/tension decreases.

Refractory period(Option A): The refractory period is the time following an action potential during which a muscle fibre cannot be re-excited, ensuring that action potentials are separated. This does not represent the initial latency period in a simple muscle twitch.

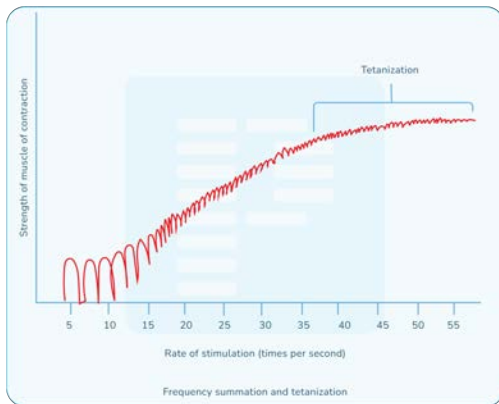
Solution for Question 19:

Correct Answer: B) Overlapping muscle contractions that increase total strength with higher frequency

Explanation: As the frequency of stimulation increases, contractions overlap, summing to increase total force, leading to smooth, continuous contractions known as tetanisation.

Frequency Summation and Tetanization:

- Low-frequency stimulation produces individual twitch contractions. (Option A)
- As frequency increases, contractions overlap, enhancing total strength.
- Contractions fuse into a smooth, continuous contraction called tetanisation at a critical frequency.
- Further frequency increases maintain maximum contraction strength without additional force increase. (Option D)
- Tetany occurs because calcium ions remain in the muscle sarcoplasm between action potentials, sustaining full contraction without relaxation.



Individual twitch contractions at low frequency (Option A) :

It describes the initial phase before frequency summation and tetanisation.

Relaxation between action potentials (Option C)

During tetanisation, there is no relaxation between action potentials, resulting in sustained contraction.

Maximum contraction strength with no further increase despite higher frequency (Option D):

This describes the plateau after tetanisation is achieved, not the process itself.

Solution for Question 20:

Correct Answer: D) Small motor units are excited first

Explanation: During motor unit recruitment, the small motor units are excited first, followed by the larger motor units, which is known as the size principle.

Recruitment:

- The recruitment of motor units follows the size principle, where smaller motor units with fewer muscle fibres and lower force thresholds are activated first.
- As the load increases, progressively larger motor units are recruited to generate the necessary force.
- This allows for smooth and precise control of muscle contractions, starting with fine, low-force movements and progressing to stronger contractions as needed.

Derecruitment:

- When the task is completed, the motor units are relaxed, known as the Derecruitment of a motor unit.
- The largest motor unit is derecruited first, followed by the smallest.

Large motor units are excited first (Option A): Large motor units are recruited later in the process, as they are responsible for generating more force but lack precision.

Small motor units are relaxed first (Option B): Motor units are not relaxed during the process of recruitment. Instead, they are sequentially activated.

Large motor units are relaxed first (Option C): While large motor units might be derecruited first when the force demand decreases, the process of initial recruitment for increasing loads starts with smaller units.

Previous Year Questions

1. Which of the following changes occurs during muscle contraction while exercising?

- A. M length increase
 - B. A length decrease
 - C. I length decrease
 - D. M length increase and I decrease
-

2. Sequence the events in neuromuscular action potential conduction: Sodium channels open in the end plate Calcium enters at the nerve end plate Release of acetylcholine

- A. 1 → 2 → 3
 - B. 1 → 3 → 2
 - C. 3 → 2 → 1
 - D. 2 → 3 → 1
-

3. How does the muscle spindle detect muscle length during muscle contraction?

- A. Stretch receptor on the tendon
 - B. Stretch receptor in the muscle belly
 - C. Golgi tendon organ
 - D. Mechanoreceptor in the joint capsule
-

4. Which of the following components is absent in smooth muscle?

- A. Actin
 - B. Myosin
 - C. Tropomyosin
 - D. Troponin
-

5. Which protein does calcium bind to during smooth muscle contraction?

- A. Calmodulin
 - B. Troponin
 - C. Topoisomerase
 - D. Actin
-

6. A man presents to the hospital with complaints of drooping eyelids (ptosis) and increasing eye irritation. He notes that his symptoms worsen as the day goes on, particularly during outdoor activities. What is the most likely cause of his symptoms?

- A. Lambert-Eaton syndrome
 - B. Myasthenia gravis
 - C. Botulism toxicity
 - D. Tetanus toxicity
-

7. Which of the following is not a calcium-binding protein?

- A. Calbindin
 - B. Calmodulin
 - C. Troponin
 - D. Clathrin
-

8. What is the primary source of energy used by skeletal muscles to replenish ATP during a 100-meter dash?

- A. Phosphocreatine
 - B. Aerobic glycolysis
 - C. Oxidation of Fatty acids
 - D. Oxidation of triglycerides
-

9. Which mechanism is responsible for the detachment of the myosin head from actin?

- A. Binding of ATP
 - B. Entry of calcium into the sarcoplasmic reticulum
 - C. Change in troponin C configuration
 - D. Release of ADP and Pi
-

10. What is the primary function of acetylcholine in the nervous system?

- A. Inducing muscle contraction
- B. Inhibiting neurotransmission
- C. Promoting bone growth
- D. Regulating blood sugar levels

11. Dynein protein is an example of?

- A. Transport protein
 - B. Cell membrane protein
 - C. Cell organelle adherent protein
 - D. Intracellular junctional protein
-

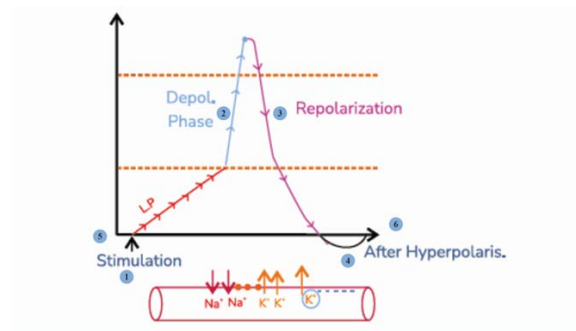
12. What nerve fibers are responsible for the transmission of proprioception?

- A. A alpha
 - B. A beta
 - C. A gamma
 - D. A delta
-

13. Which of the following A fiber subtypes is responsible for motor innervation of the ends of the muscle spindle?

- A. Alpha
 - B. Beta
 - C. Gamma
 - D. Delta
-

14. The diagram displays nerve cell physiology. The event marked by 3 to 4 is due to _____?



- A. Na⁺ entering the cell
- B. Na⁺ exiting the cell.
- C. K⁺ exiting the cell
- D. K⁺ entering the cell

15. What is the term used to describe the consistent amplitude of action potential when the stimulus is delivered at an intensity above the threshold?

- A. All or none law
- B. Local potential
- C. Absolute refractory period
- D. Relative refractory period

16. Voltage gated calcium channel present on T-tubule is?

- A. SERCA pump
- B. Troponin-myosin complex
- C. Dihydropyridine receptor
- D. Ryanodine receptor

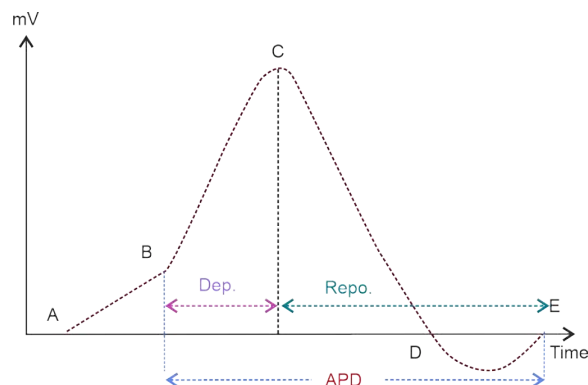
17. Which of the following options has the slowest conduction velocity?

- A. Postganglionic sympathetic fibers
 - B. Somatic motor fibers
 - C. Preganglionic autonomic fibers
 - D. Touch and pressure fibers
-

18. What is the formula used to calculate Stroke Volume (SV) in the context of cardiac physiology?

- A. $SV = \text{Heart Rate (HR)} \times \text{Ejection Fraction (EF)}$
 - B. $SV = \text{End-Diastolic Volume (EDV)} - \text{End-Systolic Volume (ESV)}$
 - C. $SV = \text{Cardiac Output (CO)} / \text{Heart Rate (HR)}$
 - D. $SV = \text{Preload} \times \text{Afterload}$
-

19. Which statement is accurate about the nerve action potential curve provided?



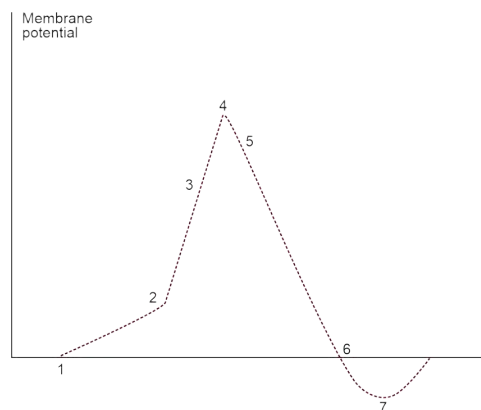
- A. The threshold point is at A
- B. At point E, the nerve is more excitable

- C. Point C to D is due to the opening of Na^+ and closure of K^+ channels
 - D. Point B to D is a refractory period
-

20. A patient came to the hospital after a road traffic accident. He had severe muscle injury, and his serum K^+ level was found to be 6.5 mEq/L . What will happen to the resting membrane potential in this patient?

- A. No change
 - B. Becomes more negative
 - C. Becomes more positive
 - D. First becomes more positive then negative
-

21. The image shows changes in membrane potential during an action potential. The region marked between 4 to 5 is due to _____.

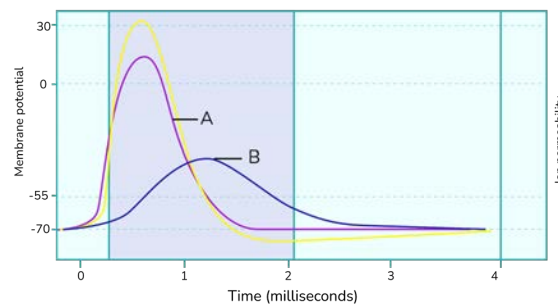


- A. Influx of Na^+
 - B. Efflux of Na^+
 - C. Influx of K^+
 - D. Efflux of K^+
-

22. What causes the detachment of myosin from actin filaments?

- A. Attachment of ATP to the myosin head
 - B. Cessation of calcium release from the sarcoplasm
 - C. Conformational change in Troponin T
 - D. Dissociation of ADP and Pi
-

23. A and B in the graph below represent which of the following ions, in terms of permeability for action potential generation?



- A. A is Na⁺, B is K⁺
 - B. A is Na⁺, B is Ca²⁺
 - C. A is K⁺, B is Ca²⁺
 - D. A is Ca²⁺, B is K⁺
-

24. Arrange the following parts of the sarcomere from the periphery to center. 1. Z line 2. M line 3. A band 4. H zone

- A. 1,3,4,2
- B. 4,2,3,1

C. 2,3,4,1

D. 3,1,4,2

25. During exercise, once a state of balance is achieved, it was observed that the body's oxygen consumption rate was 1500 ml/min. To determine the MET score, consider a baseline oxygen consumption rate of 250 ml/min.

A. 5

B. 6

C. 1/5

D. 1/6

26. What are the chloride equilibrium potentials when the extracellular concentration is 100 mmol/L, and the intracellular concentration is 10 mmol/L?

A. + 10 mV

B. – 10 mV

C. – 60 mV

D. + 60 mV

27. A person after sleeping overnight with the arm under his head now experiences paresis but no numbness in the morning. Which of the following is the best explanation for it?

A. A fibers are more sensitive to pressure than C fibers

B. A fibers are more sensitive to hypoxia than B fibers

C. A fibers are more susceptible to pressure changes than C fibers

D. A fibers are more susceptible to hypoxia than C fibers

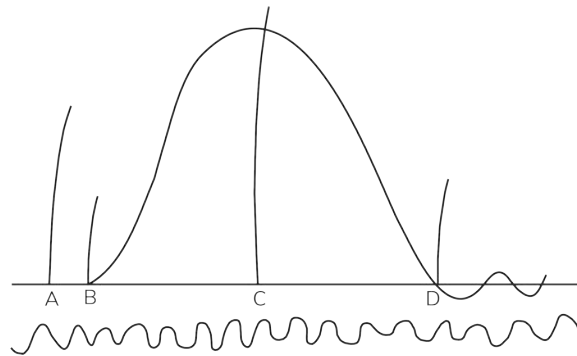
28. The resting membrane potential of nerve fiber is close to the isoelectric potential of:

- A. Potassium ions
 - B. Sodium ions
 - C. Chloride ions
 - D. Magnesium ions
-

29. In skeletal muscle, dihydropyridine, and ryanodine receptors are coupled by?

- A. Chemically
 - B. Electrically
 - C. Chemically and then electrically
 - D. Mechanically
-

30. The image below shows a simple muscle twitch along with the time trace, recorded in a frog with the minimum tetanizing frequency. (The time tracer was made with a 100 Hz tuning fork)

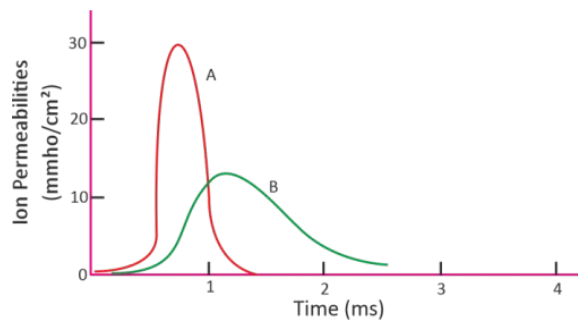


- A. 25 per second
- B. 15 per second
- C. 17 per second
- D. 20 per second

31. Which of the following statements accurately describes the mechanisms involved in depolarization, repolarization, and hyperpolarization during a skeletal muscle action potential? Depolarization due to influx of Na^+ through voltage-gated channels Depolarization due to influx of Na^+ through ligand-gated channels Repolarization due to efflux of K^+ through a voltage-gated channel Repolarization due to efflux of K^+ through a ligand-gated channel Hyperpolarisation due to efflux of K^+

- A. 2 and 4 are correct
- B. 1,3 and 5 are correct
- C. 1,4 and 5 are correct
- D. 2,4 and 5 are correct

32. Identify the ions A and B based on the conductance graph of a neuron shown below:



- A. Na^+ , K^+
 - B. K^+ , Na^+
 - C. Na^+ , Ca^{++}
 - D. Ca^{++} , Na^+
-

33. Which of the following is not a feature of an electrical synapse?

- A. Pre and postsynaptic neurons are connected through gap junctions
 - B. The direction of impulse is unidirectional
 - C. Connexons are involved in electrical synapse
 - D. The direction of impulse is bidirectional
-

34. Integrin connects with:

- A. Cadherin
 - B. Dynein
 - C. Fibronectin
 - D. Selectin
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	4
Question 3	2
Question 4	4
Question 5	1
Question 6	2
Question 7	4
Question 8	1
Question 9	1
Question 10	1
Question 11	1
Question 12	1
Question 13	3
Question 14	3
Question 15	1
Question 16	3
Question 17	1
Question 18	2
Question 19	4
Question 20	3
Question 21	4
Question 22	1

Question 23	1
Question 24	1
Question 25	2
Question 26	3
Question 27	1
Question 28	3
Question 29	4
Question 30	4
Question 31	2
Question 32	1
Question 33	2
Question 34	3

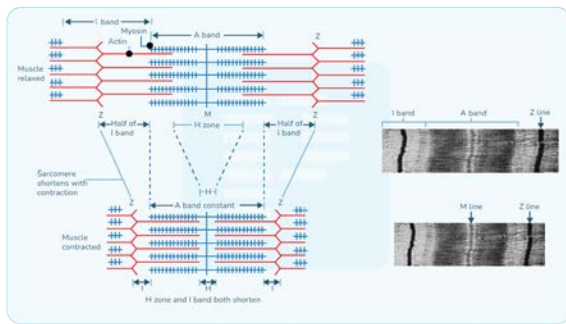
Solution for Question 1:

Answer: C) I length decrease

Explanation:

During muscle contraction, the sarcomere shortens, leading to specific changes in the myofilament structure:

- A-band: The length of the A-band remains the same during muscle contraction because it corresponds to the length of the myosin filaments, which do not change.
- H-band: The H-band decreases or disappears as the actin filaments slide over the myosin filaments during contraction, reducing the region where only myosin is present.
- I-band: The length of the I-band decreases as the sarcomere shortens and the actin filaments overlap more with the myosin filaments.
- M-line: The M-line becomes more prominent as it marks the central point of the sarcomere where the thick filaments are anchored.



M length increase (Option A) is incorrect because the M line does not change in length during contraction.

A length decrease (Option B) is incorrect because the A band remains constant (myosin filament length does not change).

M length increase and I decrease (Option D) is incorrect because while I band decreases, the M line does not change in length.

Solution for Question 2:

Correct Option D - 2 → 3 → 1:

2. Calcium enters at the nerve end plate : Calcium ions enter the nerve terminal when an action potential reaches the nerve end. This influx of calcium is crucial for the subsequent release of neurotransmitters.

3. Release of acetylcholine : The entry of calcium ions into the nerve terminal triggers the release of acetylcholine (ACh) from vesicles into the synaptic cleft.

1. Sodium channels open in the end plate : The binding of acetylcholine to receptors on the muscle end plate causes sodium channels to open, leading to an influx of sodium ions into the muscle cell and generating an action potential in the muscle.

Incorrect Options:

- Option A, B and C are incorrect as given above

Solution for Question 3:

Correct Option B - Stretch receptor in the muscle belly:

- The muscle spindle is a specialized sensory receptor located within the muscle belly. It detects changes in muscle length (stretch) and velocity to assess the joint's position.

Incorrect Options:

- Options A, C, and D do not detect muscle length during muscle contraction.
-

Solution for Question 4:

Correct Option D - Troponin:

- Troponin is absent in smooth muscle. In striated muscles, troponin is a calcium-binding protein that regulates muscle contraction. In smooth muscle, this function is performed by calmodulin, another calcium-binding protein. Calmodulin binds calcium and activates myosin light-chain kinase, which then phosphorylates myosin and facilitates contraction.

Incorrect Options:

Option A and B - Actin & Myosin: Both actin and myosin are present in smooth muscle. Unlike striated muscle, smooth muscle fibers do not have organized sarcomeres. Instead, the actin and myosin filaments are arranged in an oblique pattern. Actin filaments radiate from dense bodies, which function similarly to Z-lines in striated muscle.

Option C - Tropomyosin: Tropomyosin is present in smooth muscle and plays a role in regulating the interaction between actin and myosin.

Solution for Question 5:

Correct Option A - Calmodulin:

- In smooth muscle contraction, calcium binds to calmodulin, which then activates myosin light-chain kinase (MLCK). This activation is crucial for muscle contraction.

Incorrect Options:

Option B - Troponin: Troponin is a calcium-binding protein found in skeletal muscle, not smooth muscle. In skeletal muscle, calcium binds to troponin to initiate contraction.

Option C - Topoisomerase: Topoisomerase is an enzyme that helps manage DNA supercoiling during replication and transcription. It does not interact with calcium during muscle contraction.

Option D - Actin: Actin is a filamentous protein that forms the backbone of the thin filaments in muscle fibers. While actin interacts with myosin for muscle contraction, it does not bind directly with calcium.

Solution for Question 6:

Correct Option B - Myasthenia Gravis:

- Myasthenia gravis is an autoimmune disorder that leads to weakness of the voluntary muscles, particularly affecting the eyes and face. The condition often worsens with activity and improves with rest. Symptoms commonly include ptosis, diplopia (double vision), and generalized muscle weakness.

Incorrect Options:

Option A - Lambert-Eaton Syndrome: Lambert-Eaton syndrome is an autoimmune disorder affecting the neuromuscular junction, characterized by muscle weakness that improves with activity. It typically presents with weakness in the proximal muscles and autonomic symptoms, rather than localized ptosis and worsening symptoms with exertion.

Option C - Botulism Toxicity: Botulism is caused by botulinum toxin, which inhibits acetylcholine release at the neuromuscular junction. It results in muscle weakness, but symptoms typically include descending paralysis and can affect multiple muscle groups. While ptosis may occur, botulism generally presents with more widespread weakness and not specifically with worsening symptoms throughout the day.

Option D - Tetanus Toxicity: Tetanus is caused by the toxin of *Clostridium tetani*, leading to severe muscle spasms and rigidity. Symptoms often include lockjaw, muscle stiffness, and generalized spasms, rather than ptosis or symptoms that worsen with the day.

Solution for Question 7:

Correct Option D - Clathrin:

- Clathrin is not a calcium-binding protein. It is a structural protein that plays a crucial role in the formation of coated vesicles during endocytosis, a process by which cells internalize molecules from the extracellular environment. Clathrin helps shape the vesicles by assembling them into a lattice-like structure around the invaginated membrane.

Incorrect Options:

Option A - Calbindin: Calbindin is a calcium-binding protein that regulates calcium homeostasis in cells. It is found in various tissues, including the brain, kidney, and intestines, and helps buffer and transport calcium ions.

Option B - Calmodulin: Calmodulin is a highly conserved calcium-binding protein found in The smooth muscle cell.

Option C - Troponin: Troponin is a complex of three subunits (troponin C, troponin I, and troponin T) found in skeletal muscle cells and is a calcium binding protein.

Solution for Question 8:

Correct Option A - Phosphocreatine:

- During rapid strenuous activity like 100-meter dash the demand of ATP is covered by anaerobic glycolysis due to reduced oxygen availability.
- Phosphocreatine (anaerobic glycolysis) is a high-energy phosphate compound that is stored in skeletal muscles. During high-intensity activities like sprinting, it can rapidly donate a phosphate group to ADP (adenosine diphosphate), regenerating ATP (adenosine triphosphate). This provides a quick source of energy without the need for oxygen or extensive metabolic pathways.

Incorrect Options:

Option B - Aerobic glycolysis: Aerobic glycolysis refers to the breakdown of glucose in the presence of oxygen to produce ATP. This process occurs in the cytoplasm of cells and involves a series of enzymatic reactions. It is an important energy pathway during prolonged endurance activities when oxygen is readily available.

Option C and D: Fatty acids and tryglycerides are the primary fuel source for aerobic metabolism. During low-intensity and long-duration activities, such as distance running, the body relies on the oxidation of fatty acids to generate ATP.

Solution for Question 9:

Correct Option A - Binding of ATP:

- The detachment of the myosin head from actin occurs when ATP (adenosine triphosphate) binds to the myosin head. This binding causes a conformational change in the myosin molecule, leading to the release of myosin from actin.

Incorrect Options:

Option B - Entry of calcium into sarcoplasmic reticulum: The entry of calcium play a crucial role in initiating muscle contraction by binding to troponin, which leads to the exposure of myosin-binding sites on actin.

Option C - Change in troponin C configuration: This process is responsible for the attachment of the actin-myosin during muscle contraction by uncoiling the troponin I from the tropomysoin.

Option D - Release of ADP and Pi: The release of ADP (adenosine diphosphate) and inorganic phosphate (Pi) occurs after the myosin head has formed a cross-bridge with actin and undergone a power stroke.

Solution for Question 10:

Correct option A - Inducing muscle contraction:

- Acetylcholine is a neurotransmitter that plays a crucial role in the nervous system, particularly in the transmission of signals between nerve cells and in the communication between nerves and muscles. One of its primary functions is to stimulate muscle contraction by transmitting signals across the neuromuscular junction. Upon release from nerve endings, acetylcholine binds to receptors on muscle cells, leading to the generation of an action potential and ultimately causing muscle contraction.
- Therefore, option (a) "Inducing muscle contraction" is the correct choice.

Incorrect Options:

Option B - Inhibiting neurotransmission: Acetylcholine generally acts as an excitatory neurotransmitter in the nervous system, promoting the transmission of nerve impulses rather than inhibiting it.

Option C - Promoting bone growth: While acetylcholine receptors are present in bone cells, there is no evidence to suggest that acetylcholine plays a significant role in promoting bone growth.

Option D - Regulating blood sugar levels: Acetylcholine does not directly regulate blood sugar levels. Insulin and glucagon, produced by the pancreas, are the primary hormones involved in blood sugar regulation.

Solution for Question 11:

Correct Option A - Transport protein:

- Dynein is a motor protein that facilitates retrograde transport within cells. It moves towards the minus end of microtubules, which is typically oriented towards the center of the cell body of the nerve.
- By utilizing the energy derived from ATP hydrolysis, dynein generates mechanical force, allowing it to transport various cellular cargoes, including organelles, vesicles, and protein complexes, along microtubule tracks within the cytoplasm.

Incorrect Options:

- Options B, C, and D are incorrect: Refer to the explanation of Option A.
-

Solution for Question 12:

Correct Option A - A alpha:

- A alpha fibers are responsible for proprioception in the body.

Incorrect Options:

Option B - A beta: A beta fibers are responsible for pressure and touch.

Option C - A gamma: A gamma fibers are efferents to muscle spindles.

Option D - A delta: A delta fibers are responsible for pain and cold temperatures.

Solution for Question 13:

Correct Option C- Gamma:

A Fiber sub-type	Numerical Classification (By Lloyd and Hunt)	Functions
Alpha	Ia Ib	Proprioception and somatic motor
Beta	II	Touch and pressure
Gamma		Efferent to muscle spindle
Delta	III	Pain (fast) and temperature (cold)

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 14:

Correct Option C - K⁺ exiting the cell:

Skeletal muscle action potential:

- 1 to 2: Stimulation of neurons leads to the generation of local potential towards the threshold voltage.
- 2 to 3: It is followed by a rapid increase in the membrane potential due to influx of the Na^+ by opening of the fast type of voltage gated channel marked by depolarization phase.
- 3 to 4: Next phase of the action potential is repolarization characterized by closing of the voltage gated Na^+ channel and opening of the slow type of voltage gated K^+ channel leading to efflux of the K^+ .
- 4: The last phase is hyperpolarization, characterized by the presence of some already open K^+ channels and K^+ leak channels leading to continuous efflux of K^+ .

Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 15:

Correct Option A - All or none law:

- The all-or-none law states that once a nerve or muscle cell is stimulated beyond a certain threshold, it will generate an action potential of consistent magnitude.
- If the stimulus is strong enough, the resulting action potential will have a fixed amplitude and will propagate without loss of intensity. Below the threshold, no action potential occurs; above it, the response is maximal.

Incorrect Options:

Option B - Local potential: Local potential refers to the passive changes in membrane potential that occur during the propagation of electrical signals along a neuron.

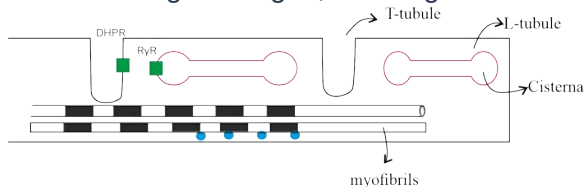
Option C - Absolute refractory period: Absolute refractory period refers to a brief period after an action potential when the neuron is unresponsive to further stimulation. During this period, another action potential cannot be generated, regardless of the strength of the stimulus.

Option D - Relative refractory period: Relative refractory period refers to a period following the absolute refractory period during which a neuron can be stimulated to generate another action potential, but only with a stronger stimulus than the previous stimulus.

Solution for Question 16:

Correct Option C - Dihydropyridine receptor:

- The dihydropyridine receptor (DHPR), also called the L-type calcium channel, is found in the T-tubule membrane near the sarcoplasmic reticulum (SR) in muscle cells.
- When an action potential travels along the T-tubule, DHPR channels open in response to voltage changes, allowing calcium ions to enter the sarcoplasm.



Incorrect Options:

Option A - SERCA pump (Sarcoplasmic/Endoplasmic Reticulum Calcium ATPase):

- The SERCA pump, situated on the sarcoplasmic reticulum (SR) membrane in muscle cells, moves calcium ions (Ca^{2+}) from the cell's cytoplasm into the SR lumen, against their concentration gradient.

Option B - Troponin-myosin complex:

- Troponin, a protein complex on muscle actin filaments, comprises three subunits: troponin C, troponin I, and troponin T.
- This complex regulates muscle contraction and relaxation by modulating actin-myosin filament interaction in response to intracellular calcium changes.

Option D - Ryanodine receptor:

- Ryanodine receptors (RyRs) are intracellular calcium release channels located on the sarcoplasmic reticulum membrane or L tubule of muscle cells. They are activated by calcium-induced calcium release (CICR) and play a key role in releasing calcium ions from the SR into the cytoplasm in response to depolarization of the T-tubule membrane during excitation-contraction coupling.

Solution for Question 17:

Correct Option A - Postganglionic sympathetic fibers:

- Postganglionic sympathetic fibers have the least conduction velocity compared to the other options as these are unmyelinated fibers. These fibers are part of the autonomic nervous system and transmit signals from the autonomic ganglia to target organs, such as smooth muscles, cardiac muscles, and glands.

Incorrect Options:

- Options B, C, and D are incorrect. These fibers have a higher conduction velocity than postganglionic sympathetic fibers of the autonomic nervous system.
-

Solution for Question 18:

Correct Option B - $SV = \text{End-Diastolic Volume (EDV)} - \text{End-Systolic Volume (ESV)}$:

- Stroke Volume (SV) refers to the amount of blood ejected by the left ventricle of the heart in a single contraction.
- The formula to calculate Stroke Volume is based on the difference between the End-Diastolic Volume (EDV) and the End-Systolic Volume (ESV) and is around 70 ml.

Incorrect Options:

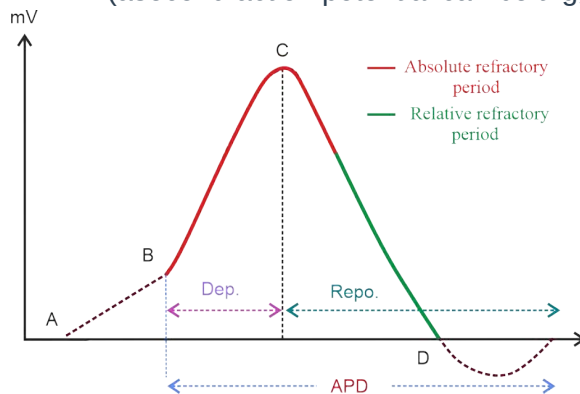
- Options A, C, and D are incorrect. Refer to Option B for an explanation.
-

Solution for Question 19:

Correct Option D - Point B to D is a refractory period:

- The given graph represents an action potential curve of a nerve.
- Point B to D represents the refractory period, a phase during which a second action potential is impossible or difficult to trigger. This period includes the absolute refractory

period (impossible to trigger a second action potential) and the relative refractory period (a second action potential can be triggered but requires a stronger stimulus).



Incorrect Options:

Option A - The threshold point is at A: Threshold point is typically represented by point B, where the depolarization is sufficient to open sodium channels and generate an action potential.

Option B - At point E, the nerve is more excitable: the point E represents the hyperpolarization state of the action potential.

Option C - Point C to D is due to the opening of Na^+ and closure of K^+ channels: Point C to D is the repolarization phase, primarily due to the opening of voltage-gated potassium channels and closure of sodium channels.

Solution for Question 20:

Correct Option C - Becomes more positive:

- In hyperkalemia, where serum potassium levels are higher than normal (as in this patient with muscle injury), the concentration gradient of potassium ions across the cell membrane changes.
- The larger extracellular potassium concentration diminishes the concentration gradient, which causes more potassium ions to stay inside the cell.
- This results in the cell's resting membrane potential becoming more positive (less negative), leading to cell depolarization.

Incorrect Options:

Option A - No change: This would not be the case in hyperkalemia; changes in serum potassium levels directly affect the resting membrane potential.

Option B - Becomes more negative: This could occur in hypokalemia (low serum potassium levels) but not hyperkalemia.

Option D - First becomes more positive than negative: It is an incorrect option.

Solution for Question 21:

Correct Option D - Efflux of K⁺:

- The region marked between 4 and 5 represents the repolarization phase of a nerve action potential.
- It occurs due to closure of fast voltage gated sodium channels and opening of slow voltage gated potassium channels leading to the efflux of K⁺ out of the cell resulting in the restoration of the resting membrane potential.

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 22:

Correct Option A - Attachment of ATP to the myosin head:

- The detachment of myosin from actin filaments occurs when ATP binds to the myosin head.
- After the power stroke (when myosin pulls on actin), ADP and inorganic phosphate (Pi) are still bound to the myosin head. When ATP binds to the myosin head, it causes the myosin head to detach from actin. This detachment allows the muscle to relax and prepares the myosin head for the next contraction cycle.

Incorrect Options:

Option B - Cessation of calcium release from the sarcoplasm: While calcium ions are crucial for initiating muscle contraction by binding to troponin and causing a conformational change that exposes myosin-binding sites on actin. Absence of calcium will result in the cessation of the cross bridge cycle at power stroke.

Option C - Conformational change in Troponin T: Troponin T is involved in regulating the position of tropomyosin on actin, which in turn controls the binding of myosin to actin.

Option D - Dissociation of ADP and Pi: Dissociation of ADP and Pi will result in the power stroke generation.

Solution for Question 23:

Correct Option A - A is Na⁺, B is K⁺:

- In the resting state, the neuronal membrane is more permeable to K⁺ ions than to Na⁺ ions, maintaining a negative resting potential.
- Upon receiving a stimulus, Na⁺ ions enter the cell, causing rapid depolarization seen as an upward slope on the membrane potential vs. time graph.
- Subsequently, voltage-gated K⁺ channels open, leading to K⁺ efflux and membrane repolarization, shown as a downward slope.
- Thus, it can be concluded that A- Na⁺, B- K⁺.

Incorrect Options:

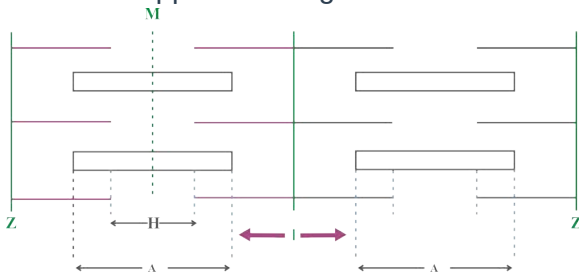
- Option B, C and D are incorrect.

Solution for Question 24:

Correct Option A - 1, 3, 4, 2:

- Z line: The Z line is located at the periphery of the sarcomere. It forms the boundary of the sarcomere and anchors the thin filaments (actin) in place.

- M line: The M line is located in the center of the sarcomere, running vertically through the middle. It provides structural support and helps to stabilize the position of the thick filaments (myosin).
- A band: The A band is a dark-staining region in the center of the sarcomere. It represents the length of the thick filaments (myosin) and includes overlapping thick and thin filaments.
- H zone: The H zone is located within the A band. It is a lighter region in the center of the sarcomere where only thick filaments (myosin) are present. The H zone gets shorter or disappears during muscle contraction.



Incorrect Options:

- Option B, C and D are incorrect.

Solution for Question 25:

Correct Option B - 2:

- MET (Metabolic Equivalent of Task) is a unit used to estimate energy expenditure during physical activity. It is defined as the ratio of the metabolic rate during the activity to the resting metabolic rate.
- To calculate the MET score, we need to determine the ratio of oxygen consumption during exercise to resting oxygen consumption.
- MET score = Oxygen consumption during exercise (ml/min) / Resting oxygen consumption (ml/min)
- In this case, the oxygen consumption during exercise is given as 1500 ml/min, and the resting oxygen consumption is 250 ml/min.
- MET score = 1500 ml/min divided by 250 ml/min
- MET score = 6

Incorrect Options:

- Option A, C & D are incorrect
-

Solution for Question 26:

Correct Option C - 60 mV:

- The equilibrium potential of an ion can be calculated using the Nernst equation: $E = (RT/zF) * \ln ([\text{outside}]/[\text{inside}])$
- $E = (RT/zF) * \ln ([\text{outside}]/[\text{inside}])$
- Where: E = equilibrium potential R = gas constant T = temperature (in Kelvin) z = valency of the ion F = Faraday constant [Outside] = concentration of the ion outside the cell [Inside] = concentration of the ion inside the cell In this case, we have: [Cl⁻] outside = 100 mmol/L [Cl⁻] inside = 10 mmol/L z = -1 (since chloride has a negative charge)
- E = equilibrium potential
- R = gas constant
- T = temperature (in Kelvin)
- z = valency of the ion
- F = Faraday constant
- [Outside] = concentration of the ion outside the cell
- [Inside] = concentration of the ion inside the cell
- In this case, we have:
- [Cl⁻] outside = 100 mmol/L
- [Cl⁻] inside = 10 mmol/L
- z = -1 (since chloride has a negative charge)
- Substituting the values into the Nernst equation: $E = (RT/zF) * \ln ([\text{outside}]/[\text{inside}])$
- $E = (RT/zF) * \ln ([\text{outside}]/[\text{inside}])$
- $E = (RT/zF) * \ln ([\text{outside}]/[\text{inside}])$
- E = equilibrium potential
- R = gas constant
- T = temperature (in Kelvin)
- z = valency of the ion
- F = Faraday constant

- [Outside] = concentration of the ion outside the cell
- [Inside] = concentration of the ion inside the cell
- In this case, we have:
- [Cl⁻] outside = 100 mmol/L
- [Cl⁻] inside = 10 mmol/L
- z = -1 (since chloride has a negative charge)
- $E = (RT/zF) * \ln ([\text{outside}]/[\text{inside}])$

$$= (8.314 * 310 / -1 * 96485) * \ln (100/10) \approx -61.54 \text{ mV}$$

- Therefore, the equilibrium potential of chloride (Cl⁻) is approximately -60 mV.

Incorrect Options:

- Option A, B & D are incorrect.

Solution for Question 27:

Correct Option A - fibers are more sensitive to pressure than C fibers:

- The person experiences paresis (muscle weakness), but no numbness, indicating that motor function is affected, but the sensory function is preserved. In this scenario, pressure on the nerves during sleep likely affects the larger, myelinated motor neurons (A fibers) more than the smaller, unmyelinated sensory neurons (C fibers). A fibers carry motor signals to muscles and are more susceptible to pressure, leading to temporary muscle weakness without sensory deficits.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 28:

Correct Option C - Chloride ions:

- The resting membrane potential of a nerve fiber is equal to the equilibrium potential of chloride ion i.e., -70 mV.

Incorrect Options:

- Options A, B and D are incorrect. Refer to Option C for an explanation.
-

Solution for Question 29:

Correct Option D - Mechanically:

- In skeletal muscle, the DHPR (dihydropyridine receptor) and ryanodine receptors work together through electro-mechanical coupling. The DHPR, located on the cell membrane, acts as a voltage-gated calcium channel, detecting membrane depolarization during an action potential. Meanwhile, the ryanodine receptor, situated on the sarcoplasmic reticulum, releases calcium ions into the muscle cell's cytoplasm during contraction.
- After the membrane depolarization, the DHPR channel will undergo conformation changes to extend one arm towards the RyR channel, which will lead to a mechanical interaction between the two channels.

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.
-

Solution for Question 30:

Correct Option D - 20 per second:

- The image displays a muscle twitch recorded with a 100 Hz tuning fork. Counting 5 muscle twitches in the trace, we calculate the frequency: $5 \text{ twitches} / 100 \text{ Hz} = 0.05 \text{ Hz}$. To express it as twitches per second, considering 20 twitches per cycle (100 Hz tuning fork), we get $0.05 \text{ Hz} * 20 = 1 \text{ twitch per second}$.
- So, the correct option is "20 per second," indicating a frequency of 20 twitches per second.

The image displays a muscle twitch recorded with a 100 Hz tuning fork. Counting 5 muscle twitches in the trace, we calculate the frequency: 5 twitches / 100 Hz = 0.05 Hz. To express it as twitches per second, considering 20 twitches per cycle (100 Hz tuning fork), we get $0.05 \text{ Hz} \times 20 = 1 \text{ twitch per second}$.

So, the correct option is "20 per second," indicating a frequency of 20 twitches per second.

Incorrect Options:

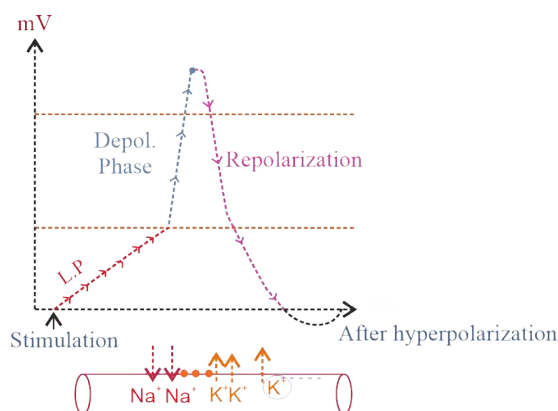
Option A - 25 per second: This option is not supported by the information given in the question. The time trace does not show 25 muscle twitches within the given time interval based on the 100 Hz tuning fork.

Option B - 15 per second: This option is also not supported by the information given in the question. The time trace does not show 15 muscle twitches within the given time interval based on the 100 Hz tuning fork.

Option C - 17 per second: Similarly, this option is not supported by the information given in the question. The time trace does not show 17 muscle twitches within the given time interval based on the 100 Hz tuning fork.

Solution for Question 31:

Correct Option B - 1,3 and 5 are correct:



Skeletal muscle action potential:

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.
-

Solution for Question 32:

Correct Option A - Na^+ , K^+ :

- Sodium and Potassium are two essential ions involved in the generation and conduction of action potentials in neurons. Sodium plays a crucial role in depolarization, while potassium contributes to repolarization and restoring the resting membrane potential.
- A - It is Na^+ ion, characterized by rapid influx of Na^+ through fast type of voltage gated Na^+ channels leading to sudden rise in action potential.
- B - It is K^+ ions, characterized by slow and steady efflux of K^+ ion through a slow type of voltage gated K^+ channel leading to slow decrease in membrane potential.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 33:

Correct Option B - The direction of impulse is unidirectional:

- In an electrical synapse, however, the impulse can flow bidirectionally, allowing for direct electrical communication between the neurons in both directions.

Characteristic	Electrical Synapse	Chemical Synapse

Illustration	Electrical Synapse Bidirectional AP 2 – 4 nm Na⁺ time Synaptic Delay Gap Jn.. cleft Delay → Nil AP	Chemical Synapse Unidirectional 20 – 40 nm Na⁺ Ca⁺⁺/Na⁺ AP cleft No gap Jn (vesicle) Delay → 0.5ms (0.3 to 5 ms)
Occurrence	Rare	Recreate this image Mostly present in the CNS
Components	Presynaptic neuron and postsynaptic neuron	Presynaptic neuron and postsynaptic neuron
Synaptic Cleft Width	Narrow (2-4 nm, average 4 nm)	Wide (20-40 nm)
Pre-synaptic Terminals	No neurotransmitter-containing vesicles	Contains neurotransmitter-containing vesicles
Channels	Gap junctions between presynaptic and postsynaptic neuron	No gap junctions
Synaptic Transmission	Action potential (Na ⁺) from the presynaptic membrane can easily pass through the gap junctions to the postsynaptic membrane without any synaptic delay.	Release of neurotransmitters from the presynaptic neuron; the neurotransmitter gets attached to the receptor on the post-synaptic membrane. This will lead to the release of either Ca ⁺⁺ or Na ⁺ ions, leading to the generation of an action potential in the post-synaptic neuron. There will be an average synaptic delay of 0.5 milliseconds (0.3 to 5 milliseconds) due to multiple steps at the level of synaptic transmission.
Transmission Direction	Bidirectional	Unidirectional (from presynaptic to postsynaptic)
Location	Mainly found in the Inferior Olivary Nucleus (ION), Thalamus, Hypothalamus, Cortex	Distributed throughout the CNS

Incorrect Options:

- Options A, C and D are incorrect.

Solution for Question 34:

Correct Option C - Fibronectin:

- Fibronectin: Fibronectin is a large glycoprotein present in the extracellular matrix. It forms a fibrous network and acts as a scaffold for cells. Fibronectin interacts with integrins, particularly integrin $\alpha 5 \beta 1$ and integrin $\alpha v \beta 3$, among others. This interaction is important for cell adhesion, migration, and signaling. Fibronectin also plays a role in tissue development, wound healing, and cell-matrix interactions.

Incorrect Options:

Option A - Cadherins: Mediate calcium-dependent cell-cell adhesion, forming adherens junctions crucial for tissue organization and cell signaling.

Option B - Vitronectin : Interacts with integrins, particularly $\alpha v \beta 3$ and $\alpha v \beta 5$, facilitating cell adhesion, migration, and signaling in various tissues.

Option D - Selectins: Aid leukocyte recruitment during inflammation by mediating rolling adhesion on endothelial cells, facilitating interaction with inflammatory molecules.

Cardiac Conducting System and Action Potentials

1. Which of the following structures is repolarized last during ventricular repolarization?

- A. Apical epicardial surface
 - B. Posterobasal portion of the left ventricle
 - C. Pulmonary conus
 - D. Endocardial surface
-

2. Which of the following best describes the effect of cyclic adenosine 3',5'-monophosphate (cAMP) on heart rate regulation through nodal tissue?

- A. cAMP decreases heart rate by increasing potassium (K^+) conductance in nodal tissue.
 - B. cAMP increases heart rate by enhancing the opening of L-type calcium channels and "funny channels" in nodal tissue.
 - C. cAMP does not affect the heart rate as it does not influence the activity of "funny channels" or calcium channels.
 - D. cAMP decreases the heart rate by inhibiting the activity of L-type calcium channels and "funny channels."
-

3. Which of the following accurately describes the unique properties of "funny channels" in the heart's nodal tissue?

- A. "Funny channels" open during depolarization and are primarily permeable to calcium (Ca^{2+}) ions.
- B. "Funny channels" close upon hyperpolarization and restrict sodium (Na^+) influx.

- C. "Funny channels" open upon hyperpolarization, allowing both sodium (Na^+) and potassium (K^+) ions to pass through.
 - D. "Funny channels" are activated by parasympathetic stimulation, speeding up the heart rate.
-

4. A 55 y/o male presents with episodes of dizziness and palpitations. He has been prescribed a new medication to help control his heart rate recently. His ECG reveals a resting heart rate of 45 bpm. Which of the following mechanisms is most likely contributing to his slower heart rate?

- A. Increased activity of L-type Ca^{2+} channels
 - B. Decreased "funny current" (I_h)
 - C. Enhanced sympathetic stimulation
 - D. Activation of T-type Ca^{2+} channels
-

5. Which of the following mechanisms is primarily responsible for the relaxation of myocardial cells?

- A. Opening of L-type calcium channels
 - B. Activity of the SERCA2 pump
 - C. Blocking the sodium-calcium exchanger
 - D. Dephosphorylation of phospholamban
-

6. During the process of calcium-induced calcium release (CICR) in cardiac muscle cells, what triggers the release of calcium from the sarcoplasmic reticulum

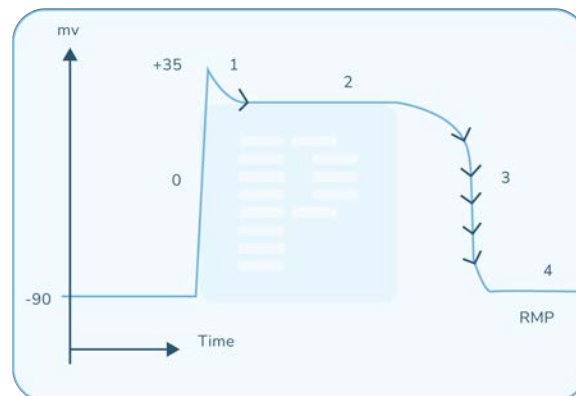
- A. Depolarization directly triggers Ca^{2+} release from SR
- B. Influx of extracellular calcium through voltage-gated channels

- C. The $\text{Na}^+/\text{Ca}^{2+}$ exchanger directly triggers Ca^{2+} release from SR
 - D. Troponin directly triggers Ca^{2+} release from SR
-

7. In which phase in a cardiac muscle cell does the closure of the Ca^{2+} channels and a Slow, delayed increase of K^+ efflux occur?

- A. Late rapid repolarization
 - B. Depolarisation
 - C. Plateau phase
 - D. Initial rapid repolarization
-

8. Given below is the action potential of a cardiac muscle fiber. Which of the following gives rise to Phase 2 depicted in the image?



- A. Closure of Na^+ channels and opening of K^+ channel
 - B. Closure of the Ca^{2+} channels and a slow, delayed increase of K^+ efflux
 - C. Opening of voltage-gated Na^+ channels.
 - D. Opening of slow L-type voltage-gated Ca^{2+} channels and efflux of K^+ through rectifier channels.
-

9. Which of the following correctly matches the resting membrane potential with the corresponding cardiac cell type in the given order: Purkinje cells, AV node, SA node, Ventricular myocytes?

- A. -40, -60, -80, -90
 - B. -60, -60, -60, -90
 - C. -90, -60, -60, -90
 - D. -60, -90, -90, -60
-

10. Which of the following statements correctly differentiates cardiac muscle from skeletal muscle?

- A. Cardiac muscle cells are multinucleated, while skeletal muscle cells have a single, centrally located nucleus.
 - B. Cardiac muscle forms a functional syncytium with intercalated discs; skeletal muscle lacks these structures.
 - C. Cardiac muscle contraction is voluntary and regulated by the somatic nervous system; skeletal muscle contraction is involuntary.
 - D. Cardiac muscle primarily utilizes glycolytic pathways for energy; skeletal muscle relies solely on oxidative metabolism.
-

11. Which of the following does NOT contribute to the high conduction velocity in Purkinje fibers?

- A. They are large diameter fibers
- B. Fast-type action potential
- C. High density of gap junctions
- D. High myofibril content

12. If the S-A node discharges at 0.00 seconds, when will the action potential normally arrive at the epicardial surface at the base of the left ventricle?

- A. 0.22sec
 - B. 0.18sec
 - C. 0.12sec
 - D. 0.09sec
-

13. Arrange the following according to the conduction times (conduction delay) in the heart in ascending order (lowest time to highest time) a. SA node to AV node b. AV node to bundle of His c. Apex to the base of the heart d. Bundle of his to interventricular septum

- A. c, b, a, d
 - B. c, a, d, b
 - C. b, a, d, c
 - D. d, a, b, c
-

14. Which of the following is the name of the middle internodal bundle between SA node and AV node in the atrium of the heart?

- A. Bachmann's bundle
 - B. Wenckebach's bundle
 - C. Thorel's bundle
 - D. Kent bundle
-

15. A 65 y/o patient with a history of complete heart block presents for follow-up. The patient's ECG shows an idioventricular rhythm with a ventricular rate of 30 beats per minute. What is the most likely location of the pacemaker driving the patient's idioventricular rhythm?

- A. SA node
 - B. AV node
 - C. Bundle of His
 - D. Purkinje Fibres
-

16. A 68y/o male presents with episodes of dizziness and fainting. O/E, his heart rate is found to be irregular and bradycardic. ECG reveals complete heart block. Further evaluation shows that his ventricular rhythm is around 30 beats per minute. What is the most likely cause of his symptoms?

- A. Atrial fibrillation
 - B. AV nodal block
 - C. Infranodal block
 - D. Sick sinus syndrome
-

17. A 55 y/o male heart transplant patient comes for a follow-up visit. He reports that he has been engaging in moderate exercise for about 30 minutes daily. Despite lacking sympathetic innervation to his transplanted heart, he notices an increase in his stamina and ability to exercise over time. Given the patient's condition, what physiological mechanism allows him to experience an increase in cardiac output during exercise?

- A. Increased heart rate through parasympathetic withdrawal.
- B. An enhanced force of contraction via the Frank-Starling mechanism.

- C. Parasympathetic stimulation reduces heart rate variability.
 - D. Increased heart rate through sympathetic stimulation to heart.
-

18. Which of the following characteristics primarily enables the SA node to act as the heart's primary pacemaker?

- A. Presence of fast sodium channels
 - B. Highest intrinsic firing rate among cardiac tissues
 - C. Lack of calcium channels
 - D. Inactivation by parasympathetic stimulation
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	3
Question 4	2
Question 5	2
Question 6	2
Question 7	1
Question 8	4
Question 9	3
Question 10	2
Question 11	4
Question 12	1
Question 13	2
Question 14	2
Question 15	4
Question 16	3
Question 17	2
Question 18	2

Solution for Question 1:

Correct Answer:D) Endocardial surface

Explanation:

Repolarization of ventricular muscle starts from the epicardial surface and ends in the endocardial surface.

Depolarization	Repolarization
Atrial depolarization starts at the SA node	Atrial repolarization starts near the SA node
Starts from the middle portion of the septum on the left side	Starts from the apex of the heart,
Ventricular depolarization: Starts at the left side of the interventricular septum Crosses to the right side through the interventricular septum Spreads down to the apex of the heart	Ventricular repolarization occurs in the reverse direction
The endocardial surface depolarizes first and the epicardial last	The epicardial surface repolarizes first and the endocardial last.
Depolarized last: Posterobasal portion of the left ventricle Pulmonary conus The uppermost portion of the septum	Repolarized last: Septum Endocardial surface

Solution for Question 2:

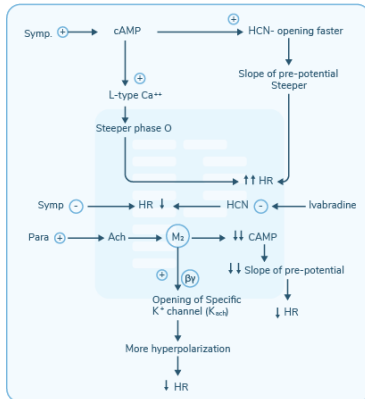
Correct Answer: B) cAMP increases heart rate by enhancing the opening of L-type calcium channels and "funny channels" in nodal tissue.

Explanation:

Elevated cAMP levels increase the opening of "funny channels" and L-type calcium channels, which speeds up depolarization and results in a higher heart rate.

- Increased cAMP levels , often triggered by sympathetic stimulation via β_1 receptors, directly facilitate the opening of L-type calcium channels. This leads to a more rapid influx of calcium ions (Ca^{2+}), speeding up the depolarization phase of the pacemaker potential and, consequently, increasing the heart rate.
- Sympathetic stimulation increases cAMP levels in nodal cells. This increase in cAMP facilitates the opening of "funny channels" and subsequently L-type calcium channels.

- The parasympathetic nervous system releases acetylcholine through the vagus nerve, which acts on the sinoatrial (SA) node. Acetylcholine binds to M2 muscarinic receptors on the SA nodal cells, leading to a reduction in cAMP levels. This decrease in cAMP slows the opening of L-type calcium channels, reducing the influx of calcium ions and slowing the depolarisation rate of the SA node. As a result, the heart rate decreases.



Increased cAMP enhances heart rate by increasing the activity of "funny channels" and L-type calcium channels, rather than increasing potassium conductance (Option A).

cAMP does influence the activity of "funny channels" and L-type calcium channels, thereby affecting heart rate (Option C).

Increased cAMP enhances the activity of these channels, leading to an increased heart rate, not a decrease (Option D).

Solution for Question 3:

Correct Answer: C) "Funny channels" open upon hyperpolarization, allowing both sodium (Na^+) and potassium (K^+) ions to pass through.

Explanation:

"Funny channels," or hyperpolarization-activated cyclic nucleotide-gated (HCN) channels, are crucial for initiating the pacemaker potential in the heart's nodal tissue. They open upon hyperpolarization, allowing sodium and potassium ions to enter the cell, gradually depolarizing the membrane and contributing to the rhythmic electrical activity that controls the heart rate.

This is incorrect because "funny channels" open during hyperpolarization and are not primarily permeable to calcium. They are mainly permeable to sodium and potassium ions (Option A).

This statement is incorrect. "Funny channels" actually open upon hyperpolarization, allowing the influx of sodium and potassium ions (Option B).

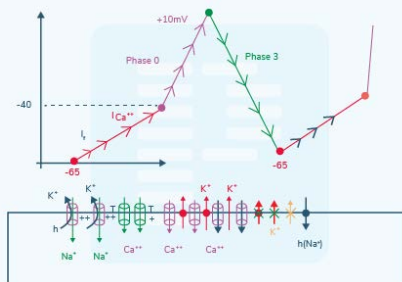
This statement is incorrect. Parasympathetic stimulation slows the heart rate by increasing potassium conductance, which hyperpolarizes the membrane and slows the rate of spontaneous depolarization (Option D).

Solution for Question 4:

Correct Answer: B) Decreased "funny current" (I_h)

Explanation:

Decreased "funny current" (I_h): The "funny current" (I_h) is crucial for the pacemaker potential (Phase 4) in nodal tissue. It gradually depolarizes the membrane potential towards the threshold. A decrease in (I_h) would slow down the rate of spontaneous depolarization, leading to a lower heart rate, which aligns with the patient's bradycardia.



Phases	Closure and Opening channels
Prepotential phase	• Opening of HCN (Na⁺ entry) channel → Slow • Opening of T type Ca²⁺ channel → Rapid • Decrease permeability of K⁺
Phase 0 (Depolarization)	• Opening of L-type Ca²⁺ channel → Slow
Phase 3 (Repolarization)	• Opening of delayed rectifier (K⁺) channel → Slow • Closure of L-type Ca²⁺ channel

The L-type Ca^{2+} channels are responsible for the slow depolarization (Phase 0) in nodal cells. Increased activity of these channels would lead to a faster and stronger depolarizing impulse, which is not consistent with the patient's symptoms of a slow heart rate (Option A).

Sympathetic stimulation increases the rate of spontaneous discharge by enhancing I_h and making the depolarization phase more rapid. This would lead to an increased heart rate, which is the opposite of what is observed in the patient (Option C).

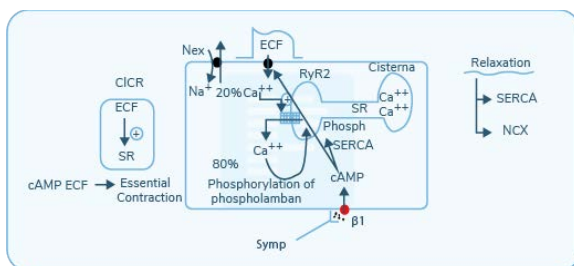
T-type Ca^{2+} channels contribute to the late Phase 4 depolarization by bringing the membrane potential closer to the threshold. Their activation would enhance the depolarization process, potentially increasing the heart rate, not decreasing it (Option D).

Solution for Question 5:

Correct Answer: B) Activity of the SERCA2 pump

Explanation:

The activity of the SERCA2 pump is the primary mechanism responsible for myocardial cell relaxation. SERCA2 actively pumps Ca^{2+} from the cytoplasm back into the sarcoplasmic reticulum, reducing cytoplasmic Ca^{2+} concentration and facilitating muscle relaxation.



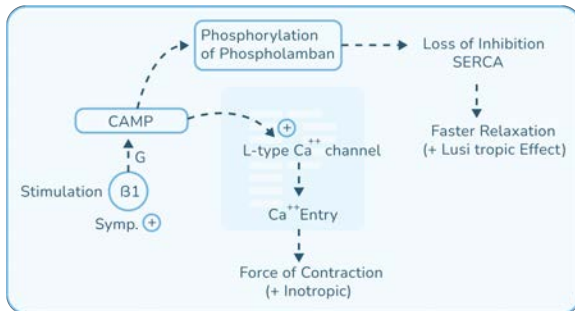
This action would lead to an increase in Ca^{2+} influx into the myocardial cells, promoting muscle contraction rather than relaxation. For relaxation to occur, these channels must close (Option A).

Blocking this exchanger would prevent the removal of Ca^{2+} from the cytoplasm, maintaining high cytoplasmic Ca^{2+} levels and preventing muscle relaxation (Option C).

Phospholamban plays a key regulatory role in myocardial cell relaxation by modulating the activity of the SERCA2 pump.

In its dephosphorylated state, it inhibits SERCA2, reducing calcium reuptake and prolonging contraction.

When phosphorylated, this inhibition is relieved, allowing for increased calcium uptake and faster relaxation of the myocardial cells (Option D).

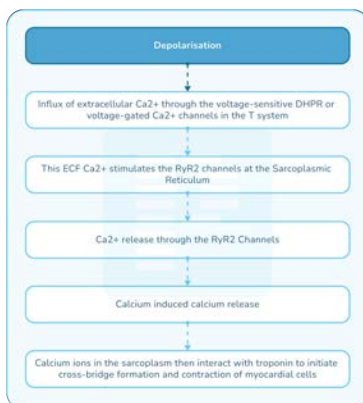


Solution for Question 6:

Correct Answer: B) Influx of extracellular calcium through voltage-gated channels

Explanation:

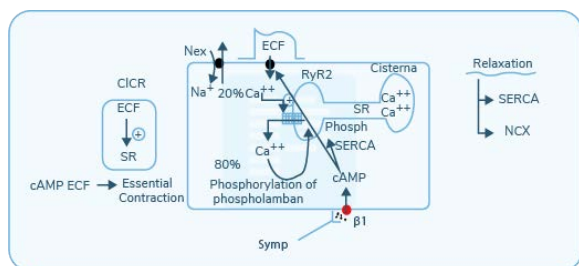
The influx of extracellular calcium through the voltage-sensitive dihydropyridine receptor (DHPR) or voltage-gated Ca²⁺ channels in the T system stimulates the ryanodine receptor (RyR2) channels at the sarcoplasmic reticulum, leading to calcium release.



Depolarization initiates the process by opening voltage-gated Ca²⁺ channels, but it is not the direct trigger for Ca²⁺ release from SR (Option A).

The Na⁺/Ca²⁺ exchanger helps in the recovery of cytosolic calcium concentrations but does not directly trigger the Ca²⁺ release from SR (Option C).

Troponin interacts with calcium in the sarcoplasm to initiate muscle contraction, but it does not directly trigger the Ca^{2+} release from SR (Option D).

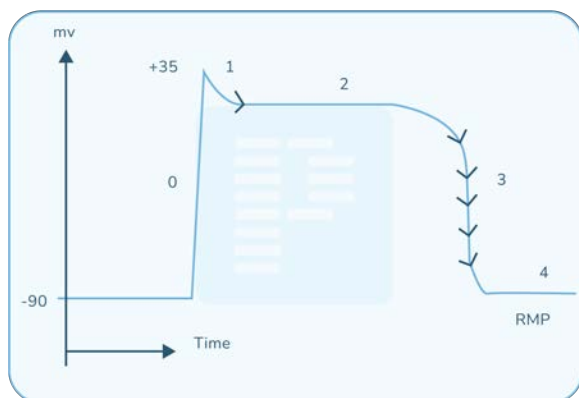


Solution for Question 7:

Correct Answer: A) Late rapid repolarization

Explanation:

Late rapid repolarization is characterized by the closure of the Ca^{2+} channels and a Slow, delayed increase of K^+ efflux which is Phase 3 of the action potential of cardiac myocytes.



PHASES	OPENING AND CLOSURE OF CHANNELS
Phase 0- Depolarization (Option B)	Opening of voltage-gated Na^+ channels
Phase 1- Initial rapid repolarization (Option D)	Closure of Na^+ channels Opening of K^+ channel
Phase 2- Plateau phase (Option C)	Slower but prolonged opening of L-type voltage-gated Ca^{2+} channels Fast Potassium channels remain open

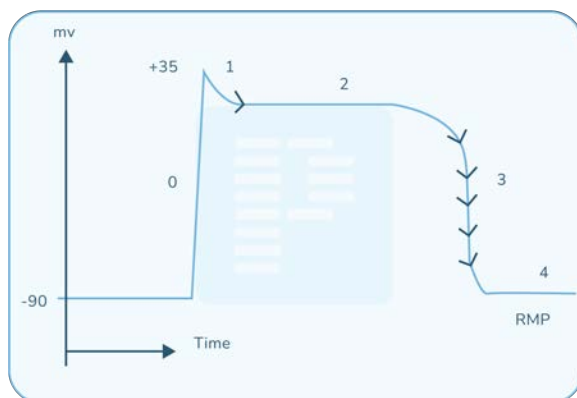
Phase 3- Late rapid repolarization	Closure of the Ca ²⁺ channels Slow, delayed increase of K ⁺ efflux
Phase 4- Baseline	Resting Membrane Potential About –80 to –90 millivolts.

Solution for Question 8:

Correct Answer: D) Opening of slow L-type voltage-gated Ca²⁺ channels and efflux of K⁺ through rectifier channels

Explanation:

The given prolonged plateau (phase 2) is due to a slower but prolonged opening of voltage-gated L-type Ca²⁺ channels and efflux of Potassium through rectifier channels .



PHASES	OPENING AND CLOSURE OF CHANNELS
Phase 0- Depolarization	Opening of voltage-gated Na ⁺ channels (Option C)
Phase 1- Initial rapid repolarization	Closure of Na ⁺ channels Opening of K ⁺ channel (Option A)
Phase 2- Plateau phase	Slower but prolonged opening of L-type voltage-gated Ca ²⁺ channels K ⁺ channel also remains open.
Phase 3- Late rapid repolarization	Closure of the Ca ²⁺ channels Slow, delayed increase of K ⁺ efflux (Option B)
Phase 4- Baseline	Resting Membrane Potential About –80 to –90 millivolts.

Solution for Question 9:

Correct Answer: C) -90, -60, -60, -90

Explanation:

Cardiac Myocytes Resting Membrane Potential Purkinje fibers and atrial and ventricular myocytes –90 mV Sinoatrial (SA) and atrioventricular (AV) nodes –60 mV

Cardiac Myocytes	Resting Membrane Potential
Purkinje fibers and atrial and ventricular myocytes	–90 mV
Sinoatrial (SA) and atrioventricular (AV) nodes	–60 mV

Solution for Question 10:

Correct answer: B) Cardiac muscle forms a functional syncytium with intercalated discs; skeletal muscle lacks these structures.

Explanation :

Aspect	Cardiac Muscle	Skeletal Muscle
Structure and Organization	Branched muscle fibers. Interconnected by intercalated discs. Forms a functional syncytium.	Elongated, cylindrical fibers. Arranged in parallel bundles. They lack branching and intercalated discs.
Cellular Structure (Option A)	Single, centrally located nucleus.	Multinucleated. Nuclei located peripherally.
Intercellular Communication	Intercalated discs with gap junctions. Enables rapid passage of ions and electrical signals.	Do not possess gap junctions.
Regulation of Contraction (Option C)	Involuntary. Regulated by the intrinsic conduction system and autonomic nervous system.	Voluntary. Controlled by the somatic nervous system.

Contraction Duration	Longer duration.	Generally, it has a shorter duration, varying with muscle fiber type.
Energy Utilization (Option D)	Primarily oxidative metabolism of fatty acids	Utilizes both oxidative and glycolytic pathways. Depends on muscle fiber type and exercise intensity.
Transverse Tubule System	Larger T tubules. Located at the Z line. Forms a diad with a single cistern.	Smaller T tubules. Located at the A-I junction. Forms a triad with two cisternae.
Calcium Handling	Relies on extracellular calcium influx through T tubules. Calcium release from the sarcoplasmic reticulum	Predominantly depends on calcium release from the sarcoplasmic reticulum.

Solution for Question 11:

Correct Answer:D) High myofibril content

Explanation:

Purkinje fibers have very few myofibrils, meaning that they contract little or not at all during impulse transmission and hence have high conduction velocity.

Conduction velocity in cardiac tissue

Cardiac tissue	Conduction velocity (m/s)
SA node	0.05
Internodal pathways	1
AV node	0.02-0.05
His bundle	1
Purkinje fibers	4
Ventricular muscles	1

Purkinje fibers- fastest impulse conduction from apex to base of the heart

AV node- slowest conduction

Reasons for high conduction in Purkinje fibers

- High level of permeability of the gap junctions at the intercalated discs between the successive cells that make up the Purkinje fibers (Option C)
- They are large cells and can transmit potentials at a high velocity (Option A)
- Very few myofibrils-minimal contractions during impulse transmission
- Fast-type action potential (Option B)

Solution for Question 12:

Correct Answer: A) 0.22sec

Explanation:

Action potential arriving at the epicardial surface at the base of the ventricle is the same as the total time taken for depolarization = 0.22 sec

The time interval for impulse conduction

SA node to AV node	0.03 seconds
AV node to bundle of His	0.10 seconds
Bundle of his to interventricular septum	0.04 seconds
Interventricular septum to the apex of the heart	0.03 seconds
Apex to the base of the heart	0.02 seconds
Total	0.22 seconds

After the S-A node discharges, the action potential travels through the atria, the A-V bundle system, and finally to the ventricular septum and the ventricle.

The last place that the impulse arrives is at the epicardial surface at the base of the left ventricle, which requires a transit time of 0.22 sec.

Solution for Question 13:

Correct Answer: B) c, a, d, b

Explanation:

The time taken for the conduction of impulses in the heart in ascending order is as follows:

Apex to the base of the heart (0.02sec) < SA node to AV node (0.03sec) < Bundle of his to interventricular septum (0.04sec) < Apex to the bundle of His (0.1sec).

The time interval for impulse conduction

Apex to the base of the heart	0.02 seconds
SA node to AV node	0.03 seconds
Interventricular septum to the apex of the heart	0.03 seconds
Bundle of his to interventricular septum	0.04 seconds
AV node to bundle of His	0.10 seconds

AV nodal Delay:

- Impulse travels through the AV node with a length of 5 mm and is very slow, known as AV nodal "Delay".
- The time required for impulse conduction from the SA node to the AV node is 0.03 seconds.
- The time taken for impulse conduction from the upper to the lower part of the AV node is 100 milliseconds
- Due to av nodal delay, Atrial depolarization is completed much before the beginning of ventricular depolarization. Therefore, when atrial systole occurs, the ventricle is in diastole. This helps the ventricular filling to occur.

His Bundle Delay:

- The time taken for impulse conduction from the upper to the lower part of His bundle is 40 milliseconds known as His bundle delay.
- His bundle is in the uppermost portion of the interventricular septum.
- When an impulse crosses the bundle of His, it causes depolarization of the septal muscle. The middle part of the interventricular septum is first depolarized.
- When the SA node produces impulse at zero time point, it takes approximately 170 ms to start ventricular depolarization.

An impulse travels through the septum depolarizing it from left to right and, reaching the heart's apex in 30 milliseconds. The total time taken to reach the apex of the heart is 200 milliseconds.

The impulse depolarizes both ventricular muscles through Purkinje fibers. Depolarization occurs from the endocardium toward the epicardium and takes around 20 milliseconds

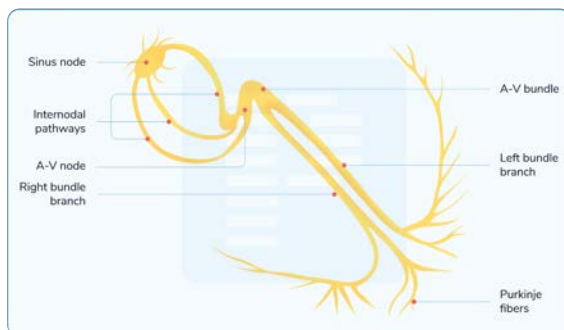
The total time taken for complete depolarization of the heart is 220 ms.

Solution for Question 14:

Correct Answer: B) Wenckebach's bundle

Explanation:

The middle internodal bundle of the heart is called the Wenckebach's bundle or the tract of Wenckebach.



Internodal Pathways:

- 3 bundles of atrial fibers that connect the SA node to the AV node
- Conduction is more rapid in these bundles
- These merge into the AV node

Anterior	Tract of Bachman (Option A)
Middle	Tract of Wenckebach
Posterior	Tract of Thorel (Option C)

Bundle of his:

- A small bundle of fibers arises from the AV node and terminates in the Purkinje system.
- Situated below the AV node and passes towards the interventricular septum
- Gives off 2 branches

Right bundle branch	Left bundle branch
---------------------	--------------------

Conducts impulse to the right ventricle	Conducts impulse to the left ventricle
Exclusively innervates right ventricle	Divides into Anterior fascicle: anterior portion of the left ventricle Posterior fascicle: posterolateral part of left ventricle

Purkinje fibers

- Network of small bundles of conducting fibers that are present throughout the sub-endocardial regions of the right and left ventricles
- Purkinje cells are the largest cells in the heart
- Gap junctions are present between the cells
- The rate of impulse conduction is highest in the Purkinje cells d/t larger diameter and low impedance cell-to-cell connection.
- Fast response type action potentials are generated

Kent's bundle: is an accessory conducting pathway between the atria and the ventricle of the heart. (Option D) .The bundle of Kent allows electrical activity to bypass the AV node, which can lead to an earlier activation of the ventricles. This can result in tachydysrhythmias

Solution for Question 15:

Correct Answer:D) Purkinje Fibres

Explanation:

In a complete heart block with a very slow ventricular rate, the pacemaker is often located in the Purkinje fibers, resulting in rates around 30 beats per minute or slower.

While the Bundle of His can be involved in conduction, Purkinje fibers are more likely at this low rate (Option C).

- Role in Heart Block : Conduction Disruptions can occur between the SA and AV nodes or within the AV node, leading to heart block. Complete Heart Block : SA node signal doesn't reach ventricles; ventricles beat independently at a slower rate driven by a ventricular pacemaker. Ventricular Pacemaker Location : Determines idioventricular rhythm rate.
- Conduction Disruptions can occur between the SA and AV nodes or within the AV node, leading to heart block.

- Complete Heart Block : SA node signal doesn't reach ventricles; ventricles beat independently at a slower rate driven by a ventricular pacemaker.
- Ventricular Pacemaker Location : Determines idioventricular rhythm rate.
- Conduction Disruptions can occur between the SA and AV nodes or within the AV node, leading to heart block.
- Complete Heart Block : SA node signal doesn't reach ventricles; ventricles beat independently at a slower rate driven by a ventricular pacemaker.
- Ventricular Pacemaker Location : Determines idioventricular rhythm rate.

Key Differences and Functions

Aspect	SA Node	AV Node
Intrinsic Rhythmicity	60-80 beats per minute(can go upto 100) (natural pacemaker of the heart) (Option A)	40-60 beats per minute (Option B)
Location	The upper wall of the right atrium	The lower wall of the right atrium
Function	The primary pacemaker; initiates the heart's rhythmic contractions	Delays impulses from atria; ensures coordinated contractions
Action Potential	Calcium-driven, slower conduction	
Prepotentials	Present - Slow, gradual depolarizations observed	
Parasympathetic Modulation	Acetylcholine decreases the firing rate	
Ach decreases the firing rate by hyperpolarizing nodal cells, increasing potassium conductance, slowing the "funny" current (I _h), and reducing cAMP levels.		
Sympathetic Modulation	Norepinephrine increases the firing rate	
Norepinephrine: Increases firing rate by binding to β_1 receptors, raising cAMP levels, and facilitating L-type calcium channel opening, accelerating depolarization.		
Role in Heart Block	Fastest intrinsic rate, so normally unaffected	It can be a site of conduction disruption leading to heart block

Solution for Question 16:

Correct Answer: C) Infranodal block

Explanation:

Infranodal block leads to a slower ventricular rate, often around 35 beats per minute or lower, consistent with the patient's rate of 30 beats per minute and symptoms of dizziness and fainting due to Stokes-Adams syndrome.

Atrial fibrillation causes an irregularly irregular heart rhythm but does not typically cause complete heart block or an idioventricular rhythm (Option A).

Sick sinus syndrome involves dysfunction of the sinoatrial node, leading to bradycardia or pauses, but does not account for the complete heart block observed in this patient (Option D).

Complete Heart Block and Idioventricular Rhythm

- Complete Heart Block : Electrical impulses from the atria fail to reach the ventricles due to a complete interruption in the conduction system. The interruption can occur at the AV node (AV nodal block) or below the AV node (infranodal block).
- Electrical impulses from the atria fail to reach the ventricles due to a complete interruption in the conduction system.
- The interruption can occur at the AV node (AV nodal block) or below the AV node (infranodal block).
- Electrical impulses from the atria fail to reach the ventricles due to a complete interruption in the conduction system.
- The interruption can occur at the AV node (AV nodal block) or below the AV node (infranodal block).
- Independent Ventricular Rhythm : Ventricles develop an independent rhythm driven by a ventricular pacemaker due to the absence of pacing signals from the atria.
- Ventricles develop an independent rhythm driven by a ventricular pacemaker due to the absence of pacing signals from the atria.
- Location and Rate of Idioventricular Rhythm : AV Nodal Block : Remaining nodal tissue becomes pacemaker, resulting in a rate of ~45 beats per minute. (Option B) Infranodal Block : Pacemaker located more peripheral, resulting in a slower rate, averaging ~35 beats per minute, but can be as low as 15 beats per minute.

- AV Nodal Block : Remaining nodal tissue becomes pacemaker, resulting in a rate of ~45 beats per minute. (Option B)
- Infranodal Block : Pacemaker located more peripheral, resulting in a slower rate, averaging ~35 beats per minute, but can be as low as 15 beats per minute.
- Ventricles develop an independent rhythm driven by a ventricular pacemaker due to the absence of pacing signals from the atria.
- AV Nodal Block : Remaining nodal tissue becomes pacemaker, resulting in a rate of ~45 beats per minute. (Option B)
- Infranodal Block : Pacemaker located more peripheral, resulting in a slower rate, averaging ~35 beats per minute, but can be as low as 15 beats per minute.
- Occurs in individuals with complete heart block, especially infranodal block.
- Characterized by episodes of dizziness and fainting (syncope).
- Caused by periods of asystole, where ventricles fail to contract, leading to temporary cessation of blood circulation and decreased blood flow to the brain.

Solution for Question 17:

Correct Answer: B) Enhanced force of contraction via the Frank-Starling mechanism

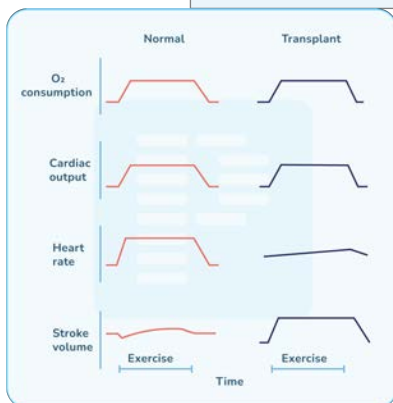
Explanation:

The Frank-Starling mechanism allows the heart to increase stroke volume by enhancing the force of contraction in response to increased venous return during exercise, thereby increasing cardiac output.

While parasympathetic withdrawal can lead to a slight increase in heart rate, it is not the primary mechanism by which cardiac output is increased in a heart transplant patient during exercise (Option A).

Aspect	Details
Frank-Starling Mechanism	More heart muscle stretch during filling leads to stronger contractions and higher stroke volume.
Sympathetic Innervation	Lost in heart transplant patients; cannot increase heart rate through the sympathetic nervous system. (Option D)
Parasympathetic Innervation	Lost in heart transplant patients; cannot decrease heart rate through the parasympathetic nervous system. (Option C)

Stroke Volume Increase	Achieved by increased force of contraction in response to increased venous return during exercise.
Cardiac Output	Increased primarily due to higher stroke volume, not heart rate.
Comparison to Healthy Individuals	Less increase in cardiac output, relying on stroke volume rather than both stroke volume and heart rate.



So, in transplant patients, the increase in cardiac output during exercise is mainly due to an increase in stroke volume, not heart rate.

Solution for Question 18:

Correct answer: B) Highest intrinsic firing rate among cardiac tissues.

Explanation:

The SA node's high intrinsic firing rate allows it to depolarize other cardiac tissues before they reach their own firing threshold, establishing its role as the primary pacemaker.

Key Features of the SA Node and Its Role as the Heart's Primary Pacemaker:

Automaticity and Unique Properties of SA Node Cells	
Intrinsic Leakiness	Less negative resting membrane potential due to sodium and calcium ion leakiness.
Funny Current (I _h)	Inward sodium current drives slow depolarization toward the action potential threshold.

Calcium Current (I _{Ca})	Voltage-gated calcium channels open at ~ -40 mV, causing rapid depolarization. (Option C)
Fast Sodium Channels	Mostly inactivated, leading to a slower depolarization rate compared to ventricular cells. (Option A)
Influence of the Autonomic Nervous System	
Sympathetic Stimulation	Norepinephrine increases pacemaker potential slope, accelerating depolarization and heart rate.
Parasympathetic Stimulation	Acetylcholine decreases pacemaker potential slope, slowing depolarization and heart rate (Option D)
Dominance	
A higher intrinsic firing rate depolarizes other cardiac tissues(AV nodes & Purkinje fibers) before they reach their own threshold.	

1. Which positive wave in a normal lead II ECG appears after the wave of atrial depolarization?

- A. R wave
 - B. Q wave
 - C. P wave
 - D. T wave
-

2. Which segment on the ECG corresponds to the period during which the ventricles remain in a state of complete depolarization?

- A. PR interval
 - B. ST segment
 - C. QRS complex
 - D. QT interval
-

3. Which of the following statements is false about the Class I antiarrhythmics on the cardiac action potential?

- A. Prolong the repolarization phase by blocking potassium channels
 - B. Class Ib decrease the QT interval
 - C. Class Ic are the most potent
 - D. Reduce the rate of depolarization during phase 0
-

4. Which among the following options primarily contributes to QT prolongation on the ECG?

- A. Opening of inward rectifying K⁺ channels in phase 4
 - B. Prolonged depolarization during Phase 1
 - C. Delayed repolarization during Phase 2 or 3
 - D. Increased sodium influx during Phase 0
-

5. If the voltage in Lead II of the Einthoven triangle is 1.5 mV and the voltage in Lead III is 0.5 mV, what is the voltage in Lead I?

- A. 2.0 mV
 - B. 1.0 mV
 - C. 2.5 mV
 - D. 1.5 mV
-

6. Which of the following statements is true regarding ECG leads?

- A. Chest leads are bipolar limb leads- horizontal plane.
 - B. Positive end of the voltmeter is connected to the Wilson terminal.
 - C. In lead II the negative end is the right arm
 - D. Augmented limb lead is bipolar
-

7. According to hexaxial reference system of ECG, the direction of voltage vector of lead aVR is:

- A. $+150^\circ$
 - B. -150°
 - C. $+30^\circ$
 - D. -30°
-

8. In a patient presenting with ST elevation in leads V1 and V2 on an ECG, which of the following is the most likely diagnosis and the corresponding coronary artery involved?

- A. Anterior wall myocardial infarction due to Right Coronary Artery (RCA) occlusion.
 - B. Inferior wall myocardial infarction due to Left Anterior Descending (LAD) artery occlusion.
 - C. Septal wall myocardial infarction due to Left Anterior Descending (LAD) artery occlusion.
 - D. Lateral wall myocardial infarction due to Left Circumflex (LCX) artery occlusion.
-

9. If a T wave spans 4 small boxes and 3 vertical boxes on an ECG recording paper moving at a standard speed of 25 mm/second, with standard voltage calibration, what is the duration and amplitude of the T wave?

- A. Duration: 0.16 seconds, Amplitude: 0.3 mV
 - B. Duration: 0.20 seconds, Amplitude: 0.4 mV
 - C. Duration: 0.16 seconds, Amplitude: 0.4 mV
 - D. Duration: 0.20 seconds, Amplitude: 0.3 mV
-

10. Which of the following can be considered as Left axis deviation?

- A. $+30^\circ$

- B. -60°
 - C. $+90^{\circ}$
 - D. -10°
-

11. In the His bundle electrogram (HBE), what does the H wave denote?

- A. Atrial depolarization
 - B. Ventricular depolarization
 - C. Depolarization of the His bundle
 - D. Repolarization of the ventricles
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	1
Question 4	3
Question 5	2
Question 6	3
Question 7	2
Question 8	3
Question 9	1
Question 10	2
Question 11	3

Solution for Question 1:

Correct Answer: A) R wave

Explanation:

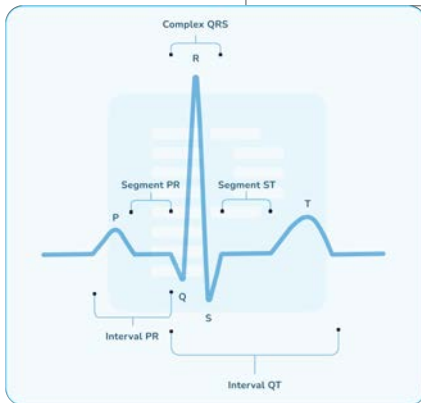
R wave is the first positive wave that appears after P wave (atrial depolarization)

ECG:

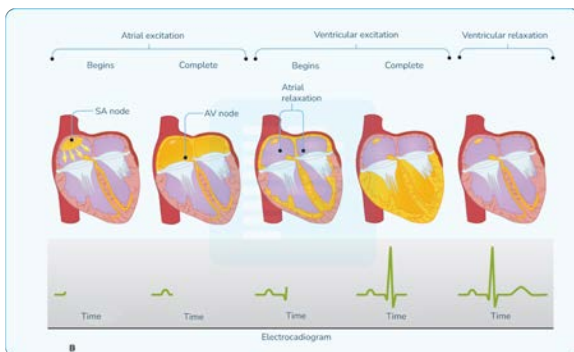
Trace of electrical conduction of the heart.

Wave	Key characteristics
P wave	First wave to appear due to atrial depolarization (option C)
Q wave	First negative wave after the P wave (option B)

R wave	First positive wave after the P wave (option A)
S wave	Immediately after R wave
T wave	After the QRS complex (option D)
U wave	Ventricular myocytes with long action potentials



Yellow color denotes the areas that are depolarized.



Interval	Duration in seconds	Events of the heart
PR	0.12 to 0.20	Atrioventricular conduction
QRS duration	0.08	Ventricular depolarization
QT	0.40	Ventricular action potential
ST	0.32	Plateau portion of the ventricular action potential

Solution for Question 2:

Correct Answer: B) ST interval

Explanation:

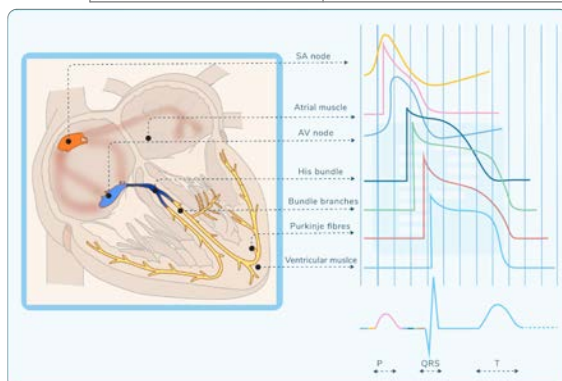
ST interval corresponds to the plateau portion of the ventricular action potential, where the action potential remains in a state of sustained depolarization.

Action potential (two types in heart tissues):

- Fast response action potential - In atrial and ventricular muscles and Purkinje fibers
- Slow response action potential - In nodal tissues (SA and AV nodes)

Fast response action potential in ventricular muscles:

The phases of action potential	Corresponding ECG segment	Ionic activity	Key features in the phase
Phase 0	The initial part of the QRS complex (Q wave)	Opening of voltage-gated sodium channels The rapid myocardial influx of sodium ions.	Rapid depolarization and overshoot.
Phase 1	The rest of the QRS complex	Closure of sodium channels Cessation of sodium influx	Initial rapid repolarization
Phase 2	ST segment (option B)	Slow and prolonged opening of voltage-gated calcium channel Slow calcium influx Efflux of potassium	Plateau phase, sustained depolarization
Phase 3	T wave	Cessation of calcium influx. Increased potassium efflux.	Final repolarization.
Phase 4	No direct representation on the ECG	Opening of inward rectifying K ⁺ channels	Restoration of membrane potential



Solution for Question 3:

Correct Answer: A) Prolong the repolarization phase by blocking potassium channels

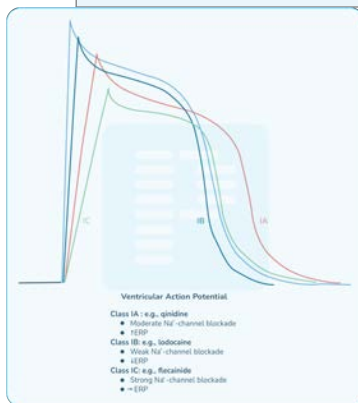
Explanation:

Class III antiarrhythmics are potassium channel blockers.

Class 1 antiarrhythmics:

Mechanism of action: Voltage-gated sodium channel blockers reduce the rate of depolarization during phase 0 . (option D)

	Class Ia	Class Ib	Class Ic
Degree of block	Moderate	Mild	Marked (option C)
QT interval	Prolong	Decrease (option B)	No effect



Class III antiarrhythmics are potassium channel blockers, which decrease potassium efflux out of the cardiac cells and prolong the QTc interval.

Solution for Question 4:

Correct Answer: C) Delayed repolarization during Phase 2 or 3

Explanation:

QT prolongation on ECG is primarily caused by delayed repolarization during Phase 2 and 3 of the cardiac action potential.

The	Corresponding ECG segment	Ionic activity	Key features in the phase
Phase 0	The initial part of the QRS complex (Q wave)	Opening of voltage-gated sodium channels A rapid myocardial influx of sodium ions. (option D)	Rapid depolarization and overshoot.
Phase 1	The rest of the QRS complex	Closure of sodium channels Cessation of sodium influx	Initial rapid repolarization
Phase 2	ST segment (option C)	Slow and prolonged opening of voltage-gated calcium channel Slow calcium influx Efflux of potassium	Plateau phase, sustained depolarization
Phase 3	T wave (option C)	Cessation of calcium influx. Increased potassium efflux.	Final repolarization.
Phase 4	No direct representation on the ECG	Opening of inward rectifying K ⁺ channels (option A)	Restoration of membrane potential

Class III antiarrhythmic drugs are potassium channel blockers that decrease potassium efflux out of the cell and prolong the QT interval.

Solution for Question 5:

Correct Answer: B) 1.0 mV

Explanation:

Voltages of lead I + lead III = lead II (Einthoven law)

Hence, from the question,

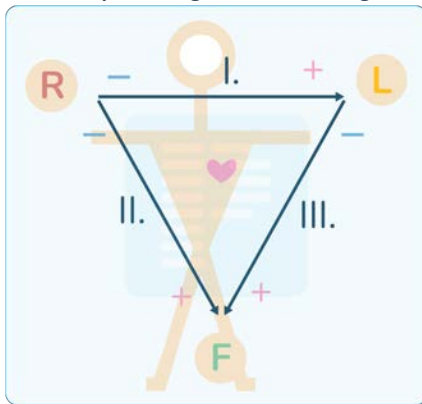
Lead I + 0.5 mV = 1.5 mV

Lead I = 1.5 mV - 0.5 mV

Lead II = 1.0 mV

Einthoven triangle:

- It is an equilateral triangle and each side represents the axis of one of the bipolar limb leads.
- It is shaped like an inverted triangle, with the base between two shoulders and the apex pointing toward the groin.



- The heart is considered to be at the center of the Einthoven triangle.
- Each corner of the triangle is the location for an electrode for two leads to record ECG. Consequently, the three limbs of the triangle represent three leads.
- Einthoven law:
 - Voltages of lead I + lead III = lead II
- Voltage calculation:
 - Voltage recorded in each lead as follows
 - Lead I: left arm - right arm
 - Lead II: Left foot - right arm
 - Lead III: Left foot - left arm

Einthoven law:

Voltage calculation:

Solution for Question 6:

Correct Answer: C) In lead II the negative end is the right arm

Explanation:

In lead II the negative end is right arm and the positive end is left foot.

ECG Leads:

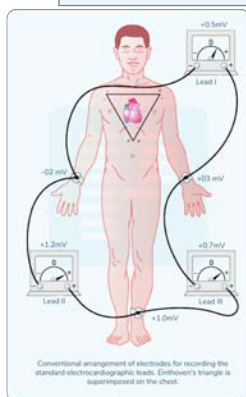
An ECG lead is a pair of electrodes used to detect the potential differences of the heart.

Leads in a standard 12-lead ECG:

Leads	Number of Leads
Bipolar limb leads	Lead I, Lead II, Lead III
Augmented unipolar limb leads (option D)	aVR, aVL, aVF
Precordial/ chest unipolar leads	V1, V2, V3, V4, V5 ,V6
Bipolar	Unipolar
Both the electrodes are active It is the difference between positive - negative Located on the frontal/ coronal plane. (option A)	One electrode is the active or recording electrode and the other one is the indifferent electrode. There are six unipolar chest leads and three unipolar limb leads(augmented limb leads) Located on the frontal/ coronal plane .(option A)

Bipolar limb leads:

Leads	Voltmeter connection	Volage vector direction	
Negative end	Positive end		
Lead I	Right arm	Left arm	Negative to positive
Lead II	Right arm (option C)	Left foot	Negative to positive
Lead III	Left arm	Left foot	Positive to negative



Chest leads:

- Unipolar, located on the horizontal plane.
- Exploring electrodes on the chest surface and indifferent electrodes are connected to the right arm, left arm and left leg through the high resistance, which is called Wilson's terminal. (option B)

Leads	Position
V1	4th intercostal space, right sternal border
V2	4th intercostal space, left sternal border
V3	Midway between V2 and V4
V4	5th intercostal space, midclavicular line
V5	5th intercostal space, anterior axillary line
V6	5th intercostal space, midaxillary line

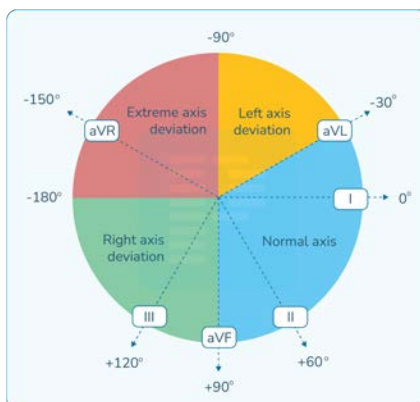
Solution for Question 7:

Correct Answer: B) -150°

Explanation:

The positive of lead aVR is at -150° and the negative of aVR is at $+30^\circ$

The axes of augmented limb leads are interspersed between bipolar limb leads as shown as in the picture. This is known as the Hexaxial system.



Augmented limb leads:

Lead	Positive Electrode	Negative Electrodes	View	Clinical Use
aVR	Right Arm (RA)	Left Arm (LA) + Left Leg (LL)	From the right shoulder	Right-sided heart abnormalities
aVL	Left Arm (LA)	Right Arm (RA) + Left Leg (LL)	From the left shoulder	Left-sided heart abnormalities
aVF	Left Leg (LL)	Right Arm (RA) + Left Arm (LA)	From the inferior (foot) position	Inferior heart abnormalities

Vector of augmented limb lead = $\frac{3}{2}$ vector of unaugmented limb lead

Solution for Question 8:

Correct Answer: C) Septal wall myocardial infarction due to Left Anterior Descending (LAD) artery occlusion.

Explanation:

Leads V1 and V2: ST elevation in these leads is indicative of damage to the septal wall of the left ventricle.

Left Anterior Descending (LAD) Artery: The LAD artery supplies the anterior and septal walls of the left ventricle. Therefore, ST elevation in V1 and V2, which view the septal wall, is typically associated with LAD artery occlusion.

ST Elevation Location	Affected Cardiac Wall	Coronary Artery
I, aVL, V5, V6	Lateral Wall of the Left Ventricle	Left Circumflex (LCX) or Left Anterior Descending (LAD)
II, III, aVF	Inferior Wall of the Left Ventricle	Right Coronary Artery (RCA)
V1, V2	Septal Wall of the Left Ventricle	Left Anterior Descending (LAD)
V3, V4	Anterior Wall of the Left Ventricle	Left Anterior Descending (LAD)
V1 to V4	Anterior Wall including Septal	Left Anterior Descending (LAD)
V7, V8, V9	Posterior Wall of the Left Ventricle	Right Coronary Artery (RCA) or Left Circumflex (LCX)

ST elevation in leads V1 and V2 is not associated with the anterior wall but with the septal wall. RCA occlusion is more likely to cause inferior wall damage (Option A)

Although LAD artery occlusion is involved, V1 and V2 are indicative of septal, not inferior, wall damage. Inferior wall damage is associated with leads II, III, and aVF (Option B)

Lateral wall myocardial infarction is associated with ST elevation in leads I, aVL, V5, and V6, not V1 and V2. LCX occlusion is involved in lateral wall damage (Option D)

Solution for Question 9:

Correct Answer: A) Duration: 0.16 seconds, Amplitude: 0.3 mV

Explanation:

Duration Calculation:

- Each small box on the ECG paper represents 0.04 seconds.
- If the T wave spans 4 small boxes, the duration of the T wave is: $4 \times 0.04 = 0.16 \text{ sec}$

Amplitude Calculation:

- Each small box represents 0.1 mV.
- If the T wave spans 3 small boxes in height, the amplitude is: $3 \times 0.1 = 0.3 \text{ mV}$

Paper Speed:

The standard paper speed is 25 mm/second. This means that the ECG paper moves at a rate of 25 millimeters per second.

Measurement for Duration:

The ECG paper is divided into small and large boxes. Each large box is 5 mm x 5 mm, and each small box within a large box is 1 mm x 1 mm.

1 Large Box: Represents 0.20 seconds (200 milliseconds).

1 Small Box: Represents 0.04 seconds (40 milliseconds)

Amplitude:

The amplitude of the ECG waveforms is measured in millivolts (mV).

Each vertical small box represents 0.1 mV.

Each large box represents 0.5 mV (since it contains 5 small boxes).

Although the duration is correct, the amplitude given is for a different number of boxes (Option B)

The duration calculation is correct, but the amplitude is not accurate for the given number of boxes (Option C)

The amplitude is correct, but the duration provided is for a different number of boxes (Option D)

Solution for Question 10:

Correct Answer: B) -60°

Explanation:

Left axis deviation falls between -90° to -30°

The normal cardiac axis value-

- Normal axis deviation: -30° to $+110^\circ$
- Left axis deviation: -90° to -30°
- Right axis deviation: $+110^\circ$ to $+180^\circ$

Methods to calculate the mean cardiac axis-

- Lead I (+) and aVF (+): Normal axis
- Lead I (-) and aVF (+): Right axis deviation
- Lead I (-) and aVF (-): Extreme right axis deviation
- Lead I (+) and aVF (-): Left axis deviation

Uses the net deflections in Leads I, II, and III to calculate the axis.

- Calculate the net QRS deflection for each of the three limb leads.
- Use vector addition to combine these deflections.
- Calculate the resultant vector and determine its angle.

Involves plotting the QRS vectors on a graph to find the axis.

- Measure the net deflection of the QRS complex in at least two leads.
- Plot these values on a coordinate system.
- Use vector addition to determine the overall direction of the QRS vector.

Identifies the isoelectric lead (lead with the smallest net QRS deflection).

- Find the lead with the most isoelectric QRS complex.
- The QRS axis is perpendicular to this lead.
- Use adjacent leads to refine the exact axis using hexaxial system.

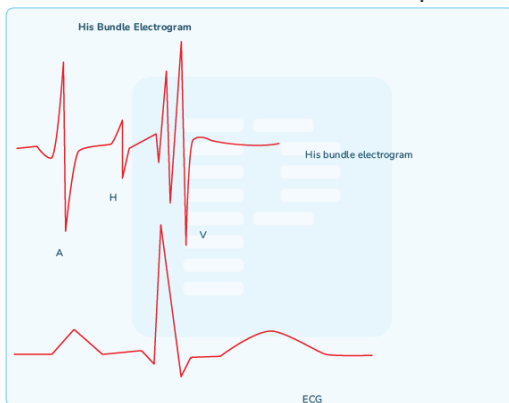
Solution for Question 11:

Correct Answer: C) Depolarization of the His bundle

Explanation: H wave in His bundle electrogram represents the depolarization of the His bundle

His Bundle Electrogram (HBE):

- Invasive ECG procedure
- A catheter containing an electrode at its tip is inserted through an arm vein into the right atrium close to the tricuspid valve. The electrode records the HBE.



	Waveforms in the His Bundle Electrogram	
	A Wave (Option A)	The A wave represents atrial depolarization.
	H Wave (Option C)	The H wave represents depolarization of the His bundle's upper part.
	V Wave: (Option B)	The V wave represents ventricular depolarization.
	Intervals in the His Bundle Electrogram	

	P-A Interval	The PA interval is the time from the onset of the P wave on the surface ECG to the onset of the A wave on the His bundle electrogram. The PA interval typically ranges from 25 to 55 milliseconds.
	A-H Interval	The A-H interval is the time between the onset of the A wave and the onset of the H wave. Typically ranges from 50 to 120 milliseconds. This interval represents conduction time through the AV node. A prolonged A-H interval suggests AV nodal conduction delay.
	H-V Interval	Typically ranges from 35 to 55 milliseconds. This interval represents conduction time through the His-Purkinje system. A prolonged H-V interval suggests an infranodal conduction delay or block.

Hemodynamics, BP Regulation, and Regional Circulation

1. A physiologist is studying blood flow in a network of four parallel blood vessels. The effective resistance of the network of vessels is 4.8 R units. The pressure at the proximal end of the network is 150 mmHg, and the pressure at the distal end is 90 mmHg. What is the total blood flow rate through the network of vessels?

- A. 8 L/s
 - B. 12.5 L/s
 - C. 10.5 L/s
 - D. 20 L/s
-

2. Which of the following statements is true?

- A. Capillaries provide maximum total peripheral resistance
 - B. Arteries contain the most blood volume in the systemic circulation
 - C. Fenestrated capillaries has gaps size ranging from 300-600 nm
 - D. A-V anastomoses found in earlobes have thick, muscular walls
-

3. Which of the following best describes the characteristics of laminar blood flow?

- A. Laminar blood flow is characterized by chaotic and irregular movements with significant mixing of fluid layers.
- B. Laminar blood flow occurs at high velocities and low fluid viscosity, leading to smooth and orderly movement.
- C. Laminar blood flow occurs at low velocities and high fluid viscosity, with fluid particles moving in parallel layers.

D. Laminar blood flow is responsible for generating Korotkoff sounds during blood pressure measurement.

4. Which of the following statements correctly applies the law of Laplace in cardiovascular physiology?

- A. Capillaries are prone to rupture due to high wall tension.
 - B. Dilated hearts have to work harder due to increased myocardial tension.
 - C. Increased flow velocity in narrow vessels reduces distending pressure.
 - D. The total energy in a tube or blood vessel remains constant.
-

5. A 45-y/o male patient presents with a blood pressure reading of 150/90 mmHg. Which of the following statements about his pulse pressure (PP) and mean arterial pressure (MAP) is correct?

- A. Pulse pressure is 40 mmHg, and MAP is 110 mmHg.
 - B. Pulse pressure is 60 mmHg, and MAP is 100 mmHg.
 - C. Pulse pressure is 60 mmHg, and MAP is 110 mmHg.
 - D. Pulse pressure is 40 mmHg, and MAP is 100 mmHg.
-

6. Which of the following is the fastest acting method of BP regulation?

- A. Capillary fluid shift
 - B. Baroreceptor reflex
 - C. Renin-angiotensin vasoconstriction
 - D. Renal body fluid mechanism
-

7. A patient undergoing PCI for myocardial infarction suddenly develops bradycardia, hypotension, and episodes of apnoea. Which of the following reflexes is most likely mediated by chemosensitive vagal C fibers in the cardiopulmonary region?

- A. Cushing reflex
 - B. Bainbridge reflex
 - C. Bezold-Jarisch reflex
 - D. Baroreceptor reflex
-

8. A patient with massive blood loss is found to have increased sympathetic nervous system activity, resulting in elevated blood pressure. Which combination of factors primarily triggers the central nervous system (CNS) ischemic response?

- A. Hypoxia and hypercapnia
 - B. Hypoxia and hypocapnia
 - C. Hypoglycemia and hypoxia
 - D. Hypoglycemia and hypercapnia
-

9.

- A. Stimulation of chemoreceptors leading to hypertension
 - B. Deactivation of chemoreceptors leading to hypotension
 - C. No response of chemoreceptors leading to hypotension
 - D. Deactivation of chemoreceptors but no response on blood pressure
-

10. A 55-year-old patient with chronic obstructive pulmonary disease (COPD) presents with acute exacerbation. ABG shows severe hypoxemia and hypercapnia. How do peripheral chemoreceptors respond to these changes, and its effect on blood pressure?

- A. Stimulation of chemoreceptors leading to hypertension
 - B. Deactivation of chemoreceptors leading to hypotension
 - C. No response of chemoreceptors leading to hypotension
 - D. Deactivation of chemoreceptors but no response on blood pressure
-

11. A 55 y/o man is undergoing a treadmill stress test. Which of the following is true about coronary perfusion?

- A. Primarily occurs during left ventricle systole
 - B. It is the difference between aortic systolic pressure and diastolic pressure.
 - C. Right ventricle perfusion decreases during systole than diastole.
 - D. Regulation of coronary blood flow involves local metabolites
-

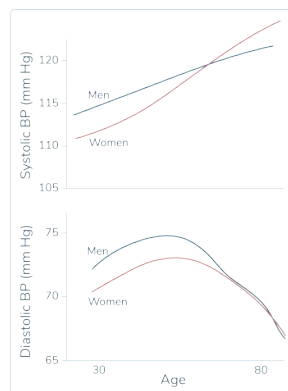
12. Which of the following statements about myocardial muscle tension is correct?

- A. Preload primarily affects the velocity of muscle shortening
 - B. Afterload does not affect the duration of systole
 - C. Increasing preload leads to decreased muscle tension
 - D. The force-velocity relationship in cardiac muscle is inverse and hyperbolic
-

13. According to the Monro-Kellie doctrine, which of the following statements is incorrect regarding the intracranial pressure (ICP) homeostasis?

- A. An increase in cerebrospinal fluid volume can be compensated by a decrease in brain tissue volume
 - B. An increase in blood volume can be offset by a decrease in cerebrospinal fluid volume
 - C. The sum of volumes of brain tissue, blood, and cerebrospinal fluid remains constant
 - D. A significant increase in one intracranial component always leads to a proportional increase in ICP
-

14. In a scientific research, large group of non-hypertensive individuals (systolic blood pressures < 120 mm Hg) were studied every 2 years throughout their adult lives. Effects of age and sex on arterial pressure components (systolic and diastolic BP) in the study are given below. Fall of diastolic pressure after the middle age is due to:



- A. Increase stiffness of arteries
 - B. Decreases stiffness of arteries
 - C. Decrease in contractile force of heart
 - D. Decrease in total peripheral resistance
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	3
Question 4	2
Question 5	3
Question 6	2
Question 7	3
Question 8	1
Question 9	1
Question 10	1
Question 11	4
Question 12	4
Question 13	1
Question 14	1

Solution for Question 1:

Explanation:

Effective Perfusion Pressure = $150 \text{ mmHg} - 90 \text{ mmHg} = 60 \text{ mmHg}$

Using the blood flow analogue of Ohm's law,

■ total = 4.8R units

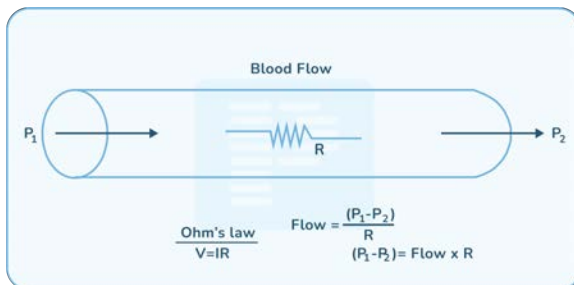
Flow = Pressure / Resistance

$$= 60/4.8 = 12.5 \text{ L/s}$$

Factors Influencing Blood Flow within vessels:

- Pressure differences
- Resistance

Flow, Pressure, and Resistance Relationship: Blood Flow Analog of Ohm's law: Blood flow (F) in any part of the circulatory system is determined by the effective perfusion pressure (P) and the resistance (R) it encounters: $F = P/R$



- The Poiseuille-Hagen formula describes the relationship between flow in a narrow tube, fluid viscosity, and tube radius. This formula is represented as:

$$F = (P_A - P_B) \times (\pi/8) \times (1/\eta) \times (r^4/L)$$

Where:

- F = flow
- $P_A - P_B$ = pressure difference between the tube's two ends
- η = viscosity
- r = tube radius
- L = tube length

Solution for Question 2:

Correct Answer: D) A-V anastomoses found in earlobes have thick, muscular walls

Explanation:

Arteriovenous anastomoses (A-V anastomoses) , also known as shunts, are direct connections between arterioles and venules that bypass the capillary bed. They have thick, muscular walls and are abundantly innervated, likely by vasoconstrictor nerve fibers. These shunts are found in the fingers, palms, and earlobes.

Other types of Blood vessels:

Blood Vessel Type	Blood Volume
Arteries Carry oxygenated blood away from the heart. Arteries have elastic tissue in their walls, allowing them to stretch and recoil with the pumping of the heart.	8%
Arterioles They are the primary site of resistance to blood flow Contain a high proportion of smooth muscle	1%
Capillaries Tiny vessels with thin walls for the exchange of gases, nutrients, and waste products between blood and tissues.	5%
Veins Veins have thinner walls allowing them to distend easily and accommodate larger volumes of blood. Contain venous valves to prevent backflow of blood	54% (Option B)

Types of capillaries:

Capillary type	Gap size	Location	
Continuous capillaries are divided into 2 types: I. Continuous capillary with tight junctions II. Continuous capillary with loose junctions	Less than 20 nm	With tight junctions	With loose junctions
		Brain capillary (minimum gap), spinal cord, enteric nervous system, and retinal capillaries.	Lungs, skin, muscles, adipose tissue, cortical bone, and GIT (mesentery)
Fenestrated capillaries	20-80 nm.	Kidneys (Glomerulus, peritubular capillaries), GIT (except mesentery), endocrine, and exocrine glands.	
Sinusoidal capillaries	300 and 600 nm.	Bone marrow, spleen, and liver (maximum gap)	

Capillaries, due to their large cross-sectional area, provide low resistance to blood flow. The arterioles actually provide the maximum total peripheral resistance (Option A)

Veins, not arteries, contain the most blood volume in the systemic circulation (Option B)

The typical pore size in fenestrated capillaries is around 60-80 nm(Option C)

Solution for Question 3:

Correct Answer: C) Laminar blood flow occurs at low velocities and high fluid viscosity, with fluid particles moving in parallel layers.

Explanation:

Laminar blood flow is where the fluid moves in smooth, parallel layers with minimal mixing, typically occurring at low velocities and high viscosity.

- Smooth, orderly flow of fluid in parallel layers with no disruption between the layers.
- Characteristics : Fluid particles move in straight, parallel paths. Velocity of fluid is highest at the center and decreases towards the vessel walls. Occurs at lower velocities and with lower fluid density or higher fluid viscosity (Option B) .
- Fluid particles move in straight, parallel paths.
- Velocity of fluid is highest at the center and decreases towards the vessel walls.
- Occurs at lower velocities and with lower fluid density or higher fluid viscosity (Option B) .
- Fluid particles move in straight, parallel paths.
- Velocity of fluid is highest at the center and decreases towards the vessel walls.
- Occurs at lower velocities and with lower fluid density or higher fluid viscosity (Option B) .
- Chaotic, irregular flow where fluid particles move in various directions, often forming eddies.
- Characteristics : Disorganized movement with mixing of fluid layers (Option A) . Velocity varies unpredictably at any point in the fluid. Higher resistance compared to laminar flow. Occurs at higher velocities, with higher fluid density, or lower fluid viscosity.
- Disorganized movement with mixing of fluid layers (Option A) .
- Velocity varies unpredictably at any point in the fluid.
- Higher resistance compared to laminar flow.
- Occurs at higher velocities, with higher fluid density, or lower fluid viscosity.
- Disorganized movement with mixing of fluid layers (Option A) .
- Velocity varies unpredictably at any point in the fluid.
- Higher resistance compared to laminar flow.
- Occurs at higher velocities, with higher fluid density, or lower fluid viscosity.
- Dimensionless numbers used to predict flow patterns in fluid dynamics.
- Interpretation : $Re < 2000$: Flow is typically laminar. $Re > 4000$: Flow is typically turbulent. $Re\ 2000-4000$: Transitional flow, where both laminar and turbulent flow may exist.
- $Re < 2000$: Flow is typically laminar.

- $Re > 4000$: Flow is typically turbulent.
- $Re\ 2000-4000$: Transitional flow, where both laminar and turbulent flow may exist.
- $Re < 2000$: Flow is typically laminar.
- $Re > 4000$: Flow is typically turbulent.
- $Re\ 2000-4000$: Transitional flow, where both laminar and turbulent flow may exist.
- Sounds heard while measuring blood pressure using a sphygmomanometer and stethoscope.
- Mechanism : Created by turbulent blood flow in the brachial artery when external pressure is between systolic and diastolic pressures. As cuff pressure drops below systolic pressure, blood begins to flow turbulently (Option D) , producing Korotkoff sounds. When cuff pressure drops below diastolic pressure, laminar flow is restored, and sounds disappear.
- Created by turbulent blood flow in the brachial artery when external pressure is between systolic and diastolic pressures.
- As cuff pressure drops below systolic pressure, blood begins to flow turbulently (Option D) , producing Korotkoff sounds.
- When cuff pressure drops below diastolic pressure, laminar flow is restored, and sounds disappear.
- Created by turbulent blood flow in the brachial artery when external pressure is between systolic and diastolic pressures.
- As cuff pressure drops below systolic pressure, blood begins to flow turbulently (Option D) , producing Korotkoff sounds.
- When cuff pressure drops below diastolic pressure, laminar flow is restored, and sounds disappear.

Solution for Question 4:

Correct Answer: B) Dilated hearts have to work harder due to increased myocardial tension.

Explanation:

According to the law of Laplace, dilated hearts require greater myocardial tension to produce a given pressure, increasing cardiac workload.

Principle	Definition	Applications
-----------	------------	--------------

Law of Laplace	The tension in the wall of a cylinder is equal to the product of transmural pressure and radius divided by wall thickness. $T = P \cdot r / 2w$ (T= Tension, P= Transmural pressure, r= Radius, w= Wall thickness)	Capillaries : Low tension in walls due to small radius, resistant to rupture. (Option A) Dilated Hearts : Increased radius requires more tension, increasing cardiac workload.(Option B)
Bernoulli Principle	The total energy in a tube or blood vessel (kinetic energy + potential energy) remains constant. (Option D)	Pressure Measurement : Lower side arm pressure in a T tube due to kinetic energy of flow. Atherosclerotic Plaques : Increased flow velocity in narrow vessels reduces distending pressure, maintaining vessel narrowing. (Option C)

Solution for Question 5:

Correct Answer: C) Pulse pressure is 60 mmHg, and MAP is 110 mmHg.

Explanation:

Pulse Pressure (PP) : Pulse pressure is calculated as the difference between systolic blood pressure (SBP) and diastolic blood pressure (DBP). $PP = SBP - DBP = 150 - 90 = 60 \text{ mmHg}$

Mean Arterial Pressure (MAP) : Mean arterial pressure is calculated using the formula: $MAP = DBP + 1/3(SBP - DBP) = 90 + 1/3(150 - 90) = 90 + 20 = 110 \text{ mmHg}$

Feature	Pulse Pressure (PP)	Mean Arterial Pressure (MAP)
Definition	Difference between systolic and diastolic blood pressure	Average pressure in arteries during one cardiac cycle
Calculation	$PP = SBP - DBP$	$MAP = DBP + 1/3(SBP - DBP)$
Normal Range	40-60 mmHg	70-100 mmHg
Clinical Significance	Indicates cardiovascular health and heart function	A better indicator of perfusion to vital organs
Implications of High Values	Risk of cardiovascular events	Can indicate a risk of hypertension and organ damage
Implications of Low Values	This may suggest poor heart function	This may indicate poor perfusion of vital organs

Solution for Question 6:

Correct Answer: B) Baroreceptor reflex

Explanation:

Time Frame	Mechanisms	Functions
Short-term (seconds to minutes)	Autonomic nervous system Baroreceptor reflex Central nervous system ischemic response Chemoreceptor mechanism	Constrict veins to increase blood flow to the heart Increase heart rate and contractility Constrict peripheral arterioles to elevate BP
Intermediate-term (minutes to hours)	Angiotensin mediated vasoconstriction (Option C) Stress relaxation Capillary fluid shift (Option A)	Renin triggers angiotensin II, constricting blood vessels and increasing BP Sustained high BP leads to gradual pressure reduction Fluid absorption or loss in capillaries adjusts blood volume and BP
Long-term (days to years)	Renal-body fluid mechanism (Option D) Aldosterone influence	Regulate salt and water excretion to maintain BP stability Aldosterone promotes salt and water retention when BP is low

Aspect	Details
Mechanism	Baroreceptors in carotid sinus and aortic arch detect BP changes Increased BP stretches receptors, sending signals to medulla via glossopharyngeal and vagus nerves Medulla inhibits vasoconstrictor center and stimulates vagal parasympathetic center
Actions	Vasodilation of blood vessels Decreased heart rate and contraction strength Reduced BP due to lower peripheral resistance and cardiac output
Importance	Maintains BP stability during activities like standing Fastest acting (Option B) Tends to reset within 1-2 days for long-term regulation

Solution for Question 7:

Correct Answer: C) Bezold-Jarisch reflex

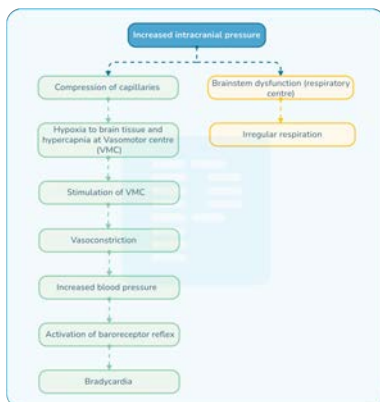
Explanation:

Myocardial ischemia and reperfusion activates chemosensitive vagal C fibers in the cardiopulmonary region causing Bezold–Jarisch reflex leading to profound bradycardia, hypotension, and a brief period of apnea followed by rapid shallow breathing.

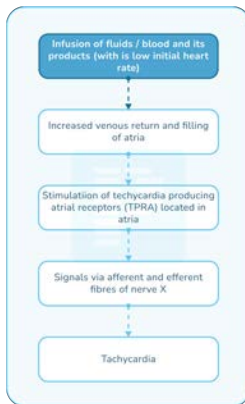
Bezold-Jarisch reflex or coronary chemoreflex:

- Activators : Capsaicin, serotonin, phenylbiguanide, veratridine and pathophysiologic conditions like reperfusion after myocardial ischemia (increased production of oxygen radicals and agents used as radio contrast for coronary angiography).
- Mechanism: Activation of chemosensitive vagal C fibers (Option C) in the cardiopulmonary region (juxtacapillary region of alveoli, ventricles, atria, great veins, and pulmonary artery).
- Causing bradycardia, hypotension, and a brief apnea period followed by rapid shallow breathing.

Cushing reflex (Option A): A combination of hypertension, bradycardia, and irregular respiration is known as the triad of the Cushing reflex.



Bainbridge reflex(Option B): Following vagotomy, the reflex is abolished as the responses are mediated by vagus nerves shown below.



Baroreceptor reflex (Option D) or sino-aortic reflex: It quickly adjusts heart rate and vascular tone in response to changes in blood pressure to maintain cardiovascular stability.

Solution for Question 8:

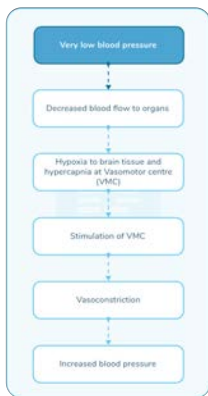
Correct Answer: A) Hypoxia and hypercapnia

Explanation:

Decreased blood flow due to massive hemorrhage here causes significant drop in blood pressure leading to severe hypoxia and hypercapnia which activates Vasomotor center, increasing blood pressure and restoring cerebral perfusion.

CNS ischemic response:

- Vasomotor center (VMC) is very sensitive to local hypoxia and local hypercapnia causing strong sympathetic system activation as follows. (Option A)
- Failure of this response leads to irreversible shock as this is the last physiological reflex to correct blood pressure.



Hypoglycemia does not trigger the central nervous system (CNS) ischemic response.

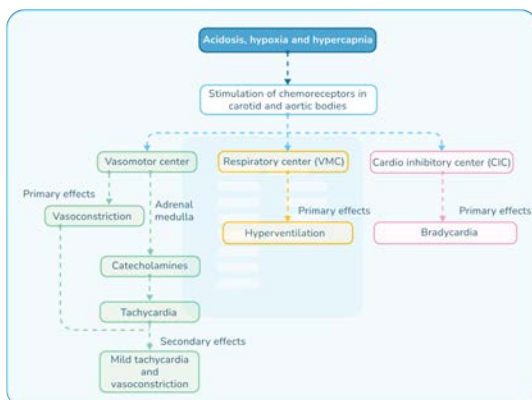
Solution for Question 9:

Correct Answer: A) Stimulation of chemoreceptors leading to hypertension

Explanation:

In the scenario, severe hypoxemia and hypercapnia stimulate peripheral chemoreceptors and afferent signals stimulate cardio inhibitory center (CIC) to produce bradycardia and vasomotor center (VMC) to produce vasoconstriction as a part of the primary effect.

Chemoreceptor reflex and BP:



- Afferent from the carotid body: IX cranial nerve
- Afferent from aortic body: X cranial nerve

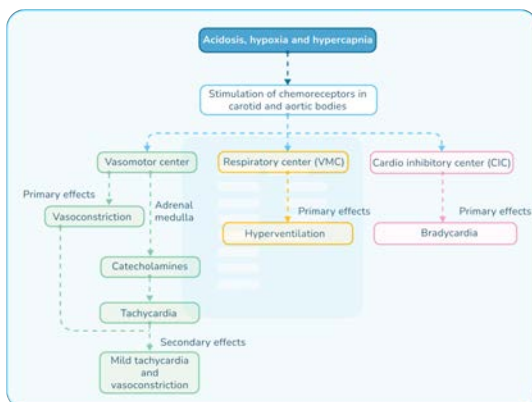
Solution for Question 10:

Correct Answer: A) Stimulation of chemoreceptors leading to hypertension

Explanation:

In the scenario, severe hypoxemia and hypercapnia, stimulates peripheral chemoreceptors and afferent signals stimulate cardio inhibitory center (CIC) to produce bradycardia and vasomotor center (VMC) to produce vasoconstriction as apart of primary effect.

Chemoreceptor reflex and BP:



- Afferent from carotid body: IX cranial nerve
- Afferent from aortic body: X cranial nerve

Deactivation of chemoreceptors leading to hypotension (Option B) – Severe hypoxemia and hypercapnia stimulate, not deactivate, chemoreceptors.

No response of chemoreceptors leading to hypotension (Option C) – Peripheral chemoreceptors actively respond to hypoxemia by increasing sympathetic output.

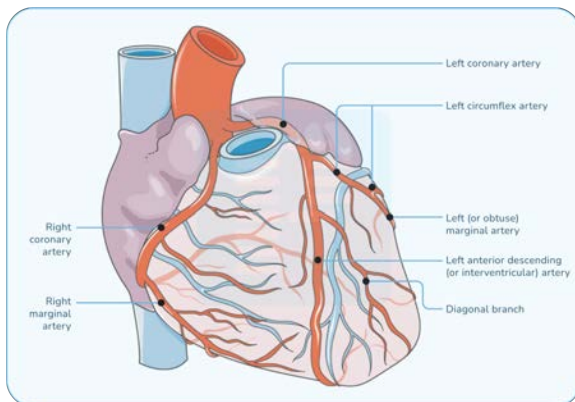
Deactivation of chemoreceptors but no response on blood pressure (Option D) – Chemoreceptors are stimulated, not deactivated, and they influence blood pressure via vasoconstriction.

Solution for Question 11:

Correct Answer: D) Regulation of coronary blood flow involves local metabolites

Explanation: Regulation of coronary blood flow involves primarily local metabolites like adenosine, hydrogen ions, lactic acid, potassium ions and certain prostaglandins by mechanism known as metabolite mediated vasodilatation.

Coronary perfusion:



Key points	Explanation
Coronary circulation	Supplies heart muscles
Major coronary arteries	Left and right coronary arteries
Rate of blood flow in myocardium	250ml/minute
Percentage of blood flow in heart	5% of cardiac output

- The right and left coronary arteries originate from the coronary sinuses at the aortic root.
- These arteries extend along the epicardium, and then branch through the myocardium to form subendocardial plexuses to perfuse the myocardium.
- Myocardial contraction during systole compresses arterial branches and prevents perfusion.
- Coronary perfusion occurs during diastole rather than systole from epicardial to endocardial regions.

Coronary perfusion pressure: (Option B)

- Coronary Perfusion Pressure (CPP) = Aortic Diastolic Pressure – Left Ventricular end-diastolic Pressure (LVEDP).
- Aortic systolic pressure; 120 mmHg
- Aortic diastolic pressure: 80 mmHg

- Blood flow to the left ventricle occurs only during the early stage of diastole (isovolumetric relaxation) (Option A)
- Blood flow in the right ventricle occurs during systole and diastole. Blood flow increases in systole than in diastole. (Option C)

Regulation of coronary perfusion:

Chemical Factors:

- Regulation of blood flow occurs primarily through vasodilation caused by local metabolites (Option D) which are formed due to 70-80% oxygen extraction of myocardium from the coronary arteries.
- Metabolites are CO₂, H⁺, K⁺, lactate, prostaglandins, adenine nucleotides, and adenosine.

Neural factors:

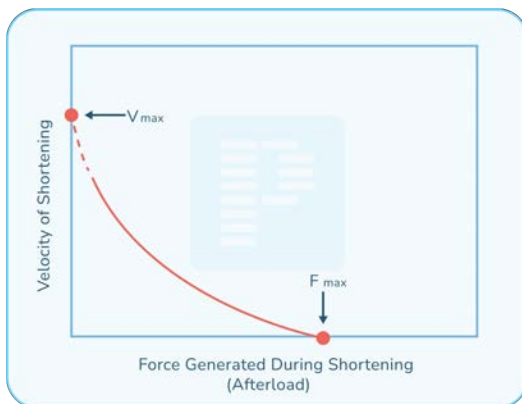
- α -adrenergic receptors on coronary arterioles: vasoconstriction
- β -adrenergic receptors on coronary arterioles: vasodilation

Solution for Question 12:

Correct Answer: D) The force-velocity relationship in cardiac muscle is inverse and hyperbolic

Explanation: The force-velocity relationship in cardiac muscle is inverse and hyperbolic. This fundamental property of cardiac muscle mechanics describes how the velocity of muscle shortening is related to the force (or tension) developed by the muscle.

- As the load (force) against which the muscle contracts increases, the velocity of shortening decreases.
- When the muscle contracts against a lighter load, it can shorten more quickly.
- This relationship is not linear but follows a hyperbolic curve.
- At zero load, the muscle shortens at its maximum velocity (V_{max}), while at maximum isometric force, the shortening velocity is zero.
- This is how the heart can maintain adequate cardiac output over a wide range of afterloads by adjusting its contractile properties.



Preload primarily affects the force of contraction, not the velocity of muscle shortening. This is based on the Frank-Starling mechanism (Option A)

Afterload does affect the duration of systole. Increased afterload can prolong the duration of systole as the ventricle requires more time to generate sufficient pressure to open the aortic valve and eject blood (Option B)

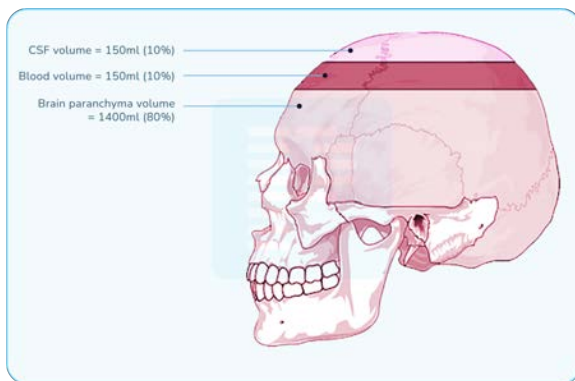
More EDV (which is the best determinant of preload) leads to increased tension in the cardiac muscle. $\uparrow$ EDV $\rightarrow$ $\uparrow$ ventricular muscle length $\rightarrow$ $\uparrow$ muscle contraction force (contractility of the heart). This is known as the Frank-Starling law of the heart (Option C)

Solution for Question 13:

Correct Answer: A) An increase in cerebrospinal fluid volume can be compensated by a decrease in brain tissue volume

Explanation:

The Monro-Kellie doctrine states that the cranial compartment is incompressible, and the volume inside the cranium is fixed hence Option A in the incorrect statement. The doctrine describes the relationship between the volumes of brain tissue, cerebrospinal fluid (CSF), and blood within the cranium.



- According to this principle, an increase in one component must be offset by a decrease in one or both of the other components to maintain a constant intracranial pressure.
- Brain tissue is not easily compressible, and its volume does not acutely decrease to compensate for increases in other components .
- Therefore, compensatory mechanisms primarily involve changes in CSF and blood volumes. During a rise in intracranial pressure, CSF can be displaced into the spinal subarachnoid space or its production can be reduced, and/ or venous blood can be displaced from the cranial cavity.
- During a rise in intracranial pressure, CSF can be displaced into the spinal subarachnoid space or its production can be reduced, and/ or venous blood can be displaced from the cranial cavity.
- During a rise in intracranial pressure, CSF can be displaced into the spinal subarachnoid space or its production can be reduced, and/ or venous blood can be displaced from the cranial cavity.

An increase in blood volume can be compensated by a decrease in CSF volume (Option B)

The Principle of the Monro-Kellie doctrine is that the sum of the volumes of brain tissue, blood, and CSF must remain constant within the rigid confines of the skull (Option C)

Small increases in the volume of one component can often be compensated without a significant change in ICP, due to the compensatory mechanisms described earlier. However, once these compensatory mechanisms are exhausted, even small volume increases can lead to dramatic rises in ICP (Option D)

Solution for Question 14:

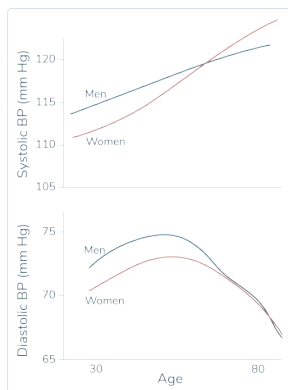
Correct answer: A) Increase stiffness of arteries

Explanation:

Fall of diastolic pressure after the middle age is due to increase in stiffness of arteries

Blood Pressure Changes with Age:

- Systolic Pressure : Increases steadily throughout life, even in individuals with normal initial levels (<120 mm Hg at ages 50–60).
- Diastolic Pressure : Rises with age but starts to decline in middle age due to increased arterial stiffness.
- Pulse Pressure : Increases as the gap between systolic and diastolic pressure widens.
- Gender Differences : Women have lower blood pressures than men until around ages 55–65. This may contribute to their longer lifespan, as lower blood pressure is linked to a reduced risk of heart attacks and strokes.



Decrease in Stiffness of Arteries (Option B) : Aging typically increases arterial stiffness.

Decrease in Contractile Force of Heart (Option C) : Diastolic pressure changes are more related to arterial stiffness than heart contractility.

Decrease in Total Peripheral Resistance (Option D) : Aging often increases peripheral resistance, not decreases it.

Cardiac Cycle and Cardiac Output

1. In the jugular venous pressure (JVP) waveform, which of the following statements accurately describes the "c wave"?

- A. The "c wave" is caused by atrial systole
 - B. The "c wave" arises from the increase in atrial pressure during isovolumetric ventricular contraction
 - C. The "c wave" corresponds to the rise in atrial pressure as blood fills the atria
 - D. The "c wave" is influenced by the fluctuations in venous pressure caused by respiration
-

2. Which of the following accurately describes the differentiation pathway of hematopoietic stem cells (HSCs) in the production of granulocytes?

- A. HSCs differentiate directly into granulocytes without intermediate stages.
 - B. HSCs differentiate into common lymphoid progenitors, which then produce granulocytes.
 - C. HSCs differentiate into common myeloid progenitors, which then give rise to granulocyte-monocyte progenitors.
 - D. HSCs differentiate into megakaryocyte-erythroid progenitors, which then produce granulocytes.
-

3. A 65-year-old male presents to the emergency department with severe chest pain and shortness of breath. His blood pressure is 80/50 mmHg, heart rate is 110 bpm, and has cold, clammy skin and elevated JVP. ECG shows ST-segment elevations in lead II, III and aVF. The patient is also oliguric. Which of the following is the most likely underlying cause of this patient's symptoms and signs?

- A. Myocardial infarction

- B. Sepsis
 - C. Adrenal insufficiency
 - D. Cardiac tamponade
-

4. Which physiological mechanism best describes the Frank-Starling law of the heart?

- A. The heart's stroke volume decreases as myocardial fiber length increases.
 - B. The heart's stroke volume is regulated primarily by sympathetic nervous system stimulation.
 - C. The heart's stroke volume increases in response to increased end-diastolic volume due to the length-tension relationship of cardiac muscle fibers.
 - D. The heart's stroke volume remains unchanged regardless of changes in venous return, due to intrinsic pacemaker activity
-

5. A 50-year-old patient with heart failure is evaluated for changes in myocardial contractility. Which of the following statements best describes the impact of decreased contractility on the Frank-Starling curve?

- A. Shifts the curve upward and to the left, resulting in a greater stroke volume at any given preload.
 - B. Shifts the curve downward, resulting in a decreased stroke volume at any given preload.
 - C. Does not affect the Frank-Starling curve but increases the heart rate.
 - D. Shifts the curve upward and to the right, resulting in a decreased stroke volume at any given preload.
-

6. A 55-year-old patient undergoes a blood transfusion, which results in a significant increase in blood volume. Which of the following best describes the

physiological response to activation of the Bainbridge reflex due to an increase in blood volume?

- A. Decreased heart rate and decreased contractility
 - B. Increase in heart rate
 - C. Decreased heart rate and increased contractility
 - D. No change in heart rate or contractility
-

7. According to the Fick principle, which of the following is essential for calculating cardiac output?

- A. Measurement of the patient's blood pressure before and after exercise
 - B. Determination of the oxygen uptake by the lungs and the arteriovenous oxygen content difference
 - C. Assessment of the patient's heart rate and electrocardiographic findings
 - D. Injection of a contrast dye and measurement of its concentration in the blood over time
-

8. A 42-year-old patient undergoes cardiac evaluation, and the following values are obtained: end-diastolic volume (EDV) of 140 mL, and ejection fraction of 48%. Which of the following best describes the patient's cardiac parameters?

- A. The patient's ejection fraction is below the normal range, indicating reduced ventricular function.
 - B. The stroke volume is higher than normal, suggesting improved cardiac output.
 - C. The cardiac output is higher than average due to the increased EDV.
 - D. The cardiac output can be calculated accurately from the given values alone.
-

9. A 50-year-old patient is undergoing evaluation of cardiac function, and a pressure-volume (PV) loop of the left ventricle is analyzed. From the PV loop, the following measurements are obtained: end-diastolic volume (EDV) is 140 mL and end-systolic volume (ESV) is 60 mL. What is the ejection fraction (EF) for this patient, and how is it calculated?

- A. 30%, calculated as $(EDV - ESV)/EDV$
 - B. 43%, calculated as $(SV/EDV) * 100$
 - C. 57%, calculated as $(EDV - ESV)/EDV$
 - D. 57%, calculated as $(SV/EDV) * 100$
-

10. What is the duration of the cardiac cycle for a heart rate of 180 beats per minute?

- A. 0.5 seconds
 - B. 0.75 seconds
 - C. 0.33 seconds
 - D. 0.66 seconds
-

11. Which factor primarily contributes to an increased pulse pressure in the aortic pressure curve?

- A. Decreased stroke volume
 - B. Increased arterial compliance
 - C. Increased stroke volume
 - D. Decreased heart rate
-

12. A 30-year-old healthy volunteer undergoes invasive hemodynamic monitoring. At which point in the cardiac cycle is the pressure gradient between the left atrium and left ventricle greatest, and what event does this gradient trigger?

- A. End of atrial systole; opening of mitral valve
 - B. Mid-ventricular systole; closure of aortic valve
 - C. Early diastole; rapid ventricular filling
 - D. End of isovolumic relaxation; opening of mitral valve
-

13. A 40-year-old patient with suspected heart failure undergoes a comprehensive cardiac evaluation. During auscultation, both S3 and S4 are heard. Regarding the correlation of these extra heart sounds with the ECG, which statement is correct?

- A. S3 occurs during the P wave; S4 coincides with the QRS complex
 - B. S3 is heard after the T wave & before next P wave; S4 occurs immediately after the P wave
 - C. S3 coincides with the U wave; S4 is heard during the PR interval
 - D. S3 occurs during the ST segment; S4 is heard at the peak of the R wave
-

14. During a cardiac examination, a 35-year-old patient's simultaneous ECG and phonocardiography are recorded. Which of the following statements most accurately describes the relationship between the second heart sound (S2) and the ECG?

- A. S2 occurs at the peak of the T wave in all cardiac cycles
 - B. S2 is heard just before the T wave
 - C. S2 coincides with the onset of the U wave when present
 - D. S2 typically occurs at the end of T wave and the subsequent P wave
-

15. Regarding the timing and correlation of the Phonocardiogram's first heart sound (S1) with the ECG, which of the following statements is CORRECT?

- A. S1 occurs immediately after the P wave and coincides with atrial systole
 - B. S1 begins during the ST segment and ends before the T wave
 - C. S1 starts 0.02-0.04 seconds after the onset of the QRS complex
 - D. S1 is heard simultaneously with the peak of the R wave in all lead
-

16. A 30-year-old patient undergoes cardiac catheterization with simultaneous ECG recording. During the isovolumic relaxation phase, which combination of events is most accurate?

- A. T wave ends; ventricular pressure falls below atrial pressure; mitral valve opens
 - B. U wave begins; ventricular pressure equals aortic pressure; all valves closed
 - C. P wave starts; ventricular volume increases; tricuspid valve opens
 - D. Second heart sound occurs; ventricular pressure drops rapidly; aortic valve closes
-

17. During the isovolumic contraction phase, which of the following combinations of events is most accurate?

- A. Second heart sound occurs; ventricular pressure exceeds atrial pressure; aortic valve opens
 - B. QRS complex ends; ventricular pressure rises but remains below aortic pressure; all valves closed
 - C. T wave begins; ventricular volume decreases slightly; mitral valve closes
 - D. Second heart sound occurs; ventricular pressure equals aortic pressure; pulmonary valve closes
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	3
Question 3	1
Question 4	3
Question 5	2
Question 6	2
Question 7	2
Question 8	1
Question 9	4
Question 10	3
Question 11	3
Question 12	4
Question 13	2
Question 14	4
Question 15	3
Question 16	4
Question 17	2

Solution for Question 1:

Correct Answer: B) The "c wave" arises from the increase in atrial pressure during isovolumetric ventricular contraction, transmitted through the closed tricuspid valve.

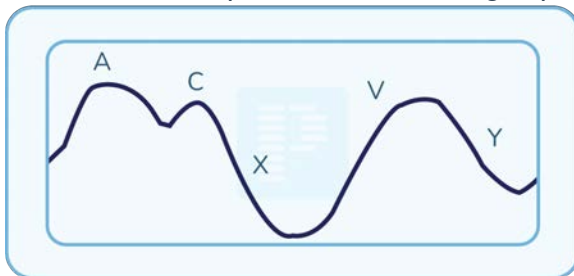
Explanation:

The "c wave" arises from the increase in atrial pressure during isovolumetric ventricular contraction, which is transmitted through the closed tricuspid valve.

a wave	Caused by atrial systole. (Option A ruled out) Atrial contraction results in a temporary halt in venous inflow. Some blood flows back into the veins, causing a rise in venous pressure.
c wave	Arises from the increase in atrial pressure during isovolumetric ventricular contraction. (Option B) Pressure is transmitted through the closed and bulging tricuspid valve.
x descent	Ventricular systole pulls the tricuspid valve down
v wave	Corresponds to the rise in atrial pressure as blood fills the atria. (Option C ruled out) Occurs during diastole while the tricuspid valve is still closed.
y descent	Opening of tricuspid valve causing emptying of right atrium into ventricle

Respiration Influence :

- Venous pressure decreases during inspiration due to negative intrathoracic pressure. (Option D ruled out)
- Venous pressure rises during expiration.



Solution for Question 2:

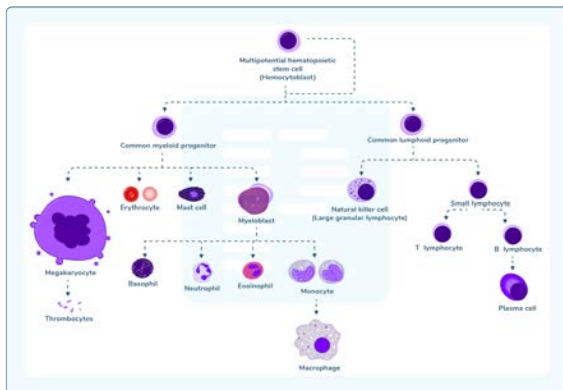
Correct Answer: C) HSCs differentiate into common myeloid progenitors, which then give rise to granulocyte-monocyte progenitors.

Explanation:

HSCs first differentiate into common myeloid progenitors (CMPs). CMPs then differentiate into granulocyte-monocyte progenitors (GMPs), which give rise to granulocytes (neutrophils, eosinophils, and basophils).

Hematopoiesis-

- Definition : Hematopoiesis is the process of blood cell formation and development from hematopoietic stem cells (HSCs) in the bone marrow.
- Location : Primarily occurs in the bone marrow, but also in the liver and spleen during fetal development.
- Hematopoietic Stem Cells (HSCs) : Multipotent stem cells that can differentiate into all types of blood cells.
- Stages : Involves differentiation from HSCs to progenitor cells, which further differentiate into various mature blood cells.
- Lineages : Myeloid Lineage : Produces red blood cells, platelets, monocytes, neutrophils, eosinophils, and basophils. Lymphoid Lineage : Produces T cells, B cells, and natural killer (NK) cells.
- Myeloid Lineage : Produces red blood cells, platelets, monocytes, neutrophils, eosinophils, and basophils.
- Lymphoid Lineage : Produces T cells, B cells, and natural killer (NK) cells.
- Myeloid Lineage : Produces red blood cells, platelets, monocytes, neutrophils, eosinophils, and basophils.
- Lymphoid Lineage : Produces T cells, B cells, and natural killer (NK) cells.



HSCs differentiate directly into granulocytes without intermediate stages. (Option A)

Hematopoietic stem cells (HSCs) do not differentiate directly into granulocytes; they go through several intermediate stages.

HSCs differentiate into common lymphoid progenitors, which then produce granulocytes. (Option B) Common lymphoid progenitors produce lymphocytes (T cells, B cells, and NK cells), not granulocytes.

HSCs differentiate into megakaryocyte-erythroid progenitors, which then produce granulocytes (Option D) Megakaryocyte-erythroid progenitors (MEPs) give rise to megakaryocytes (which produce platelets) and erythrocytes (red blood cells), not granulocytes.

Solution for Question 3:

Correct Answer: A) Myocardial infarction

Explanation: ST segment elevation in leads II, III and aVF corresponds to inferior wall Myocardial infarction, which leads to cardiogenic shock as in this scenario.

Shocks:

Types of shock	Pathophysiology	Causes	Cardiac output
Cardiogenic (Option A)	Inadequate pumping of the heart due to myocardial function.	Acute myocardial infarction Heart failure Cardiac arrhythmias	↓
Hypovolemic	Decreased amount of fluid in the vascular system	Acute hemorrhage Burn Metabolic: adrenal insufficiency (Option C ruled out) , diabetic ketoacidosis Severe diarrhea or vomiting.	↓
Distributive	Sudden increase in the vascular compartment capacity due to acute marked vasodilation	Neurogenic shock Anaphylactic shock Septic shock (Option B ruled out)	↓
Obstructive	Obstruction to the ejection of blood from the ventricle	Cardiac tamponade (Option D ruled out) Constrictive pericarditis Cardiac tumor Tension pneumothorax	↓

Solution for Question 4:

Correct Answer: C) The heart's stroke volume increases in response to increased end-diastolic volume due to the length-tension relationship of cardiac muscle fibers.

Explanation:

This option correctly describes the Frank-Starling law. It states that as the end-diastolic volume (the volume of blood filling the heart) increases, the myocardial fibers stretch more, which enhances their contraction strength due to the optimal length-tension relationship, thereby increasing stroke volume.

The Frank-Starling law of the heart describes the relationship between the volume of blood filling the heart (end-diastolic volume) and the force of contraction during the next heartbeat (stroke volume) .

Mechanism:

- Preload: Increased venous return leads to increased end-diastolic volume.
- Myocardial Stretch: Greater end-diastolic volume stretches the myocardial fibers.
- Force of Contraction: The stretched fibers generate a more forceful contraction due to the optimal overlap of actin and myosin filaments (length-tension relationship).

The heart's stroke volume decreases as myocardial fiber length increases beyond the optimal length (Option A) It is incorrect because the Frank-Starling law indicates that stroke volume increases with increased myocardial fiber length, up to an optimal point, rather than decreasing.

The heart's stroke volume is regulated primarily by sympathetic nervous system stimulation. (Option B) This option is incorrect because, while sympathetic nervous system stimulation does affect heart function, the Frank-Starling law specifically refers to the intrinsic ability of the heart to adjust stroke volume based on changes in end-diastolic volume, not primarily on external neural influences.

The heart's stroke volume remains unchanged regardless of changes in venous return, due to intrinsic pacemaker activity. (Option D) This option is incorrect because the Frank-Starling law explains that stroke volume changes in response to venous return and end-diastolic volume, not that it remains unchanged.

Solution for Question 5:

Correct Answer: B) Shifts the curve downward and to the right, resulting in a decreased stroke volume at any given preload.

Explanation:

Decreased contractility shifts the Frank-Starling curve downward and to the right, leading to a reduced stroke volume for any given preload- Negative Inotropic Effect.

Myocardial Contractility

- Definition: The intrinsic ability of the myocardium to contract and generate force, independent of preload (end-diastolic volume) and afterload (resistance against which the heart pumps). Changes in contractility shift the Frank-Starling curve (relationship between stroke volume and end-diastolic volume).
- The intrinsic ability of the myocardium to contract and generate force, independent of preload (end-diastolic volume) and afterload (resistance against which the heart pumps).
- Changes in contractility shift the Frank-Starling curve (relationship between stroke volume and end-diastolic volume).
- Impact on Frank-Starling Curve: Increased Contractility (Positive Inotropic Effects): Shifts the curve upward and to the left. Results in a greater stroke volume at any given preload. Decreased Contractility (Negative Inotropic Effects): Shifts the curve downward and to the right.
- Increased Contractility (Positive Inotropic Effects): Shifts the curve upward and to the left. Results in a greater stroke volume at any given preload.
- Shifts the curve upward and to the left.
- Results in a greater stroke volume at any given preload.
- Decreased Contractility (Negative Inotropic Effects): Shifts the curve downward and to the right.
- Shifts the curve downward and to the right.
- The intrinsic ability of the myocardium to contract and generate force, independent of preload (end-diastolic volume) and afterload (resistance against which the heart pumps).
- Changes in contractility shift the Frank-Starling curve (relationship between stroke volume and end-diastolic volume).
- Increased Contractility (Positive Inotropic Effects): Shifts the curve upward and to the left. Results in a greater stroke volume at any given preload.

- Shifts the curve upward and to the left.
- Results in a greater stroke volume at any given preload.
- Decreased Contractility (Negative Inotropic Effects): Shifts the curve downward and to the right.
- Shifts the curve downward and to the right.
- Shifts the curve upward and to the left.
- Results in a greater stroke volume at any given preload.
- Shifts the curve downward and to the right.

The Bowditch Effect

Key Points

- Alternative Names: Treppe phenomenon, staircase phenomenon, frequency-dependent activation
- Definition: An increase in heart rate enhances the force of contraction of myocardial cells with each heartbeat, independent of other influences.
- An increase in heart rate enhances the force of contraction of myocardial cells with each heartbeat, independent of other influences.
- Mechanism: Related to calcium ion handling and mishandling in cardiac cells. Involves proteins crucial for excitation-contraction coupling, such as sarcoplasmic endoplasmic reticulum calcium-ATPase (SERCA).
- Related to calcium ion handling and mishandling in cardiac cells.
- Involves proteins crucial for excitation-contraction coupling, such as sarcoplasmic endoplasmic reticulum calcium-ATPase (SERCA).
- An increase in heart rate enhances the force of contraction of myocardial cells with each heartbeat, independent of other influences.
- Related to calcium ion handling and mishandling in cardiac cells.
- Involves proteins crucial for excitation-contraction coupling, such as sarcoplasmic endoplasmic reticulum calcium-ATPase (SERCA).

Aspect	Details
Positive Bowditch Effect	Seen in healthy hearts, related to SER Ca load alterations.
Negative Bowditch Effect	Pathological, due to diminished SERCA protein expression, hallmark of heart failure.
Clinical Significance	Shift from increasing heart rate to beta-blockers for improved cardiac function in heart failure.

Solution for Question 6:

Correct Answer: B) Increase in heart rate

Explanation:

The Bainbridge reflex, also known as the atrial reflex, is triggered by the stretching of the atrial walls due to increased venous return, resulting in an increased heart rate.

- Intrinsic Regulation: Frank-Starling Mechanism Describes how increased stretching of the cardiac muscle (within physiological limits) due to more incoming blood results in a stronger contraction.
- Describes how increased stretching of the cardiac muscle (within physiological limits) due to more incoming blood results in a stronger contraction.
- Describes how increased stretching of the cardiac muscle (within physiological limits) due to more incoming blood results in a stronger contraction.
- Extrinsic Regulation: Autonomic Nervous System Sympathetic Stimulation- Increases both heart rate and contractility. Parasympathetic Stimulation- Decreases heart rate and has a minimal effect on contractility.
- Sympathetic Stimulation- Increases both heart rate and contractility.
- Parasympathetic Stimulation- Decreases heart rate and has a minimal effect on contractility.
- Temperature changes impact contractility
- Sympathetic Stimulation- Increases both heart rate and contractility.
- Parasympathetic Stimulation- Decreases heart rate and has a minimal effect on contractility.
- Overall Regulation: combines intrinsic mechanisms, neural control, and environmental factors to adjust cardiac output.

Bainbridge Reflex (Atrial Reflex)

- Trigger: Stretching of the atrial walls due to increased venous return (higher blood volume).
- Mechanism: Receptors: Low-pressure stretch receptors (atrial receptors) are activated by the increased blood volume. Signal Transmission: These receptors send signals to the medulla oblongata via the vagus nerve. Response: The medulla sends efferent signals through both vagal and sympathetic nerves. Outcome: Increased heart rate and often enhanced contraction strength, which helps to eject excess blood from the atria and prevent its accumulation.

- Receptors: Low-pressure stretch receptors (atrial receptors) are activated by the increased blood volume.
- Signal Transmission: These receptors send signals to the medulla oblongata via the vagus nerve.
- Response: The medulla sends efferent signals through both vagal and sympathetic nerves.
- Outcome: Increased heart rate and often enhanced contraction strength, which helps to eject excess blood from the atria and prevent its accumulation.
- Receptors: Low-pressure stretch receptors (atrial receptors) are activated by the increased blood volume.
- Signal Transmission: These receptors send signals to the medulla oblongata via the vagus nerve.
- Response: The medulla sends efferent signals through both vagal and sympathetic nerves.
- Outcome: Increased heart rate and often enhanced contraction strength, which helps to eject excess blood from the atria and prevent its accumulation.
- Interaction with Baroreceptors: Baroreceptor Reflex: Sensitive to changes in arterial pressure, it works to counteract the effects of the Bainbridge reflex by slowing the heart rate.
- Baroreceptor Reflex: Sensitive to changes in arterial pressure, it works to counteract the effects of the Bainbridge reflex by slowing the heart rate.
- Baroreceptor Reflex: Sensitive to changes in arterial pressure, it works to counteract the effects of the Bainbridge reflex by slowing the heart rate.

Solution for Question 7:

Correct Answer: B) Determination of the oxygen uptake by the lungs and the arteriovenous oxygen content difference

Explanation:

The Fick principle calculates cardiac output by measuring the total oxygen uptake by the lungs and relating it to the difference in oxygen content between arterial and mixed venous blood.

This method is based on the principle that the total uptake of a substance (oxygen in this case) by an organ is equivalent to the product of blood flow to that organ and the arteriovenous concentration difference of the substance.

Cardiac output is volume of blood ejected by the heart into the aorta per minute and it is measured by:

Method	Principle	Measurement Basis	Key Considerations
Fick Principle	Oxygen uptake and arteriovenous difference	Oxygen content in blood	Accurate in steady states
Indicator Dilution Method	Indicator concentration dilution	Injected dye or tracer	Requires precise indicator handling
Echocardiography	Ultrasound visualization and flow velocity	Stroke volume estimation	Non-invasive, real-time data
Thoracic Electrical Bioimpedance	Electrical impedance changes	Impedance fluctuations	Sensitive to external factors

Measurement of the patient's blood pressure before and after exercise (Option A) It is not related to the Fick principle. It provides information about blood pressure changes but does not calculate cardiac output based on oxygen uptake.

Determination of the oxygen uptake by the lungs and the arteriovenous oxygen content difference (Option C) Assessment of heart rate and electrocardiographic findings does not directly relate to the Fick principle. Cardiac output measurement via the Fick principle does not rely on these parameters.

Injection of a contrast dye and measurement of its concentration in the blood over time (Option D) This describes the Indicator Dilution Method, which involves injecting a contrast dye and measuring its concentration to determine cardiac output, not the Fick principle.

Solution for Question 8:

Correct Answer: A) The patient's ejection fraction is below the normal range, indicating reduced ventricular function.

Explanation:

An ejection fraction of 48% is below the normal range (50%-70%), which may suggest reduced ventricular function or potential heart failure.

Parameter	Description	Typical Value
-----------	-------------	---------------

Cardiac Output	The total volume of blood pumped by the heart per minute, reflecting its ability to meet the body's oxygen and nutrient demands.	Varies with demand
Stroke Volume	The volume of blood ejected from each ventricle per heartbeat.	~70-90 mL at rest
End-Diastolic Volume (EDV)	Volume of blood in the ventricle at the end of diastole, just before contraction.	~130 mL at rest
Cardiac Index	Cardiac output adjusted for body size, expressed as liters per minute per square meter of body surface area, allowing for comparison across different body sizes.	~3.2 L/min/m ²
Ejection Fraction	Percentage of EDV ejected with each heartbeat, reflecting ventricular function and efficiency.	50%-70%

Additional Factors: Sympathetic and parasympathetic nervous system activity, hormonal influences, and mechanical factors can affect these parameters.

The stroke volume is higher than normal, suggesting improved cardiac output (Option B) – Stroke volume of 70 mL is within the normal range (70-90 mL) and does not indicate improved function.

The cardiac output is higher than average due to the increased EDV (Option C) – Higher EDV does not necessarily mean higher cardiac output if ejection fraction is reduced.

The cardiac output can be calculated accurately from the given values alone (Option D) – Heart rate is also required to determine cardiac output ($CO = SV \times HR$).

Solution for Question 9:

Correct Answer: D) 57%, calculated as $(SV/EDV) \times 100$

Explanation:

The stroke volume (SV) is calculated as $EDV - ESV = 140 \text{ mL} - 60 \text{ mL} = 80 \text{ mL}$.

The ejection fraction (EF) is then $EF = (SV/EDV) \times 100\% = (80/140) \times 100\% = 57\%$.

PV Loop

- A graphical representation of pressure and volume changes in the left ventricle during the cardiac cycle.
- Used to derive stroke volume and ejection fraction.

Key Phases:

- Diastole: The period when the left ventricle fills with blood. End-Diastolic Volume (EDV): Volume of blood in the left ventricle at the end of diastole.
- End-Diastolic Volume (EDV): Volume of blood in the left ventricle at the end of diastole.
- Systole: The period when the left ventricle contracts and ejects blood. End-Systolic Volume (ESV): Volume of blood remaining in the left ventricle after contraction.
- End-Systolic Volume (ESV): Volume of blood remaining in the left ventricle after contraction.
- End-Diastolic Volume (EDV): Volume of blood in the left ventricle at the end of diastole.
- End-Systolic Volume (ESV): Volume of blood remaining in the left ventricle after contraction.

Stroke Volume (SV)

- Represents the blood ejected from the left ventricle per heartbeat.
- Determined by the horizontal distance between end-diastolic volume (EDV) and end-systolic volume (ESV).
- Formula: $SV = EDV - ESV$.

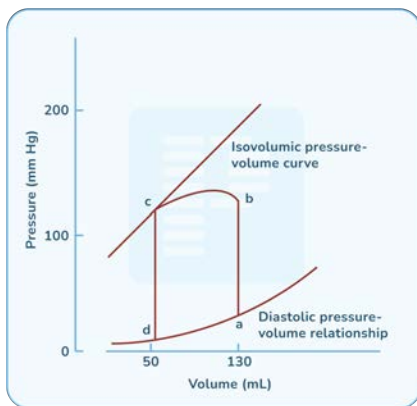
Ejection Fraction (EF)

- Indicates the percentage of blood ejected from the left ventricle relative to the end-diastolic volume.
- Formula: $EF = (SV/EDV) \times 100\%$.
- Normal value is approximately 65%.

Normal PV Loop of Left Ventricle:

a to b: Isovolumetric Contraction. ; b to c: Period of ejection.

c to d: Isovolumetric Relaxation. ; d to a: Period of filling.



Solution for Question 10:

Correct Answer: C) 0.33 seconds

Explanation:

Duration of cardiac cycle can be calculated by $60 \text{ (sec)} / \text{heart rate per min}$

So, $60 / 180 = 0.33 \text{ sec}$

Duration of a normal cardiac cycle is 0.8 sec (for 72 beats per minute)

Ventricular Systole = 0.3 sec

Ventricular Diastole = 0.5 sec

Phases of cardiac cycle -

Phase	Sub-Phase	Key Events
Systole	Isovolumetric Contraction	Ventricles begin to contract Ranges from closure of AV valves (mitral and tricuspid) till opening of semilunar valves. First heart sound appears. No blood volume change in ventricles

	Ventricular Ejection	Has 2 phases- Rapid ejection phase and Reduced ejection phase. Ventricular pressure exceeds aortic and pulmonary pressure Aortic and pulmonary valves open Blood ejected into aorta and pulmonary artery Ventricular volume decreases Ends with the closure of the aortic valve.
Diastole	Isovolumetric Relaxation	Ventricles relax Ventricular pressure drops Aortic and pulmonary valves close No blood volume change in ventricles Second heart sounds appears. Ends with the opening of AV valves.
	Rapid Filling	AV valves (mitral and tricuspid) open Blood flows rapidly from atria to ventricles Ventricular volume increases rapidly Third heart sound appears.
	Diastasis	Blood flow from atria to ventricles slows No change in the atrial and ventricular pressure. Ventricular volume increases slowly
	Atrial Systole	Atria contract Additional blood pushed into ventricles Ventricular volume increases slightly End-diastolic volume (EDV) reached Fourth heart sound is produced.

Solution for Question 11:

Correct Answer: C) Increased stroke volume

Explanation:

Increased stroke volume leads to a more pronounced increase in systolic pressure during ejection and a more substantial decrease during diastole, thereby increasing pulse pressure.

Phase	Description
Left Ventricle Contraction	Ventricular pressure increases until aortic valve opens
Period of Ejection 80-120 mmHg	Left ventricular pressure > aortic pressure; blood ejected
Isovolumic Relaxation	Aortic pressure forces blood back, closing aortic valve
Incisura	Notch on the curve due to aortic valve closure
Diastolic Decline 80mmHg diastolic	Aortic pressure declines as blood flows into systemic circulation

Factors Influencing Aortic Pressure

Stroke Volume
Arterial Compliance

Transmission of Pressure Pulses

Pressure Wave Propagation- generated by ventricular contraction travels through the arterial system
Dampening Effect- pulsations diminish in smaller arteries, becoming negligible in the capillaries.

Stroke Volume:

- Increase in Stroke Volume: When stroke volume increases, more blood is pumped into the arteries, leading to a higher systolic pressure. If diastolic pressure remains unchanged, this results in an increased pulse pressure.
- Decrease in Stroke Volume: Conversely, if stroke volume decreases, less blood is ejected into the arteries, leading to a lower systolic pressure. If diastolic pressure stays the same, this results in a decreased pulse pressure (Option A)

Arterial Compliance:

- Increased Arterial Compliance: Compliance refers to the ability of arteries to expand and accommodate the blood ejected with each heartbeat. Higher arterial compliance means arteries can stretch more easily. This reduces the rise in systolic pressure and dampens the pulse pressure, as the arteries absorb more of the pressure variation (Option B)
- Decreased Arterial Compliance: Lower arterial compliance means arteries are stiffer and less able to expand. This results in a greater increase in systolic pressure with each heartbeat, leading to a higher pulse pressure, as the arteries cannot absorb the pressure as effectively.

So, an increase in stroke volume and a decrease in arterial compliance typically raises pulse pressure.

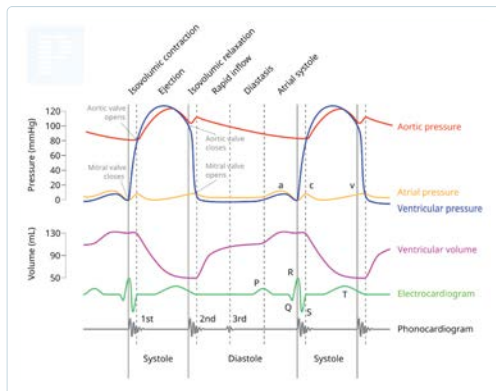
Heart Rate (Option D) - lower heart rate allows more time for diastolic runoff, leading to a lower diastolic pressure, which may slightly increase pulse pressure, but stroke volume remains the primary determinant.

Solution for Question 12:

Correct Answer: D) End of isovolumic relaxation; opening of mitral valve

Explanation:

The greatest pressure gradient between the left atrium and left ventricle occurs at the end of isovolumic relaxation. At this point, ventricular pressure has fallen below atrial pressure, creating a significant pressure gradient. This gradient triggers the opening of the mitral valve and initiates rapid ventricular filling.



End of atrial systole; opening of mitral valve (Option A): At the end of atrial systole, the pressure gradient is actually minimal, as atrial contraction has pushed blood into the ventricle, equalizing pressures.

Mid-ventricular systole; closure of aortic valve (Option B): During mid-ventricular systole, ventricular pressure exceeds atrial pressure. The aortic valve closes at the end of systole, not mid-systole.

Early diastole; rapid ventricular filling (Option C): While rapid ventricular filling occurs in early diastole, it is the result of, not the cause of, the greatest pressure gradient.

Solution for Question 13:

Correct Answer: B) S3 is heard just after the T wave; S4 occurs immediately after the P wave

Explanation:

S3 is typically heard just after the T wave on the ECG, during early diastole. S4 occurs immediately after the P wave, during late diastole, just before the QRS complex. This timing corresponds to the physiological events causing these sounds.

Phase	Heart Sound	Timing on ECG	Cause	Clinical Significance	Key Points
Early Diastole	S3	Early diastole, after T wave	Rapid ventricular filling	Normal in children and athletes; in adults may indicate heart failure	Rapid ventricular filling phase AV valves open Ventricles relax
Late Diastole	S4	Late diastole, just before P wave	Atrial contraction against a stiff ventricle	Always abnormal; may indicate hypertension, ischemia, or cardiomyopathy	Atrial systole Completes ventricular filling Precedes next ventricular systole

S3 occurs during the P wave; S4 coincides with the QRS complex (Option A): This option switches the timing of S3 and S4. S3 occurs in early diastole, after the T wave, not during the P wave. S4 occurs in late diastole, after the P wave, not during the QRS complex.

S3 coincides with the U wave; S4 is heard during the PR interval (Option C): While S3 may sometimes coincide with the U wave (if present), it's not a consistent relationship. S4 occurs after the P wave, not during the PR interval.

S3 occurs during the ST segment; S4 is heard at the peak of the R wave (Option D): S3 occurs in diastole, after the T wave, not during the ST segment which is part of systole. S4 occurs after the P wave, not at the peak of the R wave.

Solution for Question 14:

Correct Answer: D) S2 typically occurs between the T wave and the subsequent P wave

Explanation:

The second heart sound (S2) typically occurs between the T wave and the subsequent P wave on the ECG. This timing corresponds to the end of ventricular systole and the beginning of ventricular diastole when the semilunar valves (aortic and pulmonary) close.

Phase	Heart Sound	Timing on ECG	Cause	Clinical Significance	Key Points
Systole	S2	At the end of the T-wave	Closure of aortic and pulmonic valves	Normal finding; Split S2 may indicate delayed pulmonic valve closure	Ventricular ejection ends Marks beginning of diastole Aortic valve closes

S2 does not occur at the peak of the T wave. The T wave represents ventricular repolarization, which occurs before the end of mechanical systole and valve closure (Option A).

While S2 does occur after the T wave, it's not consistently 0.02-0.04 seconds after the end of the T wave. The timing can vary based on heart rate and other factors (Option B).

The U wave, when present, typically occurs after S2, not coinciding with it. The U wave is thought to represent the final repolarization of the ventricles or Purkinje fibers (Option C).

Solution for Question 15:

Correct Answer: C) S1 starts 0.02-0.04 seconds after the onset of the QRS complex

Explanation :

The first heart sound (S1) typically starts 0.02-0.04 seconds after the onset of the QRS complex. This slight delay represents the time between the electrical activation of the ventricles and the mechanical closure of the AV(atrioventricular) valves, which produces S1.

Phase	Heart Sound	Timing on ECG	Cause	Clinical Significance	Key Points
-------	-------------	---------------	-------	-----------------------	------------

Systole	S1	Just after QRS complex	Closure of mitral and tricuspid valves	Normal finding; Split S1 may indicate bundle branch block	Ventricular contraction begins Isovolumetric contraction phase Aortic valve opens
---------	----	------------------------	--	---	---

S1 does not occur immediately after the P wave. The P wave represents atrial depolarization, while S1 is associated with the onset of ventricular systole and closure of the AV valves (Option A).

S1 occurs much earlier in the cardiac cycle, not during the ST segment. The ST segment represents ventricular repolarization, which happens after ventricular systole has begun (Option B).

While S1 often occurs near the peak of the R wave, it doesn't always coincide exactly with the R wave peak, especially not in all ECG leads. The timing can vary slightly depending on conduction and mechanical factors (Option D).

Solution for Question 16:

Correct Answer: D) Second heart sound occurs; ventricular pressure drops rapidly; aortic valve closes

Explanation :

During the isovolumic relaxation phase:

- The second heart sound (S2) occurs at the beginning of this phase
- Ventricular pressure drops rapidly
- The aortic valve closes at the start of this phase
- Ventricular volume remains constant as all valves are closed

The T wave typically ends before the isovolumic relaxation phase. Ventricular pressure falling below atrial pressure and mitral valve opening marks the end of this phase and the beginning of rapid ventricular filling (Option A).

The U wave, when present, usually occurs after the T wave and before the next P wave.

Ventricular pressure is falling during this phase and is below aortic pressure. While all valves are

indeed closed, this is not a defining characteristic unique to this phase (Option B) .

The P wave occurs later in diastole, during the late ventricular filling phase. Ventricular volume remains constant during isovolumic relaxation, and the tricuspid valve remains closed throughout this phase (Option C)

Solution for Question 17:

Correct Answer: B) QRS complex ends; ventricular pressure rises but remains below aortic pressure; all valves closed

Explanation :

During the isovolumic contraction phase:

- The QRS complex typically ends just as this phase begins
- Ventricular pressure rises rapidly but remains below aortic pressure
- All valves (mitral, tricuspid, aortic, and pulmonary) remain closed

The first heart sound occurs at the beginning of systole, just before isovolumic contraction. While ventricular pressure does exceed atrial pressure, the aortic valve doesn't open during this phase (Option A).

The T wave typically occurs later in systole, during the ejection phase. Ventricular volume remains constant during isovolumic contraction. The mitral valve closes just before this phase begins (Option C) .

The second heart sound occurs at the end of systole, during isovolumic relaxation. Ventricular pressure doesn't equal aortic pressure during isovolumic contraction, and the pulmonary valve remains closed throughout this phase (Option D) .

Previous Year Questions

1. In the case of reduced contractility of the heart, which of the following changes would be observed? a) ESV (End-systolic volume) b) EDV (End-diastolic volume)

- A. Decreased end-diastolic volume and pressure
 - B. Shift of the end-systolic volume and pressure curves to the right
 - C. Increased stroke volume
 - D. Enhanced response to sympathetic stimulation
-

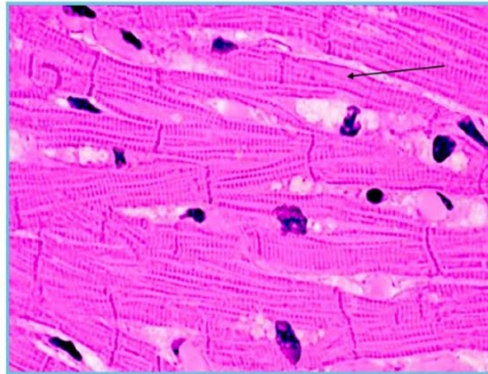
2. What is the most effective trigger for the secretion of vasopressin?

- A. Hypotonicity of extracellular fluid
 - B. Hypertension
 - C. Hypotension
 - D. Hypo-osmolality of extracellular fluid
-

3. Please arrange the following items in the order they occur after the "p wave" in an ECG. A- 'a' wave B – 1st heart sound C – Rapid filling of ventricles D – 't' wave

- A. ABCD
 - B. ABDC
 - C. CABD
 - D. ACBD
-

4. Which organelle is not present in the designated region of the cardiac muscle?



- A. Mitochondria
 - B. Lysosome
 - C. Golgi bodies
 - D. Endoplasmic reticulum
-

5. Which reflex is responsible for tachycardia when the right atrium is distended?

- A. J reflex
 - B. Cushing reflex
 - C. Bezold jarisch reflex
 - D. Bainbridge reflex
-

6. What is the cause of dicrotic notch?

- A. Passive filling of blood in ventricles
- B. Rapid ejection phase
- C. Peripheral resistance
- D. Isovolumic contraction

7. Which of the following are properties of ventricular cardiac muscle? All or none law Length tension relationship Long refractory period Can be tetanized

- A. 1 only
 - B. 2 only
 - C. 1 and 2
 - D. 1,2,3
-

8. A patient is brought to the Emergency Room following a road traffic accident (RTA) with thoracic and abdominal injuries. The patient is found to be in hypovolemic shock. What is the most probable cause of the decreased cardiac output in this case?

- A. Decrease in preload
 - B. Decrease in contractility of the heart
 - C. Increase in preload
 - D. Increase in afterload
-

9. A man presents with an irregularly irregular pulse and dyspnoea. What is the expected finding in the jugular venous pressure (JVP)?

- A. Absent a wave
 - B. Cannon a wave
 - C. Prominent X descent
 - D. Prominent Y descent
-

10. In the jugular venous pressure (JVP) waveform, the "a" wave corresponds to:

- A. Right Atrial contraction
 - B. Tricuspid valve bulging into Right atria
 - C. Right Atrial relaxation
 - D. Right atrial filling
-

11. Please arrange the following steps of ventricular action potential in the correct order. A. Activation of calcium channels regulated by changes in voltage. B. Repolarizing K⁺ currents C. Opening of voltage – gated sodium channels D. Transient outward flow of potassium

- A. B, A, C, D
 - B. D, C, B, A
 - C. C, D, A, B
 - D. A, B, C, D
-

12. If radius of a vessel is doubled then the blood flow is increased by:

- A. 32 times
 - B. 16 times
 - C. 256 times
 - D. 8 times
-

13. Which of the following is correct about isovolumetric relaxation?

- A. Semilunar valves open
 - B. C Wave of JVP
 - C. Corresponds to QT interval
 - D. AV Valves are closed
-

14. Baroreceptor reflex is a type of _____.

- A. Negative feedback
 - B. Positive feedback
 - C. Both negative and positive feedback
 - D. Feedforward control
-

15. A 65-year-old patient visits his family practitioner for a yearly examination. On examination, his BP was 190/100 mm Hg. His pulse pressure was 90 mm Hg. What would have decreased resulting in this increase in pulse pressure?

- A. Myocardial contractility
 - B. Stroke volume
 - C. Arterial wall compliance
 - D. Cardiac output
-

16. Which of the following contributes to the zero phase of the action potential in the SA node?

- A. Calcium
- B. Sodium
- C. Potassium

D. Calcium and sodium

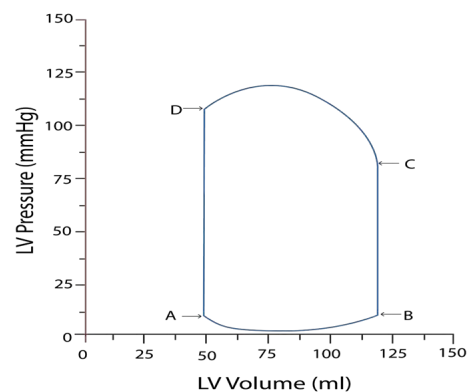
17. 'c' wave in JVP is seen in?

- A. Iso-volumetric contraction
 - B. Slow filling at end of diastole
 - C. End of systole
 - D. Start of diastole
-

18. In the cardiac muscle, the inhibition of Na⁺ K⁺ ATPase results in?

- A. Hyperpolarization and then depolarization
 - B. Depolarization
 - C. Hyperpolarization
 - D. Repolarization
-

19. In the following LV pressure-volume loop diagram, point C denotes –



- A. Mitral valve closes
 - B. Aortic valve opens
 - C. Aortic valve closes
 - D. Mitral valve opens
-

20. Which of the following events occur immediately when a person changes his/her posture from supine to standing?

- A. Decrease in firing from baroreceptor
 - B. Vagal output to cardiac fibers increase
 - C. Mesenteric Vasoconstriction
 - D. decreased cardiac contractility
-

21. The cardiac event that occurs at the end of the isometric relaxation phase is?

- A. Atrioventricular valves open
 - B. Atrioventricular valves close
 - C. Corresponds to the peak of C wave in JVP
 - D. Corresponds to T wave in ECG
-

22. C wave of JVP represents?

- A. Tricuspid bulge into right atrium during isovolumetric ventricular contraction
- B. Contraction of right atrium
- C. Passive filling of right atrium
- D. Passive emptying of right atrium

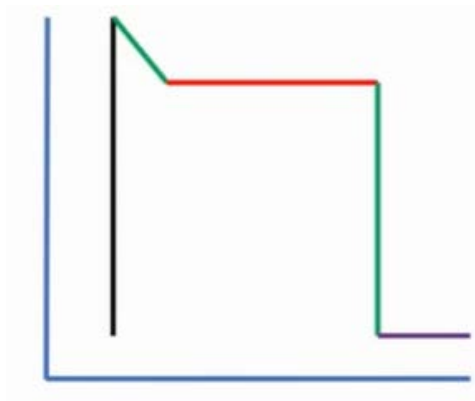
23. The first heart sound coincides with which phase of cardiac cycle?

- A. Rapid atrial filling
 - B. Aortic ejection
 - C. Isovolumetric contraction
 - D. Isovolumetric relaxation
-

24. The presence of a prominent 'v' wave of jugular venous pulse is suggestive of?

- A. Tricuspid regurgitation
 - B. Cardiomyopathy
 - C. Tricuspid stenosis
 - D. Atrial fibrillation
-

25. Identify the true statement regarding the segment marked in red on the myocardial action potential curve?



- A. Referred to as the pacemaker potential
 - B. Plateau phase of the fast-type action potential
 - C. Membrane potential is returning to the resting membrane potential
 - D. Represents the resting membrane potential
-

26. What options can be employed to determine cardiac output? A. ECHO B. Ventilation – perfusion scan C. Thermodilution D. Fick's principle

- A. A, B, C, D
 - B. B, A, D
 - C. B, C, D
 - D. A, C, D
-

27. What is the cause of fetal heart rate acceleration?

- A. Carotid baroreceptor
 - B. Aortic baroreceptor
 - C. Autonomic nervous system
 - D. Chemoreceptors
-

28. Windkessel effect is not shown by which of the following vessels?

- A. Aorta
- B. Pulmonary artery
- C. Radial artery
- D. Abdominal aorta

29. Blood pressure changes in the radial artery were measured while performing a valsalva maneuver. Which of the following is the reason for the initial rise in blood pressure?

- A. Increase in left ventricular volume
 - B. Decrease in left ventricular pressure
 - C. Increase in aortic pressure
 - D. Decrease in aortic pressure
-

30. PR interval in ECG denotes which of the following?

- A. Conduction through AV nodes
 - B. Ventricular depolarization
 - C. Ventricular repolarization
 - D. Atrial depolarization
-

31. Where do electrical impulses originate?

- A. SA node
 - B. AV node
 - C. Bundle of his
 - D. Purkinje fibers
-

32. The stroke volume depends on end-diastolic volume. This is best represented as?

- A. Length–weight relationship
 - B. Length–velocity relationship
 - C. Length-tension relationship
 - D. Length-viscosity relationship
-

33. Which of the following jugular venous pressure (JVP) waves corresponds to the isovolumetric contraction phase of the cardiac cycle?

- A. A wave
 - B. C wave
 - C. X descent
 - D. Y descent
-

34. Reflex responsible for tachycardia during right atrial distension is?

- A. Bezold-Jarisch reflex
 - B. Bain bridge reflex
 - C. Cushing reflex
 - D. Baroreceptor reflex
-

35. Pulmonary capillary wedge pressure increases in which of the following?

- A. Hypovolemic shock
- B. Septic shock

- C. Cardiogenic shock
 - D. Anaphylactic shock
-

36. What is the cause of aortic notch?

- A. Passive filling of blood in ventricles
 - B. Rapid ejection phase
 - C. Closure of semilunar valves
 - D. Isovolumic contraction
-

37. Which of the following statements accurately describes Isovolumic relaxation?

- A. AV valves are closed
 - B. Corresponds to QT interval
 - C. 'C' wave of JVP
 - D. Semilunar valves open
-

38. What is the increase in resistance of an artery if its radius is reduced to one-third?

- A. 9 times
 - B. 81 times
 - C. 27 times
 - D. 18 times
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	2
Question 4	2
Question 5	4
Question 6	3
Question 7	4
Question 8	1
Question 9	1
Question 10	1
Question 11	3
Question 12	2
Question 13	4
Question 14	1
Question 15	3
Question 16	1
Question 17	1
Question 18	2
Question 19	2
Question 20	1
Question 21	1
Question 22	1

Question 23	3
Question 24	1
Question 25	2
Question 26	4
Question 27	3
Question 28	3
Question 29	3
Question 30	1
Question 31	1
Question 32	3
Question 33	2
Question 34	2
Question 35	3
Question 36	3
Question 37	1
Question 38	2

Solution for Question 1:

Option A. Decreased end-diastolic volume and pressure

- Incorrect. Reduced contractility often leads to impaired filling during diastole, causing an increase in end-diastolic volume and pressure as the heart struggles to fill properly.

Option B. Shift of the end-systolic volume and pressure curves to the right

- Correct. Reduced contractility shifts the end-systolic volume and pressure curves to the right on the pressure-volume loop, indicating that the heart is ejecting less blood per contraction.

Option C. Increased stroke volume

- Incorrect. Reduced contractility typically results in a decreased stroke volume because the heart's ability to pump blood effectively is compromised.

Option D. Enhanced response to sympathetic stimulation

- Incorrect. Reduced contractility often indicates a weakened response to sympathetic stimulation, as the heart muscles are not contracting as forcefully as they should.
- So, the correct answer is Option B. Shift of the end-systolic volume and pressure curves to the right in the case of reduced contractility of the heart.

So, the correct answer is Option B. Shift of the end-systolic volume and pressure curves to the right in the case of reduced contractility of the heart.

Solution for Question 2:

Option A. Hypotonicity of extracellular fluid

Explanation:

Vasopressin, also known as antidiuretic hormone (ADH), is a hormone produced by the hypothalamus and released by the posterior pituitary gland. Its primary function is to regulate water balance in the body by controlling the reabsorption of water in the kidneys, thereby influencing urine concentration.

The most effective trigger for the secretion of vasopressin is the hypotonicity of extracellular fluid (option a). When the extracellular fluid becomes hypotonic (meaning it has a lower concentration of solutes), the body detects this decrease in osmolality. In response, vasopressin is released, which leads to increased water reabsorption in the kidneys, conserving water and reducing urine output. This mechanism helps the body maintain proper fluid balance.

Incorrect Options:

Option B. Hypertension: High blood pressure is not a direct trigger for vasopressin secretion. Vasopressin primarily regulates water balance rather than blood pressure.

Option C. Hypotension: Low blood pressure is not the primary trigger for vasopressin secretion. While severe hypotension can stimulate vasopressin release as a compensatory mechanism to increase blood pressure, it is not the most effective trigger.

Option D. Hypo-osmolality of extracellular fluid: This option is incorrect because it repeats the concept of hypotonicity, which is the decreased concentration of solutes in the extracellular fluid,

leading to vasopressin secretion.

Therefore, A . Hypotonicity of extracellular fluid is the correct option.

Solution for Question 3:

Correct Option B:

The correct sequence after the P wave is as follows:

A. 'a' wave: The 'a' wave represents atrial contraction. It occurs during atrial systole when the atria contract to push blood into the ventricles.

B. 1st heart sound: The 1st heart sound, also known as S1, is the closure of the atrioventricular (AV) valves (mitral and tricuspid valves) at the beginning of ventricular systole. It marks the onset of ventricular contraction.

D. 't' wave: The 't' wave is a repolarization phase of the ventricles. It represents the relaxation and resetting of the ventricles after contraction.

C. Rapid filling of ventricles: After the 't' wave, the ventricles enter the diastole phase, during which they passively fill with blood from the atria. This is known as rapid ventricular filling.

Therefore, the correct sequence is ABDC: 'a' wave, 1st heart sound, rapid filling of ventricles, and 't' wave.

Incorrect Options:

Option A: ACBD: This option incorrectly places the rapid filling of ventricles (C) before the 'a' wave (A). The correct sequence is 'a' wave (A), 1st heart sound (B), rapid filling of ventricles (C), and 't' wave (D).

Option C: CABD: This option incorrectly places the rapid filling of ventricles (C) before the 'a' wave (A) and the 1st heart sound (B). The correct sequence is 'a' wave (A), 1st heart sound (B), rapid filling of ventricles (C), and 't' wave (D).

Option D: ACBD: This option incorrectly places the rapid filling of ventricles (C) before the 'a' wave (A). The correct sequence is 'a' wave (A), 1st heart sound (B), rapid filling of ventricles (C), and 't' wave (D).

Solution for Question 4:

Correct Option B:

- Lysosomes are responsible for intracellular digestion and waste removal. They contain enzymes that break down cellular waste materials and cellular components that are no longer functional.
- The area marked is devoid of nuclei, and lysosomes are usually seen in perinuclear areas.

Incorrect Options:

Option A . Mitochondria are abundant in cardiac muscle cells due to their high energy demands. They are responsible for generating ATP through oxidative phosphorylation, which is necessary for the contraction and relaxation of the cardiac muscle.

Option C . Golgi bodies are involved in processing and packaging proteins and lipids synthesized in the endoplasmic reticulum. They modify and sort these molecules and direct them to their specific destinations within the cell or for secretion outside the cell.

Option D. The endoplasmic reticulum (ER) is a complex network of membranes involved in protein synthesis, folding, and transport. It is responsible for the synthesis of proteins that are destined for secretion, as well as the synthesis of membrane lipids.

Solution for Question 5:

Correct option is D: Bainbridge reflex.

- The Bainbridge reflex is responsible for tachycardia during right atrial distension. When the right atrium becomes stretched or distended due to an increased volume of blood, specialized stretch receptors in the atrial wall called atrial mechanoreceptors are activated. These mechanoreceptors send signals to the cardiovascular control centers in the brain, resulting in an increase in heart rate through the Bainbridge reflex. This reflex helps to regulate cardiac output by increasing heart rate to accommodate the increased venous return to the heart .

Incorrect options:

Option A: J reflex : The J reflex is a pulmonary reflex that is involved in regulating airway resistance and breathing. It is not directly related to tachycardia during right atrial distension.

Option B: Cushing reflex: The Cushing reflex refers to a physiological response that occurs in the brain in response to increased intracranial pressure. It leads to increased blood pressure and decreased heart rate, which is the opposite of tachycardia.

Option C: Bezold-Jarisch reflex: The Bezold-Jarisch reflex is a reflex that can be triggered by stimulation of specialized sensory receptors in the heart, typically in response to excessive ventricular filling or ischemia. It leads to a decrease in heart rate, vasodilation, and hypotension.

Solution for Question 6:

Correct Option C - Peripheral resistance:

- The dicrotic notch is primarily caused by the closure of the aortic valve during ventricular diastole. This closure, in the presence of peripheral resistance, creates a momentary increase in pressure, leading to the notch in the arterial pressure waveform.

Incorrect Options:

Option A - Passive filling of blood in ventricles: The dicrotic notch is not directly related to the passive filling of blood in the ventricles.

Option B - Rapid ejection phase: The dicrotic notch occurs during the diastolic phase, not the rapid ejection phase.

Option D - Isovolumic contraction: The isovolumic contraction is a phase of the cardiac cycle where ventricular pressure increases without changes in volume. It is not directly associated with the dicrotic notch.

Solution for Question 7:

Correct statements

Statement 1: All or None Law This is a fundamental property of ventricular cardiac muscle, which states that once a threshold is reached, the cardiac muscle fibers contract maximally. There is no partial contraction in response to a stimulus; the response is either full or none.

Statement 2: Length-Tension Relationship Ventricular cardiac muscle follows the length-tension relationship, where the force of contraction is dependent on the initial length of the cardiac muscle fibers. This relationship is crucial for the heart's function, allowing it to pump efficiently by adjusting the force of contraction based on the volume of blood filling the heart (preload).

Statement 3: Long Refractory Period Cardiac muscle cells have a long refractory period, which is significantly longer than that of skeletal muscle cells. This long refractory period is essential to prevent tetanization (sustained contraction) and to ensure proper rhythmic contractions of the heart.

Incorrect statements

Statement 4: Can be Tetanized Unlike skeletal muscle, ventricular cardiac muscle cannot be tetanized due to its long refractory period. This characteristic is vital to maintaining the rhythmic and cyclical nature of heartbeats, preventing continuous contraction that would impede normal heart function.

Solution for Question 8:

Correct Option A - Decrease in preload:

- In hypovolemic shock, there is a significant loss of blood or fluid volume. This results in a reduced volume of blood returning to the heart (preload). Since preload is a key determinant of stroke volume and cardiac output, a decrease in preload will lead to a decreased cardiac output.

Incorrect Options:

Option B - Decrease in contractility of the heart: While decreased contractility can impact cardiac output, in this scenario of hypovolemic shock due to fluid loss, the primary issue is reduced preload rather than contractility.

Option C - Increase in preload: This is incorrect because in hypovolemic shock, preload is actually decreased due to fluid loss, not increased.

Option D - Increase in afterload: Although increased afterload can affect cardiac output, in the context of hypovolemic shock, the immediate cause of decreased cardiac output is the reduction in preload due to the decreased blood volume.

Solution for Question 9:

Correct Option A - Absent a wave:

- In the JVP waveform, the 'a' wave represents atrial contraction. An absent 'a' wave is typically seen in atrial fibrillation, where the atria do not contract effectively due to irregular impulses.

Incorrect Options:

Option B - Cannon a wave : Cannon 'a' waves are large, prominent waves in the JVP that occur when atrial contraction is against a closed tricuspid valve, often due to arrhythmias like complete heart block. This is not typically associated with an irregularly irregular pulse.

Option C - Prominent X descent : The X descent in the JVP represents atrial relaxation and descent of the tricuspid valve during ventricular systole. Prominent X descent is generally seen in conditions like tricuspid regurgitation or right ventricular failure but not specifically in atrial fibrillation.

Option D - Prominent Y descent : The Y descent reflects the rapid emptying of the right atrium into the right ventricle. Prominent Y descent can be seen in conditions such as constrictive pericarditis or right heart failure but is not a characteristic finding in atrial fibrillation.

Solution for Question 10:

Correct Answer: A) Right Atrial contraction

Explanation:

The "a" wave in the jugular venous pressure (JVP) waveform corresponds to right atrial contraction .

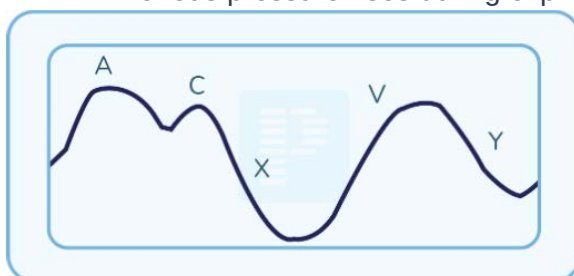
- Mechanism: During the right atrial systole , blood is pushed into the right ventricle , causing a transient increase in right atrial pressure , which is reflected as the "a" wave in the JVP. This occurs just before the first heart sound (S1) and the QRS complex on ECG.
- During the right atrial systole , blood is pushed into the right ventricle , causing a transient increase in right atrial pressure , which is reflected as the "a" wave in the JVP.
- This occurs just before the first heart sound (S1) and the QRS complex on ECG.
- During the right atrial systole , blood is pushed into the right ventricle , causing a transient increase in right atrial pressure , which is reflected as the "a" wave in the JVP.
- This occurs just before the first heart sound (S1) and the QRS complex on ECG.

Jugular Venous Pressure (JVP) Waveform:

a wave	Caused by atrial systole. Atrial contraction results in a temporary halt in venous inflow. Some blood flows back into the veins, causing a rise in venous pressure.
c wave	Arises from the increase in atrial pressure during isovolumetric ventricular contraction. Pressure is transmitted through the closed and bulging tricuspid valve.
x descent	Ventricular systole pulls the tricuspid valve down
v wave	Corresponds to the rise in atrial pressure as blood fills the atria. Occurs during diastole while the tricuspid valve is still closed.
y descent	Opening of tricuspid valve causing emptying of right atrium into ventricle

Respiration Influence :

- Venous pressure decreases during inspiration due to negative intrathoracic pressure.
- Venous pressure rises during expiration.



Tricuspid valve bulging into the right atria (Option B) is incorrect because this corresponds to the "c" wave, which occurs when the tricuspid valve bulges into the atrium during isovolumetric ventricular contraction.

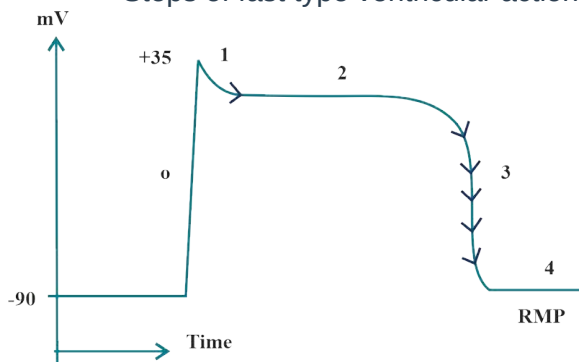
Right atrial relaxation (Option C) is incorrect because this occurs after the "a" wave and corresponds to the "x" descent in the JVP waveform.

Right atrial filling (Option D) is incorrect because this corresponds to the "v" wave, which represents venous return filling the right atrium against a closed tricuspid valve.

Solution for Question 11:

Correct Option C - C, D, A, B:

- Steps of fast type ventricular action potential in the correct order are:



Phase 0 - Opening of voltage-gated sodium channels: This step involves the rapid influx of sodium ions into the heart muscle cells, leading to cell membrane depolarization.

Phase 1 - Transient outward flow of potassium: This step involves the transient (brief) outward flow of potassium ions through K_T channel from the heart muscle cells, leading to partial cell membrane repolarization.

Phase 2 - Opening of voltage-gated calcium channels: This step involves the influx of calcium ions through L-type of calcium channels into the heart muscle cells along with efflux of the potassium channels balancing the charge flow and resulting in the plateau phase.

Phase 3 - Repolarizing K^+ currents: This step involves the outward flow of potassium ions through delayed rectifier channels from the heart muscle cells, repolarizing the cell membrane and restoring its resting state.

Incorrect Options:

- Options A, B & D are incorrect.

Solution for Question 12:

Correct Options B - 16 times:

- Flow is equal to pressure difference divided by resistance. So, flow is inversely proportional to the resistance of the vessel.
- According to the Hagen–Poiseuille equation, resistance can be directly measured by the following formula. $R = 8\eta L / \pi r^4$ Where, R = resistance. η = viscosity. L = length. r = radius of the vessel. From this equation, the resistance of a vessel is inversely proportional to its r^4 (radius). If other parameters remain constant. So, if the radius is doubled of the original, then the resistance will be: $R = 1/(2)^4$. $R = 1/16$.
- $R = 8\eta L / \pi r^4$
- Where, R = resistance. η = viscosity. L = length. r = radius of the vessel.
- R = resistance.
- η = viscosity.
- L = length.
- r = radius of the vessel.
- From this equation, the resistance of a vessel is inversely proportional to its r^4 (radius).
- If other parameters remain constant.
- So, if the radius is doubled of the original, then the resistance will be:
- $R = 1/(2)^4$.
- $R = 1/16$.
- So, the resistance will decrease by 16 times.
- So, the flow will increase by 16 times.
- $R = 8\eta L / \pi r^4$
- Where, R = resistance. η = viscosity. L = length. r = radius of the vessel.
- R = resistance.
- η = viscosity.
- L = length.

- r = radius of the vessel.
- From this equation, the resistance of a vessel is inversely proportional to its r^4 (radius).
- If other parameters remain constant.
- So, if the radius is doubled of the original, then the resistance will be:
- $R = 1/(2)^4$.
- $R = 1/16$.
- R = resistance.
- η = viscosity.
- L = length.
- r = radius of the vessel.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 13:

Correct Option D - AV Valves are closed:

- During isovolumetric relaxation, the ventricles are relaxing, and the AV valves are closed. Both the tricuspid and mitral valves prevent blood from flowing back into the atria.

Incorrect Options:

Option A - Semilunar valves open: Isovolumetric relaxation is a phase where both the AV valves (tricuspid and mitral valves) and the semilunar valves (aortic and pulmonary valves) are closed.

Option B - C Wave of JVP: The C wave of the jugular venous pulse (JVP) is associated with the bulging of the tricuspid valve into the right atrium during ventricular contraction, not with isovolumetric relaxation.

Option C - Corresponds to QT interval: The QT interval on an electrocardiogram represents the total time for ventricular depolarization and repolarization, not specifically isovolumetric relaxation.

Solution for Question 14:

Correct Option A - Negative feedback :

- The baroreceptor reflex is a negative feedback mechanism that helps regulate blood pressure. It involves stretch receptors called baroreceptors, which are located in the walls of blood vessels such as the carotid sinus and aortic arch, as well as the heart. These receptors detect changes in blood pressure and send signals to the brain, specifically the cardiovascular control center in the medulla oblongata, to initiate a response.
- When blood pressure increases, the nucleus tractus solitarius (NTS) is activated, which in turn activates the caudal ventrolateral medulla (CVLM) and inhibits the rostral ventrolateral medulla (RVLM). This results in decreased sympathetic stimulation, leading to decreased cardiac output and contractility.
- Conversely, if blood pressure decreases, the NTS and CVLM are not activated, which means they do not inhibit the RVLM. This leads to an increase in sympathetic supply, resulting in increased cardiac output.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 15:

Correct Option C - Arterial wall compliance:

- Pulse pressure is the difference between systolic blood pressure and diastolic blood pressure. Ratio of stroke volume to the compliance of the aorta is also known as pulse pressure. So, pulse pressure is inversely proportional to the compliance of the aorta.
- Aorta wall compliance refers to the elasticity and distensibility of the arterial walls. When aorta wall compliance decreases, the arteries become stiffer and less expandable with each cardiac cycle. This results in an increase in pulse pressure, as the pressure difference between systole and diastole becomes greater.

Incorrect Options:

Option A - Myocardial contractility: Myocardial contractility can affect stroke volume and indirectly affects the pulse pressure.

Option B - Stroke volume: Stroke volume is directly proportional to the pulse pressure, i.e., with decrease in stroke volume pulse pressure will also increase. The amount of blood ejected by the heart with each contraction. While changes in stroke volume can influence pulse pressure, they are not the primary cause of the increased pulse pressure in this case.

Option D - Cardiac output: Cardiac output is the volume of blood pumped by the heart per unit of time i.e., stroke volume multiplied with heart rate and indirectly affects the pulse pressure..

Solution for Question 16:

Correct Option A - Calcium:

- The action potential in SA node is a Slow-type of action potential with only 3 phases that are phase 4, 0, and 3.
- Phase 0: A large number of L-type Ca^{++} channels, which are slow in nature, open and result in the influx of calcium, leading to depolarization, and the membrane voltage slowly rises to +10 mV.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to the explanation of the correct answer.
-

Solution for Question 17:

Correct Option A - Iso-volumetric contraction:

- The 'c' wave in the JVP is typically seen during the isovolumetric contraction phase of the cardiac cycle. It occurs during early ventricular systole when the tricuspid valve bulges back into the right atrium due to the increased pressure in the right ventricle. This causes a slight rise in the JVP.
- The isovolumetric contraction occurs immediately after the closure of the atrioventricular valves (mitral and tricuspid valves) and before the onset of ventricular ejection.

Incorrect Options:

- Options B, C, and D are incorrect.
-

Solution for Question 18:

Correct Option B - Depolarization:

- In the cardiac muscle, the Na⁺ + K⁺ + ATPase is an essential membrane-bound enzyme that maintains the resting membrane potential of the cell membrane. It actively transports sodium ions (Na⁺) out of the cell and potassium ions (K⁺) into the cell against their concentration gradients, utilizing ATP as an energy source.
- When the Na⁺ + K⁺ + ATPase is inhibited, the efflux of Na⁺ and influx of K⁺ is reduced, which in turn disrupts a secondary active transporter, i.e., NCX (Sodium calcium exchanger) responsible for the efflux of the calcium out of the cell and influx of Na⁺.
- This disruption of the Na⁺ + K⁺ + ATPase results in an increase of the positive charge inside the cell leading to depolarization.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.
-

Solution for Question 19:

Correct Option B - Aortic Valve Opens:

- In a left ventricular (LV) pressure-volume loop, point C represents the transition from the filling phase to the ejection phase of the cardiac cycle. During this transition, the ventricular volume reaches its maximum, and the ventricular pressure continues to rise leading to opening of the aortic valve and marks the beginning of the ejection phase..

Incorrect Options:

Option A - Mitral valve closes: It is represented by the point B.

Option C - Aortic valve closes: Closure of aortic valve is represented by point D.

Option D - Mitral valve opens : It is represented by point A.

Solution for Question 20:

Correct Option A - Decrease in firing from baroreceptor:

- When a person transitions from supine to standing, the effect of gravity causes blood to pool in the lower extremities. This decreases the venous return to the heart and subsequently reduces cardiac output. In response, the baroreceptors detect the decrease in blood pressure and initiate the baroreceptor reflex.
- The baroreceptor reflex involves a decrease in firing from the baroreceptors, which leads to a subsequent increase in sympathetic nervous system activity and a decrease in parasympathetic (vagal) output.
- This results in vasoconstriction of peripheral blood vessels, increased heart rate, and increased cardiac contractility, and increased venous return, and maintains the blood pressure.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 21:

Correct Option A - Atrioventricular valves open:

- The cardiac event that occurs after the isometric relaxation phase is the opening of the atrioventricular valves followed by the first rapid filling phase of the ventricles.

Incorrect Options:

Option B - Atrioventricular valves close: Closure of atrioventricular valves marks the isovolumetric contraction phase.

Option C - Corresponds to the peak of the C wave in JVP: Isovolumic relaxation phase is marked by 'v' wave. The C wave represents the bulging of the tricuspid valve into the right atrium during ventricular contraction.

Option D - Corresponds to the T wave in the ECG: The T wave in the ECG represents ventricular repolarization, which occurs before the isometric relaxation phase.

Solution for Question 22:

Correct Option A - Tricuspid bulge into right atrium during isovolumetric ventricular contraction:

- The 'c' wave of the jugular venous pulse (JVP) represents the bulging of the tricuspid valve into the right atrium during isovolumetric ventricular contraction.

Incorrect Options:

Option B - Contraction of right atrium: Contraction of the right atrium is represented by the 'a' wave.

Option C - Passive filling of right atrium: Passive filling of the right atrium is represented by the 'v' wave .

Option D - Passive emptying of right atrium: Passive emptying of the right atrium is not represented by the 'y descent' wave.

Solution for Question 23:

Correct Option C - Isovolumetric contraction:

- The first heart sound, also known as S1, is a low-pitched sound that occurs due to the closure of the atrioventricular valve during the beginning of isovolumetric contraction.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 24:

Correct Option A - Tricuspid Regurgitation:

- In a jugular venous examination, the presence of a prominent 'V' wave in the jugular venous pulse can be indicative of tricuspid regurgitation. Tricuspid regurgitation is a valvular heart condition where the tricuspid valve does not close properly, allowing blood to flow backward from the right ventricle to the right atrium during systole. This regurgitant flow leads to an increased 'V' wave in the jugular venous pulse.

Incorrect Options:

Option B - Cardiomyopathy: It is characterized by prominent 'a' waves.

Option C - Tricuspid Stenosis: It is characterized by prominent 'a' waves.

Option D - Atrial Fibrillation: It is characterized by absent 'a' waves.

Solution for Question 25:

Correct Option B - Plateau phase of the fast-type action potential :

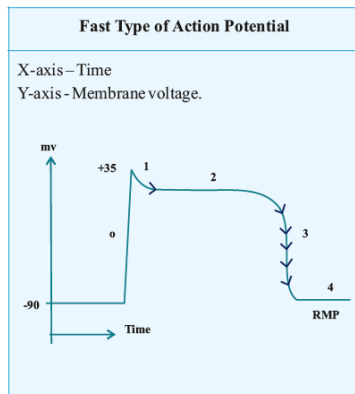
- The point marked on the myocardial action potential curve represents phase 2, the plateau phase.
- This phase is characterized by the slow but prolonged opening of L-type calcium channels, aka dihydropyridine receptor (DHPR) designated as CaV 1.2, which allows the influx of calcium ions into the cell.
- An outward potassium current balances this inward current through the slow, delayed rectifier potassium channels.
- The net effect is a plateau phase in the action potential, making the cardiac muscle action potential unique compared to other types of action potentials.

Incorrect Options:

Option A: Seen in slow type Slow Type of Action Potential

Options B: Represents phase 3

Options C: Represents phase 4



Solution for Question 26:

Correct Option D - A, C, D:

Methods used for the measurement of cardiac output are:

- ECHO : The stroke volume is measured by assessing blood velocity through the arch of the aorta and multiplying the stroke volume by the heart rate.
- Thermodilution: The thermodilution method is done by inserting a catheter at the level of pulmonary artery, hence known as the pulmonary artery catheter method. The thermodilution method is the gold-standard method. It works best for a high cardiac output state of the body.
- Fick's Principle: Fick's principle is a method used to calculate cardiac output based on the oxygen consumption of the body. By measuring the oxygen content of arterial and venous blood, along with the oxygen consumption, cardiac output can be calculated using the formula:
$$\text{Cardiac Output} = \frac{\text{Oxygen Consumption}}{(\text{Arterial Oxygen Content} - \text{Venous Oxygen Content})}$$

Incorrect Options:

Options A, B, and C are incorrect.

Solution for Question 27:

Correct Option C - Autonomic nervous system:

- The autonomic nervous system plays a crucial role in regulating various involuntary functions in the body, including heart rate. It controls the balance between the sympathetic and parasympathetic nervous systems, which are responsible for accelerating or decelerating heart rate, respectively. Fetal heart rate acceleration is primarily mediated by the autonomic nervous system in response to stimuli like fetal movement.

Incorrect Options:

Option A - Carotid baroreceptor: These receptors are involved in regulating blood pressure by sensing changes in arterial pressure.

Option B - Aortic baroreceptor: Similar to carotid baroreceptors, they also play a role in regulating blood pressure.

Option D - Chemoreceptors: Chemoreceptors are sensory receptors that detect changes in chemical concentrations, such as oxygen and carbon dioxide levels, in the blood.

Solution for Question 28:

Correct Option C - Radial artery:

- The Windkessel effect is due to arterial elasticity, which facilitates the movement of blood in arteries in forward direction during diastole.
- The radial artery is a muscular artery rather than an elastic artery, and it does not exhibit the same level of elasticity and compliance as large elastic arteries like the aorta.

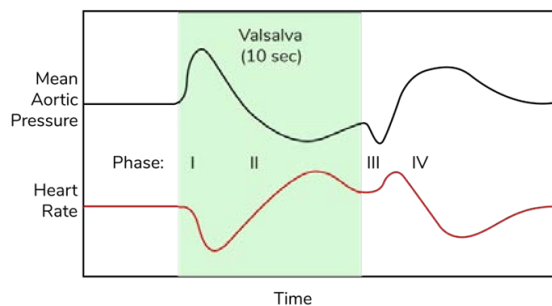
Incorrect Options:

- Options A, B, and D are incorrect. These arteries show the Windkessel effect.
-

Solution for Question 29:

Correct Option C - Increase in aortic pressure:

- The Valsalva maneuver is a technique used to assess autonomic nervous system function and cardiovascular response.



- Valsalva maneuver has four phases: Phase I (onset of strain): In this phase, there is a transient increase in blood pressure that lasts for a few seconds. This occurs due to increased intrathoracic pressure adding to the aortic pressure. Phase II (phase of straining): In this phase, venous return initially decreases, lowering cardiac output and blood pressure for 4 seconds. Later, blood pressure returns to normal due to increased peripheral resistance from sympathetic vasoconstriction. Heart rate steadily increases due to vagal withdrawal and sympathetic activation. Phase III: This phase occurs following the release of strain characterized by a transient decrease in blood pressure lasting for a few seconds. This is caused by the mechanical displacement of blood to the pulmonary vascular bed, which was under increased intrathoracic pressure. There is little change in heart rate. Phase IV: Occurs after further release of strain; blood pressure slowly increases, and heart rate decreases for about a minute. This is due to an increase in venous return, stroke volume, and cardiac output.

- Phase I (onset of strain): In this phase, there is a transient increase in blood pressure that lasts for a few seconds. This occurs due to increased intrathoracic pressure adding to the aortic pressure.

- Phase II (phase of straining): In this phase, venous return initially decreases, lowering cardiac output and blood pressure for 4 seconds. Later, blood pressure returns to normal due to increased peripheral resistance from sympathetic vasoconstriction. Heart rate steadily increases due to vagal withdrawal and sympathetic activation.

- Phase III: This phase occurs following the release of strain characterized by a transient decrease in blood pressure lasting for a few seconds. This is caused by the mechanical

displacement of blood to the pulmonary vascular bed, which was under increased intrathoracic pressure. There is little change in heart rate.

- Phase IV: Occurs after further release of strain; blood pressure slowly increases, and heart rate decreases for about a minute. This is due to an increase in venous return, stroke volume, and cardiac output.
- Phase I (onset of strain): In this phase, there is a transient increase in blood pressure that lasts for a few seconds. This occurs due to increased intrathoracic pressure adding to the aortic pressure.
- Phase II (phase of straining): In this phase, venous return initially decreases, lowering cardiac output and blood pressure for 4 seconds. Later, blood pressure returns to normal due to increased peripheral resistance from sympathetic vasoconstriction. Heart rate steadily increases due to vagal withdrawal and sympathetic activation.
- Phase III: This phase occurs following the release of strain characterized by a transient decrease in blood pressure lasting for a few seconds. This is caused by the mechanical displacement of blood to the pulmonary vascular bed, which was under increased intrathoracic pressure. There is little change in heart rate.
- Phase IV: Occurs after further release of strain; blood pressure slowly increases, and heart rate decreases for about a minute. This is due to an increase in venous return, stroke volume, and cardiac output.

Incorrect Options:

Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 30:

Correct Option A - Conduction through AV nodes:

- The PR interval represents the time it takes for the electrical impulse to travel from the atria, through the atrioventricular (AV) node, and into the ventricles of the heart. It is measured from the beginning of the P wave (which represents atrial depolarization) to the beginning of the QRS complex (which represents ventricular depolarization).
- The AV node acts as a gateway between the atria and the ventricles, slowing down the electrical impulse to allow for proper coordination and synchronization of atrial contraction before ventricular contraction. The PR interval, therefore, reflects the time it takes for the electrical signal to travel through the AV node.

Incorrect Options:

Option B - Ventricular depolarization: It is marked by the R wave of the ECG.

Option C - Ventricular repolarization: Ventricular repolarization is represented by the T wave on an electrocardiogram (ECG).

Option D - Atrial depolarization: It is represented by the P wave of the ECG.

Solution for Question 31:

Correct Option. A - SA node:

- The SA node, also known as the pacemaker of the heart, is a specialized group of cells located in the right atrium. It is responsible for initiating the electrical impulses that regulate the heart's rhythm. The SA node generates electrical impulses spontaneously, causing the atria to contract and initiating the electrical conduction system of the heart.

Incorrect Options:

- Options B ,C, and D are incorrect. These structures are latent pacemakers of the heart and help in the conduction of electrical impulse.

Solution for Question 32:

Correct Option C - Length-tension relationship:

- Accurately represents the relationship between the stretching of the cardiac muscle fibers (due to EDV) and the force of contraction (stroke volume). When the cardiac muscle fibers are stretched to an optimal length , they can generate a stronger contraction, leading to a larger stroke volume.

Incorrect Options:

- Options A, B, and D are incorrect.

Solution for Question 33:

Correct Option B - C wave:

- The jugular venous pressure (JVP) reflects the pressure in the right atrium, corresponding to different cardiac cycle phases.
- The C wave in the JVP corresponds to the isovolumetric contraction phase of the cardiac cycle. It occurs during early ventricular systole when the tricuspid valve bulges back into the right atrium due to the increased pressure in the right ventricle. This causes a slight rise in the JVP.

Incorrect Options:

Option A - A wave: The A wave in the JVP represents atrial contraction.

Option C - X descent: The X descent in the JVP represents the atrial relaxation phase.

Option D - Y descent: The Y descent in the JVP represents the rapid filling phase of the right ventricle. It occurs during early diastole when the tricuspid valve opens, and blood flows from the right atrium to the right ventricle.

Solution for Question 34:

Correct Option B - Bain bridge reflex:

- The Bainbridge reflex is a cardiovascular reflex that increases heart rate in response to increased blood volume in the right atrium.
- Special pressure sensors called baroreceptors (or veno-atrial stretch receptors) located in the right atrium of the heart detect increases in the volume and pressure of blood returned to the heart.
- These receptors transmit information along the vagus nerve (10th cranial nerve) to the central nervous system.

- This response activates sympathetic nerve pathways that increase the strength of contraction of the heart muscle and heart rate (tachycardia).

Incorrect Options:

Option A - Bezold-Jarisch reflex:

- The Bezold-Jarisch reflex is an inhibitory reflex usually denoted as a cardioinhibitory reflex defined as bradycardia, vasodilation, and hypotension resulting from the stimulation of cardiac receptors.
- The Bezold-Jarisch reflex originates from inhibitory mechanoreceptors in the left ventricle (particularly the infero-posterior wall).

Option C - Cushing reflex:

- The Cushing reflex is a physiological nervous system response to acute elevations of intracranial pressure (ICP), resulting in Cushing's triad of widened pulse pressure (increasing systolic, decreasing diastolic), hypertension, bradycardia, and irregular respirations.

Option D - Baroreceptor reflex:

- Baroreceptors are present in the carotid sinus, arch of aorta, and adventitial layer of blood vessels.

Solution for Question 35:

Correct Option C - Cardiogenic shock:

Pulmonary capillary wedge pressure (PCWP) increases in the following condition:

- Pulmonary capillary wedge pressure measures the left atrium pressure. Cardiogenic shock refers to a state of inadequate cardiac output due to severe dysfunction or failure of the heart. In this condition, PCWP typically increases. Cardiogenic shock can result from conditions such as myocardial infarction (heart attack), severe heart failure, or arrhythmias, leading to decreased cardiac output and elevated pressures within the heart chambers, including the left atrium.
- The elevated PCWP reflects the increased pressure in the left atrium, which is transmitted to the pulmonary capillaries.

Incorrect Options:

Option A - Hypovolemic shock: Hypovolemic shock occurs due to a significant loss of blood volume, such as from hemorrhage or fluid loss. In hypovolemic shock, the PCWP may be decreased.

Option B - Septic shock: Septic shock is a severe systemic infection characterized by inflammatory response causing vasodilation and increased capillary permeability, which can lead to decreased systemic vascular resistance and lower filling pressures. In septic shock, the PCWP may be normal or decreased..

Option D - Anaphylactic shock: In anaphylactic shock, the PCWP may be normal or decreased. The systemic vasodilation and increased capillary permeability associated with anaphylaxis can result in decreased filling pressures, including PCWP.

Solution for Question 36:

Correct Option C - Closure of semilunar valves:

- Closure of semilunar valves: The aortic notch is a small downward deflection that occurs in the aortic pressure waveform after the closure of the aortic valve (semilunar valve) .

Incorrect Options:

- Options A, B, and D are incorrect. Aortic notch is formed due to closure of aortic valve.
-

Solution for Question 37:

Correct Option A - Potassium ions:

- Isovolumic relaxation is marked by closure of the semilunar valves and already closed atrioventricular valves. The duration of this phase is 0.06 sec.
- It corresponds to the 'v' wave of jugular venous pulse.

Incorrect Options:

Option B - Corresponds to QT interval: Ventricular depolarization and repolarization corresponds to the QT interval.

Options C and D are incorrect. Refer to Option A for an explanation.

Solution for Question 38:

Correct Option B - 81 times:

- According to the Hagen–Poiseuille equation, resistance can be directly measured by the following formula.
- $R = 8\eta L / \pi r^4$
- Where, R = resistance. η = viscosity. L = length. r = radius of the vessel.
- R = resistance.
- η = viscosity.
- L = length.
- r = radius of the vessel.
- From this equation, the resistance of a vessel is inversely proportional to its r^4 (radius). If other parameters remain constant.
- So, if the radius is reduced by 1/3rd of the original, then resistance will be: $R = 1 / (1/3)^4$. $R = 81$.
- $R = 1 / (1/3)^4$.
- $R = 81$.
- So, the resistance will increase by 81 times.
- R = resistance.
- η = viscosity.
- L = length.
- r = radius of the vessel.
- $R = 1 / (1/3)^4$.
- $R = 81$.

Incorrect Options:

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Blood Physiology

1. Which of the following conditions is associated with a right shift in Arneth count?

- A. Active tuberculosis
 - B. Acute progressive infections
 - C. Bone marrow suppression
 - D. Bone marrow stimulation
-

2. Which of the following statements about platelets are true?

- A. Alpha granules contain clotting factors and platelet-derived growth factor (PDGF)
 - B. Splenectomy causes a decrease in the platelet count
 - C. Interleukins IL-2, IL-4, IL-5, and IL-7 are involved in stimulating platelet production
 - D. Thrombopoietin is produced exclusively by megakaryocytes and regulates platelet production
-

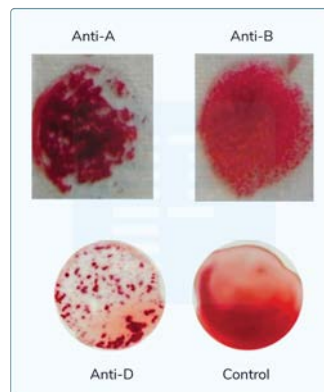
3. Which of the following coagulation factors is crucial for initiating the extrinsic pathway of blood coagulation?

- A. Factor II
 - B. Factor III
 - C. Factor V
 - D. Factor VII
-

4. Identify the incorrect statement:

- A. Platelet concentrates have a shelf life of up to 8 days at $+22^{\circ}\text{C} \pm 2^{\circ}\text{C}$
 - B. Red cell concentrates stored with CPDA-1 have a shelf life of 35 days
 - C. Plasma stored at temperatures colder than -25°C has a shelf life of up to 3 years
 - D. Red cell concentrates stored with CPD have a shelf life of 28 days
-

5. A 45-year-old patient needs an urgent blood transfusion. The image below shows the results of the patient's blood group typing test. Based on the agglutination patterns observed, identify the patient's blood group?



- A. O positive (O+ve)
 - B. A positive (A+ve)
 - C. AB positive (AB+ve)
 - D. B positive (B+ve)
-

6. A pediatrician is reviewing the development of hematopoiesis in a fetus at various stages of gestation. They are particularly interested in understanding the shifts in hematopoietic activity locations as the fetus develops. At which gestational age does the bone marrow become the primary site of hematopoiesis?

- A. 3-6 weeks
 - B. 6-8 weeks
 - C. Around 20 weeks
 - D. After birth
-

7. A researcher is studying the behavior of hematopoietic stem cells (HSCs) and wants to understand the mechanisms that regulate their self-renewal during tissue repair. Which type of stem cell division increases the stem cell pool during tissue repair?

- A. Asymmetric division
 - B. Symmetric division
 - C. Terminal differentiation
 - D. Lateral inhibition
-

8. Which of the following statements best describes the Burst Forming Unit-Erythroid (BFU-E) in the process of erythropoiesis?

- A. It is a mature red blood cell that participates in oxygen transport
 - B. It is a type of stem cell that can differentiate into any blood cell type
 - C. It is an early progenitor cell in erythropoiesis that gives rise to CFU-E
 - D. It is a precursor cell in erythropoiesis that directly produces reticulocytes
-

9. Which of the following statements regarding erythropoietin is not true?

- A. It is majorly produced by the liver
- B. Major stimulus for its production is hypoxia

- C. It acts on JAK-STAT pathway
 - D. The maximum erythropoietin receptor density is seen on CFU-E
-

10. How many oxygen atoms can one molecule of hemoglobin bind?

- A. 4
 - B. 2
 - C. 6
 - D. 8
-

11. Which of the following is the primary protein responsible for binding and transporting free heme in the bloodstream?

- A. Haptoglobin
 - B. Albumin
 - C. Hemopexin
 - D. Transferrin
-

12. A 45-year-old female with asthma presents with persistent coughing, wheezing, and shortness of breath. Which interleukin is most directly involved in the activation of eosinophils in this condition?

- A. IL-1
 - B. IL-4
 - C. IL-5
 - D. IL-10
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	2
Question 4	1
Question 5	3
Question 6	3
Question 7	2
Question 8	3
Question 9	1
Question 10	4
Question 11	3
Question 12	3

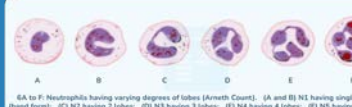
Solution for Question 1:

Correct Answer : C) Bone marrow suppression

Explanation:

The presence of older neutrophils (N4 and N5) indicates suppression of bone marrow with a shift to the right in Arneth count.

The Arneth count is a method used to classify neutrophils as N1 to N5 based on the number of lobes in their nuclei.

 6A to F: Neutrophils having varying degrees of lobes (Arneeth Count). (A and B) N1 having single band (band form); (C) N2 having 2 lobes; (D) N3 having 3 lobes; (E) N4 having 4 lobes; (F) N5 having 5 lobes	
Classification of Neutrophil	Class I : Myelocytes and neutrophils with indented or horseshoe-shaped nuclei. (5%) Class II : Neutrophils with bilobed nuclei. (35%) Class III : Neutrophils with tri-lobed nuclei. (41%) Class IV : Neutrophils with four-lobed nuclei. (17%) Class V : Neutrophils with five or more lobes in their nuclei. (2%)
Toxic Conditions	Fewer cells in Classes IV and V indicate a left-sided shift Active tuberculosis often shows a left shift. (Option A) Other infections showing a left shift may have superimposed tuberculosis
Extrinsic Factors Influencing Arneeth Count	Certain drugs, tissue extracts, and hormones. Trauma , various forms of radiation. Induction of anesthesia.
Infections and Arneeth Count	Acute progressive infections: Notable left shift. (Option B) As infection subsides: Less marked left shift, may shift to the right. Chronic infections: Other features like lymphocytosis come into range.
Normal Neutrophil Distribution	N1 and N3 constitute most neutrophils in peripheral blood.
Bone Marrow Stimulation	Presence of younger neutrophils (N1 and N2) indicates stimulation of bone marrow (shift to the left). (Option D)
Bone Marrow Suppression	Presence of older cells (N4 and N5) indicates suppression of bone marrow (shift to right) (Option C)
Hypersegmented Neutrophil	Seen in megaloblastic anemia due to folate and vitamin B12 deficiency (shift to right)

Solution for Question 2:

Correct Answer: A) Alpha granules contain clotting factors and platelet-derived growth factor (PDGF)

Explanation: Alpha granules are abundant in platelets, about 50-80 granules per platelet. They contain secreted proteins like clotting factors and platelet-derived growth factor(PDGF).

Platelets

General Description	Small, granulated, non-nucleated discs important for hemostasis
Size	1-4 µm in diameter
Concentration in Blood	150,000 - 450,000/µL
Half-life	~4 days
Origin	Formed from megakaryocytes in the bone marrow
Distribution	60-75% in circulation; the remainder mainly in the spleen
Splenectomy Effect	Causes an increase in platelet count (thrombocytosis) (Option B)
Thrombopoietin	Produced by liver and kidneys; regulates platelet production (Option D)
Membrane Receptors	For collagen, ADP, von Willebrand factor and fibrinogen
Cytoplasmic Components	Actin, myosin, glycogen, lysosomes and granules
Granules	Alpha granules: Soluble coagulation proteins (vWF, Factor V, II), adhesion molecule, growth factor (IGF, PDGF , TGF-β, platelet factor 4), Integrin, Cytokines (Option A) Dense granules: ADP, ATP, Calcium, Serotonin
Response to Injury	Adhere to exposed collagen and von Willebrand factor; release granule contents upon activation
Deficiency Effect	Thrombocytopenic purpura (easy bruising, multiple subcutaneous hemorrhages)
Interleukins	IL-1, IL-3, IL-6, IL-11 stimulate platelet production (Option C)
Factors increasing platelet synthesis	Inflammation, thrombopoietin and IL-1, IL-3, IL-6, IL-11

Solution for Question 3:

Correct Answer: B) Factor III

Explanation: Factor III, also known as Tissue thromboplastin , is responsible for the initiation of the extrinsic pathway of coagulation.

Clotting Factors:

Clotting Factor	Synonyms	Function	Half life
I	Fibrinogen	Clot formation	3-5 days
II (Option A)	Prothrombin	Converts fibrinogen to fibrin Activates platelets Also has anti clotting activity Has growth factor and cytokine like activities	60 hours
III (Option B)	Tissue thromboplastin	Initiator of extrinsic pathway of coagulation Activates factor VII	-
IV	Calcium	Convert X to Xa Convert prothrombin to thrombin	-
V (Option C)	Proaccelerin; labile factor; accelerator globulin (Ac-G)	Activate prothrombin to thrombin Participates in activation of X	12 hours
VII (Option D)	Serum prothrombin conversion accelerator (SPCA); proconvertin; stable factor	Activates factor IX and X	3.5 hours
VIII	Antihemophilic factor(AHF); antihemophilic globulin(AHG); antihemophilic factor A	Activates factor X	8 - 12 hours
IX	Plasma thromboplastin component (PTC); antihemophilic factor B ; christmas factor	Activates factor X	18 - 24 hours
X	Stuart factor; Stuart-Prower factor	Conversion of prothrombin to thrombin	34 - 40 hours
XI	Plasma thromboplastin antecedent (PTA); antihemophilic factor C	Activates factor IX	45 hours
XII	Hageman factor; glass factor	Activates factor XI	-
XIII	Fibrin- stabilizing factor; Laki-Lorand factor	Crosslinks fibrin	10 days
Prekallikrein	Fletcher factor	Contact activation of blood coagulation	35 hours

High molecular weight kininogen (HMW-K)	Fitzgerald factor	Non enzymatic cofactor in contact activation Act as thiol protease inhibitor	150 hours
Platelets phospholipid	PL	Temporary hemostasis Synthesis factor V and XI	4 days

Deficiency of Factor	Clinical Syndrome	Cause
I	Afibrinogenemia	Depletion during pregnancy with premature separation of placenta; also congenital (rare)
II	Hypoprothrombinemia (hemorrhagic tendency in liver disease)	Decreased hepatic synthesis, often due to vitamin K deficiency
V	Parahemophilia	Congenital
VII	Hypoconvertinemia	Congenital
VIII	Hemophilia A (classic hemophilia)	Congenital defect in the gene on the X chromosome coding for factor VIII; inherited as a sex-linked characteristic.
IX	Hemophilia B (Christmas disease)	Congenital
X	Stuart-Prower factor deficiency	Congenital
XI	PTA deficiency	Congenital
XII	Hageman trait	Congenital

Solution for Question 4:

Correct Answer: A) Platelet concentrates have a shelf life of up to 8 days at $+22^{\circ}\text{C} \pm 2^{\circ}\text{C}$

Explanation: Platelets can be kept for as long as 8–13 days, but blood banks are only allowed to keep them for 5 days because of bacterial contamination.

Blood constituents	Preservative/ Shelf life	Uses	Temperature
--------------------	-----------------------------	------	-------------

Red cells	Citrate-citric acid- glucose (ACD) / 21 days	Blood transfusion	+4°C ± 2°C
Citrate phosphate dextrose (CPD) / 28 days (Option D)			
CPDA-1(CPD with adenine) / 35 days (Option B)			
CPD+ Additive solution / 42 days			
Platelets	Plasma / 5-7 days (5 days preferred due to bacterial risk) (Option A)	Platelet therapy	+22°C ± 2°C
Plasma	CPD,ACD, Buffered Citrate/ 3 years if frozen/ 3 months	Stable clotting factors (FII, FVII, FIX, FX, fibrinogen)	+4°C ± 2°C
Labile clotting factors (FV, FVIII)	3 years if colder than -25°C (Option C) / 3months if stored at -18°C to -25°C		

Solution for Question 5:

Correct Answer: C) AB positive (AB+ve)

Explanation: In the above clinical scenario, the image shows agglutination with anti-A and anti-B and positive anti-D (Rh) sera, indicating the presence of A antigen, B antigen and positive Rh factor. Therefore, the blood group is AB positive (AB+ve).

A-B-O System:

Blood type	Antigens on RBC	Antibodies in Plasma	Genotypes	Frequency
O	-	Anti-A and Anti-B	OO	47%
A	A	Anti-B	OA or AA	41%
B	B	Anti-A	OB or BB	9%
AB	A and B	-	AB	3%

Landsteiner's Law states that if a particular agglutinin is present on the red cell membrane of an individual, the corresponding agglutinin must be absent in his plasma and vice versa.

Agglutination test results:

Anti-A	Anti-B	Blood group
		Agglutination in anti-A Blood group : A
		Agglutination in anti-B Blood group : B
		No agglutination in both anti-A and anti-B Blood group : O
		Agglutination in both anti-A and anti-B Blood group : AB

H Antigen:

- The H gene codes for fucose transferase, forming the H antigen, found in all blood types .
- In addition, Type A adds N-acetylgalactosamine, Type B adds galactose, Type AB has both transferases, and Type O has neither, leaving the H antigen unchanged .
- Blood group O with H antigen in plasma or saliva is called the OH group or Bombay (Oh : if H is not dominant).

Bombay Blood Group :

- Phenotype: Oh , lacking the H gene and unable to form A or B antigens.
- Identified as group O in routine tests, with serum containing anti-A, anti-B, and anti-H antibodies. Therefore transfusion can be done only with Bombay blood group.

Rh Factor :

Common Rh antigen	C, D, E, c, d and e
Detection	Only in red cells
D antigen	Most antigenic, defines Rh positive
No D antigen	Rh negative ; Forms anti-D agglutinins upon exposure to Rh+ cells (Option B)
Prevalence	85% of whites Rh+ve ,15% Rh-ve Over 99% of Asians Rh+ve
Anti-D antibody formation	Requires exposure (transfusion or feto-maternal blood transfer) to D-positive red cells
Transfusion reaction onset	Requires massive exposure to Rh antigen to produce enough agglutinins for significant transfusion reaction to

	develop
--	---------

Solution for Question 6:

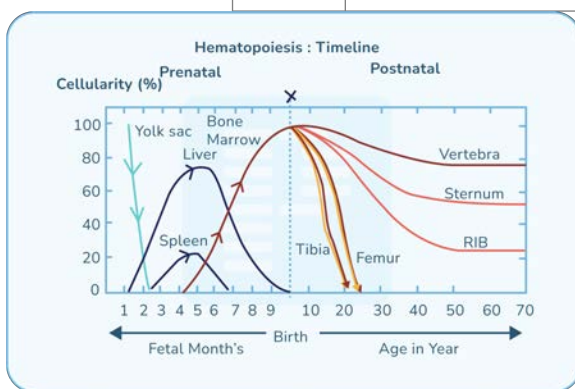
Correct Answer: C) Around 20 weeks

Explanation:

Around 20 weeks of gestation, hematopoiesis becomes centered in the bone marrow, which remains the major site for the remainder of fetal development and throughout adult life.

Stage	Location	Key Points
Mesoblastic	Yolk sac, PSP, AGM	Yolk sac: Multilineage differentiating capabilities (starts functioning by the 2nd week of gestation). Non-yolk sac regions: Para Aortic splanchnopleura (PSP) and aorta-gonad mesonephros (AGM) regions. PSP: Gives rise to B-cell progenitors. AGM: Contains pluripotent stem cells.
Hepatic	Liver, spleen	2nd trimester of pregnancy. Liver: Hematopoiesis starts at the 5th week, peaks at the 5th month, and ceases at birth. Spleen: Hematopoiesis mainly in the 2nd trimester, less than liver. Postnatal: Normally does not occur in liver and spleen unless demand is high (extramedullary hemopoiesis).

Medullary	Bone marrow (medullary hemopoiesis).	Marrow cavities appear by the 5th month (around 20 weeks) of fetal life, and become hemopoietic. (Option C) Initial phase: Bone marrow exclusive for granulocytic and megakaryocytic proliferation. Erythropoiesis starts in bone marrow towards the end of the 3rd trimester, and continues throughout life. Postnatal: Hemopoiesis confined to bone marrow, extramedullary hemopoiesis abnormal.
-----------	--------------------------------------	---



Solution for Question 7:

Correct Answer: B) Symmetric division

Explanation:

Symmetric division results in both daughter cells retaining stem cell properties, thereby increasing the stem cell pool, which is crucial during tissue repair.

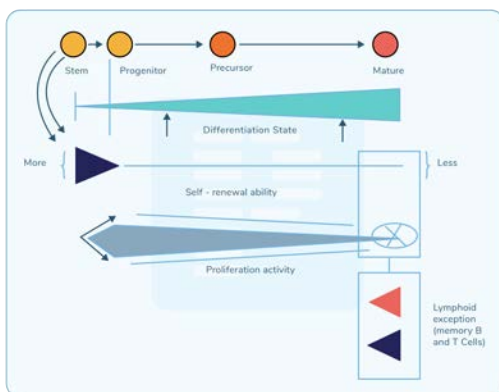
Pathway	Definition	Mechanisms / Factors	Signaling Pathways
Self-Renewal	Produces at least one daughter cell with stem cell properties	Symmetric and Asymmetric Division	Wnt, Notch, TGF- β
Differentiation	Produces specialized cell types	Transcription Factors, Microenvironmental Signals	Wnt, Notch, TGF- β

Both Pathways	Balance between self-renewal and differentiation to maintain stem cell pool and tissue function	Interplay of intrinsic and extrinsic factors	
Stem Cell Type	Description	Example	
Totipotent Stem Cells	Can generate the entire organism, including the placenta.	Single cell from the 8-cell embryo stage.	
Pluripotent Stem Cells	Can generate the entire organism except the placenta.	Cells from the inner cell mass of the blastocyst.	
Multipotent Stem Cells	Can form only one lineage of cells.	Hematopoietic stem cells that give rise to various blood cell types.	

Steps in hematopoiesis:

Multipotent Stem Cell → Progenitor Cell → Precursor Cell → Mature Cell
 Differentiated cell with decreased self-renewal and proliferation properties.

- Hematopoietic Stem Cells: Convert to multipotent stem cells, giving rise to two progenitor types.
- Memory B and T Cells are mature cells but have the ability to self-renew and proliferate.



Asymmetric division (Option A) – Maintains homeostasis by generating one stem cell and one differentiated cell, rather than expanding the pool.

Terminal differentiation (Option C) – Leads to fully specialized cells with no further division

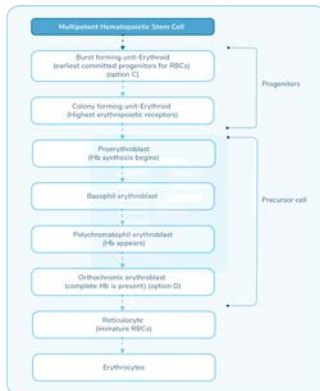
Lateral inhibition (Option D) – A process in which a cell inhibits its neighboring cells from adopting the same fate, playing a role in cell differentiation rather than expanding the stem cell pool.

Solution for Question 8:

Correct Answer: C) It is an early progenitor cell in erythropoiesis that gives rise to CFU-E

Explanation:

Erythropoiesis:



Since hemoglobin is not present in BFU-E, it can not transport oxygen. (Option A)

Multipotent hematopoietic stem cell gives rise to all the cells circulating in blood. (Option B)

Solution for Question 9:

Correct Answer: A) It is majorly produced by the liver

Explanation:

Erythropoietin:

- It is the major regulator of erythropoiesis, which increases the number of RBCs in blood by acting on the JAK STAT pathway (Option C) .
- Erythropoietin is synthesized from: Kidney (around 90%) (Option A) Liver Brain
- Kidney (around 90%) (Option A)
- Liver

- Brain
 - The major stimulus for erythropoietin production is Low oxygen levels or tissue hypoxia (Option B) .
 - When there is tissue hypoxia, erythropoietin acts on Burst forming unit-Erythroid (BFU-E), converting them into colony forming unit-Erythroid (CFU-E), which has the maximum erythropoietin receptor density (Option D) .
 - CFU-E on the continued action of erythropoietin converts into proerythrocytes and following.
 - Kidney (around 90%) (Option A)
 - Liver
 - Brain
-

Solution for Question 10:

Correct Answer: D) 8

Explanation:

Hemoglobin:

- Hb synthesis begins in the polychromatophil erythroblasts.
 - The most common form of hemoglobin found in adults is HbA, which is a combination of 2 alpha chains and 2 beta chains.
 - The heme group in one Hb molecule has 4 atoms of iron, which binds to 4 molecules of oxygen or 8 atoms of oxygen (option D)
 - Therefore, each Hb molecule carries 4 oxygen molecules to the tissues.
-

Solution for Question 11:

Correct Answer: C) Hemopexin

Explanation:

- Hemopexin is a primary protein that is responsible for binding and transporting free heme in the bloodstream.
- When RBC undergoes hemolysis, heme is released into the blood.
- Free heme is potentially toxic due to its ability to generate reactive oxygen species and cause oxidative damage.
- Hemopexin binds to free heme with high affinity, facilitating safe transport of the liver for degradation and prevention of toxicity (option D).

Haptoglobin binds free hemoglobin, but not free heme (Option A)

Albumin maintains osmotic pressure and serves as a carrier protein for various substances, including hormones, fatty acids, and drugs, but is not a carrier of heme (Option B)

Transferrin is responsible for transfer of iron in the bloodstream. (Option C)

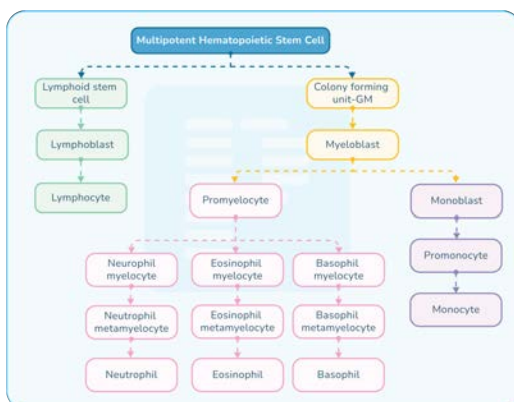
Solution for Question 12:

Correct Answer: C) IL-5

Explanation:

In an allergic reaction, the leukocytes most involved are eosinophils. The interleukin involved in eosinophilia is IL-5.

Leukopoiesis:



Factors involved in leukopoiesis:

- Colony-stimulating factors such as G-CSF and GM-CSF

- IL-5 is required for differentiation of leukocytes into eosinophils (option C)
- IL-3 & 4 are required for the development of basophils (option B)

IL-1 is responsible for the activation of T cells and macrophages and the promotion of inflammation (Option A)

IL-10 is an anti-inflammatory cytokine that regulates immune reactions. (Option D)

Previous Year Questions

1. What is the underlying cause of hypoxia in a newly arrived resident in a high-altitude region?

- A. Low hemoglobin levels
 - B. Low partial pressure of O₂
 - C. Low blood lactate levels
 - D. High partial pressure of CO₂
-

2. In the extrinsic pathway of coagulation, which clotting factor takes part?

- A. Factor VIII
 - B. Factor VII
 - C. Factor XI
 - D. Factor IX
-

3. Hemoglobin with iron in ferric form is

- A. Methemoglobin
 - B. HbA
 - C. Fetal hemoglobin
 - D. HbS
-

4. Please match the following molecules with their corresponding carriers in plasma. TRANSPORTER ORGANIC SUBSTANCE

1. Thyroxine A. Hemopexin

2. Fatty acid B. Prealbumin

3. Hemoglobin C. Albumin

4. Heme D. Haptoglobin

TRANSPORTER	ORGANIC SUBSTANCE
1. Thyroxine	A. Hemopexin
2. Fatty acid	B. Prealbumin
3. Hemoglobin	C. Albumin
4. Heme	D. Haptoglobin

- A. 1-B, 2-C, 3-D, 4-A
 - B. 1-A, 2-B, 3-C, 4-D
 - C. 1-A, 2-D, 3-B, 4-C
 - D. 1-B, 2-C, 3-A, 4-D
-

5. In patients undergoing warfarin therapy, which clotting factor would display a decrease in gamma carboxyglutamate residue?

- A. Factor 2
 - B. Factor 11
 - C. Tissue factor
 - D. Factor 5
-

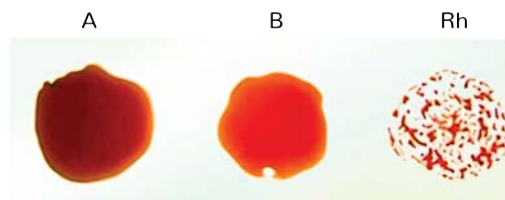
6. What makes CPDA superior than ACD for blood storage, except:

- A. The fall in 2,3 DPG is less
- B. Improved Viability of Red Blood Cells
- C. Increased Potassium Load
- D. It is less acidic

7. What would be the duration for preserving red blood cells (RBCs) using Acid Citrate Dextrose (ACD) for storage, when phosphate is added alone and when adenine and phosphate are added together to ACD?

- A. 35 days and 42 days
 - B. 21 days and 35 days
 - C. 28 days and 45 days
 - D. 35 days and 28 days
-

8. What is the blood group determined by the agglutination test depicted below?



- A. B positive
 - B. A positive
 - C. O positive
 - D. AB positive
-

9. Which of the following is not synthesized by the liver?

- A. Albumin

- B. Coagulation factors
 - C. Acute phase proteins
 - D. Immunoglobulin
-

10. cGMP is the second messenger of

- A. Somatostatin
 - B. Angiotensin II
 - C. ADH
 - D. NO (nitric oxide)
-

11. What is required for the conversion of prothrombin to thrombin?

- A. Calcium
 - B. Magnesium
 - C. Potassium
 - D. Sodium
-

12. Erythropoietin is secreted by which of the following organs? A. Liver B. Kidney C. Muscle D. Heart"

- A. AB
 - B. BC
 - C. ABC
 - D. Only B
-

13. Which of the following describes the chloride ion exchange in red blood cells, where bicarbonate ions are exchanged for chloride ions to maintain electrical neutrality?

- A. Haldane effect
 - B. Root effect
 - C. Bohr effect
 - D. Chloride shift
-

14. Which of the following factors causes a rightward shift in the oxygen-hemoglobin dissociation curve?

- A. Increase in CO_2
 - B. Increase in O_2
 - C. Decrease in CO_2
 - D. Decrease in temperature
-

15. At what stage does erythropoietin first demonstrate effectiveness in the formation of red blood cells?

- A. Burst forming units
 - B. CFU-GM
 - C. Late erythroblast stage
 - D. Normoblast stage
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	1
Question 5	1
Question 6	3
Question 7	2
Question 8	3
Question 9	4
Question 10	4
Question 11	1
Question 12	1
Question 13	4
Question 14	1
Question 15	1

Solution for Question 1:

Correct Option B:

- Low partial pressure of O₂ cause hypoxia in high altitude.
- In the given scenario, the individual has recently moved to a high-altitude area with lower atmospheric pressure than at sea level.

- The partial pressure of oxygen in the air is directly proportional to the atmospheric pressure.
- At high altitudes, the atmospheric pressure decreases, and consequently, the partial pressure of oxygen decreases.
- This results in a reduced amount of oxygen being available for inhalation, leading to hypoxia.

Incorrect Options:

Option A : Low hemoglobin levels can lead to a condition called anaemia. Anaemia can cause hypoxia, but it is not the causative factor in this scenario.

Option C: Blood lactate levels are not related to the development of hypoxia. increased lactic acid levels cause the fatigue.

Option D: High levels of carbon dioxide (CO₂) in the blood can lead to a condition called hypercapnia. Hypercapnia can cause hypoxia, but it is not the causative factor in this scenario.

Solution for Question 2:

Correct Option B:

- The extrinsic pathway of coagulation is initiated by tissue factor (TF), which is released from damaged tissues.
- Factor VII (FVII) is a vitamin K-dependent clotting factor that is activated by binding to TF.
- Once activated, the FVII-TF complex activates factor X (FX), which leads to the formation of thrombin and ultimately the formation of a blood clot.

Incorrect Options:

Option A: Factor VIII is a clotting factor that participates in the intrinsic pathway of coagulation, which is activated by damage to the endothelial lining of blood vessels.

Option C: Factor XI is also a clotting factor that participates in the intrinsic pathway of coagulation.

Option D: Factor IX is a clotting factor that participates in the intrinsic pathway of coagulation, and is activated by factor XI.

Solution for Question 3:

Correct Option A:

- Methemoglobin is a form of hemoglobin where the iron in heme is oxidized to the ferric state (Fe^{3+}).
- This alters the shape of the heme group, making it unable to bind oxygen effectively.
- Methemoglobinemia is a condition characterized by increased methemoglobin in the blood, which can result in decreased oxygen delivery to the tissues.
- Hemoglobin is a protein found in red blood cells that is responsible for carrying oxygen from the lungs to the tissues of the body.
- The iron in hemoglobin is in the ferrous (Fe^{2+}) form, which allows it to bind to oxygen.
- However, under certain conditions, the iron in hemoglobin can be oxidized to the ferric (Fe^{3+}) form, forming methemoglobin.

Incorrect Options:

Option B: Hemoglobin A (HbA) is the most common form of hemoglobin in adults. It is composed of two alpha and two beta globin chains, each containing a heme group with a ferrous (Fe^{2+}) ion.

Option C: Fetal hemoglobin (HbF) is the primary form of hemoglobin found in fetuses and newborn infants. It has a higher affinity for oxygen than adult hemoglobin, which allows for efficient transfer of oxygen from the mother's blood to the fetus. HbF has two alpha and two gamma globin chains, with each chain containing a heme group with a ferrous (Fe^{2+}) ion.

Option D: Hemoglobin S (HbS) is a variant of hemoglobin that results from a mutation in the beta globin gene. It is responsible for sickle cell disease, a condition in which red blood cells become stiff and sickle-shaped, leading to decreased oxygen delivery and various complications. HbS contains two alpha and two mutant beta globin chains, each with a heme group containing a ferrous (Fe^{2+}) ion.

Solution for Question 4:

Correct Option A - 1-B, 2-C, 3-D, 4-A:

1. Thyroxine

- Correct Match: Thyroxine is transported by Albumin (Option C) . Albumin is the most abundant plasma protein and carries various substances, including thyroxine.

2. Fatty acid

- Correct Match: Fatty acids are transported by Prealbumin (also known as Transthyretin) (Option B) . Prealbumin binds to thyroxine as well as fatty acids.

3. Hemoglobin

- Correct Match: Hemoglobin is transported by Haptoglobin (Option D) . Haptoglobin binds to free hemoglobin, preventing its loss through the kidneys and potential kidney damage.

4. Heme

- Correct Match: Heme is transported by Hemopexin (Option A) . Hemopexin binds to heme and prevents its oxidative effects and potential tissue damage.

Solution for Question 5:

Correct Option A: Factor 2

- The clotting factor that would have a decreased gamma carboxyglutamate residue vitamin in patients on warfarin therapy is Factor 2, also known as prothrombin.
- By inhibiting the regeneration of reduced vitamin K, warfarin decreases the availability of active vitamin K in the body. This results in impaired carboxylation of clotting factors, leading to decreased gamma carboxyglutamate residues. Among the clotting factors affected by warfarin therapy, Factor 2 (prothrombin) is one of them. Therefore, patients on warfarin therapy would have decreased gamma carboxyglutamate residues in Factor 2.

Incorrect Options:

Option B, C & D: Warfarin is an oral anticoagulant medication that inhibits the activity of vitamin K epoxide reductase, an enzyme required for the regeneration of reduced vitamin K. Vitamin K is essential for the carboxylation of certain clotting factors, including Factors 2, 7, 9, and 10, and proteins C and S. The carboxylation process adds gamma carboxyglutamate residues to these factors, allowing them to bind calcium ions and participate in the coagulation cascade.

Solution for Question 6:

Option C. Increased Potassium Load: Incorrect . CPDA does not cause an increased potassium load compared to ACD. Both CPDA and ACD experience a rise in potassium levels during blood storage as red blood cells leak potassium over time. However, CPDA generally preserves the structural integrity of RBCs better than ACD, which helps reduce potassium leakage. Therefore, CPDA does not result in an increased potassium load and is in fact preferable for reducing the risk of hyperkalemia in sensitive patients.

Incorrect options

Option A. The fall in 2,3 DPG is less: Correct. CPDA helps preserve 2,3-diphosphoglycerate (2,3 DPG), an important compound for oxygen delivery. Higher levels of 2,3 DPG in stored blood ensure that oxygen is released more readily to tissues when transfused.

Option B. Improved Viability of Red Blood Cells: Correct. CPDA helps maintain the viability and structural integrity of red blood cells, ensuring they function effectively after transfusion. Higher viability means the transfused cells can properly carry oxygen to the body's tissues.

Option D. It is less acidic: Correct. CPDA has a higher pH compared to ACD, making it less acidic. This less acidic environment helps preserve the structural integrity of red blood cells during storage. Acidic conditions can cause hemolysis, damaging the cells and affecting their functionality.

Solution for Question 7:

The correct answer is:

Option B. 21 days and 35 days

Explanation:

- Red Blood Cells (RBCs) Preservation using ACD (Acid Citrate Dextrose) : ACD is an anticoagulant solution used for the preservation of blood. When RBCs are preserved using

ACD alone, the duration of storage is approximately 21 days .

- Addition of Adenine and Phosphate to ACD : Adenine and phosphate are often added to the ACD solution to enhance the preservation of RBCs. With the addition of adenine and phosphate to ACD, the duration of storage is extended to approximately 35 days .

Red Blood Cells (RBCs) Preservation using ACD (Acid Citrate Dextrose) : ACD is an anticoagulant solution used for the preservation of blood. When RBCs are preserved using ACD alone, the duration of storage is approximately 21 days .

Addition of Adenine and Phosphate to ACD : Adenine and phosphate are often added to the ACD solution to enhance the preservation of RBCs. With the addition of adenine and phosphate to ACD, the duration of storage is extended to approximately 35 days .

These additives provide necessary nutrients and enhance the storage conditions, allowing for a longer preservation period of red blood cells.

Solution for Question 8:

Correct option is C: O positive.

- O positive : In this case, there is no agglutination in either A or B antigens, indicating the absence of ABO antigens on the red blood cells. Blood group O is known as the universal donor because individuals with blood group O do not have A or B antigens on their red blood cells.

Incorrect options:

Option A : B positive : Since there is no agglutination in the B antigen, it is not indicative of blood group B. Therefore, option a) is incorrect.

Option B: A positive : Similarly, since there is no agglutination in the A antigen, it is not indicative of blood group A. Therefore, option b) is incorrect.

Option D: AB positive : Agglutination in both A and B antigens would indicate the presence of both A and B antigens on the red blood cells, which corresponds to blood group AB. However, since no agglutination is observed in the A or B antigens in this case, blood group AB is not the correct option. Therefore, option d) is incorrect.

Solution for Question 9:

Correct option is D: Immunoglobulin.

- Immunoglobulins , also known as antibodies, are primarily produced by plasma cells, which are a type of white blood cell derived from B lymphocytes . While the liver is involved in immune-related functions , such as phagocytosis by Kupffer cells, it does not synthesize immunoglobulins.

Incorrect options

Option A : Albumin is a protein synthesized by the liver. It is the most abundant protein in the blood and plays a crucial role in maintaining oncotic pressure, transporting substances such as hormones and drugs, and regulating fluid balance.

Option B: Coagulation factors, such as fibrinogen, prothrombin, and factors VII, VIII, IX, X, are produced by the liver. These factors are essential for blood clotting and are involved in the coagulation cascade.

Option C : Acute phase proteins, including C-reactive protein, haptoglobin, and serum amyloid A, are synthesized by the liver. They are part of the immune response and play a role in inflammation and tissue repair.

Solution for Question 10:

Correct Option D: NO (nitric oxide)

- Nitric oxide (NO) activates the enzyme guanylate cyclase, which converts guanosine triphosphate (GTP) into cyclic guanosine monophosphate (cGMP) .
- cGMP then serves as the second messenger, mediating various cellular responses.
- NO is a key signaling molecule involved in several physiological processes, including vasodilation, neurotransmission, and immune regulation.

Incorrect Options:

Option A: Somatostatin - Somatostatin acts through G-protein-coupled receptors, but its second messenger system involves inhibitory G-proteins and intracellular pathways such as the inhibition of adenylate cyclase and decrease in cyclic adenosine monophosphate (cAMP) levels. It does not involve cGMP as the second messenger.

Option B: Angiotensin II- This option is incorrect. Angiotensin II primarily acts through the angiotensin II type 1 receptor (AT1 receptor) and utilizes the inositol trisphosphate (IP3) and diacylglycerol (DAG) signaling pathway, leading to an increase in intracellular calcium levels. It does not involve cGMP as the second messenger.

Option C: ADH (antidiuretic hormone, also known as vasopressin)- ADH acts through the V2 receptor and utilizes the cAMP signaling pathway, activating protein kinase A (PKA) and regulating water reabsorption in the kidneys. It does not involve cGMP as the second messenger.

Solution for Question 11:

Correct Option A: Calcium

- Calcium: Calcium plays a crucial role in the conversion of prothrombin to thrombin during the coagulation cascade. Calcium ions are necessary for the activation of several coagulation factors, including factor X, which leads to the conversion of prothrombin (factor II) into thrombin (factor IIa). Calcium ions act as cofactors, helping to stabilize the formation of the enzyme complex necessary for the conversion process.

Incorrect options

Option B: Magnesium: Magnesium is not directly involved in the prothrombin-to-thrombin conversion. However, magnesium ions are required for several other enzymatic reactions in the body and play a role in maintaining normal blood clotting function.

Option C: Potassium: Potassium does not directly participate in the prothrombin-to-thrombin conversion. Potassium is primarily involved in maintaining the cell membrane potential and plays a vital role in nerve and muscle function.

Option D: Sodium: Sodium is not directly involved in the conversion of prothrombin to thrombin. Sodium ions are primarily involved in maintaining fluid balance, nerve function, and muscle contraction.

Solution for Question 12:

Correct Option A – AB:

- Erythropoietin (EPO) is produced from peritubular epithelium cells of the kidney and perivenous hepatocytes of the liver.

Incorrect Options:

- Option B, C & D are incorrect.
-

Solution for Question 13:

Answer: D) Chloride shift

Explanation:

The chloride shift, also known as the Hamburger effect, is the process by which bicarbonate (HCO_3^-) exits red blood cells (RBCs) in exchange for chloride (Cl^-) ions, maintaining electrical neutrality.

- Mechanism: In peripheral tissues, CO_2 diffuses into RBCs and reacts with water, catalysed by carbonic anhydrase, forming HCO_3^- and H^+ . HCO_3^- is transported out of the RBC into the plasma via the band 3 protein in exchange for Cl^- . This shift maintains ionic balance and facilitates CO_2 transport in the blood. In the lungs, the process reverses, allowing CO_2 exhalation.
- In peripheral tissues, CO_2 diffuses into RBCs and reacts with water, catalysed by carbonic anhydrase, forming HCO_3^- and H^+ .
- HCO_3^- is transported out of the RBC into the plasma via the band 3 protein in exchange for Cl^- .
- This shift maintains ionic balance and facilitates CO_2 transport in the blood.
- In the lungs, the process reverses, allowing CO_2 exhalation.
- In peripheral tissues, CO_2 diffuses into RBCs and reacts with water, catalysed by carbonic anhydrase, forming HCO_3^- and H^+ .

- HCO_3^- is transported out of the RBC into the plasma via the band 3 protein in exchange for Cl^- .
- This shift maintains ionic balance and facilitates CO_2 transport in the blood.
- In the lungs, the process reverses, allowing CO_2 exhalation.

Chloride shift (Hamburger phenomenon) vs Reverse chloride shift:

Feature	Chloride shift	Reverse chloride shift
Location	Tissues	Lungs
Process	CO_2 enters RBCs, forming $\text{H}^+ + \text{HCO}_3^-$. HCO_3^- exits in exchange for Cl^- .	CO_2 exits RBCs. HCO_3^- re-enters RBCs in exchange for Cl^- .
Effect on RBCs	Chloride influx causes water influx, swelling RBCs	Chloride efflux balances HCO_3^- re-entry, CO_2 is released
Impact on haematocrit	Increases hematocrit due to RBC swelling	Primarily facilitates CO_2 release, indirectly affecting hematocrit

Haldane Effect (Option A) is incorrect because it describes increased CO_2 binding to haemoglobin in deoxygenated blood, enhancing CO_2 transport.

Root Effect (Option B): Incorrect because it refers to a reduction in hemoglobin's O_2 carrying capacity in acidic conditions.

Bohr Effect (Option C) is incorrect because it refers to the effect of CO_2 and H^+ on oxygen affinity, where increased CO_2 lowers haemoglobin's oxygen affinity, promoting O_2 unloading in tissues.

Solution for Question 14:

Correct Answer: A) Increase in CO_2

Explanation:

A rightward shift in the oxygen-haemoglobin dissociation curve means reduced oxygen affinity, facilitating oxygen unloading to tissues. This occurs in conditions where tissues require more oxygen, such as exercise or metabolic activity.

- Factors causing a rightward shift (Bohr effect): $\uparrow \text{CO}_2$: Increases carbonic acid ($\text{H}^+ + \text{HCO}_3^-$) formation, lowering pH and promoting oxygen release. $\uparrow \text{H}^+$ ($\downarrow \text{pH}$): Acidic conditions

reduce haemoglobin's oxygen affinity. ↑ Temperature: Higher metabolic activity increases oxygen demand. ↑ 2,3-BPG (DPG): Enhances O₂ unloading by stabilising the deoxygenated form of haemoglobin.

- ↑ CO₂: Increases carbonic acid (H⁺CO₃⁻) formation, lowering pH and promoting oxygen release.
- ↑ H⁺ (↓ pH): Acidic conditions reduce haemoglobin's oxygen affinity.
- ↑ Temperature: Higher metabolic activity increases oxygen demand.
- ↑ 2,3-BPG (DPG): Enhances O₂ unloading by stabilising the deoxygenated form of haemoglobin.
- ↑ CO₂: Increases carbonic acid (H⁺CO₃⁻) formation, lowering pH and promoting oxygen release.
- ↑ H⁺ (↓ pH): Acidic conditions reduce haemoglobin's oxygen affinity.
- ↑ Temperature: Higher metabolic activity increases oxygen demand.
- ↑ 2,3-BPG (DPG): Enhances O₂ unloading by stabilising the deoxygenated form of haemoglobin.

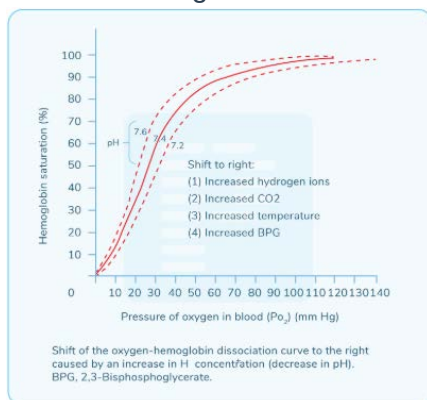


Table rendering failed

The Double Bohr Effect is a physiological phenomenon observed in the placenta. This effect involves two distinct Bohr effects occurring simultaneously—one on the fetal side and one on the maternal side.

Double Bohr effect	Details
Location	Placenta, where the maternal and fetal circulation meet.
Fetal Blood Flow	Umbilical arteries carry deoxygenated blood with high CO ₂ from fetus to placenta
CO ₂ Diffusion	CO ₂ diffuses from fetal to maternal blood
Effect on Fetal Blood	CO ₂ decreases, blood becomes alkaline Oxygen dissociation curve shifts left

Effect on Maternal Blood	CO ₂ increases, blood becomes acidic Oxygen dissociation curve shifts right
Clinical Significance	Enhances oxygen transfer from mother to fetus, improving fetal oxygenation

Increase in O₂ (Option B) is incorrect because higher O₂ levels promote oxygen binding rather than a rightward shift.

Decrease in CO₂ (Option C) is incorrect because low CO₂ leads to higher pH, causing a leftward shift (increased O₂ affinity, reduced unloading).

Decrease in Temperature (Option D) is incorrect because low temperature stabilises haemoglobin's oxygen-binding state, shifting the curve leftward.

Solution for Question 15:

Correct Option A - Burst forming units:

- BFUs are the earliest stage at which erythropoietin is effective in red blood cell formation. BFUs are pluripotent stem cells that give rise to erythroid colony-forming units (CFUs). Erythropoietin acts on BFUs to promote their differentiation into erythroid progenitor cells.

Incorrect Options:

Option B - CFU-GM: Colony forming unit GM are the progenitor cells for the monocytes and granulocytes.

Option C - Late erythroblast stage: The late erythroblast stage refers to the later stages of erythropoiesis when the cells have progressed beyond the BFU/CFU stage.

Option D - Normoblast stage: The normoblast stage is another term for the late erythroblast stage, referring to the stage where the erythroblasts have reached a normal size and shape.

Renal Physiology

1. Which of the following combinations correctly describes the effect on Glomerular Filtration Rate (GFR) given the status of the afferent and efferent arterioles?

- A. Afferent arteriole constriction- GFR increased
 - B. Efferent arteriole dilation - GFR increased
 - C. Afferent arteriole dilation - GFR decreased
 - D. Efferent arteriole mild constriction - GFR increased
-

2. A physiologist is studying the effect of food additives on GFR in order to determine what additives should be avoided in individuals with CKD. Caffeine was found to affect urine output through an effect on GFR. What is the effect and the mechanism involved?

- A. Increase urine output; afferent arteriole vasodilation
 - B. Increase urine output; efferent arteriole vasodilation
 - C. Increase urine output; afferent arteriole vasoconstriction
 - D. Decrease urine output; afferent arteriole vasodilation
-

3. A 56-year-old man is admitted to the ED with an acute myocardial infarction. The man was transferred to the CCU and was placed on 24 h intake and output. A few days later his 24 hr urinary output was decreased. An increase in which of the following contributes to the reduced urine flow in a patient with congestive heart failure?

- A. ANP

- B. Urodilatin
 - C. Renal perfusion pressure
 - D. Renal sympathetic nerve activity
-

4. Which of the following best describes what happens to the PAH excretion rate when the plasma concentration of PAH exceeds the transport maximum (T_m)?

- A. The excretion rate continues to increase linearly with increasing plasma concentration upto a certain level.
 - B. The excretion rate reaches a plateau and remains constant despite further increases in plasma concentration beyond a level..
 - C. The excretion rate decreases with increasing plasma concentration.
 - D. The excretion rate first increases linearly, then decreases after reaching the transport maximum.
-

5. Which of the following is NOT a component of the juxtaglomerular (JG) apparatus?

- A. Macula densa
 - B. Juxtaglomerular cells
 - C. Extraglomerular mesangial cells
 - D. Loop of Henle
-

6. Which of the following best explains the relationship between creatinine clearance ($CrCl$) and glomerular filtration rate (GFR)?

- A. Creatinine clearance is a direct measure of GFR.
- B. Creatinine clearance is an indirect measure of GFR and typically overestimates GFR.

C. Creatinine clearance underestimates GFR because it accounts for tubular secretion of creatinine.

D. Normal creatinine clearance is 50mL/min.

7. Which of the following statements about "free water clearance" (C_{H_2O}) is correct? Free water clearance = C_{H_2O} Urine osmolality = U_{osm} Plasma osmolality = P_{osm} Urine flow rate = V

A. Free water clearance is positive when the urine is hypotonic and negative when the urine is hypertonic.

B. Free water clearance is the same as urine flow rate (V).

C. Free water clearance is calculated using $C_{H_2O} = U_{osm} (V) / P_{osm}$.

D. Free water clearance is positive when the urine is hypertonic and negative when the urine is hypotonic.

8. A patient's hematocrit (the fraction of blood volume occupied by red blood cells) is 40%. The patient's renal plasma flow (RPF) is measured to be 600 mL/min. What is this patient's renal blood flow (RBF)?

A. 360 mL/min

B. 720 mL/min

C. 1000 mL/min

D. 1500 mL/min

9. Why is inulin clearance the gold standard for measuring glomerular filtration rate (GFR)?

A. Inulin is readily available and naturally produced in the body

- B. Inulin is freely filtered and neither reabsorbed nor secreted, providing an accurate measurement of GFR.
 - C. Inulin has a high plasma concentration that enhances measurement accuracy
 - D. Inulin is rapidly metabolised by the kidneys, making it easier to measure
-

10. A patient has a urine concentration of a substance of 10 mg/mL, a urine flow rate of 2 mL/min, and a plasma concentration of that substance of 5 mg/mL. What is the clearance rate of this substance?

- A. 4 mL/min
 - B. 5 mL/min
 - C. 10 mL/min
 - D. 20 mL/min
-

11. A 65-year-old patient with severe renal artery stenosis is admitted for hypertension management. The physician decides to prescribe a medication to control the blood pressure but must avoid drugs that could adversely affect kidney function due to the existing renal impairment. Which drug is contraindicated in this patient due to the risk of worsening renal function?

- A. Beta-blocker
 - B. Angiotensin-converting enzyme (ACE) inhibitor
 - C. Calcium channel blocker
 - D. Diuretic
-

12. Which of the following is NOT an effect of angiotensin II?

- A. Constriction of efferent arterioles in the kidneys

- B. Decreased sodium reabsorption in the renal tubules
 - C. Stimulation of aldosterone release from the adrenal cortex
 - D. Increased thirst and vasopressin (ADH) secretion
-

13. All of the following increase GFR except:

- A. Bradykinin
 - B. Nitric Oxide
 - C. Sympathetic Nerves
 - D. Prostaglandins (PGE2, PGI2)
-

14. Which of the following statements about nephrons is correct? Total renal blood flow rate is 1000 to 1260 ml/minute Tubular reabsorption is the first step in urine production Chronic kidney disease occurs when there is a 75% reduction in the original number of nephrons. The cortical region of the kidney is most vulnerable to ischemia. Nephrons can regenerate after age 40. Maintains Acid-Base balance by secreting hydrogen ions and reabsorbing bicarbonate.

- A. Statements 1, 2, and 5.
 - B. Statements 1, 3, and 6.
 - C. Statements 2, 4, and 6
 - D. Statements 3, 4, and 5.
-

15. Which of the following mechanisms is responsible for Myogenic Autoregulation of renal blood flow?

- A. Calcium influx
- B. Activation of sympathetic fibers

- C. Angiotensin release
 - D. Dopamine release
-

16. Calculate Renal Plasma Flow (RPF) if the GFR is 125 ml/min and the Filtration Fraction (FF) is 0.2.

- A. 625
 - B. 500
 - C. 600
 - D. 610
-

17. Calculate the Net Filtration Pressure from the values given below: Glomerular Hydrostatic Pressure = 60 mmHg Bowman's Capsule Pressure = 18 mmHg Glomerular Colloid Osmotic Pressure = 32 mmHg

- A. 10 mmHg
 - B. 20 mmHg
 - C. 30 mmHg
 - D. 15 mmHg
-

18. If the molecular radius of a substance is 3.0 nm, it is known as:

- A. Freely filtrable substance
 - B. Selectively permeable substance
 - C. Impermeable substance
 - D. None of the above
-

19. Which of the following diseases can be caused by the defects in proteins of the slit diaphragm?

- A. Congenital Nephrotic Syndrome
 - B. Steroid Resistant Nephrotic Syndrome
 - C. Focal Segmental Glomerulosclerosis
 - D. All of the above
-

20. Which of the following statements regarding the Glomerular Basement Membrane (GBM) is true?

- A. It is not a part of the glomerular filtration barrier
 - B. Contains type 3 collagen
 - C. Excludes substances larger than 8 nm
 - D. Charges of substances do not affect their permeability through GBM
-

21. Which of the following is the primary site of action of acetazolamide in the nephron?

- A. Proximal convoluted tubule
 - B. Loop of Henle
 - C. Distal convoluted tubule
 - D. Collecting duct
-

22. Which of the following characteristics correctly differentiates the Proximal Convoluted Tubule (PCT) from the Collecting Duct (CT)?

- A. Paracellular transport present in CT and not in PCT
 - B. Brush border of epithelium present in PCT, not in CT
 - C. Tight junctions are tight in PCT and leaky in CT
 - D. Carbonic anhydrase type 4 absent in PCT, but present in CT
-

23. A 58-year-old patient with chronic renal failure presents with symptoms of fatigue, fluid retention, and low serum potassium levels. On examination, the patient shows signs of hypertension and oedema. Which of the following types of cells being defective in the renal tubules is responsible for the following condition?

- A. Principal Cells
 - B. Alpha Intercalated Cells
 - C. Beta Intercalated Cells
 - D. Juxtaglomerular Cells
-

24. Which part of the proximal tubule is the most vulnerable and prone to ischemia?



- A. S1
 - B. S2
 - C. S3
 - D. S2 and S3
-

25. Match each component of the countercurrent system with its respective function: Components Functions

- 1. Ascending Limb of Loop of Henle** **a. Blood loses water and gains solutes as it passes through the hyperosmotic medullary interstitium.**
- 2. Descending Limb of Loop of Henle** **b. Blood gains water and loses solutes as it passes through the less concentrated interstitium.**
- 3. Ascending Vasa Recta** **c. Actively transports sodium and chloride into the interstitium, diluting the filtrate.**
- 4. Descending Vasa Recta** **d. Permeable to water; water is reabsorbed into the interstitium, concentrating the filtrate.**

Components	Functions
1. Ascending Limb of Loop of Henle	a. Blood loses water and gains solutes as it passes through the hyperosmotic medullary interstitium.
2. Descending Limb of Loop of Henle	b. Blood gains water and loses solutes as it passes through the less concentrated interstitium.
3. Ascending Vasa Recta	c. Actively transports sodium and chloride into the interstitium, diluting the filtrate.
4. Descending Vasa Recta	d. Permeable to water; water is reabsorbed into the interstitium, concentrating the filtrate.

- A. 1-b, 2-d, 3-a, 4-c
- B. 1-c, 2-d, 3-b, 4-a
- C. 1-c, 2-b, 3-a, 4-d
- D. 1-d, 2-a, 3-c, 4-b

26. Which of the following statements correctly differentiates between cortical and juxtamedullary nephrons?

- A. Cortical nephrons have longer loops of Henle.
- B. Juxtamedullary nephrons constitute approximately 85% in the kidney.
- C. Cortical nephrons usually have a higher filtration rate.

D. Juxtamedullary nephrons are located primarily in the renal cortex, whereas cortical nephrons are in the renal medulla.

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	1
Question 3	4
Question 4	2
Question 5	4
Question 6	2
Question 7	1
Question 8	3
Question 9	2
Question 10	1
Question 11	2
Question 12	2
Question 13	3
Question 14	2
Question 15	1
Question 16	1
Question 17	1
Question 18	2
Question 19	4
Question 20	3
Question 21	1
Question 22	2

Question 23	1
Question 24	3
Question 25	2
Question 26	3

Solution for Question 1:

Correct answer: D) Efferent arteriole mild constriction - GFR increased

Explanation:

Arteriole Change	Effect on Resistance	Effect on Glomerular Hydrostatic Pressure	Effect on GFR
Afferent Arteriole Constriction	Increases	Decreases	Decreases
Afferent Arteriole Dilation	Decreases	Increases	Increases
Efferent Arteriole Constriction- Mild	Increases	Increases	Slight increase
Efferent Arteriole Constriction- Severe	Increases	Increases	Initially increases GFR then ultimately decreases. Severe efferent arteriole constriction reduces renal blood flow and increases filtration fraction, concentrating plasma proteins in the glomerular capillaries and elevating glomerular colloid osmotic pressure. This counteracts the effect of increased hydrostatic pressure, ultimately reducing GFR
Efferent Arteriole Dilation	Decreases	Decreases	Decreases

Solution for Question 2:

Correct answer: A) Increase urine output; afferent arteriole vasodilation

Explanation:

Caffeine's primary effect is to increase urine output via afferent arteriole vasodilation .

- Caffeine causes relaxation of the smooth muscle in the afferent arterioles
- This vasodilation increases blood flow into the glomeruli.
- The increase in blood flow raises the hydrostatic pressure within the glomeruli, leading to an increased GFR.
- A higher GFR means that more plasma is filtered into the renal tubules.
- With a higher GFR, more filtrate is produced, which ultimately results in an increase in urine volume.

Increase urine output; efferent arteriole vasodilation (Option B) :

- This would decrease glomerular pressure and GFR, leading to lower urine output.

Increase urine output; afferent arteriole vasoconstriction (Option C) :

- Vasoconstriction would reduce blood flow into the glomeruli, decreasing GFR and urine output.

Decrease urine output; afferent arteriole vasodilation (Option D) :

- Afferent arteriole vasodilation increases GFR and urine output, not decreases it.
-

Solution for Question 3:

Correct answer: D) Renal sympathetic nerve activity

Explanation:

In a patient with congestive heart failure, reduced urine flow is primarily due to an increase in renal sympathetic nerve activity.

Renal Sympathetic Nerve Activity:

Promotes a decrease in sodium and water excretion by:

- Decreasing glomerular filtration rate (GFR).
- Increasing renin secretion, leading to further fluid retention via the renin-angiotensin-aldosterone system.
- Enhancing tubular sodium reabsorption.

ANP (Atrial Natriuretic Peptide) (Option A) : Typically promotes diuresis and sodium excretion; its levels are often lower in heart failure due to decreased atrial stretch.

Urodilatin (Option B) : A natriuretic peptide isolated from human urine, synthesized in kidney tubular cells.

- It is a 32-amino-acid peptide processed from the same precursor as ANP.
- Acts as a paracrine regulator of sodium and water homeostasis in the distal nephron.
- Increases sodium and water excretion by interacting with luminal receptors in the distal and connecting tubules, but its action may be blunted in heart failure.

Renal Perfusion Pressure (Option C) : While decreased due to heart failure, this does not directly increase urine flow; it typically leads to retention.

Solution for Question 4:

Correct Answer: B) The excretion rate reaches a plateau and remains constant despite further increases in plasma concentration.

Explanation:

Transport Maximum - The highest rate at which a substance can be reabsorbed or secreted by the renal tubules of the kidneys.

Significance:

- Diagnostic Tool: It aids in diagnosing conditions like diabetes mellitus (glycosuria) and renal tubular disorders.
- Homeostasis: T_m ensures efficient essential nutrients and electrolytes reabsorption, maintaining body homeostasis.

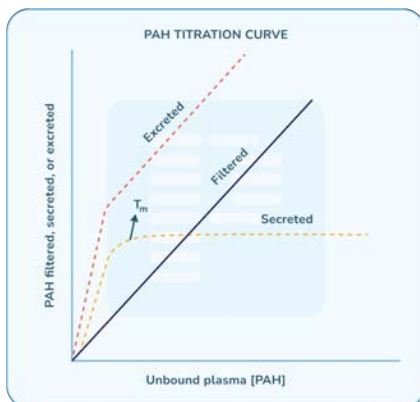
- **Drug Excretion:** It influences the excretion of drugs and toxins, impacting pharmacokinetics and dosing regimens.
- **Pathophysiology:** T_m provides insights into kidney function and the pathophysiology of diseases affecting renal transport mechanisms.
- **Clinical Monitoring:** It assists in monitoring the progression and treatment of kidney diseases and other metabolic conditions.

When the plasma concentration of PAH exceeds the transport maximum (T_m), the renal tubules are saturated and cannot secrete PAH any faster.

As a result, the excretion rate reaches a plateau and does not increase further despite increasing plasma concentrations of PAH.

PAH

Plasma conc. Vs clearance rate



The excretion rate does not continue to increase linearly with increasing plasma concentration as the transport mechanisms in the kidneys have a maximum capacity (T_m), beyond which they cannot increase the rate of PAH excretion, causing the rate to plateau. (Option A)

The excretion rate should remain at the maximum capacity of the transport mechanisms (T_m) and not decrease. (Option C)

While it correctly states that the excretion rate initially increases linearly, it inaccurately suggests that the rate decreases after reaching the transport maximum. The excretion rate plateaus and remains constant after reaching T_m , without decreasing. (Option D)

Solution for Question 5:

Correct Answer: D) Loop of Henle

Explanation:

Loop of Henle:

- A segment of the nephron in the kidney's medulla.
- It is involved in concentrating urine and reabsorbing water and solutes.
- However, it is not part of the juxtaglomerular apparatus.

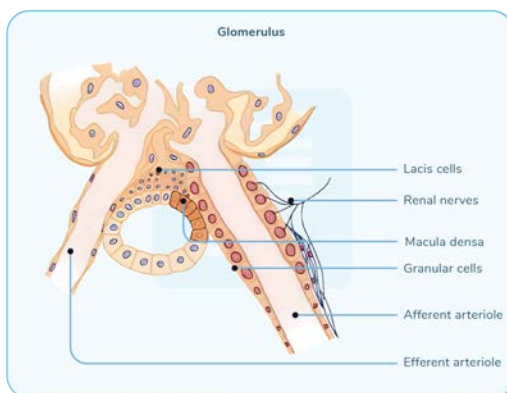
JuxtaGlomerular Apparatus(JG) : Crucial for regulating kidney function and blood pressure

Significance:

- Renin Release: Juxtaglomerular cells secrete renin in response to low blood pressure, low sodium chloride concentration, or sympathetic nervous system stimulation.
- Renin-Angiotensin-Aldosterone System (RAAS): Renin initiates the RAAS, leading to the production of angiotensin II, which increases blood pressure by constricting blood vessels and stimulating aldosterone release to retain sodium and water.
- Blood Pressure Regulation: JGA plays a vital role in maintaining systemic blood pressure and fluid balance.
- Glomerular Filtration Rate (GFR) Regulation: Macula densa cells sense changes in sodium chloride and help adjust the GFR by influencing afferent arteriole constriction or dilation.
- Homeostasis: Essential for maintaining electrolyte balance, blood volume, and overall homeostasis.

Consists of 3 components:

- Macula densa
- Juxtaglomerular cells/Granular cells
- Extraglomerular mesangial cells/Lacis cells



Macula densa:

- A specialised group of epithelial cells located in the distal convoluted tubule adjacent to the glomerulus.
- They detect sodium concentration in the filtrate and help regulate blood pressure and filtration rate.

Juxtaglomerular cells/Granular cells:

- Specialised smooth muscle cells in the walls of the afferent arterioles.
- They produce and secrete renin, crucial in regulating blood pressure and electrolyte balance.

Extraglomerular mesangial cells/Lacis cells:

- It is located between the afferent and efferent arterioles and the distal convoluted tubule.
- They are thought to assist in signalling between the macula densa and juxtaglomerular cells.

Solution for Question 6:

Correct Answer: B) Creatinine clearance is an indirect measure of GFR and typically overestimates GFR.

Explanation:

Creatinine Clearance measures how effectively the kidneys filter creatinine from the blood.

- It estimates the glomerular filtration rate (GFR), reflecting overall kidney function.
- The measurement involves collecting urine over 24 hours and taking a blood sample to determine the creatinine concentration in both urine and blood.
- Normal creatinine clearance is 140 mL/min.

Creatinine clearance is an indirect measure of GFR and typically overestimates GFR. (Option B)

- Creatinine clearance is often used to estimate GFR.
- However, it tends to overestimate GFR because creatinine is filtered by the glomeruli and secreted in small amounts by the renal tubules.

Creatinine clearance is not a direct measure of GFR. (Option A)

- It is the amount of blood filtered by the glomeruli per minute, and creatinine clearance approximates this value by measuring the clearance of creatinine, a waste product, from the

blood.

Creatinine clearance does not underestimate GFR. (Option C)

- Tubular secretion of creatinine would cause creatinine clearance to be overestimated, not underestimated, in the GFR.
- The additional creatinine secreted by the tubules leads to a higher clearance rate than the actual filtration rate.

Solution for Question 7:

Correct Answer: A) Free water clearance is positive when the urine is hypotonic and negative when the urine is hypertonic.

Explanation:

Free Water Clearance:

- Measure of the kidney's ability to excrete or conserve free water, independent of solute excretion.
- Free water clearance is positive when the urine is hypotonic and negative when the urine is hypertonic.
- Free water clearance reflects the ability of the kidneys to excrete or conserve water. When urine is hypotonic (less concentrated than plasma), the kidneys excrete more free water, making C_{H_2O} positive. When urine is hypertonic (more concentrated than plasma), the kidneys conserve water, making C_{H_2O} negative.
- When urine is hypotonic (less concentrated than plasma), the kidneys excrete more free water, making C_{H_2O} positive.
- When urine is hypertonic (more concentrated than plasma), the kidneys conserve water, making C_{H_2O} negative.
- When urine is hypotonic (less concentrated than plasma), the kidneys excrete more free water, making C_{H_2O} positive.
- When urine is hypertonic (more concentrated than plasma), the kidneys conserve water, making C_{H_2O} negative.

Free water clearance is not the same as urine flow rate. (Option B) It is derived from the difference between the urine flow rate and the osmolar clearance, indicating that it adjusts the urine flow rate by the osmotic concentration of the urine compared to plasma.

Free water clearance is calculated using the formula: (Option C)

$$C_{H_2O} = \frac{U_{osm} \times V}{P_{osm}}$$

Solution for Question 8:

Correct Answer: C) 1000 mL/min

Explanation:

Renal Blood Flow (RBF):

- The volume of blood delivered to the kidneys per unit of time.
- Typically, RBF is about 1.0 to 1.2 liters per minute in a healthy adult, accounting for approximately 20-25% of the cardiac output.

Renal Plasma Flow (RPF):

- Volume of plasma (the liquid part of blood without cells) that passes through the kidneys per unit of time.
- It reflects the effective blood flow that is available for filtration in the kidneys.
- RPF is usually about 55-60% of the total renal blood flow.

Renal blood flow (RBF) can be calculated using the formula:

$$RBF = RPF / (1 - \text{Hematocrit})$$

Given:

$$\text{Renal Plasma Flow (RPF)} = 600 \text{ mL/min}$$

$$\text{Hematocrit} = 0.40$$

$$RBF = 600 / (1 - 0.40) = 600 / 0.60 = 1000 \text{ mL/min.}$$

Solution for Question 9:

Correct answer: B) Inulin is freely filtered and neither reabsorbed nor secreted, providing an accurate measurement of GFR.

Explanation

Inulin clearance is a method for estimating glomerular filtration rate (GFR), a key indicator of kidney function. This method utilises inulin, a plant-derived polysaccharide, due to its unique properties:

- Freely Filtered: Inulin, similar to water, passes through the glomerular capillaries without restriction.
- Not Reabsorbed or Secreted: Inulin remains unchanged in the tubules, neither reabsorbed into the bloodstream nor secreted into the tubules. This ensures that the excretion rate of inulin accurately reflects the filtration rate by the glomeruli.
- Inulin remains unchanged in the tubules, neither reabsorbed into the bloodstream nor secreted into the tubules.
- This ensures that the excretion rate of inulin accurately reflects the filtration rate by the glomeruli.
- Inulin remains unchanged in the tubules, neither reabsorbed into the bloodstream nor secreted into the tubules.
- This ensures that the excretion rate of inulin accurately reflects the filtration rate by the glomeruli.

To measure inulin clearance:

Clearance Formula : $(C_s) = (U_s \times V) / P_s$

U_s : Urine concentration of the substance

V : Urine flow rate

P_s : Plasma concentration of the substance

Despite its accuracy, inulin clearance is less practical for routine use due to the need for intravenous administration. Therefore, other methods like creatinine clearance are more commonly used for estimating GFR in clinical settings.

Solution for Question 10:

Correct answer: A) 4 mL/min

Explanation:

Clearance measures the volume of plasma completely cleared of a substance by the kidneys per minute (mL/min), reflecting renal excretory efficiency. It is essential for evaluating kidney function, including filtration, reabsorption, and secretion processes.

To calculate the clearance rate (Cs) of the substance, use the formula:

Clearance Formula : $(Cs) = (Us \times V) / Ps$

Us : Urine concentration of the substance

V : Urine flow rate

Ps : Plasma concentration of the substance

Plugging in the values:

$Cs = (10 \times 2) / 5 = 4 \text{ ml /min}$

So, the clearance rate of the substance is 4 mL/min .

Types of Clearance Measurements:

Inulin Clearance	Used to measure Glomerular Filtration Rate (GFR) as it is freely filtered but not reabsorbed or secreted. Not ideal for routine use due to intravenous administration.
Creatinine Clearance	Estimates GFR using creatinine, which is primarily filtered by the glomeruli. Slightly inaccurate due to tubular secretion, but practical for clinical settings.
PAH Clearance	Estimates Renal Plasma Flow (RPF). PAH is cleared through both glomerular filtration and tubular secretion. Correction for PAH extraction ratio improves accuracy.

Solution for Question 11:

Correct answer: B) Angiotensin-converting enzyme (ACE) inhibitor

Explanation

Renal artery stenosis reduces blood flow to the affected kidney, decreasing renal perfusion pressure. This decrease is perceived as threatening blood pressure homeostasis, triggering a cascade of events within the renin-angiotensin system.

Increased angiotensin II levels help maintain GFR by constricting the efferent arteriole, preserving glomerular pressure despite reduced renal blood flow.

ACE Inhibitors block the conversion of Angiotensin I to Angiotensin II, disrupting the compensatory mechanism and potentially causing a significant drop in GFR.

ACE inhibitors can lead to acute renal failure in renal artery stenosis by reducing protective efferent arteriole constriction and worsening GFR.

Beta-blockers (Option A) reduce blood pressure by decreasing heart rate and cardiac output. They do not directly affect renal perfusion or glomerular filtration rate.

Calcium channel blockers (Option C) reduce blood pressure by relaxing blood vessels, including the arterioles. They do not significantly impair renal perfusion and are generally safe in patients with renal artery stenosis.

Diuretics (Option D) help reduce blood pressure by increasing urine production and reducing fluid volume. While they must be used cautiously to avoid dehydration, they do not directly impair renal perfusion or GFR like ACE inhibitors.

Solution for Question 12:

Correct answer: B) Decreased sodium reabsorption in the renal tubules

Explanation:

Effects of Angiotensin II:

1	Renal Tubular Reabsorption	Directly increases sodium reabsorption in proximal tubules, loops of Henle, distal tubules, and
---	----------------------------	---

		collecting tubules. (Option B)
2	Vasoconstriction	Angiotensin II constricts arterioles, especially efferent arterioles in the kidneys, increasing glomerular pressure and maintaining GFR even at low blood pressure. (Option A)
3	Aldosterone Secretion	Stimulates adrenal cortex to release aldosterone, enhancing sodium and water retention in the kidneys. (Option C)
4	Baroreflex Sensitivity	Reduces baroreflex sensitivity in the brain, boosting its effects on blood pressure.
5	Thirst and Hormone Secretion	Increases thirst, stimulates vasopressin (ADH) release for water reabsorption, and triggers ACTH secretion. (Option D)
6	Mesangial Cells	Causes contraction of mesangial cells, potentially reducing GFR by decreasing filtration surface area
7	System Interactions	Effects are influenced by sodium intake, blood volume, and other hormones. Active during low sodium or volume depletion to maintain GFR and fluid balance.

Solution for Question 13:

Correct answer: C) Sympathetic Nerves

Explanation:

Mediator	Effect on GFR	Additional Notes
Prostaglandins (PGE2, PGI2) (Option D)	Increase	Cause vasodilation; especially important in counteracting vasoconstriction from sympathetic nerves or angiotensin II
Bradykinin (Option A)	Increase	Functions similarly to prostaglandins by causing vasodilation

Nitric Oxide (Option B)	Increases	Important for maintaining normal renal excretion of sodium and water; decreased production linked to hypertension
Angiotensin II	Biphasic effect on GFR	Constricts efferent arterioles, increasing glomerular pressure; severe constriction decreases GFR
Sympathetic Nerves (Option C)	Decrease	Cause vasoconstriction, especially under stress or volume depletion
Norepinephrine, Epinephrine, Endothelin	Decreases	Vasoconstriction

Solution for Question 14:

Correct Answer: B) Statements 1, 3, and 6 .

Explanation:

Nephrons	Structural and functional units of the kidney
Number per Kidney	0.8 to 1.2 million (average of 1 million)
Length	4.5 to 6.5 cm (average of 5 cm)
Renal Blood Flow	Total Blood Flow Rate: 1000 to 1260 ml/minute (20 to 25% of cardiac output) Blood Flow Distribution: Cortex: 90%, Medulla: 10% Vulnerability to Ischemia: The Medulla is prone to ischemia; the Corticomedullary junction is more vulnerable Plasma Flow Rate: 625 to 700 ml/minute through both kidneys combined
Nephron Function	Glomerular Filtration: First step in urine formation; filters water, ions, and small molecules from blood into Bowman's capsule Tubular Reabsorption: Process where the nephron removes water and solutes from the filtrate and returns them to the blood Tubular Secretion: Transfer of materials from blood into the tubular fluid Excretion: Final elimination of urine from the body

Additional Functions	Regulation of Blood Pressure: By controlling blood volume through reabsorption and secretion mechanisms Acid-Base Balance: By secreting hydrogen ions and reabsorbing bicarbonate Electrolyte Balance: By reabsorbing and secreting electrolytes such as sodium, potassium, calcium, and magnesium Erythropoietin Production: A hormone that stimulates red blood cell production in the bone marrow, especially in response to low oxygen levels in the blood
Age-Specific Dynamics of Nephron	Nephrons cannot regenerate. Nephron count decreases by 10% every 10 year after age 40. By age 80, about 40% of nephrons remain functional. Chronic kidney disease occurs when there is a 70 to 75% reduction in the original number of nephrons.

Solution for Question 15:

Correct Answer: A) Calcium influx

Explanation:

Myogenic Autoregulation:

It maintains relatively constant renal blood flow and GFR and may protect the kidney from hypertension-induced injury.

Solution for Question 16:

Correct Answer: A) 625

Explanation:

Filtration fraction is the fraction of renal plasma flow that is filtered.

Filtration fraction = $\text{GFR} / \text{Renal Plasma Flow}$

$$0.2 = 125/\text{Renal Plasma Flow}$$

$$\text{Renal Plasma Flow} = 125/0.2 = 625$$

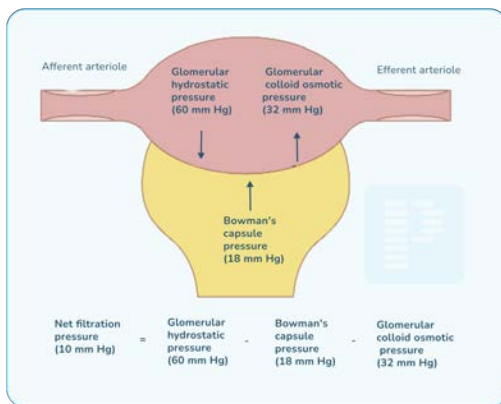
Solution for Question 17:

Correct Answer: A) 10 mmHg

Explanation:

Net filtration pressure = Glomerular hydrostatic pressure - Bowman's capsule pressure - Glomerular colloid osmotic pressure

$$= 60 - 18 - 32 = 10.$$



Forces Favoring Filtration:

- Promotes filtration by pushing fluid out of the glomerular capillaries into Bowman's capsule.

Forces Opposing Filtration:

- Opposes filtration by pushing fluid back into the glomerular capillaries.

2. Glomerular Capillary Colloid Osmotic Pressure: 32 mm Hg

- The presence of plasma proteins prevents filtration by attracting water back into the glomerular capillaries.

Net Filtration Pressure: Forces favouring filtration - Forces opposing filtration

$$= 60 \text{ mm Hg} - (18 \text{ mm Hg} + 32 \text{ mm Hg})$$

= 10 mm Hg

Thus, healthy humans have a net filtration pressure of approximately 10 mm Hg.

Solution for Question 18:

Correct Answer: B) Selectively permeable substance

Explanation:

The permeability of substances at the molecular filtration barrier is affected by their molecular diameter.

Substances with 1.8-4.2 nm molecular radius are selectively permeable; their charges decide their permeability across the glomerular filtration barrier (order of permeability: positively charged substances > neutral substances > negatively charged substances).

Type of substance	Molecular radius	Explanation	Examples
Freely filtrable	<1.8 nm	Easily pass through the barrier	Sodium Potassium Water Urea Creatinine Inulin PAH Bicarbonate
Selectively permeable	1.8-4.2 nm	The charge on them determines filtration	Albumin Haemoglobin Myoglobin Bence Jones proteins
Impermeable	>4.2 nm	It cannot be filtered	

Solution for Question 19:

Correct Answer: D) All of the above

Explanation:

- The foot processes of podocytes form the 3rd and innermost layer of the glomerular filtration barrier
- The gaps between these foot processes are called slit pores or filtration slits
- A porous filtration slit diaphragm bridges the filtration slits
- Proteins in the slit diaphragm and diseases caused by the mutations of the genes coding for it:

Protein	Disease
Nephrin	Congenital Nephrotic Syndrome of Finnish type
Podocin	Steroid-Resistant Nephrotic Syndrome
■-Actinin	Focal Segmental Glomerulosclerosis

Solution for Question 20:

Correct Answer: C) Excludes substances larger than 8 nm

Explanation:

The Glomerular Basement Membrane is the layer of the glomerular filtration barrier.

Layers of glomerular filtration barrier:

Glomerular Basement Membrane:

Aspect	Description
Structure	A specialized membrane forming a sheath shared by neighboring capillaries
Contents	Fine fibrils in a gel-like matrix The matrix contains type 4 collagen, laminin, fibronectin, and other negatively charged proteins
Permeability	Permits free passage of neutral substances up to 4 nm in diameter Excludes substances larger than 8 nm
Molecular Charge Influence	Charge on molecules affects their passage into Bowman's capsule.

Solution for Question 21:

Correct Answer: A) Proximal convoluted tubule

Explanation:

Acetazolamide primarily acts on the proximal convoluted tubule (PCT) of the nephron. It works by inhibiting carbonic anhydrase, specifically types II and IV.

Carbonic anhydrase plays a crucial role in the reabsorption of bicarbonate (HCO_3^-) in this segment. By inhibiting this enzyme, acetazolamide reduces the reabsorption of bicarbonate, leading to increased bicarbonate excretion and diuresis.

The Loop of Henle (Option B) is primarily affected by loop diuretics, such as furosemide, which inhibit the Na-K-2Cl symporter in the thick ascending limb.

The distal convoluted tubule (Option C) is the site of action for thiazide diuretics, which inhibit the Na-Cl symporter.

Potassium-sparing diuretics, such as spironolactone and amiloride, act on the collecting duct. Acetazolamide does not have its primary action in the collecting duct. (Option D)

Solution for Question 22:

Correct Answer: B) Brush border of epithelium present in PCT, not in CT

Explanation:

Feature	Proximal Convoluted Tubule (PCT)	Collecting Duct (CT)
Epithelium Type	Simple cuboidal with brush border	Simple cuboidal to columnar without brush border
Brush Border	Present	Absent
Microvilli	Dense microvilli (forming brush border)	Absent microvilli

Paracellular Transport	Significant due to leaky tight junctions	Minimal due to tight tight junctions
Tight Junctions	Leaky	Tight
Carbonic Anhydrase Type 4	Present in microvilli	Generally absent
Mitochondrial Density	High	Lower than PCT
Transport Mechanisms	Active and passive	Primarily passive
Reabsorption of Glucose	Almost complete	Minimal

Solution for Question 23:

Correct Answer: A) Principal Cells

Explanation:

Cell Type	Functions	Characteristics
Principal Cells (Option A)	Sodium reabsorption Potassium secretion Water reabsorption	Responsive to aldosterone ADH receptors Columnar epithelium
Alpha Intercalated Cells (Option B)	Secrete H ⁺ ions Reabsorb bicarbonate	Apical H ⁺ -ATPases Cuboidal epithelium
Beta Intercalated Cells (Option C)	Secrete bicarbonate Reabsorb H ⁺ ions	Apical bicarbonate transporters Cuboidal epithelium

Juxtaglomerular Cells (Option D):

- They are located in the juxtaglomerular apparatus and are responsible for the secretion of renin, which plays a role in the regulation of blood pressure and the renin-angiotensin-aldosterone system.
- They are not directly involved in sodium reabsorption, potassium secretion, or water reabsorption.

Solution for Question 24:

Correct Answer: C) S3

Explanation:

The proximal tubule is divided into three segments: S1, S2, and S3. Among these, S3 is the most vulnerable to ischemia.

This increased susceptibility is due to the following reasons:

- **High Metabolic Activity:** S3 has the highest metabolic rate and density of mitochondria, making it more dependent on oxygen.
- **Proximity to Renal Medulla:** Located closer to the renal medulla, which has lower oxygen levels.
- **Fewer Collateral Blood Vessels:** S3 has less collateral circulation, reducing its ability to compensate for decreased blood flow.

	S1	S2	S3
Location	Proximal to the glomerulus.	Between S1 and S3.	Near the loop of Henle.
Function	Reabsorption of nutrients, ions, and water.	Reabsorption of water, ions, and bicarbonate.	Reabsorption of sodium, bicarbonate, and nutrients.
Features	High brush border density.	A brush border is present but less extensive than S1.	High mitochondrial density.
Risk of ischaemia	Least prone	Intermediate	Most prone

Solution for Question 25:

Correct Answer: B) 1-c, 2-d, 3-b, 4-a

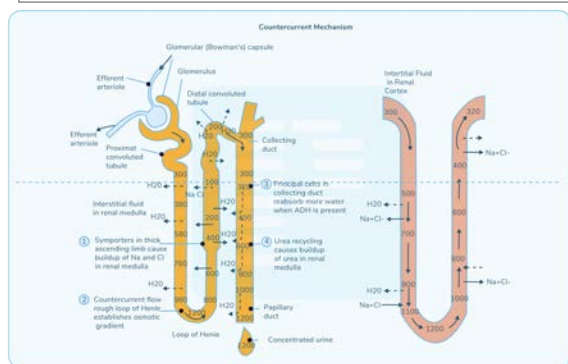
Explanation:

The countercurrent system is essential for maintaining the body's fluid and electrolyte balance, particularly through urine concentration.

This system involves two main components in the kidneys:

- Countercurrent multiplier system in the loop of henle.
- Countercurrent exchanger system in the vasa recta.

Countercurrent Multiplier System	Countercurrent Exchanger System
Location: Loop of Henle	Location: Vasa Recta
Function: Creates an osmotic gradient in the renal medulla to concentrate urine	Function: Maintains the osmotic gradient established by the countercurrent multiplier system and supplies nutrients to the medulla
Descending Limb: Permeable to water; water is reabsorbed into the interstitium, concentrating the filtrate	Descending Vasa Recta: Blood loses water, and gains solute as it descends through the hyperosmotic medullary interstitium
Ascending Limb: Impermeable to water; actively transports sodium and chloride out into the interstitium, diluting the filtrate	Ascending Vasa Recta: Blood gains water and loses solutes as it ascends through the less concentrated interstitium.
Mechanism: Countercurrent flow between descending and ascending limbs multiplies the osmotic gradient, enabling the concentration of urine	Mechanism: Countercurrent exchange of water and solutes between blood in the vasa recta and the surrounding interstitium preserves the osmotic gradient



Solution for Question 26:

Correct Answer: C) Cortical nephrons usually have a higher filtration rate.

Explanation:

Cortical nephrons and Juxtamedullary nephrons:

Feature	Cortical Nephrons	Juxtamedullary Nephrons
Location	Renal cortex.	Near the boundary between the renal cortex and medulla.

Percentage of Total Nephrons	Approximately 85%	Approximately 15%
Loop of Henle	Short and extends only into the outer medulla.	Long and extends deep into the inner medulla.
Vasa Recta	Short and less extensive.	Long vasa recta, which follows the loop of Henle into the medulla.
Glomerular Capsule	Has a larger surface area relative to the length of the loop of Henle.	Smaller surface area relative to the length of the loop of Henle.
Filtration Rate	High	Lower filtration rate but more specialised in concentration.
Presence of Macula Densa	Typically not present.	These nephrons are closely associated with the juxtaglomerular apparatus.
Function	Primarily involved in regulating blood composition and volume through reabsorption and secretion.	Concentration of urine due to the longer loops and extensive vasa recta Contributing to the medullary osmotic gradient.

Tubular Transport

1. A 45-year-old woman presents with a 6-month history of muscle weakness, fatigue, and frequent urination. Her blood pressure is 160/95 mmHg. Investigations: Serum K⁺: 2.8 mEq/L, Serum Na⁺: 148 mEq/L, Plasma aldosterone: 35 ng/dL, Plasma renin activity: 0.2 ng/mL/h. What is the most likely diagnosis?

- A. Bartter's syndrome
 - B. Gitelman's syndrome
 - C. Conn's syndrome
 - D. Liddle's syndrome
-

2. When a nephrologist studying kidney reabsorption encounters the effects of Anti-diuretic hormone (ADH) on water reabsorption, what is the most likely influence of ADH on water reabsorption?

- A. Facultative water reabsorption involves glucose transporters
 - B. Facultative reabsorption dependent on ADH
 - C. Obligatory reabsorption dependent of ADH
 - D. Facultative reabsorption occurs in proximal tubules
-

3. A 52-year-old woman with type 2 diabetes is being monitored for her renal glucose handling. During the monitoring, the clinician noticed that glucose appeared in her urine even at low plasma glucose concentrations. What is the most likely cause?

- A. Explained by Splay effect
- B. High Renal threshold for glucose

- C. Increased Transport maximum
 - D. Increased Glomerular filtration
-

4. A 63-year-old male complains of increased micturition, tingling numbness, and a non-healing ulcer. On examination, his FBS is 260 mg/dl, and he has glucose in his urine. What is the most likely cause of glucose in urine?

- A. Decreased tubular secretion of glucose.
 - B. Exceeded renal threshold of glucose.
 - C. Increased insulin sensitivity.
 - D. Increased tubular reabsorption of glucose.
-

5. A 47-year-old diabetic woman is being treated with medication that inhibits the reabsorption of glucose by the kidneys. What is the likely transporter inhibited by oral hyperglycemic medication, and what is its likely absorption mechanism?

- A. SGLT-2, Secondary active transport
 - B. GLUT-2, Facilitated diffusion
 - C. SGLT-2, Facilitated diffusion
 - D. GLUT-2, Secondary active transport
-

6. A 35-year-old female complains of muscle weakness, metabolic alkalosis and hypertension. Genetic analysis reveals a mutation of ENaC. What is the most likely condition that occurs due to a mutation of ENaC?

- A. Gordon Syndrome
- B. Barttin syndrome
- C. Gitelman's syndrome

D. Liddle's syndrome

7. Which channels are not typically found on the luminal side of principal cells in the DCT?

- A. Epithelial Sodium Channel (ENaC)
 - B. Renal Outer Medullary Potassium (ROMK) channel
 - C. Aquaporin-2 (AQP2) water channel
 - D. $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter (NKCC2)
-

8. Which of the following statements about the cells in the distal tubule and cortical collecting tubule is NOT correct?

- A. Principal cells are responsible for sodium reabsorption and potassium secretion.
 - B. Type A intercalated cells secrete hydrogen ions and reabsorb bicarbonate
 - C. Amiloride and triamterene act as potassium-sparing diuretics by blocking sodium channels in principal cells
 - D. Spironolactone and eplerenone increase sodium reabsorption and potassium secretion in principal cells
-

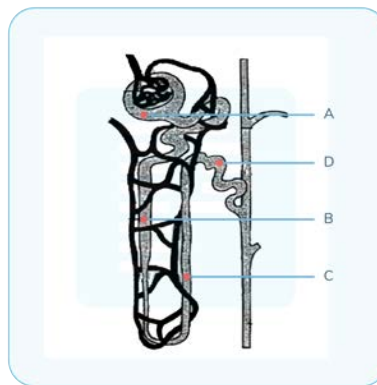
9. Which of the following is a characteristic feature of Gitelman syndrome?

- A. Hyperkalemia
 - B. Hypercalciuria
 - C. Hypomagnesemia
 - D. Metabolic acidosis
-

10. Which of the following statements regarding Bartter's syndrome is NOT true?

- A. Caused by mutations in the sodium-chloride-potassium co-transporter in the loop of Henle.
 - B. Common features include hypokalemia and metabolic alkalosis.
 - C. Typically inherited in an autosomal dominant manner.
 - D. Leads to the activation of the renin-angiotensin-aldosterone system.
-

11. While flowing through which of the following segments does the filtrate become progressively hypertonic compared to plasma?



- A. D
 - B. A
 - C. B
 - D. C
-

12. Which of the following statements is true regarding the roles of principal cells and type A intercalated cells in the distal convoluted tubule (DCT) and cortical

collecting ducts?

- A. Type B intercalated cells reabsorb bicarbonate ions and secrete hydrogen ions.
 - B. Type B intercalated cells reabsorb sodium and secrete potassium ions
 - C. Principal cells reabsorb sodium and secrete potassium ions into the tubular lumen.
 - D. Type A intercalated cells reabsorb potassium ions and secrete bicarbonate.
-

13. A 65-year-old man presents with muscle cramps and tetany for the past 2 days. Investigations reveal hypocalcemia and elevated phosphate levels. If the hypocalcemia is caused by damage to the part of the nephron responsible for the maximum reabsorption of calcium, which part of the nephron is likely affected?

- A. Loop of Henle
 - B. Distal convoluted tubule
 - C. Proximal tubule
 - D. Collecting duct
-

14. Which of the following statements regarding potassium homeostasis by the renal tubule is NOT True?

- A. Maximum K⁺ reabsorption occurs in the PCT.
 - B. Principal cells are the major site for K⁺ secretion.
 - C. Reabsorption of K⁺ in the PCT depends on the daily changes in potassium intake.
 - D. Type B intercalated cells secrete potassium into the tubule when there is excess K⁺.
-

15. At which part of the renal tubule does the maximum reabsorption of sodium occur?

- A. Proximal convoluted tubule
 - B. Thick Ascending loop of Henle
 - C. Distal convoluted tubule
 - D. Collecting duct
-

16. Where is the sodium-glucose-linked transporter (SGLT) located in the renal tubule?

- A. Early part of the Proximal convoluted tubule
 - B. Ascending Loop of Henle
 - C. Distal convoluted tubule
 - D. Later part of the Proximal convoluted tubule
-

17. A 60-year-old male with chronic kidney disease presents with electrolyte abnormalities. He undergoes renal function tests revealing impaired reabsorption in his proximal convoluted tubule (PCT). All the following ions are reabsorbed in the proximal convoluted tubule (PCT) except?

- A. Amino Acids
 - B. Phosphate
 - C. Bicarbonate
 - D. Hydrogen Ions
-

18. A 45-year-old male patient with chronic hypertension undergoes renal function tests. His glomerular filtration rate (GFR) fluctuates due to his condition, but the percentage of sodium reabsorption at the proximal tubule remains constant. This renal phenomenon ensures stable sodium reabsorption relative to GFR changes

and is primarily mediated by?

- A. Renal Autoregulation
 - B. Modulating GFR based on feedback from sodium concentrations.
 - C. Glomerulo-tubular balance
 - D. Renin-angiotensin-aldosterone system (RAAS) regulating sodium reabsorption
-

19. A 45-year-old woman presents with complaints of excessive thirst and frequent urination. Laboratory tests reveal hypernatremia (elevated serum sodium levels) and low urine osmolality. Further investigation shows undetectable levels of antidiuretic hormone (ADH). What is the location of minimum urine tonicity expected in this patient?

- A. Distal convoluted tubule (DCT)
 - B. End of the collecting duct
 - C. Descending thin limb of the loop of Henle
 - D. Ascending limb of the loop of Henle
-

20. A 40-year-old woman presents to the clinic with complaints of excessive thirst and polyuria. Further investigation reveals a diagnosis of central diabetes insipidus. What is the approximate tubular fluid-to-plasma osmolality ratio expected in this patient's collecting duct?

- A. Less than 1
 - B. Equal to 1
 - C. Greater than 1
 - D. Between 1 and 2
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	1
Question 4	2
Question 5	1
Question 6	4
Question 7	4
Question 8	4
Question 9	3
Question 10	3
Question 11	3
Question 12	3
Question 13	3
Question 14	3
Question 15	1
Question 16	1
Question 17	4
Question 18	3
Question 19	2
Question 20	1

Solution for Question 1:

Correct Answer: C) Conn's syndrome

Explanation: The likely diagnosis from the given scenario is Conn's syndrome, also known as primary hyperaldosteronism, which is due to autonomous overproduction of aldosterone, usually from an adrenal adenoma.

- It is characterised by resistant hypertension, hypokalemia, and metabolic alkalosis.
- Causes: Aldosterone-producing adenoma (Conn's adenoma) - Most common. Bilateral adrenal hyperplasia.
- Aldosterone-producing adenoma (Conn's adenoma) - Most common.
- Bilateral adrenal hyperplasia.
- Aldosterone-producing adenoma (Conn's adenoma) - Most common.
- Bilateral adrenal hyperplasia.
- Symptoms often include muscle weakness, fatigue, headaches, and frequent urination.
- Laboratory findings: Elevated plasma aldosterone levels Suppressed plasma renin activity High aldosterone-to-renin ratio
- Elevated plasma aldosterone levels
- Suppressed plasma renin activity
- High aldosterone-to-renin ratio
- Treatment involves surgical removal of the adenoma and the use of mineralocorticoid receptor antagonists, such as spironolactone and eplerenone.
- Elevated plasma aldosterone levels
- Suppressed plasma renin activity
- High aldosterone-to-renin ratio

Role of Aldosterone : It acts on the collecting duct to –

- Increase sodium reabsorption via epithelial sodium channels (ENaC).
- Enhance potassium secretion through potassium channels (ROMK).

Bartter's syndrome (Option A)

- It is a genetic disorder, childhood-onset, affecting the kidneys, leading to abnormalities in electrolyte reabsorption.
- Typically presents with low blood pressure, hypokalemia, metabolic alkalosis, elevated renin levels, and high urinary calcium excretion (hypercalciuria).

Gitelman's syndrome : (Option B)

- A genetic disorder affects the distal convoluted tubule of the kidney, leading to hypokalemia, hypomagnesemia, and metabolic alkalosis.
- Typically presents with low or normal blood pressure and low urinary calcium excretion (hypocalciuria).

Liddle's syndrome (Option D)

- It is a genetic disorder caused by the increased activity of epithelial sodium channels (ENaC), leading to sodium retention and potassium excretion.
- It has low aldosterone and renin levels and typically presents with hypertension.

Solution for Question 2:

Correct Answer: B) Facultative reabsorption independent of ADH

Explanation: Facultative Reabsorption of water occurs in the distal convoluted tubules and collecting ducts under the influence of ADH.

Facultative reabsorption of water:

- When ADH binds to receptors on tubule cell membranes, it triggers the insertion of aquaporin channels.
- This increases water reabsorption into the bloodstream, helping to conserve water and produce concentrated urine, especially during dehydration or high plasma osmolality.

Obligatory reabsorption of water:

- This occurs primarily in the proximal convoluted tubules and is driven by osmotic gradients and other factors, not directly influenced by ADH.
- It involves the reabsorption of a significant amount of water and solutes regardless of the body's hydration status.

Solution for Question 3:

Correct Answer: A) Splay effect

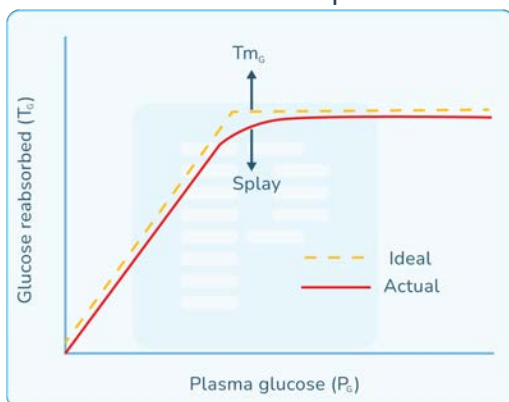
Explanation:

The splay effect describes the phenomenon where glucose begins to appear in the urine before reaching the expected renal threshold for glucose reabsorption.

This occurs due to variability in the glucose transport capacity across different nephrons and differences in the efficiency of glucose reabsorption.

Splay effect:

- It is the difference between renal threshold and transport maximum value.
- It arises because some neurons don't function properly, and few neurons don't function equally.
- Due to improper functioning of neurons, the actual renal threshold is less than the predicted threshold.
- Splay is inversely proportional to avidity, with which the transport mechanism binds the substance it transports.



Renal threshold for glucose (Option B)

It is the plasma glucose concentration at which glucose starts to appear in the urine, but it does not explain the appearance of glucose in the urine at lower concentrations.

Transport maximum (Option C)

The transport maximum (T_m) refers to the maximum rate at which the renal tubules can reabsorb glucose. Only if plasma glucose concentration exceeds this maximum, glucose will appear in the urine.

Glomerular filtration (Option D)

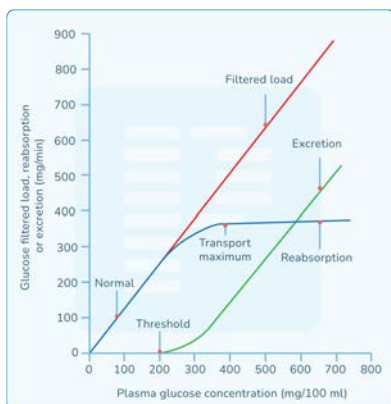
This process involves the filtering of blood by the kidneys. While important, it does not specifically account for glucose appearing in the urine at lower plasma levels.

Solution for Question 4:

Correct Answer: B) Exceeded renal threshold of glucose.

Explanation: The presence of glucose in urine in a patient with diabetes is due to exceeding the renal threshold of glucose.

- When plasma glucose is normal, all the filtered glucose is reabsorbed, but when plasma glucose levels ↑ are high, filtered glucose leads to the tubular load and appears in the urine.
- For example, when the plasma glucose concentration is 100 mg/dL, the filtered glucose load is at the normal level of 125 mg/min, and no glucose is present in the urine.
- However, when the plasma glucose concentration increases to 200 mg/dL, the filtered load rises to approximately 250 mg/min, and a small amount of glucose starts to appear in the urine.



Decreased tubular secretion of glucose- Glucose is reabsorbed but not secreted into the tubules. (Option A)

Increased insulin sensitivity leads to decreased blood glucose levels but does not affect glucose levels in urine. (Option C)

Increased tubular reabsorption of glucose leads to decreased glucose excretion. (Option D)

Solution for Question 5:

Correct Answer: A) SGLT-2, Secondary active transport

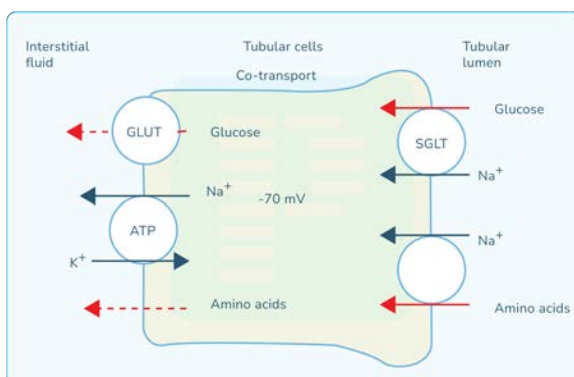
Explanation:

- In this scenario, the likely transporter inhibited by oral hyperglycemic medication is SGLT-2 (Sodium-Glucose Co-Transporter 2).
- Secondary active transport is the mechanism of glucose reabsorption inhibited by this medication.
- Examples of SGLT-2 inhibitors include canagliflozin, dapagliflozin, and empagliflozin.

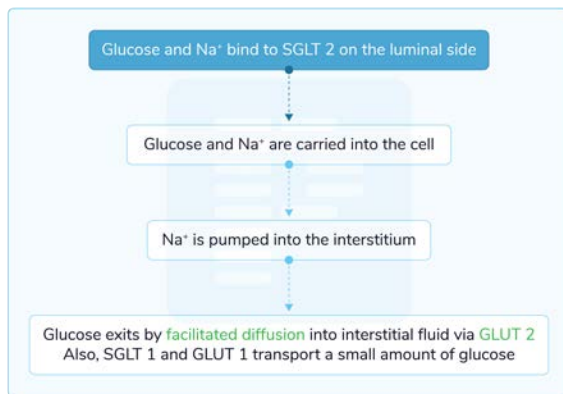
Type of glucose transporters in the Kidney:

Aspects	Luminal transporters	Basolateral transporters		
SGLT-1	SGLT-2	GLUT-1	GLUT-2	
Location in PCT	In the distal part (S3 segment)	In the later part	S3 segment	S1 segment
Inhibition by medication	No	Yes	No	No
Transport mechanism	Transport of glucose from the lumen to the cytoplasm by secondary active transporter.	Transport of glucose from the cytoplasm to blood by facilitated diffusion.		

PCT: Proximal Convoluted Tubule



Mechanism of Glucose absorption:



Solution for Question 6:

Correct Answer: D) Liddle's syndrome

Explanation:

- Liddle's syndrome is characterised by hypertension and metabolic alkalosis, which are due to a mutation of ENaC (Epithelial Sodium Channel) present in the distal and collecting tubules.
- This rare autosomal dominant disorder causes excessive ENaC activity, resulting in increased sodium and water reabsorption.
- Patients experience sodium retention, decreased renin secretion, and reduced levels of Angiotensin II, which subsequently decreases aldosterone secretion.
- Treatment: Amiloride (Blocks excessive ENaC activity).

Option A: Gordon's syndrome:

This condition is caused by mutations in the WNK1 and WNK4 genes and typically presents with hyperkalemia and metabolic acidosis.

Option B: Barttin syndrome:

- This syndrome results from mutations affecting the sodium-potassium-chloride cotransporter, potassium channels in the luminal membrane, and chloride channels in the basolateral membrane of the thick ascending loop of Henle.
- Patients present with hypokalemia and metabolic alkalosis.

Option C: Gitelman's syndrome:

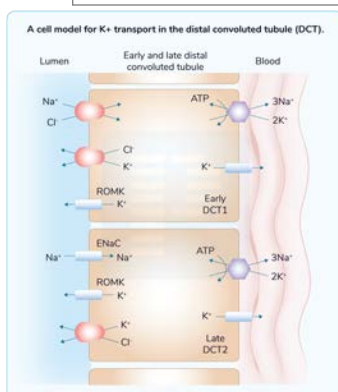
It arises from a mutation in the thiazide-sensitive sodium-chloride cotransporter in the distal tubule. Clinical manifestations include hypokalemia, hypocalciuria, and metabolic alkalosis.

Solution for Question 7:

Correct Answer: D) $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter (NKCC2)

Explanation: The $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporter (NKCC2) is not typically found on the luminal membrane of principal cells in the distal convoluted tubule. It is primarily located in the thick ascending limb of the loop of Henle.

Channel/Transporter	Location	Role
Epithelial Sodium Channel (ENaC)	Luminal membrane	Sodium reabsorption from the tubular lumen into the cell.
Renal Outer Medullary Potassium (ROMK) channel	Potassium secretion from the cell into the tubular lumen.	
Aquaporin-2 (AQP2) water channel	Water reabsorption under the influence of ADH.	
Aquaporin - Type 3/4	Basolateral membrane	Water reabsorption from cell cytoplasm to blood.
$\text{Na}^+\text{-K}^+$ ATPase pump	Active transport of sodium out of the cell into blood and potassium into the cell.	



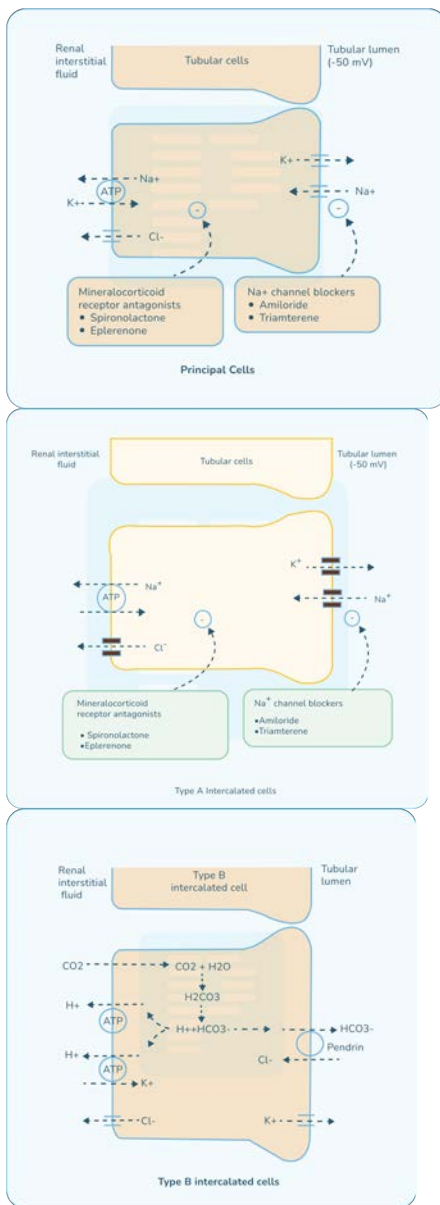
Solution for Question 8:

Correct Answer: D) Spironolactone and eplerenone increase sodium reabsorption and potassium secretion in principal cells.

Explanation: Spironolactone and eplerenone decrease sodium reabsorption and potassium secretion in principal cells.

Cells of Collecting Duct:

Characteristic	Principal Cells	Intercalated Cells
Main Function	Sodium reabsorption and potassium secretion (Option A)	Acid-base regulation
Proportion	60-70% of cells in collecting tubules and ducts.	30-40% of cells in collecting tubules and ducts.
Types	-	Type A Type B
Key Processes	Na ⁺ reabsorption via Na ⁺ channels K ⁺ secretion via K ⁺ channels Water reabsorption (regulated by ADH).	Type A: (Option B) H ⁺ secretion HCO ₃ ⁻ reabsorption Type B: HCO ₃ ⁻ secretion H ⁺ reabsorption
Key Transporters	Na ⁺ -K ⁺ ATPase pump in basolateral membrane.	Type A H ⁺ -ATPase (apical) H ⁺ -K ⁺ -ATPase (apical) Cl ⁻ -HCO ₃ ⁻ exchanger (basolateral) Type B: H ⁺ -ATPase (basolateral) Pendrin (Cl ⁻ -HCO ₃ ⁻ exchanger, apical)
Hormonal Regulation	Responsive to aldosterone	Not primarily regulated by aldosterone.
Diuretic Targets	Primary targets for potassium-sparing diuretics	Not primary targets for most diuretics.
Specific Diuretics	Spironolactone and eplerenone (aldosterone antagonists) Amiloride and triamterene (Na ⁺ channel blockers) (Option C)	Not specifically targeted by common diuretics.
Effect of K ⁺ -sparing Diuretics	Decrease Na ⁺ reabsorption and K ⁺ secretion	Minimal direct effect.



Solution for Question 9:

Correct Answer: C) Hypomagnesemia

Explanation: Gitelman syndrome is characterised by several electrolyte abnormalities, with hypomagnesemia being one of its hallmark features.

- It's caused by mutations in the SLC12A3 gene, which encodes the thiazide-sensitive sodium-chloride cotransporter in the distal convoluted tubule.
- It is an autosomal-recessive disorder with characteristics similar to Bartter syndrome but usually less severe: Salt and water loss. Mild water volume depletion. Activation of the renin-angiotensin-aldosterone system.
- Salt and water loss.
- Mild water volume depletion.
- Activation of the renin-angiotensin-aldosterone system.
- It typically presents with Hypokalemia (Option A is ruled out) Hypocalciuria (Option B is ruled out) Metabolic alkalosis (Option D is ruled out)
- Hypokalemia (Option A is ruled out)
- Hypocalciuria (Option B is ruled out)
- Metabolic alkalosis (Option D is ruled out)
- Treatment focuses on replacing losses of sodium chloride and potassium and correcting hypokalemia.
- Potential therapeutic approaches: Blockade of prostaglandin synthesis with NSAIDs
Administration of aldosterone antagonists like spironolactone
- Blockade of prostaglandin synthesis with NSAIDs
- Administration of aldosterone antagonists like spironolactone
- Salt and water loss.
- Mild water volume depletion.
- Activation of the renin-angiotensin-aldosterone system.
- Hypokalemia (Option A is ruled out)
- Hypocalciuria (Option B is ruled out)
- Metabolic alkalosis (Option D is ruled out)
- Blockade of prostaglandin synthesis with NSAIDs
- Administration of aldosterone antagonists like spironolactone

Gitelman syndrome is often described as a milder variant of Bartter syndrome but with distinct features such as hypomagnesemia and hypocalciuria.

Solution for Question 10:

Correct Answer: C) Typically inherited in an autosomal dominant manner.

Explanation: Bartter's syndrome is typically inherited in an autosomal recessive manner, not autosomal dominant

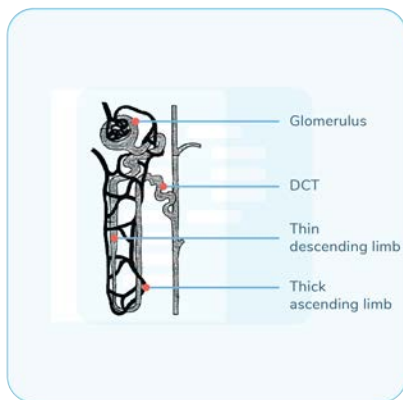
- It is a rare group of kidney disorders caused by genetic mutations affecting:
Sodium-chloride-potassium co-transporter in the loop of Henle. Potassium channels in the luminal membrane. Chloride channels in the basolateral membrane of the thick ascending loop of Henle.
 - Sodium-chloride-potassium co-transporter in the loop of Henle.
 - Potassium channels in the luminal membrane.
 - Chloride channels in the basolateral membrane of the thick ascending loop of Henle.
 - It is characterised by increased renal excretion of water, sodium, chloride, potassium, and calcium.
 - It activates the renin-angiotensin-aldosterone system (RAAS), resulting in hypokalemia (low potassium levels) and metabolic alkalosis.
 - The mechanism involves increased aldosterone and high distal tubular flow, stimulating potassium and hydrogen secretion in collecting tubules.
 - Subtypes: Based on loss-of-function mutations in the gene Type 1: Na–K–2Cl cotransporter Type 2: ROMK K⁺ channel Type 3: ClC–Kb Cl[–] channel Type 4: Barttin protein
 - Type 1: Na–K–2Cl cotransporter
 - Type 2: ROMK K⁺ channel
 - Type 3: ClC–Kb Cl[–] channel
 - Type 4: Barttin protein
 - Sodium-chloride-potassium co-transporter in the loop of Henle.
 - Potassium channels in the luminal membrane.
 - Chloride channels in the basolateral membrane of the thick ascending loop of Henle.
 - Type 1: Na–K–2Cl cotransporter
 - Type 2: ROMK K⁺ channel
 - Type 3: ClC–Kb Cl[–] channel
 - Type 4: Barttin protein
-

Solution for Question 11:

Correct Answer: C) B

Explanation:

The segment of the nephron where the filtrate becomes progressively hypertonic compared to plasma is the Descending limb of the loop of Henle (Segment B).



Descending limb of Loop of Henle:

- In the descending limb, the permeability to water allows for the passive reabsorption of water into the hypertonic medullary interstitium.
- This segment is impermeable to solutes such as sodium and chloride but permeable to water.
- As the filtrate descends into the medulla, where the interstitial osmolality is very high (up to 1200-1400 mOsm/kg), water moves out of the descending limb due to osmosis.
- This results in the filtrate becoming more concentrated (hypertonic) compared to plasma as it progresses through the descending limb.

Thick Ascending limb of LOH (Segment C)(Option C)

The ascending limb actively transports sodium, potassium, and chloride ions out of the filtrate into the interstitium, making the filtrate progressively hypotonic as ions are reabsorbed without water.

Early Distal convoluted tubule (Segment D)(Option A)

The early distal tubule has properties similar to those of the thick ascending loop of Henle, so further dilution of the tubular fluid occurs as solutes are reabsorbed while water remains in the tubule.

Glomerulus (Segment A)(Option B)

It's a filtration process. No active transport or reabsorption occurs in the glomerulus itself that would change the osmolality of the filtrate.

Solution for Question 12:

Correct Answer: C) Principal cells reabsorb sodium and secrete potassium ions into the tubular lumen.

Explanation: Principal cells reabsorb sodium and secrete potassium in the DCT.

Cell	Channel / Pump	Side	Function
Principal Cells (Option C)	Na ⁺ -K ⁺ ATPase pump	Basolateral membrane	Exchanges sodium with potassium in renal interstitial fluid.
Sodium channels	Luminal membrane	Reabsorb sodium from the lumen.	
Potassium channels	Luminal membrane	Secrete potassium into the lumen..	
Type A Intercalated Cells (Option D)	H ⁺ -ATPase pump	Luminal membrane	Secrete hydrogen ions into the lumen.
H ⁺ -K ⁺ ATPase pump	Luminal membrane	Reabsorb potassium ions from the lumen.	
Chloride-bicarbonate exchanger	Basolateral membrane	Exchanges Chloride with bicarbonate into the interstitial fluid.	
Type B Intercalated Cells (Option A and B)	H ⁺ -ATPase pump	Basolateral membrane	Reabsorb hydrogen ions.
Chloride-bicarbonate exchanger (Pendrin)	Apical membrane	Secrete bicarbonate into the lumen.	

Solution for Question 13:

Correct Answer: C) Proximal tubule

Explanation: Almost 65% of filtered calcium is reabsorbed in the PCT, mostly via the paracellular (passive) pathway.

99% of the filtered calcium is reabsorbed, and only 1% is excreted.

Sites of reabsorption

- PCT: 65% of filtered calcium is reabsorbed via a paracellular/passive pathway, and it parallels sodium and water reabsorption (Option C)
- Loop of Henle (Option A): Thick ascending limb reabsorbs 25-30% of filtered calcium. 50% via the paracellular route due to a positive lumen charge. 50% via the transcellular pathway, stimulated by PTH.
- 50% via the paracellular route due to a positive lumen charge.
- 50% via the transcellular pathway, stimulated by PTH.
- Distal and Collecting Tubules: (Option B and D) Reabsorbs 4-9% of filtered calcium. Almost entirely by active transport. Calcium enters cells via apical channels and exits via basolateral calcium-ATPase pumps and sodium-calcium exchangers.
- Reabsorbs 4-9% of filtered calcium.
- Almost entirely by active transport.
- Calcium enters cells via apical channels and exits via basolateral calcium-ATPase pumps and sodium-calcium exchangers.
- 50% via the paracellular route due to a positive lumen charge.
- 50% via the transcellular pathway, stimulated by PTH.
- Reabsorbs 4-9% of filtered calcium.
- Almost entirely by active transport.
- Calcium enters cells via apical channels and exits via basolateral calcium-ATPase pumps and sodium-calcium exchangers.

Hormonal Regulation of calcium excretion

- Parathyroid Hormone (PTH): Primary regulator. Increases calcium reabsorption in the thick ascending limb and distal tubules. Reduces urinary calcium excretion.
- Primary regulator.
- Increases calcium reabsorption in the thick ascending limb and distal tubules.
- Reduces urinary calcium excretion.

- Primary regulator.
- Increases calcium reabsorption in the thick ascending limb and distal tubules.
- Reduces urinary calcium excretion.
- Vitamin D (Calcitriol): Enhances calcium reabsorption in the distal tubule and thick ascending limb.
- Calcitonin: Stimulates calcium reabsorption in the thick ascending limb and distal tubule.

Calcium is passively absorbed in PCT and TAL (50% is passive), and magnesium is also absorbed similarly.

Solution for Question 14:

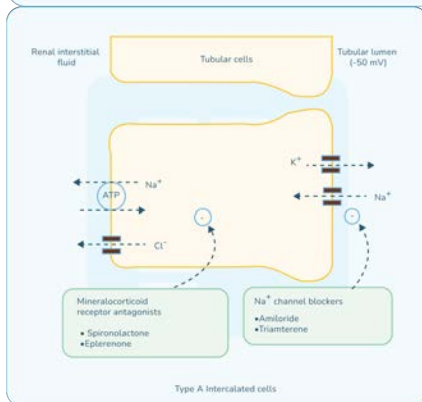
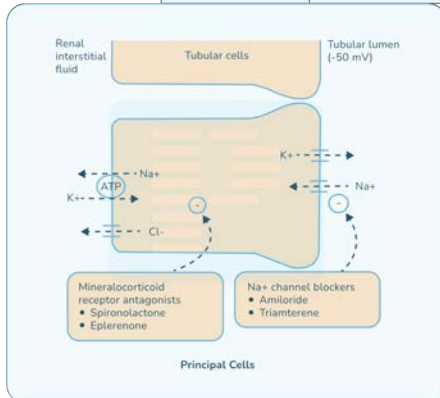
Correct Answer: C) Reabsorption of K⁺ in the PCT depends on the daily changes in potassium intake.

Explanation: Reabsorption of K⁺ in the PCT is relatively constant and not influenced by daily changes in potassium intake.

Mechanism of Potassium reabsorption and secretion at different levels:

Site	Potassium reabsorption and secretion	
PCT	Maximum potassium reabsorption (65%). Does not depend on daily K ⁺ intake.	
Loop of Henle	About 30% of K ⁺ is reabsorbed. Potassium is cotransported with sodium and chloride.	
DCT and Collecting duct	Principal cells	Major site for potassium secretion Na ⁺ -K ⁺ ATPase pump moves K ⁺ into the cell from the interstitium. Passive diffusion of K ⁺ from the cell into the tubular lumen through ROMK and BK channels.
Type A intercalate d cells.	Reabsorbs potassium during potassium depletion. Mechanism: Hydrogen-potassium ATPase transporter in the luminal	

	membrane exchanges hydrogenions for potassium	
Type B intercalated cells	Secrete potassium when there is an excess. Mechanism: Hydrogen-potassium ATPase transporter on the basolateral membrane pumps potassium into the cell, which then diffuses into the tubular lumen.	



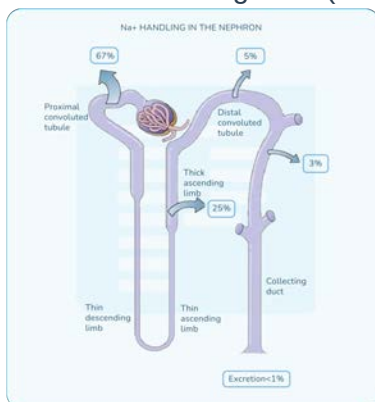
Solution for Question 15:

Correct Answer: A) Proximal convoluted tubule

Explanation:

- About 65-70% of sodium reabsorption occurs in the proximal convoluted tubule (PCT)
- Sodium is only filtered and reabsorbed by the renal tubules, and no secretion occurs.

- Only the thin descending segment of the nephron cannot reabsorb sodium as it is impermeable to all solutes, but all the other parts can reabsorb sodium.
- All segments of the nephron play a role in active and passive sodium reabsorption except for the ascending thin segment (near the JG apparatus), where only passive reabsorption occurs.
- Reabsorption of sodium at respective levels are Proximal convoluted tubule (PCT): 65-70 % Thick Ascending loop of Henle (TAL): 25-30 % (Option B) Distal Convoluted tubule (DCT): 5-7 % (Option C) Collecting Duct (CD): 2-3 % (Option D)
- Proximal convoluted tubule (PCT): 65-70 %
- Thick Ascending loop of Henle (TAL): 25-30 % (Option B)
- Distal Convoluted tubule (DCT): 5-7 % (Option C)
- Collecting Duct (CD): 2-3 % (Option D)
- Proximal convoluted tubule (PCT): 65-70 %
- Thick Ascending loop of Henle (TAL): 25-30 % (Option B)
- Distal Convoluted tubule (DCT): 5-7 % (Option C)
- Collecting Duct (CD): 2-3 % (Option D)



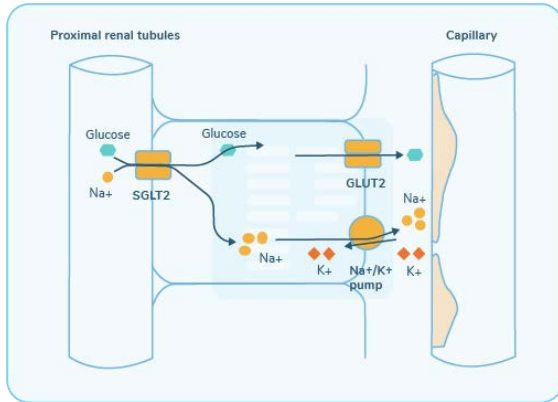
Solution for Question 16:

Correct Answer: A) Early Proximal convoluted tubule

Explanation: Sodium-glucose linked transporter (SGLT1 and 2), both are located on the brush border of the proximal convoluted tubule (Early PCT).

- It helps carry glucose against a concentration gradient.

- SGLT2 resorbs 90% of filtered glucose in the early proximal convoluted tubule (PCT), while SGLT1 reabsorbs 10% in the latter half.
- The reabsorbed glucose exits the cell via the basolateral membrane through GLUT2 and GLUT1 transporters, respectively.



Types of transporters in the different parts of the tubule:

PART of TUBULE	TRANSPORTERS
Proximal tubule	Early PCT (Option A) Sodium-Glucose Co-transporter (SGLT) Sodium-Amino Acid Cotransporter Sodium-Hydrogen Exchanger (NHE) Na ⁺ /K ⁺ ATPase Pump
Late PCT (Option D) Sodium-Chloride Cotransporter Chloride Channels	
Thin descending loop of Henle	Freely permeable to water and some solutes such as urea and sodium No specific transporter
Thick ascending loop of Henle (Option B)	Na ⁺ -K ⁺ -2Cl ⁻ Co-transporter (NKCC2) Sodium-Hydrogen Exchanger Na ⁺ /K ⁺ ATPase Pump
Distal convoluted tubule (Option C)	Early DCT Sodium-chloride cotransporter (NCC) Na ⁺ /K ⁺ ATPase Pump Chloride Channels
Late DCT Principal Cells : Epithelial Sodium Channels (ENaC) Potassium Channels Na ⁺ /K ⁺ ATPase Pump Type A Intercalated Cells: Hydrogen-ATPase Hydrogen-Potassium ATPase Chloride-Bicarbonate Exchanger	

Type B Intercalated Cells: Hydrogen-ATPase (Basolateral) Chloride-Bicarbonate Exchanger (Pendrin)	
---	--

Solution for Question 17:

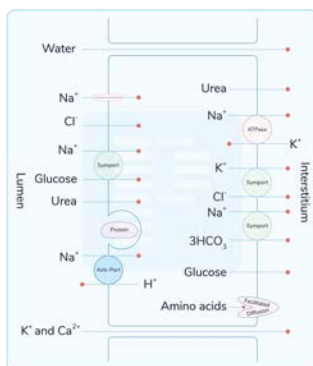
Correct Answer: D) Hydrogen Ions

Hydrogen ions (H^+) are actively secreted into the tubular lumen in the PCT (Not Reabsorbed)

Proximal Convoluted Tubule:

PCT plays a crucial role in renal function by reabsorbing approximately 65% of filtered sodium, chloride, potassium and bicarbonate, along with almost all filtered glucose and amino acids.

- Isotonic Absorption
- Passive reabsorption of Calcium (65%)



Sodium Resorption: Occurs through various Carrier Proteins

- **Na+/H+ Antiporter.**
- It provides H^+ ions and aids in the absorption of Bicarbonate
- Na/K ATPase is present on the Basolateral side, which creates the gradient for Sodium.
- Sodium Bi-Carbonate Co-transporter (NBC): Helps in Na reabsorption in the basolateral membrane.

Proximal Tubule has two parts:

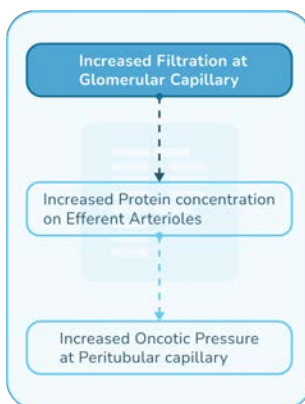
- Proximal Convoluted Tubule (PCT): It is a little impermeable for chloride reabsorption, leading to a higher concentration of chloride in the Proximal convoluted tubule.

- Proximal Straight Tubule (PST): Na^+ is reabsorbed along with chloride.

Solution for Question 18:

Correct Answer: C) Glomerulo-tubular balance

Glomerulo-tubular balance is the process by which the proximal tubule adjusts sodium reabsorption in response to changes in glomerular filtration rate (GFR). This ensures a constant percentage of sodium is reabsorbed despite variations in GFR.



Peritubular capillaries play a crucial role in reabsorbing substances from the renal tubules back into the bloodstream. These capillaries will proportionately reabsorb whatever is reabsorbed at the proximal tubule, thus depleting the Sodium in the interstitium.

This process is governed by Starling forces, which involve hydrostatic pressure (P) and oncotic pressure (π).

Renal Autoregulation (Option A) refers to the kidney's ability to maintain stable renal blood flow and GFR despite changes in systemic blood pressure.

Modulating GFR based on feedback from sodium concentrations (Option B)

Tubuloglomerular feedback is a mechanism where the macula densa cells in the distal tubule monitor sodium chloride levels and regulate afferent arteriolar resistance to control GFR.

Renin-angiotensin-aldosterone system (RAAS) regulating sodium reabsorption (Option D) The RAAS system is vital for regulating sodium and water balance in response to changes in blood pressure and blood volume by acting on distal nephron segments (e.g., distal convoluted tubule, collecting duct) to enhance sodium reabsorption.

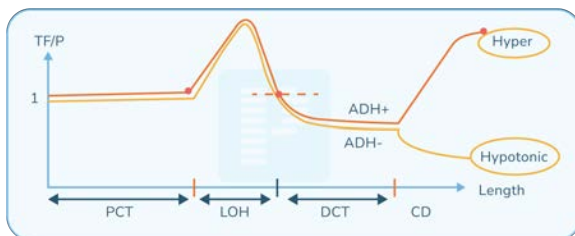
Solution for Question 19:

Correct Answer: B) End of the collecting duct

Explanation:

Without ADH, the collecting ducts cannot effectively reabsorb water, resulting in diluted urine with low osmolality despite high plasma osmolality. Therefore, the minimum urine tonicity is at the end of the collecting duct, where water reabsorption is impaired due to the lack of ADH.

The Graph explains the osmolality of Tubular Fluid with respect to plasma in the presence and absence of ADH:



RED: Presence of ADH

GREEN: Absence of ADH

Condition	Maximum Urine Tonicity	Minimum Urine Tonicity
In the Presence of ADH	End of the Collecting Duct(CD) / Tip of Loop of Henle (LOH)	Distal Convulated Tubule (DCT)
In the Absence of ADH	Tip of Loop of Henle (LOH)	End of Collecting Duct (CD)

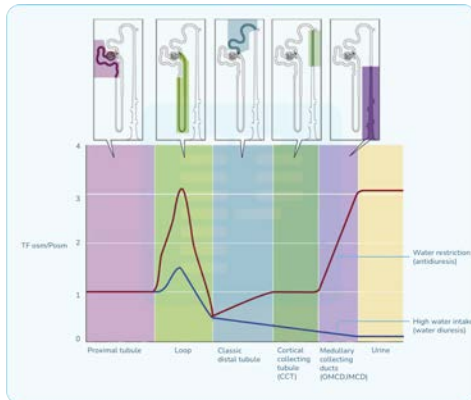
Solution for Question 20:

Correct Answer: A) Less than 1

In central diabetes insipidus, antidiuretic hormone (ADH) is deficient, which leads to impaired water reabsorption in the kidneys' collecting ducts. As a result, urine osmolality is typically much lower than plasma osmolality.

Tubular Fluid Plasma Ratio:

The graph below plots the length of the nephron and the ratio where TF is the tubular fluid osmolality and P is the plasma osmolality.



- 1 on the y-axis indicates the isotonic nature of urine
- The highest point of the graph is at the tip of the Loop of Henle (LOH)
- The red curve goes upwards in the presence of ADH due to Antidiuresis. $TF/P > 1$
- The Blue curve goes downwards in the absence of ADH, causing water diuresis $TF/P < 1$

Acid Secretion and Counter-Current Mechanism of the Kidney

1. What is the approximate daily production of non-volatile (metabolic) acids in an adult human body?

- A. 12,000 to 24,000 mmol
- B. 70 to 100 mmol
- C. 1,000 to 1,500 mmol
- D. 450 to 550 mmol

2. How does the excretion of ammonium (NH_4^+) contribute to bicarbonate (HCO_3^-) regulation in the kidneys?

- A. It directly increases the concentration of HCO_3^- in the urine.
- B. It generates HCO_3^- , which is released into the bloodstream.
- C. It decreases the overall HCO_3^- reabsorption in the kidneys.
- D. It prevents the secretion of H^+ in the renal tubules.

3. Which transport mechanism is primarily responsible for bicarbonate (HCO_3^-) reabsorption in the proximal convoluted tubule (PCT)?

- A. Na^+/K^+ ATPase pump
- B. Na^+/H^+ antiporter
- C. $\text{Cl}^-/\text{HCO}_3^-$ antiporter
- D. H^+ ATPase pump

4. What effect does aldosterone have on acid secretion in the distal convoluted tubule (DCT) and collecting duct (CD)?

- A. Decreases the activity of $H^+-ATPase$.
 - B. Inhibits the secretion of H^+ .
 - C. Increases the activity of $H^+-ATPase$
 - D. No effect on acid secretion.
-

5. What triggers the formation of ammonium ions (NH_4^+) and bicarbonate (HCO_3^-) in the ammonia buffer system?

- A. Increased sodium concentration
 - B. Alkalosis
 - C. Acidosis
 - D. Dehydration
-

6. A 40 y/o man presents with muscle weakness, bone pain, and polyuria. Lab tests show a normal anion gap metabolic acidosis, hypophosphatemia, hypokalemia, and glycosuria despite normal blood glucose levels. His urine pH is less than 5.5. Which of the following conditions is most likely responsible for his symptoms?

- A. Distal Renal Tubular Acidosis (Type 1)
 - B. Proximal Renal Tubular Acidosis (Type 2)
 - C. Hyperkalemic Renal Tubular Acidosis (Type 4)
 - D. Chronic Kidney Disease (CKD)
-

7. Which of the following is not a prerequisite for a functional counter-current mechanism in the body?

- A. Parallel and opposite flow of fluids
 - B. Close proximity
 - C. Active transport of solutes across all segments
 - D. Selective permeability of pathways
-

8. What is the primary aim of the counter-current mechanism in the kidneys?

- A. To dilute urine for better excretion of waste products
 - B. To conserve sodium and chloride ions in the body
 - C. To concentrate urine and maintain fluid and electrolyte balance
 - D. To increase blood flow to the renal cortex
-

9. Which of the following is NOT a component of the counter-current multiplier system in the kidneys?

- A. Cortical Collecting Duct (CCD)
 - B. Thick Ascending Limb (TAL) of the Loop of Henle
 - C. Thin Ascending Segment (ATS) of the Loop of Henle
 - D. Medullary Collecting Duct (MCD)
-

10. A 45 y/o patient presents with polyuria and polydipsia. Lab tests reveal a low urine osmolarity and a high plasma osmolarity. After administration of desmopressin (a synthetic analog of ADH), the patient's urine osmolarity increases significantly. Which of the following best explains the mechanism by which ADH improved the patient's condition?

- A. ADH increases the permeability of the ascending limb of the loop of Henle to water.
 - B. ADH decreases the active transport of sodium chloride in the thick ascending limb of the loop of Henle.
 - C. ADH increases the permeability of the collecting ducts to water.
 - D. ADH causes the osmolarity of the medullary interstitium to decrease rapidly.
-

11. Which of the following statements about urea transport in the kidneys is correct?

- A. UT-A1 and UT-A3 in the cortical collecting duct are regulated by aldosterone
 - B. Urea helps establish the osmotic gradient in the medullary pyramids but is not involved in urine concentration.
 - C. A high-protein diet reduces the kidneys' ability to concentrate urine.
 - D. UT-B is found in erythrocytes and the descending limbs of the vasa recta.
-

12. Which of the following substances plays the most crucial role in establishing and maintaining the hyperosmotic medullary interstitium?

- A. Sodium
 - B. Potassium
 - C. Urea
 - D. Chloride
-

13. Which of the following structures is responsible for maintaining the counter-current exchange mechanism in the kidney?

- A. Loops of Henle

B. Collecting Ducts

C. Vasa Recta

D. Glomerulus

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	3
Question 5	3
Question 6	2
Question 7	3
Question 8	3
Question 9	1
Question 10	3
Question 11	4
Question 12	3
Question 13	3

Solution for Question 1:

Answer: B) 70 to 100 mmol

Explanation:

- Non-volatile (Metabolic) or fixed acids are produced from incomplete metabolism of carbohydrates (e.g. lactate), fats (e.g. ketones), and proteins (e.g. sulfate, phosphate).
- In a typical diet, the body excretes about 40% of nonvolatile acids (around 30 mEq/day) through phosphate systems and 60% (approximately 50 mEq/day) through NH_4^+ excretion.

- The net production of these fixed acids is about 70 to 100 mmol of H^+ per day in an adult. This excludes lactate, as most of the lactate produced is metabolized and not excreted

12,000 to 24,000 mmol: Volatile (Respiratory) Acids(option A)

- The main volatile acid is carbonic acid (H_2CO_3), which is more correctly referred to as carbon dioxide (CO_2). CO_2 is the end-product of the complete oxidation of carbohydrates and fatty acids.
- Basal CO_2 production is typically 12,000 to 13,000 mmol/day. However, actual daily CO_2 production is usually higher, often in the range of 12,000 to 24,000 mmol/day, depending on metabolic activity levels.

Options C) and D) are incorrect values

Solution for Question 2:

Correct answer: B) It generates HCO_3^- , which is released into the bloodstream.

Explanation:

- The kidneys filter bicarbonate (HCO_3^-) from the blood. Approximately 85% to 90% of bicarbonate is reabsorbed in the proximal tubule, while only about 10% is reabsorbed in the distal tubules and collecting ducts.
- When NH_4^+ is excreted, it is produced from the deamination of amino acids. For each NH_4^+ ion excreted, one HCO_3^- ion is generated and released into the bloodstream, helping to maintain acid-base balance.

It directly increases the concentration of HCO_3^- in the urine: While NH_4^+ is excreted in the urine, it does not directly increase the concentration of HCO_3^- in the urine. Instead, NH_4^+ excretion is linked to HCO_3^- generation in the bloodstream (Option A).

It decreases the overall HCO_3^- reabsorption in the kidneys: The excretion of NH_4^+ does not decrease HCO_3^- reabsorption; rather, it is part of the mechanism that helps maintain HCO_3^- levels in the blood (Option C).

It prevents the secretion of H^+ in the renal tubules: The presence of NH_4^+ does not prevent the secretion of H^+ ions. In fact, H^+ ions are secreted into the tubular fluid, which is critical for HCO_3^- reabsorption (Option D).

Factors Influencing Bicarbonate Excretion

Increase H ⁺ Secretion and HCO ₃ ⁻ Reabsorption	Decrease H ⁺ Secretion and HCO ₃ ⁻ Reabsorption
↑Pco ₂	↓Pco ₂
↑H ⁺ , ↓HCO ₃ ⁻	↓H ⁺ , ↑HCO ₃ ⁻
↓Extracellular fluid volume	↑Extracellular fluid volume
↑Angiotensin II	↓Angiotensin II
↑Aldosterone	↓Aldosterone
Hypokalemia	Hyperkalemia

Solution for Question 3:

Correct Answer: B) Na⁺-H⁺ antiporter

Explanation:

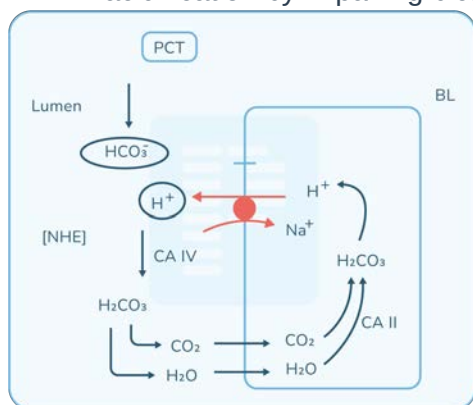
Mechanism of Bicarbonate Reabsorption

Proximal Convoluted Tubule (PCT):

- Na⁺-H⁺ Antiporter: The primary mechanism for bicarbonate reabsorption involves the Na⁺-H⁺ antiporter located on the apical surface of epithelial cells.
- Carbonic Anhydrase: CA-4 catalyzes the hydration of CO₂ to form carbonic acid (H₂CO₃), which dissociates into H⁺ and HCO₃⁻. This reaction occurs at the brush border of proximal tubular cells. CA Traps CO₂ in the tubular fluid, preventing its complete removal. Facilitates the secretion of H⁺ into the tubular lumen while retaining bicarbonate within the cell.
- CA Traps CO₂ in the tubular fluid, preventing its complete removal.
- Facilitates the secretion of H⁺ into the tubular lumen while retaining bicarbonate within the cell.
- Higher permeability to H⁺ and HCO₃⁻ leads to a tubular fluid pH of approximately 6.5.
- Bicarbonate ions diffuse into the interstitium, similar to K⁺ ions.
- Na⁺ and K⁺ Exchange: Na⁺-K⁺ ATPase pump is located in the basolateral membrane. Na⁺ enters the interstitium in exchange for K⁺ via the Na⁺-K⁺ ATPase, maintaining K⁺ concentration while decreasing Na⁺ concentration. (Option A)
- CA Traps CO₂ in the tubular fluid, preventing its complete removal.
- Facilitates the secretion of H⁺ into the tubular lumen while retaining bicarbonate within the cell.

Regulation of Acid-Base Balance

- Urine acidification in the PCT is dependent on Na^+ concentration.
- Parathyroid Hormone (PTH): PTH inhibits the Na^+/H^+ exchanger in the PCT, affecting urine pH.
- Carbonic Anhydrase Inhibitors: Drugs that inhibit carbonic anhydrase decrease urine acidification by impairing bicarbonate reabsorption.



$\text{Cl}^-/\text{HCO}_3^-$ antiporter, also known as Band 3 protein is a Chloride-bicarbonate exchanger at the basolateral membrane of the distal tubules and collecting ducts aids bicarbonate retention and urine acidification (Option C).

The H^+/ATPase pump is located at the apical surface of the tubular cells of distal tubules and collecting ducts and facilitates ATP-driven transport of H^+ into the tubular lumen (Option D).

Solution for Question 4:

Acid Secretion at the DCT and CD

- Mechanism: Intercalated Cells: Produce H^+ and HCO_3^- from CO_2 , facilitated by carbonic anhydrase. H^+ Secretion: Via H^+/ATPase or $\text{K}^+/\text{H}^+ \text{ATPase}$ into tubular fluid. HCO_3^- Transport: Into interstitium via $\text{Cl}^-/\text{HCO}_3^-$ antiporter.
- Intercalated Cells: Produce H^+ and HCO_3^- from CO_2 , facilitated by carbonic anhydrase.
- H^+ Secretion: Via H^+/ATPase or $\text{K}^+/\text{H}^+ \text{ATPase}$ into tubular fluid.
- HCO_3^- Transport: Into interstitium via $\text{Cl}^-/\text{HCO}_3^-$ antiporter.

- Acidification of Tubular Fluid: Low H^+ permeability results in more acidic fluid (pH ~4.5), crucial for urinary buffer excretion. H^+ secretion is independent of Na^+ concentration.
- Low H^+ permeability results in more acidic fluid (pH ~4.5), crucial for urinary buffer excretion.
- H^+ secretion is independent of Na^+ concentration.
- Role of Aldosterone: Enhances acid secretion by increasing proton pump activity. (Option C)
- Enhances acid secretion by increasing proton pump activity. (Option C)
- Band 3 Protein (Anion Exchanger 1 or AE1): Chloride-bicarbonate exchanger at basolateral membrane aids bicarbonate retention and urine acidification.
- Chloride-bicarbonate exchanger at basolateral membrane aids bicarbonate retention and urine acidification.
- NH_4^+ Excretion: H^+ combines with NH_3 to form NH_4^+ , which is excreted, and essential for acid-base balance.
- H^+ combines with NH_3 to form NH_4^+ , which is excreted, and essential for acid-base balance.
- Clinical Implications: Reduced H^+ secretion decreases net acid excretion and affects HCO_3^- buffering of non-volatile acids.
- Reduced H^+ secretion decreases net acid excretion and affects HCO_3^- buffering of non-volatile acids.
- Intercalated Cells: Produce H^+ and HCO_3^- from CO_2 , facilitated by carbonic anhydrase.
- H^+ Secretion: Via $H^+-ATPase$ or $K^+-H^+-ATPase$ into tubular fluid.
- HCO_3^- Transport: Into interstitium via Cl^- - HCO_3^- antiporter.
- Low H^+ permeability results in more acidic fluid (pH ~4.5), crucial for urinary buffer excretion.
- H^+ secretion is independent of Na^+ concentration.
- Enhances acid secretion by increasing proton pump activity. (Option C)
- Chloride-bicarbonate exchanger at basolateral membrane aids bicarbonate retention and urine acidification.
- H^+ combines with NH_3 to form NH_4^+ , which is excreted, and essential for acid-base balance.
- Reduced H^+ secretion decreases net acid excretion and affects HCO_3^- buffering of non-volatile acids.

A) Decreases the activity of $H^+-ATPase$: Aldosterone increases the activity of $H^+-ATPase$ but does not decrease it.

B) Inhibits the secretion of H^+ : Aldosterone enhances the secretion of H^+ , not inhibits it.

Solution for Question 5:

Acidosis stimulates the formation of ammonium ions (NH_4^+) and bicarbonate (HCO_3^-) in the ammonia buffer system. This process helps the body excrete excess acid and maintain acid-base balance.

BUFFERS	DESCRIPTION
Bicarbonate Buffer	Most powerful extracellular buffer system. Composed of H_2CO_3 and NaHCO_3 Buffers strong acids by forming H_2CO_3 and strong bases by reacting with H^+ to form H_2O and CO_2 . Effective due to regulation by kidneys and lungs.
Phosphate Buffer	Composed of H_2PO_4^- and HPO_4^{2-} . pK of 6.8, effective in slightly acidic urine. Phosphate concentration increases in tubular fluid due to water reabsorption.
Ammonia Buffer	Composed of ammonia (NH_3) and ammonium ion (NH_4^+). In the liver, Glutamine metabolized to 2NH_4^+ and 2HCO_3^- . NH_4^+ is secreted into the tubular lumen HCO_3^- reabsorbed into blood. Acidosis stimulates NH_4^+ and HCO_3^- formation; alkalosis decreases their formation. The rate of NH_4^+ excretion can increase significantly with chronic acidosis.
Others (less significant)	Urate Buffer Citrate Buffer

A) Increased sodium concentration: Sodium concentration does not directly trigger the formation of ammonium ions (NH_4^+) and bicarbonate (HCO_3^-) in the ammonia buffer system. Instead, sodium balance is primarily related to the regulation of fluid volume and blood pressure.

B) Alkalosis: In contrast, alkalosis decreases the formation of NH_4^+ and HCO_3^- .

D) Dehydration: Dehydration primarily affects fluid balance and can lead to an increase in blood osmolarity. While severe dehydration can lead to metabolic disturbances, it does not specifically trigger the formation of NH_4^+ and HCO_3^- in the ammonia buffer system.

Solution for Question 6:

Proximal RTA results from a defect in bicarbonate reabsorption in the proximal tubule. The patient's symptoms and laboratory findings match those of proximal RTA, making it the most likely diagnosis.

RTA	Features
Proximal Renal Tubular Acidosis (Type 2)	Caused by a defect in bicarbonate reabsorption in the proximal tubule. Associated with generalized proximal tubular dysfunction. Symptoms include low plasma bicarbonate, glycosuria, aminoaciduria, phosphaturia, hypophosphatemia, and hypokalemia The urine pH may be acidic when plasma bicarbonate is low.
Distal Renal Tubular Acidosis (Type 1)	Results from an inability to acidify the urine in the distal tubule. Characterized by persistently high urine pH (>5.5), hypokalemia, and a non-anion gap metabolic acidosis. Often associated with nephrolithiasis and nephrocalcinosis. (Option A)
Hyperkalemic Renal Tubular Acidosis (Type 4)	Characterized by hyperkalemia and a mild non-anion gap metabolic acidosis. Common in conditions such as diabetes mellitus and chronic kidney disease. (Option C)

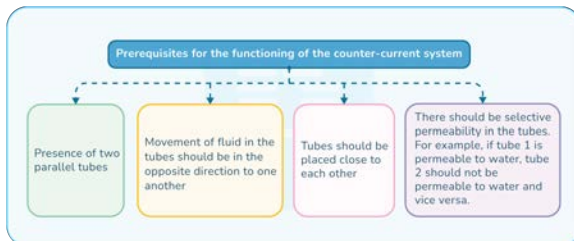
D) While CKD can lead to metabolic acidosis, it typically presents with features such as elevated serum creatinine and urea levels and a mixed acid-base disturbance that may include both metabolic acidosis and changes in electrolyte balance. The patient's specific findings are more indicative of proximal RTA rather than CKD alone.

Solution for Question 7:

Correct Answer: C) Active transport of solutes across all segments

Explanation:

This is NOT a prerequisite for a functional counter-current mechanism. While active transport is essential in some segments, such as the thick ascending limb of the loop of Henle, which actively transports Na^+ and Cl^- , it is not required across all segments of the system. Passive transport and selective permeability are critical components.



Parallel and opposite flow of fluids : This is a fundamental requirement for a counter-current mechanism. In the kidneys, the descending and ascending limbs of the loop of Henle and the vasa recta have fluid flows in opposite directions, which maximizes the efficiency of exchange processes (Option A)

Close proximity and selective permeability of pathways: This selective permeability is crucial for the system's function. In the loop of Henle, the descending limb is permeable to water but not solutes, while the ascending limb is impermeable to water but actively transports solutes (Option B)

A concentration gradient between inflow and outflow pathways : In the kidneys, the active transport of Na^+ and Cl^- from the ascending limb of the loop of Henle into the medullary interstitium creates the osmotic gradient that is essential for the counter-current mechanism to function effectively. The vasa recta helps maintain this gradient (Option D)

Solution for Question 8:

The counter-current mechanism concentrates urine by creating a hyperosmotic environment in the renal medulla. This allows the kidneys to excrete waste products with minimal water loss, conserving water and maintaining fluid and electrolyte balance.

A) To dilute urine for better excretion of waste products: The counter-current mechanism aims to concentrate, not dilute, the urine. Dilution would result in a loss of excess water, which could

lead to dehydration.

B) To conserve sodium and chloride ions in the body: While sodium and chloride reabsorption occurs during the process, the primary goal is not to conserve these ions but to create an osmotic gradient that facilitates water reabsorption.

D) To increase blood flow to the renal cortex: The counter-current mechanism primarily operates in the renal medulla and focuses on concentrating urine rather than regulating blood flow to the renal cortex.

Solution for Question 9:

The CCD is involved in the reabsorption of water and electrolytes under the influence of hormones such as aldosterone and antidiuretic hormone (ADH). However, it is not directly part of the counter-current multiplier system. The counter-current multiplier specifically refers to the mechanisms occurring within the loop of Henle and associated structures in the medulla, not the cortical regions.

COUNTERCURRENT MULTIPLIER	
Site	Function
Proximal Tubule	Delivers an isosmotic filtrate to the loop of Henle, with approximately 65% of filtered electrolytes and water reabsorbed. The osmolarity here is around 300 mOsm/L, similar to plasma.
Thin Descending Limb of the Loop of Henle	Highly permeable to water due to the presence of aquaporin-1 (AQP-1) channels and mildly permeable to solutes like urea and sodium chloride. As the filtrate moves deeper into the hyperosmotic medulla, water exits by osmosis, concentrating the tubular fluid.
Thin Ascending Limb of the Loop of Henle	Impermeable to water but permeable to solutes. NaCl passively diffuses out, and some urea enters, contributing to medullary hyperosmolarity. (Option C)
Thick Ascending Limb of the Loop of Henle (TAL)	Impermeable to water and urea, actively transports NaCl, K ⁺ out of the tubule into the interstitium using the Na ⁺ -K ⁺ -2Cl ⁻ cotransporter. This action increases interstitial osmolarity and delivers hypotonic fluid to the distal tubule. (Option B)

Medullary Collecting Duct (MCD)	Under the influence of ADH, becomes permeable to water. Water is reabsorbed from the MCD into the hyperosmotic medullary interstitium, concentrating the urine. The MCD operates within the hyperosmotic environment established by the counter-current multiplier. (Option D)
---------------------------------	---

Solution for Question 10:

ADH increases the permeability of the collecting ducts to water by promoting the insertion of aquaporin-2 water channels into the apical membranes of collecting duct cells. This allows water to be reabsorbed from the tubular fluid into the hyperosmotic medullary interstitium, concentrating the urine and reducing urine volume.

A) The ascending limb of the loop of Henle is impermeable to water, a characteristic that is essential for the creation of a hyperosmotic medullary interstitium. ADH does not alter this permeability.

B) ADH does not affect the active transport of sodium chloride in the thick ascending limb. The active transport in this segment is necessary for establishing the osmotic gradient but is not regulated by desmopressin.

D) ADH does not decrease the osmolarity of the medullary interstitium. Instead, it facilitates water reabsorption in the collecting ducts, leveraging the existing high osmolarity of the medullary interstitium to concentrate the urine.

Solution for Question 11:

UT-B is indeed found in erythrocytes and the descending limbs of the vasa recta.

Role of Urea in Kidney Function:

Functions:

- Osmotic Gradient: Urea helps establish the osmotic gradient in the medullary pyramids.
- Concentrated Urine: Contributes to forming concentrated urine in the collecting ducts. (Option B)

Urea Transport:

- Mediated by: Urea transporters (UT-A isoforms).
- Types: UT-A1 to UT-A4: Found in kidneys. UT-B: Found in erythrocytes and descending limbs of the vasa recta.
- UT-A1 to UT-A4: Found in kidneys.
- UT-B: Found in erythrocytes and descending limbs of the vasa recta.
- Key Transporters in Collecting Duct: UT-A1 and UT-A3: Found in the medullary collecting ducts. Regulated by vasopressin. (Option A)
- UT-A1 and UT-A3: Found in the medullary collecting ducts. Regulated by vasopressin. (Option A)
- UT-A1 to UT-A4: Found in kidneys.
- UT-B: Found in erythrocytes and descending limbs of the vasa recta.
- UT-A1 and UT-A3: Found in the medullary collecting ducts. Regulated by vasopressin. (Option A)

Regulation:

- Vasopressin (ADH): Increases during antidiuresis, raising urea levels in the medullary interstitium, and enhancing the kidney's urine concentrating capacity.

Impact of Diet:

- High-Protein Diet: Increases urea filtration, enhancing urine concentration ability. (Option C)
- Low-Protein Diet: Decreases urea filtration, reducing urine concentration ability.

Solution for Question 12:

Urea is the most important substance for establishing and maintaining the hyperosmotic medullary interstitium. It significantly contributes to the osmotic gradient and urine concentration by:

- Urea recycling: Urea is reabsorbed in the collecting ducts and secreted into the loop of Henle. This recycling helps to maintain a high concentration of urea in the medulla.

- **Osmotic Gradient:** The high concentration of urea in the medullary interstitium creates an osmotic gradient. This gradient allows for the reabsorption of water from the collecting ducts, concentrating the urine.
- **Water Reabsorption:** The osmotic gradient established by urea, along with sodium and other solutes, helps in reabsorbing water from the filtrate, making the medullary interstitium hyperosmotic.

A) Sodium is involved in creating the osmotic gradient but is not the most crucial substance for maintaining it.

B) Potassium plays a role in ionic transport esp. In the thick ascending limb of the loop of Henle, but is not the primary substance for the hyperosmotic gradient.

D) Chloride is involved in the osmotic gradient but is not as critical as urea for maintaining the hyperosmotic medullary interstitium.

Solution for Question 13:

The vasa recta maintains the osmotic gradient through countercurrent exchange.

A) The loops of Henle are responsible for creating the osmotic gradient (countercurrent multiplier system) but not for maintaining it.

B) The collecting ducts use the osmotic gradient to concentrate urine but do not maintain the gradient.

D) The glomerulus is involved in the filtration of blood to form urine but does not maintain the osmotic gradient in the medulla.

Counter-Current Exchange Mechanism in the Vasa Recta

Aspect of Vasa Recta	Details
Function	Maintains the osmotic gradient established by the loops of Henle, crucial for concentrated urine production.
Structure	Hairpin-loop blood vessels that descend and ascend alongside the loops of Henle.

Descending Limb	Solutes (NaCl and urea) diffuse from the interstitium into the vasa recta. Water moves out of the vasa recta into the interstitium due to the osmotic gradient. Result: Blood becomes more concentrated as it descends.
Ascending Limb	Solutes diffuse from the vasa recta into the interstitium. Water moves from the interstitium into the vasa recta. The ascending vasa recta are fenestrated to facilitate this exchange.
Net Effect	Role: Prevents dissipation of the osmotic gradient. Outcome: Maintains the hypertonicity of the medulla, essential for water reabsorption and concentrated urine production.
Conclusion:	Works with the loops of Henle to establish and maintain medullary osmotic gradient

Previous Year Questions

1. What is the typical Glomerular Filtration Rate (GFR) in a healthy adult?

- A. 125 ml/min
 - B. 200 ml/min
 - C. 100 ml/min
 - D. 150 ml/min
-

2. How does the kidney react to shock?

- A. Increasing renal blood flow
 - B. Constricting afferent arteriole
 - C. Increasing GFR
 - D. Both B and C
-

3. In the renal arterial circulation, the glomerulus is supplied with blood through an afferent arteriole. Unlike other capillary beds, the glomerular capillaries do not lead to venules but instead to efferent arterioles. The resistance in the efferent arterioles creates enough hydrostatic pressure within the glomerulus to facilitate ultrafiltration. Which of the subsequent molecules will be found in the glomerular ultrafiltrate?

- A. Polycationic large radius molecules
- B. Polycationic small radius molecules
- C. Polyanionic small radius molecules
- D. Polyanionic large radius molecules

4. As filtrate flows through PCT, the concentration of all of the following decreases except:

(or)

Except for which of the following, the concentration of all the following decreases as the filtrate passes through the proximal convoluted tubule (PCT)?

- A. Aminoacid
- B. Glucose
- C. Chloride
- D. Bicarbonate

5. In a normal physiological state, what is the cause of hemoglobin being unable to pass through the glomerulus?

- A. It is a large molecule bound with another protein
- B. It gets attracted to the negative charge of the glomerular filtration barrier
- C. Both A and B
- D. None of the above

6. Which of the following leads to a decrease in GFR?

- A. Increased RBF
- B. Increased Glomerular hydrostatic pressure
- C. Decreased Bowman's oncotic pressure
- D. Decreased Glomerular oncotic pressure

7. Which part of renal tubules reabsorbs maximum water in the presence of ADH?

- A. PCT
 - B. DCT
 - C. Loop of Henle
 - D. Collecting duct
-

8. Where does the majority of water absorption occur within the renal tubules?

- A. Proximal Convoluted Tubule (PCT)
 - B. Distal Convoluted Tubule (DCT)
 - C. Loop of Henle (LOH)
 - D. Collecting Duct
-

9. Which of the following is true regarding an increase in Glomerular Filtration Rate (GFR)?

- A. Increase in the resistance of the afferent arteriole
 - B. Decrease in the resistance of the efferent arteriole
 - C. Increase in Bowman's capsule hydrostatic pressure
 - D. Increase in renal perfusion
-

10. A patient arrives at the emergency department unconscious. An arterial blood gas (ABG) analysis reveals the following results: pH: 7.1 PCO₂: 23 mmHg PO₂: 98

mmHg HCO₃: 16 mEq/L What do these ABG results indicate about the patient's condition?

- A. Metabolic acidosis
 - B. Metabolic alkalosis
 - C. Respiratory acidosis
 - D. Respiratory alkalosis
-

11. What renal function is represented by the formula UV/P ■?

- A. Filtration
 - B. Tubular Secretion
 - C. Tubular Reabsorption
 - D. Clearance
-

12. Which segment of the nephron has the lowest osmolality under the influence of antidiuretic hormone (ADH)?

- A. Early distal tubule
 - B. Descending limb of the loop of Henle
 - C. Collecting duct
 - D. Late distal tubule
-

13. In which of the body's parts is the countercurrent mechanism found? A. Eye B. Testes C. Kidney D. Gut E. Lungs

- A. A and B are correct
- B. A, B, C are correct

C. A, B, C, D, E are correct

D. B, C, D, E are correct

14. A renal biopsy image is shown. Identify the cells marked in the histological section.



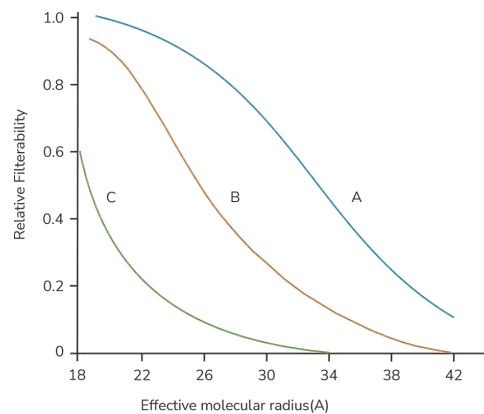
A. Macula densa

B. Juxtaglomerular (JG) cells

C. Lacis cells

D. Podocytes

15. This is the graph depicting the filterability of dextran through the kidney, with 1 representing complete filtration and 0 meaning no filtration. Choose the substance A, B, and C respectively.



- A. A - Polycationic, B - Neutral, C - Polyanionic
- B. A - Polyanionic, B - Neutral, C - Polycationic
- C. A - Neutral, B - Polycationic, C - Polyanionic
- D. A - Polyanionic, B - Polycationic, C - Neutral

16. Renin is secreted by?

- A. Lacis cells
- B. Macula densa
- C. JG cells
- D. Intraglomerular mesangial cells

17. Which of the following is the site of maximum water reabsorption?

- A. Proximal tubule
- B. Loop of Henle
- C. Distal convoluted tubule
- D. Collecting duct

18. Which of the following is the diluting segment of the kidney?

- A. PCT
 - B. Collecting duct
 - C. Ascending thick loop of Henle
 - D. Descending loop of Henle
-

19. Glucose is absorbed by?

- A. Upper part of proximal tubule
 - B. Ascending limb of LOH
 - C. Cortical Collecting duct
 - D. Medullary collecting duct
-

20. Mark the TRUE statements:

- A. Efferent arteriole dilation decreases RBF
 - B. Afferent arteriole constriction increases GFR
 - C. Efferent arteriolar constriction decreases renal blood flow
 - D. Mild contraction of efferent arteriole decreases GFR
-

21. What happens to urine after passing the thin descending loop of Henle?

- A. Hypotonic
- B. Normal

- C. Hypertonic
 - D. Remains unchanged
-

22. Glucose is reabsorbed in which part?

- A. Proximal convoluted tubule
 - B. Proximal straight tubule
 - C. Collecting duct
 - D. Distal convoluted tubule
-

23. The cause of edema in nephrotic syndrome is:

- A. Increased colloid osmotic pressure
 - B. Increased hydrostatic pressure
 - C. Decreased colloid osmotic pressure
 - D. Decreased hydrostatic pressure
-

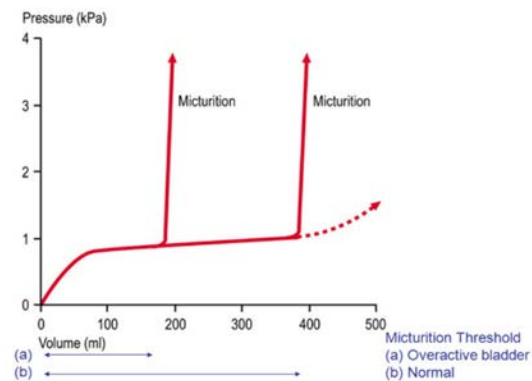
24. Normal anion gap value is ____ mEq/L?

- A. 4-6
 - B. 6-12
 - C. 12-24
 - D. 24-50
-

25. Why is sugar added to ORS?

- A. Enhance acceptability
 - B. Enhance shelf life
 - C. Enhance salt absorption
 - D. Enhance taste
-

26. In the adjacent graph, plateau phase represents which one of the following properties?



- A. Elasticity
 - B. Plasticity
 - C. Contractility
 - D. Automaticity
-

27. Which of the following statements is accurate regarding creatinine clearance?

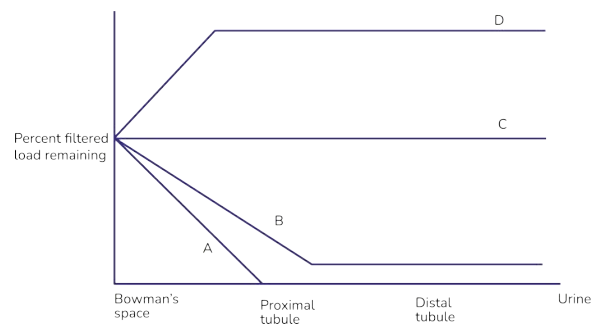
- A. Equal to inulin clearance
- B. Less than inulin clearance
- C. Greater than inulin clearance

D. Equal to renal plasma flow

28. What is the role of tubuloglomerular feedback?

- A. Regulate GFR
 - B. Decrease plasma sodium
 - C. Increase plasma volume
 - D. Decrease tubular secretion
-

29. Identify the correctly matched pair of substances with their renal clearance from the graph below.



- A. A-Glucose, B-PAH, C- Bicarbonate and D - Inulin
 - B. A-Glucose, B- Bicarbonate, C-Inulin and D - PAH
 - C. A-PAH, B-Inulin, C - Glucose and D - Bicarbonate
 - D. A - Inulin, B - Glucose, C - Bicarbonate and D - PAH
-

30. Which among the following organs has the least arteriovenous oxygen difference?

- A. Brain
 - B. Kidney
 - C. Skin
 - D. Liver
-

31. Where is macula densa located?

- A. Junction of thick descending limb and DCT
 - B. Junction of thick descending limb and PCT
 - C. Junction of thick ascending limb and DCT
 - D. Junction of DCT and collecting duct
-

32. Hyperkalemia refers to the serum potassium concentration of ____.

- A. 4.5 mEq/L
 - B. 6.0 mEq/L
 - C. 4.0 mEq/L
 - D. 5 meq/L
-

33. The protein in the glomerular basement membrane responsible for charge-dependent filtration is?

- A. Fibronectin
- B. Heparan sulfate
- C. Collagen Type IV
- D. Entactin

34. The formula for urea clearance is $U \times V/P$, where U is?

- A. Urine volume
 - B. Serum urea in mg/100 ml
 - C. Urine concentration of urea in mg/100ml
 - D. Urine concentration of urea in mg/ml
-

35. All of the following ions have decreased concentration on the luminal side of the proximal convoluted tubule except:

- A. Amino acids
 - B. Glucose
 - C. Chloride
 - D. Bicarbonate
-

36. Which one of the following is not associated with renal auto-regulation?

- A. Tubuloglomerular feedback
 - B. Myogenic reflex
 - C. Calcium
 - D. Renal sympathetic system fibers
-

37. Which of the following ions in the thick ascending limb of the loop of Henle is responsible for the reabsorption of divalent ions?

- A. H^+
 - B. HCO_3^-
 - C. K^+
 - D. Na^+
-

38. Which sensory receptors are accountable for tactile discrimination among the options provided? Merkel disc Ruffini end organ Pacinian corpuscle Meissner's corpuscle

- A. 1 and 4 are correct
 - B. 2 is correct
 - C. 3 is correct
 - D. None are correct.
 - E. Both A & D
-

39. According to myogenic hypothesis of renal autoregulation the afferent arterioles contract in response?

- A. NO
 - B. Noradrenaline
 - C. Calcium
 - D. Adenosine
-

40. Which part of the nephron forms the macula densa?

- A. PCT
- B. DCT

- C. Afferent arteriole
 - D. Bowman's capsule
-

41. Through which aquaporin channel does ADH act?

- A. 1
 - B. 2
 - C. 3
 - D. 4
-

42. As filtrate flows through PCT, the concentration of all of the following decreases except:

- A. Amino-acid
 - B. Glucose
 - C. Chloride
 - D. Bicarbonate
-

43. Which of the following components is not responsible for the countercurrent multiplier mechanism in the kidney?

- A. Sodium outflow in thick ascending limb
 - B. Water outflow in thin descending limb
 - C. Sodium outflow in thin ascending limb
 - D. Flow of tubular fluid from PCT to DCT
-

44. Which of the following is true about renal tubular transport?

- A. Maximum reabsorption in the distal tubule
 - B. Urea is absorbed 100% in PCT
 - C. H^+ is absorbed in the proximal tubule
 - D. Amino acid is reabsorbed in the proximal tubule
-

45. All of the following are true regarding the actions of endothelin-1 except_____.

- A. Bronchodilation
 - B. Vasoconstriction
 - C. Decreased GFR
 - D. Has inotropic effect
-

46. GFR is increased by all except:

- A. Increased renal blood flow
 - B. Efferent arteriolar constriction
 - C. Ureteric stone
 - D. Decreased oncotic pressure
-

47. Calculate the net filtration pressure across the GBM for a patient with Glomerular ultrafiltration coefficient = 6, Glomerular capillary hydrostatic pressure = 45, Glomerular capillary oncotic pressure = 20, Bowman's capsule hydrostatic

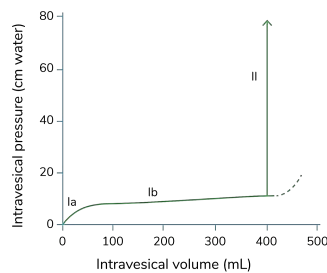
pressure = 10, Bowman's capsule oncotic pressure = 0.

- A. 90 mmHg
 - B. 60 mmHg
 - C. 30 mmHg
 - D. 15 mmHg
-

48. Normal anion gap is _____ mEq/L?

- A. 4-6
 - B. 6-12
 - C. 12-24
 - D. 24-50
-

49. Which of the following statements is true regarding the given cystometrogram?



- A. Segment Ia is due to residual urine
- B. Segment Ib is due to Laplace law
- C. Micturition fails to happen in segment II
- D. Micturition occurs in segment Ib

50. Which of the following factors is not necessary for determining creatinine clearance?

- A. Urine creatinine
 - B. Serum creatinine
 - C. Urine flow per unit time
 - D. The volume of serum passing through the kidney in 24 hours
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	2
Question 4	3
Question 5	1
Question 6	3
Question 7	4
Question 8	1
Question 9	4
Question 10	1
Question 11	4
Question 12	1
Question 13	4
Question 14	3
Question 15	1
Question 16	3
Question 17	1
Question 18	3
Question 19	1
Question 20	3
Question 21	3
Question 22	1

Question 23	3
Question 24	2
Question 25	3
Question 26	2
Question 27	3
Question 28	1
Question 29	2
Question 30	2
Question 31	3
Question 32	2
Question 33	2
Question 34	4
Question 35	3
Question 36	4
Question 37	3
Question 38	1,5
Question 39	3
Question 40	2
Question 41	2
Question 42	3
Question 43	4
Question 44	4
Question 45	1
Question 46	3
Question 47	4
Question 48	2

Question 49	2
Question 50	4

Solution for Question 1:

Correct Option A - 125 ml/min:

- This is the typical GFR for a healthy adult. GFR measures how well the kidneys are filtering blood, and 125 ml/min is considered the normal rate, reflecting optimal kidney function.

Incorrect Options:

Option B - 200 ml/min: This value is higher than the normal range and could indicate an overestimation of kidney function or an error in measurement. Such a high rate is not typical for healthy adults.

Option C - 100 ml/min: This value is slightly below the normal range. A GFR of 100 ml/min may suggest mild kidney impairment, although it can still be considered acceptable in some contexts.

Option D - 150 ml/min: This value is above the normal range, which could indicate an overestimation or abnormal kidney function. It is not a typical GFR for a healthy adult.

Solution for Question 2:

Correct Option B: Constricting afferent arteriole

- During shock, the kidney responds by constricting the afferent arteriole, which is the blood vessel that supplies blood to the glomerulus.
- this constriction helps to maintain the glomerular filtration pressure and prevent a drastic decrease in the glomerular filtration rate (GFR).
- By constricting the afferent arteriole, the kidney can redirect blood flow to vital organs and ensure adequate perfusion.

Incorrect Options:

Option A: Increasing renal blood flow- In response to shock, the kidney actually decreases renal blood flow. The constriction of the afferent arteriole reduces the amount of blood entering the glomerulus, resulting in reduced renal blood flow.

Option C: Increasing GFR- During shock, the kidney aims to maintain a stable GFR rather than increasing it. The constriction of the afferent arteriole helps to preserve the glomerular filtration pressure and prevent a significant decline in the GFR. This response is important for maintaining renal function in conditions of reduced blood volume or blood pressure.

Option D: Both B and C- This statement is incorrect because option (c) is incorrect. While the kidney does respond to shock by increasing the tone of the afferent arteriole (option B), it does not increase the GFR (option C).

Solution for Question 3:

Correct Option B:

The glomerular ultrafiltrate primarily consists of small molecules and ions that can pass through the filtration barrier of the glomerulus. The filtration barrier consists of three layers: the fenestrated endothelium of the glomerular capillaries, the basement membrane, and the filtration slits formed by podocyte foot processes. This barrier allows the passage of small molecules, such as water, electrolytes, glucose, and small proteins, while preventing the filtration of larger molecules like red blood cells and most plasma proteins.

The hydrostatic pressure within the glomerulus, generated by the resistance of the efferent arteriole, is responsible for the ultrafiltration process. This pressure forces water and small solutes through the filtration barrier into the glomerular capsule, forming the glomerular ultrafiltrate.

Polycationic small radius molecules, which are positively charged and have a small size, are capable of passing through the filtration barrier and would therefore be present in the glomerular ultrafiltrate.

Incorrect Options:

Option A. Polycationic large radius molecules: Large molecules, regardless of their charge, generally cannot pass through the filtration barrier of the glomerulus. Therefore, polycationic large radius molecules would not be present in the glomerular ultrafiltrate.

Option C. Polyanionic small radius molecules: Polyanionic small radius molecules, which are negatively charged and have a small size, may pass through the filtration barrier of the glomerulus. However, the question specifically mentions polycationic molecules, so polyanionic molecules are not relevant to the answer.

Option D. Polyanionic large radius molecules: Similar to option A, large molecules, regardless of their charge, would be hindered from passing through the filtration barrier of the glomerulus. Therefore, polyanionic large radius molecules would not be present in the glomerular ultrafiltrate.

Solution for Question 4:

Correct Option C:

- As the filtrate flows through the proximal convoluted tubule (PCT) of the nephron in the kidney, various substances are reabsorbed back into the bloodstream. This reabsorption process helps to maintain the balance of electrolytes and other solutes in the body.
- In the PCT, glucose, amino acids, and bicarbonate are actively reabsorbed from the filtrate back into the bloodstream. This reabsorption is essential for maintaining the appropriate levels of these substances in the body. Therefore, the concentrations of glucose, amino acids, and bicarbonate decrease as they are being reabsorbed.
- However, chloride (Cl^-) is primarily passively reabsorbed in the PCT. This means that it follows the movement of other solutes, particularly sodium (Na^+), through the paracellular pathway. As sodium is actively reabsorbed in the PCT, chloride ions also passively move along, resulting in their reabsorption. Therefore, the concentration of chloride does not decrease significantly as the filtrate flows through the PCT.

Incorrect Options:

Option A: Amino acid: The concentration of amino acids decreases as they are actively reabsorbed in the PCT.

Option B: Glucose: The glucose concentration decreases as it is actively reabsorbed in the PCT.

Option D: Bicarbonate: The concentration of bicarbonate decreases as it is actively reabsorbed in the PCT.

Solution for Question 5:

Correct Option A. It is a large molecule bound with another protein

The correct reason that hemoglobin cannot pass through the glomerulus in a normal physiological state is because it is a large molecule. Hemoglobin is a complex protein with a molecular weight of about 64,500 Daltons. The glomerular filtration barrier in the kidneys allows the passage of small molecules and ions but restricts the movement of large proteins and cells. Hemoglobin is too large to pass through the filtration barrier, so it is retained within the bloodstream and does not get filtered into the renal tubules.

Incorrect Options:

Option B. It gets attracted to the negative charge of the glomerular filtration barrier

This statement is not accurate. The glomerular filtration barrier does have negative charges due to the presence of glycoproteins in the glomerular basement membrane. These negative charges repel negatively charged molecules, preventing them from being filtered. However, hemoglobin is not positively charged; its inability to pass through the glomerulus is primarily due to its large size, not its charge.

Option C. Both A and B

Option C is incorrect because hemoglobin's inability to pass through the glomerulus is primarily due to its large size.

Option D. None of the above

Option D is incorrect based on the explanation provided above. Hemoglobin's large size is the main reason it cannot pass through the glomerulus in a normal physiological state.

Solution for Question 6:

Correct Option C - Decreased Bowman's oncotic pressure:

Net filtration pressure = $P_C - P_B + \pi_B - \pi_C$

- P_C - Hydrostatic pressure of glomerular capillaries
- P_B - Hydrostatic pressure of Bowman's capsule

- π_B - Oncotic pressure of Bowman's capsule
- π_C - Oncotic pressure of glomerular capillaries

Hence, a decrease in Bowman's oncotic pressure causes a decrease in GFR rather than an increase.

Incorrect Options :

Option A - Increased RBF: Increased blood flow to the kidney enhances blood delivery to the glomerulus, facilitating filtration, while decreased flow hampers filtration.

Option B - Increased Glomerular hydrostatic pressure: An increase in Glomerular hydrostatic pressure increases the net forces hence increasing the GFR.

Option D - Decreased Glomerular oncotic pressure: Glomerular capillary osmotic pressure, from plasma proteins, opposes filtration. So, low protein levels increase GFR, while dehydration reduces GFR and urine production.

Solution for Question 7:

Correct Option D - Collecting duct:

- Antidiuretic Hormone (ADH) primarily acts on the collecting ducts of the kidneys. In the presence of ADH, the collecting ducts become more permeable to water by increasing the expression of aquaporin-2 channels on the tubular cells. This allows more water to be reabsorbed from the filtrate back into the bloodstream, helping to concentrate urine and conserve water in the body.

Incorrect Options

- A. Proximal Convolute Tubule (PCT) : The PCT does reabsorb a large amount of water, but this process is independent of ADH.
- B. Distal Convolute Tubule (DCT) : The DCT reabsorbs a smaller portion of water, but the major site of ADH action is the collecting duct.
- C. Loop of Henle : The ascending limb of the Loop of Henle is impermeable to water, while the descending limb reabsorbs water passively, but this is not directly influenced by ADH.

Solution for Question 8:

Correct Option A - Proximal Convolutd Tubule (PCT):

- Approximately 65-70% of water reabsorption occurs in the PCT. The PCT plays a crucial role in reabsorbing both sodium and water from the filtrate.
- This process begins with the filtration of blood at the glomerulus, which produces a filtrate rich in sodium and other solutes. The PCT cells, connected by tight junctions, actively transport these substances from the lumen of the nephron into the surrounding peritubular capillary blood.
- The approximate contribution to sodium reabsorption in the PCT is: SGLT: 5%
Co-transporters: 45% NHE: 50%
- SGLT: 5%
- Co-transporters: 45%
- NHE: 50%
- SGLT: 5%
- Co-transporters: 45%
- NHE: 50%

Incorrect Options:

Option B - Distal Convolutd Tubule (DCT): The DCT is responsible for fine-tuning the reabsorption of sodium and water, but it does not account for the majority of water reabsorption. It primarily adjusts the concentration of urine in response to hormonal signals.

Option C - Loop of Henle (LOH): The LOH plays a significant role in establishing the medullary osmotic gradient, crucial for water reabsorption in the collecting duct, but does not absorb the majority of water.

Option D - Collecting Duct: While the collecting duct is responsible for the final adjustments in water reabsorption under the influence of antidiuretic hormone (ADH), the majority of water absorption has already occurred by this point in the PCT.

Solution for Question 9:

Correct option D - Increase in renal perfusion:

- Increasing renal perfusion (blood flow to the kidneys) enhances the amount of blood reaching the glomeruli, which typically increases the GFR.

Incorrect Options:

Option A - Increase in the resistance of the afferent arteriole: Increasing the resistance of the afferent arteriole reduces the blood flow into the glomerulus, which decreases the GFR.

Option B - Decrease in the resistance of the efferent arteriole: Decreasing the resistance of the efferent arteriole would lead to increased outflow from the glomerulus, which typically reduces the GFR, not increases it.

Option C - Increase in Bowman's capsule hydrostatic pressure: An increase in the hydrostatic pressure within Bowman's capsule would oppose filtration and thus decrease the GFR.

Solution for Question 10:

Correct Option A - Metabolic Acidosis:

To determine the type of acid-base disturbance, let's break down the ABG results:

- pH: The pH of 7.1 indicates acidemia (a lower-than-normal pH).
- HCO₃ (Bicarbonate): The bicarbonate level is 16 mEq/L, which is lower than the normal range (22-26 mEq/L). Low bicarbonate levels typically indicate a primary metabolic problem.
- PCO₂ (Partial Pressure of Carbon Dioxide): The PCO₂ level is 23 mmHg, which is lower than the normal range (35-45 mmHg). This indicates that the body is compensating for the primary metabolic disturbance by increasing respiratory rate to blow off CO₂.
- Metabolic Acidosis (Option A): The low pH (acidosis) combined with low HCO₃ suggests a primary metabolic acidosis. The decreased HCO₃ is consistent with metabolic acidosis. The low PCO₂ is a compensatory response to the metabolic acidosis, where the body is trying to reduce CO₂ to help counteract the acidosis.

Incorrect Options:

Option B - Metabolic Alkalosis: This would be characterized by a high pH and high HCO₃. This is not the case here, as the pH is low and HCO₃ is low.

Option C - Respiratory Acidosis: This would be characterized by a low pH and high PCO₂, not a low PCO₂. The bicarbonate level would not be as low in a primary respiratory acidosis.

Option D - Respiratory Alkalosis: This would show a high pH and low PCO₂, but the bicarbonate level would be normal or compensatory high. In this case, the pH is low, and HCO₃ is also low.

Solution for Question 11:

Correct Option D - Clearance:

- The formula UV/P , where U is the concentration of a substance in the urine, V is the urine flow rate, and P is the concentration of the substance in the plasma, represents the renal clearance of that substance. It quantifies the volume of plasma from which the substance is completely removed per unit time.

$$C = \left(\frac{\text{Amount in urine (ml/min)}}{\text{Plasma conc. (mg/ml)}} \right)$$

$C = \text{ml/min} = \text{volume/time}$

Incorrect Options:

Option A - Filtration: This process refers to the initial filtration of blood through the glomerulus to form filtrate. The formula ■ does not measure this process directly.

Option B - Tubular Secretion: This is the process by which substances are actively transported from the blood into the renal tubules. The formula does not specifically measure tubular secretion.

Option C - Tubular Reabsorption: This refers to the process of reabsorbing substances from the renal tubules back into the blood. The formula ■ does not measure reabsorption directly.

Solution for Question 12:

Correct Answer: A) Early distal tubule

Explanation:

The early distal tubule has the lowest osmolality under the influence of antidiuretic hormone (ADH) .

- Mechanism: ADH increases water reabsorption in the collecting duct and late distal tubule , concentrating the urine. The descending limb of the loop of Henle is highly permeable to water but not solutes , leading to an increase in osmolality . In contrast, the early distal tubule is impermeable to water , but actively reabsorbs NaCl , making the tubular fluid hypotonic. As a result, the lowest osmolality is found in the early distal tubule (~100mOsm/kg), even in the presence of ADH.
- ADH increases water reabsorption in the collecting duct and late distal tubule , concentrating the urine.
- The descending limb of the loop of Henle is highly permeable to water but not solutes , leading to an increase in osmolality .
- In contrast, the early distal tubule is impermeable to water , but actively reabsorbs NaCl , making the tubular fluid hypotonic.
- As a result, the lowest osmolality is found in the early distal tubule (~100mOsm/kg), even in the presence of ADH.
- ADH increases water reabsorption in the collecting duct and late distal tubule , concentrating the urine.
- The descending limb of the loop of Henle is highly permeable to water but not solutes , leading to an increase in osmolality .
- In contrast, the early distal tubule is impermeable to water , but actively reabsorbs NaCl , making the tubular fluid hypotonic.
- As a result, the lowest osmolality is found in the early distal tubule (~100mOsm/kg), even in the presence of ADH.

Descending limb of the loop of Henle (Option B) is incorrect because it is highly permeable to water and allows water to leave, increasing osmolality (~1200 mOsm/kg in the medullary region).

Collecting duct (Option C) is incorrect because under ADH influence, the collecting duct becomes highly water-permeable , leading to urine concentration (osmolality can reach >1200 mOsm/kg).

Late distal tubule (Option D) is incorrect because it is more permeable to water in the presence of ADH, leading to an increase in osmolality compared to the early distal tubule.

Solution for Question 13:

The countercurrent mechanism is primarily found in the kidneys . It plays a crucial role in the concentration of urine. The correct answer is:

Correct Option D - B, C , D, E:

- B. Testes: Countercurrent exchange is present in the testicular blood vessels, which helps regulate the temperature within the testes, crucial for sperm production.
- C. Kidney: Countercurrent mechanisms, such as those in the loop of Henle, are fundamental in the kidney's ability to concentrate urine and regulate water and electrolyte balance.
- D and E: The countercurrent mechanism is a significant feature in the lungs or the gut.

B. Testes: Countercurrent exchange is present in the testicular blood vessels, which helps regulate the temperature within the testes, crucial for sperm production.

C. Kidney: Countercurrent mechanisms, such as those in the loop of Henle, are fundamental in the kidney's ability to concentrate urine and regulate water and electrolyte balance.

D and E: The countercurrent mechanism is a significant feature in the lungs or the gut.

Solution for Question 14:

Answer: C) Lacis cells

Explanation:

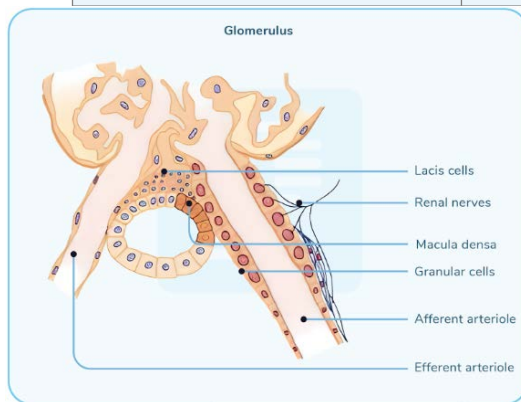
Lacis cells, also called extraglomerular mesangial cells or Goormaghtigh cells , are found within the juxtaglomerular apparatus (JGA) of the kidney.

- These cells are located between the afferent and efferent arterioles and the macula densa .
- They play a role in signal transmission between the macula densa and juxtaglomerular cells, influencing renin secretion .

- They are part of the renin-angiotensin-aldosterone system (RAAS) and contribute to blood pressure regulation.

Juxtaglomerular Apparatus (JGA) and its Components

Component	Location	Function
Macula densa	Distal tubule near afferent arteriole	Senses sodium concentration & regulates renin secretion
Juxtaglomerular (JG) cells	Afferent arteriole	Secrete renin in response to low BP
Lacis cells (Extraglomerular mesangial cells)	Between the afferent & efferent arterioles	Facilitate communication between macula densa and JG cells
Podocytes	Visceral layer of Bowman's capsule	Filtration barrier in glomerulus



Macula densa (Option A): Incorrect because macula densa cells are found in the distal tubule and sense sodium concentration.

Juxtaglomerular (JG) cells (Option B): Incorrect because JG cells are located in the afferent arteriole and primarily secrete renin.

Podocytes (Option D): Incorrect because podocytes are specialized epithelial cells covering the glomerular capillaries.

Solution for Question 15:

Correct Option A: A - Polycationic, B - Neutral, C - Polyanionic:

- The Glomerular Filtration Barrier is a negatively charged barrier.

- Depending on the nature of filtration, there are 3 different substances at the level of the barrier.
- Freely Filtered Substances: If the molecular radius < 1.8 nm. They can easily pass through irrespective of charge. E.g., Na⁺, K⁺, H₂O, Urea, Creatinine, Inulin, PAH, Bicarbonate, etc.
- Selective Permeable Substances: If the molecular radius is 1.8 nm-4.2 nm. The filtration of the substances is determined by the charge present on them. The filtration scale is Positively charged >> Neutral charge >> Negatively charged.
- The filtration scale is Positively charged >> Neutral charge >> Negatively charged.
- Impermeable substances: If the molecular radius > 4.2 nm.
- So, in graph A represents a positively charged molecule, B represents a neutral charged molecule, and C represents the negatively charged molecule.
- The filtration scale is Positively charged >> Neutral charge >> Negatively charged.

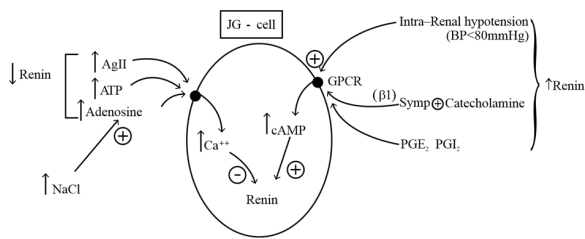
Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 16:

Correct Option C - JG cells:

- Juxtaglomerular cells are the modified smooth muscle cells of the afferent arteriole with granules containing renin.
- JG cells contain GPCR (G_s/I type), which generates a secondary messenger cyclic AMP (cAMP) and another GPCR (G_q type) generates calcium ions that are a secondary messenger.
- Increased intracellular cAMP causes degranulation of JG cells, thus stimulating renin secretion.
- Increased Calcium concentration inhibits degranulation and restricts renin secretion.
- Factors increasing renin secretion: Intrarenal hypotension (< 80 mm Hg - most potent stimulator), increased sympathetic stimulation, and prostaglandins (E₂ and I₂).
- Factors decreasing renin secretion: Increased levels of angiotensin II, ATP, and adenosine.



Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 17:

Correct Option A - Proximal tubule:

- The proximal tubule is the first segment of the renal tubule after the glomerulus. It is responsible for the reabsorption of water, electrolytes, and other essential substances from the glomerular filtrate back into the bloodstream.
- In this segment, water reabsorption occurs through aquaporin 1 and osmosis, driven by the concentration gradient created by active transport of solutes out of the tubule. 60 - 70% of water is reabsorbed at the proximal tubule.

Incorrect Options:

- Options B, C, and D are incorrect. Maximum water reabsorption occurs at proximal tubule of the nephron.

Solution for Question 18:

Correct Option C - Ascending thick loop of Henle:

- The Ascending Thick Loop of Henle is the segment of the nephron responsible for diluting the urine. It actively transports ions, such as sodium, chloride, and potassium, but is impermeable to water and urea. Hence, referred to as the diluting segment because it also dilutes the tubular fluid.

Incorrect Options:

Option A - PCT (Proximal Convoluted Tubule): The Proximal Convoluted Tubule is the first segment of the nephron following the glomerulus. It is involved in both solute and water reabsorption, but it is not specifically associated with urine dilution.

Option B - Collecting Duct: It is involved in reabsorbing water and concentrating the urine under the influence of antidiuretic hormone (ADH).

Option D - Descending Loop of Henle: It is permeable to water but not to solutes, resulting in the concentration of the tubular fluid as it descends deeper into the medulla.

Solution for Question 19:

Correct Option A - Upper part of proximal tubule:

- 100% glucose is absorbed in the proximal tubules. Proximal tubules are divided into two parts as follows.

Tubular Segment	Glucose Reabsorption (%)	Luminal Transporter(Lumen to cytoplasm)	Basolateral Transporter (Cytoplasm to blood)
Proximal Convoluted Tubules (PCT)	90%	SGLT type II (Secondary active Transporter) 1 molecule of sodium and 1 molecule of glucose from the lumen to the cell.	GLUT-2 (Facilitated diffusion transporter)
Proximal Straight Tubules (PST)	10%	SGLT type I (Secondary active Transporter) 2 molecules of sodium and 1 molecule of glucose from the lumen to the cell.	GLUT-1 (Facilitated diffusion transporter)

Incorrect options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 20:

Correct Option C - Efferent arteriolar constriction decreases renal blood flow:

- This statement is true. Constriction of the efferent arteriole leads to a decrease in renal blood flow. This constriction increases the resistance to blood flow, causing a reduction in blood flow to the kidneys.

Incorrect Option:

Option A - Efferent arteriole dilation decreases RBF: This statement is incorrect. Efferent arteriole dilation will result in increased renal plasma flow.

Option B - Afferent arteriole constriction increases GFR: This statement is incorrect. Afferent arteriole constriction leads to decrease in plasma flow resulting in decreased capillary hydrostatic pressure and reduced GFR.

Option D - Mild constriction of the efferent arteriole decreases GFR: This statement is incorrect. Mild constriction leads to less reduction in RPF. It may decrease to 500 ml. Due to the mild constriction, the plasma cannot exit the capillary easily, leading to accumulation of plasma when the second 500 ml arrives in the next minute. The subsequent 500 ml pushes the first 500 ml. Thus, in steady-state conditions, the total volume of plasma present at the capillary is approximately 1000 ml, i.e., higher than normal. This leads to a high hydrostatic pressure, which in turn leads to increased GFR.

Solution for Question 21:

Correct Option C - Hypertonic:

- The descending thin segment of the loop of henle is permeable to water and impermeable to the solutes.
- The tubular fluid will be hypertonic after passing through the descending thin segment of the Loop of Henle.

Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.
-

Solution for Question 22:

Correct Option A - Proximal convoluted tubule:

- 100% glucose is absorbed in the proximal tubules. Proximal tubules are divided into two parts as follows.

Tubular Segment	Glucose Reabsorption (%)	Luminal Transporter (Lumen to cytoplasm)	Basolateral Transporter (Cytoplasm to blood)
Proximal Convoluted Tubules (PCT)	90%	SGLT type II (Secondary active Transporter) 1 molecule of sodium and 1 molecule of glucose from the lumen to the cell.	GLUT-2 (Facilitated diffusion transporter)
Proximal Straight Tubules (PST)	10%	SGLT type I (Secondary active Transporter) 2 molecules of sodium and 1 molecule of glucose from the lumen to the cell.	GLUT-1 (Facilitated diffusion transporter)

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 23:

Correct Option C - Decreased colloid osmotic pressure:

- In nephrotic syndrome, the kidneys lose large amounts of protein, particularly albumin, through the urine.
- Since albumin helps maintain colloid osmotic pressure in the blood vessels, decreased levels of albumin result in decreased colloid osmotic pressure. Osmotic pressure is a retaining force.
- This reduction in colloid osmotic pressure leads to fluid shifting from the blood vessels into the surrounding tissues, causing edema.

Incorrect Options:

- Options A, B, and D are incorrect. Refer to the explanation of the correct answer.

Solution for Question 24:

Correct Option B - 6-12:

- The normal anion gap is between 7-16 mEq/L and is used to assess the presence of unmeasured cations and anion ions in the blood and helps in diagnosing the different causes of metabolic acidosis.

Incorrect Options:

- Options A, C, and D are incorrect.
-

Solution for Question 25:

Correct Option C - Enhance salt absorption:

- In diarrhea, only salt is lost but sugar is added as it helps to absorb salt and sugar by process of symport. The transporter in the gut is SGLT-1 and kidney is SGLT-2.
- Coupling ratio: SGLT 1: 2:1 as 2 molecules of Na are transported for 1 molecule of glucose. SGLT 2: 1:1 as 1 molecule of Na is transported for 1 molecule of glucose.
- SGLT 1: 2:1 as 2 molecules of Na are transported for 1 molecule of glucose.
- SGLT 2: 1:1 as 1 molecule of Na is transported for 1 molecule of glucose.
- SGLT 1: 2:1 as 2 molecules of Na are transported for 1 molecule of glucose.
- SGLT 2: 1:1 as 1 molecule of Na is transported for 1 molecule of glucose.

Incorrect Options:

- Options A, B, and D are incorrect.
-

Solution for Question 26:

Correct Option B - Plasticity:

- The given graph is cystometrogram (CMG).
- A CMG is a test that measures the pressure inside the bladder as it fills with fluid. It helps assess the bladder's function and its ability to store and release urine.
- Plateau Phase Explanation : The plateau phase of the cystometrogram, where the pressure remains relatively constant despite increasing volume, can be explained by the bladder's ability to stretch and accommodate increasing volumes of urine. This is a result of the unique properties of the bladder's smooth muscle.
- Law of Laplace : According to the Law of Laplace, for a spherical viscus (like the bladder), the pressure (P) inside the bladder is related to the wall tension (T) and the radius (r) of the bladder. The relationship is given by $P = 2T/r$. As the bladder fills, the radius increases, which helps keep the pressure relatively low despite the increased volume.
- Plasticity : Plasticity refers to the ability of the bladder's smooth muscle to undergo prolonged changes in length (stretch) without a significant increase in pressure. This allows the bladder to store larger volumes of urine without a significant rise in pressure.

A CMG is a test that measures the pressure inside the bladder as it fills with fluid. It helps assess the bladder's function and its ability to store and release urine.

Plateau Phase Explanation : The plateau phase of the cystometrogram, where the pressure remains relatively constant despite increasing volume, can be explained by the bladder's ability to stretch and accommodate increasing volumes of urine. This is a result of the unique properties of the bladder's smooth muscle.

Law of Laplace : According to the Law of Laplace, for a spherical viscus (like the bladder), the pressure (P) inside the bladder is related to the wall tension (T) and the radius (r) of the bladder. The relationship is given by $P = 2T/r$. As the bladder fills, the radius increases, which helps keep the pressure relatively low despite the increased volume.

Plasticity : Plasticity refers to the ability of the bladder's smooth muscle to undergo prolonged changes in length (stretch) without a significant increase in pressure. This allows the bladder to store larger volumes of urine without a significant rise in pressure.

Incorrect Options:

Option A - Elasticity: Elasticity is the property of a material to return to its original shape after being stretched or deformed.

Option C - Contractility: Contractility is the fundamental property of muscle tissue that enables it to contract when stimulated.

Option D - Automaticity: Automaticity refers to the ability of certain cells (such as cardiac pacemaker cells) to spontaneously generate electrical impulses without external stimulation.

Solution for Question 27:

Correct Option C - Greater than inulin clearance:

- Creatinine clearance is an indirect measure of GFR that is calculated from the amount of creatinine that is filtered by the glomeruli and then excreted in the urine over a given period.
- Creatinine is not as accurate as inulin in measuring GFR because creatinine is partially secreted and never reabsorbed by the renal tubules, which can result in an overestimation of GFR.
- Inulin is a freely filterable substance, neither endogenously secreted by the body nor reabsorbed by the nephron.
- The clearance of creatinine is slightly higher than that of inulin due to the tubular secretion of creatinine.
- Order of clearance of different substances across the kidney is: C PAH >>C Cr >>C inulin (125ml/min (GFR))>>C Urea >>C Na⁺ >>C Glucose/AA
- C PAH >>C Cr >>C inulin (125ml/min (GFR))>>C Urea >>C Na⁺ >>C Glucose/AA
- C PAH >>C Cr >>C inulin (125ml/min (GFR))>>C Urea >>C Na⁺ >>C Glucose/AA

C PAH >>C Cr >>C inulin (125ml/min (GFR))>>C Urea >>C Na⁺ >>C Glucose/AA

Incorrect Options:

Options A and B are incorrect. Refer to Option C for an explanation.

Option D - Equal to renal plasma flow: Renal plasma flow is a measure of para-amino hippuric acid clearance which is filtered and completely secreted in the nephron.

Solution for Question 28:

Correct Option A - Regulate GFR:

- Tubuloglomerular feedback is responsible for the autoregulation of GFR. . The juxtaglomerular apparatus, which consists of macula densa cells, has an NKCC channel in the luminal membrane which absorbs Na^+ , K^+ , and 2Cl^- . The transport of Na^+ across the basolateral membrane requires $\text{Na}^+ \text{K}^+ \text{ATPase}$ leading to increased adenosine formation.
- The lacis cells have A1 receptor for the adenosine leading to influx of Ca^{2+} into lacis cells and then diffuses into smooth muscles of afferent arteriole leading to constriction and juxtaglomerular cells leading to decrease in renin release.

Incorrect Options:

Option B: Decrease plasma sodium : The regulation of plasma sodium concentration primarily Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 29:

Correct Option B - A-Glucose, B- Bicarbonate, C-Inulin and D - PAH:

The given graph shows renal clearance of different substances.

- Substance A, with a clearance of zero, is glucose. Normally, glucose is completely reabsorbed in the proximal convoluted tubule, and none is excreted in the urine.
- Substance B is bicarbonate, with a clearance value less than the glomerular filtration rate (GFR). The kidneys reabsorb most filtered bicarbonate, resulting in a clearance lower than the GFR.
- Substance C is inulin, with a clearance value equal to the GFR. Inulin is filtered but neither reabsorbed nor secreted, making it a good measure of the GFR.
- Substance D, with a clearance value higher than the GFR, is para-amino hippuric acid (PAH). PAH is filtered and secreted, resulting in a clearance higher than the GFR. also can be creatinine.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 30:

Correct Option B - Kidney:

- The kidneys have the least arteriovenous oxygen difference, meaning that they extract a relatively small amount of oxygen from the blood as it passes through. This is because the kidneys do not have a high metabolic demand and primarily function to filter waste products from the blood rather than perform energy-intensive processes.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.
-

Solution for Question 31:

Correct Option C - Junction of thick ascending limb and DCT:

- The macula densa is located at the junction of the thick ascending limb and DCT. The macula densa is a specialized group of cells in the wall of the DCT that are involved in regulating renal blood flow and glomerular filtration rate (GFR). The cells of the macula densa sense changes in the concentration of sodium and chloride ions in the filtrate and signal the juxtaglomerular cells of the afferent arteriole to adjust renal blood flow and GFR accordingly.

Incorrect Options:

- Option A, B and D are incorrect.
-

Solution for Question 32:

Correct Option B - 6.0 mEq/L:

- Hyperkalemia refers to an elevated level of serum potassium more than 5.5 mEq/L.
- Normal value of serum potassium is 3.5 to 5.5 mEq/l.

Incorrect options

Options A, C, and D are incorrect.

Solution for Question 33:

Correct Option B - Heparan sulfate:

- The protein in the glomerular basement membrane responsible for charge-dependent filtration is Heparan sulfate.
- Heparan sulfate is a type of glycosaminoglycan (GAG) that is a key component of the glomerular basement membrane (GBM) in the kidney.
- Heparan sulfate has negative charges that repel negatively charged proteins, contributing to the selective filtration of molecules based on their charge.

Incorrect Options:

- Option A, C & D are incorrect. Heparan sulfate is the charged protein present in the glomerular basement membrane.
-

Solution for Question 34:

Correct Option D:Urine urea in mg/ml

- The formula for urea clearance is $U \times V / P$,
- Where U represents the concentration of urea in the urine in mg/ml, V represents the urine flow rate or volume in ml/min, P represents the concentration of urea in the plasma in mg/ml.
- U represents the concentration of urea in the urine in mg/ml,
- V represents the urine flow rate or volume in ml/min,
- P represents the concentration of urea in the plasma in mg/ml.

- U represents the concentration of urea in the urine in mg/ml,
- V represents the urine flow rate or volume in ml/min,
- P represents the concentration of urea in the plasma in mg/ml.

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.
-

Solution for Question 35:

Correct Option C - Chloride:

- In the proximal convoluted tubule (PCT), various substances are reabsorbed to maintain electrolyte and fluid balance in the body. Normally, glucose, amino acids, and bicarbonate are reabsorbed from the tubular lumen into the renal interstitium, leading to decreased concentrations of these substances in the luminal side of the PCT.
- Chloride, on the other hand, is primarily reabsorbed paracellularly in the PCT. It follows the movement of sodium ions, which are actively reabsorbed across the luminal membrane. Therefore, chloride concentration in the luminal side of the PCT remains relatively high, as it is not actively reabsorbed.

Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.
-

Solution for Question 36:

Correct Option D - Renal sympathetic system fibers:

- Renal auto-regulation refers to the ability of the kidneys to maintain a relatively constant glomerular filtration rate (GFR) and renal blood flow (RBF) despite changes in systemic blood pressure.
- It involves the following mechanisms that work together to adjust the diameter of the renal arterioles and maintain adequate renal perfusion. Tubuloglomerular feedback mediated by

adenosine. Myogenic reflex mediated by calcium.

- Tubuloglomerular feedback mediated by adenosine.
- Myogenic reflex mediated by calcium.
- Renal sympathetic system fibers are not involved in the autoregulation of the GFR.
- Tubuloglomerular feedback mediated by adenosine.
- Myogenic reflex mediated by calcium.

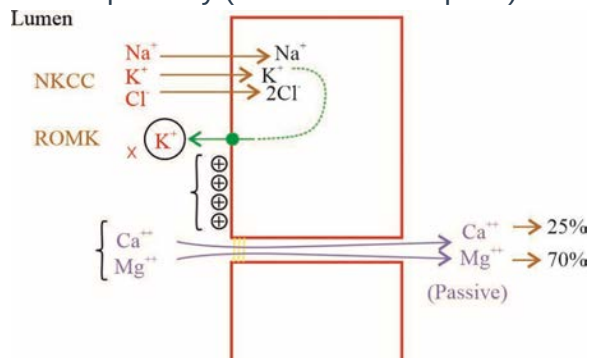
Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 37:

Correct Option C -

- In the thick ascending limb of the loop of Henle, the primary ion responsible for the reabsorption of divalent ions, such as calcium (Ca^{2+}) and magnesium (Mg^{2+}), is potassium (K^+). The thick ascending limb is impermeable to water but actively transports ions, including sodium (Na^+), potassium (K^+), and 2 chloride (Cl^-), out of the tubular lumen through the action of the apical membrane sodium-potassium-2 chloride cotransporter (NKCC2) into the thick ascending limb (TAL).
- The another channel in the luminal side of the TAL is ROMK (Renal Outer-Medullary Potassium) channel which pushes the absorbed K^+ , out into the lumen leaving a positive charge on the lumen called lumen positive charge.
- Because of this, the positive ions such as Ca^{++} and Mg^{++} are repulsed into a paracellular pathway (Passive reabsorption) through the tight junction and reabsorbed into capillary blood.



Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.
-

Solution for Question 38:

Correct Option A - 1 and 4 are correct (Merkel disc and Meissner's corpuscle):

- Tactile discrimination, which refers to the ability to discern fine details, textures, and shapes through touch, relies on superficial sensory receptors in the skin.
- Both Merkel discs and Meissner's corpuscles are involved in tactile discrimination, allowing individuals to sense and distinguish intricate textures and shapes.

Incorrect Options:

- Options B, C, and D are incorrect. Pacinian and ruffini end organs are deep sensory receptors stimulated by pressure.
-

Solution for Question 39:

Correct Option C - Calcium:

- According to the myogenic hypothesis, it suggests that when the afferent arterioles experience an increase in blood pressure or stretch, it triggers the opening of calcium channels in the smooth muscle cells of the arterioles.
- The influx of calcium ions leads to vasoconstriction, reducing the blood flow and preventing excessive pressure within the glomerulus. This response helps maintain a relatively constant blood flow and glomerular filtration rate despite changes in systemic blood pressure.

Incorrect Options:

Options A and B: These are not involved in the autoregulation of GFR.

Option D - Adenosine: Adenosine is a mediator of tubuloglomerular feedback mechanism.

Solution for Question 40:

Correct Option B - DCT:

- Macula densa cells are the columnar epithelial cells in contact with afferent and efferent arterioles, involved in the regulation of renal blood flow and glomerular filtration rate.
- These cells are located abundantly in the uppermost part of the thick ascending limb (TAL) of the loop of Henle and few in the distal convoluted tubule.
- It is the intra-renal chemoreceptor cells that sense the presence of NaCl load inside the lumen of the tubule.

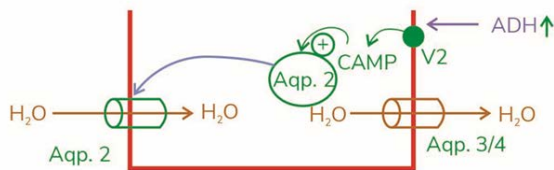
Incorrect Options:

- Options A, C, and D are incorrect. Macula densa cells are not located in these parts of the nephron.

Solution for Question 41:

Correct Option is B - 2:

- Vasopressin acts on Aquaporin 2 (AQP2) channels present in the collecting duct. When vasopressin binds to its V2 receptor on the basolateral membrane of the collecting duct cells, it triggers increased cAMP levels promoting the translocation and insertion of AQP2 channels into the apical membrane. This insertion allows for increased water reabsorption from the urine, leading to concentrated urine and water conservation.



Incorrect Options:

- Options A, C, and D are incorrect. These are ADH independent hormones.

Solution for Question 42:

Correct Option C - Chloride:

- In the proximal convoluted tubule (PCT), various substances are reabsorbed to maintain electrolyte and fluid balance in the body. Normally, glucose, amino acids, and bicarbonate are reabsorbed from the tubular lumen into the renal interstitium, leading to decreased concentrations of these substances in the luminal side of the PCT.
- Chloride, on the other hand, is primarily reabsorbed paracellularly in the PCT. It follows the movement of sodium ions, which are actively reabsorbed across the luminal membrane. Therefore, chloride concentration in the luminal side of the PCT remains relatively high, as it is not actively reabsorbed.

Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.
-

Solution for Question 43:

Correct Option D - Flow of tubular fluid from PCT to DCT:

- This component is not involved in the countercurrent multiplier mechanism of the kidney.
- The countercurrent multiplier involves the following: The high medullary tonicity is maintained by the following: Sodium and chloride outflow in thick ascending limb. Sodium and chloride outflow in the thin ascending limb. Reabsorption of urea from medullary collecting duct into the interstitium. This outflow of solutes into the medullary interstitium will result in hypertonicity which will result in the outflow of the water from isotonic fluid flowing through the thin descending limb of the loop of the henle.
- The high medullary tonicity is maintained by the following: Sodium and chloride outflow in thick ascending limb. Sodium and chloride outflow in the thin ascending limb. Reabsorption of urea from medullary collecting duct into the interstitium.
- Sodium and chloride outflow in thick ascending limb.

- Sodium and chloride outflow in the thin ascending limb.
- Reabsorption of urea from medullary collecting duct into the interstitium.
- This outflow of solutes into the medullary interstitium will result in hypertonicity which will result in the outflow of the water from isotonic fluid flowing through the thin descending limb of the loop of the henle.
- The high medullary tonicity is maintained by the following: Sodium and chloride outflow in thick ascending limb. Sodium and chloride outflow in the thin ascending limb. Reabsorption of urea from medullary collecting duct into the interstitium.
- Sodium and chloride outflow in thick ascending limb.
- Sodium and chloride outflow in the thin ascending limb.
- Reabsorption of urea from medullary collecting duct into the interstitium.
- This outflow of solutes into the medullary interstitium will result in hypertonicity which will result in the outflow of the water from isotonic fluid flowing through the thin descending limb of the loop of the henle.
- Sodium and chloride outflow in thick ascending limb.
- Sodium and chloride outflow in the thin ascending limb.
- Reabsorption of urea from medullary collecting duct into the interstitium.

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 44:

Correct Option D - Amino acid is reabsorbed in the proximal tubule:

- Co-transporters present in the luminal membrane of the proximal tubule reabsorb sodium along with other substances like amino acids (sodium-amino acid cotransporter), phosphate (sodium-phosphate cotransporter), citric acid (sodium-citric acid cotransporter).

Incorrect Options:

Option A - Maximum reabsorption in the distal tubule: Maximum reabsorption occurs in PCT.

Option B - Urea is absorbed 100% in PCT: Urea is not 100% absorbed in PCT.

Option C - H⁺ is absorbed in the proximal tubule: NHE (Sodium-Acid exchanger) reabsorbs sodium and secretes one molecule of H⁺ ion into the lumen.

Solution for Question 45:

Correct Options A - Bronchodilation:

- Endothelin 1 is the most potent vasoconstrictor known so far. Veins are more sensitive to endothelin 1 than arteries. Endothelin-1 can cause a decrease in glomerular filtration rate (GFR) by constricting the afferent arterioles of the kidney.
- It is also a potent positive chronotropic and inotropic agent (increases heart rate and myocardial contractility).

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 46:

Correct Option C - Ureteric stone:

- Ureteric stone: A ureteric stone refers to a stone that forms in the ureter, the tube that connects the kidneys to the bladder. Ureteric stones can cause obstruction and block the flow of urine, which can result in decreased urine flow from the affected kidney.
- The obstruction can lead to increased pressure in the renal pelvis and the affected kidney, potentially impairing the filtration process and reducing the GFR. Therefore, a ureteric stone is expected to decrease GFR.

Incorrect Options:

Option A - Increased renal blood flow: When there is a higher volume of blood flowing into the kidneys, it results in increased glomerular hydrostatic pressure, which promotes filtration and thus increases GFR.

Option B - Efferent arteriolar constriction: Constriction of the efferent arteriole causes an increase in glomerular hydrostatic pressure and promotes filtration.

Option D - Decreased oncotic pressure: Oncotic pressure is a retaining force and it tends to retain water in the renal capillaries. Decreased oncotic pressure will result in reduced Glomerular filtration rate.

Solution for Question 47:

Correct Option: D

- According to Starling law, the net filtration pressure at the level of glomerular capillary is the combination of hydrostatic pressure and oncotic pressure.
- Net Force = $\{(P_C - P_B) + (\pi_B - \pi_C)\}$

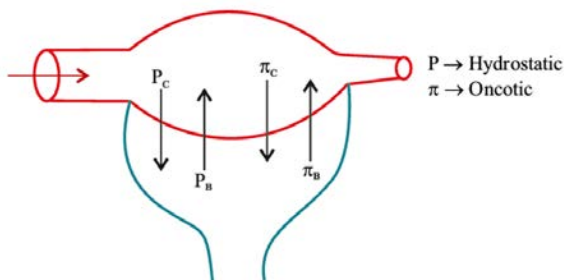
where,

P_C = Glomerular capillary hydrostatic pressure

P_B = Bowman capsule hydrostatic pressure

π_C = Capillary oncotic pressure

π_B = Bowman capsule oncotic pressure



$$\text{Net} = (P_C - P_B) - (\pi_B - \pi_C)$$

- Substituting the values, $\text{NFP} = (45 - 10) + (0 - 20) \blacksquare 45 - 30 \blacksquare 15 \text{ mmHg}$
- $\text{NFP} = (45 - 10) + (0 - 20) \blacksquare 45 - 30 \blacksquare 15 \text{ mmHg}$
- Therefore, the net filtration pressure, in this case, is 15 mmHg.
- $\text{NFP} = (45 - 10) + (0 - 20) \blacksquare 45 - 30 \blacksquare 15 \text{ mmHg}$

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 48:

Correct Option B - 6-12:

- Anion gap is used to assess the presence of unmeasured cations and anions in the blood and can help in the diagnosing the different causes of metabolic acidosis.
- The normal anion gap ranges within the 7-16 mEq/L.

Incorrect options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.
-

Solution for Question 49:

Correct Option B - Segment Ib is due to Laplace law:

- Segment Ib is due to Laplace law where volume is increased but there is no rise in pressure.
- This law states that the pressure in a spherical viscus is equal to twice the wall tension divided by the radius.
- In the case of the bladder, the tension increases as the organ fills, but so does the radius. Therefore, the pressure increase is slight until the organ is relatively full.

Incorrect Options:

Option A - Segment Ia is due to residual urine: Segment Ia represents the filling phase of the bladder.

Option C - Micturition fails to happen in segment II: Segment II represents the tremendous increase in pressure due to large volume of urine (400 ml) and results in micturition.

Option D - Micturition occurs in segment Ib: It occurs in segment II.

Solution for Question 50:

Correct Option D - The volume of serum passing through the kidney in 24 hours:

- To calculate creatinine clearance, the following formula is used: $\text{Clearance} = U \times V / P$ Where, U - concentration of creatinine in the urine in mg/ml. V - urine flow rate or volume in ml/min. P - concentration of creatinine in the plasma in mg/ml.
- $\text{Clearance} = U \times V / P$
- Where, U - concentration of creatinine in the urine in mg/ml. V - urine flow rate or volume in ml/min. P - concentration of creatinine in the plasma in mg/ml.
- U - concentration of creatinine in the urine in mg/ml.
- V - urine flow rate or volume in ml/min.
- P - concentration of creatinine in the plasma in mg/ml.
- $\text{Clearance} = U \times V / P$
- Where, U - concentration of creatinine in the urine in mg/ml. V - urine flow rate or volume in ml/min. P - concentration of creatinine in the plasma in mg/ml.
- U - concentration of creatinine in the urine in mg/ml.
- V - urine flow rate or volume in ml/min.
- P - concentration of creatinine in the plasma in mg/ml.
- U - concentration of creatinine in the urine in mg/ml.
- V - urine flow rate or volume in ml/min.
- P - concentration of creatinine in the plasma in mg/ml.

Incorrect Options:

- Options A, B, and C are incorrect. These parameters are used for the calculation of creatinine clearance.
-

Functional Anatomy, Mechanics, Ventilation-Perfusion, Compliance of RS

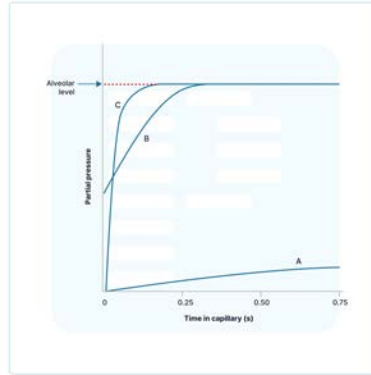
1. A 45-year-old male presents with a persistent dry cough and shortness of breath. A CT scan of the chest reveals localized airway obstruction at the level of the terminal bronchioles. Based on Weibel's classification of the respiratory system, which generation of airways is most likely affected in this patient?

- A. Generation 4
- B. Generation 8
- C. Generation 16
- D. Generation 20

2. Which of the following statements about intra-alveolar pressure changes during normal breathing is incorrect?

- A. The maximum and minimum intra-alveolar pressure is at start of expiration and start of inspiration respectively
- B. Minimum Intra-alveolar pressure is approximately -1 mm Hg during inspiration.
- C. Maximum Intra-alveolar pressure is approximately +1 mm Hg during expiration.
- D. At the end of both inspiration and expiration, intra-alveolar pressure is 0 mm Hg.

3. Uptake of various substances during their transit through a pulmonary capillary is shown here. Curve B represents:

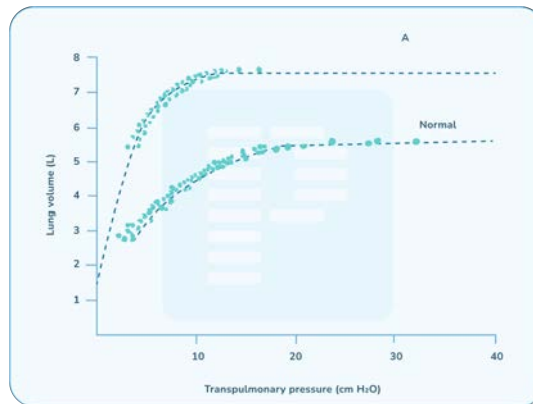


- A. N₂O
- B. O₂
- C. CO₂
- D. CO

4. A 30-year-old patient with a history of asthma presents with complaints of shortness of breath and wheezing. Pulmonary function tests are performed and indicate that the patient's dynamic compliance is significantly reduced, while static compliance remains relatively unchanged. What is the most likely explanation for these findings?

- A. Surfactant deficiency leading to increased alveolar surface tension
- B. Reduced chest wall compliance due to muscle weakness
- C. Increased airway resistance due to bronchospasm
- D. Presence of fibrotic tissue in the lungs, restricting lung expansion

5. Which of the following correctly depicts A ?



- A. Emphysema
- B. Lung fibrosis
- C. Myasthenia Gravis
- D. Kyphoscoliosis

6. Which of the following denotes the compliance curve depicting both inspiratory and expiratory phase? (Y-axis denotes volume and x-axis denotes pressure)

- A.
- B. {{caption_text}}
- C.
- D.

7. In the alveoli, surfactant primarily reduces surface tension by:

- A. Increasing the viscosity of the alveolar fluid, thereby reducing the tendency to collapse
- B. Interfering with the diffusion of water molecules across the alveolar lining.
- C. Spreading phospholipids and proteins uniformly across the alveolar surface.
- D. Forming a film that alters the behavior of water molecules at the air-liquid interface

8. Which of the following statements is false?

- A. Old-age lung has high compliance
 - B. Elastic tissue recoil and surface tension oppose each other during expansion of alveoli
 - C. Hyaline membrane disease has low compliance and high surface tension
 - D. Low compliance of the lungs is seen in Atelectasis
-

9. A patient's lung volume increases by 500 ml when their transpulmonary pressure increases from 3 cm H₂O to 5.5 cm H₂O. What is the compliance of the patient's lungs?

- A. 100 ml/cm
 - B. 150 ml/cm
 - C. 200 ml/cm
 - D. 250 ml/cm
-

10. Match the following zones of the lungs with their corresponding pressure relationships and blood flow characteristics. Zone Pressure Relationship Blood Flow Characteristic

1 A) Pulmonary artery pressure > Pulmonary venous pressure > Alveolar pressure i) Continuous blood flow

2 B) Alveolar pressure > Pulmonary artery pressure > Pulmonary venous pressure ii) No blood flow

3 C) Pulmonary artery pressure > Alveolar pressure > Pulmonary venous pressure iii) Intermittent blood flow (only during systole)

Zone	Pressure Relationship	Blood Flow Characteristic
1	A) Pulmonary artery pressure > Pulmonary venous pressure >	i) Continuous blood flow

	Alveolar pressure	
2	B) Alveolar pressure > Pulmonary artery pressure > Pulmonary venous pressure	ii) No blood flow
3	C) Pulmonary artery pressure > Alveolar pressure > Pulmonary venous pressure	iii) Intermittent blood flow (only during systole)

- A. 1 - A - i, 2 - B - ii, 3 - C - iii
- B. 1 - B - ii, 2 - C - iii, 3 - A - i
- C. 1 - C - iii, 2 - B - ii, 3 - A - i
- D. 1 - B - ii, 2 - A - i, 3 - C - iii

11. Two separate patients are being considered for respiratory distress. Imaging reveals an obstruction in the right bronchus in one patient and an embolism in the pulmonary arteriole in another patient. Based on these findings, what are the expected changes in the ventilation-perfusion (V/Q) ratio and the partial pressures of oxygen (PO₂) and carbon dioxide (PCO₂) in the affected alveoli?

- A. In the obstructed bronchus, the V/Q ratio is infinite, and the partial pressures are PO₂ = 149 mmHg and PCO₂ = 0.3 mm Hg.
- B. In the embolized vessel, the V/Q ratio is zero, and the partial pressures are PO₂ = 40 mmHg and PCO₂ = 46 mm Hg.
- C. In the obstructed bronchus, the V/Q ratio is zero, and the partial pressures are PO₂ = 40 mmHg and PCO₂ = 46 mm Hg.
- D. In the embolized vessel, the V/Q ratio is infinite, and the partial pressures are PO₂ = 40 mmHg and PCO₂ = 46 mm Hg.

12. A 65-year-old male patient with a history of shortness of breath on exertion presents to the OPD. During the examination, it is noted that his ventilation-perfusion (V/Q) ratio varies significantly in different regions of the

lungs. What are the expected partial pressures of oxygen (pO₂) and carbon dioxide (pCO₂) at the apex, middle, and base of the lungs in the upright position?

- A. At the apex: pO₂ is lowest and pCO₂ is highest.
 - B. At the base: pO₂ is highest and pCO₂ is lowest.
 - C. At the middle: pO₂ is highest and pCO₂ is lowest.
 - D. At the base: pO₂ is lowest and pCO₂ is highest.
-

13. A 45-year-old female patient is evaluated for differences in ventilation-perfusion (V/Q) ratios in various positions. In the upright position, the V/Q ratios at the apex, middle, and base of the lungs differ due to gravitational effects. Which of the following statements correctly describes the V/Q ratio at these lung regions in the upright vs. supine position?

- A. In the upright position, the V/Q ratio is highest at the base of the lungs.
 - B. In the upright position, the V/Q ratio is highest at the apex of the lungs.
 - C. In the supine position, the V/Q ratio difference between the apex and base is more pronounced than in the upright position.
 - D. In the supine position, the V/Q ratio is highest at the middle of the lungs.
-

14. A 55-year-old male patient presents with shortness of breath and fatigue. His respiratory rate is 16 breaths per minute, and his tidal volume is 400 mL. The dead space is estimated to be 150 mL. Which of the following statements about ventilation and perfusion is/are correct? 1) Pulmonary ventilation refers to the amount of air entering the lungs in a minute, while alveolar ventilation is the volume of air reaching the alveolar zone for gas exchange. 2) Pulmonary ventilation for this patient is approximately 6.4L/min. 3) Alveolar ventilation for this patient is approximately 4L/min. 4) Perfusion is the blood flow at the level of the alveoli which is approximately equal to cardiac output from the right ventricle.

- A. 1 and 4

- B. 1, 2 and 4
 - C. 1, 2, 3 and 4
 - D. 1, 3 and 4
-

15. Which statement correctly describes normal inspiration?

- A. Diaphragm relaxation leads to decreased intrapleural pressure, allowing air to enter the lungs.
 - B. Diaphragm contraction creates negative intrapleural pressure, causing air to enter the lungs.
 - C. Increased intra-alveolar pressure from diaphragm contraction draws air into the lungs.
 - D. External intercostal muscle relaxation decreases intrapleural pressure, expanding the alveoli.
-

16. A 35-year-old male presents with severe shortness of breath and chest pain after being involved in a motor vehicle accident. On examination, he is tachycardic and hypotensive, with decreased breath sounds on the right side and tracheal deviation to the left. Which of the following pressure changes most accurately explains the underlying mechanism of his condition?

- A. Increased intrapulmonary pressure
 - B. Increased intrapleural pressure
 - C. Decreased intrapleural pressure
 - D. Decreased intrapulmonary pressure
-

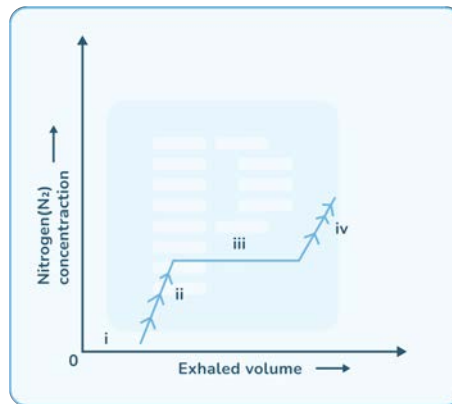
17. A 35-year-old male holds his breath at the end of normal expiration. Which of the following accurately describes the condition of his lungs at this moment?

- A. The alveolar pressure is higher than the atmospheric pressure.
 - B. The intrapleural pressure is higher than the atmospheric pressure.
 - C. The functional residual capacity is at its minimum.
 - D. The elastic recoil forces of the lungs and chest wall are in equilibrium.
-

18. A 55-year-old woman with a history of asthma presents to the emergency department with severe dyspnea. On examination, she is using accessory muscles to breathe and her oxygen saturation is reduced. Arterial blood gas analysis shows a high level of carbon dioxide (PaCO_2) despite increased ventilation. A bedside pulmonary function test reveals that her physiological dead space has increased compared to her last visit. Which of the following is the most likely explanation for the increase in physiological dead space in this patient?

- A. Increased alveolar dead space due to poorly ventilated but well-perfused alveoli, leading to reduced effective gas exchange.
 - B. Anatomical dead space has increased due to an obstruction in the conducting airways, causing less air to reach the alveoli.
 - C. Increased alveolar dead space due to poorly perfused but well-ventilated alveoli, reducing the amount of air available for gas exchange.
 - D. A reduction in anatomical dead space due to inflammation and narrowing of the airways, leading to a decrease in overall physiological dead space
-

19. In the Nitrogen Washout Method for measuring lung volumes, what does the phase III observed in the graph represent?



- A. The volume of air in the conducting zone.
- B. The mixed air from the conducting zone and alveoli.
- C. The pure alveolar air coming from the alveolar region.
- D. The closing volume of the lungs.

20. A 45-year-old woman undergoes elective surgery under general anesthesia. During the procedure, her respiratory parameters are closely monitored. Which of the following statements best explains how general anesthesia typically affects anatomical dead space?

- A. General anesthesia increases alveolar ventilation by reducing dead space.
- B. General anesthesia decreases anatomical dead space by promoting bronchoconstriction.
- C. General anesthesia does not affect anatomical dead space but increases alveolar dead space.
- D. General anesthesia increases anatomical dead space due to the loss of bronchial tone, leading to bronchial dilation.

21. A 60-year-old man with a history of smoking presents with a persistent cough and occasional wheezing. During a physical examination, you notice that the

patient has increased breath sounds over the central airways. In which part of the respiratory system is the velocity of airflow is high, leading to turbulent airflow in this patient?

- A. Alveoli
 - B. Trachea
 - C. Terminal bronchioles
 - D. Respiratory bronchioles
-

22. A 55-year-old male with a history of chronic bronchitis presents with increased difficulty in breathing. Based on Poiseuille's law and the distribution of airway resistance, which part of the respiratory system contributes the most to the total resistance to airflow?

- A. Alveolar ducts
 - B. Small airways (bronchioles)
 - C. Medium to large-size bronchi (lobar and segmental bronchi)
 - D. Upper airways (pharynx and larynx)
-

23. A premature infant presents with severe respiratory distress shortly after birth. The neonate is diagnosed with neonatal respiratory distress syndrome (NRDS). Which type of alveolar cell is most likely deficient in this condition, leading to inadequate surfactant production?

- A. Type I pneumocytes
 - B. Type II pneumocytes
 - C. Alveolar macrophages
 - D. Endothelial cells
-

24. All of the following statements about Clara cells of the respiratory tract are true except:

- A. They are found in the bronchioles.
 - B. They stabilise surfactant proteins
 - C. They have cilia.
 - D. They detoxify harmful substances.
-

25. A biopsy from a segment of the respiratory tree reveals pseudostratified ciliated columnar epithelium with goblet cells. From which part of the respiratory tract was this sample most likely taken?

- A. Alveoli
 - B. Terminal bronchioles
 - C. Trachea
 - D. Respiratory bronchioles
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	2
Question 4	3
Question 5	1
Question 6	4
Question 7	4
Question 8	2
Question 9	3
Question 10	2
Question 11	3
Question 12	4
Question 13	2
Question 14	3
Question 15	2
Question 16	2
Question 17	4
Question 18	3
Question 19	3
Question 20	4
Question 21	2
Question 22	3

Question 23	2
Question 24	3
Question 25	3

Solution for Question 1:

Correct Answer: C) Generation 16

Explanation: Weibel's classification of the respiratory system organizes the airways into 23 generations, starting from the trachea (Generation 0) down to the alveolar sacs (Generation 23).

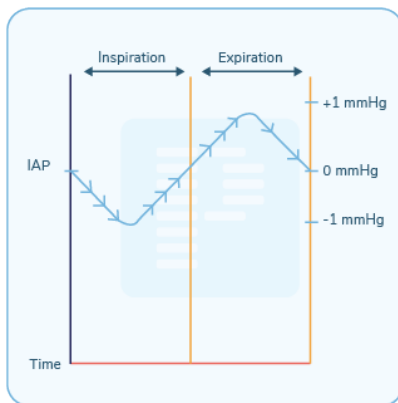
- The terminal bronchioles are found at generation/division 16, marking the end of the conducting zone and the beginning of the respiratory zone. This level is critical as it is the last portion of the airways involved in conducting air before gas exchange begins in the respiratory bronchioles.
- This level is critical as it is the last portion of the airways involved in conducting air before gas exchange begins in the respiratory bronchioles.
- Division number 0 to division number 16 is known as the conducting zone of the airway . Division number 1 to 10 are known as Bronchi. Division numbers 11 to 16 are known as terminal bronchioles.
- Division number 1 to 10 are known as Bronchi.
- Division numbers 11 to 16 are known as terminal bronchioles.
- Division number 17 to division number 23 are responsible for gas exchange and are known as the exchange zone of alveoli.
- This level is critical as it is the last portion of the airways involved in conducting air before gas exchange begins in the respiratory bronchioles.
- Division number 1 to 10 are known as Bronchi.
- Division numbers 11 to 16 are known as terminal bronchioles.

Solution for Question 2:

Correct Answer: A) The maximum and minimum intra-alveolar pressure is at the start of expiration and start of inspiration respectively.

Explanation: The maximum and minimum intra-alveolar pressure are during mid-expiration and mid-inspiration respectively.

Changes in Intra-alveolar pressure during respiration



Phase of Respiration	Intra-Alveolar Pressure (IAP)
Resting Condition	0 mm Hg
Start of Inspiration	0 mm Hg
Middle of Inspiration	-1 mm Hg
End of Inspiration	0 mm Hg
Start of Expiration	0 mm Hg
Middle of Expiration	+1 mm Hg
End of Expiration	0 mm Hg

Solution for Question 3:

Correct answer: B) O₂

Explanation:

Curve B represents O₂

capillary " data-author="na" data-hash="536" data-license="na" data-source="na" data-tags="January2025" src="https://image.prepladder.com/notes/PulREAui57Hk9Pff6Oyv1735921405.png" />

Substances moving from alveoli to capillary blood can reach equilibrium based on their interactions with blood components during the 0.75 seconds blood spends in the pulmonary capillaries:

- Nitrous Oxide (Option A) :Reaches equilibrium quickly (about 0.1 seconds) as it does not react with blood components. Its uptake is flow-limited because diffusion is not the limiting factor.
- Carbon Monoxide (Option D) : Binds very strongly to hemoglobin, keeping its partial pressure in the capillaries very low. As a result, equilibrium is not reached in the 0.75 seconds blood spends in the capillaries, making its transfer diffusion-limited .
- Oxygen (Option B) : Intermediate between N₂O and CO. Binds to hemoglobin, but not as avidly as CO. It reaches equilibrium in about 0.3 seconds, which makes its transfer primarily perfusion-limited

Solution for Question 4:

Answer: C) Increased airway resistance due to bronchospasm

Explanation:

Static Compliance :

- This refers to lung compliance measured when there is no airflow in the lungs. It provides insights into the condition of the lung parenchyma, alveoli, and the chest wall.
- Surfactant deficiency primarily affects static compliance (Option A)

Dynamic Compliance :

- This refers to lung compliance measured during continuous airflow, not when breathing is paused. It reflects the additional force needed to overcome airway resistance during breathing.
- Dynamic compliance considers both airway resistance and the factors influencing static compliance.
- It is always lower than static compliance due to the added resistance of airflow.

- Conditions like bronchospasm can reduce dynamic compliance by increasing airway resistance, even when static compliance remains normal.
- In restrictive lung diseases like fibrosis, both dynamic and static compliance are lower due to increased lung stiffness. (Option D)

Specific Compliance :

- This measure is calculated by dividing static compliance (C) by the functional residual capacity (FRC).
- It is particularly useful for comparing lung compliance between individuals of different sizes, as lung size influences overall compliance

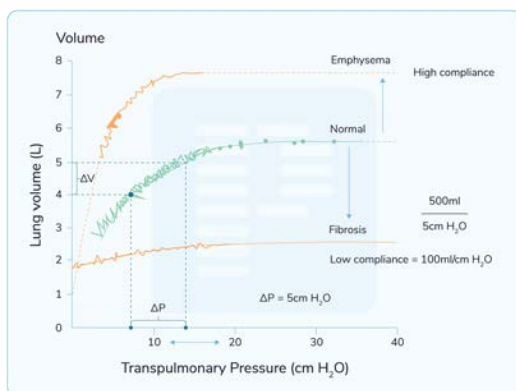
Reduced chest wall compliance due to muscle weakness can decrease both static and dynamic compliance. (Option B)

Solution for Question 5:

Answer: A) Emphysema

Explanation:

In emphysema, the walls of the alveoli are damaged and lose their elasticity, making it harder for the lungs to recoil during exhalation. This would cause increased lung compliance.



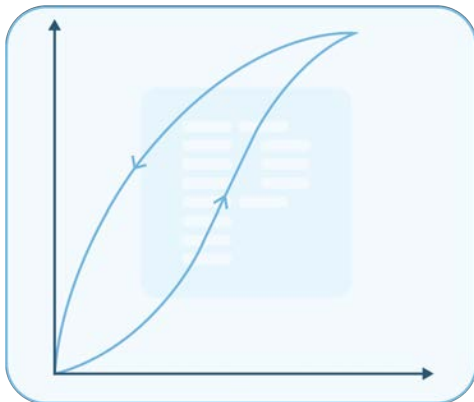
- A compliance curve shown above, with transpulmonary pressure plotted on the X-axis and lung volume on the Y-axis.
- The normal curve depicted above is the expiratory curve.

- The compliance is calculated from this curve by plotting two perpendiculars to the exponential part of the curve from the X-axis and Y-axis.
- The difference between the two points is ΔV on the Y-axis and ΔP on the X-axis.
- Also compliance can be compared among various lung conditions in this graph.

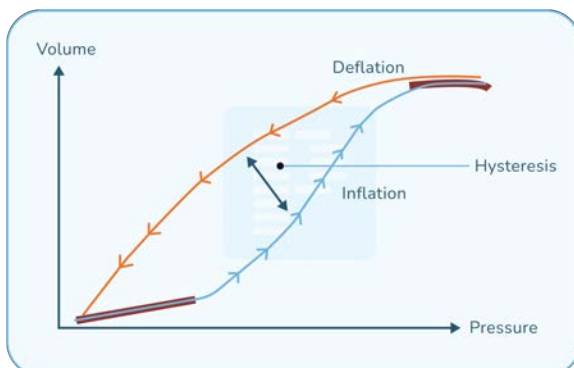
Options (B, C, D): All these conditions have a reduced lung compliance and hence have an expiratory graph below the normal curve as shown.

Solution for Question 6:

Correct Answer: D)



Explanation:



HYSTERESIS LOOP:

Inhalation :

- During inhalation, the inspiratory muscles (primarily the diaphragm) contract, expanding the thoracic cavity. This expansion reduces pleural pressure, creating a more negative pressure compared to the alveolar pressure.
- The difference between alveolar pressure and pleural pressure, known as transpulmonary pressure, increases, causing the lungs to expand.
- The inspiratory compliance curve on the compliance diagram illustrates this relationship, showing how lung volume increases as pleural pressure decreases (becomes more negative).
- The shape of the inspiratory compliance curve is influenced by the elastic properties of the lungs, with steeper curves indicating greater compliance (easier lung expansion).

Exhalation :

- During exhalation, the inspiratory muscles relax, and the elastic recoil of the lungs and chest wall causes the thoracic cavity to decrease in volume. This decrease in volume increases the pleural pressure, making it less negative.
- As pleural pressure rises, the transpulmonary pressure decreases, leading to lung deflation.
- The expiratory compliance curve on the compliance diagram depicts this relationship, showing lung volume decreasing as pleural pressure increases (becomes less negative).

Solution for Question 7:

Answer: D) Forming a film that alters the behavior of water molecules at the air-liquid interface

Explanation:

Surface tension is a force that arises at the interface of air and liquid, causing the liquid surface to contract.

- In the lungs, this interface exists between the alveolar fluid and the air inside the alveoli. Surfactant is a surface-active agent in water that significantly reduces the surface tension of water.
- Surfactant is produced by type II alveolar epithelial cells, which comprise about 10% of the alveolar surface area. Type II alveolar epithelial cells are granular and contain lipid inclusions that are secreted as surfactant into the alveoli.
- Surfactant is a mixture of phospholipids, proteins, and ions.

- Dipalmitoyl phosphatidylcholine, surfactant apoproteins, and calcium ions are the most important components. They are primarily responsible for reducing surface tension by partially dissolving in the fluid lining the alveolar surface while the rest spreads over the water's surface in the alveoli.

Importance of Surfactant

- Surfactant reduces the surface tension in the alveoli, preventing their collapse.
- The pressure generated by surface tension in an alveolus is inversely proportional to its radius – smaller alveoli experience greater pressure.
- The absence of surfactant in premature infants with small alveoli can lead to respiratory distress syndrome of the newborn, a potentially fatal condition

Increasing the viscosity of the alveolar fluid, thereby reducing the tendency to collapse (Option A): Surfactant reduces surface tension, but it does not increase viscosity of alveolar fluid.

Interfering with the diffusion of water molecules across the alveolar lining (Option B): Surfactant does not interfere with water diffusion; it acts at the air-liquid interface to reduce surface tension.

Spreading phospholipids and proteins uniformly across the alveolar surface. (Option C): Surfactant is not uniformly spread; it dynamically adjusts its distribution to stabilize alveoli of different sizes.

Solution for Question 8:

Correct Answer: B) Elastic tissue recoil and surface tension oppose each other during expansion

Explanation:

Elastic tissue recoil and surface tension are the two opposing forces to lung compliance and are not opposing forces to each other.

Two primary factors influence lung compliance:

- Elastic forces of the lung tissue: These forces are primarily determined by the presence of elastin and collagen fibers within the lung parenchyma (the functional tissue of the lungs). In a deflated lung, these fibers are contracted and kinked. As the lung expands during inspiration, these fibers stretch and unkink, generating an elastic force that opposes further expansion.

- These forces are primarily determined by the presence of elastin and collagen fibers within the lung parenchyma (the functional tissue of the lungs).
- In a deflated lung, these fibers are contracted and kinked. As the lung expands during inspiration, these fibers stretch and unkink, generating an elastic force that opposes further expansion.
- Surface tension : A thin layer of fluid lines the inside of the alveoli. At the air-fluid interface within the alveoli, surface tension arises from the stronger attraction between water molecules compared to the attraction between water molecules and air molecules. This surface tension creates an inward pull, attempting to minimize the surface area of the fluid lining the alveoli. Consequently, this inward pull contributes to the elastic recoil of the lungs, making them tend to collapse
- A thin layer of fluid lines the inside of the alveoli. At the air-fluid interface within the alveoli, surface tension arises from the stronger attraction between water molecules compared to the attraction between water molecules and air molecules.
- This surface tension creates an inward pull, attempting to minimize the surface area of the fluid lining the alveoli. Consequently, this inward pull contributes to the elastic recoil of the lungs, making them tend to collapse
- These forces are primarily determined by the presence of elastin and collagen fibers within the lung parenchyma (the functional tissue of the lungs).
- In a deflated lung, these fibers are contracted and kinked. As the lung expands during inspiration, these fibers stretch and unkink, generating an elastic force that opposes further expansion.
- A thin layer of fluid lines the inside of the alveoli. At the air-fluid interface within the alveoli, surface tension arises from the stronger attraction between water molecules compared to the attraction between water molecules and air molecules.
- This surface tension creates an inward pull, attempting to minimize the surface area of the fluid lining the alveoli. Consequently, this inward pull contributes to the elastic recoil of the lungs, making them tend to collapse

Option A) Old-age lung has loss of elastic tissue and hence have reduced elastic recoil force, causing easy expansion of lung, i.e high compliance

Option C) Hyaline membrane disease, normally seen in premature babies has low compliance and high surface tension due to reduced production of surfactants.

Option D) Low compliance is seen in atelectasis due to a decrease in lung volume and requires higher pressure to inflate the alveoli.

Solution for Question 9:

Correct Answer: C) 200 ml/cm

Explanation:

Lung compliance refers to the lungs' ability to stretch and expand in response to changes in transpulmonary pressure.

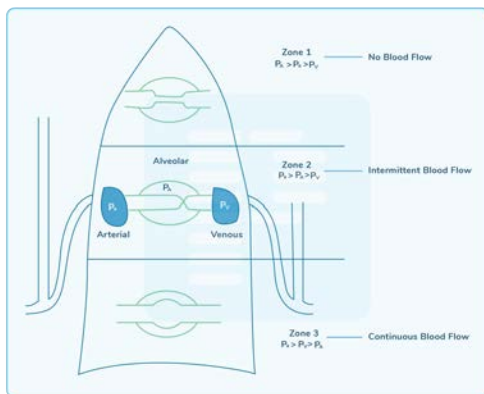
- Compliance = Change in Lung Volume / Change in Transpulmonary Pressure
 - In this scenario, the change in lung volume is 500 ml, and the change in transpulmonary pressure is 2.5 cm H₂O (5.5 cm H₂O - 3 cm H₂O)
 - Therefore , Compliance = 500 ml / 2.5 cm H₂O = 200 ml/cm H₂O
-

Solution for Question 10:

Correct Answer: B) 1 - B - ii, 2 - C - iii, 3 - A - i

Explanation:

Zone	Pressure Relationship	Blood Flow Characteristic
1	Alveolar pressure > Pulmonary artery pressure > Pulmonary venous pressure	No blood flow, as alveolar pressure occludes the capillaries.
2	Pulmonary artery pressure > Alveolar pressure > Pulmonary venous pressure	Intermittent blood flow occurs during systole.
3	Pulmonary artery pressure > Pulmonary venous pressure > Alveolar pressure	Continuous blood flow.



Note that the zones discussed are not the anatomical zones of the lungs.

Solution for Question 11:

Correct Answer: C) In the obstructed bronchus, the V/Q ratio is zero, and the partial pressures are $PO_2 = 40$ mmHg and $PCO_2 = 46$ mm Hg.

Explanation: Obstruction:

- V/Q Ratio: Zero
- Partial Pressures: The alveoli receive no ventilation, but perfusion continues. Consequently, the partial pressures of gases in the alveoli equilibrate with the venous blood, resulting in $PO_2 = 40$ mmHg and $PCO_2 = 46$ mmHg.

Embolism:

- V/Q Ratio: Infinite
- Partial Pressures: The alveoli receive ventilation, but no perfusion. The partial pressures of gases in the alveoli match those of the inspired air, resulting in $PO_2 = 149$ mmHg and $PCO_2 = 0.3$ mmHg.

Solution for Question 12:

Correct Answer: D) At the base: pO_2 is lowest and pCO_2 is highest.

Explanation:

- The base of the lungs has a low V/Q ratio and ventilation is drastically less in comparison to perfusion, so less oxygen is going to enter the alveoli and will be easily delivered to the venous blood. Thus, remaining oxygen at the level of basal alveoli is the lowest ($PO_2 = 89$ mm Hg).
- Also, carbon dioxide is very high at the level of basal alveoli because of the huge amount of venous blood and the exit of carbon dioxide into alveoli is incomplete. Therefore, the PCO_2 is the highest at the base ($PCO_2 = 42$ mm Hg).

Region of Lungs	V/Q Ratio	PO_2 (mm Hg)	PCO_2 (mm Hg)	pH of blood
Apex	Highest (3.3)	Highest (132)	Lowest (28)	7.51 (slightly alkaline)
Middle	Moderate (0.8)	100-104	40	7.4
Base	Lowest (0.6)	Lowest (89)	Highest (42)	7.39 (slightly acidic)

Solution for Question 13:

Correct Answer: B) In the upright position, the V/Q ratio is highest at the apex of the lungs.

Explanation:

- Both ventilation (V) and perfusion (Q) are higher at the base than at the apex, in upright position.
- However, the relative change in blood flow from the apex to the base is greater than the relative change in ventilation, so the ventilation/perfusion (V/Q) ratio is low at the base and high at the apex.
- In supine position, the apex to base differences in V and Q are negligible.

Solution for Question 14:

Answer: C) 1, 2, 3 and 4 Explanation:

Statement	Description	Explanation
1) Pulmonary ventilation refers to the amount of air entering the lungs in a minute, while alveolar ventilation is the volume of air reaching the alveolar zone for gas exchange.	Correct	Pulmonary ventilation = $TV \times RR$; Alveolar ventilation = $(TV - \text{Dead Space}) \times RR$.
2) Pulmonary ventilation for this patient is approximately 6.4 L/min.	Correct	Pulmonary ventilation = $400 \text{ mL} \times 16 = 6.4 \text{ L/min}$.
3) Alveolar ventilation for this patient is approximately 4 L/min.	Correct	Alveolar ventilation = $(400 \text{ mL} - 150 \text{ mL}) \times 16 = 4.0 \text{ L/min}$.
4) Perfusion is the blood flow at the level of the alveoli which is approximately equal to cardiac output from the right ventricle.	Correct	Perfusion is typically equal to the cardiac output of the right ventricle, approximately 5.4 L/min.

Solution for Question 15:

Correct Answer: B) Diaphragm contraction creates negative intrapleural pressure, causing air to enter the lungs.

Explanation:

Mechanism of Normal Respiration:

Inspiration

- The diaphragm, which contracts during inspiration and contributes to 75% of the inspiratory effort, and the external intercostal muscles, account for 25% of the inspiratory effort.
- During normal inspiration, the diaphragm contracts and moves downward.
- Inspiration results in an increase in intrapleural volume, leading to a decrease in intrapleural pressure from -2.5 mm Hg to -6 mm Hg. This negative intrapleural pressure is the force that expands the alveoli. Greater negative intrapleural pressure causes the alveoli to stretch.
- As alveolar volume increases, intra-alveolar pressure drops from 0 to -1 mm Hg.

- This pressure drop below atmospheric levels draws air into the lungs until intra-alveolar pressure reaches equilibrium at 0 mm Hg which marks the end of air entry, concluding inspiration.
- At the end of inspiration, the alveoli are expanded, and intra-alveolar pressure is 0 mm Hg.

Expiration

- Expiration is a passive process that occurs when the muscles contracted during inspiration relax.
- As the diaphragm relaxes, it moves upward, compressing the alveolar space.
- This compression reduces the volume of the alveoli.
- As a result, intra-alveolar pressure increases to +1 mmHg, which is higher than atmospheric pressure, pushing air out of the lungs.
- By the end of expiration, intra-alveolar pressure returns to equilibrium at 0 mmHg.

Option A) Diaphragm relaxation would cause an increase in intra-alveolar pressure, not a decrease in intrapleural pressure.

Option C) Diaphragm contraction decreases intra-alveolar pressure (making it negative relative to atmospheric pressure), which allows air to flow into the lungs.

Option D) During normal inspiration, the external intercostal muscles contract, not relax. Their contraction along with diaphragm contraction, helps decrease intrapleural pressure and expand the alveoli.

Solution for Question 16:

Correct Answer: B) Increased intrapleural pressure

Explanation:

Based on the provided scenario, the patient most likely has tension pneumothorax, which occurs following a laceration to the lung.

Intrapleural Pressure/Intrathoracic Pressure

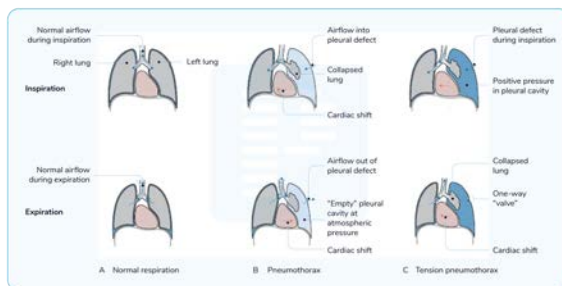
- The intrapleural pressure is typically -2.5 mmHg or -5 cm H₂O, which helps keep the alveoli open.

- Alveoli tend to collapse unless maintained by a stretching force from inside or negative pressure from outside.
- Intrapleural fluid is secreted by capillaries in the parietal pleura and absorbed by lymphatics in the visceral pleura. The net force pushing fluid out of capillaries is 9 cm H₂O, and the absorption pressure by lymphatics is 10 cm H₂O. This creates a net negative pressure in the intrapleural space, maintained by lymphatic drainage.
- The pressure in the alveoli at rest is equal to atmospheric pressure (760 mmHg), and this intra-alveolar pressure is considered 0 mmHg relative to atmospheric pressure.
- Intrapleural pressure is -2.5 mmHg relative to atmospheric pressure, meaning it is below the atmospheric pressure by 2.5 mmHg.

Changes in Tension Pneumothorax:

- Some injuries create a one-way valve effect at the site of the damaged visceral pleura.
- During inspiration, air enters the pleural space, but the valve closes during expiration, trapping the air inside. This repetitive trapping of air with each breath leads to tension pneumothorax.
- As air continues to accumulate in the pleural space, the intrapleural pressure increases, becoming positive relative to atmospheric pressure (greater than 760 mmHg). This increased pressure compresses the alveoli, which are already prone to collapse.
- The result is a significant collapse of the lung and a shift of the mediastinum.

In contrast, a simple pneumothorax involves air entering the pleural space without the one-way valve effect, leading to a collapse of the lung without the progressive increase in intrapleural pressure or mediastinal shift seen in tension pneumothorax.



Option A) Increased intrapulmonary pressure generally refers to pressure within the airways and alveoli during conditions like forced expiration or positive-pressure ventilation. This pressure would lead to barotrauma or air trapping.

Option C) Decreased intrapleural pressure does not cause the symptoms observed in tension pneumothorax.

Option D) Tension pneumothorax is characterized by increased intrapleural pressure due to trapped air, not decreased intrapulmonary pressure or fluid accumulation.

Solution for Question 17:

Correct Answer: D) The elastic recoil forces of the lungs and chest wall are in equilibrium.

Explanation:

Resting Condition of the Lungs:

- At the end of a normal exhalation, the lungs are in a state called Functional Residual Capacity (FRC).
- FRC comprises the Residual Volume (RV) and the Expiratory Reserve Volume (ERV). In this state, the respiratory muscles are relaxed.
- The elastic recoil forces of the lungs, which pull inward, and the chest wall, which pulls outward, are balanced.
- This equilibrium creates a negative intrapleural pressure, helping to keep the lungs partially inflated and preventing the alveoli from collapsing.

At the end of normal expiration, alveolar pressure is equal to atmospheric pressure, not higher. This equilibrium allows the air inside the lungs to stay at rest without any flow in or out. Option A)

The intrapleural pressure is actually lower than the atmospheric pressure. This negative pressure is crucial for keeping the lungs partially inflated and preventing collapse. (Option B)

Functional Residual Capacity (FRC) is not at its minimum at the end of normal expiration. FRC is the volume of air remaining in the lungs after a normal exhalation and represents a resting volume where no additional effort is required by the respiratory muscles. (Option C)

Solution for Question 18:

Correct Answer: C) Increased alveolar dead space due to poorly perfused but well-ventilated alveoli, reducing the amount of air available for gas exchange.

Explanation:

This option best explains the increase in physiological dead space by noting that elevated alveolar dead space (poor perfusion of ventilated alveoli) would lead to ineffective gas exchange, thereby increasing the physiological dead space.

Alveolar Dead Space

- Definition : Alveolar dead space is the portion of the air in the alveoli that is ventilated but not perfused with blood, meaning it does not participate in gas exchange.
- This occurs when certain alveoli receive air but lack adequate blood flow. Conditions such as pulmonary embolism can cause this imbalance.
- The presence of alveolar dead space results in inefficient gas exchange, as some of the air that reaches the alveoli cannot be used to oxygenate blood or remove carbon dioxide.

Physiological Dead Space

- Definition : Physiological dead space is the total volume of air in the respiratory system that does not participate in gas exchange. It includes both anatomical and alveolar dead space.
- Components : Anatomical Dead Space : It is the air in the conducting airways (trachea, bronchi, etc.) where no gas exchange occurs. It is typically about 150 mL in a healthy adult.
Alveolar Dead Space : This is the air in alveoli that are ventilated but not perfused.
- Anatomical Dead Space : It is the air in the conducting airways (trachea, bronchi, etc.) where no gas exchange occurs. It is typically about 150 mL in a healthy adult.
- Alveolar Dead Space : This is the air in alveoli that are ventilated but not perfused.
- Significance : An increase in physiological dead space means that a greater portion of the inhaled air is not being used effectively for gas exchange. This can result from increases in either anatomical or alveolar dead space.
- Anatomical Dead Space : It is the air in the conducting airways (trachea, bronchi, etc.) where no gas exchange occurs. It is typically about 150 mL in a healthy adult.
- Alveolar Dead Space : This is the air in alveoli that are ventilated but not perfused.

In this case, alveolar dead space would not be increased; instead, the problem would be related to low ventilation-perfusion ratio. The scenario described actually involves increased alveolar dead space due to well-ventilated but poorly perfused alveoli. (Option A)

An increase in anatomical dead space would result from obstructions or increased airway size, not from the alveolar dead space problem described. In this patient, the issue is with the alveoli themselves, not the conducting airways. (Option B)

The patient's problem is with an increase in physiological dead space, which is primarily due to an increase in alveolar dead space rather than anatomical dead space. Inflammation and narrowing of airways are not consistent with the scenario of increased physiological dead space. (Option D)

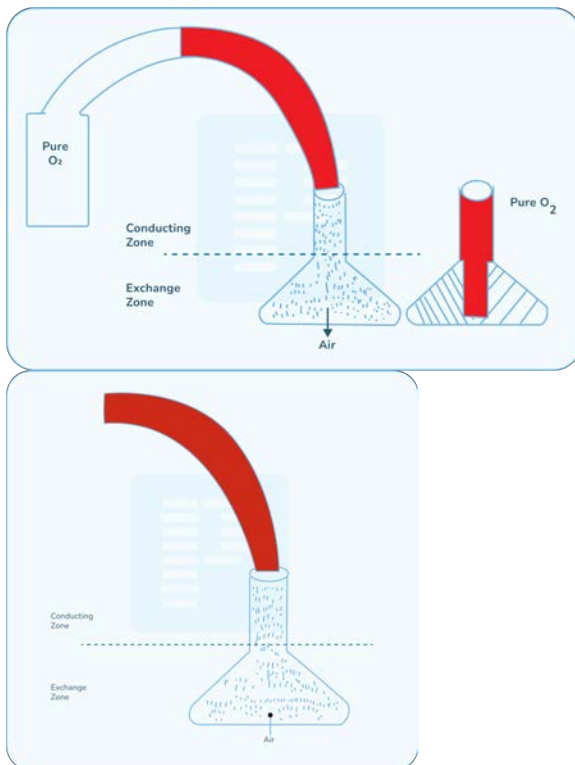
Solution for Question 19:

Correct Answer: C) The pure alveolar air coming from the alveolar region.

Explanation:

Nitrogen washout method is used to calculate the volume of anatomical dead space and for measuring the closing volume or closing capacity of the lung.

The procedure of Nitrogen Washout Method:

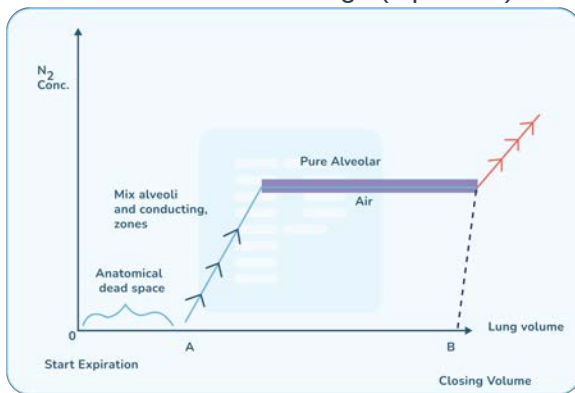


- **Inhalation Phase:** The subject inhales pure oxygen from a cylinder, which replaces the normal atmospheric air in the conducting zone and enters the alveoli. By the end of inhalation, the conducting zone is filled with pure oxygen, while the alveoli have a mix of pure oxygen and residual alveolar air.
- The subject inhales pure oxygen from a cylinder, which replaces the normal atmospheric air in the conducting zone and enters the alveoli.

- By the end of inhalation, the conducting zone is filled with pure oxygen, while the alveoli have a mix of pure oxygen and residual alveolar air.
- Expiration Phase: As the subject exhales, the air passing through the nitrogen meter initially contains pure oxygen from the conducting zone, resulting in a zero nitrogen concentration reading. As exhalation progresses, air from the alveolar region—containing a constant nitrogen concentration—begins to mix with the air from the conducting zone.
- As the subject exhales, the air passing through the nitrogen meter initially contains pure oxygen from the conducting zone, resulting in a zero nitrogen concentration reading.
- As exhalation progresses, air from the alveolar region—containing a constant nitrogen concentration—begins to mix with the air from the conducting zone.
- Graph of Nitrogen washout method: The resulting graph of nitrogen concentration versus lung volume will show: An initial zero reading (representing anatomical dead space) (Option A) A gradual increase as alveolar air mixes with the conducting zone air (Option B) A plateau at a fixed nitrogen concentration, which indicates the pure alveolar air (Option C) . This is followed by a sudden increase in nitrogen concentration known as the Closing Volume of the Lungs (Option D).
- The resulting graph of nitrogen concentration versus lung volume will show: An initial zero reading (representing anatomical dead space) (Option A) A gradual increase as alveolar air mixes with the conducting zone air (Option B) A plateau at a fixed nitrogen concentration, which indicates the pure alveolar air (Option C) . This is followed by a sudden increase in nitrogen concentration known as the Closing Volume of the Lungs (Option D).
- An initial zero reading (representing anatomical dead space) (Option A)
- A gradual increase as alveolar air mixes with the conducting zone air (Option B)
- A plateau at a fixed nitrogen concentration, which indicates the pure alveolar air (Option C) .
- This is followed by a sudden increase in nitrogen concentration known as the Closing Volume of the Lungs (Option D).
- The subject inhales pure oxygen from a cylinder, which replaces the normal atmospheric air in the conducting zone and enters the alveoli.
- By the end of inhalation, the conducting zone is filled with pure oxygen, while the alveoli have a mix of pure oxygen and residual alveolar air.
- As the subject exhales, the air passing through the nitrogen meter initially contains pure oxygen from the conducting zone, resulting in a zero nitrogen concentration reading.
- As exhalation progresses, air from the alveolar region—containing a constant nitrogen concentration—begins to mix with the air from the conducting zone.
- The resulting graph of nitrogen concentration versus lung volume will show: An initial zero reading (representing anatomical dead space) (Option A) A gradual increase as alveolar air mixes with the conducting zone air (Option B) A plateau at a fixed nitrogen concentration,

which indicates the pure alveolar air (Option C) . This is followed by a sudden increase in nitrogen concentration known as the Closing Volume of the Lungs (Option D).

- An initial zero reading (representing anatomical dead space) (Option A)
- A gradual increase as alveolar air mixes with the conducting zone air (Option B)
- A plateau at a fixed nitrogen concentration, which indicates the pure alveolar air (Option C) .
- This is followed by a sudden increase in nitrogen concentration known as the Closing Volume of the Lungs (Option D).
- An initial zero reading (representing anatomical dead space) (Option A)
- A gradual increase as alveolar air mixes with the conducting zone air (Option B)
- A plateau at a fixed nitrogen concentration, which indicates the pure alveolar air (Option C) .
- This is followed by a sudden increase in nitrogen concentration known as the Closing Volume of the Lungs (Option D).



Solution for Question 20:

Correct Answer: D) General anesthesia increases anatomical dead space due to the loss of bronchial tone, leading to bronchial dilation.

Explanation:

Anatomical Dead Space Volume is the volume of air that occupies the conducting zone of the airways and does not participate in the gaseous exchange.

Key Points:

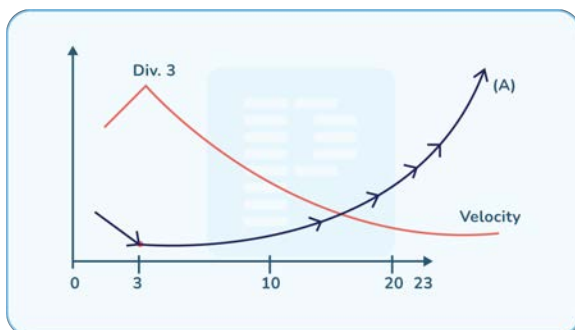
- Any kind of bronchodilation experienced by a person will increase the anatomical dead space volume because bronchus and bronchioles will be dilated, which will entrap more dead volume.
- If general anesthesia is given to a person, there will be a loss of bronchial tone, and dilation will be more, which will be more anatomical dead space. (Option A).
- In inspiration the length of the anatomical dead space zone is increasing, hence dead space volume will also increase.
- Extension of the neck and protrusion of the jaw, all done during anesthesia will result in increased anatomical dead space.

Solution for Question 21:

Correct Answer: B) Trachea

Explanation:

- Velocity is inversely proportional to cross-sectional area. In the respiratory system, consider the cross-sectional area of different segments like a cross-section of the trachea, both the bronchus and bronchioles, etc,
- The total cross-sectional area is minimum at division number 3. Hence velocity is highest.



- Velocity will be maximum at division number 3 and then velocity progressively decreases towards the alveoli, and minimum velocity is seen at the level of alveoli.

Alveoli(Option A): The alveoli have the largest total cross-sectional area, leading to minimal airflow velocity and predominantly laminar flow.

Terminal bronchioles(Option C): Terminal bronchioles have a larger cumulative cross-sectional area than the trachea, leading to lower airflow velocity and smoother flow

Respiratory bronchioles (Option D): Respiratory bronchioles, being part of the gas exchange zone, have an even greater cross-sectional area, causing minimal velocity and mostly laminar flow.

Solution for Question 22:

Correct Answer : C) Medium to large-size bronchi (lobar and segmental bronchi)

Explanation: The medium to large-size bronchi, specifically the lobar and segmental bronchi, are the site of maximum resistance to airflow. This is because these bronchi have a relatively larger radius but are fewer in number, resulting in higher individual resistance.

- According to Poiseuille's law, resistance to airflow (R) is inversely proportional to the fourth power of the radius (r) of the airway.
- Resistance: $R = 8\eta L / \pi r^4$
- Therefore, small changes in the radius result in significant changes in resistance. Despite the smaller radius of the bronchioles, their parallel arrangement results in a lower total resistance.

The upper airways (pharynx and larynx) also contribute significantly to airflow resistance but less than medium to large-sized bronchi. (Option D) Only about 20% of the resistance occurs at the level of the small airways. (Option B) Alveoli have a large cross-sectional area which significantly reduces the resistance according to Poiseuille's law. Option C)

Solution for Question 23:

Correct Answer: B) Type II pneumocytes

Explanation: Neonatal respiratory distress syndrome (NRDS) is primarily caused by a deficiency in surfactant, a substance crucial for reducing surface tension in the alveoli and preventing their collapse during exhalation. Surfactant is produced by Type II pneumocytes.

- Type I pneumocytes (Option A) are thin, flat cells that cover the majority of the alveolar surface and are involved in gas exchange.
 - Alveolar macrophages (Option C) are immune cells that ingest and remove debris and pathogens.
 - Endothelial cells (Option D) line the capillaries surrounding the alveoli and are involved in the exchange of gases between the alveoli and blood.
-

Solution for Question 24:

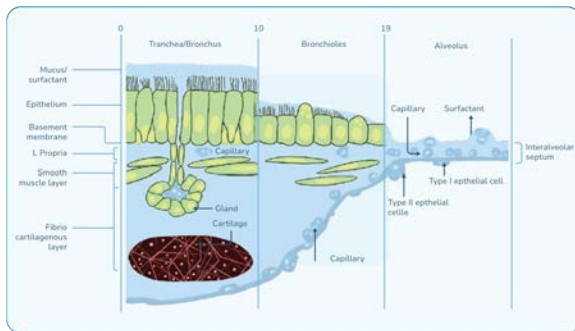
Correct Answer: C) They have cilia.

Explanation:

- Clara cells, also known as club cells, are non-ciliated, dome-shaped cells found in the bronchioles of the respiratory tract. (Option A ruled out)
 - They play a crucial role in maintaining the integrity of the bronchiolar epithelium by stabilising surfactant proteins that reduce surface tension. (Option B ruled out)
 - They are involved in detoxifying harmful substances through their enzymatic activity. (Option D ruled out)
 - They are not ciliated, differentiating them from other epithelial cells in the respiratory tract that do have cilia, such as the pseudostratified ciliated columnar epithelial cells found in the trachea and larger bronchi.
-

Solution for Question 25:

Correct Answer: C) Trachea Explanation: The epithelial lining of trachea and large bronchi are composed of pseudostratified ciliated columnar epithelium.



- Ciliated epithelium is present at the level of the trachea, bronchus, and bronchioles.
- Cartilaginous rings are present up to the bronchi.
- Smooth muscles are present up to respiratory bronchioles (bronchial tone). (Option D ruled out)
- Epithelial lining Trachea and Large Bronchi: Pseudostratified columnar ciliated epithelium. Bronchioles: Cuboidal-ciliated epithelium. (Option B ruled out) Alveoli: Simple squamous epithelium. (Option A ruled out)
- Trachea and Large Bronchi: Pseudostratified columnar ciliated epithelium.
- Bronchioles: Cuboidal-ciliated epithelium. (Option B ruled out)
- Alveoli: Simple squamous epithelium. (Option A ruled out)
- Trachea and Large Bronchi: Pseudostratified columnar ciliated epithelium.
- Bronchioles: Cuboidal-ciliated epithelium. (Option B ruled out)
- Alveoli: Simple squamous epithelium. (Option A ruled out)

Pulmonary Function Tests

1. Which of the following parameters belong to static lung volumes and capacities?
Maximum Voluntary Ventilation Functional Residual Capacity (FRC) Dead space
Residual Volume (RV) Forced Expiratory Volume in 1 second (FEV1)

- A. 1, 3, and 4
 - B. 2, 3, and 4
 - C. 1, 2, and 5
 - D. 2, 4, and 5
-

2. Which of the following parameters cannot be measured by spirometry?

- A. Vital capacity
 - B. Residual volume
 - C. Inspiratory reserve volume
 - D. Tidal volume
-

3. Which of the following formulas correctly represents the Vital Capacity (VC) of the lungs?

- A. $TV + IRV + ERV + RV$
 - B. $TV + IRV + ERV$
 - C. $TV + IRV$
 - D. $ERV + RV$
-

4. What is the true statement regarding Closing Volume (CV)?

- A. CV is the lung volume at which large airways begin to close during inspiration.
 - B. CV measurement is typically performed using a Multi-Breath Helium Dilution Test.
 - C. Increased CV is seen in conditions such as ageing, obstructive lung diseases, obesity, and smoking.
 - D. A normal CV indicates small airway dysfunction and abnormal lung mechanics.
-

5. A 60-year-old man with a history of smoking presents with shortness of breath. During spirometry, dynamic compression of the airways is observed during forceful expiration. Which of the following statements best describes the point at which dynamic compression occurs?

- A. Early expiration, high lung volume.
 - B. Mid expiration, FRC reached.
 - C. Late expiration, pleural pressure > airway pressure.
 - D. End expiration, near RV.
-

6. A healthy 35-year-old male undergoes pulmonary function tests as part of a routine health check-up. Which of the following findings is most likely to be observed?

- A. Tidal volume 400 ml
 - B. FEV₁ /FVC 0.65
 - C. Residual volume 1200 ml
 - D. Vital capacity 3500 ml
-

7. A 65-year-old male, lifelong smoker, presents with chronic shortness of breath and a persistent cough. Imaging studies confirm hyperinflation and destruction of alveolar walls. He was diagnosed with emphysema. What are the expected results of his Pulmonary Function Test?

- A. Increased Residual Volume
 - B. Increased Total Lung Capacity
 - C. Increased FEV/FVC
 - D. Both A and B
-

8. Which of the following is false on the findings of restrictive lung disease?

- A. Decreased TLC
 - B. Decreased VC
 - C. Preservation of Forced Expiratory Flow Rates
 - D. All are true
-

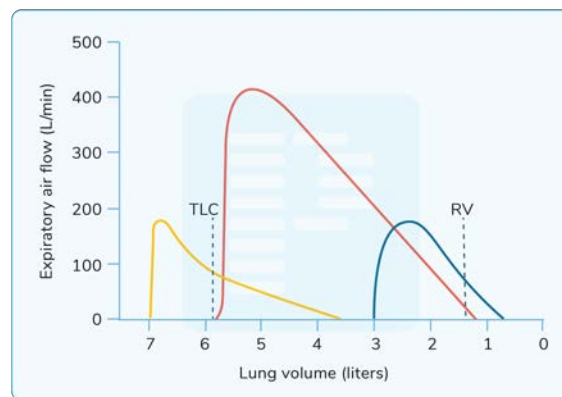
9. Which of the following is false about Peak Expiratory Flow Rate (PEFR)?

- A. Maximum velocity of flow in litres per minute with air being forced out of the lungs
 - B. Recorded by using the peak-flow meter
 - C. The normal value is 400 to 600 L/min
 - D. All are true
-

10. Which of the following is the most sensitive indicator of small airway diseases?

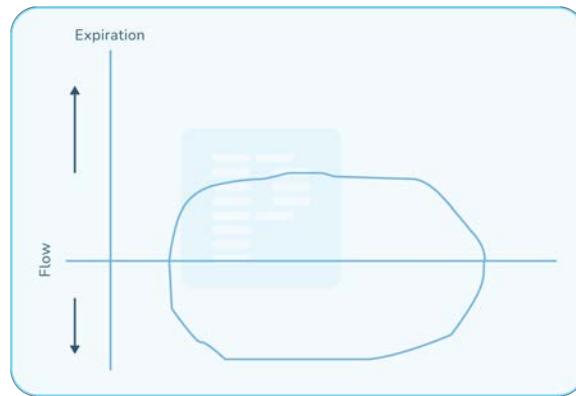
- A. PEFR
 - B. FEV1
 - C. FEV1/FVC
 - D. MMEFR
-

11. In the image given below what does the yellow curve represent?



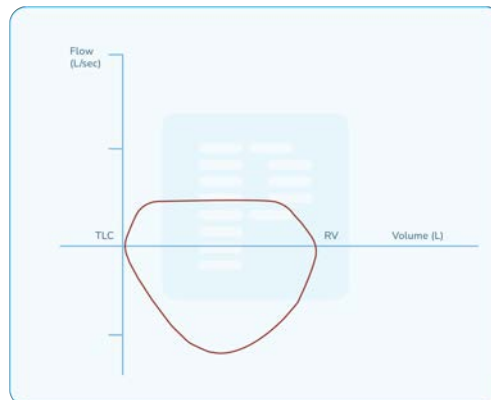
- A. Normal lung function
 - B. Obstructive lung disease
 - C. Restrictive lung disease
 - D. Respiratory failure
-

12. A patient underwent pulmonary function testing, and the flow-volume loop is shown below. What does it show?



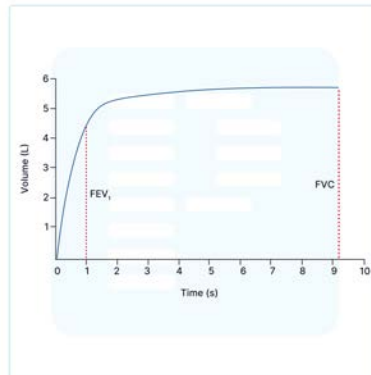
- A. Fixed upper airway obstruction
- B. Variable intra-thoracic obstruction
- C. Normal lung functioning
- D. None of the above

13. Analyse the following flow volume loop diagram. Which of the following statements is NOT true regarding this flow volume loop?



- A. The loop demonstrates flatness in the expiratory portion
- B. This loop occurs in variable intrathoracic obstruction
- C. The "box pattern" indicates fixed upper airway obstruction.
- D. The loop shows a reduction in peak expiratory flow (PEF) due to the obstruction.

14. The volume of gas expired by a normal adult man during a forced expiration, demonstrating the FEV1 and the FVC. Calculate the FEV1/FVC of this person.



- A. 60%
- B. 70%
- C. 80%
- D. 90%

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	2
Question 4	3
Question 5	3
Question 6	3
Question 7	4
Question 8	4
Question 9	4
Question 10	4
Question 11	2
Question 12	1
Question 13	3
Question 14	3

Solution for Question 1:

Correct Answer: B) 2, 3, and 4 are correct

Explanation:

Aspect	Static Lung Volumes & Capacities	Dynamic Lung Volumes & Capacities
Measurement Focus	Lung volume without the recording of time	Lung volume to time

Units of Measurement	Measured in millilitres (ml) or litres	Measured in millilitres per second (ml/second)
Examples	Volume: Tidal Volume (TV) Inspiratory Reserve Volume (IRV) Expiratory Reserve Volume (ERV) Residual Volume (RV) Anatomical Dead Space Capacity: Total Lung Capacity (TLC) Vital Capacity (VC) Inspiratory Capacity (IC) Functional Residual Capacity (FRC)	Forced Vital Capacity (FVC) Forced Expiratory Volume in 1 second (FEV1) FEV1/FVC Ratio Peak Expiratory Flow Rate (PEFR) Inspiratory Flow Rate Expiratory Flow Rate
Clinical Significance	Assess the maximum air volume the lungs can hold and the different capacities during various breathing phases. Diagnosis of restrictive lung diseases such as fibrosis and sarcoidosis	Assess the rate of airflow and lung function over time. Diagnosis of obstructive lung diseases such as asthma and COPD

Solution for Question 2:

Correct Answer: B) Residual volume

Explanation:

Respiratory parameters that cannot be measured by spirometry:

Parameter	Method for Measurement
Residual Volume (RV)	Body Plethysmography (Best). Gas Dilution (Helium dilution Method). Nitrogen Washout
Functional Residual Capacity (FRC)	
Total Lung Capacity (TLC)	Body Plethysmography. Gas Dilution (Helium Dilution Method)
Alveolar Dead Space	Single-Breath Tests. Imaging Studies.

Diffusing Capacity (DLCO)	DLCO Test (Carbon Monoxide Diffusion Test)
---------------------------	--

Solution for Question 3:

Correct Answer: B) TV + IRV + ERV

Explanation:

Vital Capacity (VC) is the maximum amount of air that can be exhaled after a maximum inhalation and is calculated by summing Tidal Volume (TV), Inspiratory Reserve Volume (IRV), and Expiratory Reserve Volume (ERV).

$$VC = TV + IRV + ERV$$

Lung Volume/Capacity	Description	Typical Volume (mL)	Formula
Tidal Volume (TV)	Amount of air inhaled or exhaled during normal breath	500	-
Inspiratory Reserve Volume (IRV)	Additional air that can be inhaled after a normal inhalation	3000	-
Expiratory Reserve Volume (ERV)	Additional air that can be exhaled after a normal exhalation	1200	-
Residual Volume (RV)	Air remaining in the lungs after a maximal exhalation	1200	-
Anatomical Dead Space	Volume of air in conducting airways where no gas exchange occurs	150	-
Total Lung Capacity (TLC)	Maximum amount of air the lungs can hold	6000	$TLC = TV + IRV + ERV + RV$
Vital Capacity (VC)	Maximum air exhaled after a maximal inhalation	4800	$VC = TV + IRV + ERV$
Inspiratory Capacity (IC)	Maximum air inhaled after a normal exhalation	3500	$IC = TV + IRV$
Functional Residual Capacity (FRC)	Air remaining in the lungs after a normal exhalation	2400	$FRC = ERV + RV$

Solution for Question 4:

Correct Answer: C) Increased CV is seen in conditions such as ageing, obstructive lung diseases, obesity, and smoking.

Explanation:

Closing Volume (CV)

Definition:

- Closing Volume (CV) is the lung volume at which small airways begin to close during expiration, particularly in the dependent parts of the lungs. (Option A ruled out)
- $CV = \text{Closing capacity} - \text{Residual volume}$.

Mechanism:

- During expiration, as lung volume decreases, pleural pressure becomes less negative.
- At the closing volume, airway pressure drops below pleural pressure, causing small airways to close.

Measurement:

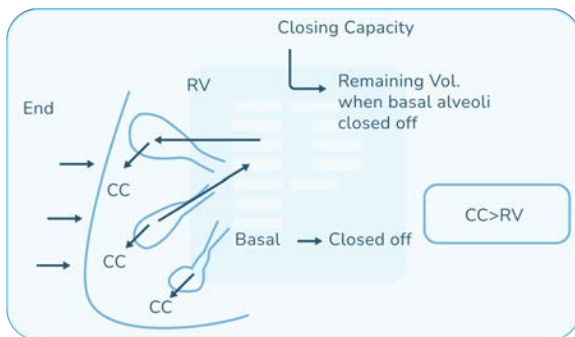
- Single-Breath Nitrogen Washout Test: This test measures CV by observing the washout of nitrogen from the lungs during a single breath.

Clinical Implications:

Increased CV: Indicates small airway dysfunction and can be seen in: (Option D ruled out)

- Ageing: Decreased elastic recoil and increased lung compliance.
- Obstructive Lung Diseases: COPD and asthma.
- Obesity and Pregnancy: Reduced lung volumes and diaphragmatic changes.
- Smoking: Indicates early small airway dysfunction.

Closing Capacity (CC):



- Closing Capacity (CC) is the lung volume at which the basal alveoli begin to close off during expiration.
- It is the sum of the closing and residual volumes ($CC = CV + RV$).
- During forceful expiration, the volume at which only the basal alveoli are closed off is known as closing capacity.
- Further expiration will eventually lead to all alveoli being closed off, reaching the residual volume.
- Ageing: Functional residual capacity (FRC) is greater than CC in a normal adult. By around 66 years of age, FRC is nearly equal to CC, indicating that CC increases gradually with age.
- Pathological Implications: If CC significantly exceeds FRC, it indicates highly collapsible lung diseases, such as hyaline membrane disease.

Measurement :

Single-Breath Nitrogen Washout Method (Fowler's) measures CC and anatomical dead space by observing nitrogen washout during a single breath. (Option B ruled out)

Solution for Question 5:

Dynamic compression of the airways occurs primarily during the late phase of forceful expiration when intrathoracic pressure (pleural pressure) exceeds airway pressure.

This leads to airway narrowing or collapse, particularly in chronic obstructive pulmonary disease (COPD), with increased resistance and decreased elastic recoil.

Mechanism:

- During forceful expiration, elastic tissues in the alveoli help in passive expiration by recoiling inward. As air moves through the bronchi and bronchioles, airway resistance

causes the pressure to drop.

- An external force from the chest wall also acts on the bronchi and bronchioles. At the end of expiration, the pressure inside the bronchioles equals the external pressure, known as the equal pressure point.

Dynamic compression does not occur during the early expiration phase when the lung volume is still relatively high; the surrounding lung tissue generally supports the airways well. (Option A)

Dynamic compression is not associated with functional residual capacity (FRC) but occurs later when airway and pleural pressures interact. (Option B)

When reaching the residual volume (RV) by the end of expiration, dynamic compression is less of a concern, as the airflow has largely ceased. (Option D)

Solution for Question 6:

Correct Answer: C) Residual volume 1200 ml

Explanation:

A residual volume of 1200 ml is normal for a healthy 35-year-old male.

Residual volume remains stable unless there is a pathological condition affecting lung function, and it helps differentiate between obstructive and restrictive lung diseases:

- Obstructive lung diseases (e.g., COPD) increase residual volume due to air trapping.
- Restrictive lung diseases (e.g., pulmonary fibrosis) decrease residual volume due to reduced lung compliance.

Additionally, residual volume can differentiate between types of restrictive lung diseases:

- Parenchymal restrictive lung disease (e.g., pulmonary fibrosis) shows highly reduced residual volume due to stiff and scarred lung tissue.
- Extra-parenchymal restrictive lung disease (e.g., neuromuscular diseases, chest wall deformities) may show normal or slightly reduced residual volume, as the issue is outside the lung tissue, affecting breathing mechanics rather than the lung parenchyma.

Key parameters of pulmonary function tests:

Spirometry

Forced Vital Capacity (FVC)	Total air exhaled forcefully after deep breath. Approximately 4-6 liters (depending on age, sex, height, and weight)
Forced Expiratory Volume in 1 Second (FEV 1)	Air exhaled in the first second of the FVC manoeuvre. Approximately 80% of FVC
FEV 1 /FVC Ratio	Percentage of FVC exhaled in the first second. Approximately 70-80%
Peak Expiratory Flow (PEF)	Maximum speed of expiration. Varies by age, sex, and height (e.g., 400-600 L/min for adults)
Lung Volumes	
Total Lung Capacity (TLC)	The total volume of air in the lungs after maximal inhalation Approximately 6-8 liters
Residual Volume (RV)	Air remaining in the lungs after maximal exhalation Approximately 1.2-1.5 litres
Vital Capacity (VC)	Air that can be exhaled after maximal inhalation (FVC minus RV) Approximately 4-5 liters
Inspiratory Reserve Volume (IRV)	Additional air inhaled after normal inhalation Approximately 2-3 liters
Expiratory Reserve Volume (ERV)	Additional air exhaled after normal exhalation Approximately 1-1.5 liters
Diffusing Capacity (DLCO)	
Lung's ability to transfer gases (e.g., CO) Approximately 20-30 mL/min/mmHg	

The other options (A, B, and D) are also within normal ranges for healthy adults but are less specific to routine findings in pulmonary function tests compared to residual volume.

Solution for Question 7:

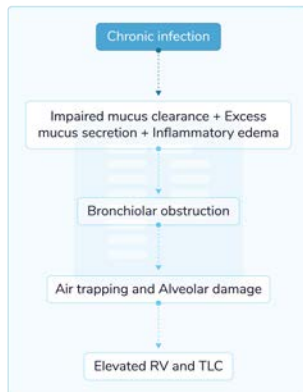
Correct Answer: D) Both A and B

Explanation:

Emphysema is an obstructive lung disease. Increased Residual Volume and Total Lung Capacity are the findings in obstructive lung disease.

Features of Obstructive Diseases:

Emphysema:



Examples of obstructive lung disease:

- Asthma
- COPD (Chronic bronchitis and emphysema)
- Bronchiectasis
- Bronchiolitis
- Cystic fibrosis

Solution for Question 8:

Correct Answer: D) All are true

Explanation:

Definition	Lung disorders are characterised by a reduced distensibility of the lungs, compromising lung expansion, and, in turn, reduced lung volumes
Features	Decreased TLC (Total Lung Capacity) Decreased VC (Vital Capacity) Decreased RV (Residual Volume) Preservation of Forced Expiratory Flow Rates.
Pathology	1. Intrinsic Conditions:

	Destruction of Distal Lung Parenchyma Inflammatory Changes in the Alveolar Interstitium Interstitial Lung Diseases Alterations from Within the Alveolus(oedema or haemorrhage) 2. Extrinsic Conditions: Limitations in Neuromuscular Function Chest Wall Movement Impairments (pleural disorders, obesity, costosternal or costovertebral fusion, thoracic vertebral fusion or deviation)
Parenchymal Restrictive Dysfunction	Pulmonary fibrosis Pneumoconiosis Radiation-induced interstitial lung disease
Extra-parenchymal Restrictive Dysfunction	1. Neuromuscular Diseases: Myasthenia gravis Paralysis of diaphragm Muscular dystrophy 2. Chest Wall Diseases Kyphoscoliosis Obesity Ankylosing spondylitis

Solution for Question 9:

Correct Answer: D) All are true

Explanation:

- Peak expiratory flow rate (PEFR) is the maximum flow velocity in litres per minute with air being forced out of the lungs.(Option A)
 - PEFR is recorded by using the peak-flow meter.(Option B)
 - The normal value of PEFR is 400 to 600 L/min or 6 to 10 L/s.(Option C)
 - PEFR is an index of patency of airways.
-

Solution for Question 10:

Correct Answer: D) MMEFR

Explanation:

Maximum Mid-expiratory Flow Rate (MMEFR):

- Maximum flow achieved during the middle third of the total expired volume.
- Forced expiratory flow at 25% to 75% of the lung volume.
- Indicates the patency of small airways.
- More sensitive marker of airway disease (particularly for the obstruction of smaller-sized airways) than traditional parameters like FEV₁, FEV₁/FVC ratio, or PEF.

PEFR (Option A) – Less sensitive as it reflects large airway function and can remain normal in early small airway disease.

FEV₁ (Option B) – Commonly used but detects moderate-to-severe airway obstruction, making it less sensitive for early disease.

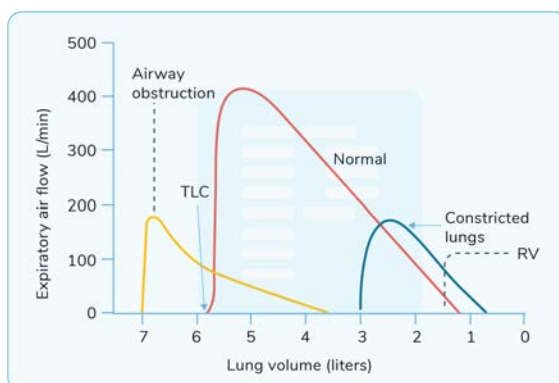
FEV₁/FVC Ratio (Option C) – Helps diagnose obstructive lung diseases, but may not detect early small airway dysfunction as well as MMEFR.

Solution for Question 11:

Correct Answer: B) Obstructive lung disease

Explanation:

This is a Peak Expiratory Air Flow-Lung Volume Curve.



Condition	Curve Characteristics
Restrictive diseases (Constricted lung)	Curve is smaller and shifted to the right (Option C ruled out) Peak flow is lower, and the volume is less due to reduced TLC and RV
Obstructive diseases	Curve is smaller and shifted to the left Lower peak flow Scooped-out appearance during expiration

Normal lung function (Option A) – Normal curve would have a smooth, symmetrical peak and descent without a scooped-out appearance or shift in volume.

Respiratory failure (Option D) – Respiratory failure is a clinical condition rather than a distinct spirometric pattern, and it can involve both obstructive and restrictive features depending on the cause

Solution for Question 12:

Correct Answer: A) Fixed upper airway obstruction

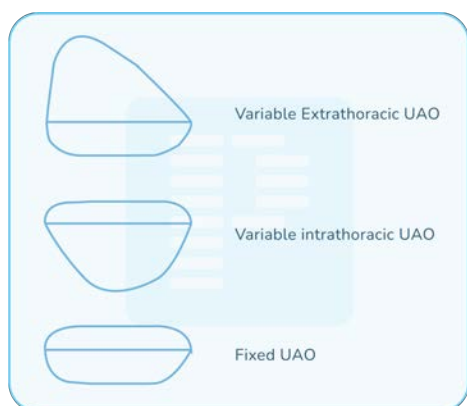
Explanation:

Given is a flow-volume loop of a fixed upper airway obstruction.

Fixed upper airway obstruction:

- Both inspiration and expiration affected.
- Examples: Foreign body Scarring Enlarged thyroid gland
- Foreign body
- Scarring
- Enlarged thyroid gland
- Foreign body
- Scarring
- Enlarged thyroid gland

The flow-volume loops in variable extrathoracic, variable intrathoracic, and fixed upper airway obstruction:



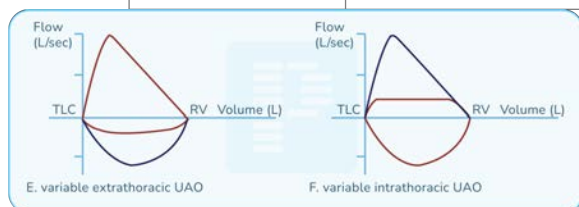
Solution for Question 13:

Correct answer: C) The "box pattern" indicates fixed airway obstruction.

Explanation:

A "box pattern" on the flow volume loop, characterised by flattening both inspiratory and expiratory portions, can occur in both fixed and variable intrathoracic obstructions and is not specific to one.

Feature	Variable Intrathoracic Obstruction	Variable Extrathoracic Obstruction
Location	Within the thoracic cavity (trachea, bronchi)	Outside the thoracic cavity (larynx, pharynx)
Causes	Tracheobronchomalacia, mainstem bronchial obstruction, lower airway obstructions	Laryngeal oedema, vocal cord dysfunction, tumours in the upper airway
Flow-Volume Loop Characteristics	Flattened expiratory portion (Option A ruled out) ; reduced peak expiratory flow (PEF) (Option D ruled out)	Flattened inspiratory portion; expiration may remain relatively unaffected



Solution for Question 14:

Correct answer: C) 80%

Explanation:

FEV1 (Forced Expiratory Volume in 1 second): Volume of air that can be forcibly exhaled in the first second of expiration.

FVC (Forced Vital Capacity): Total volume of air that can be forcibly exhaled after taking a deep inhalation.

Here,

- FEV1 = 4L
- FVC = 5L

$$= \frac{4}{5} \times 100 = 80\%$$

Interpretation of Values:

- Normal Ratio (70%-80% or higher): Indicates normal lung function.
- Low Ratio (<70%): Suggests obstructive lung disease.
- Normal or High Ratio with Reduced Volumes : Suggests restrictive lung disease.

Normal Ratio (70%-80% or higher): Indicates normal lung function.

Low Ratio (<70%): Suggests obstructive lung disease.

Diffusion of Gases, O₂ and CO₂ Transport

1. Which of the following formulas gives the oxygen extraction ratio (OER) of a substance?

- A. $(\text{Arterial Concentration} + \text{Venous Concentration}) / \text{Arterial Concentration}$
 - B. $(\text{Arterial Concentration} - \text{Venous Concentration}) / \text{Arterial Concentration}$
 - C. $\text{Venous Concentration} / \text{Arterial Concentration}$
 - D. $(\text{Venous Concentration} - \text{Arterial Concentration}) / \text{Venous Concentration}$
-

2. Which of the following contributes to the higher hematocrit in venous blood than arterial blood?

- A. Lower Plasma Volume in venous blood
 - B. Chloride shift causing red blood cell swelling
 - C. Greater carbon dioxide content in venous blood
 - D. All of the above
-

3. What is the significance of the Haldane effect?

- A. Oxygenation promotes CO₂ release in the lungs.
 - B. Deoxygenation enhances oxygen uptake in tissues.
 - C. CO₂ binding boosts oxygen release in tissues.
 - D. Haemoglobin buffers blood pH by binding hydrogen ions.
-

4. Which phenomenon explains the release of oxygen from haemoglobin in response to high carbon dioxide concentrations at the tissue level?

- A. Haldane Effect
 - B. Bohr Effect
 - C. Chloride Shift
 - D. Root Effect
-

5. In which form is carbon dioxide predominantly transported in the blood?

- A. Dissolved CO₂
 - B. Carbaminohemoglobin
 - C. Bicarbonate ions (HCO₃⁻)
 - D. Protein bound
-

6. Which of the following solutions is considered best for blood storage?

- A. ACD (Acid Citrate Dextrose)
 - B. CPDA (Citrate Phosphate Dextrose Adenine)
 - C. CPD (Citrate Phosphate Dextrose)
 - D. EDTA (Ethylenediaminetetraacetic Acid)
-

7. What is the shape of the oxygen dissociation curve for myoglobin?

- A. Rectangular Hyperbolic
- B. Sigmoidal
- C. Linear

D. Exponential

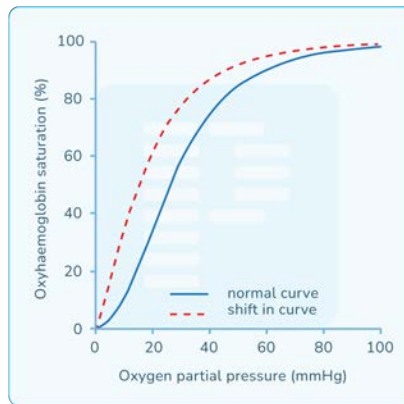
8. Which of the following statements is true regarding the Luebering-Rapoport pathway?

- A. It reduces the affinity of hemoglobin for oxygen by producing 2,3-bisphosphoglycerate (2,3-BPG).
 - B. It generates NADPH as a key product and decreases the affinity of hemoglobin for oxygen
 - C. It functions to increase the ATP yield of glycolysis in red blood cells.
 - D. It involves the enzyme hexokinase to initiate the pathway.
-

9. A patient has a haemoglobin concentration of 15 g/dL. Calculate the oxygen-carrying capacity of the patient's blood.

- A. 20.1 mL O₂/g
 - B. 17.5 mL O₂/g
 - C. 19.8 mL O₂/g
 - D. 15.0 mL O₂/g
-

10. What is the cause of this shift in the hemoglobin-oxygen (Hb-O₂) dissociation curve given below?



- A. Increased pH
- B. Increased temperature
- C. Increased metabolic rate
- D. Increased carbon dioxide levels

11. Select the correct statements from the following The binding of one oxygen molecule to Hb decreases the affinity of the remaining heme groups for oxygen. Hb shifts from the Tense (T) state to the Relaxed (R) state upon oxygen binding, increasing oxygen affinity. At low PO₂, small changes in PO₂ lead to large changes in oxygen saturation. At moderate PO₂, the dissociation curve is steep, meaning small increases in PO₂ results in large increase in oxygen saturation.

- A. 1 and 2
- B. 2 and 3
- C. 2 and 4
- D. 3 and 4

12. All of the following are true regarding the oxygen-hemoglobin (O₂-Hb) dissociation curve except?

- A. It is a sigmoid-shaped curve.
 - B. The plateau phase ranges from 10-60 mm Hg.
 - C. The steep phase ensures efficient oxygen delivery to tissues even with small drops in PO_2 .
 - D. P 50 assesses the binding affinity of haemoglobin for oxygen.
-

13. Which of the following is not a component of the respiratory unit?

- A. Respiratory bronchiole
 - B. Alveolar duct
 - C. Terminal bronchiole
 - D. Alveolus
-

14. What is the major mode of transport for oxygen in the blood?

- A. Dissolved directly in plasma
 - B. Bound to haemoglobin in red blood cells
 - C. As bicarbonate ions in plasma
 - D. Bound to plasma proteins
-

15. Which of the following describes Henry's law?

- A. The volume of a gas is inversely proportional to its pressure at constant temperature.
- B. A gas's effusion rate is inversely proportional to the square root of its molar mass.
- C. The solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid at a constant temperature.
- D. The volume of a gas is directly proportional to its temperature at constant pressure

16. Which of the following statements is true regarding the effect of high altitude on gas diffusion?

- A. Diffusion capacity increases
 - B. Diffusion capacity decreases
 - C. Diffusion capacity remains constant
 - D. Any of the above, depending on the type of gas
-

17. A 55-year-old female patient with a history of chronic obstructive pulmonary disease (COPD) is admitted to the hospital with severe shortness of breath. Arterial blood gas analysis shows an oxygen content of 16 ml/dl, while the mixed venous blood sample shows an oxygen content of 12 ml/dl. The pulmonary capillary oxygen content is determined to be 20 ml/dl. Calculate the percentage of cardiac output shunt for this patient.

- A. 25%
 - B. 10%
 - C. 15%
 - D. 50%
-

18. Nitrous oxide is classified as a perfusion-limited gas. What does this imply about its diffusion properties in the lungs?

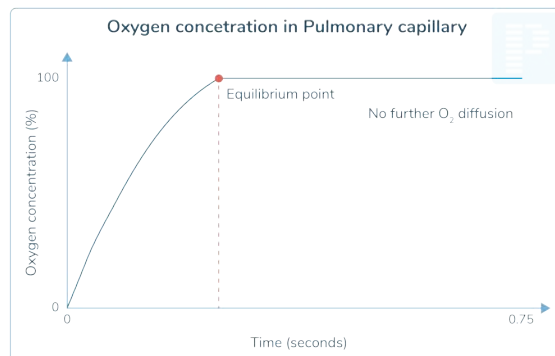
- A. N₂O uptake into pulmonary capillaries is dependent on the amount of blood flowing through it
- B. N₂O uptake into pulmonary capillaries is dependent on temperature
- C. N₂O uptake into pulmonary capillaries is dependent on atmospheric pressure

D. None of the above

19. Calculate the alveolar oxygen tension (PAO₂) for a patient with a barometric pressure (P_{atm}) of 760 mmHg, a water vapour pressure (P_{H₂O}) of 47 mmHg, an inspired oxygen fraction (F_iO₂) of 0.21, and an arterial carbon dioxide tension (PACO₂) of 40 mmHg. Assume the respiratory quotient (R.Q) is 0.8.

- A. 80 mmHg
 - B. 90 mmHg
 - C. 100 mmHg
 - D. 70 mmHg
-

20. During a pulmonary function test, it is noted that the diffusion equilibrium for oxygen is reached at 1/4 rd of the transit time in the pulmonary capillary. Based on the information provided in the diagram, at what point during the red blood cell's transit through the pulmonary capillary is diffusion equilibrium for oxygen reached? {{caption_text}}



- A. 0.75 second
- B. 1 second
- C. 0.25 second

D. 0.1 second

21. Carbon dioxide (CO₂) diffuses more rapidly than oxygen (O₂) primarily due to which of the following properties?

- A. Higher molecular weight
 - B. Higher solubility in water
 - C. Smaller molecular size
 - D. Higher partial pressure in the alveoli
-

22. A 65-year-old male has had progressive difficulty breathing over the past year. He reports a persistent dry cough and fatigue. He has no history of smoking or exposure to occupational hazards. On examination, he has fine crackles at the lung bases and clubbing of the fingers. PFT reveals a restrictive pattern. An HRCT scan of the chest shows honeycombing and reticular opacities, predominantly in the lower lobes. This condition primarily affects which aspect of the respiratory membrane?

- A. Alveolar epithelium
 - B. Capillary endothelium
 - C. Interstitial space
 - D. Surfactant layer
-

23. A 65-year-old male with a 40-year history of heavy smoking presents with progressive shortness of breath and a chronic cough. On examination, he has a barrel-shaped chest and decreased breath sounds bilaterally. Pulmonary function tests show an FEV₁/FVC ratio of 0.55 and a significantly decreased DLCO. Which of the following is the most likely reason for the decreased DLCO in this patient?

- A. Loss of alveolar surface area
 - B. Increased thickness of the alveolar-capillary membrane
 - C. Decreased pulmonary blood flow
 - D. Hyperinflation of the lungs
-

24. Which of the following factors does NOT affect the rate of diffusion in the lung according to Fick's Law? Thickness of the alveolar-capillary membrane Diameter of the bronchiole Surface area of the alveolar-capillary membrane Partial pressure gradient of the gases Temperature Solubility of the Gas

- A. 1 and 2
 - B. 2 and 5
 - C. 4 and 5
 - D. 2 only
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	4
Question 3	1
Question 4	2
Question 5	3
Question 6	2
Question 7	1
Question 8	1
Question 9	1
Question 10	1
Question 11	3
Question 12	2
Question 13	3
Question 14	2
Question 15	3
Question 16	1
Question 17	4
Question 18	1
Question 19	3
Question 20	3
Question 21	2
Question 22	3

Question 23	1
Question 24	4

Solution for Question 1:

Correct answer: B) (Arterial Concentration–Venous Concentration) / Arterial Concentration

Explanation: The Oxygen Extraction Ratio (OER) is defined as the ratio of oxygen consumption (VO_2) to oxygen delivery (DO_2), expressed mathematically as:

Oxygen Extraction Ratio (OER) = oxygen consumption (VO_2) / oxygen delivery (DO_2)

$\text{VO}_2 = \text{Arterial Concentration} - \text{Venous Concentration}$

$\text{DO}_2 = \text{Arterial Concentration}$

In a healthy adult, OER is about 25%. During intense physical activity, OER can rise to 70-80%.

Factors affecting OER:

Factor	Key Influences	Impact on OER
Oxygen Delivery (DO_2)	Cardiac output, Hemoglobin, SpO_2	Decrease leads to increased OER
Oxygen Consumption (VO_2)	Metabolic demands (exercise, sepsis)	Increase without DO_2 increase elevates OER
Blood Flow	Vascular diseases, Shock states	Altered blood flow dynamics affect OER
Physiological Conditions	Hypoxia, Anemia, Hypercapnia	Hypoxia increases OER
Dialysis	Hemodynamic changes during dialysis	Higher OER linked to intradialytic hypotension
Age and Health Status	Age-related cardiovascular/respiratory changes	Reduced cardiac output and hemoglobin levels impact OER
Medications/Treatments	Vasodilators, Inotropes, Mechanical ventilation	Influence blood flow and oxygen delivery, affecting OER

Solution for Question 2:

Correct answer: D) All of the above

Explanation:

Higher hematocrit in venous blood compared to arterial blood can be attributed to several factors:

- Fluid Shifting: A small amount of fluid from arterial blood returns to circulation via the lymphatic system, reducing plasma volume in veins and increasing hematocrit.
- CO₂ and Bicarbonate: Venous blood has higher CO₂ levels, which result in increased bicarbonate formation within red blood cells. The chloride shift facilitates this process, contributing to red blood cell swelling and higher hematocrit.
- Venous blood has higher CO₂ levels, which result in increased bicarbonate formation within red blood cells.
- The chloride shift facilitates this process, contributing to red blood cell swelling and higher hematocrit.
- Venous blood has higher CO₂ levels, which result in increased bicarbonate formation within red blood cells.
- The chloride shift facilitates this process, contributing to red blood cell swelling and higher hematocrit.

Chloride shift (Hamburger phenomenon) vs Reverse chloride shift:

Feature	Chloride shift	Reverse chloride shift
Location	Tissues	Lungs
Process	CO ₂ enters RBCs, forming H ⁺ CO ₃ ⁻ → H ⁺ + HCO ₃ ⁻ . HCO ₃ ⁻ exits in exchange for Cl ⁻	CO ₂ exits RBCs. HCO ₃ ⁻ re-enters RBCs in exchange for Cl ⁻
Effect on RBCs	Chloride influx causes water influx, swelling RBCs	Chloride efflux balances HCO ₃ ⁻ re-entry, CO ₂ is released
Impact on haematocrit	Increases hematocrit due to RBC swelling	Primarily facilitates CO ₂ release, indirectly affecting hematocrit

Solution for Question 3:

Correct answer: A) Oxygenation promotes CO₂ release in the lungs.

Explanation:

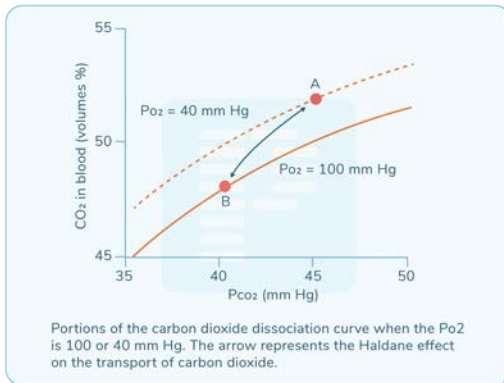


Table rendering failed

Solution for Question 4:

Correct answer: B) Bohr Effect

Explanation:

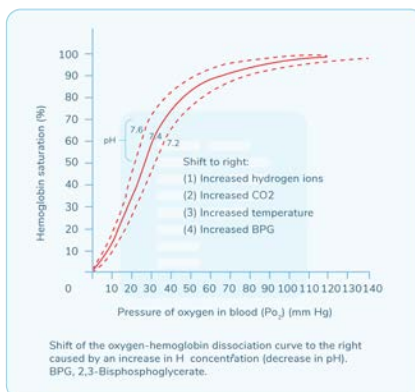


Table rendering failed

The Double Bohr Effect is a physiological phenomenon observed in the placenta. This effect involves two distinct Bohr effects occurring simultaneously—one on the fetal side and one on the maternal side .

Double bohr effect	Details
Location	Placenta where maternal and fetal circulation meet.
Fetal Blood Flow	Umbilical arteries carry deoxygenated blood with high CO ₂ from the fetus to placenta
CO ₂ Diffusion	CO ₂ diffuses from fetal to maternal blood
Effect on Fetal Blood	CO ₂ decreases, blood becomes alkaline Oxygen dissociation curve shifts left
Effect on Maternal Blood	CO ₂ increases, blood becomes acidic Oxygen dissociation curve shifts right
Clinical Significance	Enhances oxygen transfer from mother to fetus, improving fetal oxygenation

Solution for Question 5:

Correct answer: C) Bicarbonate ions (HCO₃⁻)

Explanation:

- Bicarbonate Ions (HCO₃⁻) (70%): CO₂ reacts with water to form carbonic acid, which dissociates into H⁺ and HCO₃⁻. HCO₃⁻ moves to plasma, and Cl⁻ enters red blood cells. This is the main CO₂ transport method contributing 70%.
- Dissolved CO₂ (7%)(Option A ruled out) : A small percentage of CO₂ dissolves directly into the plasma. About 2.7 ml/dl of CO₂ dissolves in plasma at 45 mmHg, making up 7% of CO₂ transport.
- Carbaminohemoglobin (23%)(Option B ruled out) : CO₂ binds to hemoglobin to form carbaminohemoglobin which is a reversible process, releasing CO₂ in the alveoli.
- Protein Bound (Option D ruled out): CO₂ binds to plasma proteins (mainly albumin). Similar to carbamino formation with hemoglobin. Minor contribution to overall CO₂ transport

Solution for Question 6:

Correct Answer: B) CPDA (Citrate Phosphate Dextrose Adenine)

Explanation: CPDA (Citrate Phosphate Dextrose Adenine) is best for blood storage for the following reasons:

Preservation Properties:

CPDA-1 is commonly used to store whole blood and red blood cells due to its superior preservation properties:

Benefits:

The combination of these components in CPDA-1 allows it to preserve blood for up to 35 days, which is longer than other anticoagulants and preservative solutions like ACD, EDTA, and CPD. This makes CPDA-1 an ideal solution for storing blood for medical use.

Solution for Question 7:

Correct Answer: A) Rectangular Hyperbolic

Explanation: Myoglobin-O₂ dissociation curve:

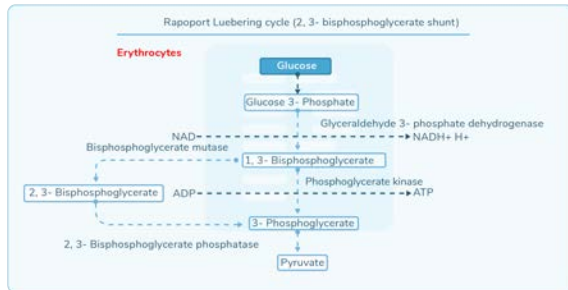
dissociation curve" data-author="na" data-hash="551" data-license="na" data-source="na" data-tags="January2025" height="600" src="https://image.prepladder.com/notes/4MwulaJsCTJ7LckFdZxW1735978703.png" width="600" />

- The oxygen dissociation curve for myoglobin is rectangular hyperbolic .
 - This is because, in the case of myoglobin, one molecule of haemoglobin binds with one molecule of oxygen with high affinity, and complete saturation will occur.
 - Unlike haemoglobin, which has a sigmoidal curve due to cooperative binding among its four subunits, myoglobin's curve does not exhibit cooperativity.
 - This means myoglobin can efficiently bind and release oxygen at varying partial pressures, making it well-suited for oxygen storage in muscle tissues.
-

Solution for Question 8:

Correct Answer: A) It reduces the affinity of haemoglobin for oxygen by producing 2,3-bisphosphoglycerate (2,3-BPG).

Explanations: Luebering- Rapoport Pathway



- The Luebering-Rapoport pathway produces 2,3-BPG, which binds to haemoglobin and decreases its affinity for oxygen, facilitating oxygen release to tissues. (Option B ruled out)
- Bisphosphoglycerate mutase enzyme initiates the Luebering-Rapoport pathway (Option D ruled out)
- The balance between bisphosphoglycerate mutase and bisphosphoglycerate phosphatase enzyme determines the amount of 2,3-BPG in blood.
- 2,3-BPG causes the right shifting of OHDC by stabilising the lower affinity of haemoglobin for oxygen.
- One molecule of haemoglobin binds with one molecule of 2,3-BPG which is a byproduct of the glycolytic pathway
- Chronic acidosis maintained for a long time inhibits the bisphosphoglycerate mutase enzyme, which decreases the blood's 2,3-BPG content, leading to left shifting of OHDC.
- The Luebering-Rapoport pathway diverts from the ATP-producing steps of glycolysis, resulting in a net decrease in ATP yield because it consumes ATP. (Option C ruled out)

Solution for Question 9:

Correct Answer: A) 20.1 mL O₂/g

Explanation:

Oxygen-carrying capacity = Hemoglobin concentration * Oxygen bound per gram of hemoglobin

Given the hemoglobin concentration of 15 g/dL and knowing that each gram of hemoglobin can bind 1.34 mL of oxygen, we can calculate:

Oxygen-carrying capacity = 15 g/dL * 1.34 mL O₂ /g hemoglobin

Oxygen-carrying capacity = 20.1 mL O₂ /g Hb

Solution for Question 10:

Correct Answer: A) Increased pH

Explanation: Given image in the question is a left shift in the Hb-O₂ dissociation curve.

Hb-O₂ dissociation curve shift

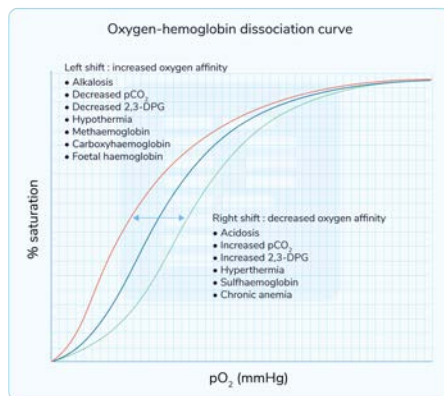


Table rendering failed

Solution for Question 11:

Correct Answer: C) 2 and 4

Explanation: Statements 1 and 3 are wrong as binding of one oxygen molecule to Hb increases the affinity of the remaining heme groups for oxygen. At low PO₂, small PO₂ changes lead to

small changes in oxygen saturation.

Cooperative Binding of Oxygen to Haemoglobin (Hb):

- Cooperative Process: The binding of one oxygen molecule to Hb increases the affinity of the remaining heme groups for oxygen, reflected in the sigmoid shape of the oxygen–haemoglobin dissociation curve.
- Conformational Change: Tense (T) State: Low affinity for oxygen. Relaxed (R) State: High affinity for oxygen. Oxygen binding to the first heme group shifts Hb from the T state to the R state, exposing more oxygen-binding sites.
- Tense (T) State: Low affinity for oxygen.
- Relaxed (R) State: High affinity for oxygen.
- Oxygen binding to the first heme group shifts Hb from the T state to the R state, exposing more oxygen-binding sites.
- As the partial pressure of oxygen (PO_2) increases, more oxygen molecules are available to bind to Hb, resulting in a higher percentage of oxygen saturation
- Tense (T) State: Low affinity for oxygen.
- Relaxed (R) State: High affinity for oxygen.
- Oxygen binding to the first heme group shifts Hb from the T state to the R state, exposing more oxygen-binding sites.

PO ₂ Levels	Oxygen Saturation	Affinity for Oxygen
Low	Slow increase	Low (T state)
Moderate	Steep increase	High (cooperative)
High	Plateau	Approaching full saturation

- Low PO_2 : Small PO_2 changes lead to small changes in oxygen saturation.
- Steep Curve (Moderate PO_2): Small PO_2 increases lead to large oxygen saturation increases.
- High PO_2 : Hb nearing full saturation; large PO_2 increases result in small saturation changes.

Solution for Question 12:

Correct answer: B) The plateau phase ranges from 10-60 mm Hg.

Explanation: The plateau phase ranges from 60-100 mm Hg.

Oxygen-Haemoglobin Disassociation Curve:

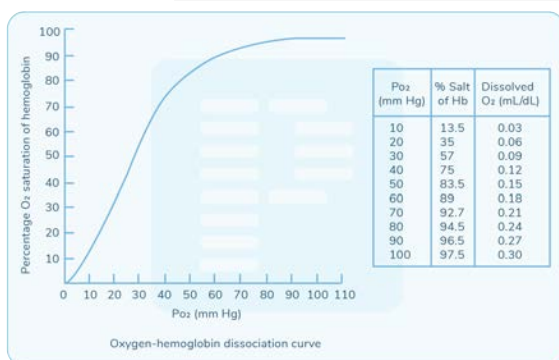
The oxygen-haemoglobin dissociation curve represents the relationship between the partial pressure of oxygen (PO_2) and the percentage of haemoglobin saturated with oxygen (SO_2).

Key Points:

- Sigmoid Shape: Reflects varying Hb affinity for oxygen at different PO_2 levels. (Option A ruled out)
- Steep Phase: Rapid increase in Hb saturation with PO_2 .
- Plateau Phase: Minimal change in Hb saturation despite large changes in PO_2 .

Phases:

	Steep phase	Plateau phase
Range	10-60 mm Hg	60-100 mm Hg
Characteristics	A rapid increase in Hb saturation. Hb ~90% saturated at PO_2 of 60 mm Hg.	Curve flattens. Small increase in Hb saturation despite PO_2 rise.
Significance	The steep phase (10-60 mm Hg PO_2) ensures efficient oxygen unloading, where small PO_2 drops cause significant O_2 release, crucial for active muscles and hypoxic conditions (Option C ruled out)	It provides a safety buffer. It maintains high Hb saturation (~90%) despite moderate PO_2 decreases. It ensures stable oxygen delivery under varying conditions.



P_{50} : P_{50} is the level of PO_2 at which 50% of haemoglobin is saturated with oxygen.

Solution for Question 13:

Correct Answer: C) Terminal bronchiole

Explanation:

Terminal bronchiole (Option C) is a component of the respiratory system's conducting zone and does not participate directly in gas exchange. It leads to the respiratory bronchioles but is not considered part of the respiratory unit.

respiratory unit" data-author="na" data-hash="547" data-license="na" data-source="na" data-tags="January2025" height="459" src="https://image.prepladder.com/notes/ZrpMZPrmti2JpEgYgCKY1735977823.png" width="500" />

The respiratory unit, also known as the acinus, is the functional unit of the lung where gas exchange occurs. It consists of several structures:

- Respiratory bronchiole (Option A) : These are the airways where gas exchange begins and leads to alveolar ducts.
 - Alveolar duct (Option B) : These passages lead from respiratory bronchioles to the alveolar sacs and provide the pathway for air to reach the alveoli.
 - Alveolus (Option D) : The alveoli are the tiny air sacs where gas exchange occurs. They are the final part of the respiratory unit and essential for respiration.
-

Solution for Question 14:

Correct answer: B) Bound to haemoglobin in red blood cells

Explanation: Oxygen Transport from Lungs to Body Tissues:

Hemoglobin Transport (97%):

- Binding: Oxygen binds reversibly to the iron atom in the heme portion of haemoglobin molecules within the red blood cells.
- Capacity: Each haemoglobin molecule can bind up to four oxygen molecules.
- Significance of Haemoglobin in Oxygen Transport: Oxygen Buffer: Haemoglobin acts as an oxygen buffer, ensuring that tissue PO₂ remains stable (between 15-40 mm Hg) despite fluctuations in environmental oxygen levels. Exercise Adaptation: During exercise, increased

levels of CO₂, higher temperatures, and lower pH (Bohr effect) cause a rightward shift in the dissociation curve, which enhances the release of oxygen to active muscles, ensuring adequate oxygen delivery.

- Oxygen Buffer: Haemoglobin acts as an oxygen buffer, ensuring that tissue PO₂ remains stable (between 15-40 mm Hg) despite fluctuations in environmental oxygen levels.
- Exercise Adaptation: During exercise, increased levels of CO₂, higher temperatures, and lower pH (Bohr effect) cause a rightward shift in the dissociation curve, which enhances the release of oxygen to active muscles, ensuring adequate oxygen delivery.
- Oxygen Buffer: Haemoglobin acts as an oxygen buffer, ensuring that tissue PO₂ remains stable (between 15-40 mm Hg) despite fluctuations in environmental oxygen levels.
- Exercise Adaptation: During exercise, increased levels of CO₂, higher temperatures, and lower pH (Bohr effect) cause a rightward shift in the dissociation curve, which enhances the release of oxygen to active muscles, ensuring adequate oxygen delivery.

Dissolved Oxygen Transport (3%):

- Mechanism: A small amount of oxygen is dissolved in the water component of plasma and blood cells.
- Amounts: At a PO₂ of 95 mm Hg, approximately 0.29 ml of oxygen is dissolved per 100 ml of blood. At a PO₂ of 40 mm Hg, approximately 0.12 ml of oxygen is dissolved.
- At a PO₂ of 95 mm Hg, approximately 0.29 ml of oxygen is dissolved per 100 ml of blood.
- At a PO₂ of 40 mm Hg, approximately 0.12 ml of oxygen is dissolved.
- While the amount of dissolved oxygen is minor under normal conditions, it becomes more significant at high alveolar PO₂ levels, such as during hyperbaric oxygen therapy.
- However, this mechanism alone cannot meet the body's oxygen demands.
- At a PO₂ of 95 mm Hg, approximately 0.29 ml of oxygen is dissolved per 100 ml of blood.
- At a PO₂ of 40 mm Hg, approximately 0.12 ml of oxygen is dissolved.

Solution for Question 15:

Correct answer: C) The solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid at a constant temperature.

Explanation: Henry's law states that the amount of gas that dissolves in a liquid is directly proportional to the partial pressure of that gas in contact with the liquid.

Partial Pressure = Concentration of Dissolved Gas / Solubility Coefficient

Solubility Coefficients for Respiratory Gases at Normal Body Temperature:

Gas	Solubility Coefficient
Oxygen	0.024
Carbon Dioxide	0.57
Carbon Monoxide	0.018
Nitrogen	0.012
Helium	0.008

- Carbon Dioxide (CO₂) has a higher solubility coefficient than oxygen, meaning more CO₂ can dissolve in water at the same partial pressure as oxygen.
- The difference in solubility coefficients plays a crucial role in transporting respiratory gases within the body.

Boyle's law states that the volume of a gas is inversely proportional to its pressure at constant temperature ($PV = nRT$) (Option A)

Graham's law states that a gas's diffusion rate is inversely proportional to molecular weight. (Option B)

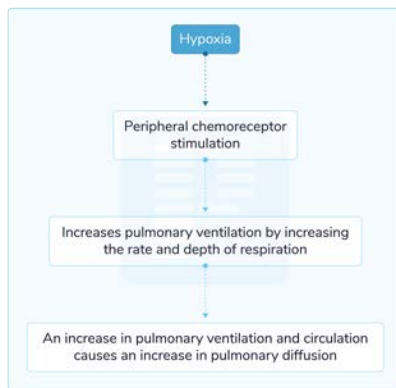
Charles's law states that at a constant pressure, the volume occupied by a fixed mass of gas is proportional to temperature ($V/T = \text{constant}$). (Option D)

Solution for Question 16:

Correct Answer: A) Diffusion capacity increases

Explanation: Pulmonary diffusion of gases increases in high altitudes.

Respiratory changes in high altitude:



Solution for Question 17:

Correct Answer: D) 50%

Explanation:

Shunt Fraction:

- The proportion of cardiac output bypassing the alveoli and not participating in gas exchange.
- Frequently calculated in critically ill patients to monitor pulmonary oxygenation effectiveness.

$$\text{Shunt fraction} = (\text{CcO}_2 - \text{CaO}_2) / (\text{CcO}_2 - \text{CvO}_2)$$

CcO₂ : Oxygen content of pulmonary capillary blood = 20 ml/dl

CaO₂ : Arterial oxygen content = 16 ml/dl

CvO₂ : Mixed venous oxygen content = 12 ml/dl

$$\text{Shunt fraction} = (20-16)/(20-12)$$

$$= 4/8 = 1/2$$

$$= 50\%$$

Solution for Question 18:

Correct Answer: A) N₂O uptake into pulmonary capillaries is dependent on the amount of blood flowing through it

Explanation: N₂O is a perfusion-limited gas. Its uptake is limited by the amount of blood flowing through the pulmonary capillaries.

Gas Type	Explanation	Example
Diffusion-Limited Gas	The diffusion process limits the transfer Diffusing capacity for CO (DLCO) is used to measure the diffusing capacity of the lungs	CO
Perfusion-Limited Gas	The amount of gas taken up is dependent on blood flow (perfusion) through pulmonary capillaries	N ₂ O
Intermediate Case	Uptake is primarily perfusion-limited under normal conditions.	O ₂

Solution for Question 19:

Correct Answer: C) 100 mmHg

Explanation:

Alveolar gas equation: $PAO_2 = (P_{atm} - P_{H_2O}) \times FiO_2 - PaCO_2 / RQ$

$$= (760 - 47) \times 0.21 - (40 / 0.8)$$

$$= 713 \times 0.21 - 50$$

$$= 99.73 \approx 100 \text{ mm Hg}$$

Solution for Question 20:

Correct Answer: C) 0.25 second

Explanation:

Diffusion equilibrium for oxygen and carbon dioxide will be reached at 0.25 seconds.

- Transit time: The total time the RBC takes to travel the whole pulmonary capillary = 0.75 seconds.
 - Air inspiration is diffused from the alveolus to the pulmonary capillary.
 - Diffusion equilibrium of oxygen and carbon dioxide will be reached when RBC has reached 1/3rd of transit time = $\frac{1}{3} \times 0.75 = 0.25$ seconds.
 - No diffusion occurs in the remaining 2/3rd of the transit time.
-

Solution for Question 21:

Correct Answer: B) Higher solubility in water

Explanation: CO₂ is about 20 times more soluble in water (and thus in blood) than O₂. This higher solubility allows CO₂ to diffuse more rapidly across the alveolar and capillary membranes.

Higher molecular weight (Option A) : Although CO₂ has a higher molecular weight (44 g/mol) compared to O₂ (32 g/mol), diffusion rate is influenced more by solubility and other factors.

Smaller molecular size (Option C): O₂ is smaller in molecular size than CO₂. However, molecular size is not the primary factor determining diffusion rates in this context.

Higher partial pressure in the alveoli (Option D) : While the partial pressures of gases affect their diffusion, O₂ generally has a higher partial pressure gradient in the alveoli than CO₂.

Solution for Question 22:

Correct Answer: C) Interstitial space

Explanation:

- The clinical vignette suggests pulmonary fibrosis (restrictive lung disease). In pulmonary fibrosis, the interstitial space of the lungs becomes thickened and scarred due to abnormal fibrotic processes. This disrupts the normal lung architecture, impairing gas exchange and restrictive lung disease.
- The thickening of the interstitium increases the diffusion distance across the respiratory membrane, hindering the transfer of oxygen and carbon dioxide. Consequently, this increased diffusion distance results in hypoxemia and contributes to the patient's difficulty in breathing.

Alveolar epithelium (Option A) can be affected in pulmonary fibrosis, the primary pathological changes occur in the interstitial space, leading to thickening and scarring.

Capillary endothelium (Option B) may be secondarily involved, but the primary issue in pulmonary fibrosis is the interstitial space, where fibrosis and scarring occur.

Surfactant layer (Option D) reduces surface tension within the alveoli, and is not the primary site affected by pulmonary fibrosis. The main problem lies in the interstitial space.

Solution for Question 23:

Correct Answer: A) Loss of alveolar surface area.

Explanation: In emphysema, the destruction of alveolar walls leads to a significant reduction in the surface area available for gas exchange. This is the primary reason for the decreased DLCO in patients with emphysema, as the reduced alveolar surface area limits the ability of gases, such as carbon monoxide, to diffuse across the alveolar-capillary membrane.

DL CO (Diffusing Capacity of the Lungs for Carbon Monoxide)

- It measures carbon monoxide (CO) transfer from the alveolar gas to the pulmonary capillary blood.
- Carbon monoxide is used because it has a high affinity for hemoglobin, and its uptake is limited primarily by the rate of diffusion rather than by perfusion.

Procedure	Inhale gas mixture (CO and helium) Hold breath for ~10 seconds Exhale and measure CO concentration
-----------	--

Factors Affecting DL CO	Lung Volume: Lower volumes decrease DL CO (e.g., restrictive lung diseases) Hemoglobin Level: Anemia lowers DL CO ; polycythemia increases DL CO Carboxyhemoglobin Levels: Elevated in smokers, reducing DL CO Alveolar Surface Area: Reduced in emphysema, decreasing DL CO Pulmonary Blood Flow: Reduced flow (e.g., pulmonary hypertension) decreases DL CO
Clinical Interpretation	Reduced DL CO : Interstitial lung disease, emphysema, pulmonary fibrosis, pulmonary vascular disease, anemia Increased DL CO : Polycythemia, asthma, left-to-right cardiac shunts
Significance in Clinical Practice	Differentiate obstructive vs restrictive lung diseases Assess disease extent in idiopathic pulmonary fibrosis, sarcoidosis

Increased thickness of the alveolar-capillary membrane can decrease DLCO, it is more characteristic of interstitial lung diseases such as pulmonary fibrosis, where fibrosis thickens the membrane. (Option B)

Decreased pulmonary blood flow can contribute to a reduced DLCO, it is not the main mechanism in emphysema. In emphysema, the primary issue is the loss of alveolar structures. (Option C)

Hyperinflation of the lungs is a common feature of emphysema, it does not directly cause a decreased DLCO. Hyperinflation results from the loss of elastic recoil and air trapping, contributing to airflow obstruction but not directly to the decrease in diffusing capacity. (Option D)

Solution for Question 24:

Correct Answer: D) 2 only

Explanation:

Fick's Law of diffusion states that the rate of gas diffusion is directly proportional to the surface area of the membrane, the partial pressure gradient, and the gas's diffusion coefficient and inversely proportional to the membrane's thickness.

Diffusion Capacity (D_L) : It measures how effectively gases like oxygen (O_2) and carbon dioxide (CO_2) move from the alveoli into the blood.

Diffusion Capacity Formula:

Fick's Law describes the rate of diffusion of a gas across a membrane. For gases diffusing across the alveolar-capillary membrane in the lungs, the law can be expressed as

$$D_L = \frac{\Delta P \times A \times K}{d}$$

ΔP : Partial pressure.

A: Total surface area.

K: Constant (Diffusion Constant of gases)

d: Diffusion distance or the thickness of the membrane.

D_L : Total diffusion capacity of the lung.

Factors Affecting the Rate of Diffusion:

Surface Area of the Membrane	The greater the surface area available for diffusion, the higher the rate of gas exchange. In diseases like emphysema, where alveolar surface area is reduced, diffusion capacity decreases.
Concentration Gradient	A steeper gradient (difference in partial pressures between the alveoli and the blood) increases the diffusion rate. For example, if the alveolar partial pressure of oxygen is high compared to the blood, oxygen will diffuse more rapidly.
Thickness of the Membrane	The thinner the membrane (alveolar-capillary membrane), the faster the diffusion rate. Conditions like pulmonary fibrosis increase membrane thickness and reduce diffusion capacity.
Diffusion Coefficient of the Gas	Different gases diffuse at different rates. For example, CO_2 diffuses more readily than O_2 due to its higher diffusion coefficient.
Temperature and Pressure	Higher temperatures increase the diffusion rate, and barometric pressure changes can also influence diffusion rates.

Solubility of the Gas	Gases more soluble in blood (like CO ₂) will diffuse more readily than gases with lower solubility (like O ₂).
-----------------------	--

Regulation of respiration

1. The true statement regarding central chemoreceptors is?

- A. Detects H^+ in brain interstitial fluid
 - B. Gets stimulated to decrease respiration
 - C. Located above the surface of the ventrolateral medulla
 - D. All are true
-

2. All of the following are true regarding the J receptor reflex except?

- A. They are located near the pulmonary capillaries.
 - B. Activation of J receptors causes rapid breathing.
 - C. They are stimulated by hyperinflation of the lungs.
 - D. They are exclusively involved in regulating airway smooth muscle tone.
-

3. A 45-year-old patient with a history of chronic obstructive pulmonary disease (COPD) is undergoing a respiratory assessment in the clinic. The patient is asked to take a deep breath and hold it for an extended period. During this maneuver, the clinician observes that the patient's inspiratory phase extends rather than transitioning to expiration. Which reflex is most likely responsible for this observation?

- A. Hering-Breuer reflex
- B. Head's paradoxical reflex
- C. Cough reflex
- D. Sneeze reflex

4. Which of the following is the receptor for Herring-Breuer reflexes?

- A. Chemoreceptors
 - B. Baroreceptors
 - C. Pulmonary stretch receptors
 - D. Proprioceptors
-

5. What happens if a person inhales a gas mixture containing 5% carbon dioxide (CO₂) during voluntary hyperventilation?

- A. The person will experience a significant drop in arterial PCO₂, leading to apnea.
 - B. The person will maintain a normal arterial PCO₂ and prevent respiratory alkalosis.
 - C. The person will experience increased alveolar and arterial PCO₂, preventing apnea.
 - D. The person will experience decreased ventilation due to the stimulation of medullary chemoreceptors.
-

6. In which of the following conditions does the sensitivity of chemoreceptors decrease?

- A. Emphysema
 - B. Asthma
 - C. Acute bronchitis
 - D. Pulmonary fibrosis
-

7. Arrange the following in the correct sequence regarding the mechanism of action of a glomus cell: Depolarisation Calcium influx Reduced K⁺ conductance Increased ventilation Dopamine release

- A. 1, 3, 2, 5, 4
 - B. 3, 1, 2, 5, 4
 - C. 2, 1, 3, 4, 5
 - D. 4, 3, 1, 2, 5
-

8. Decreased arterial blood pressure leads to?

- A. Increased ventilation
 - B. Decreased ventilation
 - C. No effect at all
 - D. Depends on the temperature
-

9. A 65-year-old patient, Mr Kumar, is admitted to the hospital with chronic obstructive pulmonary disease (COPD) exacerbation. He appears short of breath and has a low oxygen saturation of 88% on room air. Arterial blood gas (ABG) analysis reveals: pH: 7.32 PaCO₂: 55 mm Hg PaO₂: 55 mm Hg HCO₃⁻: 24 mmol/L What will most likely activate Mr. Kumar's peripheral chemoreceptors?

- A. pH: 7.32
 - B. PaCO₂ : 55 mm Hg
 - C. PaO₂ : 55 mm Hg
 - D. All of the above
-

10. Where is the pneumotaxic centre, which regulates respiratory rate, located in the brain?

- A. Medulla oblongata
 - B. Pons
 - C. Midbrain
 - D. Cerebellum
-

11. The apneustic centre is inhibited by?

- A. Dorsal respiratory group
 - B. Ventral respiratory group
 - C. Central pattern generator
 - D. Pneumotaxic center
-

12. Which group of neurons is involved in controlling forceful expiration?

- A. Dorsal respiratory group of neurons
 - B. Ventral respiratory group of neurons
 - C. Both A and B
 - D. None of the above
-

13. What is the structure responsible for the voluntary control of respiration?

- A. Cerebral cortex
- B. Medulla
- C. Pons

D. Cerebellum

14. A 45-year-old male was involved in a severe motor vehicle accident resulting in multiple injuries, including a traumatic brain injury. Imaging studies revealed a complete transection of the brain stem between the pons and medulla. Given this anatomical disruption, what would be the most likely effect on the patient's respiratory activity?

- A. Cessation of spontaneous breathing
 - B. Hyperventilation
 - C. Apneustic breathing
 - D. Cheyne-Stokes respiration
-

15. Ondine's curse, or Ondine's syndrome, is a rare disorder of the respiratory control centre. Which of the following is true about Ondine's curse?

- A. Loss of voluntary, but intact autonomic control
 - B. Loss of autonomic, but intact voluntary control
 - C. Loss of both autonomic and voluntary control
 - D. Loss of reflective control only
-

16. Which of the following is considered the primary pacemaker for respiration in the brainstem?

- A. Rostral Ventral Respiratory Group (VRG)
- B. Botzinger Complex
- C. Pre-Bötzinger Complex
- D. Caudal Ventral Respiratory Group (VRG)

17. Which of the following statements accurately describes the functions of the dorsal respiratory group (DRG) of neurons in the medulla? The DRG primarily regulates the rhythmic pattern of breathing. The DRG receives sensory input from peripheral chemoreceptors and baroreceptors. The DRG mainly influences the expiratory phase of respiration. The DRG integrates information from the vagus nerve to adjust breathing depth. The DRG controls voluntary breathing movements during speech.

- A. Statements 1 and 2 are correct.
 - B. Statements 2 and 4 are correct.
 - C. Statements 3 and 5 are correct.
 - D. Statements 1 and 4 are correct.
-

18. Which of the following controls the depth of breathing?

- A. Pneumotaxic center
 - B. Apneustic center
 - C. Dorsal respiratory group
 - D. Ventral respiratory group
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	4
Question 3	2
Question 4	3
Question 5	3
Question 6	1
Question 7	2
Question 8	1
Question 9	4
Question 10	2
Question 11	4
Question 12	2
Question 13	1
Question 14	1
Question 15	2
Question 16	3
Question 17	2
Question 18	2

Solution for Question 1:

Correct Answer: A) Detects H⁺ in brain interstitial fluid

Explanation:

The central chemoreceptors detect H⁺ in brain interstitial fluid.

Central chemoreceptors:

	Details
Location	Bilaterally below (Option C ruled out) the surface of the ventrolateral medulla, immediately caudal to the pontomedullary junction.
Synonyms	Medullary chemoreceptors. Chemosensitive areas.
Mechanism of Stimulation	Arterial PCO ₂ increases ↓ H ⁺ ions in the brain interstitial fluid increases. ↓ Central chemoreceptors get stimulated to increase respiration (Option B ruled out)

Solution for Question 2:

Correct Answer: D) They are exclusively involved in regulating airway smooth muscle tone

Explanation:

J-receptor reflex:

- The pulmonary C fibres are located adjacent to the alveoli and are accessible from the pulmonary capillary circulation (Option A ruled out)
- They are also called juxtapulmonary capillary or J receptors (described by Indian scientist Dr A. S. Paintal).
- The pulmonary C fibres are more sensitive to mechanical processes like edema, congestion, embolism, and intravenous or intracardiac administration of chemicals like capsaicin P.
- Effect: On stimulation, they cause rapid shallow breathing, bronchoconstriction, increased airway secretion, and cardiovascular depression. (Option B ruled out)
- Acute stimulation results in apnea and a marked hypotension.

- Skeletal muscle tone is also abruptly reduced.

Solution for Question 3:

Correct Answer: B) Head's paradoxical reflex

Explanation:

In this scenario, the observation of prolonged inspiration extending rather than transitioning to expiration suggests the involvement of Head's paradoxical reflex.

Head's paradoxical reflex:

Aspect	Details
Receptors	Rapidly adapting pulmonary stretch receptors (RARs)
Response	Brief burst of action potentials in response to lung inflation
Feedback Mechanism	Enhances inspiration, especially during vagus nerve recovery
Clinical Significance	Initiates first breath in newborns , generates augmented breaths, prevents lung collapse
Implications	Relevant for respiratory physiology, neonatal resuscitation, and respiratory control mechanisms

Cough reflex (Option C) is triggered by irritation in the airways. It involves a sudden, forceful expulsion of air from the lungs to clear the airways of irritants or obstructions.

Sneeze reflex (Option D) rapidly expulses air from the lungs to clear nasal passages. It is triggered by irritation in the nasal passages and involves a deep inspiration followed by a forceful exhalation.

Solution for Question 4:

Correct Answer: C) Pulmonary stretch receptors

Explanation:

- Hering and Breuer in 1868 found that lung inflation decreases tidal volume and increases respiratory frequency.
- There are two Hering- Breuer reflexes: Hering-Breuer inflation & deflation reflexes.

Hering-Breuer inflation reflex:

- Steady increase in lung volume (lung inflation) results in an increase in the duration of expiration.
- Receptors: Slowly adapting
- Marked inflation of the lungs with sustained pressure may even abruptly terminate inspiration in progress, prolonging expiration.

Hering-Breuer deflation reflex:

- Marked deflation of the lung results in a decrease in expiration duration.
- As described above, the receptors are slowly adapting stretch receptors in the lung parenchyma and airways, and the vagus nerve provides the afferent and efferent pathways.

Significance of Hering-Breuer Reflex:

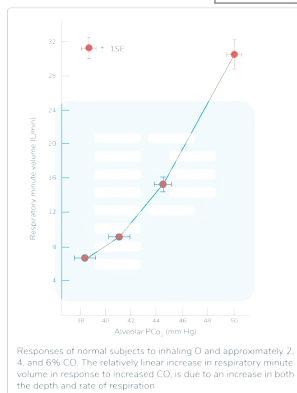
- It protects the lung from overinflation while maintaining normal alveolar ventilation.
- It is more useful in infants than in adults.
- It controls tidal volume during eupnea (normal breathing) in infants.
- This reflex does not operate in adults unless the tidal volume is much greater than the normal inspiration.
- Especially in the awake state in adults, this reflex is overwhelmed by more prominent central control.

Solution for Question 5:

Correct Answer: C) The person will experience increased alveolar and arterial PCO₂, preventing apnea.

Explanation:

Physiological Change	Description
Increase in Alveolar CO_2	Inhalation of 5% CO_2 increases the alveoli's partial pressure of CO_2 (PCO_2). Elevated alveolar PCO_2 leads to an increase in arterial PCO_2 .
Stimulation of Ventilation	A rise in arterial PCO_2 stimulates the medullary chemoreceptors. This results in an increase in ventilation (hyperventilation continues) to eliminate excess CO_2 .
Avoidance of Apnea	Without additional CO_2 , hyperventilation-induced hypocapnia can lead to apnea (temporary cessation of breathing). Inhaling 5% CO_2 prevents a significant drop in arterial PCO_2 , avoiding apnea.
Maintaining Blood pH	Inhaled CO_2 helps maintain blood pH closer to normal by preventing respiratory alkalosis. This balance is crucial to avoid adverse effects on cellular and enzymatic functions.
New Equilibrium	A new equilibrium is established with slightly elevated alveolar and arterial PCO_2 . The person continues to hyperventilate as long as CO_2 is inhaled.



Solution for Question 6:

Correct Answer: A) Emphysema

Explanation:

Emphysema impairs chemoreceptor function, affecting respiratory drive, increasing dyspnea, and causing potential complications.

Impaired Chemoreceptor Function:

- ↓ Detection of gas exchange abnormalities
- ↓ Response to CO₂ levels

Consequence	Description	Complications
Dyspnea	Difficulty breathing due to impaired chemoreceptor function	Occurs even with non-critically abnormal blood gas levels
Acidosis and Mental Status Changes	Respiratory acidosis from inadequate CO ₂ expulsion	Mental status changes, need for ventilatory support
Ventilation-Perfusion Mismatch	Mismatch due to structural lung changes reduced compensatory mechanism	Complicates breathing regulation

Asthma (Option B) generally increases chemoreceptor sensitivity due to intermittent hypoxia and hypercapnia.

Acute bronchitis (Option C) is often associated with increased respiratory irritability and sensitivity.

Pulmonary fibrosis (Option D) can lead to hypoxia and increased sensitivity of chemoreceptors, though the overall impact on sensitivity can vary.

Solution for Question 7:

Correct Answer: B) 3, 1, 2, 5, 4

Explanation:

- Under hypoxic conditions, the reduced oxygen levels lead to decreased conductance of potassium (K⁺) channels in the glomus cell membrane.
- This is due to the hypoxia causing the K⁺ channels to close or become less active.
- The reduced K⁺ conductance decreases K⁺ efflux, resulting in the depolarisation of the glomus cell membrane.
- This change in membrane potential makes the inside of the cell less negative.

- Depolarisation opens voltage-gated L-type calcium (Ca^{2+}) channels, allowing calcium ions to flow into the cell.
- The influx of Ca^{2+} further depolarises the cell and initiates intracellular signalling.
- The increased intracellular Ca^{2+} triggers the release of neurotransmitters, such as dopamine, from the glomus cell.
- Dopamine is released into the synaptic cleft of the afferent nerve endings.
- Dopamine acts on the central nervous system by binding to D2 receptors in the medulla oblongata.
- This signalling increases respiratory drive and ventilation (breathing rate and depth).

Solution for Question 8:

Correct Answer: A) Increased ventilation

Explanation:

Decreased arterial blood pressure leads to increased ventilation by central and peripheral chemoreceptors.

	Peripheral Chemoreceptors	Central Chemoreceptors
P a t h w a y	Decreased arterial pressure → decreased perfusion of peripheral tissues, including the carotid and aortic bodies → hypoxia (decreased oxygen levels) → stimulation of peripheral chemoreceptors.	Decreased arterial pressure → reduced removal of carbon dioxide (CO_2) → CO_2 retention → increased arterial P_{CO_2} → increased concentration of hydrogen ions (H^+) in cerebrospinal fluid (CSF) due to the conversion of CO_2 to HCO_3^- and H^+ → stimulation of central chemoreceptors in the medulla.
R e s u l t	The peripheral chemoreceptors send signals to the respiratory centres in the brainstem to increase the rate and depth of breathing to improve oxygen delivery and restore homeostasis.	The central chemoreceptors also stimulate the respiratory centres to enhance ventilation, thereby increasing the elimination of CO_2 and improving blood gas levels.

This dual mechanism ensures that the body responds promptly to a drop in arterial blood pressure by increasing ventilation, thereby improving oxygenation and CO_2 clearance to maintain physiological balance.

Decreased ventilation (Option B) – Incorrect, as hypoxia and hypotension trigger compensatory mechanisms to increase, not decrease, ventilation.

No effect at all (Option C) – Incorrect, as the body responds to hypotension with increased respiratory drive via chemoreceptor activation.

Depends on the temperature (Option D) – Incorrect, as ventilation regulation is primarily controlled by blood gases and pH, not temperature.

Solution for Question 9:

Correct Answer: D) All of the above

Explanation:

Peripheral chemoreceptors respond to hypoxia ($\text{PaO}_2 < 60 \text{ mm Hg}$), hypercapnia ($\text{PaCO}_2 > 45 \text{ mm Hg}$), and acidosis ($\text{pH} < 7.35$).

Peripheral chemoreceptors:

Topic	Details
Location	Carotid bodies: At bifurcations of common carotid arteries Aortic bodies: Along the ascending aorta and arch of the aorta
Function	Send signals to the dorsal respiratory group (DRG) in the medulla to increase ventilation
Carotid Bodies	Type I (glomus) cells and Type II (sustentacular) cells
Type I Cells (Glomus Cells)	Release dopamine in response to hypoxia, triggering an action potential in the carotid sinus nerve
Aortic Bodies	Contain glomus cells similar to carotid bodies but are less effective in regulating respiration
Responses	Hypoxia: Increased firing rate of afferent nerves when $\text{PaO}_2 < 60 \text{ mm Hg}$ Hypercapnia: Linear increase in chemoreceptor discharge rate with increased PaCO_2 Acidosis: Stimulates ventilation independently of hypoxia and hypercapnia

Solution for Question 10:

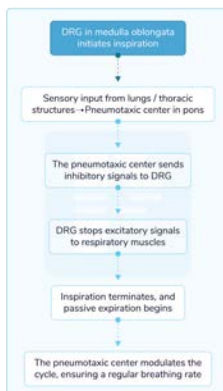
Correct Answer: B) Pons

Explanation:

The pneumotaxic centre is part of the brain's respiratory control system and is located in the upper part of the pons. It maintains the rate of breathing.

The primary function of the pneumotaxic centre is to limit the duration of inspiration, thereby controlling the respiratory rate. Sending inhibitory signals to the inspiratory centre in the medulla oblongata helps prevent the lungs from overinflating.

Mechanism:



Solution for Question 11:

Correct Answer: D) Pneumotaxic center

Explanation:

Apneustic center:

- Located in the caudal pons

- Stimulation results in prolonged inspiratory effort interrupted by brief expiration (apneusis)
 - Strongly activates the medullary inspiratory centre
 - Tonically inhibited by the pneumotaxic centre. It ensures that the inspiratory drive originating in the apneustic centre is regulated and checked.
-

Solution for Question 12:

Correct Answer: B) Ventral respiratory group of neurons

Explanation: The Ventral Respiratory Group of neurons (VRG) in the medulla controls forceful expiration.

Ventral Respiratory Group of neurons (VRG):

- When the need for increased pulmonary ventilation arises, signals spill over into the VRG neurons.
 - It contributes to the extra respiratory drive which is required.
 - Electrical stimulation of some neurons causes inspiration, while stimulation of others causes expiration.
 - It is crucial for providing powerful expiratory signals to the abdominal muscles during forceful expiration.
 - This is important during activities requiring high levels of pulmonary ventilation (heavy exercise).
-

Solution for Question 13:

Correct Answer: A) Cerebral cortex

Explanation:

- The cerebral cortex is the outermost layer of the brain, involved in various higher-order brain functions such as sensation, voluntary muscle movement, thought, reasoning, and memory.

- The primary motor cortex, located in the precentral gyrus of the frontal lobe, plays a key role in voluntary movements, including the voluntary control of breathing.
- The voluntary control of respiration originates in the motor cortex and involves the corticospinal tract. This pathway directly sends signals from the cerebral cortex to the spinal cord, bypassing the brainstem respiratory centres.
- Through this pathway, individuals can consciously control their breathing, such as holding their breath, taking a deep breath, or altering the breathing pattern during speech and singing.
- While the brainstem (specifically the medulla and pons) is primarily responsible for the automatic, rhythmic breathing pattern, the cerebral cortex can override these automatic signals when exercising voluntary control.
- This voluntary control is critical during activities requiring conscious modulation of breathing, such as talking, singing, playing wind instruments, or deliberate breath-holding.

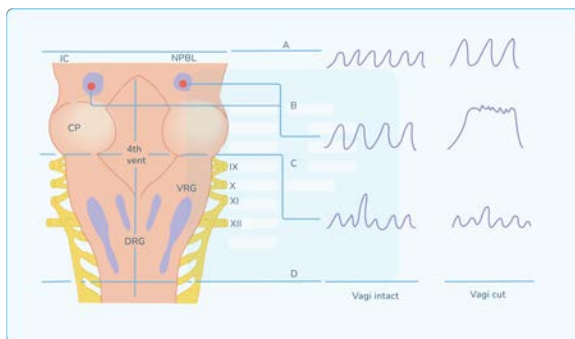
Solution for Question 14:

Correct Answer: A) Spontaneous but irregular respiration

Explanation:

Effects of various lesions and brainstem transections:

	Level of transection	Vagus intact	Vagus cut
A	Complete transection above the pons	Regular breathing continues	Regular breathing continues; the depth of inspiration increases.
B	Mid-pontine level section	Apneusis	
C	Transection midway between the pons and the medulla	Irregular and rhythmic respiration	
D	Complete transection below the medulla	Cessation of spontaneous respiration	



Hyperventilation (Option B) is typically associated with lesions in the pons, particularly affecting the pneumotaxic centre, which helps regulate the rate and pattern of breathing.

Apneustic breathing (Option C) is characterised by prolonged inspiratory phases, is typically caused by damage to the apneustic centre in the lower pons. Since the transection is below the pons, it is unlikely to result in apneustic breathing.

Cheyne-Stokes respiration (Option D) involves periodic breathing with cycles of increasing and decreasing tidal volumes followed by periods of apnea. This pattern is usually associated with bilateral cortical or diencephalic damage, congestive heart failure, or high altitude.

Solution for Question 15:

Correct Answer: B) Loss of autonomic, but intact voluntary control

Explanation:

Ondine's curse, also known as congenital central hypoventilation syndrome (CCHS), involves a failure of the autonomic regulation of breathing, leading to episodes of hypoventilation or hypoxia and requiring the individual to remember to breathe consciously.

Cause: Failure in autonomic regulation of breathing, often congenital or acquired due to brainstem damage.

Symptoms: Difficulty breathing during sleep, leading to sleep apnea or hypoventilation. Symptoms may not be present while awake.

Diagnosis: Sleep studies (polysomnography) to monitor and identify disruptions in breathing during sleep.

Treatment: Positive pressure ventilation devices like CPAP or BiPAP to maintain adequate ventilation during sleep.

Solution for Question 16:

Correct Answer: C) Pre-Bötzinger Complex

Explanation:

The Ventral Respiratory Group consist of 4 clear zones.

Zones	Location	Function
Pre-Bötzinger Complex	Ventrolateral medulla	A primary pacemaker for respiration that generates the basic rhythmic pattern of breathing.
Bötzinger Complex (Option B ruled out)	Medulla oblongata, dorsal	Modulates respiratory rhythm; involved in expiratory phase regulation and inhibition of inspiration.
Rostral Ventral Respiratory Group (VRG) (Option A ruled out)	Medulla oblongata, rostral	It contributes to rhythm generation and integrates respiratory signals; it is involved in the inspiratory and expiratory phases.
Caudal Ventral Respiratory Group (VRG) (Option D ruled out)	Medulla oblongata, caudal	Primarily involved in controlling the expiratory phase of respiration.

Solution for Question 17:

Correct Answer: B) Statements 2 and 4 are correct

Explanation:

Feature	Dorsal Respiratory Group (DRG)	Ventral Respiratory Group (VRG)
---------	--------------------------------	---------------------------------

Location	Medulla oblongata, primarily in the nucleus tractus solitarius (NTS)	Medulla oblongata, ventrolateral region, including nucleus ambiguus and nucleus retroambiguus
Primary Function	Controls basic rhythmic breathing; initiates and regulates respiration	Modulates rhythm and force of breathing; involved in complex breathing patterns
Breathing Pattern	Generates and maintains rhythmic breathing; primarily active during inspiration	Modifies breathing patterns; active during both inspiration and expiration
Connection with Other Centers	Integrates sensory input from peripheral chemoreceptors and baroreceptors to adjust breathing	Coordinates with pneumotaxic and apneustic centres; adjust breathing for varying demands
Sensory Input	Receives input from lung stretch receptors and chemoreceptors detecting CO ₂ and O ₂ levels and integrates information from the vagus nerve to adjust breathing depth	Receives input from higher brain centres
Effect on Breathing	Establishes and regulates basic breathing rhythm; adjusts rate and depth	Fine-tune breathing patterns and force; adjust breathing during different activities

Statement 1 (Incorrect): The DRG primarily regulates the rhythmic breathing.

- This function is primarily associated with the ventral respiratory group (VRG).

Statement 3 (Incorrect): The DRG mainly influences the expiratory phase of respiration.

- The DRG is more involved in the inspiratory than the expiratory phase.

Statement 5 (incorrect): The DRG controls voluntary breathing movements during speech.

- Voluntary breathing movements, such as those involved in speech, are primarily controlled by higher brain centres.

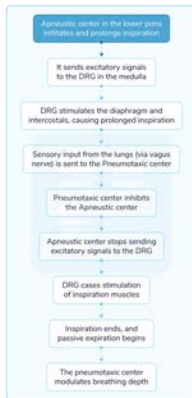
Solution for Question 18:

Correct Answer: B) Apneustic center

Explanation:

The apneustic center, located in the lower part of the pons, sends signals to the dorsal respiratory group to promote deep, prolonged inspirations, thus influencing the depth of each breath.

Mechanism:



Hypoxia, Altitude, and Deep sea pressures

1. A 32 year old woman presents with severe headache and dizziness after using a faulty gas heater. Blood tests reveal normal arterial PO₂ but significantly elevated carboxyhemoglobin levels, indicating impaired oxygen transport. Which of the following types of hypoxia does not stimulate chemoreceptors?

- A. Hypoxic hypoxia
 - B. Anemic hypoxia
 - C. Stagnant hypoxia
 - D. Histotoxic hypoxia
-

2. A 35 year old female hiker ascends to 5,000m in 2 days. Within 24 hours, she develops severe headaches, nausea, coordination difficulties, confusion, and occasional hallucinations. On examination, she is ataxic, with a GCS score of 10 and signs of papilledema. What is the next best step in managing this patient?

- A. Immediate descent to a lower altitude and start dexamethasone
 - B. Start oxygen therapy and monitor for improvement
 - C. Provide hydration, antiemetics, and continue the ascent
 - D. Administer acetazolamide and maintain current altitude for the next 24 hours at least.
-

3. A 30 year old hiker arrives at a clinic complaining of severe headache, nausea, and fatigue 12 hours after a rapid ascent to 3,500 meters. Physical examination reveals signs of mild confusion. Based on these symptoms, what is the most likely underlying mechanism contributing to his condition?

- A. Increased cerebral blood flow leading to cerebral edema

- B. Decreased capillary pressure in the brain due to hypoxia
 - C. Increased oxygen levels in the brain causing vasoconstriction
 - D. Decreased blood-brain barrier permeability due to hypoxia
-

4. A 55 year old man who recently relocated to a high-altitude region presents with worsening breathlessness and signs of right-sided heart failure. His examination reveals elevated pulmonary arterial pressure and symptoms of pulmonary edema. Given these findings, which vascular response to hypoxia is most likely responsible for his condition?

- A. Systemic vasoconstriction causing increased blood pressure
 - B. Pulmonary vasoconstriction leading to elevated pulmonary arterial pressure
 - C. Systemic vasodilation resulting in decreased oxygen delivery
 - D. Decreased pulmonary blood flow due to systemic vasoconstriction
-

5. Which adaptation is associated with long-term high-altitude exposure and helps to enhance oxygen delivery?

- A. Elevated systemic blood pressure and reduced capillary surface area
 - B. Increased production of erythropoietin and enhanced hemoglobin levels
 - C. Decreased number of red blood cells and lower 2,3-DPG concentration
 - D. Increased pulmonary blood flow and decreased oxygen affinity of hemoglobin
-

6. What primary physiological change sustains increased ventilation over several days of high-altitude exposure?

- A. Initial rapid increase in breathing rate.
- B. Compensatory renal excretion of bicarbonate.

- C. Decrease in cerebrospinal fluid pH.
 - D. Decrease in respiratory rate over time.
-

7. When a person ascends to high altitudes, their body undergoes several adaptive changes to cope with lower oxygen levels, a process known as acclimatization. Which of the following correctly describes these physiological changes?

- A. Increased pulmonary ventilation with respiratory acidosis, elevated EPO levels, higher cardiac output, and angiogenesis in skeletal muscles.
 - B. Increased pulmonary ventilation with respiratory alkalosis, increased RBC production, higher cardiac output, and increased mitochondrial content in skeletal muscles.
 - C. Decreased pulmonary ventilation with respiratory acidosis, decreased RBC production, lower cardiac output, and decreased capillary density in skeletal muscles.
 - D. Unchanged pulmonary ventilation, no changes in blood pH, constant RBC production, and unchanged skeletal muscle characteristics.
-

8. A 28-year-old climber at 4,500 meters (14,764 feet) develops severe headache, nausea, and breathlessness after 36 hours. Even with supplemental oxygen, his symptoms worsen with confusion and coordination loss, prompting his team to descend. What likely caused his symptoms?

- A. Acute Mountain Sickness (AMS).
 - B. High-Altitude Pulmonary Edema (HAPE).
 - C. High-Altitude Cerebral Edema (HACE).
 - D. An increase in alveolar PCO₂.
-

9. On a hike, a healthy 35 year old mountaineer starts feeling severe dizziness, confusion, and shortness of breath at a high altitude (12,000 meters or 39,370 feet)

despite using supplemental oxygen. His team measures his alveolar oxygen levels and finds them significantly low. They choose to descend to a lower altitude promptly. What is the likely reason for his symptoms and the team's decision to descend?

- A. The barometric pressure at 12,000meters is too low to maintain adequate PO₂ even with supplemental oxygen.
 - B. The barometric pressure at 12,000 meters causes body fluids to boil.
 - C. The composition of the air changes drastically at high altitudes.
 - D. The mountaineer's alveolar PCO₂ increases due to increased ventilation.
-

10. A 50 year old man is found unconscious after a fire in a closed space. Upon investigation of the scene, it was discovered that there was a cyanide gas attack and the patient is severely hypoxic The doctor suspects that he is suffering from histotoxic hypoxia. Which of the following is NOT seen in this condition?

- A. Inhibition of cellular metabolism
 - B. High venous PO₂
 - C. Normal arterial PO₂
 - D. Cyanosis
-

11. A 50-year-old man experiences severe shortness of breath and dizziness while climbing a mountain. Despite receiving supplemental oxygen, he continues to show signs of hypoxia. Which of the following is the most common form of hypoxia?

- A. Hypoxic hypoxia
- B. Stagnant hypoxia
- C. Anemic hypoxia
- D. Histotoxic hypoxia

12. Which of the following statements is true?

- A. Stagnant hypoxia is where tissues cannot utilize O₂ despite a normal supply
 - B. Heart failure causes histotoxic hypoxia
 - C. Stagnant hypoxia is due to the inhibition of tissue oxidative processes
 - D. Artery venous oxygen difference increases in stagnant hypoxia
-

13. A diver experiences joint pain, dizziness, and motor weakness after rapidly ascending from a deep dive. Which of the following best explains the pathophysiology of his symptoms and outlines the immediate management?

- A. Nitrogen bubbles formed during rapid ascent obstruct small blood vessels; treatment is high-flow oxygen therapy
 - B. High atmospheric pressure increases nitrogen dissolution in tissues; immediate recompression followed by slow decompression is essential.
 - C. Oxygen toxicity due to high partial pressures; reducing oxygen concentration and administering IV fluids is required.
 - D. Dehydration from prolonged diving; rehydration and analgesics for joint pain are the primary treatments
-

14. A diver at 150 feet depth begins to exhibit signs of drowsiness and impaired coordination, likely experiencing nitrogen narcosis. Which of the following statements is incorrect about nitrogen narcosis at this depth?

- A. Nitrogen narcosis can cause euphoria at depths around 120 feet.
- B. Symptoms of nitrogen narcosis worsen progressively with increasing depth.

C. The effects of nitrogen narcosis are irreversible even after ascending to a shallower depth.

D. Nitrogen narcosis occurs due to nitrogen dissolving in fatty neuronal membranes.

15. A 40 year old deep-sea diver experiences shortness of breath and dizziness after prolonged use of a rebreathing apparatus. His symptoms worsen despite increased breathing efforts, indicating possible CO₂ buildup in his equipment. What is the tolerable level up to which alveolar CO₂ pressure (Pco₂) can go for a diver?

A. 40mm Hg

B. 60mm Hg

C. 80mm Hg

D. 100mm Hg

16. A 38-year-old diver, after a deep dive with a partial pressure of oxygen of 5 atm for 50 minutes, develops sudden muscle twitching, visual disturbances, and seizures upon surfacing, necessitating urgent medical attention. What is the most likely mechanism responsible for the diver's symptoms?

A. Excessive production of oxygen-free radicals leading to cellular damage

B. Direct damage to lung tissue from high partial pressure of oxygen

C. Hypoxia caused by inadequate oxygen supply to the brain

D. Barotrauma due to pressure changes affecting air-filled spaces

17. How does the SCUBA demand system ensure that air is supplied to the diver only when needed?

- A. The air is continuously supplied to the mask at a constant pressure.
 - B. The first-stage reducing valve maintains a constant high pressure on the mask.
 - C. Air is supplied to the mask only during inhalation through a demand valve.
 - D. Exhaled air is redirected back into the tank for reuse.
-

18. A diver descends to a depth of 300 feet in the sea. If the atmospheric pressure at sea level is 760 mmHg (1 atm), what is the total pressure experienced by the diver at this depth?

- A. 31 atm
 - B. 30 atm
 - C. 11 atm
 - D. 10 atm
-

19. A 45 year old female climber, who reached 5,200m in 36 hours, presents with severe dyspnea, persistent cough with pink, frothy sputum, fatigue, cyanosis, and oxygen saturation of 75% on room air. Despite supplemental oxygen, her condition does not improve. The team is considering interventions. Which treatment option is least effective for severe HAPE in this scenario?

- A. Immediate descent to a lower altitude and administration of supplemental oxygen.
 - B. Use of a portable hyperbaric chamber and provide high-flow oxygen.
 - C. Providing aggressive hydration and using diuretics to reduce pulmonary edema.
 - D. Starting nifedipine and preparing for emergency evacuation
-

20. A 45 year old woman presents with fatigue and dyspnea. Her blood tests reveal a low hemoglobin concentration and elevated carboxyhemoglobin levels despite

normal arterial oxygen saturation. Which type of hypoxia is most consistent with patients' findings?

- A. Hypoxic hypoxia
 - B. Stagnant hypoxia
 - C. Anemic hypoxia
 - D. Histotoxic hypoxia
-

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	1
Question 3	1
Question 4	2
Question 5	2
Question 6	2
Question 7	2
Question 8	3
Question 9	1
Question 10	4
Question 11	1
Question 12	4
Question 13	2
Question 14	3
Question 15	3
Question 16	1
Question 17	3
Question 18	4
Question 19	3
Question 20	3

Solution for Question 1:

Correct Answer: B) Anemic hypoxia

Explanation:

Anemic hypoxia has normal PaO₂ levels and does not stimulate chemoreceptors.

Peripheral chemoreceptors:

- Peripheral chemoreceptors are located in the carotid and aortic bodies
- The carotid bodies are very sensitive and can detect even minor changes in PO₂, PCO₂, and pH of the blood.
- Hypoxia is the major stimulus for the activation of peripheral chemoreceptors.
- They have 2 types of cells: Type 1 -Glomus cells- the chemosensitive cells Type 2 - Sustentacular cells
- Type 1 -Glomus cells- the chemosensitive cells
- Type 2 - Sustentacular cells
- The response of peripheral chemoreceptors to oxygen depends on PaO₂
- Type 1 -Glomus cells- the chemosensitive cells
- Type 2 - Sustentacular cells

Type of Hypoxia	Features	Chemoreceptor stimulation
Hypoxic hypoxia	Decreased PO ₂ Dissolved O ₂ is low	Chemoreceptor stimulated (+) (Option A)
Anemic hypoxia	PO ₂ normal Dissolved O ₂ is normal	Chemoreceptor is not stimulated
Stagnant hypoxia	Stagnation of flow (Low blood flow) to glomus cells	Chemoreceptor stimulated (++) (Option C)
Histotoxic hypoxia	Tissue unable to extract oxygen	Chemoreceptor stimulated (++) due to impaired tissue oxygen extraction (Option D)

Solution for Question 2:

Correct Answer: A) Immediate descent to a lower altitude and start dexamethasone

Explanation:

The patient's signs and symptoms are suggestive of a condition known as High-altitude cerebral edema (HACE).

- Immediate descent to a lower altitude is the most crucial step in the management of HACE to reduce hypoxic stress.
- Starting dexamethasone can help reduce cerebral edema and inflammation, complementing the descent. This combined approach is most effective for severe HACE.

Oxygen therapy can provide temporary symptomatic relief, but it does not address the need for reducing cerebral hypoxia and edema. Immediate descent is required for effective treatment (Option B) .

While hydration and antiemetics can provide symptomatic relief, continuing the ascent is highly dangerous in the context of severe HACE. The priority is to descend to a lower altitude (Option C) .

Acetazolamide is useful for preventing and treating mild to moderate altitude sickness, but it is not sufficient as a standalone treatment for severe HACE. Maintaining the current altitude can worsen the patient's condition (Option D) .

HIGH-ALTITUDE CEREBRAL EDEMA (HACE)	
Category	Details
Cause	Capillary leakage leads to brain swelling, often due to rapid ascent to high altitudes.
Symptoms	Severe headache, nausea, vomiting, ataxia (lack of coordination), disorientation, confusion, hallucinations, papilledema, and potentially coma due to herniation of the brain through tentorium.
Mechanism	Increased capillary pressure is caused by hypoxia-induced local vasodilation of cerebral blood vessels, leading to fluid leakage into brain tissues. Chemical factors like vascular endothelial growth factor and inflammatory cytokines may also increase endothelial permeability.
Onset	Typically within 8-24 hours after rapid ascent to high altitude; symptoms can progress quickly over the next 4-8 days if untreated.

Treatment	First Step - Immediate descent to a lower altitude Administration of oxygen High-dose dexamethasone to reduce cerebral edema and inflammation Acetazolamide to promote acclimatization and reduce CSF formation. Portable Hyperbaric Chamber: Symptoms will return when the patient leaves the chamber, but they may improve briefly, allowing the patient to help with evacuation.
-----------	---

Solution for Question 3:

Correct Answer: A) Increased cerebral blood flow leading to cerebral edema

Explanation:

The symptoms point to Acute Mountain Sickness (AMS). In AMS, the body's response to reduced oxygen includes increased cerebral blood flow to enhance oxygen delivery. This response can lead to elevated capillary pressure in the brain, resulting in fluid leakage and cerebral edema.

Acute Mountain Sickness	
Symptoms	Headache Nausea Fatigue Insomnia Appear hours to days after rapid ascent
Pathophysiology	Associated with cerebral edema, caused by increased capillary pressure in the brain due to dilation of blood vessels in response to low oxygen levels. The increased permeability of the blood-brain barrier contributes to fluid leakage into brain tissue.
Treatment	Descending to a lower altitude Supplemental oxygen, and Acetazolamide to facilitate acclimatization. In severe cases, corticosteroids may be used to reduce cerebral edema.

Hypoxia typically causes increased, not decreased, capillary pressure in the brain due to the dilation of blood vessels, which can contribute to cerebral edema (Option B) .

At high altitudes, oxygen levels are low, not high. The primary issue is increased blood vessel dilation, not constriction, which leads to cerebral edema (Option C) .

Hypoxia generally increases, rather than decreases, the permeability of the blood-brain barrier, contributing to fluid leakage and cerebral edema (Option D) .

Solution for Question 4:

Correct Answer: B) Pulmonary vasoconstriction leading to elevated pulmonary arterial pressure

Explanation:

At high altitudes, widespread pulmonary vasoconstriction due to hypoxia increases pulmonary arterial pressure, which can strain the right side of the heart and contribute to symptoms like breathlessness and pulmonary edema.

Hypoxia Type	Pulmonary Vasculature	Systemic Vasculature
Illustration		
Smooth Muscle Cells	Contain O ₂ -sensitive K ⁺ channels.	Contain ATP-sensitive K ⁺ channels.
Channel Behaviour in Normal Conditions	K ⁺ channels remain open, allowing easy K ⁺ release.	ATP-sensitive K ⁺ channels remain closed with adequate ATP levels.
Effect of Hypoxia	Closure of O ₂ -sensitive K ⁺ channels in hypoxia.	Hypoxia causes ATP level decrease, leading to ATP-sensitive K ⁺ channel opening.
Cell Charge Changes	Potassium trapped inside cells increases positive charge, leading to depolarization.	Large K ⁺ release leads to negative charge accumulation, causing hyperpolarization.
Electrophysiology of smooth muscle cell	Depolarization opens the voltage-gated calcium channel.	Hyperpolarization induces the relaxation of smooth muscle cells.
Result	Vasoconstriction of pulmonary vasculature.	Vasodilation of systemic vasculature.

Hypoxia causes systemic vasodilation which increases the blood flow and oxygen supply to the tissues. Elevated systemic blood pressure is less likely to be the primary cause of the symptoms described (Options A and D) .

While hypoxia does lead to systemic vasodilation, this response is intended to enhance oxygen delivery to tissues, not decrease it. The symptoms described are more related to pulmonary rather than systemic changes (Option C) .

Solution for Question 5:

Correct Answer: B) Increased production of erythropoietin and enhanced hemoglobin levels

Explanation:

Long-term exposure to high altitude leads to increased erythropoietin production, stimulating red blood cell production and thus enhancing hemoglobin levels. This adaptation improves the blood's oxygen-carrying capacity.

Changes to Hypoxia	
Cardiovascular	Pulmonary vasoconstriction Systemic vasodilation Increased Heart Rate, Cardiac Output, Blood Pressure Angiogenesis
RBC	Increased Production (EPO, RBC) Increased Hematocrit and Hemoglobin Increased 2,3 - DPG Note: Significant Polycythemia→ increased blood viscosity→ increased workload of heart→ cardiac hypertrophy.

While systemic blood pressure may initially rise, long-term exposure leads to increased capillary surface area (angiogenesis), not reduced (Option A) .

Hypoxia causes an increase in red blood cells and 2,3-DPG levels, to enhance oxygen release in tissues, not a decrease (Option C) .

While hypoxia can lead to increased pulmonary blood flow initially, long-term adaptation involves increased hemoglobin affinity due to higher 2,3-DPG, which facilitates oxygen release in tissues rather than decreasing oxygen affinity (Option D) .

Solution for Question 6:

Correct Answer: B) Compensatory renal excretion of bicarbonate.

Explanation:

As acclimatization continues, the kidneys adjust to the respiratory alkalosis induced by hyperventilation. They do so by excreting bicarbonate, which helps to counteract the elevated blood pH (alkalosis) caused by the increased rate and depth of breathing.

This renal compensation helps to normalize the blood pH and allows for a sustained increase in ventilation over several days.



When first exposed to high altitude, the body responds to the lower oxygen levels by increasing the breathing rate rapidly (Option A).

- This initial response is driven by peripheral chemoreceptors sensing low oxygen levels.
- However, this initial increase is not sufficient to sustain increased ventilation over several days because excessive breathing leads to a decrease in CO₂ levels (hypocapnia), which can counteract the drive to breathe more.

Initially, increased breathing lowers CO₂ levels, making the CSF more alkaline. Over time, CSF pH normalizes with gradual bicarbonate adjustment. However, sustained ventilation isn't solely maintained by CSF pH changes alone; renal compensation plays a crucial role alongside other adjustments (Option C).

With acclimatization, the respiratory rate stays elevated initially and stabilizes at a higher level than before acclimatization. This sustained increase in ventilation is supported by adaptations like renal compensation, rather than a decrease in respiratory rate over time (Option D).

Solution for Question 7:

Correct Answer: B) Increased pulmonary ventilation with respiratory alkalosis, increased RBC production, higher cardiac output, and increased mitochondrial content in skeletal muscles.

Explanation:

Changes during Acclimatization	
Respiratory	At high altitudes, lower oxygen levels stimulate peripheral chemoreceptors, increasing pulmonary ventilation. This leads to enhanced CO ₂ removal, causing respiratory alkalosis. Kidneys respond by excreting more bicarbonate to stabilize blood pH, causing a sustained increase in ventilation and supporting prolonged acclimatization.
Hematological	At high altitudes, lower PO ₂ boosts erythropoietin production, increasing red blood cell count. This also raises 2,3-diphosphoglycerate (2,3-DPG) levels in red blood cells. Initially, alkalosis shifts the oxyhemoglobin dissociation curve to the left but increased 2,3-DPG shifts it to the right, improving oxygen release to tissues.
Cardiovascular	The heart rate, cardiac output, and blood pressure increase due to the activation of the sympathoadrenal axis by hypoxia. Increased cardiac output leads to an increase in blood flow, and hypoxia causes vasodilation in the systemic circulation, further contributing to increased blood flow and oxygen supply to the tissues.
Tissue	In acclimatization, skeletal muscles undergo several changes, including an increase in the number of capillaries and mitochondria and an increase in myoglobin content. Further, chemicals released from the hypoxic tissues stimulate angiogenesis.

- There is an occurrence of respiratory alkalosis, not respiratory acidosis (Option A) .
- There is an increase in pulmonary ventilation, RBC production, Cardiac output, and new capillary formation (Options C and D).

Solution for Question 8:

Correct Answer: C) High-Altitude Cerebral Edema (HACE)

Explanation:

The climber is experiencing High-Altitude Cerebral Edema (HACE). Severe headache, confusion, loss of coordination, and persistence of symptoms despite supplemental oxygen indicate HACE.

At high altitudes, ventilation increases, reducing alveolar PCO₂ (Option D) . The issue is low oxygen and potential cerebral edema.

Common problems of High Altitude:

- Due to extreme cold (hypothermia, frostbite, trench foot, chilblains).
- Effects due to low oxygen concentration (pulmonary edema, cerebral edema).
- Due to extreme cold (hypothermia, frost bites, trench foot, chilblains).
- Due to exposure to snow (snow blindness).
- Due to solar radiation (sunburn).
- Psychological problems.

High Altitude Physiology

- Decrease in barometric pressure with elevation.
- Hypoxic hypoxia occurs: As altitude increases, alveolar (lung) PO₂ decreases, hence the PaO₂ of blood is reduced, leading to decreased O₂ delivery to the tissues.

Symptoms and Effects of High Altitude

Condition	Description	Symptoms
Acute Mountain Sickness (AMS) (Option A)	A common condition, developing within 8-24 hours of ascent, lasting 4-8 days.	Headache, irritability, insomnia, breathlessness, nausea, vomiting.
High-Altitude Cerebral Edema (HACE)	Develops within 1-3 days. Progression of AMS with significant brain swelling due to capillary leakage.	Ataxia, disorientation, potential coma, death.
High-Altitude Pulmonary Edema (HAPE) (Option B)	Patchy edema in lungs from increased pulmonary arterial pressure.	Fluid accumulation, and increased capillary pressure- causing cough, dyspnea, and chest pain.

Solution for Question 9:

Correct Answer: A) The barometric pressure at 12,000 meters is too low to maintain adequate PO₂ even with supplemental oxygen.

Explanation:

At extreme altitudes, the barometric pressure is so low that even breathing 100% oxygen cannot maintain a normal alveolar PO₂, leading to severe hypoxia.

As you ascend altitude, the barometric pressure decreases because the weight of the atmosphere above you lessens. Although the composition of the air remains consistent, the partial pressure of oxygen (PO₂) decreases proportionally with the decrease in barometric pressure. (Option C)

- 1) Decrease in Barometric Pressure and PO₂:
- 2) Hypoxic Stimulation and Ventilation:
- 3) Breathing 100% Oxygen:
- 4) Limits of Oxygen Breathing at Higher Altitudes:

Solution for Question 10:

Correct Answer: D) Cyanosis

Explanation:

Cyanosis is not typically seen in histotoxic hypoxia because the arterial blood oxygen levels and hemoglobin saturation with oxygen remain normal.

Cyanosis:

- Bluish discoloration of skin.
- Cause: Excessive amounts of deoxygenated hemoglobin in the capillaries.
- Deoxygenated hemoglobin has an intense dark blue-purple color that is transmitted through the skin
- Cyanosis occurs whenever the arterial blood contains more than 5 grams of deoxygenated hemoglobin in each 100 ml of blood

Type of Hypoxia	Features	Cyanosis
Hypoxic hypoxia	Low O ₂ saturation in arterial and venous blood	Cyanosis ++

Anemic hypoxia	Anemia Methemoglobinemia	Cyanosis seen rarely Cyanosis-like appearance
Stagnant hypoxia	Reduced O ₂ saturation in venous blood	Cyanosis ++
Histotoxic hypoxia	O ₂ saturation is normal in arterial blood and high in venous blood	Cyanosis never seen

The most sensitive areas of the Brain in case of hypoxia are:

- Hippocampus
- The Purkinje cell of the cerebellum
- Striatal region of the basal ganglia

Solution for Question 11:

Correct Answer: A) Hypoxic hypoxia

Explanation:

Hypoxic hypoxia is the most common form of hypoxia because of ventilation-perfusion mismatch

Hypoxia Deficiency of O₂ at the tissue level due to decreased O₂ supply or failure to utilize available O₂.

Types of Hypoxia

Hypoxic Hypoxia

- Definition : Reduced PO₂ of arterial blood.
- Mechanism : Decreased PO₂ in arterial blood reduces O₂ delivery to tissues.

Causes:

- Low PO₂ in Inspired Air High altitude (e.g., 3000m/10,000ft, 5000m/15,000ft).
- Hypoventilation Airway obstruction, respiratory muscle paralysis, respiratory center depression, scoliosis.
- Diffusion Defect Pulmonary edema, pulmonary fibrosis, lung collapse.
- Ventilation-Perfusion Mismatch Large A-V shunts, atelectasis, lung collapse, cyanotic congenital heart disease.

Impact:

- Partial pressure is a determinant of the saturation of hemoglobin in the arterial blood.
- Reduced partial pressure, will also lead to a proportional decrease of the SaO₂ as well.
- In the case of hypoxic hypoxia, both the dissolved oxygen and hemoglobin-bound oxygen will be low.
- Ultimately resulting in a low supply of oxygen to the tissue resulting in hypoxia.

Stagnant hypoxia/ischemic hypoxia (Option B) occurs when blood flow is restricted, leading to inadequate oxygen delivery to tissues. It is frequently seen in conditions like heart failure or shock but is not as broadly encountered as hypoxic hypoxia.

Anemic hypoxia (Option C) arises from reduced hemoglobin levels or impaired hemoglobin function, limiting the blood's oxygen-carrying capacity. It is less common compared to hypoxic hypoxia and is not associated with altitude changes.

Histotoxic hypoxia (Option D) occurs when tissues are unable to utilize oxygen effectively, often due to toxins like cyanide. It is less common in general populations compared to hypoxic hypoxia

Solution for Question 12:

Correct Answer: D) Artery venous oxygen difference increases in stagnant hypoxia.

Explanation:

The arterial and venous blood oxygen difference will increase in stagnant hypoxia because the tissue will keep extracting more and more oxygen from the same blood leading to a low oxygen level in the venous blood.

Aspect	Stagnant Hypoxia	Histotoxic hypoxia
Definition	Hypoxia due to decreased blood flow to tissues	Hypoxia where tissues cannot utilize O ₂ despite normal O ₂ supply (Option A)
Mechanism	Normal arterial O ₂ content, hypoxia due to stagnation of blood circulation	Inhibition of tissue oxidative processes (Option C) .
Conditions associated	Heart failure (decreased cardiac output) (Option B) Shock Vascular obstruction (decreased blood flow)	Cyanide poisoning - Cyanide inhibits cytochrome oxidase. Tissue oxidation is paralyzed. Severe diphtheria- Diphtheria toxin inhibits cytochrome synthesis.

Consequences	Local acidosis from lactate accumulation	Inhibition of cellular metabolism
Treatment	Address underlying cause (e.g., improve cardiac output)	Cyanide poisoning: Methylene blue or nitrites, Hyperbaric O2 therapy
Arterial and Venous blood oxygen	With no adequate blood supply, the tissue will keep extracting more and more oxygen from the same blood leading to a low oxygen level in the venous blood. The arterial and venous blood oxygen difference will increase.	Tissues cannot extract oxygen from the blood hence level of venous blood oxygen will be higher than normal. The artery venous oxygen difference will decrease.

Solution for Question 13:

Correct Answer: B) High atmospheric pressure increases nitrogen dissolution in tissues; immediate recompression followed by slow decompression is essential.

Explanation:

Decompression sickness due to rapid ascent after prolonged exposure to high pressure leads to nitrogen bubble formation in tissues. The immediate and correct management is recompression therapy in a hyperbaric chamber, followed by slow decompression to safely eliminate the bubbles and prevent further damage.

Decompression sickness

Aspect	Description
Definition	Occurs when subjects exposed to high atmospheric pressures are suddenly brought to low atmospheric pressures Include deep-sea divers or workers in caissons returning to the surface rapidly or rapid ascent in an unpressurized airplane cabin.
Pressure Changes	Sea Level: 1 atmosphere. Depth: Pressure increases by 1 atmosphere for every 10 meters Nitrogen at Sea Level: 1 liter dissolved in the body (half in water, half in fat).
Mechanism	Nitrogen Dissolution : Divers breathe 80% nitrogen from gas mixtures. Nitrogen dissolves in body fluids and lipid structures of tissues, proportional to the depth of

	descent. Slow Ascent : Gases diffuse into the blood and are eliminated by the lungs. Rapid Ascent : Rapid escape of gases forms bubbles in tissues and blood, causing damage.
Symptoms	Onset : Symptoms usually appear within half an hour of surfacing. Mild : Severe pain in muscles and joints (decompression sickness or caisson disease)- bends Severe : Bubbles decrease blood flow to the brain and spinal cord causing sensory and motor deficits (paresthesia, motor weakness, paralysis, respiratory failure). Bubbles in pulmonary vessels cause dyspnea and respiratory compression.- chokes Bubbles in coronary vessels cause myocardial ischemia
Treatment	Immediate recompression in a pressurized cabin followed by slow decompression. Hyperbaric oxygen therapy is helpful. Recovery is usually complete, though some residual neural deficits may persist.

- Slow Ascent : Prevents decompression sickness by allowing nitrogen to be eliminated via the lungs.
 - Decompression Tables : Detailed procedures for safe decompression by the US Navy.
- Example: Schedule for 190 feet depth for 60 minutes:

Decompression Time	Depth
10 minutes	50ft
17 minutes	40ft
19 minutes	30ft
50 minutes	20ft
84 minutes	10ft

High-flow oxygen therapy alone is not sufficient as immediate treatment. The correct treatment involves immediate recompression in a hyperbaric chamber followed by slow decompression to safely eliminate the nitrogen bubbles (Option A) .

Symptoms are due to nitrogen bubbles, not oxygen toxicity . Therefore, reducing oxygen concentration and administering IV fluids are not the appropriate immediate treatments (Option B) .

The critical issue here is decompression sickness, not dehydration . The immediate treatment should focus on recompression therapy to address the nitrogen bubbles causing the severe symptoms (Option D) .

Solution for Question 14:

Correct Answer: C) The effects of nitrogen narcosis are irreversible even after ascending to a shallower depth.

Explanation:

The effects of nitrogen narcosis are reversible and can reverse within a few minutes upon ascending to a shallower depth.

Nitrogen Toxicity

- About four-fifths of air is nitrogen.
- Nitrogen has no significant effect on bodily function at sea level pressure.
- At high pressures, nitrogen can cause varying degrees of narcosis.

Nitrogen Narcosis

- Symptoms: The first symptoms of mild narcosis appear at about 120 feet depth after 1 hour or more underwater, causing euphoria (similar to mild alcohol intoxication) (Option A) . At 150 to 250 feet: the Diver becomes drowsy. Significant loss of strength and clumsiness, hindering work performance. Beyond 250 feet: hallucinations, stupor, coma, and even death (Option B) .
 - At 150 to 250 feet: the Diver becomes drowsy. Significant loss of strength and clumsiness, hindering work performance.
 - Beyond 250 feet: hallucinations, stupor, coma, and even death (Option B) .
 - Mechanism: Similar to alcohol intoxication and called "raptures of the depths"; believed to dissolve in fatty neuronal membranes, altering ionic conductance and reducing neuronal excitability (Option D) .
 - Reversal: Narcosis reverses within a few minutes upon ascent to a shallower depth, with no known long-term effects if the ascent is not too rapid.
 - At 150 to 250 feet: the Diver becomes drowsy. Significant loss of strength and clumsiness, hindering work performance.
 - Beyond 250 feet: hallucinations, stupor, coma, and even death (Option B) .
-

Solution for Question 15:

Correct Answer: C) 80mm Hg

Explanation:

Divers can tolerate up to an alveolar pressure of 80mm Hg by increasing the respiratory volume 8-11 fold

Carbon dioxide toxicity in deep-sea diving

- Proper Gear prevents CO₂ toxicity as depth alone doesn't increase alveolar CO₂ pressure. Normal breathing maintains CO₂ levels.
 - Certain diving helmets and rebreathing apparatus can cause CO₂ buildup, leading to rebreathing of CO₂.
 - Up to 80 mm Hg alveolar CO₂ (twice normal), divers can tolerate it by increasing respiratory volume 8- to 11-fold.
 - Above 80 mm Hg, CO₂ depresses the respiratory center, causing the respiratory failure-critical threshold
 - High CO₂ levels lead to severe respiratory acidosis, lethargy, narcosis, and eventually anesthesia.
-

Solution for Question 16:

Correct Answer: A) Excessive production of oxygen-free radicals leading to cellular damage

Explanation:

At elevated partial pressures of oxygen, the production of oxygen free radicals increases significantly.

- These radicals can overwhelm the body's antioxidant defenses, causing oxidative damage to cells, especially in the nervous system.
- This mechanism explains symptoms like muscle twitching, visual disturbances, and seizures observed in the diver.

Direct damage to lung tissue from high partial pressure of oxygen (Option B): Direct lung damage typically manifests as pulmonary symptoms, such as coughing and shortness of breath, rather than neurological symptoms like seizures and muscle twitching. The diver's symptoms are more indicative of nervous system involvement due to high levels of oxygen-free radicals.

Hypoxia caused by inadequate oxygen supply to the brain (Option C): Hypoxia can lead to symptoms like confusion, dizziness, and shortness of breath, but not typically to seizures or muscle twitching. In the scenario, the diver is exposed to a high partial pressure of oxygen, not a lack of it.

Barotrauma due to pressure changes affecting air-filled spaces (Option D): Barotrauma results from pressure changes that damage air-filled spaces in the body, such as the lungs or ears, and usually causes symptoms related to physical injury rather than neurological symptoms like those described.

Aspect	Acute Oxygen Toxicity	Chronic Oxygen Toxicity
Pressure	Breathing O ₂ at ■4 atm	Exposure to 1 atm pressure of O ₂ for about 12 hours
Symptoms	Seizures, coma, nausea, muscle twitching, dizziness, visual disturbances, irritability, disorientation	Lung congestion, pulmonary edema, atelectasis
Mechanism	Overproduction of oxygen free radicals (superoxide, hydrogen peroxide) overwhelms enzyme systems	Direct exposure of lung tissues to high O ₂ pressure damages bronchi and alveoli linings
Affected Tissues	Nervous system (high lipid content)	Lungs (direct exposure)
Onset	Symptoms appear within 30 to 60 minutes, often without warning	Develops after prolonged exposure (around 12 hours)
Severity with Exercise	Exercise increases the severity and earlier onset of symptoms	Exercise does not significantly affect the development of symptoms
Treatment	In gas mixtures intended for inhalation at high pressures, the oxygen concentration is reduced to approximately 20% to prevent oxygen toxicity	

Solution for Question 17:

Correct Answer: C) Air is supplied to the mask only during inhalation through a demand valve.

Explanation:

The SCUBA demand system works by having the demand valve open during inhalation. When the diver inhales, the slight negative pressure pulls the valve open, allowing air from the tank to enter the mask and lungs. This ensures air is supplied only when needed.

The air is continuously supplied to the mask at a constant pressure (Option A): In the SCUBA demand system, air is NOT continuously supplied to the mask. Instead, air is delivered only when the diver inhales, thanks to the demand valve.

The first-stage reducing valve maintains a constant high pressure on the mask (Option B): The first-stage reducing valve lowers the high pressure from the tanks to a lower pressure suitable for breathing, but it does not maintain a constant high pressure on the mask.

Exhaled air is redirected back into the tank for reuse (Option D): Exhaled air is expelled into the water and is not redirected back into the tank. This prevents the buildup of CO₂ in the diver's lungs.

Table rendering failed

Solution for Question 18:

Correct Answer: D) 10 atm

Explanation:

- At sea level, the atmospheric pressure is 760 mmHg (1 atmosphere) and the pressure in the lungs is equal to this pressure
- The pressure increases by 1 atmosphere for every 33 feet of descent.
- For 300 feet, the pressure increase is calculated as follows: Number of atmospheres = $300 \text{ feet} / 33 \text{ feet/atmosphere} = 9.09 \text{ atmospheres}$
- Number of atmospheres = $300 \text{ feet} / 33 \text{ feet/atmosphere} = 9.09 \text{ atmospheres}$
- Number of atmospheres = $300 \text{ feet} / 33 \text{ feet/atmosphere} = 9.09 \text{ atmospheres}$

- Total pressure = 1 atmosphere (sea level) + 9.09 atmospheres (increase due to depth) = 10.09 atmospheres. = 10 atmospheres approx.

Solution for Question 19:

Correct Answer: C) Providing aggressive hydration and using diuretics to reduce pulmonary edema.

Explanation:

Aggressive hydration and diuretics are not effective treatments for HAPE. Diuretics do not address the underlying hypoxia and pulmonary edema adequately and may worsen the condition by increasing fluid loss.

HIGH-ALTITUDE PULMONARY EDEMA (HAPE)	
Category	Details
Cause	Marked pulmonary hypertension that develops at high altitude due to severe hypoxia causing uneven vasoconstriction of pulmonary arterioles.
Symptoms	Patchy edema of the lungs, Severe pulmonary dysfunction, Cough Dyspnea Chest pain
Mechanism	Severe hypoxia causes powerful constriction of pulmonary arterioles, but the constriction is uneven, leading to increased blood flow and high capillary pressure in unconstricted areas, resulting in capillary wall disruption (stress failure).
Onset	From a few hours to up to 5 days after rapid ascent to high altitude.
Prophylaxis	Only started for people with a previous history of HAPE, specifically with recurrence. Gradual Ascent. Extended-release Nifedipine, 30 mg every 12 hours, starting the day before ascent and continuing for 4-7 days after reaching the target elevation or until descent.

Treatment	Immediate descent to a lower altitude (Option A) Administration of oxygen Use of a hyperbaric chamber, if available (Option B) Nifedipine (a Ca²⁺ channel blocker that lowers pulmonary artery pressure) (Option D) Diuretics and Acetazolamide are NOT RECOMMENDED for HAPE. (Option C)
-----------	--

Solution for Question 20:

Correct Answer: C) Anemic hypoxia

Explanation:

Anemic hypoxia results from insufficient or impaired hemoglobin thus leading to decreased oxygen transport and causing hypoxia Anemic Hypoxia

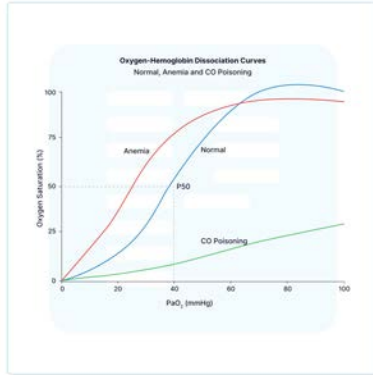
Definition: Normal PO₂ of arterial blood but inadequate hemoglobin (Hb) to carry O₂ .

Causes:

- Anemia (decreased Hb concentration)
- Carbon monoxide (CO) poisoning
- Altered Hb (e.g., Methemoglobin)

Mechanism:

- Normal PO₂: PO₂ in inspired air and arterial blood is normal.
- Reduced O₂ Transport: Inadequate Hb reduces O₂ supply to tissues.
- CO Poisoning:
 - CO binds to Hb with 210 times greater affinity than O₂, forming carboxyhemoglobin (COHb).
 - Slow release of CO from COHb prevents O₂ binding.
- Normal Hb content, but Hb is unavailable for O₂ transport.
- Hb-dissociation curve shifts left, decreasing O₂ release to tissues.



Hypoxic hypoxia (Option A) is caused by low arterial PO_2 due to factors like high altitude or respiratory disease, not related to hemoglobin levels

Stagnant hypoxia (Option B) occurs due to reduced blood flow or cardiac output, causing inadequate tissue perfusion despite normal arterial PO_2

Histotoxic hypoxia (Option D) Involves impaired cellular utilization of oxygen, often due to toxins, despite normal oxygen supply and delivery

Previous Year Questions

1. A patient presents with respiratory difficulty and a history of chronic smoking. Pulmonary function testing shows a decrease in FEV1%. What is the most probable diagnosis?

- A. Chronic Obstructive Pulmonary Disease (COPD)
 - B. Pulmonary Fibrosis
 - C. Interstitial Lung Disease
 - D. Asthma
-

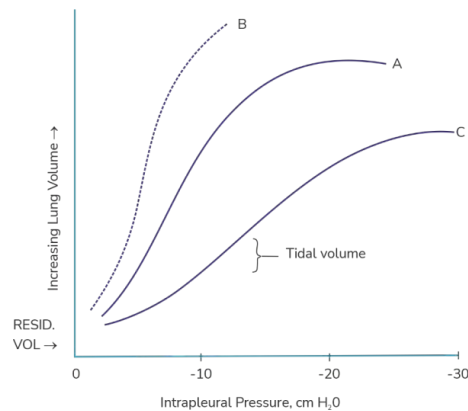
2. Which of the following secretes surfactant?

- A. Type I pneumocyte
 - B. Type II pneumocyte
 - C. Sertoli cell
 - D. Leydig cell
-

3. What would be the resulting lung volume if the chest wall is excluded and the lung is allowed to undergo elastic recoil?

- A. Minimum volume
 - B. Zero
 - C. Functional residual capacity
 - D. Residual volume
-

4. The diagram below shows the pressure–volume curves for 3 different lungs. Which of the following is the best representation of compliance between the three lungs?



- A. $A < B < C$
- B. $B > A > C$
- C. $B > C > A$
- D. $C > A > B$

5. In which of the subsequent situations does oxygen therapy prove to be ineffective?

- A. Asthma
- B. Acute myocardial infarction
- C. Pulmonary edema
- D. Central respiratory depression

6. In peripheral oxygen sensing chemoreceptors, which ion is implicated?

- A. Potassium
 - B. Calcium
 - C. Sodium
 - D. Chlorine
-

7. In which of the following situations is the equation $V/Q = \text{infinity}$ accurate?

- A. Foreign body obstruction in the bronchus
 - B. Equal to dead space
 - C. When O_2 and CO_2 ratio is equal
 - D. The PO_2 of alveolar air is 159 mmHg and PCO_2 is 40 mmHg
-

8. Hypoxic Hypoxia is seen in which of the following?

(or)

Hypoxic Hypoxia is seen in which of the following?

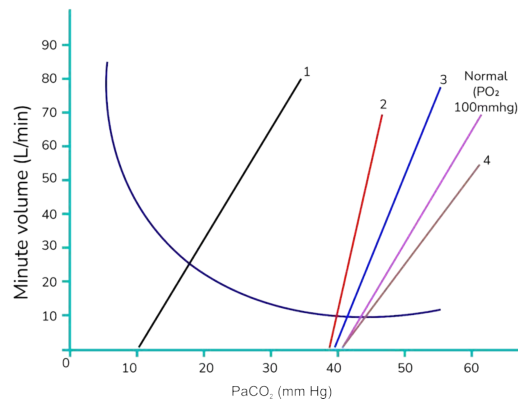
- A. CO poisoning
 - B. HCN poisoning
 - C. Ischemia
 - D. A-V shunt
-

9. Comment on the breathing pattern of a patient with prolonged inspiratory spasms that resemble breath holding?

- A. Cheyne stokes breathing
- B. Kussmaul breathing

- C. Apneustic breathing
 - D. Biot breathing
-

10. If the pH is reduced from 7.4 to 7.3 with the PO₂ being constant at 100 mmHg, which line would PaCO₂ follow?



- A. 1
 - B. 2
 - C. 3
 - D. 4
-

11. Which type of ion channel is affected by mutations in the CFTR gene?

- A. Chloride
 - B. Sodium
 - C. Potassium
 - D. Calcium
-

12. Adequate oxygen delivery at cellular level is present in which type of hypoxia?

- A. Anemic
 - B. Stagnant
 - C. Hypoxic
 - D. Histotoxic
-

13. Determine the respiratory quotient (RQ) of a patient weighing 50 kg who exhales 200 ml/min of CO₂ and consumed 250 ml/min O₂.

- A. 0.8
 - B. 1
 - C. 1.25
 - D. 1.5
-

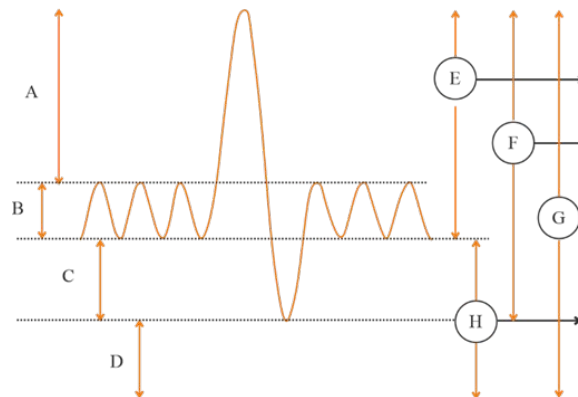
**14. Which among the following statements best describes the appropriate mechanism involved in coughing? Forceful expiration against a closed glottis
Expiration against an open glottis Normal inspiration Removing harmful substances.**

- A. 1,4
 - B. 1,2,3
 - C. 2,4
 - D. 2,3
-

15. What would be the least likely condition to happen if a group of colleagues took an excursion to a high altitude-mountain region?

- A. Increase PaO₂
- B. Increase in RBC & Hb
- C. Increased mitochondria
- D. Hyperventilation

16. In the presented spirometry image, what term corresponds to the point labeled F?



- A. Vital capacity
- B. Total lung capacity
- C. Expiratory reserve volume
- D. Inspiratory capacity

17. In the pressure-volume curve, what does a reading of zero pressure imply?

- A. Residual volume
- B. Functional residual capacity
- C. Tidal volume
- D. Inspiratory reserve volume

18. Single breath nitrogen method is used for measuring _____

- A. Physiological dead space volume
 - B. Residual volume
 - C. Tidal volume
 - D. Anatomical dead space volume
-

19. Pacemaker generating the rhythm for breathing is _____

- A. Pneumotaxic centre
 - B. Dorsal group of neurons in the medulla
 - C. Pre-Botzinger complex
 - D. Apneustic centre
-

20. During forceful expiration, which type of neurons primarily exhibit increased firing?

- A. Dorsal group of neurons
 - B. Caudal Ventral group of neurons
 - C. Pneumotaxic center
 - D. Apneustic center
-

21. Which of the following methods is used for calculation of anatomical dead space ?

- A. Multiple breath nitrogen test
 - B. Bohr's method
 - C. Spirometry
 - D. Single breath nitrogen test
-

22. Carbon monoxide poisoning causes which type of hypoxia?

- A. Anemic hypoxia
 - B. Hypoxic hypoxia
 - C. Stagnant hypoxia
 - D. Histotoxic hypoxia
-

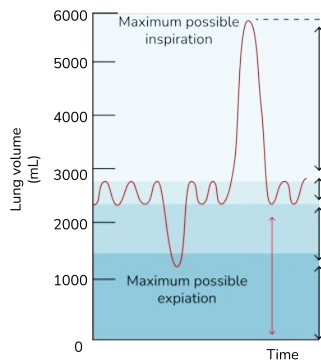
23. Adequate O₂ oxygen delivery at cellular level occurs in which type of hypoxia?

- A. Hypoxic
 - B. Anaemic
 - C. Stagnant
 - D. Histotoxic
-

24. Respiratory distress syndrome is due to the deficiency of?

- A. Dipalmitoyl Phosphatidylcholine
 - B. Phosphatidylserine
 - C. Phosphatidylethanolamine
 - D. Phosphatidylinositol
-

25. Identify the marked region in the given spirometry.



- A. Functional residual capacity
 - B. Tidal volume
 - C. Expiratory reserve volume
 - D. Inspiratory reserve volume
-

26. A preterm baby who was delivered at 28 weeks developed respiratory distress syndrome. Which of the following is true about surface tension and compliance in this baby?

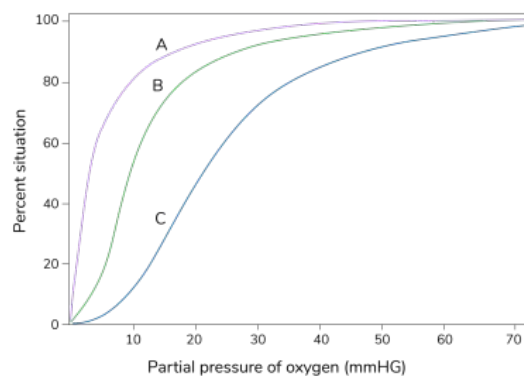
- A. Surface tension - decreased; Compliance - increased
 - B. Surface tension - increased; Compliance - decreased
 - C. Both surface tension and compliance decreased
 - D. Both surface tension and compliance increased
-

27. Which physiological adaptation does not happen at high altitudes?

- A. Alkaline urine

- B. Polycythemia
 - C. Respiratory acidosis
 - D. Hypoxia
-

28. What is indicated by the curve labeled as 'A' in the given oxygen dissociation curves graph?



- A. Myoglobin
 - B. Carboxyhemoglobinemia
 - C. Methemoglobin
 - D. Fetal hemoglobin
-

29. 2, 3-BPG causes the unloading of oxygen to tissues by _____.

- A. Increasing ATP production
 - B. Binding to the beta subunit of Hb
 - C. Inhibiting taut structure
 - D. Decreasing ATP production
-

30. Which channels are affected by the defect caused by cystic fibrosis?

- A. Na^+
 - B. Ca^{2+}
 - C. Cl^-
 - D. K^+
-

31. A person develops headache and breathlessness on a trekking expedition following a rapid ascent to over 3000 meters above the sea level. Which of the following is not used in the treatment of this condition?

- A. IV digoxin
 - B. Immediate descent
 - C. Administration of oxygen
 - D. Tablet acetazolamide
-

32. Respiratory distress syndrome is caused due to defective?

- A. Type 1 pneumocytes
 - B. Type 2 pneumocytes
 - C. Alveolar capillary endothelial cells
 - D. Bronchial mucosal epithelial cells
-

33. Which of the subsequent factors experiences an elevation in chronic obstructive pulmonary disease (COPD)? FRC TLC VC FEV1/FVC

- A. 1 only
 - B. 1 and 4
 - C. 1 and 3
 - D. 1 and 2
-

34. Which of the following conditions will shift the oxygen-Hb dissociation curve to the left?

- A. Hypoxia
 - B. Hypercarbia
 - C. Alkalosis
 - D. Increase in 2,3-DPG
-

35. Which among the following has the highest airway resistance?

- A. Alveolar duct
 - B. Bronchi
 - C. Respiratory bronchioles
 - D. Small bronchioles
-

36. Which of the following does not cause a right shift of the oxygen-hemoglobin dissociation curve?

- A. Increased HbF
- B. Raised 2,3 DPG
- C. Acidosis
- D. Increased CO₂.

37. In the lungs, which type of cells are responsible for the secretion of surfactant?

- A. Type 1 pneumocyte
 - B. Alveolar macrophage
 - C. Dust cells
 - D. Type 2 pneumocyte
-

38. Functional residual capacity is:

- A. Volume expired after normal inspiration
 - B. Volume that remains after forced expiration
 - C. Expiratory reserve volume + residual volume
 - D. Tidal volume + inspiratory reserve volume
-

39.

- A. Partial pressure of O₂ becomes zero
 - B. No exchange of O₂ & CO₂
 - C. CO₂ alone equilibrates with the venous blood
 - D. Partial pressure of both CO₂ and O₂ remain normal
-

40. What is the affinity of carbon monoxide to hemoglobin in comparison to oxygen?

- A. 100
 - B. 4
 - C. 250
 - D. 150
-

41. Which of the following breathing patterns have prolonged inspiratory spasms that resemble breath-holding?

- A. Biot breathing
 - B. Apneustic breathing
 - C. Cheyne stokes breathing
 - D. Kussmaul breathing
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	2
Question 3	1
Question 4	2
Question 5	4
Question 6	1
Question 7	2
Question 8	4
Question 9	3
Question 10	1
Question 11	1
Question 12	4
Question 13	1
Question 14	1
Question 15	1
Question 16	1
Question 17	2
Question 18	4
Question 19	3
Question 20	2
Question 21	4
Question 22	1

Question 23	4
Question 24	1
Question 25	1
Question 26	2
Question 27	3
Question 28	1
Question 29	2
Question 30	3
Question 31	1
Question 32	2
Question 33	4
Question 34	3
Question 35	2
Question 36	1
Question 37	4
Question 38	3
Question 39	2
Question 40	3
Question 41	2

Solution for Question 1:

Correct Answer: A) Chronic Obstructive Pulmonary Disease (COPD)

Explanation:

A history of chronic smoking and decreased FEV₁% (forced expiratory volume in 1 second as a percentage of forced vital capacity) are hallmarks of obstructive lung disease, with COPD being

the most probable diagnosis.

- Key Features of COPD: Chronic smoking history is the most significant risk factor. Obstructive pattern on spirometry: $\downarrow$ FEV₁ (forced expiratory volume in 1 sec) $\downarrow$ FEV₁/FVC ratio ($<70\%$) $\rightarrow$ hallmark of airflow obstruction Progressive dyspnoea, chronic cough, sputum production Irreversible airflow limitation
- Chronic smoking history is the most significant risk factor.
- Obstructive pattern on spirometry: $\downarrow$ FEV₁ (forced expiratory volume in 1 sec) $\downarrow$ FEV₁/FVC ratio ($<70\%$) $\rightarrow$ hallmark of airflow obstruction
- $\downarrow$ FEV₁ (forced expiratory volume in 1 sec)
- $\downarrow$ FEV₁/FVC ratio ($<70\%$) $\rightarrow$ hallmark of airflow obstruction
- Progressive dyspnoea, chronic cough, sputum production
- Irreversible airflow limitation
- Chronic smoking history is the most significant risk factor.
- Obstructive pattern on spirometry: $\downarrow$ FEV₁ (forced expiratory volume in 1 sec) $\downarrow$ FEV₁/FVC ratio ($<70\%$) $\rightarrow$ hallmark of airflow obstruction
- $\downarrow$ FEV₁ (forced expiratory volume in 1 sec)
- $\downarrow$ FEV₁/FVC ratio ($<70\%$) $\rightarrow$ hallmark of airflow obstruction
- Progressive dyspnoea, chronic cough, sputum production
- Irreversible airflow limitation
- $\downarrow$ FEV₁ (forced expiratory volume in 1 sec)
- $\downarrow$ FEV₁/FVC ratio ($<70\%$) $\rightarrow$ hallmark of airflow obstruction

Features of Obstructive Diseases:

Emphysema:

Chronic infection

$\downarrow$

Impaired mucus clearance + Excess mucus secretion + Inflammatory edema

Bronchiolar obstruction

Air trapping and Alveolar damage

Elevated RV and TLC

Examples of obstructive lung disease:

- Asthma
- COPD (Chronic bronchitis and emphysema)

- Bronchiectasis
- Bronchiolitis
- Cystic fibrosis

Pulmonary Fibrosis (Option B) is incorrect because pulmonary fibrosis is a restrictive lung disease, which shows decreased FVC but normal or increased FEV₁/FVC ratio.

Interstitial Lung Disease (Option C) is incorrect because interstitial lung diseases (ILDs) cause a restrictive pattern, with normal or increased FEV₁/FVC rather than a decreased ratio.

Asthma (Option D) is incorrect because while asthma also causes obstructive airway disease with a low FEV₁%, it is typically reversible with bronchodilators, unlike COPD, which is progressive and largely irreversible.

Solution for Question 2:

Correct Option B: Type II pneumocyte

- Type II pneumocytes: Type II pneumocytes are specialized cells in the alveoli that produce and secrete surfactant. Surfactant is a mixture of lipids and proteins that lines the inner surface of the alveoli. It reduces the surface tension within the alveoli, preventing their collapse during expiration and promoting efficient gas exchange.

Incorrect Options:

Option A: Type I pneumocyte: Type I pneumocytes are responsible for the structure and integrity of the alveoli. They are thin and flat cells that form the majority of the alveolar surface area for gas exchange. However, they do not secrete surfactants

Option C: Sertoli cell: Sertoli cells are found in the seminiferous tubules of the testes and play a crucial role in supporting and nourishing developing sperm cells. They are not involved in the secretion of surfactants.

Option D: Leydig cell: Leydig cells are found in the interstitial tissue of the testes and are responsible for the production and secretion of testosterone. They are not involved in the secretion of surfactants.

Solution for Question 3:

Correct option:

Option A.

- Minimum volume: The minimum volume of the lung refers to the volume of the lung when it is allowed to undergo maximum elastic recoil without any opposing forces from the chest wall. It is the smallest volume the lung can reach during expiration.

Incorrect options

Option B. Zero: If the lung undergoes elastic recoil without the chest wall, it does not mean that the lung volume would reach zero. The lungs still have residual air and do not collapse completely.

Option C. Functional residual capacity (FRC): FRC is the volume of air remaining in the lungs after a normal expiration, when the forces of lung recoil and chest wall expansion are in equilibrium. It is the volume at which the lung and chest wall exert equal but opposite forces.

Option D. Residual volume: The residual volume is the volume of air that remains in the lungs after a maximal expiration. It represents the air that cannot be expelled from the lungs and serves to keep the alveoli open.

Solution for Question 4:

Correct Option B:

- The pressure-volume curve of the lungs represents the relationship between the pressure applied to the lungs and the corresponding volume of air in the lungs. This curve helps us understand the behavior of the lungs during inspiration and expiration.
- The pressure-volume curve of the lungs typically exhibits two phases: inflation and deflation.

During the inflation phase:

- Initially, as the lung volume increases, the compliance of the lungs is high, meaning that the lungs easily expand with minimal changes in pressure.
- As the lung volume continues to increase, the compliance gradually decreases. This is due to the stretching of the lung tissue and the increasing resistance to expansion.

- At a certain point, the compliance reaches its lowest value, which is known as the lower inflection point. Beyond this point, the compliance increases again, indicating easier expansion of the lungs.

During the deflation phase:

- Initially, as the lung volume decreases, the compliance remains relatively high, allowing the lungs to recoil passively.
- As the lung volume further decreases, the compliance starts to decrease again, indicating increasing resistance to deflation.
- At a certain point, the compliance reaches its highest value, which is known as the upper inflection point. Beyond this point, the compliance decreases again, indicating increasing resistance to deflation.
- The pressure-volume curve of the lungs can provide valuable information about lung mechanics and can be used to assess lung diseases or evaluate the effects of therapeutic interventions. Changes in the shape and position of the curve can indicate alterations in lung compliance and resistance.

Solution for Question 5:

Correct Option D: Central respiratory depression

- In cases of central respiratory depression, oxygen therapy may not be beneficial or effective.
- Central respiratory depression refers to a decrease or cessation of breathing due to a suppressed respiratory drive in the central nervous system. In such cases, providing oxygen alone may not stimulate the respiratory centers enough to overcome the underlying cause of respiratory depression.

Incorrect Options:

Option A: Asthma- Oxygen therapy can be beneficial in asthma exacerbations. In asthma, there is bronchoconstriction and inflammation of the airways, which can lead to reduced oxygen levels. Supplemental oxygen can help improve oxygenation and alleviate symptoms in individuals with asthma. Option B: Acute myocardial infarction- Oxygen therapy is commonly used in the management of acute myocardial infarction (heart attack). It helps to increase oxygen supply to the heart muscle and improve oxygenation in the body. Providing supplemental oxygen during an acute myocardial infarction can help reduce myocardial damage and improve outcomes.

Option C: Pulmonary edema- This option is incorrect. Oxygen therapy is typically used in the management of pulmonary edema. Pulmonary edema is characterized by fluid accumulation in the lungs, leading to impaired gas exchange. Supplemental oxygen can help improve oxygenation and relieve respiratory distress in individuals with pulmonary edema.

Solution for Question 6:

Correct option is A: Potassium.

- Potassium ions (K^+) are involved in peripheral oxygen-sensing chemoreceptors. These chemoreceptors, known as peripheral chemoreceptors or arterial chemoreceptors, are primarily located in the carotid bodies and aortic bodies. They detect changes in oxygen levels in the blood and play a crucial role in regulating respiratory and cardiovascular responses.

Incorrect options

Option B: Calcium ions (Ca^{2+}) are not directly involved in peripheral oxygen-sensing chemoreceptors. However, calcium ions are important for various cellular processes and signaling pathways within the body.

Option C: Sodium ions (Na^+) are not directly involved in peripheral oxygen-sensing chemoreceptors. Sodium ions play a critical role in maintaining the body's fluid balance, nerve impulse transmission, and muscle contraction.

Option D: Chlorine ions (Cl^-) are not directly involved in peripheral oxygen-sensing chemoreceptors. Chlorine ions primarily function as an anion to maintain electrical neutrality and are involved in maintaining the osmotic balance and acid-base balance in the body.

Solution for Question 7:

Correct Option B: Equal to dead space

Incorrect Options:

Option A: Foreign body obstruction in the bronchus - V/Q is 0 here

Option C: When O_2 and CO_2 ratio is equal V/Q is not Infinite in this case.

Option D: The PO_2 of alveolar air is 159 mmHg and PCO_2 is 40 mm Hg V/Q is not Infinite in this case.

Solution for Question 8:

Correct option:

Option D.

A-V shunt : An A-V (arteriovenous) shunt refers to an abnormal connection between an artery and a vein, bypassing the capillary network. It is associated with hypoxic hypoxia

Incorrect options:

Option A. CO poisoning: Carbon monoxide (CO) can lead to anemic hypoxia. CO has a strong affinity for hemoglobin, binding to it more readily than oxygen. This reduces the oxygen-carrying capacity of the blood, resulting in decreased oxygen delivery to the tissues.

Option B. HCN poisoning: Hydrogen cyanide (HCN) poisoning can cause Histotoxic hypoxia. HCN inhibits cellular respiration by interfering with the electron transport chain, impairing the utilization of oxygen by the cells.

Option C. Ischemia: Ischemia refers to inadequate blood flow to tissues, which can result in stagnant hypoxia. Reduced blood flow means less oxygen is delivered to the affected tissues, leading to hypoxia.

Solution for Question 9:

Correct Option C - Apneustic breathing:

- Apneustic breathing is marked by prolonged inspiratory spasms, often resembling breath holding. It results from damage to the pons in the brainstem, disrupting normal respiratory

control.

Incorrect Options:

Option A - Cheyne-Stokes breathing: This breathing pattern involves alternating periods of deep, rapid breathing followed by brief periods of apnea. It doesn't match the prolonged inspiratory spasms described.

Option B - Kussmaul breathing: Kussmaul breathing is characterized by deep, rapid, and labored breathing often associated with metabolic acidosis. It doesn't involve prolonged inspiratory spasms resembling breath holding.

Option D - Biot breathing: Biot breathing is characterized by irregular breathing with periods of apnea and hyperpnea. It doesn't specifically involve prolonged inspiratory spasms resembling breath holding.

Solution for Question 10:

Correct Option A - 1:

- The reduction in pH from 7.4 to 7.3 would trigger a compensatory response by the respiratory system to increase the excretion of carbon dioxide, which would lead to a decrease in PaCO₂ (partial pressure of carbon dioxide in arterial blood). This response is known as respiratory compensation.
- On a minute ventilation and PaCO₂ graph, the line representing PaCO₂ would follow a downward trend, indicating a decrease in PaCO₂. The exact slope and magnitude of the change would depend on various factors such as baseline minute ventilation, respiratory rate, and dead space ventilation.
- Note that the relationship between minute ventilation and PaCO₂ is not linear, and many factors can affect this relationship, including metabolic rate, alveolar ventilation, and pulmonary perfusion.

Solution for Question 11:

Correct Option A - Chloride:

- The CFTR (Cystic Fibrosis Transmembrane Conductance Regulator) gene codes for a chloride ion channel. Mutations in this gene lead to a defective CFTR protein, which disrupts chloride ion transport across cell membranes, contributing to the symptoms of cystic fibrosis.

Incorrect Options:

Option B - Sodium: Sodium ion channels are not directly affected by CFTR mutations. However, changes in chloride transport can indirectly impact sodium transport and balance in the body.

Option C - Potassium: Potassium ion channels are unrelated to the CFTR gene. Mutations in the CFTR gene do not affect potassium transport.

Option D - Calcium: Calcium ion channels are not connected to CFTR gene mutations. The CFTR gene specifically affects chloride ion channels, not calcium ones.

Solution for Question 12:

Correct Option D - Histotoxic:

- Histotoxic hypoxia occurs when there is a disruption in the utilization of oxygen at the level of the tissue by the cells, despite adequate oxygen delivery.
- It can be caused by cyanide poisoning and beriberi that interfere with the cellular metabolic processes, preventing the cells from effectively utilizing the oxygen delivered to them.

Incorrect Options:

Option A - Anemic: Anemic hypoxia results from a decrease in the oxygen-carrying capacity of the blood due to reduced hemoglobin levels or abnormal hemoglobin function.

Option B - Stagnant: Stagnant hypoxia arises from the impaired circulation of blood, leading to inadequate oxygen delivery to tissues, as seen in hemorrhage and heart failure.

Option C - Hypoxic: Hypoxic hypoxia is a condition where the partial pressure of oxygen in the arterial blood is too low.

Solution for Question 13:

Correct Option A - 0.8:

- To calculate the respiratory quotient (RQ), we need to determine the ratio of carbon dioxide production (VCO₂) to oxygen consumption (VO₂). The formula for RQ is: $RQ = VCO_2 / VO_2$
Given the values: CO₂ exhaled (VCO₂) = 200 ml/min O₂ consumed (VO₂) = 250 ml/min
Substituting these values into the formula: $RQ = 200 \text{ ml/min} / 250 \text{ ml/min}$ $RQ = 0.8$
- $RQ = VCO_2 / VO_2$
- Given the values: CO₂ exhaled (VCO₂) = 200 ml/min O₂ consumed (VO₂) = 250 ml/min
- Substituting these values into the formula: $RQ = 200 \text{ ml/min} / 250 \text{ ml/min}$ $RQ = 0.8$
- $RQ = 200 \text{ ml/min} / 250 \text{ ml/min}$ $RQ = 0.8$
- $RQ = VCO_2 / VO_2$
- Given the values: CO₂ exhaled (VCO₂) = 200 ml/min O₂ consumed (VO₂) = 250 ml/min
- Substituting these values into the formula: $RQ = 200 \text{ ml/min} / 250 \text{ ml/min}$ $RQ = 0.8$
- $RQ = 200 \text{ ml/min} / 250 \text{ ml/min}$ $RQ = 0.8$
- $RQ = 200 \text{ ml/min} / 250 \text{ ml/min}$ $RQ = 0.8$

Incorrect Options:

- Options B, C, and D are incorrect.
-

Solution for Question 14:

Correct Option A - 1 and 4 are correct statements:

- Forceful expiration against a closed glottis: This describes the initial phase of a cough. The cough reflex begins with a deep inhalation, followed by the closure of the glottis (the opening between the vocal cords). The expiratory muscles then contract forcefully, increasing the pressure in the chest. When the glottis suddenly opens, the built-up pressure is released, producing a forceful expulsion of air and any irritants in the airway.
- This describes the initial phase of a cough. The cough reflex begins with a deep inhalation, followed by the closure of the glottis (the opening between the vocal cords). The expiratory muscles then contract forcefully, increasing the pressure in the chest. When the glottis

suddenly opens, the built-up pressure is released, producing a forceful expulsion of air and any irritants in the airway.

- Removing harmful substances: Coughing serves to clear the respiratory tract of irritants, such as mucus, dust, smoke, or other harmful substances. This is a protective reflex mechanism aimed at keeping the airways clear and preventing the aspiration of foreign materials.
- Coughing serves to clear the respiratory tract of irritants, such as mucus, dust, smoke, or other harmful substances. This is a protective reflex mechanism aimed at keeping the airways clear and preventing the aspiration of foreign materials.

Forceful expiration against a closed glottis:

- This describes the initial phase of a cough. The cough reflex begins with a deep inhalation, followed by the closure of the glottis (the opening between the vocal cords). The expiratory muscles then contract forcefully, increasing the pressure in the chest. When the glottis suddenly opens, the built-up pressure is released, producing a forceful expulsion of air and any irritants in the airway.

Removing harmful substances:

- Coughing serves to clear the respiratory tract of irritants, such as mucus, dust, smoke, or other harmful substances. This is a protective reflex mechanism aimed at keeping the airways clear and preventing the aspiration of foreign materials.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 15:

Correct Option A - Increase PaO₂:

- PaO₂ is the least likely condition to happen during an excursion to a high-altitude mountain region. At higher altitudes, the partial pressure of oxygen (PaO₂) decreases due to the lower atmospheric pressure. This can lead to a hypoxia, where there's a reduced availability of oxygen for the body's cells and tissues.

Incorrect Options:

Option B - Increase in RBC & Hb: When exposed to lower oxygen levels, the body can adapt by producing more red blood cells and hemoglobin to enhance oxygen-carrying capacity, improving the overall oxygen delivery to tissues.

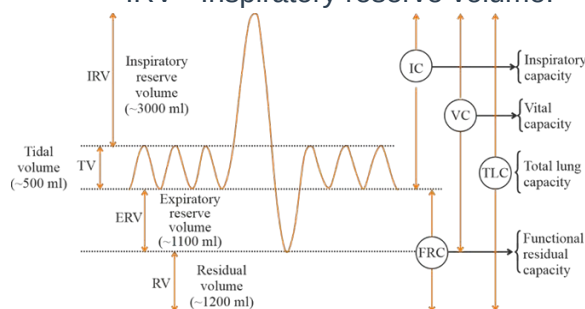
Option C - Increased mitochondria: To extract the increased amount of oxygen by the tissue, the number of mitochondria and cytochrome oxidase enzymes are increased.

Option D - Hyperventilation: It's the body's attempt to compensate for lower oxygen levels by increasing the respiratory rate to take in more oxygen. This can help temporarily alleviate the effects of hypoxia.

Solution for Question 16:

Correct Option A - Vital capacity:

- 'F' represents the vital capacity which is the maximum amount of air a person can expel from the lungs after a forced inspiration (≈ 4600 ml).
- $VC = ERV + TV + IRV$.
- Where, ERV - Expiratory reserve volume. TV - Tidal volume. IRV - Inspiratory reserve volume.
- ERV - Expiratory reserve volume.
- TV - Tidal volume.
- IRV - Inspiratory reserve volume.
- ERV - Expiratory reserve volume.
- TV - Tidal volume.
- IRV - Inspiratory reserve volume.



Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 17:

Correct Option B - Functional residual capacity:

- The zero-pressure point on the pressure-volume curve represents the Functional residual capacity (FRC), which is the point at which the lung's tendency to collapse is balanced by the chest wall's tendency to expand.
- FRC is the volume of air that remains in the lungs after a normal expiration. It is the balance point between the elastic recoil of the lungs and the chest wall.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.
-

Solution for Question 18:

Correct Option D - Anatomical dead space volume:

- The single breath nitrogen test, aka fowler's method is used to estimate anatomical dead space. It involves inhaling pure oxygen and then exhaling slowly and steadily.
- The alveolar oxygen is mixed with the nitrogen whereas the air in the anatomical dead space is free of nitrogen containing only oxygen thus measuring the anatomical dead space.

Incorrect Options:

Option A - Physiological dead space volume: Bohr's method is used to measure physiological dead space volume.

Option B - Residual Volume: It is calculated by multiple breath nitrogen washout method.

Option C - Tidal volume: It can be calculated by using a spirometer.

Solution for Question 19:

Correct Option C - Pre-Botzinger complex:

- The Pre-Botzinger complex, located in the medulla oblongata, is considered the primary pacemaker for spontaneous breathing. It consists of I (inspiratory) and E (Expiratory) neurons that spontaneously generate rhythmic signals, initiating the pattern of breathing.

Incorrect Options:

Option A - Pneumotaxic centre: The pneumotaxic center, located in the pons of the brainstem, helps regulate the duration and depth of each breath. It controls the switching off of the inspiratory phase, allowing for smoother transitions between inhalation and exhalation.

Option B - Botzinger complex: It contains the expiratory group of neurons, and their main role is to control the expiration.

Option D - Apneustic center: The apneustic center, located in the lower pons of the brainstem, is involved in regulating the intensity and duration of inhalation through inhalatory neuron.

Solution for Question 20:

Correct Option B - Caudal Ventral group of neurons:

- Caudal ventral respiratory group neurons mainly contain the expiratory group of neurons, which is dominantly discharged during forceful expiration.
- During forceful expiration, one takes the help of the accessory muscles of respiration, for example, the abdominal muscles and internal intercostal muscles.
- These are all contracted during forceful expiration. This control is exerted by the caudal ventral respiratory group neurons.

Incorrect Options:

Option A - Dorsal group of neurons: The dorsal group of neurons, also known as the dorsal respiratory group (DRG), is responsible for integrating sensory signals from the peripherals, such as the lungs or the muscles.

Option C - Pneumotaxic center: The pneumotaxic center, also known as the pontine respiratory group, is a part of the brainstem that regulates the rate of breathing.

Option D - Apneustic center: Apneustic center is also a pontine respiratory group that regulates the depth of breathing.

Solution for Question 21:

Correct Option D - Single breath nitrogen test:

- The single breath nitrogen test is a method used to estimate anatomical dead space. It involves inhaling pure oxygen and then exhaling slowly and steadily.
- The alveolar oxygen is mixed with the nitrogen whereas the air in the anatomical dead space is free of nitrogen containing only oxygen thus measuring the anatomical dead space

Incorrect Options:

Option A - Multiple breath nitrogen test: It is used for measuring the residual volume.

Option B - Bohr's method: it is used for measuring the physiological dead space.

Option C - Spirometry: Spirometry is a commonly used test to assess lung function and measure various parameters such as forced vital capacity (FVC), forced expiratory volume in one second (FEV1), and peak expiratory flow rate (PEFR).

Solution for Question 22:

Correct Option A - Anemic hypoxia:

CO poisoning causes:

- Anemic hypoxia occurs due to a decrease in the oxygen-carrying capacity of the blood, which can happen in conditions such as anemia and CO poisoning.
- In the case of carbon monoxide poisoning, CO binds strongly to hemoglobin, forming carboxyhemoglobin. This reduces the availability of hemoglobin for oxygen binding, resulting

in a decrease in the oxygen-carrying capacity of the blood.

- Carboxyhemoglobin has a higher affinity for oxygen compared to normal hemoglobin, which impairs the release of oxygen to the tissues. As a result, despite the presence of oxygen in the bloodstream, there is a reduced delivery of oxygen to the tissues. This reduction in oxygen supply to the tissues is characteristic of anemic hypoxia.

Incorrect Options:

Options B, C, and D are incorrect. Anemic hypoxia is caused by carbon monoxide poisoning.

Solution for Question 23:

Correct Option D - Histotoxic:

- Histotoxic hypoxia occurs when there is a disruption in the utilization of oxygen at the level of the tissue, despite adequate oxygen delivery.
- It can be caused by cyanide poisoning and beriberi that interfere with the cellular metabolic processes, preventing the cells from effectively utilizing the oxygen delivered to them. In histotoxic hypoxia, the oxygen delivery to tissues is adequate, but the cells are unable to utilize it properly.

Incorrect Options:

Option A - Hypoxic hypoxia: Hypoxic hypoxia occurs when there is a decrease in the oxygen tension (partial pressure of oxygen) in the arterial blood.

Option B - Anemic hypoxia: Anemic hypoxia results from a decrease in the oxygen-carrying capacity of the blood due to reduced hemoglobin levels or abnormal hemoglobin function.

Option C - Stagnant hypoxia: Stagnant hypoxia arises from the impaired circulation of blood, leading to inadequate oxygen delivery to tissues.

Solution for Question 24:

Correct Option A - Dipalmitoyl Phosphatidylcholine:

- Respiratory distress syndrome (RDS) is caused by the deficiency of surfactant in the lungs which helps in reducing the surface tension.
- Surfactant is composed of the following constituents: Lipids (90%) - Mainly dipalmitoyl choline (60%), phosphatidylglycerol (6%), and other phospholipids (10%). Proteins (8%) - Albumin and immunoglobulin A. Carbohydrates (2%).
- Lipids (90%) - Mainly dipalmitoyl choline (60%), phosphatidylglycerol (6%), and other phospholipids (10%).
- Proteins (8%) - Albumin and immunoglobulin A.
- Carbohydrates (2%).
- So, among the given options the deficiency of dipalmitoyl choline , a primary constituent of surfactant will result in respiratory deficiency.
- Lipids (90%) - Mainly dipalmitoyl choline (60%), phosphatidylglycerol (6%), and other phospholipids (10%).
- Proteins (8%) - Albumin and immunoglobulin A.
- Carbohydrates (2%).

Incorrect Options:

- Options B, C & D are incorrect. Refer to Option A for an explanation.

Solution for Question 25:

Correct Option A - Functional residual capacity:

- Spirometry measures airflow in and out of the lungs, which is useful in diagnosing and monitoring lung diseases.
- The marked region in the given spirometry represents the Functional Residual Capacity (FRC).
- FRC is the lung air volume after a normal, passive exhalation. It is calculated as the sum of the Expiratory Reserve Volume (ERV) and the Residual Volume (RV) and is around 2.3 L.

Incorrect Options:

Option B - Tidal volume: This is the air that moves in and out of the lungs with each regular breath, normal value is 500 to 750ml..

Option C - Expiratory reserve volume: This is the maximum volume of air that can be voluntarily exhaled after a normal exhalation. It is around 1.1 L.

Option D - Inspiratory reserve volume: This is the maximum volume of air that can be inhaled after a normal inhalation. It is around 2L.

Solution for Question 26:

Correct Option B - Surface tension - increased; Compliance - decreased:

- This statement is accurate regarding surface tension and compliance in a preterm baby with respiratory distress syndrome (RDS).
- Surface tension: RDS is characterized by a deficiency in pulmonary surfactant, a substance produced by type II pneumocytes in the alveoli due to premature delivery. Surfactant reduces surface tension in the alveoli, preventing their collapse during expiration. In RDS, the lack of surfactant leads to increased surface tension, making it difficult for the alveoli to remain open and causing them to collapse.
- Compliance: Compliance refers to the ease with which the lungs can expand. In the absence of surfactant, the alveoli become stiff and less compliant, requiring greater effort for inspiration. Therefore, compliance is decreased in RDS due to the decreased ability of the lungs to expand easily.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation
-

Solution for Question 27:

Correct Option C - Respiratory acidosis:

- The body undergoes respiratory alkalosis instead of respiratory acidosis at high altitudes. This is because the respiratory rate and depth increases to compensate for the reduced oxygen pressure in the air, resulting in depletion of CO₂ levels in the bloodstream, causing

respiratory alkalosis rather than acidosis.

Incorrect Options:

Option A - Alkaline urine: To correct respiratory alkalosis, the kidney secrete HCO_3^- in the urine, thus urine is alkaline in nature.

Option B - Polycythemia - Hypoxia stimulates the production of erythropoietin resulting in polycythemia to increase the oxygen-carrying capacity of the blood.

Option D - Hypoxia- Hypoxia is a physiological adaptation that occurs at high altitudes due to the decrease in oxygen levels in the air. Hypoxia can lead to a range of symptoms, including shortness of breath, fatigue, and dizziness.

Solution for Question 28:

Correct Option A - Myoglobin:

- Myoglobin is a protein found in muscle cells that is responsible for storing and delivering oxygen to the muscle tissue. The oxygen dissociation curve for myoglobin is hyperbolic, because in the case of myoglobin, one molecule of hemoglobin binds with one molecule of oxygen, and complete saturation will occur. Therefore, a sudden increase in saturation can be seen in a myoglobin curve.
- Thus, A in the graph indicates myoglobin.

Incorrect Options:

Option B - Carboxyhemoglobinemia: Carboxyhemoglobinemia shifts the oxygen dissociation curve to the left and markedly reduces the oxygen dissociation.

Option C - Methemoglobin: Methemoglobin is a form of hemoglobin that has been oxidized and can no longer bind to oxygen effectively due to ferric state or iron.

Option D - Fetal hemoglobin: Fetal hemoglobin has a higher affinity for oxygen than adult hemoglobin. This shifts the oxygen dissociation curve to the left as marked by B.

Solution for Question 29:

Correct Option B - Binding to the beta subunit of Hb:

- 2, 3-BPG is a small molecule that can bind to a pocket in the beta subunit of hemoglobin.
- 2,3-DPG carries approximately 3.5 negative charges that interact with eight positively charged amino acids in the central cavity. This interaction destabilizes the binding of Hb with oxygen.
- As a result, when 2,3-DPG binds with Hb, it causes the Oxy-Hb dissociation curve to shift to the right, promoting the release of oxygen from red blood cells into the tissues.

Incorrect Options:

- Option A, C and D are incorrect.
-

Solution for Question 30:

Correct Option C - Cl⁻ (Chloride):

- Cystic fibrosis is an autosomal recessive monogenic disorder, in which the major defect is in the chloride channel.
- The disease occurs due to mutation in the CFTR gene (cystic fibrosis transmembrane-conductance regulator gene), located on chromosome 7.
- The lung is the most common organ affected, and decreased nasociliary clearance is the common pathology.

Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.
-

Solution for Question 31:

Correct Option A - IV digoxin:

- IV digoxin is not used in the treatment of the condition described. Digoxin is a medication commonly used for heart conditions such as heart failure and atrial fibrillation.
- Altitude sickness is managed by Immediate descent greatly improves the symptoms. Diuretics: Acetazolamide is the preferred diuretic. It inhibits carbonic anhydrase and increases bicarbonate excretion in urine, thus decreasing alkali load. Steroids: Glucocorticoid decreases cerebral edema. Oxygen therapy: Hyperbaric oxygen is very useful if pulmonary edema is present. Nifedipine: Calcium channel blockers like nifedipine help to reduce pulmonary arterial pressure.
- Immediate descent greatly improves the symptoms.
- Diuretics: Acetazolamide is the preferred diuretic. It inhibits carbonic anhydrase and increases bicarbonate excretion in urine, thus decreasing alkali load.
- Steroids: Glucocorticoid decreases cerebral edema.
- Oxygen therapy: Hyperbaric oxygen is very useful if pulmonary edema is present.
- Nifedipine: Calcium channel blockers like nifedipine help to reduce pulmonary arterial pressure.
- Immediate descent greatly improves the symptoms.
- Diuretics: Acetazolamide is the preferred diuretic. It inhibits carbonic anhydrase and increases bicarbonate excretion in urine, thus decreasing alkali load.
- Steroids: Glucocorticoid decreases cerebral edema.
- Oxygen therapy: Hyperbaric oxygen is very useful if pulmonary edema is present.
- Nifedipine: Calcium channel blockers like nifedipine help to reduce pulmonary arterial pressure.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.

Solution for Question 32:

Correct Option B - Type 2 pneumocytes:

- Respiratory distress syndrome (RDS), also known as neonatal respiratory distress syndrome or infant respiratory distress syndrome, is primarily due to a defect in Type 2 pneumocytes.

- Type 2 pneumocytes, also called surfactant-producing cells, play a crucial role in the production and secretion of pulmonary surfactant. Pulmonary surfactant is a mixture of phospholipids, proteins, and other substances that coats the alveoli within the lungs. It reduces surface tension, preventing the alveoli from collapsing during exhalation and promoting efficient gas exchange.

Incorrect Options:

- Options A, C, and D are incorrect. These cells are not responsible for the production of the surfactant.

Solution for Question 33:

Correct Option D. 1 and 2

FRC (Functional Residual Capacity): FRC is increased in Chronic Obstructive Pulmonary Disease (COPD). In COPD, there is difficulty exhaling air due to airway obstruction and decreased lung elasticity, leading to air trapping in the lungs during expiration. This increased residual volume results in an elevated FRC.

TLC (Total Lung Capacity): TLC is often increased in COPD. This increase is primarily due to the elevated residual volume and FRC. In COPD like emphysema, the lungs are hyperinflated, leading to an increase in TLC.

VC (Vital Capacity): VC is reduced in COPD. The vital capacity, which is the maximum amount of air a person can exhale after a maximal inhalation, is reduced due to the airway obstruction and decreased lung elasticity in COPD.

FEV1/FVC (Forced Expiratory Volume in 1 second/Forced Vital Capacity ratio): This ratio is reduced in COPD. The reduced FEV1/FVC ratio indicates the presence of airflow obstruction. In COPD, the airflow limitation results in a decreased FEV1/FVC ratio.

Solution for Question 34:

Correct Option C - Alkalosis:

- Conditions associated with shift of the oxygen-hemoglobin dissociation curve to the left are:
Decreased temperature Increased pH Decreased PCO₂ Decreased 2,3-DPG Fetal Hb
Carbon monoxide
- Decreased temperature
- Increased pH
- Decreased PCO₂
- Decreased 2,3-DPG
- Fetal Hb
- Carbon monoxide
- Therefore among the given options alkalosis will result in left shift of the
oxygen-hemoglobin dissociation curve.
- Decreased temperature
- Increased pH
- Decreased PCO₂
- Decreased 2,3-DPG
- Fetal Hb
- Carbon monoxide

Incorrect Options:

- Options A, B, and D are incorrect. These conditions will result in the right shift of the
oxygen-hemoglobin dissociation curve.

Solution for Question 35:

Correct Option B - Bronchi:

- The main site of airway resistance is the medium-sized bronchi and bronchi down to
approximately the seventh generation.
- According to Poiseuille's law, airway resistance increases with the fourth power of airway
radius. However, despite this, small airways contribute only about 10 to 15% to the total
airway resistance. This is due to the fact that most small airways are aligned in parallel, and
the total surface area of small airways is greater.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.
-

Solution for Question 36:

Correct Option A - Increased HbF:

- Factors causing a right shift of the oxygen-hemoglobin dissociation curve are: Increased temperature Decreased pH Increased PCO₂ Increased 2,3-DPG Hypoxia
- Increased temperature
- Decreased pH
- Increased PCO₂
- Increased 2,3-DPG
- Hypoxia
- Instead, increased HbF causes a left shift of the oxygen hemoglobin dissociation curve.
- Increased temperature
- Decreased pH
- Increased PCO₂
- Increased 2,3-DPG
- Hypoxia

Incorrect Options :

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 37:

Correct Option D - Type 2 pneumocyte:

- Type 2 pneumocytes, also called surfactant-producing cells, play a crucial role in the production and secretion of pulmonary surfactant.

- Pulmonary surfactant is a mixture of phospholipids, proteins, and other substances that coat the alveoli within the lungs. It reduces surface tension, preventing the alveoli from collapsing during exhalation and promoting efficient gas exchange.

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option A for an explanation.
-

Solution for Question 38:

Correct Option C - Expiratory reserve volume + residual volume:

- Functional residual capacity (FRC) is the volume of air that remains in the lungs after a normal tidal expiration.
- It is calculated by adding the expiratory reserve volume (the amount of air that can still be exhaled after a normal tidal expiration) to the residual volume and is equal to the 2.3L.

Incorrect Options:

Option A - Volume expired after normal inspiration: The volume expired after a normal inspiration is referred to as the tidal volume and is around 500 ml..

Option B - Volume that remains after forced expiration: The volume that remains after a forced expiration is the residual volume and is around 1.2 L.

Option D - Tidal volume + inspiratory reserve volume: The sum of the tidal volume and the inspiratory reserve volume represents inspiratory capacity and is around 3.5L.

Solution for Question 39:

Correct Option B - No exchange of O₂ & CO₂:

- When the ventilation-perfusion ratio (V/Q) is infinite, it means that there is adequate ventilation and no perfusion due to an emboli leading to no exchange of oxygen (O₂) and carbon dioxide (CO₂) occurring in the lungs.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 40:

Correct Option C - 250:

- The affinity of carbon monoxide for hemoglobin is approximately 250 times greater than the affinity of oxygen for hemoglobin. This means that when both gases are present, carbon monoxide will bind to hemoglobin preferentially, reducing the amount of oxygen that can bind and be transported to the body's tissues. This can lead to hypoxia (low oxygen levels) and can be life-threatening.

Incorrect Options:

- Options A, B, and D are incorrect.

Options A, B, and D are incorrect.

Solution for Question 41:

Correct Option B - Apneustic breathing:

- Apneustic breathing is characterized by prolonged inspiratory spasms or gasping breaths that resemble breath-holding with a brief expiration.
- The apneustic center results in prolonged inspiration by activating inspiratory neurons leading to deeper inspiration.

Incorrect Options:

Option A - Biot breathing: Biot breathing is characterized by normal 3 to 4 cycles of breathing followed by abrupt pauses or periods of apnea.

Option C - Cheyne-Stokes breathing: Cheyne-Stokes breathing is characterized by a cyclical pattern of apnea and shallow breathing due to an underlying pathology like congestive heart

failure.

Option D - Kussmaul breathing: Kussmaul breathing is a deep and labored breathing pattern often seen in individuals with diabeticketoacidosis.

GIT Hormones, Secretions, and Absorption Mechanisms

1. Which of the following secretions is NOT isotonic?

- A. Salivary secretion
 - B. Gastric secretion
 - C. Pancreatic secretion
 - D. Intestinal secretion
-

2. Which of the following statements about iron absorption is TRUE?

- A. Iron absorption starts in the jejunum.
 - B. Phytic acid in cereals enhances iron absorption.
 - C. Ferrous (Fe^{2+}) form of iron is absorbed more readily than the ferric (Fe^{3+}) form.
 - D. Hepcidin directly binds to transferrin to regulate iron levels.
-

3. Which vitamin is primarily absorbed in the ileum of the small intestine and requires an intrinsic factor for its absorption?

- A. Vitamin D
 - B. Vitamin C
 - C. Vitamin B12
 - D. Vitamin K
-

4. Which of the following is true regarding the process of protein digestion?

- A. Pepsin initiates the digestion of proteins in the small intestine.
 - B. Pepsin accounts for 90% of total protein digestion.
 - C. Collagen digestion starts in the small intestine.
 - D. The final step of protein digestion occurs inside the cytosol of Enterocytes.
-

5. A newborn presents with severe diarrhoea and dehydration shortly after birth. Stool analysis reveals undigested and unabsorbed carbohydrates. Which of the following mechanisms is most likely impaired in this patient?

- A. SGLT1-mediated absorption of glucose and galactose
 - B. GLUT5-mediated absorption of fructose
 - C. Peptidase-mediated breakdown
 - D. Lipase mediated breakdown
-

6. Which of the following intestinal cells are likely to have a high zinc concentration?

- A. Goblet cells
 - B. Enteroendocrine cells
 - C. Enterocytes
 - D. Paneth cells
-

7. A 50-year-old patient with obstructive jaundice from a common bile duct stone, confirmed by elevated bilirubin and alkaline phosphatase, underwent successful stone extraction via ERCP. Post-procedure, the patient received education on bile flow regulation. Which of the following is not associated with bile synthesis?

- A. Cholecystokinin

- B. Secretin
 - C. Cholesterol
 - D. Bile salts
-

8. Which hormone is primarily responsible for stimulating the secretion of pancreatic digestive enzymes in response to the presence of peptides and fatty acids in the small intestine?

- A. Insulin
 - B. Cholecystokinin (CCK)
 - C. Secretin
 - D. Gastrin
-

9. A 45-year-old male patient complains of frequent heartburn and bloating after meals, which worsens with large or spicy foods. He experiences relief with antacids, indicating excess gastric acid secretion. Which phase of gastric secretion is likely responsible for these symptoms?

- A. Cephalic Phase
 - B. Gastric Phase
 - C. Intestinal Phase
 - D. Basal Phase
-

10. Which of the following factors is associated with increased gastric acid secretion?

- A. Somatostatin
- B. Secretin

- C. Histamine
 - D. Calcitonin
-

11. Which of the following is true regarding the mechanism of gastric acid secretion?

- A. Release of K^+ into the lumen by proton pump.
 - B. Secretion by the parietal cells in the stomach lining.
 - C. Intake of H^+ by proton pump.
 - D. H^+ and Cl^- are released together into the lumen.
-

12. A 35-year-old female complains of excessive salivation irrespective of any sight or smell of food. Which of the following is true considering the autonomic regulation of salivary secretion?

- A. Parasympathetic increases secretion, and Sympathetic decreases secretion.
 - B. Parasympathetic decreases secretion, and Sympathetic increases secretion.
 - C. Both Sympathetic and Parasympathetic increase secretion.
 - D. Both Sympathetic and Parasympathetic decrease secretion.
-

13. Which cells in the duodenum and jejunum secrete the hormone that stimulates bile release and pancreatic enzyme secretion?

- A. G cells
- B. I cells
- C. S cells
- D. K cells

14. A 55-year-old woman presents with symptoms of acid reflux and indigestion due to high acid content in her stomach. Which of the following is the most alkaline secretion in the GIT that can neutralise the acid content in her stomach?

- A. Small intestinal juice
 - B. Pancreatic juice
 - C. Saliva
 - D. Gastric juice
-

15. A 25-year-old man presents with acute diarrhoea after returning from a trip from southeast Asia. Laboratory tests confirm an infection with enterotoxigenic Escherichia coli (ETEC). Which of the following peptides is most likely influenced by this bacterial toxin?

- A. Motilin
 - B. Guanylin
 - C. Vasoactive intestinal peptide (VIP)
 - D. Calcitonin gene receptor peptide (CGRP)
-

16. Which of the following GI hormones is not responsible for insulin secretion?

- A. GLP-1
 - B. GIP
 - C. VIP
 - D. CCK
-

17. A 52-year-old patient presents with symptoms of indigestion and steatorrhea. Laboratory tests suggest decreased bicarbonate secretion from the pancreas. Which Gastrointestinal hormone is most likely deficient in this patient?

- A. Cholecystokinin
 - B. Gastrin
 - C. Secretin
 - D. Motilin
-

18. Which of the following is NOT true regarding the effects of Cholecystokinin on GI motility?

- A. Inhibition of gastric motility
 - B. Contraction of sphincter of Oddi
 - C. Stimulation of intestinal peristalsis
 - D. Contraction of pyloric sphincter
-

19. Cholecystokinin (CCK) is a gastrointestinal hormone that acts on the pancreatic acinar cells to release all of the following except:

- A. Bicarbonate ions
 - B. Pancreatic lipase
 - C. Trypsinogen
 - D. Chymotrypsinogen
-

20. Which of the following is the primary stimulus for secretin release?

- A. Fatty acids in the small intestine
 - B. Amino acids in the small intestine
 - C. Acidic chyme entering the duodenum
 - D. Distention of the stomach
-

21. A 65-year-old woman presents with fatigue, shortness of breath, and a sore tongue. Blood tests reveal macrocytic anaemia with low vitamin B12 levels. Further testing shows the presence of anti-intrinsic factor antibodies. What would you expect regarding this patient's serum gastrin levels?

- A. Normal gastrin levels
 - B. Fluctuating gastrin levels
 - C. Elevated gastrin levels
 - D. Low gastrin levels
-

22. Which of the following is NOT a primary function of a hormone responsible for gastric acid production?

- A. Stimulation of gastric acid secretion
 - B. Trophic action on gastric mucosa
 - C. Relaxation of the lower esophageal sphincter
 - D. Stimulation of gastric motility
-

23. Which of the following does NOT stimulate the secretion of Gastrin hormone?

- A. Presence of proteins in the stomach

- B. Parasympathetic stimuli
 - C. Distension of the stomach
 - D. Low pH in the stomach
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	3
Question 4	4
Question 5	1
Question 6	4
Question 7	1
Question 8	2
Question 9	2
Question 10	3
Question 11	2
Question 12	3
Question 13	2
Question 14	2
Question 15	2
Question 16	3
Question 17	3
Question 18	2
Question 19	1
Question 20	3
Question 21	3
Question 22	3

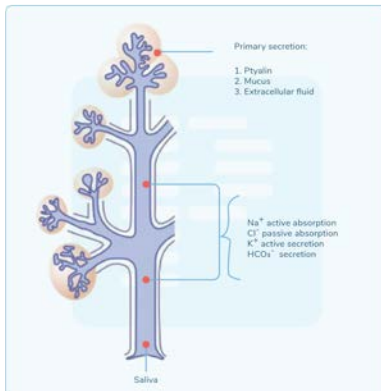
Solution for Question 1:

Correct Answer: A) Salivary secretion

Explanation:

Saliva is a hypotonic secretion, not an isotonic one.

- Saliva contains high concentrations of potassium (K^+) and bicarbonate (HCO_3^-) ions and lower concentrations of sodium (Na^+) and chloride (Cl^-) ions, making it a hypotonic solution.
- Except for saliva, all other gastrointestinal secretions, such as Gastric (Option B), Pancreatic (Option C) and intestinal secretions (Option D), have osmolality similar to osmolality plasma, i.e., isotonic.
- Saliva is secreted by acinar cells of salivary glands.



Solution for Question 2:

Correct Answer: C) Ferrous (Fe^{2+}) form of iron is absorbed more readily than the ferric (Fe^{3+}) form.

Explanation:

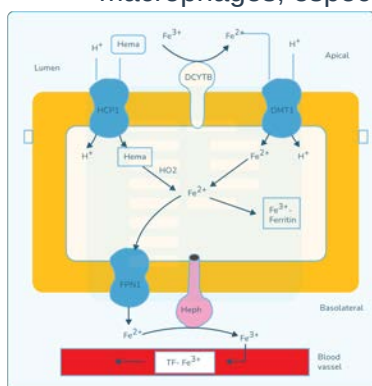
Most dietary iron is in the ferric (Fe^{3+}) form and must be reduced to the ferrous (Fe^{2+}) form by gastric secretions and reductase enzymes in the brush borders of enterocytes.

Absorption of Iron:

- Most dietary iron is ferric (Fe^{3+}), and the absorbed form is ferrous (Fe^{2+}).
- Fe^{3+} is reduced to Fe^{2+} by gastric secretions and reductase activity in enterocytes.
- Iron absorption occurs mainly in the duodenum via divalent metal transporter 1 (DMT1).
- Some Fe^{2+} is stored in ferritin within enterocytes, and the remaining is transported out by ferroportin 1.
- Hephaestin (Hp) aids basolateral transport.
- In plasma, Fe^{2+} converts to Fe^{3+} and binds to transferrin, a transport protein that delivers iron to various tissues.

Iron is stored in the body primarily as:

- Ferritin: A soluble protein complex that stores iron safely in the liver, spleen, bone marrow, and intestinal mucosa.
- Hemosiderin: A less soluble and more stable iron-storage complex, mainly found in macrophages, especially under iron overload conditions.



Iron is primarily absorbed in the jejunum (Option A): Most iron absorption occurs in the duodenum, not the jejunum. The duodenum is the first part of the small intestine immediately beyond the stomach. The acidic environment helps in the reduction of Fe^{3+} to Fe^{2+} , which is then absorbed.

Phytic acid in cereals enhances iron absorption (Option B): Phytic acid in cereals inhibits iron absorption. It binds to iron and forms insoluble complexes that the intestines cannot absorb. Oxalates and phosphates also inhibit iron absorption.

Hepcidin directly binds to transferrin to regulate iron levels (Option C): Hepcidin regulates iron levels by binding to ferroportin, causing its degradation. This decreases iron transport from enterocytes into the bloodstream.

Solution for Question 3:

Correct Answer: C) Vitamin B12

Explanation:

Vitamin B12 is primarily absorbed in the ileum of the small intestine and requires an intrinsic factor for its absorption.

Absorption of Vitamins:

Type	Vitamin	Primary Absorption Site	Mechanism of Absorption
Fat-Soluble	A, D, E, K (Options A and D)	Small intestine (Duodenum, mainly jejunum).	Require bile salts for emulsification. Incorporated into mixed micelles. Absorbed by passive diffusion into enterocytes. Packaged into chylomicrons for transport.
Water-Soluble	B Vitamins	Jejunum and Ileum. (B12 specifically in ileum)	Various mechanisms include: Active transport Passive diffusion Carrier-mediated transport Sodium-dependent transport B12 requires an intrinsic factor
C (Option B)	Sodium-dependent active transport at low concentrations. Passive diffusion at high concentrations.		

Solution for Question 4:

Correct Answer: D) The final step of protein digestion occurs inside the cytosol of Enterocytes.

Explanation:

The final step of protein digestion takes place inside the cytosol of Enterocytes, where they break down remaining dipeptides and tripeptides into amino acids.

Process of Protein Digestion:

1. In Stomach:

- Enzyme: Pepsin (Optimal pH: 2.0 to 3.0)
- Function: Pepsin initiates protein digestion (Option A) by breaking proteins into proteoses, peptones, and smaller polypeptides.
- Collagen Breakdown: Pepsin is crucial for digesting collagen in meats (Option C), allowing other digestive enzymes to access and break down meat proteins.
- Contribution: Pepsin accounts for 10-20% of total protein digestion. (Option B).

2. In Small Intestine: (Duodenum and Jejunum)

- Most of the Protein Digestion occurs in SI.
- Enzymes: Pancreatic proteolytic enzymes (trypsin, chymotrypsin, carboxypolypeptidase, elastase)
- Function: Trypsin and Chymotrypsin: Split proteins into small polypeptides. Carboxypolypeptidase: Cleaves individual amino acids from polypeptides. Elastase: Digests elastin fibres in meats.
- Trypsin and Chymotrypsin: Split proteins into small polypeptides.
- Carboxypolypeptidase: Cleaves individual amino acids from polypeptides.
- Elastase: Digests elastin fibres in meats.
- Outcome: Proteins are mostly reduced to dipeptides and tripeptides.
- Most amino acids are absorbed by sodium through a co-transporter molecule, which is part of secondary active transport. They can also be absorbed in the form of di or tripeptides by the PepT1 Transporter.
- Trypsin and Chymotrypsin: Split proteins into small polypeptides.
- Carboxypolypeptidase: Cleaves individual amino acids from polypeptides.
- Elastase: Digests elastin fibres in meats.

3. Final Digestion and Absorption: (Brush border of enterocytes in the small intestine villi)

- Enzymes: Aminopolypeptidase and dipeptidases
- Function: Brush Border Peptidases Intracellular Peptidases

- Brush Border Peptidases
- Intracellular Peptidases
- Brush Border Peptidases
- Intracellular Peptidases

Break down remaining dipeptides and tripeptides into amino acids within the enterocyte cytosol.
(Final Step)

- Absorption: Over 99% of absorbed protein digestion products are individual amino acids. Peptides and whole protein molecules are rarely absorbed, and they can cause allergic reactions.

Solution for Question 5:

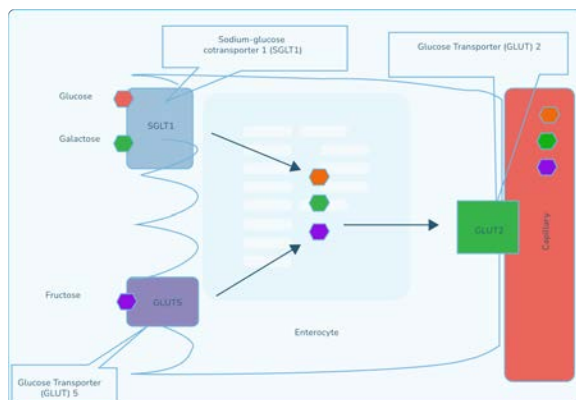
Correct Answer: A) SGLT1-mediated absorption of glucose and galactose

Explanation:

Glucose Galactose Malabsorption (GGM) is a rare genetic disorder caused by mutations in the SGLT1 gene, which encodes for the sodium-glucose cotransporter 1 (SGLT-1).

When glucose and galactose are not absorbed, they remain in the intestinal lumen, which draws water due to their osmotic properties resulting in osmotic diarrhoea and dehydration, as seen in this newborn.

Mechanism of Carbohydrate Absorption:



Transporters involved in the Mechanism of absorption of Carbohydrates:

- Glucose and Galactose Absorption (SGLT1): SGLT1 is a transporter found on the surface of intestinal cells. It actively transports glucose and galactose from the intestine into these cells using energy derived from sodium ions moving down their concentration gradient. (Secondary active transport).
- SGLT1 is a transporter found on the surface of intestinal cells.
- It actively transports glucose and galactose from the intestine into these cells using energy derived from sodium ions moving down their concentration gradient. (Secondary active transport).
- SGLT1 is a transporter found on the surface of intestinal cells.
- It actively transports glucose and galactose from the intestine into these cells using energy derived from sodium ions moving down their concentration gradient. (Secondary active transport).
- Fructose Absorption (GLUT5): GLUT5 (Option B) is another transporter located on the surface of intestinal cells. It facilitates the diffusion of fructose from the intestine into these cells without requiring energy (Facilitated diffusion). Fructose is transported independently of sodium.
- GLUT5 (Option B) is another transporter located on the surface of intestinal cells.
- It facilitates the diffusion of fructose from the intestine into these cells without requiring energy (Facilitated diffusion).
- Fructose is transported independently of sodium.
- GLUT5 (Option B) is another transporter located on the surface of intestinal cells.
- It facilitates the diffusion of fructose from the intestine into these cells without requiring energy (Facilitated diffusion).
- Fructose is transported independently of sodium.
- GLUT 2: It transports glucose, galactose, and fructose across the basolateral membrane of intestinal cells into the bloodstream. This ensures that absorbed glucose, galactose, and fructose enter the bloodstream for cell use throughout the body.
- It transports glucose, galactose, and fructose across the basolateral membrane of intestinal cells into the bloodstream.
- This ensures that absorbed glucose, galactose, and fructose enter the bloodstream for cell use throughout the body.
- It transports glucose, galactose, and fructose across the basolateral membrane of intestinal cells into the bloodstream.
- This ensures that absorbed glucose, galactose, and fructose enter the bloodstream for cell use throughout the body.

Peptidases (Option C): Peptidases are enzymes that break down proteins into peptides and amino acids, not carbohydrates.

Lipases (Option D): Lipases are associated with the breakdown of Fats. An impairment in lipase activity would lead to fat malabsorption and steatorrhea (fatty stools), not undigested carbohydrates.

Solution for Question 6:

Correct Answer: D) Paneth cells

Explanation:

Zinc plays a critical role in immune function and is particularly concentrated in Paneth cells within the small intestine.

- Paneth cells are specialised immune cells found in the intestinal crypts of Lieberkuhn, primarily in the small intestine.
- They are known for secreting antimicrobial peptides, such as defensins and lysozymes, which help protect the gut against infections.

Goblet cells (Option A):

- Goblet cells are specialised epithelial cells throughout the intestines that secrete mucus.
- This mucus forms a protective layer over the epithelial surface, safeguarding the intestinal lining from mechanical and chemical damage.

Enteroendocrine cells (Option B):

- Enteroendocrine cells are scattered throughout the gastrointestinal tract and produce hormones like serotonin, gastrin, and cholecystokinin. These hormones are involved in digestion, appetite regulation, and gut motility.

Enterocytes (Option C):

- Enterocytes are the absorptive cells lining the intestinal villi and are responsible for absorbing nutrients such as carbohydrates, proteins, fats, vitamins, and minerals.
-

Solution for Question 7:

Correct Answer: A) Cholecystokinin

Explanation:

CCK primarily stimulates the contraction of the gallbladder, which releases stored bile into the duodenum to aid in fat digestion. It also enhances pancreatic enzyme secretion. However, CCK does not directly regulate the synthesis of bile in the liver.

Regulation of Bile:

Hormonal Regulation	Secretin (Option B): Stimulates the secretion of a watery, bicarbonate-rich solution by epithelial cells in the bile ducts. Cholecystokinin (CCK): Induces gallbladder contraction and relaxation of the sphincter of Oddi, facilitating the release of concentrated bile into the duodenum in response to fatty foods.
Neural Regulation	Vagal Stimulation: Promotes weak gallbladder contraction. Enteric Nervous System: Nerve fibres in the intestine stimulate gallbladder motility and secretion.
Feedback Regulation	It is influenced by the presence of fats and acids in the duodenum. Bile salts are the most potent regulators of Bile synthesis, which is regulated by the feedback mechanism. (Option D)

Cholesterol (Option C): Cholesterol is a precursor for bile acids. In the liver, it is converted into primary bile acids (such as cholic acid and chenodeoxycholic acid), which are essential components of bile and are thus involved in bile synthesis.

Solution for Question 8:

Correct Answer: B) Cholecystokinin (CCK)

Explanation: When peptides and fatty acids enter the small intestine, Cholecystokinin (CCK) is the hormone primarily responsible for stimulating the secretion of pancreatic digestive enzymes.

Pancreatic secretion is regulated by three key stimuli:

- Acetylcholine, mainly released from vagal nerve endings in the enteric nervous system, plays a minor role in stimulating both acinar and duct cells.
- Cholecystokinin (CCK), secreted by the duodenal and upper jejunal mucosa in response to amino acids and fatty acids, is the primary stimulus for acinar cells to produce enzymes.
- Secretin, secreted by the duodenal and jejunal mucosa in response to acidic food in the small intestine, is the most important stimulus for duct cells to produce bicarbonate.

Pancreatic Digestive Enzymes:

Pancreatic secretion contains multiple enzymes for digesting all of the three major types of food (proteins, carbohydrates, fats)

- Proteins: Trypsin, chymotrypsin, carboxypolypeptidase
- Carbohydrates: Pancreatic Amylase
- Fats: Pancreatic Lipase, Cholesterol Esterase, Phospholipase
- Acidity Regulation: Bicarbonate

Solution for Question 9:

Correct Answer: B) Gastric Phase

Explanation:

This patient's symptoms of heartburn and bloating after meals, exacerbated by large or spicy foods, suggest excessive gastric acid production. The Gastric phase of secretion accounts for about 60% of total gastric secretion.

Phases of Gastric secretion:

Cephalic Phase	Trigger: Initiated by sensory stimuli such as sight, smell, thought, or taste of food. Origin: Neurogenic signals originate in the cerebral cortex, amygdala, and hypothalamus and are transmitted via vagus nerves to stimulate gastric secretion. Contribution: Accounts for approximately 30% of gastric secretion Associated with meal anticipation and preparation.
----------------	--

Gastric Phase	Trigger: Begins when food enters the stomach. Mechanisms: - Long vasovagal reflexes between the stomach and brain. - Local enteric reflexes within the stomach. - Gastrin release, which stimulates gastric juice secretion. Duration: It lasts several hours while food remains in the stomach. Contribution: Accounts for about 60% of total gastric secretion.
Intestinal phase	Trigger: Initiated by food reaching the duodenum of the small intestine. Mechanism: Small amounts of gastrin released by duodenal mucosa continue to stimulate gastric juice secretion. Contribution: Contributes to approximately 10% of the total acid response to a meal, ensuring continued digestive support as food progresses through the digestive tract.

Solution for Question 10:

Correct Answer: C) Histamine

Explanation:

Histamine increases gastric acid secretion by acting on H₂ receptors in the parietal cells of the stomach.

Regulation of gastric acid secretion:

1. Histamine:

- It is released from the stomach's enterochromaffin-like (ECL) cells and binds to H₂ receptors on parietal cells.
- This activates the G_s protein, raising cAMP levels and activating protein kinase A (PKA), which enhances the proton pump activity and increases gastric acid secretion.

2. Acetylcholine:

- It is released from the vagus nerve and acts on M₃ receptors of parietal cells.
- This interaction activates the G_q protein, leading to an increase in intracellular calcium.
- The rise in calcium activates protein kinase C (PKC) and calmodulin, which stimulate the proton pump, increasing acid secretion.

3. Gastrin:

- It is produced by G cells in the stomach's antrum and binds to CCK-B2 receptors on parietal and Enterochromaffin (ECL) cells.
- On parietal cells, this activates the Gq protein, raising intracellular calcium levels and activating protein kinase C (PKC), which stimulates the proton pump, increasing acid secretion.
- Gastrin stimulates ECL cells to release histamine, further enhancing acid secretion.

Factors influencing gastric acid secretion:

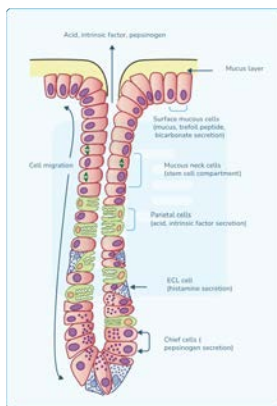
	Factors increasing gastric acid secretion	Factors inhibiting gastric acid secretion
Luminal	Peptides and amino acids. Stomach distention	Acidic pH Somatostatin (Option A)
Neural	Increased vagal discharge via GRP (Gastrin-Releasing Peptide)	—
Blood burn	Calcium Epinephrine	Secretin (Option B) GIP (Gastric Inhibitory Peptide) VIP (Vasoactive Intestinal Peptide) Glucagon Calcitonin (Option D).

Solution for Question 11:

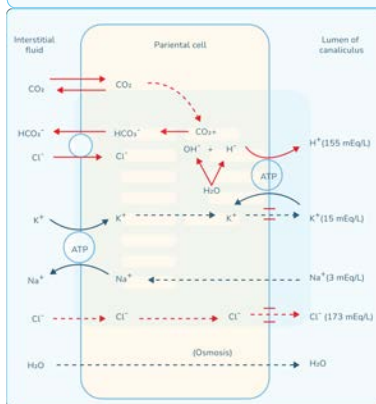
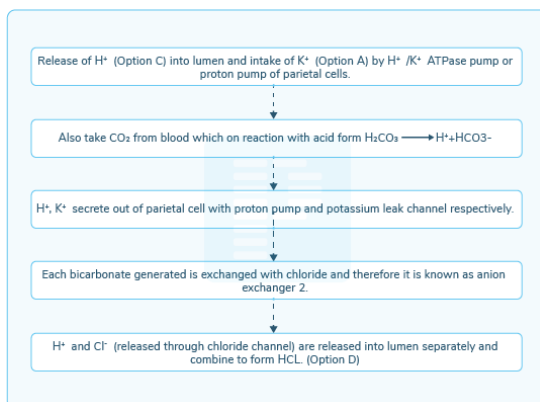
Correct Answer: B) Secretion by the parietal cells in the stomach lining

Explanation: Parietal cells or oxyntic cells, which are present in the stomach mucosa, primarily secrete gastric acid, i.e., Hydrochloric acid (HCl).

- HCl helps kill bacteria and digest proteins.
- Stomach mucosa has two significant types of tubular glands



Mechanism of Gastric acid secretion:



After meals, H⁺ and Chloride ion secretion is enhanced, and bicarbonate molecules enter the blood, making it alkaline. This is known as postprandial alkaline tide.

Solution for Question 12:

Correct Answer: C) Both Sympathetic and Parasympathetic increase secretion.

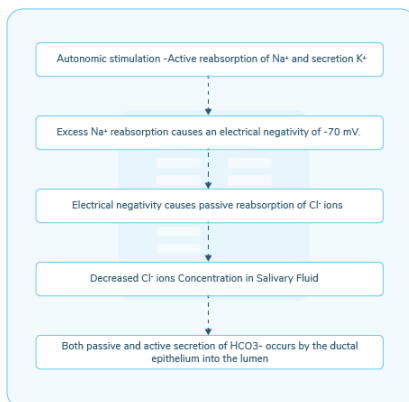
Explanation:

Both Sympathetic and Parasympathetic increase salivary secretion in different pathways.

Autonomic regulation of salivary secretion

Aspect	Sympathetic System	Parasympathetic System
Salivary Secretion	Produces less saliva	Produces a larger volume of saliva.
Saliva Characteristics	Mucus-rich and thick.	Watery and less viscous.
Effect on Blood Vessels	No significant effect.	Dilates blood vessels, increasing gland nutrition
Saliva Composition	Less bicarbonate and K ⁺	Higher concentrations of K ⁺ and bicarbonate
Saliva Tonicity	Generally more viscous and less hypotonic.	Hypotonic

Mechanism of Salivary secretion:



Solution for Question 13:

Correct Answer: B) I cells

Explanation:

Hormone	Secreting Cells	Location	Function
CCK (Cholecystokinin)	I cells (Option B)	Duodenum Jejunum	Stimulates: Pancreatic enzyme secretion Pancreatic bicarbonate secretion Gallbladder contraction (Release of bile) Growth of exocrine pancreas Inhibits: Gastric emptying
Gastrin	G cells (Option A)	Stomach (antrum) Duodenum	Stimulates: Gastric acid secretion Mucosal growth
Secretin	S cells (Option C)	Duodenum	Stimulates: Pepsin secretion Pancreatic bicarbonate secretion Biliary bicarbonate secretion Growth of exocrine pancreas Inhibits: Gastrin release and gastric acid secretion
GIP (Glucose-dependent Insulinotropic Peptide)	K cells (Option D)	Duodenum Jejunum	Stimulates: Insulin release from pancreatic β -cells β -cell proliferation and survival Glucose uptake in adipose tissue Inhibits: Gastric acid secretion Gastric motility

Solution for Question 14:

Correct Answer: B) Pancreatic juice

Explanation:

Among the given options, Pancreatic juice is more alkaline as its secretions contain high concentrations of bicarbonate ions.

But in general, Brunner's gland secretion is more alkaline than pancreatic juice.

Type of secretion	pH
Saliva	6.0-7.0 (Option C)
Gastric secretion	1.0-3.5 (Option D)
Pancreatic secretion	8.0-8.3 (Option B)
Bile	7.8
Small intestine secretion	7.5-8.0 (Option A)
Brunner's gland secretion	8.0-8.9
Large intestinal secretion	7.5-8.0

Solution for Question 15:

Correct Answer: B) Guanylin

Explanation:

Guanylin is the peptide hormone that is influenced by Enterotoxigenic Escherichia coli (ETEC).

- Guanylin promotes fluid absorption in the small and large intestines.
- Serum ProGuanylin levels indicate intestinal guanylin levels and reflect intestinal fluid balance.
- Heat-Stable Enterotoxin of E. coli: Activates guanylyl cyclase enzyme, thus increasing cGMP levels. It reduces intestinal guanylin levels. This results in excessive water and electrolyte secretion into the GIT lumen, leading to secretory diarrhoea.
- Activates guanylyl cyclase enzyme, thus increasing cGMP levels.
- It reduces intestinal guanylin levels.
- This results in excessive water and electrolyte secretion into the GIT lumen, leading to secretory diarrhoea.
- Activates guanylyl cyclase enzyme, thus increasing cGMP levels.
- It reduces intestinal guanylin levels.

- This results in excessive water and electrolyte secretion into the GIT lumen, leading to secretory diarrhoea.

Motilin (Option A): This is involved in stimulating gastric and small intestine motility during fasting. It helps initiate and regulate migratory motor complexes and is not typically influenced by *E. coli*.

Vasoactive intestinal peptide (VIP) (Option C): VIP can induce intestinal water and electrolyte secretion, but it is not specifically targeted by *E. coli* enterotoxins. VIP-induced diarrhoea is seen in VIPomas.

Calcitonin gene receptor peptide (CGRP) (Option D): CGRP is involved in somatostatin release, which is an inhibitor of GIT secretion; it also causes smooth muscle contraction

Solution for Question 16:

Correct Answer: C) VIP

Explanation: Vasoactive intestinal peptide is not responsible for insulin secretion from the pancreas.

GI Hormones associated with the secretion of insulin from the pancreas:

HORMONE	STIMULI	CELLS
GIP (Gastric inhibitory peptide) (A.k.a Glucose-dependent insulinotropic peptide)	Protein, Fats, Carbohydrates.	K cells of duodenum and jejunum.
Glucagon-like peptide	L cells in the small intestine.	
CCK	Protein, Fats, Acid	I cells in duodenum and jejunum.
Gastrin	Protein, Gastric distension, Vagal stimulation	G cells in gastric antrum, duodenum and jejunum.

Solution for Question 17:

Correct Answer: C) Secretin

Explanation:

- Secretin is released from the S cells in response to acid and fat in the duodenum and is responsible for bicarbonate secretion in the GIT.
- Functions: Pepsin secretion - helps in protein digestion (patient is suffering from indigestion because of deficiency of this function) Pancreatic and biliary bicarbonate secretion - neutralises the acid in the duodenum Stimulates hepatic ductal secretion Growth of exocrine pancreas Inhibition of gastrin release and direct gastric acid secretion It also inhibits gastric emptying and small intestinal motility.
- Pepsin secretion - helps in protein digestion (patient is suffering from indigestion because of deficiency of this function)
- Pancreatic and biliary bicarbonate secretion - neutralises the acid in the duodenum
- Stimulates hepatic ductal secretion
- Growth of exocrine pancreas
- Inhibition of gastrin release and direct gastric acid secretion
- It also inhibits gastric emptying and small intestinal motility.
- Pepsin secretion - helps in protein digestion (patient is suffering from indigestion because of deficiency of this function)
- Pancreatic and biliary bicarbonate secretion - neutralises the acid in the duodenum
- Stimulates hepatic ductal secretion
- Growth of exocrine pancreas
- Inhibition of gastrin release and direct gastric acid secretion
- It also inhibits gastric emptying and small intestinal motility.

Solution for Question 18:

Correct Answer: B) Contraction of sphincter of Oddi

Explanation:

Cholecystokinin is the major stimulus for gallbladder emptying and pancreatic enzyme secretion; thus, it causes relaxation of the sphincter of Oddi.

Effects of Cholecystokinin on GI motility:

- Inhibition of gastric motility (Option A).
 - Contraction of pyloric sphincter (Option D) . This allows adequate time for the fatty food in the upper SI to get emulsified and digested.
 - This allows adequate time for the fatty food in the upper SI to get emulsified and digested.
 - Stimulation of gallbladder emptying. This expels bile into the small intestine, helping to emulsify fat and aid in fat digestion.
 - This expels bile into the small intestine, helping to emulsify fat and aid in fat digestion.
 - Relaxation of sphincter of Oddi.
 - Stimulation of intestinal peristalsis (Option C) .
 - This allows adequate time for the fatty food in the upper SI to get emulsified and digested.
 - This expels bile into the small intestine, helping to emulsify fat and aid in fat digestion.
-

Solution for Question 19:

Correct Answer: A) Bicarbonate ions

Explanation:

Cholecystokinin causes the release of pancreatic enzymes and not electrolytes, i.e. bicarbonate.

Pancreatic secretion of bicarbonate is stimulated by Secretin and not by CCK.

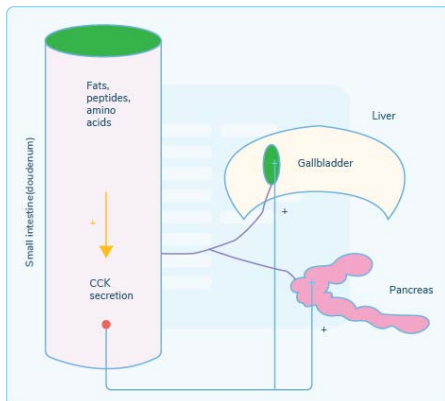
Pancreatic secretion is regulated by 3 basic stimuli:

- Acetylcholine: Released from the vagus nerve and other cholinergic nerves in the enteric system.
- Cholecystokinin: Released from the duodenum and upper jejunal mucosa when food enters the duodenum.
- Secretin: Released by duodenal and upper jejunal mucosa when acidic food enters the duodenum.

Actions of CCK:

- Stimulation of Pancreatic enzyme secretion-Amylase, Lipase and Trypsinogen (Options B, C & D) Growth of exocrine pancreas. Gallbladder contraction.
- Pancreatic enzyme secretion-Amylase, Lipase and Trypsinogen (Options B, C & D)
- Growth of exocrine pancreas.

- Gallbladder contraction.
- Relaxation of sphincter of Oddi.
- Inhibition of gastric emptying.
- Pancreatic enzyme secretion-Amylase, Lipase and Trypsinogen (Options B, C & D)
- Growth of exocrine pancreas.
- Gallbladder contraction.



Solution for Question 20:

Correct Answer: C) Acidic chyme entering the duodenum

Explanation:

Secretin's release is primarily triggered by the low pH (high acidity) of the chyme.

Acidic chyme (pH < 4.5) enters the duodenum → S cells in duodenal mucosa detect low pH → Secretin is released into the bloodstream.

Stimulus for Secretin and CCK:

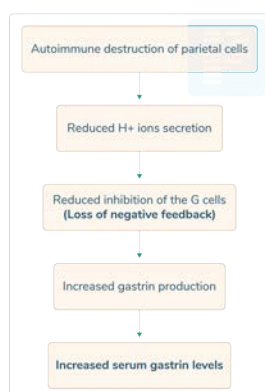
Hormones	Stimuli
Secretin	Primary: Acidic chyme entering the duodenum (pH < 4.5) Secondary: Fatty acids in the duodenum Peptides
Cholecystokinin (CCK)	Partially digested proteins (peptides) Fatty acids in the small intestine. Amino acids in the small intestine.

Solution for Question 21:

Correct Answer: C) Elevated gastrin levels

Explanation:

Pernicious anaemia is an autoimmune condition in which the body produces antibodies against parietal cells. These cells in the stomach produce intrinsic factors and hydrochloric acid.



In pernicious anaemia, gastrin levels are typically severely elevated, often 10-20 times the normal range.

Clinical implications:

Severely elevated gastrin levels cause hyperplasia of enterochromaffin-like cells in the stomach, which can potentially progress to gastric carcinoid tumours in some cases.

Solution for Question 22:

Correct Answer: C) Relaxation of the lower esophageal sphincter

Explanation:

Gastrin causes the contraction (not relaxation) of the lower oesophageal sphincter, which helps prevent the reflux of stomach contents into the oesophagus.

Functions of Gastrin:

Gastric Secretion	Stimulates gastric acid secretion from parietal cells. Promotes pepsinogen secretion from chief cells.
Gastric Motility	Enhances gastric motility and Promotes gastric emptying.
Trophic Effects	Stimulates growth and proliferation of gastric mucosa.
Sphincter Control	It causes contraction of the lower oesophageal sphincter (LES) and prevents reflux.

Solution for Question 23:

Correct Answer: D) Low pH in the stomach

Explanation:

Increased acidity (Low pH) doesn't stimulate the secretion of Gastrin; rather, it inhibits the secretion through a negative feedback mechanism.

Stimuli for Gastrin Release	
Chemical Stimuli	Proteins and amino acids in the stomach (Option A) Peptides Calcium ions
Mechanical Stimuli	Distension of the stomach (Option C)
Neural Stimuli	Vagal nerve stimulation Parasympathetic nervous system activation (Option B)
Other Factors	Caffeine Alcohol Certain medications (e.g., antacids)
Absence of Inhibitory Factors	Decreased acidity in the stomach (high pH)

GIT Motility

1. All the factors inhibit gastric emptying except?

- A. Secretin
 - B. A large volume of solid meals
 - C. Hypertonic solution
 - D. Hypotonic solution
-

2. Which of the following correctly describes the roles of the myenteric (Auerbach's) plexus and the submucosal (Meissner's) plexus in the enteric nervous system?

- A. Myenteric: enzyme secretion; Submucosal: motility
 - B. Myenteric: motility; Submucosal: enzyme secretion and blood flow
 - C. Both: motility
 - D. Submucosal: rhythmic contractions; Myenteric: mucosal surface area
-

3. A 3-day-old newborn has not passed meconium since birth. The baby has a distended abdomen and frequent vomiting, which sometimes contains bile. Upon physical examination, there was a lack of stool in the rectal vault. An abdominal X-ray reveals a dilated colon with a narrowed segment near the rectum. What is the cause of this condition?

- A. Meconium ileus
- B. Increased enteric ganglion cells
- C. Intestinal atresia
- D. Endothelial B receptor gene mutation

4. Which of the following is responsible for the basal electrical rhythm in the gastrointestinal tract (GIT)?

- A. Parietal cells
 - B. Chief cells
 - C. Interstitial cells of Cajal
 - D. Enteroendocrine cells
-

5. What is the movement responsible for the movement of food bolus in the gastrointestinal tract?

- A. Segmentation
 - B. Peristalsis
 - C. Tonic contractions
 - D. Mass movement
-

6. Which reflex is responsible for initiating mass movements in the colon?

- A. Enterogastric reflex
 - B. Gastrocolic reflex
 - C. Rectocolic reflex
 - D. Colorectal reflex
-

7. What type of motility retards the movement of intestinal contents to provide time for digestion and absorption?

- A. Peristalsis
 - B. Segmentation
 - C. Mass movement
 - D. Propulsion
-

8. Which of the following statements about the migrating motor complex (MMC) is correct?

- A. The MMC occurs during periods of digestion to facilitate nutrient absorption.
 - B. The MMC is initiated by motilin and consists of three phases.
 - C. The MMC contractions migrate orally at about 10 cm/min.
 - D. The MMC occurs at intervals of approximately 50 minutes in the interdigestive state.
-

9. Which of the following increases intestinal motility?

- A. Dehydration
 - B. High fibre diet
 - C. Acid reflex
 - D. Opioids
-

10. Which segment of the gastrointestinal tract has the longest transit time?

- A. Duodenum
- B. Jejunum

C. Colon

D. Ileum

11. Which of the following mediates the receptive relaxation of the stomach?

A. Sympathetic nervous system

B. Vagus nerve

C. Enteric nervous system

D. Hormonal control

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	4
Question 4	3
Question 5	2
Question 6	2
Question 7	2
Question 8	2
Question 9	2
Question 10	3
Question 11	2

Solution for Question 1:

Correct Answer: D) Hypotonic solution in duodenum

Explanation:

Hypotonic solutions do not inhibit gastric emptying. In fact, they may even promote faster gastric emptying due to their lower osmolarity.

Factors Promoting Gastric Emptying:

Factor	Mechanism
Vagal stimulation	Stimulates antral contraction, promoting gastric emptying
Liquid meal	Faster gastric emptying due to lower viscosity and faster transit through the stomach

Motilin	Stimulates antral contraction and gastric emptying, especially during the interdigestive state
---------	--

Factors Inhibiting Gastric Emptying:

Factor	Mechanism
Acid in the duodenum	Releases secretin, inhibiting antral contraction and stimulating pyloric sphincter contraction
Products of fat digestion	Releases CCK and GIP, inhibiting gastric emptying
Hypertonic solutions in the duodenum	Releases hormones inhibiting gastric emptying
Products of protein digestion	Releases gastrin, CCK, and GIP, inhibiting gastric emptying
Large meal volume (solids)	Prolongs gastric emptying time
Stretching of duodenum	Initiates enterogastric reflex, inhibiting gastric emptying
Sympathetic stimulation	Decreased gastric motility (by norepinephrine) and increased pyloric sphincter tone.
Hormones (CCK, GIP, secretin)	-

Solution for Question 2:

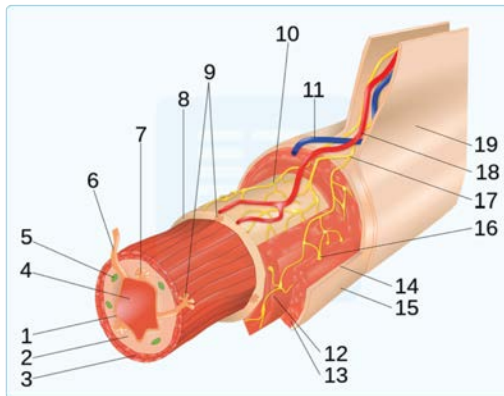
Answer: B) Myenteric: motility; Submucosal: enzyme secretion and blood flow

Explanation:

The enteric nervous system is composed of two well-organised neural plexus:-

Feature	Myenteric (Auerbach's) Plexus	Submucosal (Meissner's) Plexus
Location	Between the longitudinal and circular muscle layers	Between the circular muscle and luminal mucosa
Primary Function	Regulates gastrointestinal motility	Controls enzyme secretion & blood flow
Control Over	Peristalsis & muscle contractions	Secretion, absorption, and blood flow
Involvement in the digestive Process	Movement of food through the GI tract	Regulation of the internal environment of the GI tract

The image below shows the layers of the GI tract.



1. Mucosa: Epithelium	6. Duct of gland outside tract	11. Vein	16. Auerbach's myenteric plexus
2. Mucosa: Lamina propria	7. Gland in mucosa	12. Muscularis: Circular muscle	17. Nerve
3. Mucosa: Muscularis mucosae	8. Submucosa	13. Muscularis: Longitudinal muscle	18. Artery
4. Lumen	9. Glands in submucosa	14. Serosa: Areola connective tissue	19. Mesentery
5. Lymphatic tissue	10. Meissner's submucosal plexus	15. Serosa: Epithelium	

Solution for Question 3:

Correct Answer: D) Endothelial B receptor gene mutation

Explanation:

Hirschsprung's disease (Aganglionic megacolon):

It is due to congenital absence of enteric ganglion cells in both myenteric and meissner's plexus in segment of distal colon due to failure of cranial to caudal migration of neural crest cells during development.

Causes:

- Endothelial B receptor gene mutation.
- Neural crest cell migration defect.

- RET proto-oncogene mutation.

Symptoms:

- Failure to pass meconium within the first 48 hours of life.
- Abdominal distention.
- Bilious vomiting.
- Constipation.
- Failure to thrive.

Clinical Features:

- Lack of stool in the rectal vault on digital rectal examination.
- Explosive release of gas and stool upon withdrawal of the examining finger (squirt sign).
- Chronic constipation in older children.

Solution for Question 4:

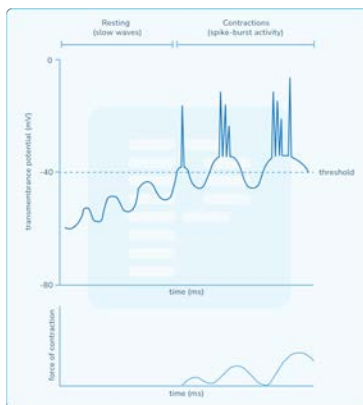
Correct Answer: C) Interstitial cells of Cajal

Explanation:

Basal Electrical Rhythm:

- Except in the oesophagus and proximal portion of the stomach, the smooth muscle of the GIT has spontaneous rhythmic fluctuations in membrane potential between -65 and -45 mV.
- The interstitial cells of Cajal are stellate mesenchymal pacemaker cells with smooth muscle-like features that send long, multiply branched processes into the intestinal smooth muscle, generating the basal electrical rhythm (BER).
- The function of BER is to coordinate peristaltic and other motor activity, such as setting the rhythm of segmentation and contraction only during the depolarising part of the wave.
- Spike potentials superimposed on the most depolarising portions of the BER waves do increase muscle tension (figure below).
- The depolarising portion of each spike is due to calcium influx, and the repolarising portion is due to potassium efflux.
- Acetylcholine increases the number of spikes and tension of the smooth muscle, whereas Epinephrine decreases them.

- The rate of BER: Stomach: 4/min. Distal ileum: 8/min. Duodenum: 12/min.
- Stomach: 4/min.
- Distal ileum: 8/min.
- Duodenum: 12/min.
- The BER rate in the colon is about 2/min at the cecum and about 6/min at the sigmoid colon.
- Stomach: 4/min.
- Distal ileum: 8/min.
- Duodenum: 12/min.



Solution for Question 5:

Correct Answer: B) Peristalsis

Explanation:

Peristalsis:

- Reflex response is initiated by the stretching of the gut wall due to the contents of the lumen.

Location:

- It occurs throughout the gastrointestinal tract, from the esophagus to the rectum.

Mechanism:

- Stretching triggers circular contraction behind the stimulus and relaxation in front of it.

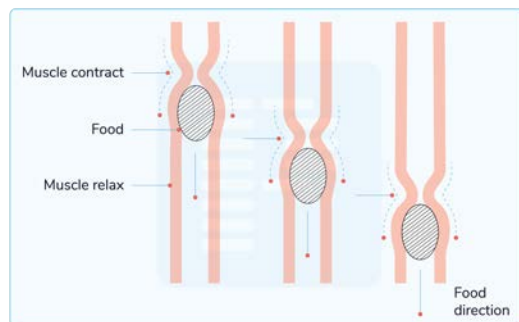
- The wave of contraction moves in an oral-to-caudal direction.
- Propels the contents of the lumen forward at rates of 2 to 25 cm/s.

Regulation:

- Peristaltic activity is influenced by autonomic input but occurs independently of extrinsic innervation.
- Local stretch releases serotonin, activating sensory neurons that stimulate the myenteric plexus.
- Cholinergic neurons in the myenteric plexus: In retrograde direction: release substance P and acetylcholine, causing contraction behind the bolus. In anterograde direction: activate the neurons releasing nitric oxide (NO) and vasoactive intestinal polypeptide (VIP), causing relaxation ahead of the bolus (Figure below).
- In retrograde direction: release substance P and acetylcholine, causing contraction behind the bolus.
- In anterograde direction: activate the neurons releasing nitric oxide (NO) and vasoactive intestinal polypeptide (VIP), causing relaxation ahead of the bolus (Figure below).
- In retrograde direction: release substance P and acetylcholine, causing contraction behind the bolus.
- In anterograde direction: activate the neurons releasing nitric oxide (NO) and vasoactive intestinal polypeptide (VIP), causing relaxation ahead of the bolus (Figure below).

Functional Integration:

- Peristalsis is an example of the integrated activity of the enteric nervous system.



Solution for Question 6:

Correct Answer: B) Gastrocolic reflex

Explanation:

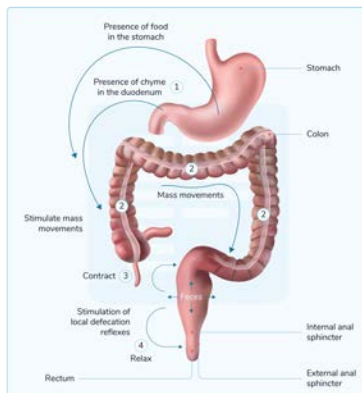
Mass Movement in Colon:

Coordinated contractions of the colon that propel faecal material towards the rectum.

Gastrocolic reflex:

The gastrocolic reflex is initiated when food accumulates in the stomach.

- This reflex increases the motility of the colon in response to the stretching of the stomach walls after food intake.
- Therefore, the urge for defecation is usually enhanced after eating a large meal.
- The gastrocolic reflex can be more active in children, often leading to a more immediate urge to defecate after eating.
- While gastrin, a hormone secreted by the stomach in response to gastric distension, plays a role, neural factors also contribute to the reflex.
- The involvement of the enteric and autonomic nervous systems is significant in mediating this reflex.



Solution for Question 7:

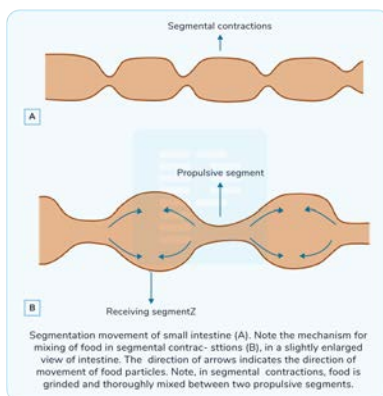
Correct Answer: B) Segmentation

Explanation:

Segmentation: The enteric nervous system promotes a motility pattern that is related to peristalsis but retards the movement of intestinal contents along the length of the intestinal tract to provide time for digestion and absorption.

This is the most common movement of the small intestine.

- It is characterised by closely spaced contractions of the circular muscle layer. These contractions divide the small intestine into many segments.
- These contractions divide the small intestine into many segments.
- The rate of segmental contraction is the same as the frequency of slow waves. Duodenum: 18/min. Jejunum: 15/min. Ileum: 12/min.
- Duodenum: 18/min.
- Jejunum: 15/min.
- Ileum: 12/min.
- Segmentation contractions effectively mix chyme with pancreatic and intestinal secretions.
- Two nearby propulsive segments force the chyme toward each other into the receiving segment, which efficiently helps in mixing.
- They also bring the fresh chyme into contact with the mucosal surface.
- Therefore, segmentation movements are also called mixing movements. They help in the digestion and absorption of nutrients.
- These contractions divide the small intestine into many segments.
- Duodenum: 18/min.
- Jejunum: 15/min.
- Ileum: 12/min.



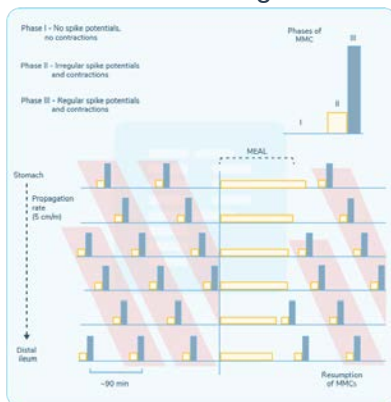
Solution for Question 8:

Correct Answer: B) The MMC is initiated by motilin and consists of three phases.

Explanation:

Migrating Motor Complex (MMC):

- Cyclic, recurring motility pattern occurs in the stomach and small intestine during fasting between meals.
- The MMC occurs during fasting periods between digestion, not during active digestion.
- Its primary role is to clear the stomach and small intestine of luminal contents in preparation for the next meal rather than facilitating nutrient absorption.
- The hormone motilin initiates the MMC.
- It consists of three distinct phases. Quiescent period (phase I) Period of irregular electrical and mechanical activity (phase II) Burst of regular activity (phase III).
- Quiescent period (phase I)
- Period of irregular electrical and mechanical activity (phase II)
- Burst of regular activity (phase III).
- The MMC contractions migrate aborally (away from the mouth, towards the rectum) at about 5 cm/min, not orally.
- The MMC occurs at intervals of approximately 90-100 minutes in the interdigestive state.
- Gastric secretion, bile flow and pancreatic secretion increased during each MMC.
- When a meal is ingested, motilin secretion is suppressed, and MMC is abolished until digestion and absorption are complete.
- Quiescent period (phase I)
- Period of irregular electrical and mechanical activity (phase II)
- Burst of regular activity (phase III).



Solution for Question 9:

Correct Answer: B) High fibre diet

Explanation:

Factors Promoting vs Inhibiting Intestinal Motility:

Factor	Promoting intestinal motility	Inhibiting intestinal mobility
Dietary	High fibre diet	Low fibre diet
Hydration	Adequate hydration	Dehydration
Hormones	Gastrin and motilin	Secretin and somatostatin
Neural	Parasympathetic	Sympathetic
Physical activity	Regular exercise	Sedentary lifestyle
Medications	Prokinetics	Opioids and anticholinergics
GI reflexes	Gastrocolic reflex	Enterogastric reflex
Food intake	Ingestion	Fasting
Electrolyte balance	Adequate potassium levels	Electrolyte imbalances
Health conditions	Infections (e.g. gastroenteritis)	Intestinal obstructions and post-surgical states

Solution for Question 10:

Correct Answer: C) Colon

Explanation:

The large intestine (colon) has the longest transit time in the gastrointestinal tract.

It is responsible for absorbing water and electrolytes from the remaining indigestible food matter and compacting it into feces. Water absorption and stool formation take longer than other gastrointestinal tract segments.

Transit Time in Small Intestine and Colon:

- The First part of the meal reaches the cecum in about 4 hours.
 - All undigested portions have entered the colon in 8 or 9 hours.
 - First remnants of the meal traverse the first third of the colon in 6 hours.
 - The second third of the colon in 9 hours and reach the terminal part of the colon (Sigmoid colon) in 12 hours.
 - From the sigmoid colon to the anus, transport is much slower.
 - 70% of the meal is recovered in stool in 72 hours, but total recovery requires more than a week.
 - Transit time can be observed by monitoring the progress of a small pill that contains sensors and a miniature radio transmitter.
-

Solution for Question 11:

Correct Answer: B) Vagus nerve

Explanation:

Receptive relaxation is a physiological process in which the stomach, specifically the fundus and upper portion of the body, relaxes and accommodates food without significantly increasing internal pressure.

Mechanism:

- This process is vagally mediated and triggered by the movement of the pharynx and oesophagus.

Neural Pathway:

- The brain responds by sending signals back through vagal efferent fibres, releasing neurotransmitters like nitric oxide (NO) and vasoactive intestinal peptide (VIP).
- They relax the stomach muscles, allowing them to expand with minimal pressure increase and aiding digestion.

Subsequent Events:

- Peristalsis begins in the lower portion of the stomach, mixing and grinding the food before it passes through the pylorus and enters the duodenum.
- Contraction of the distal stomach caused by each wave is called antral systole and can last up to 10 seconds. Waves occur 3-4 times per minute.
- Contraction of the antrum is followed by sequential contraction of the pyloric region and duodenum.
- In the antrum, partial contraction ahead of the advancing gastric content prevents solid masses from entering the duodenum.
- Regurgitation from the duodenum does not normally occur because the contraction of the pyloric segment tends to persist slightly longer than that of the duodenum.

Adaptive relaxation:

- The dilation of the stomach accommodates the food particles that have already entered the stomach.
 - The local vagovagal reflex regulates it.
-

Previous Year Questions

1. A patient who has undergone gastrectomy (surgical removal of part or all of the stomach) is at risk of deficiencies in certain nutrients. Which of the following vitamins is most likely to require supplementation in such a patient?

- A. Vitamin C
 - B. Vitamin D
 - C. Vitamin B12
 - D. Vitamin A
-

2. Which substance inhibits the iron transporter on the basal side of the enterocyte?

- A. Hepcidin
 - B. Ferritin
 - C. Transferrin
 - D. Ferroportin
-

3. Least frequency of slow waves is seen in?

- A. Duodenum
 - B. Ileum
 - C. Jejunum
 - D. Stomach
-

4. Which of the following functions is NOT attributed to the intestinal epithelium?

- A. Absorption of the ions
 - B. Bile salts secretion
 - C. Mucous secretion
 - D. Secretion of digestive enzyme
-

5. Which co-transporter is primarily responsible for facilitating glucose absorption across the basolateral membrane of an enterocyte in this scenario?

- A. SGLT1
 - B. GLUT2
 - C. GLUT5
 - D. SGLT2
-

6. Which of the following cells is responsible for destroying bacterial foreign bodies in the liver sinusoids?

- A. Hepatocytes
 - B. Ito cells
 - C. Sinusoidal endothelial cells
 - D. Kupffer cells
-

7. Cells lining the isthmus of the gastric pits:

- A. Chief cells

- B. Stem cells
 - C. Parietal cells
 - D. Mucous cells
-

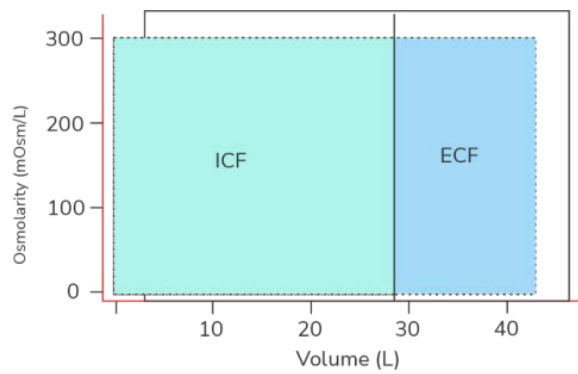
8. Which hormone among the following does not exert its effects on the pancreas?

- A. Secretin
 - B. Gastrin
 - C. Cholecystokinin
 - D. GIP
-

9. On which receptor does Hepcidin act to hinder the absorption of iron?

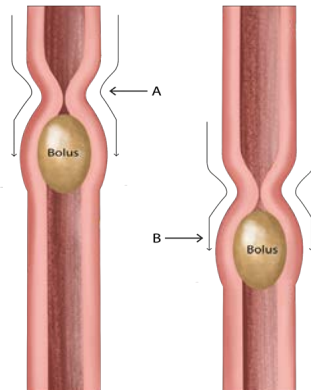
- A. DMT-1
 - B. Ferroportin
 - C. Hephaestin
 - D. Transferrin
-

10. The following Darrow Yannet diagram is suggestive of which condition? Note: The solid line is the original fluid compartment and the color shaded compartment with the dotted change caused by the condition.



- A. Excessive sweating
- B. Adrenal insufficiency
- C. Diarrhea
- D. SIADH

11. Identify this movement that takes place in the intestine.



- A. Segmented movement
- B. Peristalsis
- C. Intersegmental contraction
- D. Diffusion

12. Which of the following factors is responsible for increasing gastric motility?

- A. Presence of fatty food in the stomach
 - B. Distension of stomach
 - C. Presence of tryptophan in the stomach
 - D. The presence of acid in the stomach
-

13. Enteropeptidase enzyme is secreted by?

- A. Stomach
 - B. Duodenum
 - C. Jejunum
 - D. Ileum
-

14. Which of the following acts as a secondary messenger in the mechanism of gastric acid secretion?

- A. cGMP
 - B. cAMP
 - C. CCK-B receptor
 - D. Ach
-

15. Which part of the liver stores vitamin A?

- A. Ito cells

- B. Hepatocytes
 - C. Endothelial cells
 - D. Kupffer cells
-

16. All of the following increases gastric acid secretion except?

- A. Gastrin
 - B. Histamine
 - C. Acetylcholine
 - D. Serotonin
-

17. What is the cause of the relaxation of the intestinal segment following the segment containing the food bolus during peristalsis?

- A. VIP
 - B. Substances P
 - C. Dopamine from adrenergic fibers
 - D. Acetylcholine from myenteric fibers
-

18. What is the most probable outcome for a patient who has undergone complete ileal and partial jejunal resection?

- A. Folic acid deficiency
- B. Vitamin B12 deficiency
- C. Gastric ulcer
- D. Constipation

19. In which of the following reflexes, gastric emptying is delayed when acidic content enters the duodenum?

- A. Gastrocolic
 - B. Intestino intestinal
 - C. Gastroileal
 - D. Enterogastric
-

20. Following complete ileal and partial jejunal resection, the patient is most likely to develop?

- A. Folic acid deficiency
 - B. Vitamin B12 deficiency
 - C. Iron deficiency
 - D. Reduced absorption of carbohydrates
-

21. What are the effects of dietary fibers?

- A. Reduces intestinal transit time
 - B. No role in metabolism
 - C. Causes constipation
 - D. Increased blood Glucose
-

22. Which of the following has the maximum thermogenic effect?

- A. Fat
 - B. Protein
 - C. Carbohydrates
 - D. All have same effect
-

23. Which of the following is an incorrect match?

- A. Gastrin - G cells from stomach
 - B. Motilin - Mo cells from duodenum and jejunum
 - C. CCK - D cells of the duodenum and pancreas
 - D. Secretin - S cells of duodenum and jejunum
-

24. The second phase of saliva secretion involves:

- A. Active Na^+ reabsorption, passive Cl^- reabsorption, passive K^+ secretion, active secretion of HCO_3^-
 - B. Passive Na^+ reabsorption, active Cl^- reabsorption, passive K^+ secretion, active secretion of HCO_3^-
 - C. Passive Na^+ secretion, active Cl^- reabsorption, active K^+ secretion, passive secretion of HCO_3^-
 - D. Active Na^+ reabsorption, passive Cl^- reabsorption, active K^+ secretion, passive secretion of HCO_3^-
-

25. Transport of iron into enterocytes is done by which of the following proteins:

- A. Transferrin
- B. Ferroportin

C. Hepcidin

D. Divalent metal transporter – 1

26. The correct sequence of events during vomiting are? Relaxation of lower esophageal sphincter Strong contractions in the stomach Inspiration against a closed glottis Relaxation of the pyloric sphincter Relaxation of upper esophageal sphincter Reverse peristalsis in the small intestine

A. 6,4,2,5,1,3

B. 6,4,3,2,1,5

C. 4,6,2,1,3,5

D. 4,6,2,5,3,1

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	1
Question 3	4
Question 4	2
Question 5	2
Question 6	4
Question 7	2
Question 8	2
Question 9	2
Question 10	2
Question 11	2
Question 12	2
Question 13	2
Question 14	2
Question 15	1
Question 16	4
Question 17	1
Question 18	2
Question 19	4
Question 20	2
Question 21	1
Question 22	2

Question 23	3
Question 24	4
Question 25	4
Question 26	2

Solution for Question 1:

Correct Option C - Vitamin B12:

- A patient who has undergone gastrectomy is at risk of vitamin B12 deficiency due to the reduced production of intrinsic factor, which is necessary for vitamin B12 absorption. Therefore, vitamin B12 supplementation is typically required to prevent deficiency in such patients.

Incorrect Options:

Option A - Vitamin C: Although vitamin C is important for immune function and skin health, deficiency is not typically a direct result of gastrectomy. Patients may get enough vitamin C through their diet or supplements without specific concern related to gastrectomy.

Option B - Vitamin D: Vitamin D is crucial for bone health and calcium absorption. While vitamin D deficiency can occur in various conditions, it is not specifically linked to gastrectomy. Patients may need to monitor vitamin D levels, but it is not the primary concern post-gastrectomy.

Option D - Vitamin A: Vitamin A is important for vision, skin health, and immune function. Deficiency in vitamin A is less commonly associated with gastrectomy compared to vitamin B12.

Solution for Question 2:

Correct Option A - Hepcidin:

- This is a peptide hormone produced by the liver that regulates iron homeostasis. It inhibits ferroportin, which is the iron transporter on the basal side of the enterocyte, thereby

decreasing iron absorption and release into the bloodstream.

Incorrect Options:

Option B - Ferritin: This is a protein that stores iron within cells, including enterocytes. It does not inhibit iron transport but rather stores excess iron.

Option C - Transferrin: This is a blood plasma protein that binds and transports iron throughout the bloodstream. It does not have a direct role in inhibiting iron transport at the enterocyte level.

Option D - Ferroportin: This is the iron transporter located on the basal side of the enterocyte that releases iron into the bloodstream. Hepcidin acts to inhibit this transporter, not the other way around.

Solution for Question 3:

Correct Option D - Stomach:

- Slow waves are rhythmic electrical oscillations in the smooth muscle cells of the GI tract.
- The stomach has the lowest frequency (3-5 waves/min) to regulate gastric emptying among the given options.
- Frequency of slow waves in other parts of the gastrointestinal tract are as follows.

GI Tract Region	Frequency of Slow Waves (per minute)
Stomach	3-5
Duodenum	12-20
Jejunum	8-12
Ileum	6-9
Colon	6 -10

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.
-

Solution for Question 4:

Correct Option B - Bile salts secretion:

- Bile salts are produced in the liver and stored in the gallbladder. They are released into the small intestine when needed to aid in the digestion and absorption of fats. Bile salts emulsify large fat molecules, breaking them down into smaller droplets, which increases the efficiency of fat digestion by enzymes.

Incorrect Options:

- Options A, C, and D are incorrect. All these functions are carried out by intestinal epithelium.
-

Solution for Question 5:

Correct Option B - GLUT2:

- Glucose Transporter Type 2 (GLUT2) is a protein that helps with glucose transport across cell membranes. In the small intestine, GLUT2 is found on the basolateral membrane of enterocytes (intestinal cells), where it facilitates the transport of glucose from these cells into the bloodstream.

Incorrect Options:

Option A - SGLT1: Sodium-Glucose Transport Protein 1 (SGLT1) is a protein that aids in sodium and glucose co-transport in the apical membrane of the enterocytes.

Option C - GLUT5: Glucose Transporter Type 5 (GLUT5) facilitates the fructose absorption by facilitated diffusion from interstitium to enterocytes.

Option D - SGLT2: Sodium-Glucose Transport Protein 2 (SGLT2) is another sodium glucose co-transporter that aids in glucose absorption in the kidney.

Solution for Question 6:

Correct Option D - Kupffer cells:

- Kupffer cells are scattered flat cells belonging to the reticuloendothelial system and function as macrophages in the liver.
- They are located within the liver sinusoids and prevent liver injury or infection by breaking down and engulfing toxic substances from the blood.
- They also help in the regulation of iron metabolism within the liver.
- Kupffer cells play a major role in maintaining the structural integrity of the liver by releasing certain growth factors and enzymes that build the extracellular matrix of the liver.

Incorrect Options:

Option A - Hepatocytes:

- Hepatocytes are the functional cells of the liver that help in the secretion and excretion of bile, detoxification of harmful substances like drugs or alcohol, metabolism of lipids, proteins, and carbohydrates, and the manufacture of important plasma proteins like albumin, fibrinogen, and prothrombin.

Option B - Ito cells:

- Ito cells are fat-storing cells (containing vitamin A) located within the space of Disse, which is the space between hepatocytes and sinusoidal lining endothelial cells.

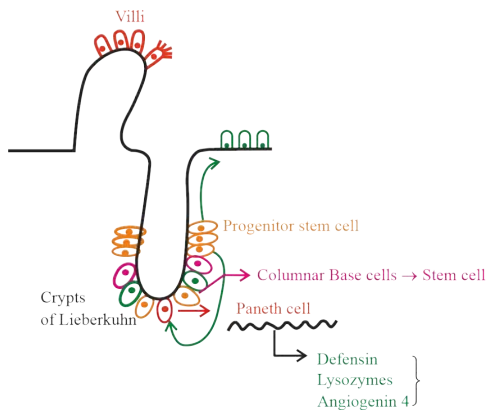
Option C - Sinusoidal endothelial cells:

- They are specialized cells that line the sinusoids.
- They have small pores called fenestrations that help in the exchange of nutrients and other various small molecules between the liver and the blood.
- They also release cytokines that help regulate the liver's immune system.

Solution for Question 7:

Correct Option B - Stem cells:

- The cells lining the isthmus of the gastric pits are known as stem cells.
- Stem cells proliferate and differentiate to form goblet and Paneth cells.
- These stem cells also ascend upward to form epithelium cells of GIT.



Incorrect Options:

Option A - Chief cells: Chief cells are found in the base of the gastric glands and are responsible for the production and secretion of pepsinogen, an inactive form of the enzyme pepsin.

Option C - Parietal cells: Parietal cells are found in the upper region of the gastric glands and are responsible for the production and secretion of hydrochloric acid and intrinsic factor.

Option D - Mucous cells: Mucous cells are found in the neck and surface regions of the gastric pits and are responsible for producing and secreting mucus, which helps protect the gastric mucosa from the acidic environment of the stomach.

Solution for Question 8:

Correct Option B - Gastrin:

- This option suggests that gastrin does not act on the pancreas. However, gastrin does have an indirect effect on the pancreas. Gastrin is released by the stomach in response to the presence of food and stimulates the secretion of gastric acid. The release of gastric acid can indirectly affect pancreatic function by modulating the release of other hormones involved in pancreatic secretion.

Incorrect Options:

Option A - Secretin: Secretin is a hormone that acts on the pancreas. It is released by the duodenum in response to low pH in the stomach. Secretin stimulates the pancreas to secrete bicarbonate-rich pancreatic juice, which helps neutralize the acidic chyme entering the small intestine.

Option C - Cholecystokinin: Cholecystokinin (CCK) is a hormone that acts on the pancreas. It is released by the duodenum in response to the presence of fatty acids and amino acids. CCK stimulates the secretion of pancreatic enzymes, such as lipase and amylase, to aid in the digestion of fats and carbohydrates.

Option D - GIP (Glucose-dependent insulintropic peptide): GIP is a hormone that acts on the pancreas. It is released by the small intestine in response to the ingestion of glucose and fatty acids. GIP stimulates the release of insulin from pancreatic beta cells, which helps regulate blood glucose levels.

Solution for Question 9:

Correct Option B - Ferroportin:

- Ferroportin is a transmembrane protein facilitating iron release from cells like enterocytes and macrophages.
- Hepcidin, produced by the liver, regulates iron levels by binding to ferroportin, triggering its internalization and degradation, thereby reducing iron release into the bloodstream. This inhibition decreases iron absorption from the intestines and iron release from macrophages, resulting in decreased iron levels in circulation.

Incorrect Options:

Option A - DMT-1: DMT-1 (divalent metal transporter 1) is a transmembrane protein responsible for transporting iron into cells.

Option C - Hephaestin: Hephaestin is an enzyme involved in iron metabolism, particularly in the release of iron from enterocytes into the bloodstream.

Option D - Transferrin: Transferrin is a protein that binds to iron in the bloodstream and transports it to cells.

Solution for Question 10:

Correct Option B - Adrenal insufficiency:

- Adrenal insufficiency, or Addison's disease, results from insufficient cortisol production by the adrenal glands. Symptoms include fatigue, weight loss, low blood pressure, and electrolyte imbalances.
- In the Darrow Yannet diagram, it leads to decreased extracellular fluid (ECF) and increased intracellular fluid (ICF) due to hormonal imbalances.

Incorrect Options:

Option A - Excessive sweating, or hyperhidrosis, involves uncontrollable sweating in specific body areas. However, it's not typically represented in a Darrow Yannet diagram.

Option C - Diarrhea causes fluid and electrolyte losses, leading to a decrease in the extracellular fluid compartment. However, without more context, it's unclear if the diagram specifically represents diarrhea.

Option D -: Syndrome of inappropriate antidiuretic hormone secretion (SIADH) results in excess ADH production, leading to water retention and expansion of the extracellular fluid compartment, as shown in the Darrow Yannet diagram.

Solution for Question 11:

Correct Option B - Peristalsis:

- Peristalsis is a coordinated muscular movement that occurs in the gastrointestinal tract, including the intestines. It involves rhythmic contractions and relaxations of the smooth muscle to propel the contents of the intestine forward.
- During peristalsis, there is a wave-like movement of contraction that begins in one section of the intestine and progresses along its length. The contraction of the circular muscles behind the food bolus (oral side) pushes the contents forward, while the relaxation of the circular muscles in front of the bolus (aboral side) allows for the forward movement.

Incorrect Options:

Option A - Segmented movement: Segmented movement refers to the localized contractions and relaxations that occur in various sections of the intestine. It helps mix the intestinal contents but does not involve the overall forward movement of the contents as seen in peristalsis.

Option C - Intersegmental contraction: Intersegmental contraction is not a specific movement that occurs in the intestine.

Option D - Diffusion: Diffusion is a process of passive movement of molecules from an area of higher concentration to an area of lower concentration.

Solution for Question 12:

Correct Option B - Distension of stomach:

- Stomach distension, caused by food or fluids, triggers gastric motility by activating stretch receptors in the gastric smooth muscle.
- This stimulation prompts the release of hormone serotonin which activates the myenteric plexus. The activation of myenteric plexus results in activation of cholinergic neurons in the anterograde and retrograde direction of the peristalsis.
- In anterograde direction neurons secrete VIP, NO and ATP resulting in the relaxation of the smooth muscles.
- In the retrograde direction (i.e., behind the food bolus), acetylcholine (ACh) and substance P are released, which cause contraction of the smooth muscles, pushing the contents forward.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.
-

Solution for Question 13:

Correct Option B - Duodenum:

- Enteropeptidase, also known as enterokinase, is an enzyme that is primarily secreted by the duodenum. It is produced by the epithelial cells lining the duodenal mucosa.
- Enteropeptidase plays a crucial role in the digestion and activation of pancreatic enzymes. It converts trypsinogen, an inactive form of the enzyme trypsin, into its active form trypsin.

Trypsin then activates other pancreatic enzymes involved in the digestion of proteins.

Enteropeptidase, also known as enterokinase, is an enzyme that is primarily secreted by the duodenum. It is produced by the epithelial cells lining the duodenal mucosa.

Enteropeptidase plays a crucial role in the digestion and activation of pancreatic enzymes. It converts trypsinogen, an inactive form of the enzyme trypsin, into its active form trypsin. Trypsin then activates other pancreatic enzymes involved in the digestion of proteins.

Incorrect Options:

- Options A, C, and D are incorrect. Refer to Option B for an explanation.

Options A, C, and D are incorrect. Refer to Option B for an explanation.

Solution for Question 14:

Correct Option B - cAMP:

- cAMP and calcium act as secondary messengers for the hormones acting on the H⁺ K⁺ ATPase in the parietal cell.
- cAMP and calcium activates protein kinase A, which phosphorylates the H⁺/K⁺ ATPase pump, increasing acid secretion into the lumen.
- Hormones acting through cAMP as a secondary messenger are prostaglandin E₂ and histamine.
- Hormones acting through calcium as a secondary messenger are gastrin and acetylcholine.

Incorrect Options:

Option A - cGMP: While cGMP is involved in the regulation of some gastrointestinal functions, such as smooth muscle relaxation.

Option C - CCK-B receptor: The CCK-B receptor is involved in the regulation of pancreatic enzyme secretion and gallbladder contraction.

Option D - Ach: While Ach is involved in the activation of parietal cells, it is not a secondary messenger in the mechanism of gastric acid secretion.

Solution for Question 15:

Correct Option A - Ito cells:

- Ito cells, also known as hepatic stellate cells, are responsible for storing vitamin A in the liver. These cells are found in the space of Disse, which is located between the hepatocytes and endothelial cells in the liver sinusoids. Ito cells store vitamin A as lipid droplets within their cytoplasm.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation
-

Solution for Question 16:

Correct Option D - Serotonin:

- Serotonin (5-hydroxytryptamine or 5-HT) plays a role in the gastrointestinal motility.
- During peristalsis the stretching of the intestinal wall leads to the release of serotonin which activates myenteric plexus and governs the peristalsis.

Incorrect Options:

- Options A, B, and C are incorrect. Gastrin, acetylcholine, and histamine increases the HCl secretion.

Options A, B, and C are incorrect. Gastrin, acetylcholine, and histamine increases the HCl secretion.

Solution for Question 17:

Correct Option A - VIP:

- Vasoactive Intestinal Peptide (VIP) is a neurotransmitter released when the cholinergic neuron from the myenteric plexus moves in the anterograde direction of the peristalsis..
- It acts as a relaxant on smooth muscles in the proximal segment of the segment containing chyme, resulting in relaxation of the proximal segment of the chyme and movement of the content towards the rectum.

Incorrect Options:

Option B - Substance P: Substance P is a neurotransmitter that is released when cholinergic neuron from the myenteric plexus moves in the retrograde direction of the peristalsis and results in the contraction of the smooth muscle.

Option C- Dopamine from adrenergic fibers: Dopamine is a neurotransmitter that can have both inhibitory and excitatory effects, depending on the receptors it acts on. It is not released during the peristalsis.

Option D - Acetylcholine from myenteric fibers: Acetylcholine is the primary neurotransmitter released by cholinergic neurons in the retrograde direction of the peristalsis and results in the contraction of the smooth muscle.

Solution for Question 18:

Correct Option B - Vitamin B12 deficiency:

- Massive ileal resection can lead to vitamin B12 deficiency due to impaired absorption in the ileum, the main absorption site for vitamin B12. This results in symptoms such as diarrhea, fatigue, headaches, and cognitive changes. Additionally, resection may cause increased defecation frequency and loose stools due to decreased absorption of fluid and jejunal secretions.

Incorrect Options:

Option A - Folic acid deficiency may occur due to resection of the distal small intestine, affecting absorption, but dietary intake and compensation from the remaining intestine mitigate severe deficiency.

Option C - Gastric ulcers, including duodenal ulcers, are not directly affected by ileum and jejunal resection, as it doesn't impact stomach acid production, thus reducing the risk.

Option D - Ileum and partial jejunal resection can lead to constipation due to slowed peristalsis and excessive water absorption in the colon, but it's not the most common symptom.

Solution for Question 19:

Correct Option D - Enterogastric:

- The Enterogastric reflex delays gastric emptying when acidic content enters the duodenum by inhibiting the release of gastrin from G cells in the antrum of the stomach inhibiting gastric emptying and further secretion of the acid in the stomach..

The Enterogastric reflex delays gastric emptying when acidic content enters the duodenum by inhibiting the release of gastrin from G cells in the antrum of the stomach inhibiting gastric emptying and further secretion of the acid in the stomach..

Incorrect Options:

Option A - Gastrocolic: The Gastrocolic reflex is a reflex that stimulates colonic motility and promotes the movement of feces from the colon to the rectum when food enters in the stomach leading to distension.

Option B - Intestino intestinal: When a part of the intestine is over-distended, the rest of the intestine relaxes. This is called intestine intestinal reflex.

Option C - Gastroileal: The Gastroileal reflex is a reflex that stimulates the emptying of the ileum, the final portion of the small intestine, by relaxing the ileocecal valve. It is not directly related to gastric emptying or the entry of acidic content into the duodenum.

Solution for Question 20:

Correct Option B - Vitamin B12 deficiency:

- Following complete ileal and partial jejunal resection, the patient is most likely to have vitamin B12 deficiency.

- The maximum absorption of vitamin B12 takes place in the ileum.

Incorrect Options:

Option A - Folic acid deficiency: The maximum absorption of folic acid takes place in the duodenum and to some extent in the proximal part of the jejunum.

Option C - Iron deficiency: Iron is also absorbed maximally in the duodenum and to some extent in the proximal part of the jejunum.

Option D - Reduced absorption of carbohydrates: Carbohydrates are absorbed maximally in the duodenum.

Solution for Question 21:

Correct Option A - Reduces intestinal transit time:

- Dietary fibers are known to increase stool bulk and promote regular bowel movements, thus reducing intestinal transit time. They add bulk to the stool, making it easier to pass through the digestive system and preventing constipation.

Incorrect Options:

Option B - No role in metabolism:

- This is incorrect because dietary fibers play a significant role in metabolism. They are not digested by human enzymes, but they are fermented by gut bacteria in the colon, producing short-chain fatty acids that can be utilized as an energy source.

Option C - Causes constipation:

- This is incorrect because dietary fibers actually help prevent constipation. They add bulk to the stool, soften it, and promote regular bowel movements.

Option D - Increased blood Glucose:

- This is incorrect because dietary fibers have the opposite effect on blood glucose levels. Soluble fibers, in particular, can slow down the absorption of glucose, leading to a more gradual and controlled rise in blood glucose levels after a meal.

Solution for Question 22:

Correct Option B - Protein:

- Protein has the highest thermogenic effect among macronutrients, estimated at 20-30% of its consumed energy.

Incorrect Options:

Option A - Fat: Fat has the lowest thermogenic effect among macronutrients, estimated at 0-3% of its consumed energy.

Option C - Carbohydrates: Carbohydrates have a moderate thermogenic effect. The thermogenic effect of carbohydrates is estimated to be around 5-10% of the energy obtained from their consumption.

Option D - All have same effect: This option is incorrect.

Solution for Question 23:

Correct Option C - CCK- D cells of duodenum and pancreas:

- Cholecystokinin (CCK) is a hormone involved in the regulation of digestion and appetite. It is primarily produced by specialized cells called I cells, which are located in the duodenum.
- D or delta cells secrete somatostatin.

Incorrect Options:

Option A - Gastrin- G cells from stomach:

- Gastrin, produced by G cells primarily in the stomach's antrum, regulates gastric acid secretion and motility in response to food and various signals.

Option B - Motilin- M cells from duodenum and jejunum:

- This statement is correct. Motilin is primarily produced by Mo cells and enterochromaffin cells of all parts of GIT except esophagus and rectum.

Option D - Secretin- S cells of duodenum and jejunum:

- Secretin is a hormone that regulates the pH and digestive functions of the gastrointestinal tract. It is primarily produced by specialized cells called S cells, which are located in the duodenum and jejunum.
-

Solution for Question 24:

Correct Option D - Active Na⁺ reabsorption, passive Cl⁻ reabsorption, active K⁺ secretion, passive secretion of HCO₃⁻:

The second phase of saliva secretion, i.e., secretion in ducts, following events take place:

- Active Na⁺ reabsorption occurs.
- Passive Cl⁻ reabsorption follows.
- Active K⁺ secretion takes place.
- Passive HCO₃⁻ secretion neutralizes saliva acidity.

Incorrect Options:

Option A, B and C are incorrect.

Solution for Question 25:

Correct Option D - Divalent metal transporter – 1:

- Divalent metal transporter-1 (DMT1) is a transmembrane protein located in the apical membrane of enterocytes. It is responsible for transporting ferrous forms of iron from the intestinal lumen into the enterocyte.
- DMT1 is expressed in the duodenum, the first part of the small intestine where iron is absorbed from the diet.

Incorrect Options:

Option A, B and C are not directly involved in the transport of iron into the enterocytes.

Solution for Question 26:

Correct Option B - 6,4,3,2,1,5:

The sequence of events during vomiting can be explained as follows:

Step 1: Reverse peristalsis in the small intestine - The small intestine undergoes reverse peristalsis, pushing the intestinal contents back into the stomach.

Step 2: Relaxation of the pyloric sphincter: The pyloric sphincter, the muscular valve at the lower end of the stomach, relaxes to allow the content from the intestine to enter in the stomach.

Step 3: Inspiration against a closed glottis - Forced inspiration occurs against the closed glottis. Lowering of the diaphragm increases intra-abdominal pressure.

Step 4: Strong contractions in the stomach - Vigorous contraction of abdominal muscles increases intra-abdominal pressure sharply, which forces the gastric contents to enter the esophagus.

Step 5: Relaxation of LES - Relaxation of the lower esophageal sphincter allows gastric content to enter the esophagus.

Step 6: Relaxation of UES - With a sharp rise in intra abdominal pressure, the upper esophageal sphincter relaxes, resulting in the passage of chyme through the mouth.

Previous Year Questions

1. Caffeine impairs with sleep by which of the following:

- A. Blocks adenosine action and wakefulness
 - B. Activate locus coeruleus and wakefulness
 - C. No role in maintaining wakefulness
 - D. Activates histamine release and prevent sleep.
-

2. Where do the afferent fibers from baroreceptors, responsible for blood pressure regulation, go to?

- A. Nucleus ambiguous
 - B. Rostral-ventro lateral medulla
 - C. Nucleus Tractus Solitariu
 - D. Caudal ventrolateral medulla
-

3. What is the term used to describe increased pain in response to a minor stimulus?

- A. Hypersensitivity
 - B. Causalgia
 - C. Hyperalgesia
 - D. Allodynia
-

4. What specific region of the thalamus is associated with the regulation of motor control?

- A. Ventrolateral thalamus
 - B. Ventral posteromedial
 - C. Ventral posterolateral
 - D. Superior and inferior colliculi
-

5. What mediates Inverse stretch reflex?

- A. Golgi tendon
 - B. Muscle spindle
 - C. Unmyelinated C fibres
 - D. Dorsal Column
-

6. Reward centre in the brain is?

- A. Amygdala
 - B. Thalamus
 - C. Hippocampus
 - D. Ventral tegmental area
-

7. What is the connecting structure between Broca's area and Wernicke's area?

- A. Arcuate fasciculus
- B. Anterior commissure
- C. Corpus callosum

D. Fornix

8. What is the term used to describe the opposition to passive stretching?

- A. Tone
 - B. Spasticity
 - C. Rigidity
 - D. Paratonia
-

9. Maximum area in homunculus signifies

- A. Low control and skilled movements
 - B. More muscle fibers are present
 - C. Increased preciseness of movements
 - D. None of the above
-

10. A person is having increased sexual activity ,increased appetite, anger and no fear , where is structure injured?

- A. Amygdala
 - B. Thalamus
 - C. Cingulate gyrus
 - D. Hypothalamus
-

11. A 55-year-old male presents with progressive weakness and spasticity in his right leg. He also notices increased muscle tone, hyperreflexia, and a positive Babinski sign on the same side. His left leg and upper limbs appear unaffected. MRI of the brain reveals a lesion involving the left motor cortex. What is the most likely diagnosis?

- A. Parkinson's disease
 - B. Amyotrophic lateral sclerosis (ALS)
 - C. Multiple sclerosis (MS)
 - D. Right-sided upper motor neuron lesion
-

12. A 40-year-old patient presented to the hospital with a complaint of chronic headache. Upon further investigation, a tumor was identified in the hypothalamus region, which was exerting pressure on the supraoptic nucleus. This structural alteration was found to be affecting hormonal regulation. Which hormone imbalance is most likely associated with this condition?

- A. ACTH
 - B. GH
 - C. ADH
 - D. TSH
-

13. In a young patient who experienced a head injury from a road traffic accident, his family reports a change in his behavior. They describe him as being excessively curious and constantly touching objects. He also exhibits rapid forgetfulness and attempts to put various things in his mouth, including plates. Additionally, they have observed him squeezing his genitals. Based on these symptoms, which structure is most likely to be affected in this patient?

- A. Gray matter around the aqueduct of Sylvius
- B. Amygdala

- C. Lateral nuclei of the hypothalamus
 - D. Anterior cingulate gyri
-

14. A 55-year-old man presents with a peculiar speech. On further evaluation, a defect in the Wernicke's area of his brain is identified. Which of the following condition is associated with the given defect?

- A. He can make the decision what he speaks
 - B. Senseless continuous speaking
 - C. He is able to understand words and speak properly
 - D. He is understanding language but unable to speak
-

15. Which of the following statements regarding GFR is accurate? GFR stands for Glomerular Filtration Rate, while RPF refers to Renal Plasma Flow.

- A. An increase in sympathetic function increases GFR
 - B. Afferent arteriolar constriction increases GFR
 - C. An increase in RPF increases GFR
 - D. A decrease in RPF increases GFR
-

16. A patient is experiencing phantom limb pain after amputation of right limb . What is observed on a PET scan in a patient with phantom limb pain?

- A. Expansion of right somatosensory fibers for the hand into neighboring areas
- B. Neighboring fibers of the right somatosensory area extending into the right-hand area
- C. Expansion of right sensory cortex
- D. Expansion of Left sensory cortex

17. Which of the following fibers reach the Purkinje cell directly?

- A. Inferior olivary nucleus
 - B. Raphe nucleus
 - C. Vestibular nucleus
 - D. Locus coeruleus
-

18. Which of the following oxygen-sensitive channels is present in peripheral chemoreceptors?

- A. K^+
 - B. Na^+
 - C. Ca^{++}
 - D. Cl^-
-

19. In response to increased blood pressure which of the following is released at the level SA node?

- A. Epinephrine
 - B. NE
 - C. Dopamine
 - D. Ach
-

20. What is the length of time for which short-term memory lasts?

- A. 3-6 Hours
 - B. 12-18 hours
 - C. 30-300 seconds
 - D. 3-4 seconds
-

21. What are the consequences of experimental muscle spindle sectioning in frogs?

- A. Voluntary Axn & stretch reflex present
 - B. Voluntary Axn lost but stretch reflex present
 - C. Both are lost
 - D. Voluntary Axn present, stretch reflex is lost.
-

22. Spinal nucleus of trigeminal is associated with which sensation?

- A. Vibration
 - B. Conscious proprioception
 - C. Pain
 - D. Two point discrimination
-

23. The function of the Golgi tendon organ is to _____

- A. Sense changes in muscle direction
 - B. Sense changes in muscle tension
 - C. Sense changes in position of the muscle
 - D. Sense changes in length of the muscle
-

24. Which type of nerve fiber is most likely to be affected in a 35-year-old female who experienced a tingling sensation over her arm after watching TV for an extended period of time with her hands placed under her head?

- A. A
 - B. B
 - C. C
 - D. Sympathetic
-

25. Which of the following are typically associated with Lower Motor Neuron (LMN) lesions? Select all that apply. Atrophy Fasciculations DTR increased Rigidity

- A. 1 and 2
 - B. 2 and 3
 - C. 1, 2 and 3
 - D. 2, 3 and 4
-

26. Weber-Fechner law relates the magnitude of sensation to _____.

- A. Strength of muscle contraction
 - B. Intensity of stimulus
 - C. The amplitude of action potential
 - D. Duration of stimulus
-

27. A patient presented to the ER with a history of a road traffic accident. On examination his vitals were stable, but he could not speak. He had normal handwriting skills, and could understand what was being asked of him to which he responded by nodding. Where could be the lesion for this clinical condition?

- A. Broca's area
 - B. Wernicke's area
 - C. Primary motor cortex
 - D. Occlusion of left internal carotid artery
-

28. In the homunculus, which of the following body parts exhibits the highest degree of representation?

- A. Low control and skilled movements
 - B. Crude touch
 - C. The precision of movements
 - D. The intensity of perception
-

29. A neuron receiving 400 inputs from 150 sensory neurons can be best described as?

- A. Divergence
 - B. Convergence
 - C. Feedback inhibition
 - D. Feedforward System
-

30. Which of the following regulate body temperature?

- A. Hypothalamus
 - B. Basal ganglia
 - C. Thalamus
 - D. Limbic system
-

31. How is cerebral perfusion pressure calculated?

- A. Systolic blood pressure + Intracranial pressure
 - B. Systolic blood pressure – intracranial pressure
 - C. Mean arterial pressure/intracranial pressure.
 - D. Mean arterial pressure – intracranial pressure.
-

32. Which of the following is responsible for relieving pain when pressure is applied to the painful area.

- A. Raphe nucleus
 - B. Corticospinal tract
 - C. Medial lemniscus
 - D. Medial longitudinal fasciculus
-

33. Which sensory receptors are accountable for tactile discrimination among the options provided? Merkel disc Ruffini end organ Pacinian corpuscle Meissner's corpuscle

- A. 1 and 4 are correct
- B. 2 is correct
- C. 3 is correct

D. None are correct.

E. Both A & D

34. What is the underlying mechanism of action for pain reduction by transcutaneous electrical nerve stimulation (TENS)?

A. Central inhibition

B. Desensitization

C. Gate control theory

D. Descending inhibition

35. Which receptors are responsible for mediating the stretch reflex among the options provided?

A. Golgi tendon organ

B. Merkel's disc

C. Muscle spindle

D. Meissner's corpuscles

36. Which of the following is not a characteristic of decerebrate rigidity according to the Herrington classification?

A. Rigidity occurs in all the muscles of the body

B. Increase in the rate of discharge of the γ efferent neuron

C. Increased excitability of the motor neuron pool

D. Mid Collicular lesion can lead to decerebrate rigidity

37. In a patient who experienced a stroke 2 days ago, the current symptoms include uncontrollable, forceful, and jerking motions of the limbs on one side. Where is the probable location of the lesion in this individual?

- A. Subthalamic nuclei
 - B. Substantia nigra
 - C. Putamen
 - D. Caudate nucleus
-

38. What findings would be observed in a patient who has suffered a cervical spine injury due to a road traffic accident, whereby the fracture fragment has penetrated the lateral aspect of the dorsal column tract?

- A. Absence of ipsilateral lower limb proprioception
 - B. Absence of fine motor movement of fingers
 - C. Absence of ipsilateral arm proprioception
 - D. Absence of contralateral lower limb proprioception
-

39. Which of the following substances causes a bitter taste that deters us from consuming toxic substances?

- A. Aldehyde
 - B. Amino acids
 - C. Alkaloids
 - D. Hydrogen ions
-

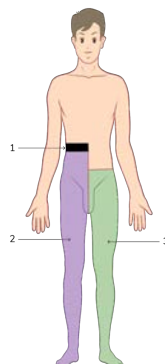
40. Which cells are impacted in multiple sclerosis?

- A. Microglia
 - B. Oligodendrocytes
 - C. Astrocytes
 - D. Ependymal cells
-

41. Which fibre is responsible for High-frequency vibration, pressure sensation?

- A. Pacinian corpuscles
 - B. Free nerve endings
 - C. Meissner corpuscles
 - D. Merkel discs
-

42. The patient exhibits symptoms depicted in the image provided below. Shows the site of the lesion Experiences decreased proprioception, vibration, and discriminative touch on the affected side. Has loss of pain and temperature beginning one or two segments below the side of the lesion on the contralateral side. What is the most likely diagnosis?



- A. Weber's syndrome

- B. Dejerine syndrome
 - C. Brown Sequard syndrome
 - D. Wallenberg syndrome
-

43. What causes the stimulation of the Pacinian corpuscle?

- A. Pain
 - B. Cold Temperature
 - C. Hot temperature
 - D. Pressure
-

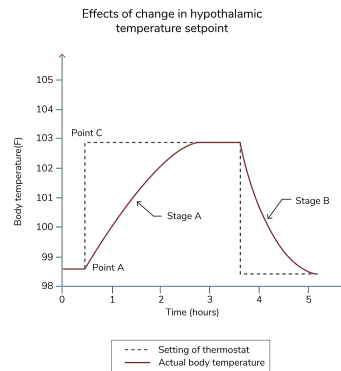
44. Which lobe is affected by a lesion that results in a deficit in hand-eye coordination?

- A. Parietal lobe
 - B. Frontal lobe
 - C. Temporal lobe
 - D. Occipital lobe
-

45. Corticospinal tract crosses at?

- A. Midbrain
 - B. Pons
 - C. Upper medulla
 - D. Lower medulla
-

46. What changes do not occur in stage A as compared to stage B when the hypothalamic thermostat in a patient is reset from point A to point C, as depicted below?



- A. Shivering
- B. Vasoconstriction
- C. Sweating
- D. Epinephrine secretion

47. Which of the options accurately describes decorticate rigidity?

- A. It is produced by the removal of cerebral cortex and basal ganglia.
- B. Flexion of lower limbs and extension of upper limbs occurs.
- C. Rigidity is pronounced.
- D. It is characterized by flexion of upper limbs and extension of lower limbs.

48. What is the effect of a lesion in the medial lemniscus at the level of the pons?

- A. Pain and temperature loss in the same side

- B. Pain and temperature loss in the opposite side
 - C. Loss of tactile and proprioception on the same side
 - D. Loss of tactile and proprioception on the opposite side
-

49. Which specific neurotransmitter is accountable for causing dilation of the small intestine in a forward direction?

- A. VIP
 - B. Acetylcholine
 - C. Substance P
 - D. Serotonin
-

50. What is the defining characteristic of Brown-Séquard syndrome?

- A. Fine touch lost on the opposite side
 - B. Pain and temperature lost on the same side
 - C. Pain and temperature loss on the opposite side
 - D. Only pain lost on the same side
-

51. What is Prosopagnosia?

- A. Impairment of consciousness
 - B. Being unaware of one's problems
 - C. Difficulty in identifying known faces
 - D. Failure to identify objects
-

52. What is the mediator of fast vibration sense?

- A. Merkel's disc
 - B. Meissner's corpuscles
 - C. Pacinian corpuscles
 - D. Ruffini's end organ
-

53. What are the factors that can lead to a person drinking water as a result of the stimulation of the thirst mechanism?

- A. Dehydration
 - B. Hemorrhage
 - C. Vomiting
 - D. All of the above
-

54. Center controlling vomiting?

- A. Apneustic center
 - B. Pneumotaxic centre
 - C. Area Postrema
 - D. Pre Botzinger complex
-

Correct Answers

Question	Correct Answer
Question 1	1
Question 2	3
Question 3	3
Question 4	1
Question 5	1
Question 6	4
Question 7	1
Question 8	2
Question 9	3
Question 10	4
Question 11	4
Question 12	3
Question 13	2
Question 14	2
Question 15	3
Question 16	2
Question 17	1
Question 18	1
Question 19	4
Question 20	3
Question 21	3
Question 22	3

Question 23	2
Question 24	1
Question 25	1
Question 26	2
Question 27	1
Question 28	3
Question 29	2
Question 30	1
Question 31	4
Question 32	1
Question 33	1,5
Question 34	3
Question 35	3
Question 36	1
Question 37	1
Question 38	3
Question 39	3
Question 40	2
Question 41	1
Question 42	3
Question 43	4
Question 44	1
Question 45	4
Question 46	3
Question 47	4
Question 48	4

Question 49	1
Question 50	3
Question 51	3
Question 52	3
Question 53	4
Question 54	3

Solution for Question 1:

Correct A - Blocks adenosine action and wakefulness:

- Caffeine works by blocking adenosine receptors. Adenosine is a neurotransmitter that promotes sleep by slowing down nerve activity. By blocking adenosine, caffeine prevents the brain from sensing sleepiness, thereby promoting wakefulness.

Incorrect Options:

Option B - Activates locus coeruleus and wakefulness:

- The locus coeruleus is a nucleus in the brainstem involved in arousal and wakefulness. While it does play a role in promoting wakefulness, caffeine's primary mechanism of action is not through direct activation of the locus coeruleus.

Option C - No role in maintaining wakefulness:

- Caffeine is well-known for its role in maintaining wakefulness by counteracting sleep-promoting mechanisms in the brain.

Option D - Activates histamine release and prevents sleep:

- Histamine is involved in promoting wakefulness, but caffeine does not directly cause histamine release. Its effect on sleep is primarily through adenosine receptor blockade rather than histamine pathways.
-

Solution for Question 2:

Correct Option:

Option C. Nucleus Tractus Solitarius: The Nucleus Tractus Solitarius (NTS) is a nucleus in the medulla oblongata that serves as the primary site for receiving afferent signals from baroreceptors. Baroreceptors, located in the walls of the carotid sinus and aortic arch, sense changes in blood pressure. Afferent fibers from these baroreceptors transmit signals to the NTS, which plays a vital role in regulating blood pressure by modulating autonomic outputs.

Incorrect Option:

Option A. Nucleus Ambiguus: The Nucleus Ambiguus is a motor nucleus in the medulla oblongata that primarily innervates the muscles involved in swallowing and phonation. It is not directly involved in receiving afferent signals from baroreceptors for blood pressure control.

Option B. Rostral-Ventro Lateral Medulla: The Rostral-Ventro Lateral Medulla (RVLM) is a region in the medulla oblongata that plays a crucial role in the regulation of sympathetic outflow and blood pressure. It is involved in the efferent pathway for blood pressure control, but it does not directly receive afferent signals from baroreceptors.

Option D. Caudal Ventrolateral Medulla: The Caudal Ventrolateral Medulla (CVLM) is a region in the medulla oblongata involved in the regulation of sympathetic outflow and blood pressure. It primarily serves as an inhibitory center, tonically suppressing sympathetic activity. It is not directly involved in receiving afferent signals from baroreceptors.

Solution for Question 3:

Correct option. C

- In case a small stimulus causes more pain, it is called "Hyperalgesia."
- Hyperalgesia is a condition where there is an increased sensitivity to painful stimuli, resulting in an exaggerated pain response. It can occur in response to various factors, including tissue injury, inflammation, nerve damage, or certain medical conditions.
- Hyperalgesia can be categorized into two types: Primary Hyperalgesia: This occurs in the area surrounding the site of tissue injury or inflammation. The affected area becomes more sensitive to painful stimuli, even in response to mild or non-painful stimuli. Secondary Hyperalgesia: This occurs in areas surrounding the primary hyperalgesic zone. It is characterized by an expanded area of increased pain sensitivity beyond the immediate injury

site.

- Primary Hyperalgesia: This occurs in the area surrounding the site of tissue injury or inflammation. The affected area becomes more sensitive to painful stimuli, even in response to mild or non-painful stimuli.
- Secondary Hyperalgesia: This occurs in areas surrounding the primary hyperalgesic zone. It is characterized by an expanded area of increased pain sensitivity beyond the immediate injury site.
- Hyperalgesia can manifest as an increased intensity or prolonged duration of pain in response to stimuli that would normally be perceived as mildly painful or non-painful. It is believed to involve changes in the peripheral and central nervous systems, including altered neuronal excitability and enhanced pain transmission.
- Primary Hyperalgesia: This occurs in the area surrounding the site of tissue injury or inflammation. The affected area becomes more sensitive to painful stimuli, even in response to mild or non-painful stimuli.
- Secondary Hyperalgesia: This occurs in areas surrounding the primary hyperalgesic zone. It is characterized by an expanded area of increased pain sensitivity beyond the immediate injury site.

Incorrect options:

Option A: Hypersensitivity refers to an exaggerated or abnormal response of the immune system to a normally harmless substance, resulting in an allergic or autoimmune reaction.

Option C: Causalgia is a type of chronic pain syndrome characterized by severe burning pain, typically following a peripheral nerve injury.

Option D: Allodynia refers to the perception of pain in response to a stimulus that is not normally painful, such as light touch or a gentle stroke.

Solution for Question 4:

Correct option. A

- The ventrolateral thalamus (VL) is a region of the thalamus that is closely associated with motor control. It is involved in the transmission of motor signals from the basal ganglia and cerebellum to the motor cortex.

- The ventrolateral thalamus receives input from the basal ganglia, which is responsible for motor planning and coordination, as well as the cerebellum, which plays a crucial role in motor learning and fine-tuning of movements. The information processed in the VL thalamus is then relayed to the motor cortex, where it contributes to the execution of voluntary movements.

Incorrect options:

Option B. Ventral posteromedial (VPM) and option (c) Ventral posterolateral (VPL) are regions of the thalamus that are primarily involved in somatosensory processing. They receive sensory information from the body, including touch, temperature, and pain, and relay it to the somatosensory cortex for further processing.

Option D. Superior and inferior colliculi are structures located in the midbrain and are involved in visual and auditory processing, respectively. They play important roles in orienting responses to visual and auditory stimuli but are not directly related to motor control.

Solution for Question 5:

Correct option. A

- The inverse stretch reflex, also known as the inverse myotatic reflex, is mediated by the Golgi tendon organs (GTOs). Golgi tendon organs are specialized sensory receptors located in the tendons near the junction of muscles and tendons.
- When the muscle contracts and applies tension to the tendon, the GTOs are activated. They sense changes in tension and send inhibitory signals to the spinal cord, which in turn inhibits the motor neurons innervating the muscle. This reflex helps prevent excessive tension and potential damage to the muscle and tendon by inhibiting further contraction.

Incorrect option:

Option B Muscle spindle is incorrect in this context. The muscle spindle is responsible for the stretch reflex, also known as the myotatic reflex, which detects changes in muscle length and initiates a reflex contraction of the muscle.

Option C Unmyelinated C fibers are associated with pain perception and are not specifically involved in the inverse stretch reflex.

Option D Dorsal Column refers to a pathway in the spinal cord involved in transmitting sensory information related to touch, proprioception, and vibration. It is not directly involved in mediating

the inverse stretch reflex.

Solution for Question 6:

Correct option. D

- The reward center in the brain is primarily associated with the ventral tegmental area (VTA). The VTA is a region located in the midbrain, specifically in the mesolimbic pathway. It is known for its role in the brain's reward system, which is involved in the experience of pleasure, motivation, and reinforcement of certain behaviors.
- The VTA contains dopamine-producing neurons that project to various regions of the brain, including the nucleus accumbens, prefrontal cortex, and amygdala. Dopamine is a neurotransmitter closely associated with reward and pleasure. When the VTA neurons release dopamine in response to rewarding stimuli, it can reinforce behaviors and create a sense of pleasure and motivation.

Incorrect options:

Option A. Amygdala: The amygdala is a key structure in the brain involved in emotional processing, particularly in the formation and regulation of fear responses. While it plays a role in processing reward-related stimuli, it is not considered the primary reward center in the brain. The amygdala is interconnected with the reward system and can influence emotional responses to rewarding stimuli, but it does not serve as the central hub for reward processing.

Option B. Thalamus: The thalamus is a critical relay station in the brain that processes and relays sensory information to different areas of the cerebral cortex. It serves as a gateway for sensory inputs but is not specifically associated with the reward system or the primary reward center in the brain.

Option C. Hippocampus: The hippocampus is primarily involved in memory formation, spatial navigation, and learning. While it is interconnected with other brain regions involved in reward processing, such as the VTA and nucleus accumbens, it is not considered the primary reward center itself. The hippocampus is crucial for memory and contextual associations but is not primarily responsible for the experience of pleasure or motivation related to rewards.

Solution for Question 7:

Correct option: A

- The arcuate fasciculus is a white matter tract that connects two important language areas of the brain: Broca's area and Wernicke's area. Broca's area, located in the frontal lobe of the dominant hemisphere (usually the left hemisphere in right-handed individuals), is involved in the production of speech and language expression. Wernicke's area, located in the posterior part of the superior temporal gyrus, is involved in language comprehension.
 - The arcuate fasciculus serves as a major pathway for communication between these two language areas. It allows for the coordination of language production and comprehension. Damage or disruption to the arcuate fasciculus can result in language deficits, such as conduction aphasia, where there is difficulty repeating words or phrases.
-

Solution for Question 8:

Correct option: B. Spasticity

- Explanation: Spasticity is characterized by increased muscle tone that is velocity-dependent, meaning the resistance to passive movement increases with higher speed or sudden stretching. This directly describes the opposition to passive stretching, as spastic muscles resist being stretched quickly.

Incorrect option:

Option A. Tone

- Explanation: Tone refers to the general tension or resistance in a muscle at rest. It can be increased (hypertonic) or decreased (hypotonic) depending on the condition. Tone itself does not specifically describe the opposition to passive stretching but rather the overall state of muscle tension.

Option C. Rigidity:

- Rigidity involves increased resistance to passive movement that is independent of velocity. It is a constant, uniform resistance and is not specifically related to the speed or force of stretching, thus not accurately describing opposition to passive stretching.

Option D. Paratonia:

- Paratonia is characterized by variable resistance to passive movement, which fluctuates between increased and decreased muscle tone. It involves an impaired ability to modulate muscle tone but does not specifically refer to opposition to passive stretching in a consistent manner.
-

Solution for Question 9:

Correct Option C.

- Increased preciseness of movements: This option is correct. The maximum area in the homunculus signifies increased preciseness of movements. The larger the area dedicated to a specific body part in the homunculus, the more neural resources are allocated to that body part, indicating a higher level of precision and fine motor control.

Incorrect Options:

Option A. Low control and skilled movements: This option is incorrect. The maximum area in the homunculus does not signify low control and skilled movements. In fact, the maximum area in the homunculus represents areas of the body that require more precise and skilled movements.

Option B. More muscle fibers are present: This option is incorrect. The size of the area in the homunculus does not indicate the number of muscle fibers present. The homunculus is a visual representation of the somatosensory and motor cortices in the brain, where the size of each body part in the homunculus corresponds to the amount of neural representation dedicated to that body part, not the number of muscle fibers.

Option D. None of the above: This option is incorrect, as option C (Increased preciseness of movements) is the correct answer.

Solution for Question 10:

The correct answer is 4. Hypothalamus.

Explanation of Options:

Amygdala: The amygdala plays a role in processing emotions, particularly fear and aggression. If there were an injury to the amygdala, it might result in altered emotional responses, including decreased fear and potentially increased aggression, but it would not likely be the primary cause of increased sexual activity and appetite.

Thalamus: The thalamus serves as a relay center for sensory and motor signals. It is not directly responsible for regulating emotions, sexual activity, appetite, or fear responses.

Cingulate Gyrus: The cingulate gyrus is involved in emotional processing and regulation, but it is not the primary structure responsible for increased sexual activity, increased appetite, anger, and decreased fear.

Hypothalamus: This is the correct answer. The hypothalamus is a key structure in the brain responsible for regulating a variety of physiological functions, including hunger, thirst, sexual behavior, and emotions. It plays a vital role in maintaining homeostasis and orchestrating the body's responses to various stimuli. An injury to the hypothalamus could lead to the described changes in behavior, including increased sexual activity, appetite, and altered emotional responses.

Solution for Question 11:

The correct answer is D) Right-sided upper motor neuron lesion.

In this clinical scenario, the patient presents with progressive weakness, spasticity, increased muscle tone, hyperreflexia, and a positive Babinski sign on the right side of the body. These clinical findings suggest an upper motor neuron lesion affecting the right side of the body. The lesion is localized to the left motor cortex, leading to disruption of the descending motor pathways that control the right side of the body.

Options A, B, and C are not the correct diagnoses because they do not match the clinical presentation described. Parkinson's disease typically presents with resting tremor, rigidity, bradykinesia, and a shuffling gait. ALS is characterized by both upper and lower motor neuron signs and can affect multiple limbs, not just one side. Multiple sclerosis (MS) is associated with multifocal demyelination, leading to various neurological symptoms that are not limited to one side of the body.

Solution for Question 12:

Correct Option C - ADH (Antidiuretic Hormone):

- ADH, also known as vasopressin, is produced in the supraoptic nucleus of the hypothalamus and released by the posterior pituitary gland. It plays a key role in regulating water balance by controlling water reabsorption in the kidneys. A tumor in the hypothalamus region that exerts pressure on the supraoptic nucleus can disrupt the production and release of ADH, leading to a condition called diabetes insipidus. In diabetes insipidus, there is an inability to concentrate urine, resulting in excessive thirst and urination.

Incorrect Options:

Option A - ACTH (Adrenocorticotropic Hormone): ACTH is produced by the anterior pituitary gland and regulates the release of cortisol from the adrenal cortex. It plays a role in the body's response to stress and helps maintain proper immune and metabolic functions. A tumor in the hypothalamus region affecting the supraoptic nucleus is unlikely to directly influence ACTH secretion.

Option B - GH (Growth Hormone): GH is also produced by the anterior pituitary gland and is responsible for promoting growth and cell reproduction. While hypothalamic tumors can sometimes cause hormonal imbalances, a tumor affecting the supraoptic nucleus is not likely to have a direct impact on GH secretion.

Option D - TSH (Thyroid Stimulating Hormone): TSH is produced by the anterior pituitary gland and controls the release of thyroid hormones (T3 and T4) from the thyroid gland. It is responsible for regulating the body's metabolism. A tumor in the hypothalamus region affecting the supraoptic nucleus is unlikely to directly influence TSH secretion.

Solution for Question 13:

Correct Answer: B

- The amygdala is an almond-shaped structure that lies in the temporal lobe, lying just beneath the uncus. The amygdala is diverse and complex in structure and comprises

approximately 13 nuclei. They further subdivide into extensive internuclear and intranuclear connections. These nuclei functionally sort into five major groups: basolateral nuclei, cortical-like nuclei, central nuclei, other amygdaloid nuclei, and extended amygdala.

- Amygdala is one of the components of the limbic system, which is responsible for the control of emotions and behavior besides memory formation. It is involved in processing emotions, including fear, anger, and pleasure. Damage to the amygdala can lead to changes in behavior, including increased aggression, impulsivity, and poor decision-making. Amygdala manages the processing of information between the prefrontal-temporal association cortices and the hypothalamus.
- Amygdala has neural circuits to carry out its different functions with two major output pathways; the Dorsal route via stria terminalis that projects to the septal area and hypothalamus, and the ventral route via the ventral amygdalofugal pathway, which terminates in the septal area, hypothalamus, and the medial dorsal thalamic nucleus. The patient's behavior is most consistent with damage to the amygdala. This is because the amygdala is involved in processing emotions, and the patient's behavior is characterized by changes in emotion.

Incorrect Choices:

Option A. Gray matter around the aqueduct of Sylvius: Periaqueductal gray (PAG) matter is a column of gray matter that surrounds the cerebral aqueduct of Sylvius in the midbrain. The periaqueductal gray matter extends from the superior border of the midbrain (which forms the posterior aspect of the floor of the third ventricle), inferiorly towards the fourth ventricle, and the superior medullary velum. The gray matter around the aqueduct of Sylvius is involved in the control of movement. Damage to this area can lead to paralysis or weakness on one side of the body.

Option C. Lateral nuclei of the hypothalamus: In the hypothalamus's lateral posterior part, the lateral tuberal nucleus is a circumscribed cell mass that houses approximately 60000 neurons. Many researchers have investigated how the lateral hypothalamus (LH), and more especially the hypothalamus, controls pain and inflammation. There are connections between the parasympathetic part of the LH and all of the main brain areas.

- It governs eating behavior, waking cycles, and the autonomic nervous system (ANS). It is a component of the reward system. It also contains a variety of neurons, the most significant of which are orexin neurons. Despite being small in number, these neurons are crucial for blocking the transmission of pain and have an impact on the reward system, food patterns, and the hypothalamus-pituitary axis (HPA), among other things.
- The horizontal cores of the nerve center are associated with managing internal heat levels, thirst, and hunger. These functions may be affected by damage to these nuclei, but it is unlikely that the symptoms described in the case will occur.

Option D. Anterior cingulate gyri: The frontal portion of the cingulate cortex is called the anterior cingulate cortex (ACC). The corpus callosum, which transmits brain information between the right and left hemispheres, is encircled by the ACC. The ACC is crucial for cognitive processes, including anticipating rewards, making decisions, empathetic behavior, and emotion. It also appears to have a role in a wide range of autonomic processes, such as controlling blood pressure and heart rate. Numerous bidirectional connections exist between the cingulate cortex and the frontal, temporal, and occipital cortex in both cerebral hemispheres.

Solution for Question 14:

Correct Option B - Senseless continuous speaking:

- A defect in the Wernicke's area, which is located in the left hemisphere of the brain's temporal lobe, can result in a condition known as Wernicke's aphasia. Wernicke's aphasia is characterized by fluent speech that lacks meaningful content. The person may produce sentences that are grammatically correct but don't make much sense. This is often referred to as "word salad" and is a hallmark of damage to the Wernicke's area.

Incorrect Options:

Option A - He can make the decision what he speaks: This option refers to intact executive function and conscious decision-making in speech. If this were the case, the person would have control over their speech and would be able to choose their words appropriately. This is not associated with a defect in the Wernicke's area.

Option C - He is able to understand words and speak properly: This option describes intact language comprehension and production abilities. If this were the case, the individual would be able to understand words and their meanings, as well as communicate their thoughts effectively. Wernicke's area is involved in language comprehension, so a defect in this area would likely impact the ability to understand words properly.

Option D - He is understanding language but unable to speak: This option describes a condition known as Broca's aphasia. Broca's aphasia is characterized by difficulty in producing speech despite relatively preserved language comprehension. Individuals with Broca's aphasia often struggle to form grammatically correct sentences and may speak in short, telegraphic phrases. This condition is associated with damage to the Broca's area, which is located in the left frontal lobe of the brain, typically in the left hemisphere.

Solution for Question 15:

Correct Option C. An increase in RPF increases GFR:

- Correct. An increase in renal plasma flow (RPF) generally increases the glomerular hydrostatic pressure, enhancing the filtration process and thus increasing GFR. When more plasma flows into the kidneys, it elevates the pressure in the glomerular capillaries, thereby increasing the filtration rate.

Option A. An increase in sympathetic function increases GFR:

- Incorrect. Increased sympathetic function generally causes vasoconstriction of the renal arterioles, particularly the afferent arterioles, which reduces renal blood flow and decreases GFR.

Option B. Afferent arteriolar constriction increases GFR:

- Incorrect. Constriction of the afferent arteriole reduces the blood flow into the glomerulus, decreasing the glomerular hydrostatic pressure, which in turn decreases GFR.

Option D. A decrease in RPF increases GFR:

- Incorrect. A decrease in renal plasma flow reduces the amount of blood reaching the glomeruli, leading to a lower glomerular hydrostatic pressure, which decreases GFR.
-

Solution for Question 16:

Correct Option B - Neighboring fibers of the right somatosensory area extending into the right-hand area:

- After amputation, the brain undergoes cortical reorganization. In this case, the neighboring somatosensory fibers that previously represented adjacent body parts extend into the region that formerly represented the right hand. This reorganization is associated with the perception of phantom limb pain.

Incorrect Options:

Option A - Expansion of right somatosensory fibers for the hand into neighboring areas:

- This option suggests that the somatosensory fibers specifically responsible for the right hand expand into neighboring regions. However, it is not the fibers themselves that expand; rather, it is the cortical representation that changes.

Option C - Expansion of right sensory cortex:

- This option implies a general enlargement of the right sensory cortex. However, the expansion is not global; rather, it involves the neighboring cortical areas invading the region corresponding to the amputated limb.

Option D - Expansion of left sensory cortex:

- The left sensory cortex is responsible for processing sensory input from the right side of the body, but this option inaccurately describes the phenomenon. The reorganization occurs in the sensory cortex corresponding to the amputated limb (in this case, the right cortex, which processes left-sided sensations).

Solution for Question 17:

Correct Option A - Inferior olivary nucleus

Incorrect Options:

Option B - Raphe nucleus::

- The Raphe nucleus in the brainstem synthesizes and releases serotonin, regulating mood. Although it connects with various brain regions, including the cerebellum, it doesn't directly reach Purkinje cells.

Option C - Vestibular nucleus::

- The Vestibular nucleus, in the brainstem, processes balance and spatial orientation signals from the vestibular system.
- It coordinates motor responses but doesn't directly connect to Purkinje cells, though it may have connections with the cerebellum.

Option D - Locus coeruleus::

- The Locus coeruleus in the brainstem releases norepinephrine, regulating attention, arousal, and stress responses.

- It has extensive brain connections but doesn't directly reach Purkinje cells in the cerebellum.
-

Solution for Question 18:

Correct Option A - K⁺:

- Potassium channels are oxygen-sensitive ion channels located in the peripheral chemoreceptors.

Incorrect Options:

- Options B, C, and D are not oxygen-sensitive.
-

Solution for Question 19:

Correct Option D – Ach:

- When there is an increase in blood pressure it stimulates the baroreceptor reflex which then stimulates the nucleus tractus solitarius (NTS) which in turn stimulates the Cardiac inhibitory center (CIC), the center for parasympathetic activity. This causes release Acetylcholine at the level of nodal tissue.

Incorrect Options:

Options A, B & C are incorrect

Solution for Question 20:

- Short-term memory, also known as working memory, refers to the temporary storage and manipulation of information for a brief period. It is responsible for holding and processing

information needed for immediate tasks. The duration of short-term memory is limited and typically lasts from 30 to 300 seconds (or 0.5 to 5 minutes).

Incorrect Choices:

- Option a. 3-6 hours: This duration falls into the range of intermediate-term memory, which can retain information for several hours to a few days.
 - Option b. 12-18 hours: This duration falls into the range of long-term memory, which can retain information for an extended period, ranging from hours to years or even a lifetime.
 - Option d. 3-4 seconds: This duration is too short for short-term memory and is more in line with sensory memory, which holds sensory information for a brief moment, usually less than a second.
-

Solution for Question 21:

Correct option:

Option C:

Both are lost: This option is correct. Cutting the muscle spindle disrupts the sensory input from the spindle, leading to the loss of both voluntary Axn and the stretch reflex.

Incorrect Option :

Option A: Voluntary Axn (Action) and stretch reflex present: This option is incorrect. When the muscle spindle is cut, the sensory input from the spindle is disrupted, leading to the loss of the stretch reflex. As a result, the stretch reflex would not be present.

Option B: Voluntary Axn lost but stretch reflex present: This option is incorrect. Cutting the muscle spindle interrupts the sensory input to the spinal cord, resulting in the loss of both voluntary Axn and the stretch reflex. Therefore, neither the voluntary Axn nor the stretch reflex would be present.

Option D: Both voluntary Axn present, stretch reflex is lost: This option is incorrect. When the muscle spindle is cut, the sensory input necessary for the stretch reflex is disrupted, resulting in the loss of the stretch reflex. Therefore, the stretch reflex would not be present.

Solution for Question 22:

Correct Option C – Pain:

- The spinal nucleus of the trigeminal nerve is primarily associated with the sensation of pain and temperature.

Incorrect Options:

Option A - Vibration: Vibration sensation is primarily transmitted through the dorsal column-medial lemniscus pathway, not the spinal nucleus of the trigeminal nerve.

Option B - Conscious proprioception: Conscious proprioception is also transmitted through the dorsal column-medial lemniscus pathway, not the trigeminal nerve.

Option D - Two-point discrimination: Two-point discrimination is mediated by the dorsal column-medial lemniscus pathway and the dorsal root ganglia, not the trigeminal nerve.

Solution for Question 23:

Correct Option B - Sense Changes in Muscle Tension:

- The Golgi tendon organ, located within the muscle tendon, senses changes in muscle tension or force generated during muscle contraction. It provides information to the central nervous system about the level of tension in the muscle.

Incorrect Options:

- Option A, C, and D: They are not the main role of the Golgi tendon organ.
-

Solution for Question 24:

The tingling sensation experienced by the woman suggests compression or irritation of a nerve, which can lead to symptoms like tingling, numbness, or paresthesia (abnormal sensations). In

this case, the most likely nerve fiber type affected is:

Option A. A β fibers

Explanation:

A β fibers are large, myelinated nerve fibers that are responsible for transmitting signals related to touch, pressure, and proprioception (sense of body position) from the skin and joints to the central nervous system. When these fibers are compressed or irritated, they can cause sensations like tingling or numbness.

The scenario described, with the woman placing her hands under her head while watching TV for an extended period of time, likely compressed the nerves in her arms. Prolonged pressure can lead to temporary compression of A β fibers, resulting in the tingling sensation experienced.

Options B (B fibers) and C (C fibers) are not the primary fibers responsible for transmitting sensations related to touch and pressure. B fibers are associated with autonomic nervous system functions, and C fibers are associated with pain and temperature sensations.

Option D (Sympathetic fibers) is not a specific type of nerve fiber. Sympathetic fibers can be A, B, or C fibers and are part of the autonomic nervous system, controlling involuntary bodily functions such as heart rate and blood pressure.

Therefore, the correct answer is Option A. A β fibers.

Solution for Question 25:

Correct Option A - 1 and 2:

- When lower motor neurons are disrupted, the muscle loses its innervation, resulting in significant muscle atrophy due to the cessation of nerve growth factor secretion and loss of muscle function. Denervation potentials such as fibrillation or fascicular twitches are observed in EMG recordings as a result of muscle denervation.

Incorrect Options:

Options B, C & D:

Solution for Question 26:

Correct Option B - Intensity of stimulus:

- Weber-Fechner's law states that the magnitude of sensation felt is proportionate to the log of the intensity of the stimulus.
- $I = K \log (S/S_0)$
- Where: S_0 : Threshold amplitude K: Constant S: Stimulus strength I: Intensity of sensation
- S_0 : Threshold amplitude
- K: Constant
- S: Stimulus strength
- I: Intensity of sensation
- S_0 : Threshold amplitude
- K: Constant
- S: Stimulus strength
- I: Intensity of sensation

Incorrect Options:

- Options A, C, and D are incorrect.
-

Solution for Question 27:

Correct Option A - Broca's area:

- Broca's area is a region located in the frontal lobe of the dominant hemisphere (usually the left hemisphere) of the brain. It is responsible for the production of speech and the coordination of the muscles involved in speech production. Lesions or damage to Broca's area can result in Broca's aphasia or expressive aphasia.
- In Broca's aphasia, the primary deficit is in language output or speech production, with relative preservation of comprehension.

Incorrect Options:

Option B - Wernicke's area: Lesion at Wernicke's area results in fluent aphasia or receptive aphasia. It is characterized by impairment of comprehension of speech and relatively fluent but

paraphasic speech (use of malformed and inappropriate words).

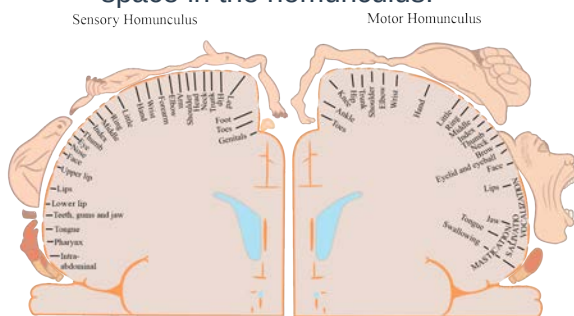
Option C - Primary motor cortex: Lesions in this area can cause motor deficits, such as weakness or paralysis in specific body parts.

Option D - Occlusion of left internal carotid artery: It will result in global aphasia characterized by features of both Broca's and Wernicke's aphasia.

Solution for Question 28:

Correct Option C - The precision of movements:

- Representation of body image in homunculus is based on the fact that the size of the cortical receiving area for impulses from a particular part of the body is proportionate to the use of the part. The more the area of a body part, the more the receptors for carrying out function carried out by those body parts. The homunculus is mainly covered by hand, face, and tongue, which carry out precise movements.
- Therefore, the part of the body responsible for precise movements occupies the maximum space in the homunculus.



Incorrect Options:

- Options A, B, and D are incorrect.

Solution for Question 29:

Correct Option B - Convergence:

- Convergence occurs when multiple neurons synapse onto a single neuron. In this case, the 150 sensory neurons are converging their signals onto a single neuron, resulting in 400 inputs.
- Each sensory neuron can transmit information from different stimuli or locations, and when their signals converge onto a single neuron, it allows for the integration of multiple inputs.

Incorrect Options:

Option A - Divergence: Divergence is the opposite of convergence, where a single neuron sends signals to multiple downstream neurons.

Option C - Feedback inhibition: Feedback inhibition refers to a regulatory mechanism where the output of a process inhibits or reduces the activity of the process itself.

Option D - Feedforward system: Feedforward system is an anticipatory system. When the control system predicts that some change is about to happen and corrective measures are being taken before the change occurs.

Solution for Question 30:

Correct Option A - Hypothalamus:

The hypothalamus plays a central role in regulating body temperature through a complex mechanism involving:

- Sensing temperature changes via thermoreceptors in the skin, deep tissue, spinal cord, and hypothalamus.
- Integrating areas are: Anterior hypothalamus (preoptic area), response to heat. Posterior hypothalamus, response to cold.
- Anterior hypothalamus (preoptic area), response to heat.
- Posterior hypothalamus, response to cold.
- Initiating effector responses, such as Activation of sudomotor sympathetic nerves to promote sweating, which helps dissipate heat through evaporative cooling. Activation of skin sympathetic nerves to induce cutaneous vasodilation, leading to increased blood flow to the skin's surface and heat loss through radiation and convection.

- Activation of sudomotor sympathetic nerves to promote sweating, which helps dissipate heat through evaporative cooling.
- Activation of skin sympathetic nerves to induce cutaneous vasodilation, leading to increased blood flow to the skin's surface and heat loss through radiation and convection.
- The combined effect of the cooling or warming mechanisms allows for the body temperature to return to the normal range (around 98.6°F/37°C). Once the body temperature stabilizes, the activity of thermoreceptors decreases, and the hypothalamus reduces its signaling accordingly.
- Anterior hypothalamus (preoptic area), response to heat.
- Posterior hypothalamus, response to cold.
- Activation of sudomotor sympathetic nerves to promote sweating, which helps dissipate heat through evaporative cooling.
- Activation of skin sympathetic nerves to induce cutaneous vasodilation, leading to increased blood flow to the skin's surface and heat loss through radiation and convection.

Incorrect Options:

Option B - Basal ganglia:

- The basal ganglia are primarily involved in the control of voluntary movements, motor coordination, and procedural learning, rather than temperature regulation.

Option C - Thalamus:

- The thalamus serves as a relay center for sensory information, transmitting signals to different regions of the brain. While it plays a role in processing sensory inputs related to temperature, it does not directly regulate body temperature.

Option D - Limbic system:

- The limbic system is involved in emotions, memory, and behavior. While certain components of the limbic system, such as the amygdala and hypothalamus (which is part of the limbic system), may influence physiological responses to emotions and stressors that can affect body temperature, the limbic system itself does not directly regulate body temperature.

Solution for Question 31:

Correct Option D - Mean arterial pressure – intracranial pressure:

- Cerebral perfusion pressure is defined as the pressure gradient that drives blood flow to the brain. It is calculated by subtracting the intracranial pressure (ICP) from the mean arterial pressure (MAP). This formula accurately measures the pressure gradient that drives cerebral blood flow and is an essential parameter to consider in managing patients with neurological conditions.

Incorrect Options:

Option A, B and C are incorrect.

Solution for Question 32:

Correct Option A - Raphe nucleus:

- The Raphe nucleus present in the medulla is responsible for profound analgesia. These nuclei are serotonergic and end in the dorsal horn in the spinal cord.
- The Pain-carrying fiber (Adelta/C) creates a synapse with a second-order neuron, which brings the sensation up to the level of the thalamus and perceives pain.
- Serotonin activates enkephalin-producing interneurons and inhibits the pain pathway, acting at the levels of presynaptic (1st order) and postsynaptic neurons (2nd order).
- Other areas responsible for analgesics are RVLM (Rostral Ventrolateral Medullary Region) and Locus coeruleus region secretes norepinephrine.

Incorrect Options:

Option B - Corticospinal tract: The corticospinal tract is primarily involved in voluntary motor control and does not play a direct role in pain relief.

Option C - Medial lemniscus: The medial lemniscus is a pathway in the central nervous system that carries sensory information related to touch, vibration, and proprioception.

Option D - Medial longitudinal fasciculus: The medial longitudinal fasciculus is a neural pathway involved in coordinating eye movements and connecting various brainstem nuclei.

Solution for Question 33:

Correct Option A - 1 and 4 are correct (Merkel disc and Meissner's corpuscle):

- Tactile discrimination, which refers to the ability to discern fine details, textures, and shapes through touch, relies on superficial sensory receptors in the skin.
- Both Merkel discs and Meissner's corpuscles are involved in tactile discrimination, allowing individuals to sense and distinguish intricate textures and shapes.

Incorrect Options:

- Options B, C, and D are incorrect. Pacinian and ruffini end organs are deep sensory receptors stimulated by pressure.
-

Solution for Question 34:

Correct Option C - Gate Control Theory:

- TENS works based on the gate control theory of pain occurring at the level of spinal cord, which suggests that non-painful sensory input (pressure) can block pain signals from reaching the brain.
- By stimulating large-diameter nerve fibers ($A\beta$ fibers) via pressure or vibrations that stimulate the interneuron, i.e., G neuron where $A\beta$, $A\delta$, and C fibers synapse.
- Stimulation of $A\beta$ fibers inhibits interneuron and stops the transmission from $A\delta$, and C fibers carrying pain perception. This is called gate control theory.

Incorrect Options:

- Options A, B and D: Central inhibition, desensitization, and descending inhibition are not the primary mechanisms of action for TENS.

Options A, B and D: Central inhibition, desensitization, and descending inhibition are not the primary mechanisms of action for TENS.

Solution for Question 35:

Correct Option C - Muscle Spindle:

- Muscle spindles are responsible for mediating the stretch reflex. They detect changes in muscle length and initiate a reflex response that leads to muscle contraction.

Incorrect Options:

Option A - Golgi tendon organs: Golgi tendon organs detect muscle tension.

Option B - Merkel's discs: Merkel's discs are involved in light touch sensation in the skin.

Option D - Meissner's corpuscles: Meissner's corpuscles perceive tactile discrimination.

Solution for Question 36:

Correct Option A - Rigidity occurs in all the muscles of the body:

- Decerebrate rigidity primarily affects the extensor muscles of the limbs, leading to the characteristic posture of rigid extension. It does not affect all the muscles of the body equally.
- According to the Herrington classification, decerebrate rigidity is a form of abnormal muscle tone that occurs due to damage or dysfunction in the medulla resulting in increased activity of motoneurons that facilitate stretch reflexes.
- Facilitation of stretch reflexes occurs due to two mechanisms: Increased excitability of the motor neuron pool in general. Increased rate of γ motor neuron discharge.
- Increased excitability of the motor neuron pool in general.
- Increased rate of γ motor neuron discharge.
- Producing a mid-collicular lesion in an experimental animal results in decerebration.
- Increased excitability of the motor neuron pool in general.
- Increased rate of γ motor neuron discharge.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 37:

Correct Option A - Subthalamic nuclei:

- The clinical presentation of uncontrollable, forceful, and jerking motions of the limbs on one side and a history of stroke suggest the diagnosis of hemiballismus, a movement disorder characterized by involuntary and violent flinging movements of the extremities on one side of the body.
- Hemiballismus is typically caused by damage to the subthalamic nucleus, a part of the basal ganglia involved in regulating movement.

Incorrect Options:

Option B - Substantia nigra: Lesions of the substantia nigra lead to extremely severe disease of rigidity, akinesia, and tremors known as Parkinson's disease

Option C - Putamen: Lesion in putamen will result in flicking movements in the hands, face, and other parts of the body, called chorea.

Option D - Caudate nucleus: Damage to the caudate nucleus can also lead to deficit in performance and chorea.

Solution for Question 38:

Correct Option C - Absence of ipsilateral arm proprioception:

- The dorsal column tract of the spinal cord is responsible for proprioception, vibration sensation, and fine touch.
- The tract is divided into two parts: the fasciculus gracilis, which carries information from the lower body (sacral, lumbar, and lower 6 thoracic regions T7 to T12), and the fasciculus cuneatus, which carries information from the upper body (cervical and upper 6 thoracic region T1 to T6) and upper extremities. The tract of cuneatus lies lateral to the tract of gracilis.

- In both tracts, the sensory fibers ascend ipsilaterally (on the same side) in the spinal cord. Therefore, an injury to the lateral aspect of the dorsal column tract would affect the fasciculus cuneatus, leading to loss of proprioception in the ipsilateral arm.

Incorrect Options:

Option A - Absence of ipsilateral lower limb proprioception: This would occur if the medial aspect of the dorsal column tract, which contains the fasciculus gracilis, were injured.

Option B - Absence of fine motor movement of fingers: Fine motor movements are governed by the corticospinal tract, not the dorsal column tract.

Option D - Absence of contralateral lower limb proprioception: Proprioception ascends ipsilaterally in the spinal cord, so an injury would not affect the contralateral side.

Solution for Question 39:

Correct Option C - Alkaloids:

- Alkaloids are a class of naturally occurring organic compounds that often have bitter tastes. They are produced by various plants as a defense mechanism against herbivores and insects. Many alkaloids are toxic or have pharmacological effects on the human body. Some examples of alkaloids include caffeine, nicotine, morphine, and quinine.

Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 40:

Correct Option B - Oligodendrocytes:

- In multiple sclerosis (MS), the cells primarily affected are oligodendrocytes. Oligodendrocytes are a type of glial cell in the central nervous system (CNS) that play a crucial role in producing and maintaining the protective myelin sheath around nerve fibers.

The myelin sheath acts as an insulating layer, allowing for efficient transmission of electrical signals between nerve cells.

- In MS, the immune system mistakenly attacks and damages the myelin sheath, leading to demyelination. This demyelination disrupts the normal transmission of nerve impulses, resulting in a wide range of neurological symptoms such as motor weakness, nuchal rigidity, swallowing difficulties, and impaired daily activities.

Incorrect Options:

Option A - Microglia: Microglia are immune cells in the CNS that act as the resident macrophages, responsible for immune surveillance and response in the brain.

Option C - Astrocytes: Astrocytes are another type of glial cell found in the CNS. They have various functions, including providing structural support, maintaining the blood-brain barrier, and regulating neurotransmitter levels.

Option D - Ependymal cells: Ependymal cells line the ventricles of the brain and the central canal of the spinal cord. They play a role in the production and circulation of cerebrospinal fluid (CSF).

Solution for Question 41:

Correct Option A - Pacinian corpuscles:

- Pacinian Corpuscles are the largest and most sensitive tactile receptor present in hairy and non-hairy skin.
- It is responsible for high-frequency vibration detection of 200 Hz or more.

Incorrect Options:

Option B - Free nerve endings: Free nerve endings are slowly adapting receptor. It is responsible for itching and tickling kind of sensations.

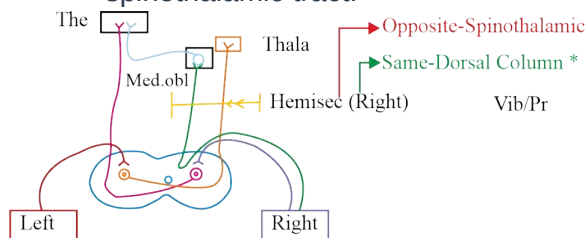
Option C - Meissner corpuscles: Meissner corpuscles are fast adapting receptor. It is responsible for the vibration detection with a frequency lower than 50 Hz.

Option D - Merkel discs: Merkel's disc are slowly adapting receptor responsible for low-frequency (5 Hz) vibrations detection and sharp edge or corner points stimulus.

Solution for Question 42:

Correct Option C - Brown Sequard syndrome:

- Brown Sequard syndrome occurs due to the hemisection of the spinal cord.
- It is characterized by: On the side of the lesion: Loss of touch, proprioception, vibration, and tactile discrimination are lost due to damage to the dorsal column pathway. On the opposite side: Pain and thermal sensations are lost due to damage to the spinothalamic tract.
- On the side of the lesion: Loss of touch, proprioception, vibration, and tactile discrimination are lost due to damage to the dorsal column pathway.
- On the opposite side: Pain and thermal sensations are lost due to damage to the spinothalamic tract.
- On the side of the lesion: Loss of touch, proprioception, vibration, and tactile discrimination are lost due to damage to the dorsal column pathway.
- On the opposite side: Pain and thermal sensations are lost due to damage to the spinothalamic tract.



Incorrect Options:

Option A: Weber's syndrome - Weber syndrome, classically described as a midbrain stroke syndrome characterized by drooping of the eyelid, diplopia associated with contralateral partial or complete weakness of upper limb and lower limb.

Option B: Dejerine syndrome - Central post-stroke pain, also known as Dejerine Roussy syndrome or thalamic pain syndrome, is a rare central neuropathic pain syndrome that occurs after infarction of the ventroposterolateral thalamus.

Option D: Wallenberg syndrome - Wallenberg's syndrome is caused by a stroke in the brain stem, specifically in one of the arteries that provide blood to the cerebellum. It is characterized by dizziness with vertigo, loss of balance with gait instability, hoarseness of voice, and difficulty swallowing.

Solution for Question 43:

Correct Option D - Pressure:

- Pacinian corpuscles are sensory receptors found in the skin and deeper tissues of the body. They are specialized to detect mechanical stimuli, particularly vibrations and pressure changes. When pressure is applied to the Pacinian corpuscle, it undergoes deformation, which triggers the generation of electrical signals and stimulates the associated sensory nerve fibers.

Incorrect Options:

- Options A, B, and D are incorrect. Pacinian corpuscles are stimulated by pressure.

Options A, B, and D are incorrect. Pacinian corpuscles are stimulated by pressure.

Solution for Question 44:

Correct Option A - Parietal lobe:

- Impairment of hand-eye coordination is commonly associated with lesions in the brain's parietal lobe especially in the brodmann area 5 and 7.
- The parietal lobe integrates visual input from the occipital lobe and motor output from the frontal lobe and acts as an association cortex.. It helps process and integrate sensory inputs to guide motor movements and coordinate hand-eye interactions and is the most important area.

Incorrect Options:

Option B - Frontal lobe: The frontal lobe is responsible for motor output and is involved in the hand-eye movements, but the parietal lobe is the most important.

Option C - Temporal lobe: The temporal lobe is mainly associated with auditory processing, memory, and language functions, and lesions in this lobe typically do not cause significant impairments in hand-eye coordination.

Option D - Occipital lobe: The occipital lobe is primarily responsible for visual processing and is involved in the hand-eye movements, but the parietal lobe is the most important.

Solution for Question 45:

Correct Option D - Lower medulla:

- 80% of the corticospinal tract after crossing the pyramid of medulla oblongata present in the lower medulla decussate to the other side forming lateral spinothalamic tract.
- Rest of the 20% of fibres continues as anterior spinothalamic tract ipsilaterally and decussate at the level of the spinal cord.

Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.
-

Solution for Question 46:

Correct Option C - Sweating:

- The hypothalamic thermostat controls the body's core temperature by initiating either heat production or heat loss mechanisms to maintain a set point. When the set point is changed, the body responds by adjusting its temperature through various mechanisms.
- When thermostat pressure is changed from point A to point C it increases the thermostat pressure all the mechanisms for raising the body temperature are brought into play, that are: Vasoconstriction. Piloerection. Epinephrine release. Shivering.
- Vasoconstriction.
- Piloerection.
- Epinephrine release.
- Shivering.
- Sweating and vasodilation is observed when thermostat temperature is set to a low temperature.

- Vasoconstriction.
- Piloerection.
- Epinephrine release.
- Shivering.

Incorrect Options:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 47:

Correct Option D - It is characterized by flexion of upper limbs and extension of lower limbs:

- Injury to the cerebral cortex results in decorticate rigidity, characterized by upper extremity flexion at the elbow and increased extension in the lower extremities.
- The flexion of the upper extremities is explained by rubrospinal excitation of flexor muscles, while the hyperextension of the lower extremities is due to dysfunction in the medulla, resulting in increased activity of motoneurons that facilitate stretch reflexes.
- Dysfunction in the descending pathways that control muscle tone and movement.

Incorrect Options:

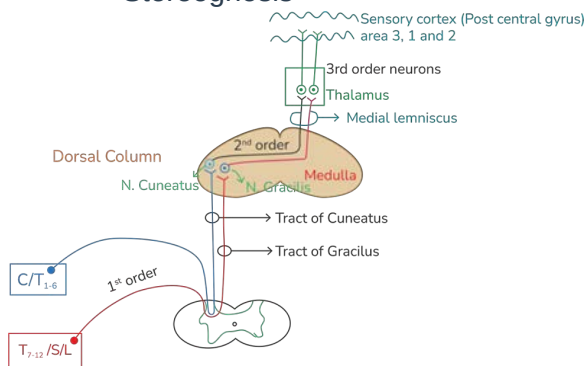
- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 48:

Correct Option D - Loss of tactile and proprioception on the opposite side:

- A lesion in the medial lemniscus at the level of the pons would lead to damage in the dorsal column pathway as the second order neurons pass from it to reach thalamus.
- The second order neurons arise from nucleus gracilis and cuneatus of the contralateral side.. These are responsible for the transmitting the following sensations: Fine touch
Vibration Proprioception Tactile localization Tactile discrimination Stereognosis

- Fine touch
- Vibration
- Proprioception
- Tactile localization
- Tactile discrimination
- Stereognosis
- So, damage to the dorsal column pathway at medial lemniscus will result in loss of proprioception and tactile on the opposite side.
- Fine touch
- Vibration
- Proprioception
- Tactile localization
- Tactile discrimination
- Stereognosis



Incorrect Options:

- Options A, B, and C are incorrect. Refer to Option D for an explanation.

Solution for Question 49:

Correct Option A - VIP (Vasoactive Intestinal Peptide):

- Vasoactive Intestinal Peptide (VIP) is a neurotransmitter released when the cholinergic neuron from the myenteric plexus moves in the anterograde direction of the peristalsis.

- It acts as a relaxant on smooth muscles in the proximal segment of the segment containing chyme, resulting in relaxation of the proximal segment of the chyme and movement of the content towards the rectum.

Incorrect Options:

Option B - Acetylcholine: Acetylcholine is the primary neurotransmitter released by cholinergic neurons in the retrograde direction of the peristalsis and results in the contraction of the smooth muscle.

Option C - Substance P: Substance P is a neurotransmitter that is released when cholinergic neuron from the myenteric plexus moves in the retrograde direction of the peristalsis and results in the contraction of the smooth muscle.

Option D - Serotonin: Serotonin is a neurotransmitter that is secreted by the gastric epithelial cells in response to stretch due to food. It is responsible for the activation of myenteric plexus resulting in coordinated movements in peristalsis.

Solution for Question 50:

Correct Option C - Pain and temperature loss on the opposite side:

- Brown Sequard syndrome occurs due to the hemisection of the spinal cord.
- It is characterized by: On the side of the lesion: Loss of touch, proprioception, vibration, and tactile discrimination are lost due to damage to the dorsal column pathway. On the opposite side: Pain and thermal sensations are lost due to damage to the spinothalamic tract.
- On the side of the lesion: Loss of touch, proprioception, vibration, and tactile discrimination are lost due to damage to the dorsal column pathway.
- On the opposite side: Pain and thermal sensations are lost due to damage to the spinothalamic tract.
- On the side of the lesion: Loss of touch, proprioception, vibration, and tactile discrimination are lost due to damage to the dorsal column pathway.
- On the opposite side: Pain and thermal sensations are lost due to damage to the spinothalamic tract.

Incorrect Option:

- Options A, B, and D are incorrect. Refer to Option C for an explanation.

Solution for Question 51:

Correct Option C - Difficulty in identifying known faces:

- Prosopagnosia, also known as face blindness, is a neurological disorder characterized by the inability or difficulty to recognize and identify familiar faces, including those of friends, family members, and even one's own face.
- It is caused due to lesion in the inferior temporal lobe.
- People with prosopagnosia may have normal vision and intellectual abilities but struggle specifically with facial recognition.

Incorrect Options:

- Options A, B, and D are incorrect.
-

Solution for Question 52:

Correct Option C - Pacinian corpuscles:

- Pacinian Corpuscles is the largest and sensitive tactile receptor. Responsible for vibration detection 200 Hz or more. It is responsible for the mediation of fast vibration sense.

Incorrect Options:

Option - Merkel's disc: Merkel's Disk detects low-frequency vibrations at 5 Hz.

Option B - Meissner's corpuscles: Meissner is responsible for vibration detection with a frequency lower than 50 Hz.

Option D - Ruffini's end organ: The best stimulus is a stretch on the skin by some large object for Ruffini's End Organ.

Solution for Question 53:

Correct Option D - All of the above:

- All the given options stimulate the thirst mechanism.
- Dehydration, hemorrhage, and vomiting will lead to loss of isotonic fluid from the blood leading to increased renin secretion. Renin leads to an increase in the concentration of angiotensin 2 (Ag II), which acts as a stimulus for the subfornical nucleus (SFO) osmoreceptor center in the anterior hypothalamus.

Incorrect Options:

- Option A, B & C are incorrect.
-

Solution for Question 54:

Correct Option C - Area Postrema:

- The area postrema is a specialized region located in the medulla oblongata at the base of the fourth ventricle in the brainstem. It is commonly referred to as the "vomiting center" or the "chemoreceptor trigger zone" (CTZ).
- The area postrema is responsible for inducing vomiting activated by toxic substances like drugs or due to hormones as seen in pregnancy.

Incorrect Options:

Option A - Apneustic center: The apneustic center is responsible to control the depth of respiration.

Option B - Pneumotaxic center: The pneumotaxic center, also located in the pons, is involved in controlling the rate of respiration.

Option D - Pre Botzinger complex: Pre Botzinger complex is known as the pacemaker of respiration.

Hormone Mechanism & Pituitary hormone

1. Which statement correctly describes the role of the supraoptic and paraventricular nuclei in pituitary hormone secretion?

- A. The supraoptic nucleus primarily produces oxytocin, while the paraventricular nucleus primarily produces ADH.
 - B. The supraoptic and paraventricular nuclei contain neurons that release hormones from axon terminals in the anterior pituitary gland.
 - C. Magnocellular neurons in these nuclei synthesize ADH and oxytocin and release them from the posterior pituitary gland.
 - D. The supraoptic and paraventricular nuclei are located in the anterior pituitary gland and control its hormone secretion directly.
-

2. A 52-year-old man has had chronic polyuria & polydipsia for the past 6 months. He has a history of long-term lithium use for bipolar disorder. O/E, he is slightly dehydrated. Laboratory tests show normal blood glucose and ADH levels, and urinalysis indicates dilute urine with low osmolality. Kidney function tests are normal. What is the most likely diagnosis?

- A. Neurogenic DI
 - B. Nephrogenic DI
 - C. SIADH
 - D. Diabetes Mellitus
-

3. Which of the following statements about ADH receptors and their functions is FALSE?

- A. V2 receptors activate the Gs protein, which increases cyclic AMP levels to facilitate water reabsorption in the kidneys.
 - B. V1A receptors are involved in increasing blood pressure by causing vasodilation in blood vessels.
 - C. V1B receptors are located in the pituitary gland and enhance the release of ACTH through phosphatidylinositol hydrolysis.
 - D. V2 receptors increase the permeability of kidney tubules to water by promoting the insertion of aquaporins into the cell membrane.
-

4. Which of the following statements is false regarding prepro-pressophysin and prepro-oxyphysin?

- A. Prepro-pressophysin is the precursor for vasopressin, while prepro-oxyphysin is the precursor for oxytocin.
 - B. Both prepro-pressophysin and prepro-oxyphysin are synthesized in the hypothalamus.
 - C. Prepro-oxyphysin contains neurophysin II, which is involved in the transport of oxytocin.
 - D. Prepro-pressophysin and prepro-oxyphysin are transported to the posterior pituitary via Herring bodies.
-

5. A 35-year-old woman presents with irregular menstrual periods, spontaneous lactation, and elevated prolactin levels. She is not pregnant or breastfeeding. She is under work-related stress and uses opioids for chronic pain. Which of the following is NOT likely contributing to her elevated prolactin levels?

- A. Angiotensin-II
 - B. Thyrotropin-releasing hormone (TRH)
 - C. Dopamine agonists
 - D. Stress
-

6. What is the likely diagnosis for a 45 y/o man presenting with increasing shoe size, prominent facial features, frequent headaches, enlarged hands and feet, joint pain, coarse facial features, protruding jaw, MRI findings of an enlarged sella turcica compressing the optic chiasm, and elevated levels of growth hormone (GH) and prolactin?

- A. Acromegaly
 - B. Gigantism
 - C. Pituitary Dwarfism
 - D. Hypothyroidism
-

7. Which Insulin-like Growth Factor (IGF) is known to be more potent in promoting cell growth?

- A. IGF-I
 - B. IGF-II
 - C. Insulin
 - D. Growth Hormone (GH)
-

8. A 45 y/o woman is found to have elevated growth hormone (GH) levels during routine screening. Despite these high levels, her Insulin-like Growth Factor-I (IGF-1) levels are also elevated, suggesting a failure in the feedback mechanism. Which condition could most likely explain the persistent high GH levels?

- A. Pituitary adenoma
- B. Hypothalamic lesion impairing somatostatin release
- C. Reduced IGF-1 receptor sensitivity in the pituitary
- D. Excessive GHRH secretion from an ectopic source

9. Researchers are studying the effects of various factors on growth hormone (GH) secretion. They observed a significant increase in GH levels in participants who fasted overnight before the study. Which of the following best explains this observation?

- A. Increased somatostatin secretion
 - B. Elevated blood glucose levels
 - C. Starvation or protein deficiency
 - D. Reduced ghrelin levels
-

10. A 30-year-old woman presents with galactorrhea (inappropriate milk secretion) and menstrual irregularities. After excluding other causes, the physician considers a potential pituitary dysfunction. Which hypothalamic hormone contributes to increased prolactin levels in this patient?

- A. Corticotropin-releasing hormone (CRH)
 - B. Dopamine
 - C. Thyrotropin-releasing hormone (TRH)
 - D. Gonadotropin-releasing hormone (GnRH)
-

11. Which cell type constitutes the majority in the anterior pituitary cell population?

- A. Corticotropes
- B. Somatotropes
- C. Lactotropes
- D. Thyrotropes

12. A researcher observes that certain cancer cells in a tissue sample are proliferating rapidly due to communication involving transforming growth factor α (TGF α) present on their surfaces. Which of the following cellular communication methods is being used?

- A. Endocrine
 - B. Paracrine
 - C. Autocrine
 - D. Juxtacrine
-

13. Which of the following statements is true regarding the neurohypophysis (posterior pituitary) and the adenohypophysis (anterior pituitary)?

- A. The adenohypophysis is derived from neural tissue.
 - B. The adenohypophysis is linked to the hypothalamus through the hypothalamic-hypophyseal portal system.
 - C. The neurohypophysis contains five types of hormone-secreting cells, whereas the adenohypophysis does not produce hormones.
 - D. The neurohypophysis secretes growth hormone and thyroid-stimulating hormone
-

14. Which of the following correctly describes a key difference between Type 1 and Type 2 steroidal or intracellular hormone receptors?

- A. Type 1 receptors are located in the nucleus, while Type 2 receptors are found in the cytoplasm.
- B. Type 1 receptors are bound to heat shock proteins in their inactive state, whereas Type 2 receptors can bind DNA even without a ligand.

- C. Type 1 receptors function as heterodimers, while Type 2 receptors function as homodimers.
 - D. Type 1 receptors do not have intrinsic enzymatic activity, whereas Type 2 receptors do.
-

15. What is the primary mechanism of action for receptor tyrosine kinases (RTKs)?

- A. Activation through intracellular ligand binding
 - B. Dimerization upon extracellular hormone-binding followed by phosphorylation
 - C. Formation of ion channels in the membrane
 - D. Direct activation of G proteins inside the cell
-

16. Which of the following best describes the activation mechanism of transmembrane guanylyl cyclases (pGCs)?

- A. Ligand binding induces a conformational change, activating cGMP synthesis.
 - B. Phosphorylation directly activates cGMP synthesis.
 - C. The intracellular domain initiates gene transcription.
 - D. Intracellular proteins bind to activate cGMP synthesis.
-

17. Which of the following statements about the adenylyl cyclase-cAMP pathway and the phospholipase C pathway is INCORRECT?

- A. In the adenylyl cyclase-cAMP pathway, hormone binding to a Gs-coupled GPCR leads to activation of adenylyl cyclase, which converts ATP to cAMP.
- B. In the phospholipase C pathway, phospholipase C converts PIP2 into inositol triphosphate (IP3) and diacylglycerol (DAG)
- C. Adenylyl cyclase activation in the adenylyl cyclase-cAMP pathway leads to the inhibition of cAMP-dependent protein kinase.

D. In the phospholipase C pathway, IP3 mobilizes calcium ions from internal stores, and DAG activates protein kinase C to mediate cellular responses.

18. Which of the following correctly describes a mechanism involving GPCRs?

- A. GPCRs have eight transmembrane segments.
 - B. The α subunit of G proteins binds GDP when activated.
 - C. Prolonged hormone exposure can decrease active GPCR numbers.
 - D. The α subunit permanently inactivates the GPCR.
-

19. Which type of hormone receptor binds to hydrophobic hormones and acts primarily as a transcription factor within the nucleus?

- A. G protein-coupled receptor (GPCR)
 - B. Insulin receptor
 - C. Estrogen receptor
 - D. Ion channel receptor
-

20. Which of the following statements correctly describes the activity of these hormones?

- A. Protein hormones diffuse through the cell membrane and act on nuclear receptors.
- B. Steroidal hormones bind to cell surface receptors and initiate second messenger cascades.
- C. Protein hormones bind to cell surface receptors and initiate intracellular signaling cascades.
- D. Steroidal hormones are hydrophilic and cannot penetrate the cell membrane.

21. Which of the following hormones can diffuse through the cell membrane and primarily act inside the cell by binding to nuclear receptors?

- A. Insulin
 - B. Catecholamines
 - C. Cortisol
 - D. Vasopressin
-

22. A 35 y/o woman reports feeling tired during the day and struggles to stay awake at work. Her sleep study reveals that her sleep-wake cycle is disrupted, pointing to an irregular circadian rhythm. Which of the following is not an example of a circadian rhythm?

- A. Sleep-wake cycle
 - B. Body temperature fluctuations.
 - C. Menstrual cycle
 - D. Peaks of cortisol levels in serum
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	2
Question 4	3
Question 5	3
Question 6	1
Question 7	1
Question 8	3
Question 9	3
Question 10	3
Question 11	2
Question 12	4
Question 13	2
Question 14	2
Question 15	2
Question 16	1
Question 17	3
Question 18	3
Question 19	3
Question 20	3
Question 21	3
Question 22	3

Solution for Question 1:

Magnocellular neurons in the supraoptic and paraventricular nuclei synthesize ADH and oxytocin, and these hormones are released from the posterior pituitary gland.

- Supraoptic and Paraventricular Nuclei: Located in the hypothalamus. (Option D) Contain magnocellular neurons that synthesize and secrete posterior pituitary hormones: ADH and oxytocin. The supraoptic nucleus primarily produces ADH. (Option A) The paraventricular nucleus primarily produces oxytocin. (Option A) Each nucleus can also synthesize smaller amounts of the other hormone. These neurons release hormones from axon terminals in the posterior pituitary. (Option B)
- Located in the hypothalamus. (Option D)
- Contain magnocellular neurons that synthesize and secrete posterior pituitary hormones: ADH and oxytocin.
- The supraoptic nucleus primarily produces ADH. (Option A)
- The paraventricular nucleus primarily produces oxytocin. (Option A)
- Each nucleus can also synthesize smaller amounts of the other hormone.
- These neurons release hormones from axon terminals in the posterior pituitary. (Option B)
- Located in the hypothalamus. (Option D)
- Contain magnocellular neurons that synthesize and secrete posterior pituitary hormones: ADH and oxytocin.
- The supraoptic nucleus primarily produces ADH. (Option A)
- The paraventricular nucleus primarily produces oxytocin. (Option A)
- Each nucleus can also synthesize smaller amounts of the other hormone.
- These neurons release hormones from axon terminals in the posterior pituitary. (Option B)
- Magnocellular neurons are also found in the suprachiasmatic nuclei and extend from paraventricular nuclei to the brainstem and spinal cord.

Median Eminence

- Not a nucleus but is crucial for hormone transport.
- Links the hypothalamus to the anterior pituitary.
- Facilitates communication between the hypothalamus and anterior pituitary.
- Hypothalamic neurons secrete releasing and inhibitory hormones into the median eminence.

- Hormones travel through hypothalamic-hypophyseal portal vessels to the anterior pituitary.

Arcuate Nucleus

- Houses neurons that secrete growth hormone-releasing hormone (GHRH).
- GHRH stimulates the anterior pituitary to secrete growth hormone.
- Indirectly contributes to pituitary hormone secretion.

Solution for Question 2:

- Nephrogenic Diabetes Insipidus occurs when the kidneys are unable to respond to ADH despite its normal secretion.
- The patient's long-term use of lithium, a known cause of nephrogenic DI, aligns with the symptoms of polyuria and polydipsia and the laboratory findings of dilute urine.
- The presence of normal kidney function tests suggests that the primary issue is with ADH action in the kidneys rather than an intrinsic kidney disease.

A) Neurogenic DI results from a deficiency of ADH due to problems in the hypothalamus or posterior pituitary. This is not consistent with the patient's history of lithium use, which is more commonly associated with nephrogenic DI. Additionally, neurogenic DI would typically present with a different clinical scenario, such as a recent head injury or central nervous system issues.

Type	Description	Causes	Key Findings	Treatment
Nephrogenic Diabetes Insipidus (DI)	The kidney fails to respond to ADH despite its normal secretion.	- Acquired: - Due to drugs such as Lithium, demeclocycline, rifampicin, aminoglycosides, cisplatin, amphotericin B - Genetic: X-linked recessive (V2 receptor deficiency), autosomal (aquaporin gene deficiency)	-Polyuria -Dilute urine with low osmolality -Normal blood glucose levels	Chlorpropamide (increases renal response to ADH)

Neurogenic/ Central Diabetes Insipidus	Deficiency of ADH secretion due to central issues (hypothalamus or posterior pituitary).	-Head injury -Tumors (craniopharyngioma, pituitary tumors) -Infections (meningitis, encephalitis) -Vascular lesions (Sheehan's syndrome, carotid aneurysm) -Congenital defects	- Polyuria -Dilute urine with low osmolality -Normal blood glucose levels	DDAVP (synthetic ADH analog), Vasopressin injection, Clofibrate therapy
Syndrome of Inappropriate ADH Secretion (SIADH)	Excessive ADH secretion leads to water retention and dilutional hyponatremia.	-Head injury -Ectopic ADH production (e.g., tumors: lung, pancreas, ovary, bladder) -Neurologic diseases (multiple sclerosis, Guillain-Barré syndrome, brain abscess, meningitis, encephalitis) -Drugs (desmopressin, chlorpropamide, oxytocin, vincristine, phenothiazines, carbamazepine)	- Hyponatremia (low sodium) - Dilute serum osmolality - Concentrated urine with high osmolality (Option C)	- Fluid restriction - Sodium replacement - Address the underlying cause

D) Diabetes Mellitus causes polyuria and polydipsia but is associated with elevated blood glucose levels and hyperosmolar urine due to glucose. The patient's normal blood glucose levels and dilute urine do not fit the profile of diabetes mellitus.

Feature	Diabetes Insipidus	Diabetes Mellitus
Urine Osmolality	<300 mOsm/L (dilute urine)	>1200 mOsm/L (concentrated urine)
Urine Taste	Tasteless	Sweet due to glucose
Cause	ADH deficiency or resistance	Glucose loss in urine

Solution for Question 3:

V1A receptors are involved in increasing blood pressure by causing vasoconstriction, not vasodilation.

Receptor Type	Location	Main Functions	Mechanism by G-Protein Coupling
V1A	Smooth muscles of blood vessels	Causes vasoconstriction	Gq - phosphatidylinositol hydrolysis Increases calcium concentration by activating phospholipase C
V1B	Pituitary gland (corticotrophs)	Increases CRH and ACTH secretion	Gq - phosphatidylinositol hydrolysis Stimulates CRH and ACTH release via V1B receptors
V2	Kidney tubules	Increases water reabsorption	Gs - cyclic AMP increase Leading to the insertion of aquaporins into the membrane

Other functions of ADH:

- Increases Water Reabsorption in kidneys.
- Raises Blood Pressure d/t vasoconstriction
- Supports Memory: Acts as a neurotransmitter in memory-related brain areas.
- Aids Ejaculation by contracting smooth muscles of the spermatic cord
- Boosts ACTH Production from the pituitary gland.
- Reduces Cardiac Output by acting on a specific brain area.
- Stimulates Glycogen Breakdown in the liver.
- Neurotransmitter Function in some parts of the brain and spinal cord.

Solution for Question 4:

Prepro-oxypheysin contains neurophysin I, not neurophysin II. Neurophysin I is responsible for the transport of oxytocin, while neurophysin II is associated with vasopressin transport.

Feature	Prepro-pressophysin	Prepro-oxypheysin
---------	---------------------	-------------------

Precursor for (Option A)	Vasopressin	Oxytocin
Components	Signal Peptide Vasopressin Neurophysin II	Signal Peptide Oxytocin Neurophysin I
Function of Active Hormone	Regulates water retention in kidneys, affects blood pressure	Plays a key role in childbirth and lactation, influences social bonding
Processing Location	Golgi apparatus	
Transport to Posterior Pituitary	Via secretory vesicles called Herring Bodies (Option D)	
Synthesis Location	Magnocellular neurons in the hypothalamus (Option B)	
Gene Location	Chromosome 20	

Solution for Question 5:

Dopamine inhibits prolactin secretion. Increased dopamine activity due to dopamine agonists would decrease prolactin levels, making it unlikely to contribute to elevated prolactin levels.

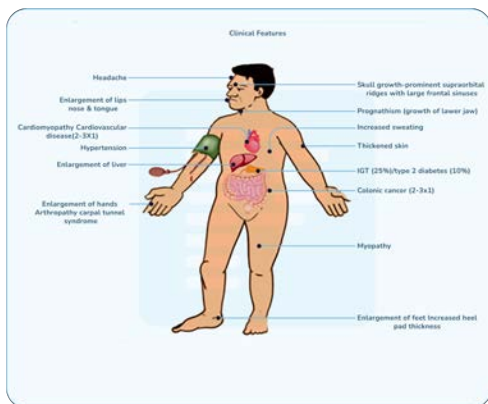
Table rendering failed

Solution for Question 6:

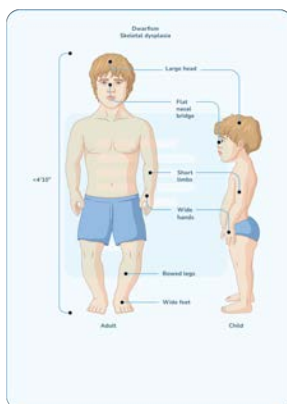
The clinical scenario describes symptoms typical of acromegaly, including enlarged hands and feet, coarse facial features, joint pain, headaches, and an enlarged sella turcica with a mass effect on the optic chiasma. The elevated GH and prolactin levels further support this diagnosis.

Table rendering failed

The image below shows gigantism.



The image below shows pituitary dwarfism.



D) Hypothyroidism: Hypothyroidism may present with symptoms such as fatigue, weight gain, and cold intolerance, but it does not cause the physical changes described (enlargement of hands, feet, and facial features) or elevated GH levels. Thus, hypothyroidism is an unlikely diagnosis in this scenario.

Solution for Question 7:

IGF-I (Somatomedin-C) is the most potent in promoting cell growth and mediating the effects of growth hormone (GH). It plays a crucial role in skeletal and cartilage growth especially during postnatal development.

B) IGF-II: IGF-II is important for fetal development but is generally less potent than IGF-I in promoting postnatal cell growth. Its actions are not as closely linked to GH regulation as IGF-I.

C) Insulin: While insulin shares structural similarities with IGFs and can promote cellular growth and metabolism, it primarily regulates glucose uptake and metabolism rather than directly mediating growth hormone effects.

D) Growth Hormone (GH): Growth Hormone (GH) stimulates the production of IGFs but is not an IGF itself. IGFs, particularly IGF-I, are responsible for directly promoting cell growth.

- Somatomedins, also known as Insulin-like Growth Factors (IGFs), are polypeptide growth factors primarily produced in the liver, although they are also formed in cartilage and other tissues.
- These factors play a critical role in mediating the effects of growth hormone (GH) on growth, cartilage, and protein metabolism.
- The two most important circulating somatomedins in humans are IGF-I (somatomedin C) and IGF-II.

Features	IGF - I (Somatomedin C)	IGF - II
Primary Function	Skeletal and cartilage growth	Crucial for fetal growth
Regulation by GH	Strongly regulated after birth	Not significantly regulated by GH postnatally
Potency	More potent in growth promotion	Less potent compared to IGF-I
Plasma Concentration	Mirrors GH secretion	Higher concentration than IGF-I
Role in Development	Important for both fetal and postnatal growth	Mainly involved in fetal development

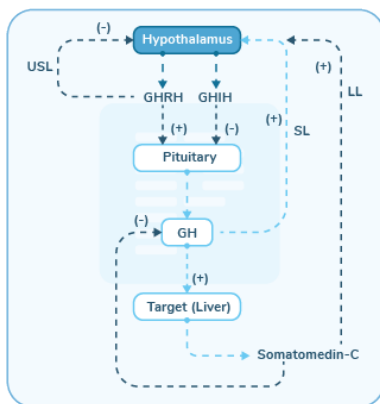
Solution for Question 8:

Reduced IGF-1 receptor sensitivity in the pituitary: If the pituitary gland's response to IGF-1 (also called somatomedins) is diminished, it would fail to inhibit GH secretion effectively, resulting in persistently high GH levels despite high IGF-1.

A) Pituitary adenoma: This could cause elevated GH levels due to tumor overproduction, but the scenario focuses on feedback failure despite high IGF-1.

B) Hypothalamic lesion impairing somatostatin release: This would decrease inhibition of GH secretion, potentially leading to elevated GH, but does not explain the elevated IGF-1 levels.

D) Excessive GHRH secretion from an ectopic source: While this could increase GH levels, it does not directly address the feedback failure involving IGF-1 mentioned in the scenario.



Factors Controlling GH Secretion:

- Direct Feedback from GH to Hypothalamus: GH can inhibit its production by reducing GHRH secretion from the hypothalamus.
- Indirect Feedback via IGF-1: GH stimulates the liver to produce IGF-1, which then inhibits GH secretion by acting directly on the pituitary and indirectly by stimulating somatostatin release from the hypothalamus.
- Negative Feedback through Somatostatin: Somatostatin, a potent inhibitor of GH secretion, is released in response to elevated GH or IGF-1 levels, contributing to the negative feedback loop.

Solution for Question 9:

Prolonged fasting or protein deficiency can lead to elevated GH levels as the body attempts to conserve proteins and mobilize other energy stores, such as fats.

A) Increased somatostatin secretion: This would typically decrease GH levels, not increase them, as somatostatin inhibits GH release.

B) Elevated blood glucose levels: High blood glucose levels are associated with reduced GH secretion, not an increase.

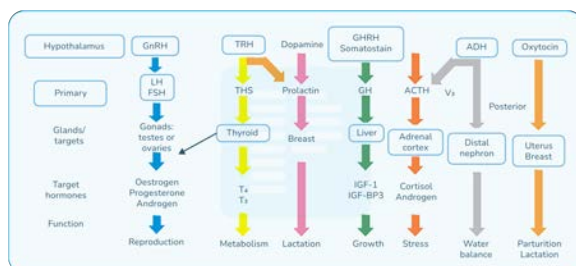
D) Reduced ghrelin levels: Ghrelin stimulates GH release, so reduced ghrelin levels would likely lead to a decrease in GH, not an increase.

Factors controlling GH secretion and their effects

Hypothalamic Control	Growth Hormone-Releasing Hormone (GHRH): Increases GH synthesis and secretion. Somatostatin (GHIH): Decreases GH secretion.
Nutritional Factors	Hypoglycemia: Promotes GH secretion to increase blood glucose levels and mobilize energy. Amino Acids: Stimulate GH release, supporting protein synthesis and growth. Starvation or Protein Deficiency: This leads to elevated GH levels to conserve body proteins and utilize fats as an energy source.
Physiological Factors	Exercise: Enhances GH secretion Sleep: GH surges during deep sleep are vital for growth and tissue repair processes. Stress: Physical and emotional stressors can increase GH secretion
Hormonal Factors	Ghrelin: Increases GH secretion and stimulates appetite. Sex Hormones (Estrogen and Testosterone): Contribute to increased GH secretion during puberty, supporting growth and development. Negative Feedback Control: GH and IGF-1 levels feedback to inhibit further GH release, maintaining homeostasis and preventing excess GH production.

Solution for Question 10:

Thyrotropin-releasing hormone (TRH), in addition to its primary role in stimulating TSH release from the anterior pituitary, can also increase prolactin secretion. This dual effect may contribute to conditions like galactorrhea when prolactin levels are elevated.



- The pituitary gland plays a pivotal role in regulating various endocrine functions, including growth, metabolism, and reproduction.
- The activity of the pituitary gland is primarily governed by the hypothalamus, which releases specific hormones, either stimulating or inhibiting, that reach the pituitary.

- The regulation of pituitary hormone secretion is a finely tuned process, primarily controlled by feedback mechanisms. Negative feedback loops are the most common, where the end products of a hormone's action inhibit further hormone release to maintain homeostasis. An example is cortisol, which inhibits its own secretion through feedback on both the hypothalamus and pituitary. Positive feedback, although less common, occurs in processes like childbirth, where oxytocin release enhances uterine contractions, leading to further oxytocin release.
- Negative feedback loops are the most common, where the end products of a hormone's action inhibit further hormone release to maintain homeostasis. An example is cortisol, which inhibits its own secretion through feedback on both the hypothalamus and pituitary.
- Positive feedback, although less common, occurs in processes like childbirth, where oxytocin release enhances uterine contractions, leading to further oxytocin release.
- Negative feedback loops are the most common, where the end products of a hormone's action inhibit further hormone release to maintain homeostasis. An example is cortisol, which inhibits its own secretion through feedback on both the hypothalamus and pituitary.
- Positive feedback, although less common, occurs in processes like childbirth, where oxytocin release enhances uterine contractions, leading to further oxytocin release.

A) Corticotropin-releasing hormone (CRH): This hormone primarily stimulates the release of adrenocorticotrophic hormone (ACTH) from the anterior pituitary, which in turn influences cortisol production by the adrenal glands. CRH does not directly affect prolactin levels.

B) Dopamine: This neurotransmitter serves as the primary inhibitor of prolactin secretion. Increased dopamine levels typically decrease prolactin release, the opposite of what is described in the scenario.

D) Gonadotropin-releasing hormone (GnRH): This hormone primarily regulates the release of luteinizing hormone (LH) and follicle-stimulating hormone (FSH), which are involved in reproductive function. GnRH does not significantly influence prolactin secretion.

Solution for Question 11:

Somatotropes are the most abundant cells in the anterior pituitary, making up 50% of the total. They secrete growth hormone (GH), crucial for growth and metabolism. Somatotropes are also known as acidophils due to their staining properties with acidic dyes.

Cell Type - Staining	Cell Type	Hormone	Percentage of Total Secretory Cells (%)
Acidophilic	Somatrope	GH	50
Lactotrope	Prolactin	10-30	
Basophilic	Gonadotrope	FSH, LH	20
Corticotrope	ACTH	10	
Thyrotrope	TSH	5	

A) Corticotropes: These cells produce adrenocorticotrophic hormone (ACTH), representing about 10% of the anterior pituitary cell population. They are involved in stimulating the adrenal cortex to secrete cortisol.

C) Lactotropes: Also known as mammotropes, these cells secrete prolactin, which is important for milk production in breastfeeding mothers. They do not constitute the majority of the anterior pituitary cells.

D) Thyrotropes: These cells secrete thyroid-stimulating hormone (TSH), which stimulates the thyroid gland to produce thyroid hormones and are not the predominant cell type in the anterior pituitary.

Solution for Question 12:

Juxtacrine communication is the communication through cell adhesion to growth factors expressed on the cell surface. Cells having receptors for TGF α interact with each other by attaching themselves to the TGF α present on their cell surface.

The chemical signaling of hormones takes place through the following pathways

Type of cell signaling	Endocrine	Paracrine	Autocrine	Juxtacrine

M e c h a n i s m	Hormones are secreted from endocrine glands and travel through the bloodstream to reach distant target tissues.	Hormones diffuse into the extracellular space and signal neighboring tissues.	Hormones bind to receptors on the same cell that secretes them, modifying the function of the parent cell.	Communication occurs through cell adhesion to growth factors expressed on cell surfaces.
Ex a m p l e	Most hormones signal target tissues far from their gland of origin, facilitating intercellular communication across the body.	Somatostatin from D cells of pancreatic islets influences insulin and glucagon secretion from neighboring islet cells.	The platelet-activating factor secreted by platelets activates the same platelets.	Cells with receptors for transforming growth factor α (TGF α) interact by attaching to TGF α present on their cell surfaces

Solution for Question 13:

The adenohypophysis is connected to the hypothalamus through a specialized vascular network called the hypothalamic-hypophyseal portal system, which allows for hormonal regulation of hormone secretion.

In contrast, the neurohypophysis is connected via nerve fibers, resulting in direct neural control of hormone release.

Structure of the Pituitary Gland

The pituitary gland has two distinct parts: the neurohypophysis (posterior pituitary) and the adenohypophysis (anterior pituitary).

Feature	Adenohypophysis (Anterior Pituitary)	Neurohypophysis (Posterior Pituitary)
Embryological Origin	Epithelial tissue (Rathke's pouch). (Option A)	Neural tissue

Cellular Composition	Composed of Epithelioid cells Contains five types of hormone-secreting cells: (Option C&D) Somatotropes (secrete growth hormone) Corticotropes (secrete adrenocorticotrophic hormone) Thyrotropes (secrete thyroid-stimulating hormone) Gonadotropes (secrete follicle-stimulating hormone and luteinizing hormone) Lactotropes (secrete prolactin)	Made up of pituicytes (glial-like cells). Contains nerve fibers from the supraoptic and paraventricular nuclei of the hypothalamus. Nerve fibers form bulbous structures near capillaries for hormone secretion.
Hormones Produced	GH, ACTH, TSH, FSH, LH, Prolactin	ADH, Oxytocin (Option D) Produced in the hypothalamus, transported via neurophysins, stored in secretory granules, and released into capillaries upon stimulation.
Connection to Hypothalamus	Hypothalamic-hypophyseal portal system	Nerve fibres
Mechanism of Hormone Secretion	Hormonal regulation via the hypothalamus hypophyseal portal system	Direct neural control via nerve fibers

Solution for Question 14:

Type 1 receptors are bound to heat shock proteins (HSPs) in their inactive state, while Type 2 receptors can bind DNA even without a ligand, often repressing transcription until activated by a ligand.

Steroid or intracellular hormone receptors are classified into two main types: Type 1 and Type 2. These receptors mediate the effects of steroid hormones and other lipophilic molecules.

Aspect	Type 1 Receptors	Type 2 Receptors
Location:	Cytoplasm (Option A)	Nucleus (Option A)
Examples:	Glucocorticoid, mineralocorticoid, androgen, and progesterone receptors	Thyroid hormone and retinoic acid receptors

Activation Mechanism:	Inactive State: Bound to chaperone proteins- heat shock proteins (HSPs) like HSP90 Hormone Binding: Causes dissociation from HSPs Dimerization: Forms homodimers (Option C) Nuclear Translocation: Homodimers move to the nucleus and bind to hormone response elements (HREs) on DNA Transcription: Initiates activation of target genes	DNA Binding: Can bind DNA even without a ligand, often repressing transcription. The hormone receptor is attached to a molecule RXR which forms a heterodimer. (Option C) Ligand Binding: Enhances or represses transcription depending on coactivators or corepressors Transcriptional Regulation: Modulates gene expression via interaction with HREs
-----------------------	--	--

(Option D) Both Type 1 and Type 2 receptors do not have intrinsic enzymatic activity. Instead, they function as transcription factors that regulate gene expression in response to hormonal signaling.

Solution for Question 15:

RTKs are activated when a hormone binds to the extracellular part of the receptor, inducing dimerization. This leads to the activation of the cytoplasmic kinase domains through trans-phosphorylation, initiating downstream signaling.

A) RTKs are activated by hormone binding to the extracellular domain, not by intracellular ligands.

C) RTKs do not form ion channels. This mechanism is characteristic of other types of receptors, such as ionotropic receptors.

D) RTKs do not directly activate G proteins. This mechanism is associated with G protein-coupled receptors (GPCRs), not RTKs.

Receptor tyrosine kinases (RTKs)

- Enzyme-linked receptors located on cell membranes.
- Pass through the membrane once; hormone-binding site is extracellular, catalytic site is intracellular.

Mechanism of Action:

- Hormone Binding: Hormone binds to the extracellular part of the RTK.

- **Enzyme Activation:** Binding activates or inactivates an intracellular enzyme. Some RTKs have intrinsic enzymatic activity, while others depend on associated enzymes.

Example: Leptin Receptor

- **Family:** Cytokine receptors.
- **Hormone:** Leptin (regulates appetite and energy balance).
- **Associated Enzyme:** JAK2 (Janus kinase family).

Signaling Process:

- **Dimerization:** Leptin binding causes receptor dimerization.
- **Phosphorylation & Conformation Change:** Leptin binding alters receptor conformation, phosphorylating the intracellular part of the JAK receptor.

Intracellular Signaling:

- Activated JAK2 phosphorylates tyrosine residues within the receptor-JAK2 complex, activating STAT proteins, which lead to target gene activation and protein synthesis.
- JAK2 also activates additional pathways, including MAPK and PI3K.
- **Rapid:** Due to activation of the above-said intracellular enzymes.
- **Slower:** Requires new protein synthesis.
- Fibroblast Growth Factor
- Growth Hormone
- Hepatocyte Growth Factor
- Insulin
- Insulin-like Growth Factor-1
- Leptin
- Prolactin
- Vascular Endothelial Growth Factor

Solution for Question 16:

Ligand binding to the extracellular domain causes a conformational change that activates the cyclase domains, leading to cGMP synthesis.

B) Phosphorylation within the kinase homology domain (KHD) is important but does not directly activate cGMP synthesis; ligand-induced conformational change is the primary mechanism.

C) The intracellular domains of pGCs do not bind to DNA or initiate gene transcription; they are involved in cGMP synthesis.

D) While some pGCs are regulated by intracellular proteins, the primary activation mechanism for most pGCs involves ligand binding to the extracellular domain.

Guanylyl Cyclase Receptor:

- Transmembrane guanylyl cyclases (pGCs) are homodimers (~1,050 amino acids each).
- Composed of a short transmembrane domain (21-25 hydrophobic amino acids) separating extracellular and intracellular regions.
- Extracellular domain acts as a receptor for specific peptide hormones.

Domains:

- Extracellular Domain: Binds specific ligands.
- Juxtamembranous Protein Kinase-Homology Domain (KHD): ~250 residues, lacks kinase activity.
- Cyclase Domain: Catalytic region for cGMP formation at the COOH-terminal end.

Activation Mechanism:

- Ligand Binding: Specific ligands (e.g., ANP) bind, causing a conformational change.
- Conformational Change: Rotation within the dimer activates cGMP synthesis by approximating cyclase domains.
- Phosphorylation: Activation is modulated by phosphorylation in the KHD; ligand binding may expose phosphorylated sites to phosphatases for dephosphorylation and inactivation.
- Desensitization: Prolonged ligand exposure can lead to dephosphorylation of the KHD, reducing pGC activity.

Intracellular Signalling:

- cGMP Production: Activated cyclase domain produces cGMP from GTP, serving as a second messenger.
- cGMP Targets: cGMP-Dependent Protein Kinases (cGKs): Phosphorylate proteins to elicit cellular responses. Phosphodiesterases (PDEs): Regulate cGMP levels by hydrolyzing it.
- cGMP-Gated Channels: Involved in phototransduction and olfactory signaling
- cGMP-Dependent Protein Kinases (cGKs): Phosphorylate proteins to elicit cellular responses.
- Phosphodiesterases (PDEs): Regulate cGMP levels by hydrolyzing it.
- cGMP-Gated Channels: Involved in phototransduction and olfactory signaling

- cGMP-Dependent Protein Kinases (cGKs): Phosphorylate proteins to elicit cellular responses.
 - Phosphodiesterases (PDEs): Regulate cGMP levels by hydrolyzing it.
 - cGMP-Gated Channels: Involved in phototransduction and olfactory signaling
-

Solution for Question 17:

Adenylyl cyclase activation in the adenylyl cyclase-cAMP pathway leads to the activation of cAMP-dependent protein kinase.

Adenylyl Cyclase-cAMP Pathway

- Activation: (Option A) Hormone Binding: Hormone binds to a GPCR coupled to a Gs protein on the cell membrane. G Protein Activation: Gs protein stimulates adenylyl cyclase. cAMP Formation: Adenylyl cyclase converts ATP to cyclic AMP (cAMP).
- Hormone Binding: Hormone binds to a GPCR coupled to a Gs protein on the cell membrane.
- G Protein Activation: Gs protein stimulates adenylyl cyclase.
- cAMP Formation: Adenylyl cyclase converts ATP to cyclic AMP (cAMP).
- The action of cAMP: Activation: cAMP activates cAMP-dependent protein kinase. Phosphorylation: This kinase phosphorylates specific proteins in the cell. Effects: Varies by cell type: Thyroid Cells: Stimulates production of thyroxine and triiodothyronine. Adrenocortical Cells: Stimulates secretion of adrenocortical steroid hormones.
- Activation: cAMP activates cAMP-dependent protein kinase.
- Phosphorylation: This kinase phosphorylates specific proteins in the cell.
- Effects: Varies by cell type: Thyroid Cells: Stimulates production of thyroxine and triiodothyronine. Adrenocortical Cells: Stimulates secretion of adrenocortical steroid hormones.
- Thyroid Cells: Stimulates production of thyroxine and triiodothyronine.
- Adrenocortical Cells: Stimulates secretion of adrenocortical steroid hormones.
- Hormone Binding: Hormone binds to a GPCR coupled to a Gs protein on the cell membrane.
- G Protein Activation: Gs protein stimulates adenylyl cyclase.
- cAMP Formation: Adenylyl cyclase converts ATP to cyclic AMP (cAMP).

- Activation: cAMP activates cAMP-dependent protein kinase.
- Phosphorylation: This kinase phosphorylates specific proteins in the cell.
- Effects: Varies by cell type: Thyroid Cells: Stimulates production of thyroxine and triiodothyronine. Adrenocortical Cells: Stimulates secretion of adrenocortical steroid hormones.
- Thyroid Cells: Stimulates production of thyroxine and triiodothyronine.
- Adrenocortical Cells: Stimulates secretion of adrenocortical steroid hormones.
- Thyroid Cells: Stimulates production of thyroxine and triiodothyronine.
- Adrenocortical Cells: Stimulates secretion of adrenocortical steroid hormones.

Phospholipase C Pathway

- Activation: Hormone Binding: Hormone binds to a GPCR. Enzyme Activation: Activates phospholipase C attached to the receptor. Breakdown of PIP2: Phospholipase C converts phosphatidylinositol biphosphate (PIP2) into inositol triphosphate (IP3) and diacylglycerol (DAG). (Option B)
- Hormone Binding: Hormone binds to a GPCR.
- Enzyme Activation: Activates phospholipase C attached to the receptor.
- Breakdown of PIP2: Phospholipase C converts phosphatidylinositol biphosphate (PIP2) into inositol triphosphate (IP3) and diacylglycerol (DAG). (Option B)
- The action of IP3 and DAG: IP3: Mobilises calcium ions from mitochondria and the endoplasmic reticulum. (Option D) Effects: Influences smooth muscle contraction and cell secretion. DAG: Activates protein kinase C. Phosphorylation: Protein kinase C phosphorylates various proteins, affecting cellular responses.
- IP3: Mobilises calcium ions from mitochondria and the endoplasmic reticulum. (Option D) Effects: Influences smooth muscle contraction and cell secretion.
- Effects: Influences smooth muscle contraction and cell secretion.
- DAG: Activates protein kinase C. Phosphorylation: Protein kinase C phosphorylates various proteins, affecting cellular responses.
- Phosphorylation: Protein kinase C phosphorylates various proteins, affecting cellular responses.
- Hormone Binding: Hormone binds to a GPCR.
- Enzyme Activation: Activates phospholipase C attached to the receptor.
- Breakdown of PIP2: Phospholipase C converts phosphatidylinositol biphosphate (PIP2) into inositol triphosphate (IP3) and diacylglycerol (DAG). (Option B)
- IP3: Mobilises calcium ions from mitochondria and the endoplasmic reticulum. (Option D) Effects: Influences smooth muscle contraction and cell secretion.

- Effects: Influences smooth muscle contraction and cell secretion.
 - DAG: Activates protein kinase C. Phosphorylation: Protein kinase C phosphorylates various proteins, affecting cellular responses.
 - Phosphorylation: Protein kinase C phosphorylates various proteins, affecting cellular responses.
 - Effects: Influences smooth muscle contraction and cell secretion.
 - Phosphorylation: Protein kinase C phosphorylates various proteins, affecting cellular responses.
-

Solution for Question 18:

Prolonged hormone exposure can lead to down-regulation of GPCRs.

- Membrane-Bound Receptors: GPCRs are integral membrane proteins vital for signal transduction, especially for protein, peptide, and catecholamine hormones.
- Structure: Characterised by seven transmembrane segments that span the cell membrane. (Option A)
- Binding Sites: Extracellular: Hormone-binding site. Intracellular: Interacts with G proteins.
- Extracellular: Hormone-binding site.
- Intracellular: Interacts with G proteins.
- Extracellular: Hormone-binding site.
- Intracellular: Interacts with G proteins.
- Heterotrimeric: Consist of α , β , and γ subunits.
- Inactive State: G protein complex binds GDP on the α subunit.
- Interaction with Effectors: α subunit interacts with enzymes or ion channels.
- Example: Activation of adenylyl cyclase, converting ATP to cAMP.
- Intrinsic GTPase Activity: α subunit hydrolyzes GTP to GDP, inactivating the α subunit.
- Dynamic Receptor Numbers: Receptor numbers on target cells can change.
- Down-Regulation: Prolonged hormone exposure reduces receptor numbers, decreasing sensitivity. Mechanisms include: Receptor inactivation. Receptor internalization and sequestration. Receptor degradation by lysosomes. Reduced receptor production.
- Receptor inactivation.

- Receptor internalization and sequestration.
- Receptor degradation by lysosomes.
- Reduced receptor production.
- Up-Regulation: Increases receptor numbers or sensitivity, enhancing responsiveness.
- Receptor inactivation.
- Receptor internalization and sequestration.
- Receptor degradation by lysosomes.
- Reduced receptor production.

(Option D) The α subunit does not permanently inactivate the GPCR. Instead, it initiates downstream signaling cascades when activated by exchanging GDP for GTP. After activation, the GTP-bound α subunit dissociates from the $\beta\gamma$ subunits and interacts with effector proteins to transmit signals within the cell. The GPCR itself can undergo desensitization and internalization after activation, but this is not caused by the α subunit permanently inactivating it.

Solution for Question 19:

The estrogen receptor is a nuclear receptor that binds to hydrophobic hormones, translocates to the nucleus, and acts as a transcription factor to regulate gene expression.

B) The insulin receptor is a tyrosine kinase receptor located on the cell surface.

D) Ion channel receptors are cell surface receptors that allow ions to pass through the membrane in response to ligand binding, not involved in transcriptional regulation.

Hormone receptors are categorized based on their cellular localization and signaling mechanisms, influenced by the chemical properties of the hormones they bind.

Membrane-bound Receptors:

- Associated Hormones: Hydrophilic hormones (peptide hormones, catecholamines)
- Properties: Located on the cell surface Bind extracellular, water-soluble hormones Cannot pass through the lipid bilayer
- Located on the cell surface
- Bind extracellular, water-soluble hormones
- Cannot pass through the lipid bilayer

- Signaling Mechanism: Initiate intracellular signaling cascades Common example: G protein-coupled receptors (GPCRs) GPCRs interact with intracellular heterotrimeric G proteins Trigger downstream events altering cellular functions, such as enzyme activity or ion channel gating
- Initiate intracellular signaling cascades
- Common example: G protein-coupled receptors (GPCRs)
- GPCRs interact with intracellular heterotrimeric G proteins
- Trigger downstream events altering cellular functions, such as enzyme activity or ion channel gating
- Types: G-protein coupled receptors Guanylyl cyclase receptors Tyrosine kinase receptors Cytokine receptors Serine kinase receptors
- G-protein coupled receptors
- Guanylyl cyclase receptors
- Tyrosine kinase receptors
- Cytokine receptors
- Serine kinase receptors
- Located on the cell surface
- Bind extracellular, water-soluble hormones
- Cannot pass through the lipid bilayer
- Initiate intracellular signaling cascades
- Common example: G protein-coupled receptors (GPCRs)
- GPCRs interact with intracellular heterotrimeric G proteins
- Trigger downstream events altering cellular functions, such as enzyme activity or ion channel gating
- G-protein coupled receptors
- Guanylyl cyclase receptors
- Tyrosine kinase receptors
- Cytokine receptors
- Serine kinase receptors

Nuclear Receptors

- Associated Hormones: Hydrophobic hormones (steroid hormones, thyroid hormones)
- Properties: Located intracellularly, typically in the cytoplasm Can passively diffuse across the cell membrane
- Located intracellularly, typically in the cytoplasm

- Can passively diffuse across the cell membrane
- Signaling Mechanism: Translocate to the nucleus upon ligand binding Function as transcription factors Form homodimers or heterodimers Bind to hormone response elements on DNA Modulate gene expression by influencing transcriptional coactivators or corepressors
- Translocate to the nucleus upon ligand binding
- Function as transcription factors
- Form homodimers or heterodimers
- Bind to hormone response elements on DNA
- Modulate gene expression by influencing transcriptional coactivators or corepressors
- Located intracellularly, typically in the cytoplasm
- Can passively diffuse across the cell membrane
- Translocate to the nucleus upon ligand binding
- Function as transcription factors
- Form homodimers or heterodimers
- Bind to hormone response elements on DNA
- Modulate gene expression by influencing transcriptional coactivators or corepressors

Solution for Question 20:

Protein hormones are hydrophilic and bind to receptors located on the cell surface, initiating intracellular signaling cascades involving second messengers like cAMP and calcium ions.

Aspect	Protein Hormones	Steroid Hormones
Nature	Hydrophilic	Hydrophobic (Option D)
Cell Entry	Cannot penetrate the cell membrane (Option A)	Freely diffuse through the cell membrane (Option D)
Receptor Location	Cell surface receptors (Option A)	Nuclear receptors (primarily in the cytoplasm)
Receptor Family	GPCRs	Nuclear receptor family
Signaling Mechanism	Initiate intracellular signaling cascades	Receptor-hormone complex moves to the nucleus
Intracellular Activity	Second messengers like cAMP and calcium ions	Form dimers, bind to DNA, regulate gene transcription (Option B)

--	--	--

Solution for Question 21:

Cortisol is a steroid hormone, which is hydrophobic and can diffuse through the cell membrane. Once inside the cell, it binds to nuclear receptors to regulate gene transcription.

Category	Type	Action Mechanism	Examples
Peptide/Protein Hormones	Water-Soluble	Bind to cell surface receptors	Insulin, Vasopressin (Options A&D)
Amine Hormones	Water-Soluble	Bind to cell surface receptors	Catecholamines (Option B)
Steroid Hormones	Hydrophobic	Diffuse into cells, bind nuclear receptors	Cortisol, Mineralocorticoids
Thyroid Hormones	Hydrophobic	Diffuse into cells, bind nuclear receptors	Thyroid Hormone

Detailed Mechanism of Hydrophobic Hormones:

- They bind to a family of large cytoplasmic proteins known as nuclear receptors.
- Nuclear receptors: Found in the cytoplasm. Bind to hormones, forming receptor-ligand complexes. Complex moves to the nucleus. Forms homodimers or heterodimers. Binds to DNA to regulate gene transcription.
- Found in the cytoplasm.
- Bind to hormones, forming receptor-ligand complexes.
- Complex moves to the nucleus.
- Forms homodimers or heterodimers.
- Binds to DNA to regulate gene transcription.
- Found in the cytoplasm.
- Bind to hormones, forming receptor-ligand complexes.
- Complex moves to the nucleus.
- Forms homodimers or heterodimers.
- Binds to DNA to regulate gene transcription.

Solution for Question 22:

The menstrual cycle is an example of an infradian rhythm, which lasts longer than 24 hours, typically around 28 days

Types of Biological Rhythm

Bio logi cal Rh yth m	Ultradian	Circadian	Infradian
Me ch ani sm	All types of 'short-term rhythms' with a frequency of fewer than 24h but commonly with periods in the range of 20 min to 6h	Circadian rhythm describes the 24-hour oscillations of biological processes associated with the dark/light cycle and the earth's daily rotation	The period of infradian rhythms is longer than the circadian rhythms with ranges from days to years
Ex am ple	The episodic secretion of insulin from the beta-cell of the pancreas regulates blood glucose levels. GnRH from the hypothalamus regulates reproductive function.	Melatonin is secreted by the pineal gland and the liver and it regulates the circadian rhythms of various physiological activities. Its biosynthesis increases at night and is inhibited during daytime by the light detected by the retina. Serum concentrations of cortisol show marked variation within the day with the highest level detected in the early morning (Option D) Sleep-wake cycle (Option A) Body temperature fluctuations over 24 hrs (Option B)	Menstrual cycle- Pituitary gonadotropins (follicle-stimulating and luteinizing hormones) and Ovarian hormones (estrogen and progesterone) show infradian rhythms.

Adrenal Physiology and Calcium Regulation

1. A 35-year-old woman presents with hypertension and hypokalemia. Genetic testing reveals a mutation in the genes coding for epithelial sodium channels (ENaCs). Which of the following conditions is most likely associated with her symptoms?

- A. Conn syndrome
 - B. Liddle syndrome
 - C. Glucocorticoid-remediable aldosteronism (GRA)
 - D. Secondary hyperaldosteronism
-

2. Which layer of the adrenal cortex is correctly paired with its primary hormone?

- A. Zona fasciculata - Androgens
 - B. Zona glomerulosa - Aldosterone
 - C. Zona reticularis - Cortisol
 - D. None of the above
-

3. A 50-year-old patient has been on long-term corticosteroid therapy for rheumatoid arthritis. The physician advises tapering the dose gradually rather than stopping it abruptly. What is the reason for this advice?

- A. To avoid the risk of adrenal insufficiency
- B. To avoid the risk of Cushing's syndrome
- C. To prevent a sudden increase in blood glucose levels
- D. To prevent sudden increase in blood pressure

4. Which among the following statements are true regarding the action of aldosterone to enhance sodium reabsorption at the molecular level in renal cells?
(a) Inhibits the excretion of K^+ to preserve Na^+ (b) Activation of sgk gene (serum and glucocorticoid- regulated kinase) (c) Increases ENaC (Epithelial Na^+ Channels) activity (d) Increases activity of Na^+-K^+ exchanger

- A. a, b, and c
 - B. a and c
 - C. b, c and d
 - D. c and d
-

5. Which of the following is the primary physiological trigger for the suppression of aldosterone secretion?

- A. Increased serum potassium levels
 - B. Expansion of the extracellular fluid (ECF) volume
 - C. Increased renin release
 - D. Decreased blood pressure
-

6. Which of the following describes the mechanism by which glucocorticoids regulate metabolic reactions and ACTH secretion?

- A. Directly catalyzing metabolic reactions and stimulating ACTH secretion
- B. Binding to glucocorticoid receptors and acting as transcription factors
- C. Inhibiting catecholamine action and decreasing mRNA synthesis
- D. Influence cellular processes by directly altering gene expression.

7. A 45-year-old male with a history of type 2 diabetes presents with difficulty managing his blood glucose levels. Which of the following effects of glucocorticoids best explains the patient's difficulty in managing his diabetes?

- A. Glucocorticoids enhance the action of insulin in peripheral tissues
 - B. Glucocorticoids decrease gluconeogenesis in the liver
 - C. Glucocorticoids promote protein anabolism
 - D. Glucocorticoids counteract the effects of insulin in peripheral tissues
-

8. A 45-year-old patient presents with symptoms of hypertension and hypokalemia. Laboratory tests reveal elevated levels of aldosterone. Which of the following statements is true regarding the enzyme and the zone responsible for producing aldosterone?

- A. Aldosterone is produced in the zona fasciculata, and the key enzyme involved is 17 α -Hydroxylase
 - B. Aldosterone is synthesized from pregnenolone via progesterone in the zona glomerulosa
 - C. In the zona reticularis, pregnenolone is converted to aldosterone by Aldosterone Synthase
 - D. The zona glomerulosa produces aldosterone directly from cholesterol, bypassing pregnenolone and progesterone
-

9. A 3-year-old child presents with signs of early virilization and ambiguous genitalia. Laboratory tests show elevated levels of ACTH and 17-hydroxyprogesterone. Which enzyme deficiency is most likely responsible for this condition?

- A. 11 β -hydroxylase deficiency
 - B. 21 β -hydroxylase deficiency
 - C. 3 β -hydroxysteroid dehydrogenase deficiency
 - D. 17 α -hydroxylase deficiency
-

10. A 45-year-old woman she exhibits positive Chvostek's and Trousseau's signs. Which of the following mechanisms best explains the patient's symptoms?

- A. Increased ionized calcium levels due to hyperventilation
 - B. Decreased ionized calcium levels due to albumin binding in alkalosis
 - C. Increased total calcium levels due to chronic kidney diseases
 - D. Decreased total calcium levels due to vitamin D overdose
-

11. A 25-year-old woman is admitted to the emergency department with symptoms of muscle spasms and difficulty breathing. She has recently been hyperventilating due to anxiety and reports a tingling sensation in her fingers. On examination, she exhibits a positive Chvostek sign and Trousseau sign. Which of the following best explains the underlying mechanism of tetany in this patient?

- A. Increased calcium absorption due to high levels of parathyroid hormone
 - B. Decreased ionized calcium leading to increased excitability of nerve and muscle cells
 - C. Increased protein ionization due to low pH causing more free Ca²⁺ availability
 - D. Enhanced bone calcium mobilization from decreased parathyroid hormone levels
-

12. Which of the following statements regarding parathyroid hormone (PTH) and calcium regulation is TRUE?

- A. PTH promotes calcium reabsorption and phosphate absorption in the renal tubules

- B. PTH directly enhances calcium absorption in the intestines by increasing the expression of calcium-binding proteins
 - C. PTH directly activates osteoclasts, which break down bone tissue and release calcium into the extracellular fluid
 - D. PTH is essential for the conversion of inactive vitamin D to its active form
-

13. Which of the following best describes the primary role of vitamin D in calcium regulation?

- A. Vitamin D directly stimulates osteoclast activity to release calcium from bones
 - B. Vitamin D enhances calcium reabsorption in the kidneys, reducing its excretion
 - C. Vitamin D increases the production of parathyroid hormone (PTH), which mobilizes calcium
 - D. Vitamin D facilitates the intestinal absorption of calcium, increasing its uptake from food
-

14. Which of the following statements best describes the primary mechanism by which calcitonin lowers blood calcium levels?

- A. Stimulates osteoblast activity to build new bone tissue
 - B. Increases the absorption of calcium in the intestines
 - C. Inhibits osteoclast activity to reduce bone resorption
 - D. Enhances the reabsorption of calcium in the kidneys
-

15. Which of the following markers is primarily used to assess bone resorption?

- A. Osteocalcin
- B. Bone-Specific Alkaline Phosphatase (BSAP)

C. C-terminal propeptide of type I procollagen

D. Cathepsins K

Correct Answers

Question	Correct Answer
Question 1	2
Question 2	2
Question 3	1
Question 4	3
Question 5	2
Question 6	2
Question 7	4
Question 8	2
Question 9	2
Question 10	2
Question 11	2
Question 12	4
Question 13	4
Question 14	3
Question 15	4

Solution for Question 1:

Liddle syndrome is characterized by hypertension and hypokalemia due to a mutation in the genes coding for epithelial sodium channels (ENaCs), leading to increased sodium reabsorption and potassium excretion.

- Rare genetic disorder causing high blood pressure and low potassium levels.

- Caused by a mutation in genes coding for epithelial sodium channels (ENaCs) in kidneys.
- Leads to increased sodium reabsorption and potassium excretion.
- Characterized by excessive aldosterone secretion.
- Also a cause of high blood pressure and low potassium levels.
- Classified into primary and secondary types.

Feature	Primary Hyperaldosteronism (Option A)	Secondary Hyperaldosteronism (Option D)
Cause	Adrenal gland disorders	High plasma renin activity conditions
Examples	Adenomas, adrenal hyperplasia, carcinoma, Glucocorticoid-Remediable Aldosteronism (Option C)	Cirrhosis, heart failure, nephrosis, salt-losing adrenogenital syndrome
Renin Activity	Suppressed	Elevated
GRA Mechanism	Hybrid gene sensitive to ACTH	Not applicable
ECF Volume	Normal or increased	Decreased

Solution for Question 2:

Zona Glomerulosa produces mineralocorticoids mainly aldosterone and small amounts of deoxycorticosterone.

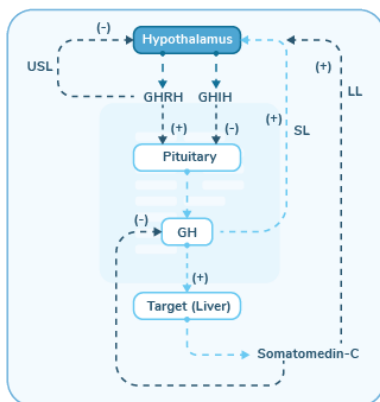
Layers of Adrenal gland			
Adrenal Cortex:			
Zones of the cortex	Hormones produced	Main function	Morphology of Zone
Zona Glomerulosa (Option B)	Mineralocorticoid Aldosterone(main) Deoxycorticosterone	Maintenance of sodium balance	Small clumps of cells
Zona fasciculata (Option A)	Glucocorticoids Cortisol Corticosterone	Maintenance of sugars	Columnar cells that form long cords

Zona reticularis (Option C)	Sex steroids Dehydroepiandrosterone Androstenedione	Maintenance of sexual functioning	Network of intercalated cells
Adrenal medulla:			
Medulla	Catecholamines: Norepinephrine Epinephrine Small amounts of dopamine	Carbohydrate and fat metabolisms Fight or flight responses	Specialized sympathetic ganglion Clumps and strands of chromaffin cells

Solution for Question 3:

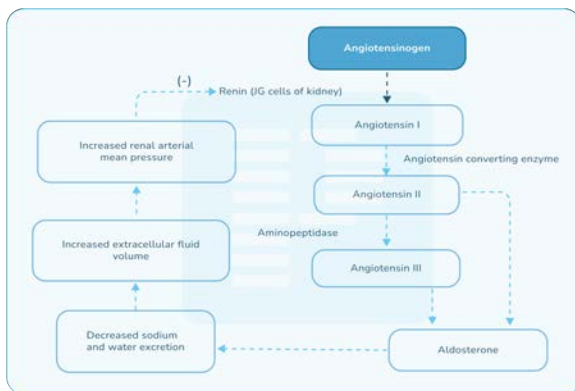
Pituitary-adrenocortical axis takes about 10 months to recover from the suppressive effect of prolonged corticosteroid therapy, which can lead to adrenal insufficiency if steroids are stopped abruptly.

Corticosteroid feedback mechanism:



- Circadian rhythm: Diurnal pattern with peak in morning and low in evening, independent of external factors
- The adrenal cortex atrophies and becomes unresponsive to the ACTH for about six months after a prolonged treatment with steroids.
- Adrenal insufficiency is precipitated as a result. (Option A).

Regulation of aldosterone:



(Option B) Cushing's syndrome, (Option C) high blood sugar and (Option D) high blood pressure occur in higher levels of steroid levels in plasma and not in insufficiency.

Solution for Question 4:

The gene activated by aldosterone is the sgk gene (serum and glucocorticoid- regulated kinase), which increases ENaC (Epithelial Na⁺ Channels) activity. Aldosterone also stimulates the activity of Na⁺-K⁺ exchanger, which increases intracellular accumulation of Na⁺.

Aldosterone Mechanism of action:

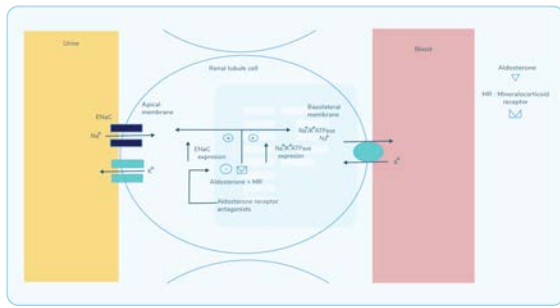
- It has both genomic and nongenomic actions.
- Aldosterone binds with the cytoplasmic receptor, and the complex moves to the nucleus where it induces transcription of mRNA leading to new protein synthesis that alters cell function.

Genomic Action:

- The gene activated by aldosterone is the sgk gene (serum and glucocorticoid- regulated kinase). (Statement b)
- Increases ENaC (Epithelial Na⁺ Channels) activity (Statement c)
- Increases insertion of these channels on cell membrane and also increases their activity

Nongenomic Action:

- It is due to stimulation of the activity of Na⁺-K⁺ exchanger (Statement c), which increases intracellular accumulation of Na⁺.



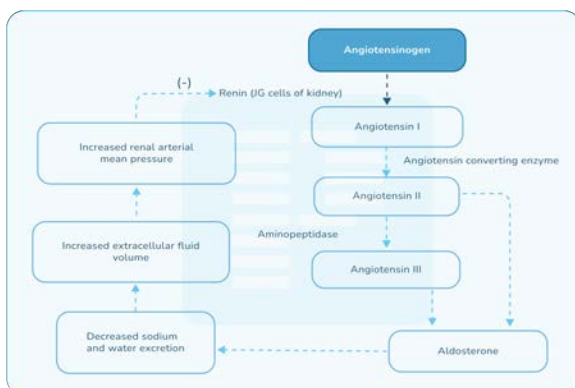
(Statement a) aldosterone increases amounts of Na^+ concentration in exchange for K^+ and H^+ in the renal tubules, producing a K^+ diuresis.

Solution for Question 5:

Expansion of the extracellular fluid (ECF) volume increases the mean arterial pressure in renal arteries, which inhibits renin thus decreasing the aldosterone secretion.

Regulation of aldosterone:

- Renin-angiotensin-aldosterone system (RAS) plays a crucial role in the regulation of aldosterone levels in the plasma.
- ACTH stimulates aldosterone synthesis and secretion, but the quantity of ACTH required to stimulate aldosterone secretion is more than the concentration required to stimulate cortisol secretion.



Physiological Actions of Aldosterone:

- The main function of aldosterone is to increase Na⁺ and water reabsorption and to promote K⁺ and H⁺ excretion from the kidney (Option A).
 - It acts primarily on P cells (principal cells) of collecting the duct and distal tubule of the kidney to increase the sodium reabsorption.
 - Chlorine is transported in the same direction to maintain electroneutrality due to Na⁺ reabsorption, causing osmotic reabsorption of water which leads to ECF expansion (Option B) and increases blood pressure (Option D).
 - (Option D) Increased renin release stimulates aldosterone secretion.
-

Solution for Question 6:

Glucocorticoids bind to their receptors, forming complexes that act as transcription factors, promoting the transcription of specific DNA segments, and leading to the synthesis of enzymes that alter cell function.

- Production: Steroid hormones produced by the adrenal cortex.
- Mechanism of Action: Bind to glucocorticoid receptors. Form steroid-receptor complexes acting as transcription factors. Promote transcription of specific DNA segments. Lead to synthesis of enzymes altering cell function via mRNA generation. (Option C)
- Bind to glucocorticoid receptors.
- Form steroid-receptor complexes acting as transcription factors.
- Promote transcription of specific DNA segments.
- Lead to synthesis of enzymes altering cell function via mRNA generation. (Option C)
- Nongenomic Actions: Influence cellular processes without directly affecting gene expression. (Option D)
- Bind to glucocorticoid receptors.
- Form steroid-receptor complexes acting as transcription factors.
- Promote transcription of specific DNA segments.
- Lead to synthesis of enzymes altering cell function via mRNA generation. (Option C)
- Role in Metabolism: Presence necessary for certain metabolic reactions, though not directly causing them. Calorigenic Effects: Enable glucagon and catecholamines. Lipolytic Effects: Allow catecholamines. Pressor Responses & Bronchodilation: Facilitate actions of catecholamines.

- Calorigenic Effects: Enable glucagon and catecholamines.
- Lipolytic Effects: Allow catecholamines.
- Pressor Responses & Bronchodilation: Facilitate actions of catecholamines.
- Calorigenic Effects: Enable glucagon and catecholamines.
- Lipolytic Effects: Allow catecholamines.
- Pressor Responses & Bronchodilation: Facilitate actions of catecholamines.

Solution for Question 7:

Glucocorticoids elevate plasma glucose levels by promoting gluconeogenesis and glycogenesis in the liver and counteracting insulin effects in peripheral tissues, making diabetes management more difficult.

Effects of Glucocorticoids:

Intermediary Metabolism:
 Promote protein catabolism. (Option C)
 Stimulate liver glycogenesis and gluconeogenesis. (Option B)
 Enhance glucose-6-phosphatase activity.
 Elevates plasma glucose level.
 Counteract insulin effects in peripheral tissues, worsening diabetes. (Option A) (Option D)
 In diabetes, elevate plasma lipid levels and promote ketone body formation. In healthy individuals, insulin secretion counters this effect.

Permissive Action:
 Essential for actions of other hormones (e.g., glucagon and catecholamines).
 Necessary for catecholamine-induced lipolysis, pressor responses, and bronchodilation.

Effects on ACTH Secretion:
 Regulate adrenocorticotrophic hormone (ACTH) secretion via negative feedback.

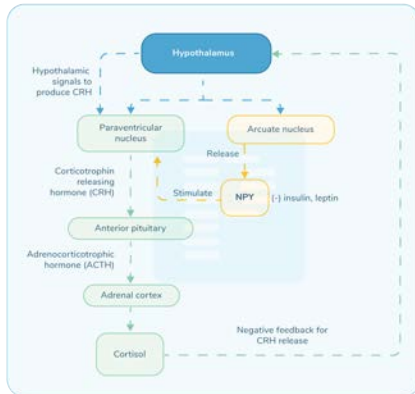
Vascular Reactivity:
 Maintain blood vessel responsiveness to norepinephrine and epinephrine.
 Deficiency leads to decreased vascular responsiveness, capillary dilation, increased permeability, and potential vascular collapse.

Water Metabolism:
 Regulate water balance.
 Adrenal insufficiency impairs water excretion, increasing water intoxication risk.

Blood Cells and Lymphatic Organs:
 Decrease eosinophils and basophils.
 Increase neutrophils, platelets, and red blood cells.
 Decrease lymphocytes and reduce lymph node and thymus size.

Stress Resistance:
Crucial for stress response.
Stress induces ACTH secretion, increasing glucocorticoid levels essential for survival.

Regulation of Glucocorticoid Secretion:



Solution for Question 8:

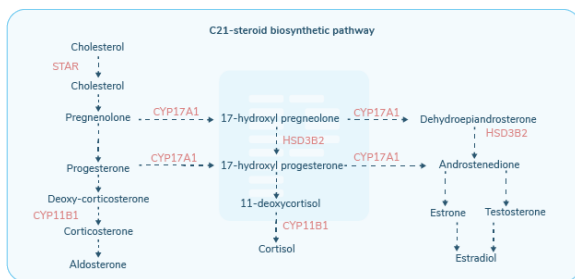
In the zona glomerulosa, pregnenolone is first transformed into progesterone, which is then further converted into aldosterone through several enzymatic steps. The final conversion to aldosterone is catalyzed by aldosterone synthase (P450c11AS or CYP11B2), an enzyme specific to this zona.

Steroid Biosynthesis in the Adrenal Cortex:

Overview:

- Location: Adrenal Cortex
- Starting Material: Cholesterol

End Products: Mineralocorticoids, Glucocorticoids, Androgens



Zona	Conversion Pathway	Key Enzymes	Steroid Hormones Produced
Zona Glomerulosa	Pregnenolone → Progesterone → Aldosterone	Aldosterone Synthase (P450c11AS)	Aldosterone
Zona Fasciculata	Pregnenolone → Progesterone or 17α-Hydroxypregnenolone	17α-Hydroxylase	Cortisol, Corticosterone
Zona Reticularis	Pregnenolone → 17α-Hydroxypregnenolone	17α-Hydroxylase	DHEA, Androstenedione

Enzyme Specificity:

Adrenal Cortex Zone	Enzymes Present	Steroid Hormones Produced
Zona Glomerulosa	Aldosterone Synthase (P450c11AS)	Aldosterone (Option A) (Option B)
Zona Fasciculata	17α-Hydroxylase, others (Option A)	Cortisol, Corticosterone
Zona Reticularis	17α-Hydroxylase, others	DHEA, Androstenedione (Option C)

Enzyme Absence:

- Zona Glomerulosa: Lacks 17α-Hydroxylase (No cortisol or sex hormones)
- Zona Fasciculata and Zona Reticularis: Lacks Aldosterone Synthase (No aldosterone)

Solution for Question 9:

21β-hydroxylase deficiency leads to reduced cortisol and aldosterone production, causing elevated ACTH and increased production of androgens, resulting in virilization and ambiguous genitalia.

Congenital Adrenal Hyperplasia (CAH)

- Definition: Group of genetic disorders affecting adrenal glands, causing enzyme deficiencies in steroid biosynthesis.
- Effects: Disruption in cortisol and aldosterone production, leading to various symptoms.
- Prevalence: Most common deficiency is 21 β -hydroxylase deficiency (90%+ cases).
- Impact: Reduced cortisol and aldosterone production.
- Consequence: Increased ACTH secretion. (Option B)
- Result: Increased precursor steroids converted to androgens → Virilization.

Enzyme Deficiency	Description	Symptoms/Consequences
Cholesterol desmolase deficiency	Fatal in utero, prevents placental progesterone synthesis.	Death before birth.
Steroidogenic acute regulatory (StAR) protein deficiency	Known as congenital lipid adrenal hyperplasia. Prevents cholesterol movement into mitochondria.	Low steroid production, female genitalia regardless of genetic sex.
3 β -Hydroxysteroid dehydrogenase deficiency	Increases DHEA secretion (weak androgen).	Masculinization in females, hypospadias in males. (Option C)
17 α -Hydroxylase deficiency	Caused by mutated CYP17 gene. Prevents sex hormone production.	Female external genitalia, hypertension, hypokalemia due to high 11-deoxycorticosterone. (Option D- not early virilization)
11 β -Hydroxylase deficiency	Increased 11-deoxycortisol and 11-deoxycorticosterone secretion.	Virilization, hypertension due to salt and water retention. (Option A)

- Glucocorticoid treatment: Recommended for all virilizing forms of CAH. Purpose: Addresses glucocorticoid deficit. Effect: Inhibits ACTH secretion, reducing androgen and steroid production.
- Purpose: Addresses glucocorticoid deficit.
- Effect: Inhibits ACTH secretion, reducing androgen and steroid production.
- Purpose: Addresses glucocorticoid deficit.
- Effect: Inhibits ACTH secretion, reducing androgen and steroid production.

Solution for Question 10:

Alkalosis, which can occur due to hyperventilation or other factors, leads to more calcium binding to albumin, thereby decreasing the levels of free-ionized calcium. This reduction in ionized

calcium increases neuromuscular excitability, resulting in symptoms like muscle cramps and tingling sensations and positive Chvostek's and Trousseau's signs.

Total Body Calcium Regulation:

- Primary Reservoir: Bone acts as the main calcium reservoir.

Serum calcium normal range = 8.5-10.5 mg/dL

Forms of ECF Calcium:

- Ionized Calcium (50%): Most physiologically important form, influencing heart, nervous system, and bone formation.
- Protein-bound Calcium (41%): Bound to plasma proteins like albumin.
- Anion-bound Calcium (9%): Bound to anions in the ECF.

Binding Dynamics of Calcium and Hydrogen Ions:

Acidosis:

Competition with Hydrogen Ions: Hydrogen ions compete with calcium for binding sites on albumin, increasing levels of free calcium.

Alkalosis: (option B)

Increased Binding of Calcium: More calcium binds to albumin, reducing free calcium levels and increasing excitability.

Other options:

Option A and C) Increased calcium would not lead to symptoms of neuromuscular excitability

Option D) Vitamin D overdose would cause increased calcium levels

Solution for Question 11:

Tetany is primarily caused by a decrease in extracellular ionized calcium (Ca^{2+}). This reduction increases the excitability of nerve and muscle cells, leading to symptoms such as muscle spasms and laryngospasm.

Tetany:

- It is characterized by extensive skeletal muscle spasms

- Cause: Decrease in extracellular ionized calcium (Ca^{2+}), leading to increased excitability of nerve and muscle cells.
- Mechanism:

Increased Excitability: Reduced extracellular Ca^{2+} lowers its binding to voltage-gated sodium channels in nerve and muscle cells.

Action Potential Generation: Lower Ca^{2+} binding decreases the activation threshold of these channels, making them more likely to open and trigger action potentials.

- Complication: Laryngospasm can obstruct the airway and potentially cause fatal asphyxia.

Calcium and Tetany:

- Normal Total Calcium Levels: Total calcium might appear normal, but a decrease in ionized calcium can still cause tetany.
- Influence of pH: Increased pH (alkalosis) from hyperventilation increases plasma protein ionization, binding more Ca^{2+} and reducing free Ca^{2+} availability.

Signs of Tetany:

- Chvostek Sign: Contraction of facial muscles upon tapping the facial nerve.
- Trousseau Sign: Spasm of upper extremity muscles, causing wrist and thumb flexion with finger extension; elicited by occluding circulation with a blood pressure cuff.

Solution for Question 12:

PTH acts as a key regulator, preventing calcium levels from falling too low and maintaining a stable calcium concentration in the extracellular fluid, even in the face of fluctuations in calcium intake or loss.

PTH and Calcium regulation:

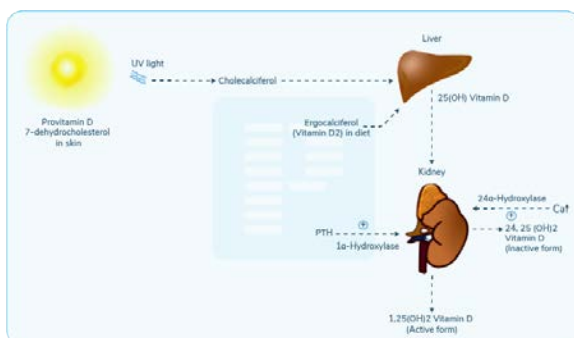
Factor	Description	Effect on Calcium Levels
PTH	A hormone secreted by the parathyroid glands in response to low extracellular calcium levels.	Increases
Bone Resorption (Option C)	PTH stimulates osteoblasts, which signal osteoclasts to break down bone tissue, releasing calcium and	Increases

	phosphate into the bloodstream.	
Renal Excretion (Option A)	PTH increases calcium reabsorption in the renal tubules, reducing calcium loss through urine. PTH decreases phosphate reabsorption in the kidneys.	Increases
Intestinal Absorption (Option B)	PTH indirectly enhances calcium and phosphate absorption in the intestines. This active form of vitamin D promotes the formation of calbindin, a calcium-binding protein, in intestinal cells.	Increases
Vitamin D	While not directly regulated by PTH, vitamin D plays a crucial role in calcium regulation. Vitamin D must be converted to its active form, 1,25-dihydroxycholecalciferol, in the kidneys. PTH is essential for this conversion. (Option D)	Increases

Solution for Question 13:

The main role of vitamin D in calcium regulation is to enhance calcium absorption in the intestines along with potassium reabsorption. This increases the serum calcium and potassium concentration.

Active Vitamin D production:



Actions of 1,25-Dihydroxycholecalciferol

Action site	Key concepts
Intestines	Increases the production of calbindin (calcium-binding protein) in intestinal epithelial cells which increases calcium absorption.
Kidneys	Increases calcium and phosphate reabsorption in the renal tubules, reducing their excretion in urine. This effect is relatively weak
Bones	Excessive vitamin D can cause bone resorption. Smaller quantities actually promote bone calcification.

Solution for Question 14:

Calcitonin is a hormone produced by the parafollicular cells (C cells) of the thyroid gland that plays a crucial role in regulating calcium homeostasis. Its primary mechanism for lowering blood calcium levels involves inhibiting the activity of osteoclasts, which are the cells responsible for breaking down bone tissue (bone resorption)

Calcitonin:

- A hormone produced by the parafollicular cells (C cells) of the thyroid gland.
- Its primary function is to lower blood calcium levels (hypocalcemic effect).

Calcium Regulation:

- Calcitonin helps to regulate calcium levels in the blood, working in opposition to parathyroid hormone (PTH), which increases blood calcium levels.

Mechanism of Action	
Action site	Key concepts
Bone	Inhibition of Osteoclast Activity by binding to calcitonin receptors on osteoclast (Option A) and (Option C) Decreased release of calcium and phosphate from bone
Kidneys	Increased Calcium Excretion (Option D) Reduces the reabsorption of calcium and phosphate in the renal tubules.
Intestine	Decreased Calcium Absorption (Minor Role) (Option B) Effect is less compared to kidney

Secretion Triggers:

- High blood calcium levels
- Gastrin and pentagastrin (hormones involved in digestion)

Solution for Question 15:

Cathepsin K primarily in osteoclast plays an essential role in osteoclast-mediated bone matrix degradation.

Markers	
Bone resorption	Bone formation
Tartrate-resistant acid phosphatase Cathepsins K (Option D) Collagen-related markers Aminoterminal cross-linked telopeptide of type I collagen	Osteocalcin (Option A) Bone-specific alkaline phosphatase (Option B) C-terminal propeptide of type I procollagen (Option C)

Thyroid Hormones & Endocrine Pancreas

1. Which of the following statements about the pancreatic islets is correct?

- A. A-cells are the most prevalent and primarily secrete insulin.
 - B. B cells constituting 60–75% of islet cells primarily secrete glucagon
 - C. F cells, which are the most abundant, primarily secrete pancreatic polypeptide.
 - D. B cells are the most prevalent and secrete insulin.
-

2. Which conditions would most likely result in a high insulin-glucagon molar (I/G ratio)?

- A. Prolonged fasting
 - B. Adopting a low-carbohydrate diet
 - C. Starvation
 - D. Consumption of a large carbohydrate meal
-

3. Which of the following actions is NOT a physiological effect of glucagon?

- A. Stimulating glycogen synthesis in the liver
 - B. Promoting gluconeogenesis
 - C. Increasing glycogen breakdown in the liver
 - D. Enhancing lipolysis in the liver
-

4. Which of the following does not stimulate insulin secretion?

- A. α -Adrenergic receptor activation
 - B. Activation of parasympathetic nerves
 - C. Stimulation of β -Adrenergic receptors
 - D. Release of GLP-1
-

5. Which of the following statements regarding the mechanisms of action of oral hypoglycemic drugs is FALSE?

- A. GLP-1 analogues enhance insulin release only at high glucose concentrations and reduce glucagon release
 - B. Sulfonylureas bind to the sulfonylurea receptor (SUR1) on pancreatic β -cells, stimulating insulin secretion even at low glucose concentrations.
 - C. GLP-1 analogues increase gastric emptying by stimulating specific GLP-1 receptors.
 - D. DPP-4 inhibitors increase endogenous GLP-1 levels by preventing its degradation.
-

6. What is the primary mechanism by which glucose stimulates insulin secretion in pancreatic β -cells?

- A. Glucose stimulates insulin release directly by binding to its receptor on β -cells, triggering immediate secretion.
 - B. Glucose is metabolized to pyruvate, increasing ATP, which inhibits ATP-sensitive K^+ channels, causing Ca^{2+} influx and insulin release.
 - C. Glucose lowers intracellular glutamate, promoting insulin secretion by altering pH in secretory granules and affecting their release.
 - D. Glucose enhances the uptake of calcium into the β -cells without affecting ATP levels, thereby directly increasing the secretion of insulin.
-

7. Match the following actions of insulin with the correct category: Categories: a) Increased by insulin b) Decreased by insulin
Insulin Action: Cell growth
Reabsorption of K⁺, Na⁺, and phosphate from the kidney
Food intake
Entry of phosphate and magnesium into the cell
Body weight
Lipolysis
Potassium uptake into cells

- A. a)-2,3,5,6,7 & b)-1,4
 - B. a)-4,5,6 & b)-1,2,3,7
 - C. a)-1,3,4,6 & b)-2,5,7
 - D. a)-1,2,4,5,7 & b)-3,6
-

8. Which of the following are the actions of insulin? 1) Increased lipogenesis 2) Increased ketogenesis 3) Increased glycogen synthesis 4) Increased glucose uptake 5) Increased gluconeogenesis 6) Increased protein synthesis

- A. 3,4,5
 - B. 2,5,6
 - C. 1,3,4,6
 - D. 1,3,6
-

9. A medical student curious about the physiological effects of insulin learns that insulin has rapid, intermediate, and delayed actions on various metabolic processes. Which of the following is an intermediate action of insulin?

- A. Increased transport of K⁺ into insulin-sensitive cells
 - B. Stimulation of protein synthesis
 - C. Increase in mRNAs for lipogenic enzymes
 - D. Increased transport of amino acids into insulin-sensitive cells
-

10. Which of the following statements about insulin receptors, GLUT, and SGLT is correct?

- A. The insulin receptor's β subunits are responsible for insulin binding.
 - B. GLUT-4 transporters increase glucose uptake in response to insulin.
 - C. SGLT transporters primarily function in muscle and adipose tissue.
 - D. SGLT-2 inhibitors work by increasing glucose absorption in the kidneys.
-

11. Which of the following statements about insulin synthesis is correct?

- A. Insulin synthesis begins in the Golgi apparatus with the formation of proinsulin.
 - B. The C peptide is removed during the conversion of preproinsulin to proinsulin.
 - C. The insulin gene is located on the long arm of chromosome 11.
 - D. Insulin and C peptides are released from B cells in equimolar amounts.
-

12. A 45-year-old female presents with a painless, enlarging neck mass. Ultrasound reveals multiple thyroid nodules. Which of the following histological findings is most consistent with an inactive thyroid gland?

- A. Small follicles with cuboidal cells and minimal colloid
 - B. Large follicles with abundant colloid
 - C. Numerous granular endoplasmic reticulum, lysosomes, and Golgi complexes
 - D. Parafollicular cells secreting calcitonin
-

13. Which of the following statements regarding the treatment of thyroid disorders is correct?

- A. Propylthiouracil is used to treat hypothyroidism.
 - B. Radioactive iodine therapy is used for the treatment of Graves disease.
 - C. Levothyroxine helps treat hyperthyroidism.
 - D. Hashimoto thyroiditis causes a long-term increase in thyroid hormone production.
-

14. What is the primary metabolic consequence of fasting on thyroid hormone levels?

- A. ↑T4 to T3 conversion, hence ↑ T3
 - B. ↓T4 to T3 conversion, ↓plasma T3 levels, and ↑ RT3 levels.
 - C. ↓T3 and normal RT3.
 - D. ↑ BMR
-

15. A 28 y/o woman visits her physician with complaints of weight loss, increased appetite, and anxiety. She feels warmer than usual and has noticed an increased frequency of bowel movements. O/E, her heart rate is elevated, and lab tests reveal elevated levels of thyroid hormones. Which of the following best describes the effects of thyroid hormones on metabolism in this patient?

- A. Decrease oxygen consumption and promote energy storage in tissues.
 - B. Stimulate lipogenesis and reduce the turnover of glucose in the body.
 - C. Increase the rate of glucose absorption & lipid oxidation
 - D. Inhibit proteolysis, thereby conserving muscle mass and reducing metabolic activity.
-

16. A 35 y/o male presents with symptoms of lethargy, hair loss, and cold intolerance. His lab results show elevated T4 levels but low T3 levels. Which of the following best explains the relationship between T4 and T3 in this patient?

- A. T4 is the primary thyroid hormone and is converted to the more active T3 in peripheral tissues.
 - B. T3 has a longer half-life than T4, allowing it to maintain steady levels in the bloodstream.
 - C. Elevated T4 levels automatically indicate that T3 levels should also be elevated, due to their synchronous production.
 - D. T4 is biologically active, while T3 is inactive and serves primarily as a storage form of thyroid hormone.
-

17. A 50 y/o male presents with fatigue, weight gain, and feeling cold. Blood tests show normal T4, low T3, and elevated RT3 levels. Which of the following best explains the patient's symptoms and lab findings?

- A. Increased activity of 5'-deiodinase in the liver and kidney
 - B. Decreased activity of 5-deiodinase
 - C. Increased activity of 5-deiodinase
 - D. Reduced secretion of thyroxine (T4) from the thyroid gland
-

18. A 30y/o pregnant woman has elevated thyroid-binding globulin (TBG) but shows no symptoms of thyroid dysfunction. Her endocrinologist explains this is common during pregnancy and that her thyroid function remains normal. Which of the following best explains why the pregnant woman is experiencing this?

- A. TSH secretion is decreased due to elevated TBG levels
- B. Elevated TBG levels reduce the production of thyroid hormones
- C. A temporary fall in free thyroid hormones stimulates TSH secretion, restoring equilibrium
- D. Increased TBG levels compensate for the reduced free hormone levels by decreasing the metabolic clearance rate of thyroid hormones

19. A 38y/o woman presents with symptoms of fatigue, weight gain, and dry skin. Laboratory tests reveal low free T4 and elevated TSH levels. Further investigation shows increased levels of thyroxine-binding globulin (TBG). Which of the following is most likely to be associated with increased levels of thyroxine-binding globulin (TBG)

- A. Phenytoin
 - B. Androgens
 - C. Glucocorticoids
 - D. Estrogen
-

20. A patient with a history of Graves' disease is treated with a high dose of iodide. Initially, there is a reduction in thyroid hormone synthesis due to the Wolff-Chaikoff effect. However, after a few days, the patient's thyroid function returns to normal. Which of the following mechanisms explains this "escape" phenomenon?

- A. Increased degradation of thyroglobulin
 - B. Upregulation of the sodium-iodide symporter (NIS)
 - C. Downregulation of the sodium-iodide symporter (NIS)
 - D. Increased conversion of T4 to T3 in peripheral tissues
-

21. A patient presents with symptoms of hypothyroidism despite adequate iodine intake. Upon investigation, it is found that there is a defect in the enzyme responsible for a key step in thyroid hormone synthesis. Which of the following steps in thyroid hormone synthesis is not facilitated by thyroid peroxidase (TPO)?

- A. Oxidation of iodide to iodine

- B. Coupling of iodotyrosines
 - C. Iodide trapping
 - D. Binding of iodine to thyroglobulin
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	4
Question 3	1
Question 4	1
Question 5	3
Question 6	2
Question 7	4
Question 8	3
Question 9	2
Question 10	2
Question 11	4
Question 12	2
Question 13	2
Question 14	2
Question 15	3
Question 16	1
Question 17	3
Question 18	3
Question 19	4
Question 20	3
Question 21	3

Solution for Question 1:

Correct Answer: D) B cells are the most prevalent and secrete insulin.

Explanation:

B cells are the most prevalent in the islets of Langerhans and they secrete insulin to lower blood glucose levels.

Pancreas Overview:

- Location: Abdominal cavity
- Functions: Endocrine: Hormone secretion (Islets of Langerhans) Exocrine: Digestive enzyme secretion
- Endocrine: Hormone secretion (Islets of Langerhans)
- Exocrine: Digestive enzyme secretion
- Islets of Langerhans: Size: 76–175 μm Distribution: Throughout the pancreas, concentrated in the tail Volume: ~2% of pancreas volume Blood Supply: Rich, drains into hepatic portal vein
- Size: 76–175 μm
- Distribution: Throughout the pancreas, concentrated in the tail
- Volume: ~2% of pancreas volume
- Blood Supply: Rich, drains into hepatic portal vein
- Endocrine: Hormone secretion (Islets of Langerhans)
- Exocrine: Digestive enzyme secretion
- Size: 76–175 μm
- Distribution: Throughout the pancreas, concentrated in the tail
- Volume: ~2% of pancreas volume
- Blood Supply: Rich, drains into hepatic portal vein

Islet Cell Types:

Cell Type	Percentage of Islet Cells	Location	Hormone Secreted	Hormone Function
B cells	60-75%	Center	Insulin (Option B)	Lowers blood glucose
A cell (Option A)	20%	Surround B cells	Glucagon	Increases blood glucose
D cells	Less common	Encircle B cells	Somatostatin	Regulates islet cell secretion

F cells	Less common (Option C)	Encircle B cells	Pancreatic polypeptide	Regulates ion transport in the intestine
---------	------------------------	------------------	------------------------	--

Solution for Question 2:

Correct Answer: D) Consumption of a large carbohydrate meal

Explanation:

Eating a meal rich in carbohydrates significantly increases insulin secretion to promote the storage of glucose as glycogen, thus resulting in a high I/G ratio.

Prolonged fasting (Option A) decreases insulin levels while increasing glucagon secretion to mobilize stored energy, resulting in a low I/G ratio.

A low-carbohydrate diet (Option B) reduces insulin levels and increases glucagon secretion to maintain blood glucose levels, leading to a lower I/G ratio.

During starvation (Option C), the body relies heavily on glucagon to mobilize energy stores, resulting in a very low I/G ratio.

Insulin Glucagon Ratio			
Condition	Hepatic Glucose Storage/ Production	I/G Ratio	Features
Large Carbohydrate Meal	Storage	70	Indicates energy abundance Promotes glycogenesis, lipogenesis, and protein synthesis
IV Glucose	Storage	25	
Small Meal	Storage	7	
Overnight Fasting	Production	2.3	Indicates the need to mobilise energy stores Promotes glycogenolysis, gluconeogenesis, and ketogenesis

Low Carbohydrate Diet	Production	1.8	
Starvation	Production	0.4	

Solution for Question 3:

Correct Answer: A) Stimulating glycogen synthesis in the liver

Explanation:

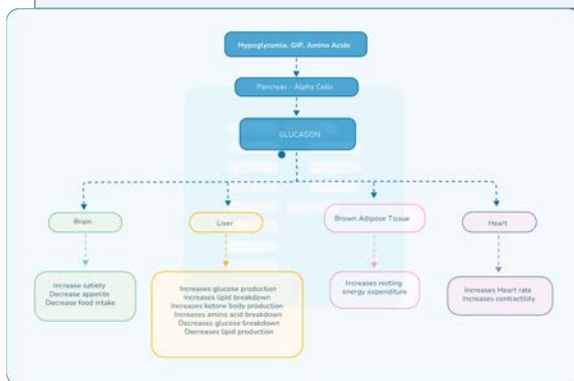
Glucagon inhibits glycogen synthesis in the liver by inactivating glycogen synthase. It prevents the storage of glucose as glycogen, instead promoting glucose release into the bloodstream.

Promoting gluconeogenesis (Option B): Glucagon activates enzymes involved in gluconeogenesis, increasing glucose production from non-carbohydrate sources, such as amino acids.

Increasing glycogen breakdown in the liver (Option C): Glucagon stimulates glycogenolysis, breaking down glycogen stores in the liver to increase blood glucose levels.

Enhancing lipolysis in the liver (Option D): Glucagon promotes lipolysis, breaking down triglycerides in the liver to free fatty acids and glycerol, which can be used for energy production.

Glucagon	
Stimulators	Inhibitors
Amino acids (alanine, arginine) Cortisol Exercise Infections	High blood glucose levels Somatostatin Insulin Free fatty acids and ketones



Solution for Question 4:

Correct Answer: A) α -Adrenergic receptor activation

Explanation:

Activation of α -adrenergic receptors, typically by norepinephrine during sympathetic stimulation, inhibits insulin secretion.

Parasympathetic activation enhances insulin secretion, through the release of acetylcholine, which acts on M3 receptors in pancreatic beta cells. (Option B)

Stimulation of β 2-adrenergic receptors in the sympathetic system can promote insulin secretion, especially when the inhibitory effects of α -adrenergic receptors are blocked. (Option C)

GLP-1 is an incretin hormone that stimulates insulin secretion in response to meals, aiding in the regulation of blood glucose levels. (Option D)

INSULIN	
Stimulators of Insulin Release	Inhibitors of Insulin Release
Glucose Amino acids (e.g., arginine, lysine) Gastrointestinal hormones (e.g., GLP-1, GIP) Parasympathetic stimulation (via acetylcholine) β -Adrenergic receptor activation Sulfonylurea drugs	Low Blood Glucose Somatostatin α -Adrenergic receptor activation Leptin Galanin

Solution for Question 5:

Correct Answer: C) GLP-1 analogues increase gastric emptying by stimulating specific GLP-1 receptors.

Explanation:

GLP-1 analogues slow gastric emptying and suppress appetite by activating specific GLP-1 receptors.

ORAL HYPOGLYCEMIC DRUGS	
Sulphonylureas (Option B)	Mechanism of action: Sulphonylureas bind to the sulphonylurea receptor (SUR1) on the membrane of pancreatic β-cells, which stimulates insulin secretion regardless of glucose concentration. This results in a significant and often unregulated release of insulin, potentially leading to severe hypoglycemia, especially at low glucose levels. Examples: Tolbutamide, Glipizide, Glibenclamide.
GLP-1 Analogues (Option A)	Mechanism of Action: GLP-1 analogues activate GLP-1 receptors, which are GPCRs located on pancreatic β-cells, α-cells, central and peripheral neurons, and gastrointestinal mucosa. This activation promotes insulin secretion specifically in response to high glucose levels, inhibits glucagon release, slows gastric emptying, and suppresses appetite. GLP-1 analogues also contribute to pancreatic β-cell health by reducing islet cell apoptosis. Examples: Exenatide, Liraglutide.
DPP-4 Inhibitors (Option D)	Mechanism of Action: DPP-4 inhibitors block the enzyme DPP-4, which normally breaks down endogenous GLP-1. By inhibiting this enzyme, these drugs increase the levels of GLP-1, thereby enhancing insulin secretion indirectly and improving glycemic control. Examples: Sitagliptin, Saxagliptin, Vildagliptin.

Solution for Question 6:

Correct Answer: B) Glucose is metabolized to pyruvate, increasing ATP, which inhibits ATP-sensitive K⁺ channels, causing Ca²⁺ influx and insulin release.

Explanation:

- Glucose metabolism produces ATP, which inhibits ATP-sensitive K⁺ channels. This inhibition leads to depolarization of the β -cell membrane, causing Ca²⁺ influx through voltage-gated channels, and triggering insulin release.

- Insulin secretion is regulated through a biphasic response to plasma glucose levels.

Biphasic response of Insulin secretion	
First Rapid Phase	Second Prolonged Phase
Glucose enters pancreatic β-cells via GLUT-2 transporters. Inside the cell, glucose is phosphorylated by glucokinase and converted to pyruvate. Pyruvate is metabolized in the mitochondria, producing ATP. (Option D) Increased ATP inhibits ATP-sensitive K^+ channels, leading to reduced K^+ efflux. This depolarizes the β-cell membrane, causing voltage-gated Ca^{2+} channels to open. Ca^{2+} influx triggers the exocytosis of a pool of insulin-containing granules, Results in a rapid, short-lived increase in insulin secretion. Peak within 1–2 minutes and then decreases to basal level in another 2 to 3 minutes	Pyruvate metabolism increases intracellular glutamate levels. (Option C) Glutamate acts as an intracellular second messenger, priming a second pool of secretory granules for release. It may also lower the pH in granules, aiding in their maturation. This leads to a prolonged phase of insulin secretion. Peak in about 60 minutes and then remain elevated for 3–5 hours.

Solution for Question 7:

Correct Answer: D) a)-1,2,4,5,7 & b)-3,6

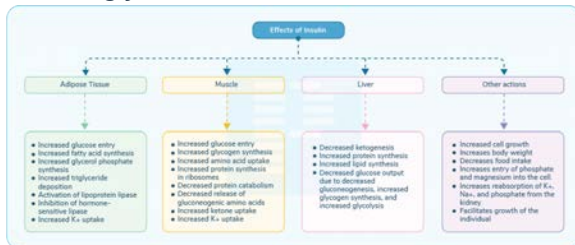
Explanation:

a) Increased by insulin

- Cell growth: Insulin promotes cell growth and proliferation.
- Reabsorption of K^+ , Na^+ , and phosphate from the kidney: Insulin enhances the reabsorption of these electrolytes in the kidneys.
- Entry of phosphate and magnesium into the cell: Insulin facilitates the uptake of phosphate and magnesium into cells.
- Body weight: Insulin can increase body weight due to its effects on fat storage increasing adipose tissue mass.
- Potassium uptake into cells: Insulin promotes the movement of potassium into cells, which can lower blood potassium levels. Therefore is very useful in the management of acute renal failure. Insulin is considered a physiological regulator of plasma K^+ concentration.

b) Decreased by insulin

- Food intake: Insulin can promote satiety by increasing leptin levels in plasma, leading to decreased food intake. Also partly mediated by its inhibitory effect on neuropeptide Y release from the hypothalamus.
- Lipolysis: Insulin inhibits lipolysis, which is the breakdown of fats into fatty acids and glycerol.

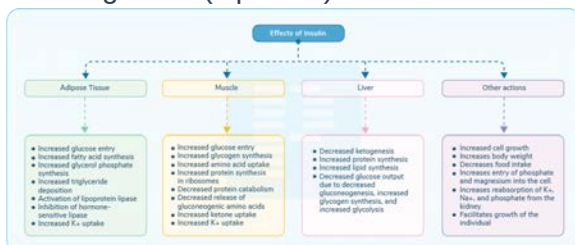


Solution for Question 8:

Correct Answer: C) 1,3,4,6

Explanation:

- Lipogenesis: Insulin promotes the synthesis of fatty acids and triglycerides, especially in adipose tissue.
- Glycogen synthesis: Insulin stimulates glycogen synthesis in both muscle and liver tissues.
- Glucose uptake: Insulin increases glucose uptake into insulin-sensitive cells such as muscle and adipose tissue.
- Protein synthesis: Insulin promotes the uptake of amino acids into cells, enhancing protein synthesis.
- Ketogenesis: Insulin decreases ketogenesis. It reduces the breakdown of fatty acids and the formation of ketone bodies in the liver. (Option 2)
- Gluconeogenesis: Insulin inhibits gluconeogenesis in the liver, reducing the production of glucose (Option 5)



Solution for Question 9:

Correct Answer: B) Stimulation of protein synthesis

Explanation:

Stimulation of protein synthesis is an intermediate action of insulin that occurs within minutes.

PRINCIPAL ACTIONS OF INSULIN	
Rapid (seconds)	Increased transport of Glucose, Amino acids (Option D) K ⁺ (Option A) into insulin-sensitive cells
Intermediate (minutes)	Stimulation of protein synthesis Inhibition of protein degradation Activation of glycolytic enzymes and glycogen synthase Inhibition of phosphorylase and gluconeogenic enzymes
Delayed (hours)	Increase in mRNAs for lipogenic and other enzymes (Option C)

Solution for Question 10:

Correct Answer: B) GLUT-4 transporters increase glucose uptake in response to insulin.

Explanation:

Insulin triggers the translocation of GLUT-4 transporters to the cell membrane, increasing glucose uptake.

Insulin Receptor, GLUT, and SGLT Overview:

Insulin Receptor:

- Structure: Tetramer of two α and two β glycoprotein subunits

- Synthesis: Single mRNA, separated via proteolysis, linked by disulfide bonds
- Gene Location: Chromosome 19, with 22 exons
- Subunit Functions: α Subunits: External, insulin-binding (Option A) β Subunits: Span the membrane, intracellular segments have tyrosine kinase activity Both subunits are glycosylated
- α Subunits: External, insulin-binding (Option A)
- β Subunits: Span the membrane, intracellular segments have tyrosine kinase activity
- Both subunits are glycosylated
- α Subunits: External, insulin-binding (Option A)
- β Subunits: Span the membrane, intracellular segments have tyrosine kinase activity
- Both subunits are glycosylated
- Mechanism: Insulin Binding: Triggers tyrosine kinase activity in β subunits
Autophosphorylation: Of β subunits at tyrosine residues
Signal Cascade: Phosphorylation/dephosphorylation of cytoplasmic proteins, mainly serine and threonine residues, mediated by insulin receptor substrate (IRS-1) and other effectors
- Insulin Binding: Triggers tyrosine kinase activity in β subunits
- Autophosphorylation: Of β subunits at tyrosine residues
- Signal Cascade: Phosphorylation/dephosphorylation of cytoplasmic proteins, mainly serine and threonine residues, mediated by insulin receptor substrate (IRS-1) and other effectors
- Insulin Binding: Triggers tyrosine kinase activity in β subunits
- Autophosphorylation: Of β subunits at tyrosine residues
- Signal Cascade: Phosphorylation/dephosphorylation of cytoplasmic proteins, mainly serine and threonine residues, mediated by insulin receptor substrate (IRS-1) and other effectors

GLUT (Glucose Transporters):

- Function: Facilitate glucose movement into cells
- Types: Seven known types have tissue-specific functions.
- GLUT-4: Primary Location: Muscle and adipose tissue (Option C) Regulation: Directly regulated by insulin Mechanism: Absence of Insulin: GLUT-4 in intracellular vesicles Presence of Insulin: Vesicles translocate and fuse with the cell membrane, increasing glucose uptake
- Primary Location: Muscle and adipose tissue (Option C)
- Regulation: Directly regulated by insulin
- Mechanism: Absence of Insulin: GLUT-4 in intracellular vesicles Presence of Insulin: Vesicles translocate and fuse with the cell membrane, increasing glucose uptake
- Absence of Insulin: GLUT-4 in intracellular vesicles

- Presence of Insulin: Vesicles translocate and fuse with the cell membrane, increasing glucose uptake
- Exercise: Induces insulin-independent GLUT-4 translocation, aiding in blood glucose regulation
- Primary Location: Muscle and adipose tissue (Option C)
- Regulation: Directly regulated by insulin
- Mechanism: Absence of Insulin: GLUT-4 in intracellular vesicles Presence of Insulin: Vesicles translocate and fuse with the cell membrane, increasing glucose uptake
- Absence of Insulin: GLUT-4 in intracellular vesicles
- Presence of Insulin: Vesicles translocate and fuse with the cell membrane, increasing glucose uptake
- Absence of Insulin: GLUT-4 in intracellular vesicles
- Presence of Insulin: Vesicles translocate and fuse with the cell membrane, increasing glucose uptake

SGLT (Sodium Glucose Linked Transporters):

- Function: Reabsorption of glucose in the kidneys (Option C)
- Mechanism:
- SGLT2 Location: Primarily in the proximal tubule of the nephron
- Co-transport Process: Utilises the sodium gradient created by the Na⁺/K⁺ ATPase pump to transport sodium and glucose from the renal tubular lumen into the cells
- Clinical Relevance: SGLT2 Inhibitors: Drugs like canagliflozin and empagliflozin Mechanism of Action: Prevent glucose reabsorption in the kidneys, leading to increased glucose excretion in urine (Option D) Therapeutic Use: Lower blood glucose levels in type 2 diabetes without causing hypoglycemia
- SGLT2 Inhibitors: Drugs like canagliflozin and empagliflozin
- Mechanism of Action: Prevent glucose reabsorption in the kidneys, leading to increased glucose excretion in urine (Option D)
- Therapeutic Use: Lower blood glucose levels in type 2 diabetes without causing hypoglycemia
- SGLT2 Inhibitors: Drugs like canagliflozin and empagliflozin
- Mechanism of Action: Prevent glucose reabsorption in the kidneys, leading to increased glucose excretion in urine (Option D)
- Therapeutic Use: Lower blood glucose levels in type 2 diabetes without causing hypoglycemia

Note: Unlike other tissues, brain cells can take up and use glucose without the need for insulin.

Solution for Question 11:

Correct Answer: D) Insulin and C peptides are released from B cells in equimolar amounts.

Explanation:

Insulin and C peptides are released in equimolar amounts from B cells.

Insulin Synthesis in B Cells:

- Location: B cells of pancreatic islets
- Gene: Chromosome: Short arm of chromosome 11 (Option C) Structure: Three exons, two introns
- Chromosome: Short arm of chromosome 11 (Option C)
- Structure: Three exons, two introns
- Chromosome: Short arm of chromosome 11 (Option C)
- Structure: Three exons, two introns

Key Steps in Insulin Synthesis:

- Synthesis: Preproinsulin is synthesized in the rough endoplasmic reticulum (RER). (Option A)
 - Folding: Preproinsulin folds and forms disulfide bonds to become proinsulin.
 - C Peptide Role: C peptide links the A and B chains and is crucial for folding.
 - Transport: Proinsulin moves to the Golgi apparatus.
 - Packaging: It is packaged into membrane-bound granules.
 - Proteolysis: Two proteases in the granules cleave the C peptide, producing insulin. (Option B)
 - Storage: Insulin is stored within the granules.
 - Granule Movement: Granules move to the plasma membrane via microtubules.
 - Exocytosis: Granules release insulin and C peptide into the bloodstream.
 - Release Ratio: 90–97% of the released product is insulin, with an equimolar amount of C peptide (Option D)
-

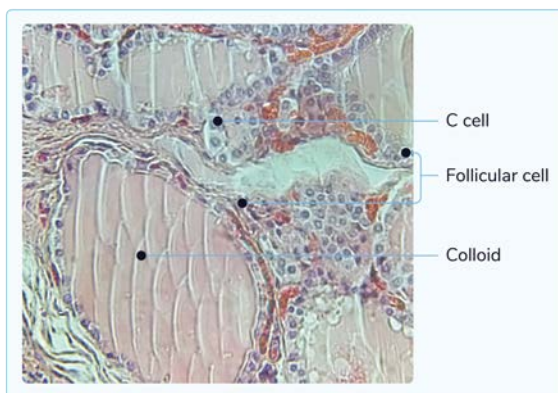
Solution for Question 12:

Correct Answer: B) Large follicles with abundant colloid

Explanation:

Large follicles with abundant colloids are characteristic of an inactive thyroid gland.

Histology of Thyroid gland



- Formed by acini or thyroid follicles

Thyroid Follicles:

- Shape: Spherical
- Consists of a single layer of epithelial cells surrounding a central colloid.
- Colloid is a viscous gel-like substance containing thyroglobulin.
- Inactive Gland: Follicles are large with abundant colloids.
- Active Gland: Follicles are small, cells are cuboidal, and colloid is present in small quantities. (Option A)

Follicular cells	Parafollicular cells
The Apical Membrane of follicular cells faces the colloid and is covered with microvilli. The cytoplasm contains numerous granular endoplasmic reticulum, lysosomes, and Golgi complexes.	Also known as C cells Located close to the follicles.
Basal Membrane is in close contact with numerous capillaries in the interfollicular space.	Secrete calcitonin.

Numerous granular endoplasmic reticulum, lysosomes, and Golgi complexes (Option C) are present in follicular cells but do not differentiate between active and inactive states.

Parafollicular cells secrete calcitonin (Option D) but are not indicative of the activity state of the thyroid gland.

Solution for Question 13:

Correct Answer: B) Radioactive iodine therapy is used for the treatment of Graves disease.

Explanation:

Radioactive iodine therapy destroys thyroid tissue, reducing hormone production, and is commonly used in Graves disease.

Hypothyroidism and hyperthyroidism represent the two extremes of thyroid dysfunction, resulting from under-activity and overactivity of the thyroid gland, respectively. These conditions have distinct causes, symptoms, and treatments.

- Causes: Iodine deficiency: Inadequate iodine intake inhibits thyroid hormone synthesis, leading to decreased secretion and compensatory thyroid hypertrophy eg. goiter.
Autoimmune destruction: eg. Hashimoto thyroiditis (Option D) involves antibodies and immune cells attacking the thyroid gland, causing hypothyroidism. Pituitary or hypothalamic failure: Disruptions in the pituitary gland or hypothalamus → ↓TSH or TRH→ ↓ thyroid stimulation. Drug-induced hypothyroidism: eg. Amiodarone, Lithium Wolff-Chaikoff Effect: A mild and transient inhibition of iodide organic binding and hormone synthesis caused by large doses of iodide acting directly on the thyroid gland in healthy individuals.
- Iodine deficiency: Inadequate iodine intake inhibits thyroid hormone synthesis, leading to decreased secretion and compensatory thyroid hypertrophy eg. goiter.
- Autoimmune destruction: eg. Hashimoto thyroiditis (Option D) involves antibodies and immune cells attacking the thyroid gland, causing hypothyroidism.
- Pituitary or hypothalamic failure: Disruptions in the pituitary gland or hypothalamus → ↓TSH or TRH→ ↓ thyroid stimulation.
- Drug-induced hypothyroidism: eg. Amiodarone, Lithium
- Wolff-Chaikoff Effect: A mild and transient inhibition of iodide organic binding and hormone synthesis caused by large doses of iodide acting directly on the thyroid gland in healthy individuals.

- Clinical features: Reduced cognitive function, fatigue, lethargy. Cold intolerance Weight gain despite poor appetite Dry, yellowish skin (carotenemia) and puffiness (myxedema) due to protein complex accumulation. Constipation: Slowed digestive processes. Thinning hair on the scalp and/or other areas.
- Reduced cognitive function, fatigue, lethargy.
- Cold intolerance
- Weight gain despite poor appetite
- Dry, yellowish skin (carotenemia) and puffiness (myxedema) due to protein complex accumulation.
- Constipation: Slowed digestive processes.
- Thinning hair on the scalp and/or other areas.
- Treatment: Addressing underlying causes: Iodine supplementation, hormone replacement therapy, and medication adjustments. Levothyroxine: Synthetic T4 administered to replace deficient thyroid hormone. (Option C)
- Addressing underlying causes: Iodine supplementation, hormone replacement therapy, and medication adjustments.
- Levothyroxine: Synthetic T4 administered to replace deficient thyroid hormone. (Option C)
- Iodine deficiency: Inadequate iodine intake inhibits thyroid hormone synthesis, leading to decreased secretion and compensatory thyroid hypertrophy eg. goiter.
- Autoimmune destruction: eg. Hashimoto thyroiditis (Option D) involves antibodies and immune cells attacking the thyroid gland, causing hypothyroidism.
- Pituitary or hypothalamic failure: Disruptions in the pituitary gland or hypothalamus → ↓TSH or TRH→ ↓ thyroid stimulation.
- Drug-induced hypothyroidism: eg. Amiodarone, Lithium
- Wolff-Chaikoff Effect: A mild and transient inhibition of iodide organic binding and hormone synthesis caused by large doses of iodide acting directly on the thyroid gland in healthy individuals.
- Reduced cognitive function, fatigue, lethargy.
- Cold intolerance
- Weight gain despite poor appetite
- Dry, yellowish skin (carotenemia) and puffiness (myxedema) due to protein complex accumulation.
- Constipation: Slowed digestive processes.
- Thinning hair on the scalp and/or other areas.
- Addressing underlying causes: Iodine supplementation, hormone replacement therapy, and medication adjustments.

- Levothyroxine: Synthetic T4 administered to replace deficient thyroid hormone. (Option C)
- Causes: Graves disease: Autoimmune condition where antibodies stimulate the TSH receptor, leading to excessive T4 and T3 secretion and goiter. Toxic adenoma or multinodular goiter: Benign thyroid nodules autonomously produce thyroid hormones. Early stages of Hashimoto thyroiditis: Initial inflammation can cause a transient phase of hyperthyroidism. TSH-secreting pituitary tumor: Rare tumors secrete excessive TSH, overstimulating the thyroid gland.
- Graves disease: Autoimmune condition where antibodies stimulate the TSH receptor, leading to excessive T4 and T3 secretion and goiter.
- Toxic adenoma or multinodular goiter: Benign thyroid nodules autonomously produce thyroid hormones.
- Early stages of Hashimoto thyroiditis: Initial inflammation can cause a transient phase of hyperthyroidism.
- TSH-secreting pituitary tumor: Rare tumors secrete excessive TSH, overstimulating the thyroid gland.
- Symptoms: Nervousness and irritability: Increased CNS stimulation. Weight loss: Despite increased appetite (hyperphagia), elevated metabolism causes weight loss. Rapid heartbeat (tachycardia) and palpitations: Thyroid hormones stimulate cardiac function. Tremor: Fine tremor of the outstretched fingers. Eye changes: Protruding eyeballs (exophthalmos) in Graves disease.
- Nervousness and irritability: Increased CNS stimulation.
- Weight loss: Despite increased appetite (hyperphagia), elevated metabolism causes weight loss.
- Rapid heartbeat (tachycardia) and palpitations: Thyroid hormones stimulate cardiac function.
- Tremor: Fine tremor of the outstretched fingers.
- Eye changes: Protruding eyeballs (exophthalmos) in Graves disease.
- Treatment: Antithyroid medications: Thioureylenes like propylthiouracil (Option A) and methimazole inhibit thyroid hormone synthesis. Radioactive iodine therapy: Destroys thyroid tissue, reducing hormone production.
- Antithyroid medications: Thioureylenes like propylthiouracil (Option A) and methimazole inhibit thyroid hormone synthesis.
- Radioactive iodine therapy: Destroys thyroid tissue, reducing hormone production.
- Graves disease: Autoimmune condition where antibodies stimulate the TSH receptor, leading to excessive T4 and T3 secretion and goiter.
- Toxic adenoma or multinodular goiter: Benign thyroid nodules autonomously produce thyroid hormones.

- Early stages of Hashimoto thyroiditis: Initial inflammation can cause a transient phase of hyperthyroidism.
- TSH-secreting pituitary tumor: Rare tumors secrete excessive TSH, overstimulating the thyroid gland.
- Nervousness and irritability: Increased CNS stimulation.
- Weight loss: Despite increased appetite (hyperphagia), elevated metabolism causes weight loss.
- Rapid heartbeat (tachycardia) and palpitations: Thyroid hormones stimulate cardiac function.
- Tremor: Fine tremor of the outstretched fingers.
- Eye changes: Protruding eyeballs (exophthalmos) in Graves disease.
- Antithyroid medications: Thioureylenes like propylthiouracil (Option A) and methimazole inhibit thyroid hormone synthesis.
- Radioactive iodine therapy: Destroys thyroid tissue, reducing hormone production.

Surgery: Removal of the thyroid gland, especially if the goiter is large and causing compressive symptoms.

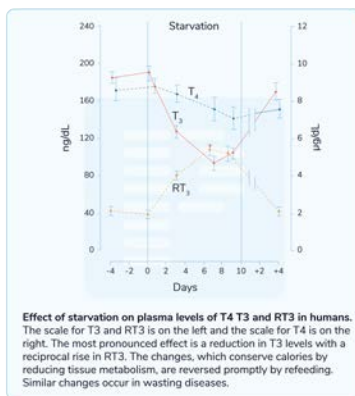
Solution for Question 14:

Correct Answer: B) ↓T4 to T3 conversion, ↓plasma T3 levels, and ↑ RT3 levels.

Explanation:

Prolonged fasting leads to decreased conversion of T4 to T3, leading to reduced plasma T3 levels and increased RT3 levels.

Effect of diet on conversion of T4 to T3:



Fasting:

- Decreases conversion of T4 to T3, reducing plasma T3 concentration. (Option A)
- Plasma T3 reduces by 10-20% within 24 hours and about 50% in 3-7 days.
- Facilitates conversion of T4 to RT3, increasing RT3 levels.
- Free and bound T4 levels remain normal.
- During prolonged starvation (severe, chronic fasting), RT3 returns to normal but T3 remains depressed. (Option C)
- Basal metabolic rate (BMR) falls. (Option D)
- Urinary nitrogen excretion, an index of protein breakdown, decreases.
- The decline in T3 conserves calories and protein

Overfeeding:

- Increases T3.
- Decreased RT3.

Solution for Question 15:

Correct Answer: C) Increase the rate of glucose absorption & lipid oxidation

Explanation:

Thyroid hormones promote glucose absorption, stimulate lipid oxidation, and hence increase overall energy expenditure.

- They increase BMR in every tissue except the anterior pituitary, brain, gonads, spleen, and lymph nodes.

Functions of thyroid hormones:

Effect of Thyroid hormone on metabolism

Carbohydrate Metabolism	Increase absorption of glucose from the intestine. Induce hepatic gluconeogenesis and glycogenolysis, raising plasma glucose. Increase glucose turnover by enhancing its uptake and oxidation. (Option B)
Fat Metabolism	Stimulate lipolysis, releasing free fatty acids and glycerol. Increase lipid turnover and facilitate oxidation. Stimulate cholesterol synthesis, oxidation, and biliary excretion. Enhance hepatic uptake of cholesterol by increasing LDL receptor expression. Net effect: a significant decrease in plasma cholesterol and total lipid levels.
Protein metabolism (Option D)	Promote proteolysis in skeletal muscle, releasing amino acids. Increase protein turnover by enhancing protein degradation.

Effects of Thyroid Hormone on CVS

Heart Rate and Cardiac Output	Increase the heart rate by upregulating and sensitizing β -receptors in the SA and AV nodes, leading to tachycardia in hyperthyroidism. Enhance myocardial contractility by increasing stroke volume hence increasing cardiac output (CO).
Blood Pressure	Systolic blood pressure: ■heart rate and stroke volume results in increased CO Diastolic pressure: decreases due to calorogenic-induced thermogenic cutaneous vasodilation, resulting in decreased peripheral resistance. The overall effect is an increased pulse pressure, which is prominent in hyperthyroidism.
Circulation Dynamics	Enhanced cardiac output and vasodilation create hyperdynamic circulation, leading to shortened circulation time.

Effects of Thyroid Hormone on other systems:

Target system	Functions
---------------	-----------

CNS	Important for the maturation of CNS especially during infancy and childhood Essential for overall brain functions including memory, learning, alertness, and reflexes. Lack of thyroid hormones during infancy can cause irreversible CNS development retardation.
Skeletal muscle	Thyroid hormones increase MHC gene expression in skeletal muscle, leading to muscle cramps and weakness in hypothyroidism. Hyperthyroidism also causes muscle weakness due to increased protein catabolism, with chronic cases resulting in thyrotoxic myopathy.
Bones	It is essential for normal growth, and musculoskeletal maturation, and promotes the expression of growth hormone (GH) genes. Increase osteoid activity, bone growth, and bone remodeling, and promote the growth of teeth, epidermis, nails, and hair. In hypothyroid children, bone growth slows, and epiphyseal closure is delayed.
GIT	Stimulates GI motility, ↑ appetite, ↑ Carbohydrate absorption
Reproduction	Follicular maturation and ovulation
Respiration	During hypercapnia & hypoxia, they increase oxygen consumption in tissues by : (Option A) ↑ RR ↑ arterial PO ₂ ↑ erythropoietin synthesis

Solution for Question 16:

Correct Answer: A) T₄ is the primary thyroid hormone and is converted to the more active T₃ in peripheral tissues.

Explanation:

T₄ is the primary thyroid hormone secreted by the thyroid gland, and it is converted to the more active T₃ in peripheral tissues.

Difference between T₄ & T₃

Aspect	T4	T3
Secretion	More (35% of total)	Less (7% of total)
Plasma conc.	8 mg/dL	0.15 mg/dL
Protein-bound	99.8%	99.8%
Free concentration in plasma	2 ng/dL	0.3 ng/dL
Half-life (t _{1/2}) (Option B)	7 days	1 day
Action	Slower	Faster
Potency	Less	More (3-5 times)
Binding to thyroid hormone receptors	Low affinity	High affinity

T4 and T3 (Option C) can be produced independently, and high T4 does not guarantee high T3 levels and depends on its peripheral conversion.

T3 is the biologically active form of thyroid hormone, not T4 (Option D)

Solution for Question 17:

Correct Answer: C) Increased activity of 5-deiodinase

Explanation:

Increased activity of 5-deiodinase leads to increased conversion of T4 to RT3 by 5-deiodinase leading to higher levels of the physiologically inert RT3 and lower levels of the active T3

Peripheral conversion of T4 to T3

- The peripheral conversion of T4 to T3 is a crucial process that regulates the active thyroid hormone T3 levels, ensuring its availability to target cells.
- The major product secreted by the thyroid gland is thyroxine (T4)

Conversion to T3	35% of secreted T4 is converted to T3 in the circulation by the enzyme 5'-deiodinase, mainly found in the liver and kidneys. This process produces about 85% of circulating T3, the most active hormone that primarily affects target cells.
------------------	---

Conversion to RT3	45% of T4 is also converted to reverse T3 (RT3) by the enzyme 5-deiodinase. RT3 is physiologically inert and accounts for 95% of circulating RT3.
-------------------	--

- The remaining 20% of T4 is conjugated with sulfates and glucuronides.
- Deiodinases are enzymes responsible for the metabolism of thyroid hormones.
- They play a crucial role in regulating the levels of active thyroid hormones in various tissues.
- All deiodinases contain selenocysteine, an amino acid essential for their enzyme activity, making them potent catalysts for oxidoreductive reactions.

There are three types of deiodinases:

Type of deiodinase	Location	Function
D1	Liver Kidney Thyroid Pituitary	Function as 5'-deiodinases, converting thyroxine (T4) to triiodothyronine (T3) Maintaining intracellular T3 levels in target tissues
D2	Skeletal muscle Brain Pituitary Placenta Brown fat	Its function is the same as D1 It maintains a steady supply of T3 to neurons in the brain.
D3	Brain Reproductive tissue	Converts T4 to reverse T3 (RT3), a biologically inactive form.

Metabolism of T3 and T4:

- T3 is degraded to diiodothyronines in the liver and kidney.
- In the liver, T4 and T3 undergo conjugation with sulfates and glucuronic acid, which are then secreted in bile and re-enter the enterohepatic circulation, minimizing iodine loss.
- T4 is also metabolized to tetraiodoacetic acid (tetrac) in various tissues.

Increased activity of 5'-deiodinase (Option A) would lead to higher levels of T3, not lower, which contradicts the patient's lab findings.

Decreased activity of 5-deiodinase (Option B) would result in reduced levels of RT3, not elevated levels as seen in the patient.

The patient has normal T4 levels, indicating that thyroxine secretion is not the issue. (Option D)

Solution for Question 18:

Correct Answer: C) A temporary fall in free thyroid hormones stimulates TSH secretion, restoring equilibrium

Explanation:

TBG levels are increased in pregnancy which binds to the thyroid hormone and reduces the levels of free thyroid hormone in the blood, this temporary fall in thyroid hormone stimulates TSH secretion, thereby restoring equilibrium.

Consequences of Increased TBG Due to Pregnancy

- **TBG Elevation During Pregnancy:** TBG levels are elevated in estrogen-treated patients and during pregnancy, necessitating the above compensatory mechanisms to maintain thyroid hormone balance.
- **Initial Impact: Decrease in Free Thyroid Hormones:** A sudden, sustained increase in thyroid-binding proteins like TBG in the plasma causes a temporary fall in the concentration of free thyroid hormones
- **Increased TBG levels do bind more thyroid hormones,** reducing the levels of free hormones temporarily. Overall binding capacity increases, but it does not necessarily compensate by decreasing the metabolic clearance rate significantly. The primary compensatory mechanism is the increased TSH secretion, which restores free hormone levels. (Option D)
- **Feedback Mechanism: TSH Secretion:** The drop in free thyroid hormones stimulates the secretion of Thyroid Stimulating Hormone (TSH). (Option A) **Increased Production:** TSH increases the production of free thyroid hormones. **New Equilibrium:** A new equilibrium is reached where the total quantity of thyroid hormones in the blood is elevated (Option B), but the concentration of free hormones, their metabolism rate, and the TSH secretion rate return to normal.

Euthyroid State: Despite the increased total thyroid hormone levels, patients remain euthyroid, meaning they are neither hyperthyroid nor hypothyroid.

Solution for Question 19:

Correct Answer: D) Estrogen

Explanation:

Estrogen is seen to cause an increase in levels of thyroxine-binding globulin

Thyroid Binding Proteins

Thyroid hormones bind with 3 types of plasma proteins

Thyroxine Binding Globulin (TBG)	Nature: Glycoprotein synthesized by the liver with a molecular weight of 54,000. Binding Capacity: Each TBG molecule has a single binding site for thyroid hormones. Binding Proportion: About 70% of T4 and 45% of T3 bind to TBG. Plasma Level Influences: Increase: During pregnancy and in patients receiving estrogen, major tranquilizers, and clofibrate. Decrease: In patients receiving glucocorticoids, androgens, danazol, L-asparaginase, salicylates, phenytoin, mitotane, and 5-fluorouracil (Option A, B, C)
Thyroxine Binding Prealbumin (TBPA) /Transthyretin	Nature: A globulin, not an albumin, recently designated as transthyretin. Binding Proportion: T3: Less than 1% binds to TBPA. T4: About 20% binds to TBPA.
Albumin	Binding Proportion: T3: About 55% binds to albumin. T4: About 10% binds to albumin.

- Protein-bound forms of T3 and T4 serve as a reservoir, replenishing circulating hormones when secretion decreases or metabolism increases temporarily.
- Protein binding protects the hormones against metabolic degradation, extending their half-lives (1 day for T3 and 7 days for T4).
- Free-form concentrations provide feedback effects on hormone regulation. Changes in plasma protein levels temporarily affect thyroid hormone levels, maintaining a euthyroid state despite alterations in binding protein levels.

Solution for Question 20:

Correct Answer: C) Downregulation of the sodium-iodide symporter (NIS)

Explanation:

Downregulation of the sodium-iodide symporter (NIS) reduces the intrathyroidal concentration of iodide, allowing normal thyroid hormone synthesis to resume despite high serum iodide levels.

Wolff-Chaikoff Effect

The Wolff-Chaikoff effect describes the thyroid's response to high doses of iodide, resulting in temporary inhibition of thyroid hormone synthesis.

Mechanism:	Inhibition of Organic Binding: Large doses of iodide inhibit the organic binding of iodide within the thyroid gland, reducing hormone synthesis.
TSH Effect Inhibition:	High iodide levels decrease cAMP production in thyroid cells, inhibiting the effect of thyroid-stimulating hormone (TSH). (Wolff-Chaikoff effect) This reduces the proteolysis of thyroglobulin, further decreasing hormone synthesis.
Purpose	The Wolff-Chaikoff effect serves to prevent the thyroid from synthesizing excessive amounts of thyroid hormones in the presence of large quantities of iodide.

Thyroid Escape Phenomenon: The Wolff-Chaikoff effect is transient and lasts for only a few days. After the acute phase, the thyroid gland resumes normal hormone synthesis through the escape phenomenon. This is achieved by:

- Down-regulation of NIS: The sodium iodide symporter (NIS) is down-regulated, reducing intrathyroidal inorganic iodine concentration.
- Resumption of TPO-H₂O₂ System: Normal activity of the thyroid peroxidase (TPO) and hydrogen peroxide system resumes, allowing the normal synthesis of thyroxine (T₄) and triiodothyronine (T₃).

Increased degradation of thyroglobulin (Option A) does not explain the escape phenomenon

Upregulation of the sodium-iodide symporter (NIS) (Option B) would increase iodide uptake, which is contrary to the escape mechanism.

Increased conversion of T₄ to T₃ (Option C) in peripheral tissues is unrelated to the escape phenomenon in the context of thyroid hormone synthesis regulation.

Solution for Question 21:

Correct Answer: C) Iodide trapping

Explanation:

Iodide trapping involves the active transport of iodide into the thyroid follicular cells via the sodium-iodide symporter (NIS) and is not facilitated by thyroid peroxidase.

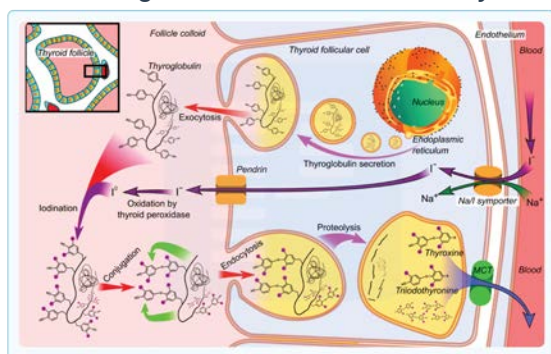
Thyroid Hormone Synthesis

1. Iodide Trapping	Mechanism: Iodide is actively transported into thyroid follicles (iodide pump). Process: Na⁺ and I⁻ co-transported into the cell by Na⁺-I⁻ symporter (NIS). TSH induces expression and retention of NIS. Na⁺ is pumped out by the Na⁺-K⁺ pump. Iodide is pumped against electrical and chemical gradients using ATP. Other Tissues: Salivary glands, placenta, choroid plexus, mammary gland, gastric mucosa, and ciliary body also accumulate iodide but do not form thyroid hormones. Iodide trapping is stimulated by TSH.
2. Conversion of Iodide into Iodine	Process: Iodide is converted to iodine by oxidation. Oxidation (organification) facilitated by thyroid peroxidase. (Option A)
3. Thyroglobulin Synthesis	Thyroglobulin: Glycoprotein with two subunits and 123 tyrosine residues. Synthesis: Synthesized in the endoplasmic reticulum of thyroid cells. Packaged in Golgi apparatus and secreted into colloid by exocytosis. Facilitated by thyroid peroxidase. (Option D) Binding: Reactive iodine binds to tyrosine on thyroglobulin, forming monoiodotyrosine (MIT) and diiodotyrosine (DIT).
4. The Coupling Reaction	Formation: Two DIT molecules condense to form thyroxine (T₄). MIT and DIT condense to form triiodothyronine (T₃). Reverse-T₃ (RT₃) formed by condensation of DIT with MIT. Mechanism: Intramolecular coupling (both DIT attached to thyroglobulin). Intermolecular coupling (DIT detached from thyroglobulin). Facilitation: Coupling reaction facilitated by thyroid peroxidase (Option B).
5. Proteolysis of Thyroglobulin	Process: Colloid ingested by thyroid cells via endocytosis. Lysosomal enzymes digest peptide bonds in thyroglobulin.

	Free T4, T3, DIT, and MIT formed in the cytoplasm. Iodotyrosine deiodinase deiodinates MIT and DIT. T3 and T4 are released into circulation. Iodine from deiodination is reutilized by the gland.
6. Secretion of Thyroid Hormones	Release: T3 and T4 are secreted into general circulation. MIT and DIT are not secreted by the thyroid gland.

Functions of Thyroid Cells:

- Accumulate iodide from blood.
- Synthesize and secrete thyroglobulin.
- Digest colloid and release thyroid hormones.



Previous Year Questions

1. Which hormone is specifically produced by the adrenal gland but not by the ovaries in females?

- A. Progesterone
 - B. DHEA
 - C. Estrogen
 - D. DHEA-S
-

2. Which hormones have a permissive function in the achievement of puberty?

- A. GnRH
 - B. Leptin
 - C. Insulin
 - D. GH
-

3. Which of the subsequent options is not a contributing factor to the development of secondary hyperparathyroidism?

- A. Parathyroid adenoma
 - B. Lithium intake
 - C. Vitamin D deficiency
 - D. Chronic kidney disease
-

4. Insulin acts on glucose metabolism by

(or)

Insulin acts on glucose metabolism by

- A. ↑ Facilitated diffusion of glucose
 - B. ↓ Permeability of glucose across cell membrane
 - C. ↑ Reabsorption of glucose from renal tubules
 - D. ↑ Glucose reabsorption from intestine
-

5. Inhibin is secreted by?

- A. Leydig cells
 - B. Sertoli cells
 - C. Interstitial cells
 - D. Peg cells
-

6. The adrenal medulla is derived from neural crest cells. These neural crest cells originate from the dorsal aorta. Clusters of chromaffin cells become the distinct medulla after birth. The adrenal gland first appears 28-30 days post-conception. Adrenal medulla secrete which hormones?

- A. Sex hormones
 - B. Thyroid stimulating hormones
 - C. Epinephrine
 - D. Glucocorticoids
-

7. For its action, which hormone is able to cross the plasma membrane?

- A. Epinephrine
 - B. Thyroxine
 - C. Glucagon
 - D. Insulin
-

8. Which hormones possess receptors within the cell?

- A. Thyroxine
 - B. Epinephrine
 - C. Glucagon
 - D. Parathormone
-

9. In regards to regulating local concentrations, the degradation of hormones is observed in _____.

- A. 11 β -hydroxysteroid dehydrogenase inactivation of glucocorticoids
 - B. Thyroid hormone deiodinases inactivating T3
 - C. Degradation of retinoic acid by Cyp26b1 in primordial germ cell of males
 - D. All of the above
-

10. Enterokinase is an activator of?

- A. Trypsinogen
- B. Trypsin
- C. Chymotrypsin

D. Antitrypsin

11. What is the duration for which sperm stays in the epididymis?

- A. 24hours
 - B. 48 hours
 - C. 20 days
 - D. 96hours
-

12. All are Glucogenic hormones except?

- A. ADH
 - B. Glucagon
 - C. Thyroxine
 - D. Glucocorticoids
-

13. Active form of Vitamin D3 is?

- A. Calcitriol
 - B. Calciferol
 - C. Calcdiol
 - D. Ergocalciferol
-

14. Which hormone binds to tyrosine kinase receptors and has 4 subunits and 2 units for receptor binding?

- A. Insulin
 - B. Glucagon
 - C. T3
 - D. ADH
-

15. Which hormone positively controls the blood levels of 1,25-OH-cholecalciferol among the options provided?

- A. Thyroxine
 - B. Parathormone
 - C. Insulin
 - D. Calcitonin
-

16. Determine the diagnosis indicated by the provided image.



- A. Gigantism
- B. Acromegaly
- C. Down's syndrome
- D. Dwarfism

17. All of the body organs or tissues demonstrate insulin-independent glucose uptake, except:

- A. RBC
 - B. Pancreas
 - C. Adipose tissue
 - D. Brain
-

18. What is the primary mechanism by which parathyroid hormone (PTH) elevates blood calcium levels?

- A. Increased renal reabsorption of Calcium.
 - B. Stimulation of osteoclast activity.
 - C. Stimulates Vitamin D synthesis.
 - D. All are True.
-

19. Thyroid hormone acts via which type of receptor?

- A. Cytoplasmic
 - B. Tyrosine kinase
 - C. Nuclear
 - D. GPCR
-

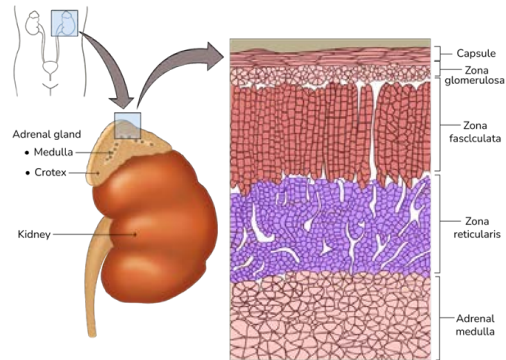
20. A 38-year-old person arrives at the ER with complaints of tachycardia, hypertension, and hypokalemia. On further investigation, the person's serum aldosterone levels are elevated, while renin levels are suppressed. What would be the most probable diagnosis for this given condition?

- A. Conn's syndrome
 - B. Cushing disease
 - C. Pheochromocytoma
 - D. Diabetes Mellitus
-

21. Which of the following factors were found to have an inhibitory effect on insulin secretion in the study conducted on a group of individuals investigating insulin regulation?

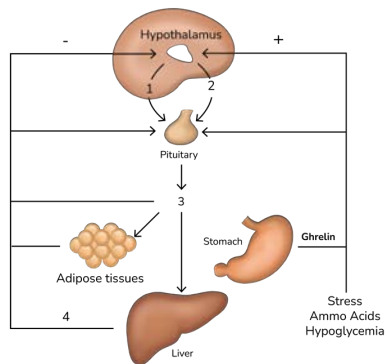
- A. Epinephrine
 - B. Gastric Inhibitory polypeptide
 - C. Acetylcholine
 - D. None of the above
-

22. Which of the following hormones are secreted from each layer of the structure marked in the image given below? a. Glomerulosa- Aldosterone b. Fasciculata- Glucocorticoids c. Reticulata- Androgens d. Medulla- Epinephrine



- A. a & b are correct
- B. a, b & c are correct
- C. b & c are correct
- D. a, b c & d are correct

23. Label the given Image:



- A. 1. SST, 2. GHRH, 3. GH, 4. IGF-1
- B. 1. IGF 1, 2. GH, 3. GHRH, 4. SST
- C. 1. GHRH, 2. GH, 3. IGF-1, 4. SST
- D. 1. GH, 2. IGF-1, 3. SST, 4. GHRH

24. Which hormone increases during periods of stress, starvation, and deep sleep?

- A. Growth Hormone (GH)
 - B. Glucocorticoids
 - C. Cortisol
 - D. Prolactin
-

25. What are the characteristic laboratory findings in primary hypothyroidism?

- A. Elevated TSH, low T4 levels
 - B. Elevated TSH, low T3 levels
 - C. Elevated TSH, elevated T4 levels
 - D. Low TSH, low T4 levels
-

26. Which receptor helps in the improvement of insulin resistance in Type 2 Diabetes Mellitus (DM2) with regular exercise and physical activities?

- A. GLUT 1
 - B. GLUT 2
 - C. GLUT 3
 - D. GLUT 4
-

27. Which of the following is responsible for the transport of iodine into the thyroid follicle?

- A. Na/I symporter

- B. Pendrin
 - C. Cl/I symporter
 - D. None of the above
-

28. A hypertensive patient's lab results reveal hypernatremia, hypokalemia, and metabolic alkalosis. What is the most probable diagnosis?

- A. Conn's Syndrome
 - B. Addison's Disease
 - C. Diabetes Insipidus
 - D. Cushing's Disease
-

29. Beta 2 receptor action leading to hyperglycemia/hypoglycemia is mediated by which of the following enzymes?

- A. Adenylyl cyclase
 - B. Guanylyl cyclase
 - C. Phospholipase C
 - D. Phospholipase A2
-

30. What type of pancreatic neuroendocrine tumor is responsible for recurrent episodes of secretory diarrhea?

- A. VIPoma
- B. Glucagonoma
- C. Insulinoma
- D. Zollinger Ellison disease

31. In the peripheral tissue, which enzyme is accountable for converting T4 to T3?

- A. Iodothyronine deiodinase I
 - B. Iodothyronine deiodinase III
 - C. Iodothyronine deiodinase II
 - D. Iodotyrosine deiodinase
-

32. Which of the following effects of puberty are not influenced by estrogen?

- A. facial hair growth
 - B. Cervical mucus secretion
 - C. Vaginal cornification
 - D. Menstruation
-

33. What is true about oxytocin among the options provided?

- A. It is responsible for galactokinesis
 - B. It can be given by oral route
 - C. It is secreted by anterior pituitary
 - D. In higher doses, it exerts diuretic like action
-

34. What hormone is responsible for the breast cramps experienced by a mother while breastfeeding her child?

- A. Progesterone
 - B. Estrogen
 - C. Prolactin
 - D. Oxytocin
-

35. Which of the following mechanisms is similar to cervical dilation and delivery occurring during uterine contraction?

- A. Milk ejection by oxytocin
 - B. Water retention by ADH
 - C. FSH secretion by anterior pituitary
 - D. None of the above
-

36. Which of the subsequent effects of puberty are not influenced by estrogen?

- A. Hair growth
 - B. Cervical mucus secretion
 - C. Vaginal cornification
 - D. Menstruation
-

37. A 12-year-old girl is undergoing pubertal changes. Her tanner stage is stage 4 for breast development and pubic hair scale. Which of the following pubertal changes is not solely due to estrogen?

(or)

Which of the following pubertal changes in a 12-year-old girl, specifically breast development and pubic hair scale at stage 4, is not exclusively caused by

estrogen?

- A. Vaginal cornification
 - B. Pubic and axillary hair growth
 - C. Cervical mucus production
 - D. Menstruation
-

38. Which one of the following hormones acts as anorexigenic agent?

- A. Melanin-concentrating hormone
 - B. Endorphins
 - C. Ghrelin
 - D. Leptin
-

39. For glucose uptake, which tissues rely on insulin?

- A. Liver
 - B. Brain
 - C. Skeletal muscle and adipose tissue
 - D. Kidneys
-

40. What is the primary protein that binds to thyroxine?

- A. Follistatin
- B. Transthyretin
- C. Thyroxine-binding globulin
- D. Transferrin

41. In your OPD, a child with loose stools for a day came to you. During examination, you observed signs of dehydration and provided the child with ORS. Through which receptors does ORS aid in glucose absorption?

- A. SGLT 1
 - B. SFLT 2
 - C. GLUT 4
 - D. Na-Ca transporter
-

42. What is found in skeletal muscle among the options provided?

- A. GLUT-4
 - B. GLUT-5
 - C. GLUT-7
 - D. GLUT-2
-

43. Prolactin level is highest during?

- A. 24 hours after ovulation
 - B. 24 hours after delivery
 - C. REM sleep
 - D. After running four 1 hours
-

44. What is the accurate order of muscle utilization during ejaculation?

- A. Ischiocavernosus contraction-external urethral sphincter relaxation-Bulbospongiosus contraction
 - B. Ischiocavernosus relaxation-External urethral sphincter contraction-Bulbospongiosus relaxation
 - C. Ischiocavernosus contraction – external urethral sphincter relaxation – Bulbospongiosus relaxation
 - D. Ischiocavernosus contraction – External urethral sphincter relaxation – Bulbospongiosus relaxation
-

45. Oxytocin plays a role in –

- A. Milk production
 - B. Milk ejection
 - C. Milk storage
 - D. Osmoregulation
-

46. The predominant organ producing insulin-like growth factor I is

- A. Liver
 - B. Pituitary gland
 - C. Pancreas
 - D. Adrenal glands
-

47. Which hormone is responsible for the breast cramps experienced by a mother during breastfeeding her child?

- A. Progesterone
- B. Estrogen

- C. Prolactin
 - D. Oxytocin
-

48. The primary source of testosterone production is primarily contributed by _____.

- A. Leydig cells
 - B. Sertoli cells
 - C. Seminiferous tubules
 - D. Epididymis
-

49. Which of the following are responsible for regulating the levels of calcium in the bloodstream?

- A. Calcitonin
 - B. Parathormone
 - C. Vitamin D
 - D. All the above
-

50. Which of the following does not have the presence of the sodium iodide symporter?

- A. Thyroid gland
 - B. Placenta
 - C. Salivary gland
 - D. Pituitary gland
-

51. To which biochemical moiety is beta hCG structurally similar?

- A. LH
 - B. FSH
 - C. ACTH
 - D. Oxytocin
-

52. What hormone is accountable for the breast cramps experienced by a mother during breastfeeding her child, as reported?

- A. Progesterone
 - B. Estrogen
 - C. Prolactin
 - D. Oxytocin
-

53. Which hormones employ signaling through receptor tyrosine kinase?

- A. Insulin
 - B. TSH
 - C. TRH
 - D. MSH
-

54. Among the given options, which condition is expected to exhibit the greatest levels of ACTH and cortisol?

- A. Early morning

- B. Normal individuals with dexamethasone administration
 - C. Addison's disease
 - D. Cushing's disease
-

55. Which specific antibody is detected during a recent infection?

- A. Ig M
 - B. Ig G
 - C. Ig E
 - D. Ig A
-

56. Histamine is secreted by?

- A. Enterochromaffin cell
 - B. Parietal cell
 - C. Oxyntic cell
 - D. Chief cell
-

57. Inhibitory control by the hypothalamus is exerted on:

- A. Prolactin only
 - B. Prolactin and growth hormone
 - C. Growth hormone only
 - D. Neither
-

58. A 45-year-old person presents to the outpatient department with a 6-month history of increased urine output and increased water intake. On investigation, the following results are obtained: HbA1C - 5.5%, Blood glucose - 80 mg/dL, Postprandial glucose - 130 mg/dL, and decreased urine osmolarity. What would be the most probable diagnosis for this given condition?

- A. DM
 - B. Diabetes Insipidus
 - C. Conn's Disease
 - D. Addison disease
-

59. Which of the options below can be used to elucidate the mechanism by which ANP (atrial natriuretic peptide) exerts its effects?

- A. cAMP
 - B. cGMP
 - C. IP3-DAG
 - D. JAK/STAT
-

60. What is the cause of "Vasopressin escape" in SIADH?

- A. Downregulation of aquaporin 2 production
 - B. Upregulation of aquaporin 3 production
 - C. Decrease in ADH secretion
 - D. Upregulation of aquaporin 2 production
-

61. In serous acinar cells, what is the typical location of the Golgi apparatus?

- A. Supranuclear
 - B. Basal
 - C. Adjacent to nucleus
 - D. Bottom of the cell
-

Correct Answers

Question	Correct Answer
Question 1	4
Question 2	2
Question 3	1
Question 4	1
Question 5	2
Question 6	3
Question 7	2
Question 8	1
Question 9	4
Question 10	1
Question 11	3
Question 12	1
Question 13	1
Question 14	1
Question 15	2
Question 16	2
Question 17	3
Question 18	4
Question 19	3
Question 20	1
Question 21	1
Question 22	4

Question 23	1
Question 24	1
Question 25	1
Question 26	4
Question 27	1
Question 28	1
Question 29	1
Question 30	1
Question 31	1
Question 32	1
Question 33	1
Question 34	4
Question 35	1
Question 36	1
Question 37	2
Question 38	4
Question 39	3
Question 40	3
Question 41	1
Question 42	1
Question 43	2
Question 44	3
Question 45	2
Question 46	1
Question 47	4
Question 48	1

Question 49	4
Question 50	4
Question 51	1
Question 52	4
Question 53	1
Question 54	4
Question 55	1
Question 56	1
Question 57	2
Question 58	2
Question 59	2
Question 60	1
Question 61	1

Solution for Question 1:

Correct Option:

Option D. DHEA-S (Dehydroepiandrosterone sulfate): DHEA-S is produced predominantly by the adrenal glands. The sulfation of DHEA to DHEA-S occurs mainly in the adrenal glands, and it is not produced by the ovaries.

Incorrect Option:

Option A. Progesterone: Progesterone is produced by both the adrenal glands and the ovaries (primarily by the corpus luteum during the menstrual cycle and by the placenta during pregnancy).

Option B. DHEA (Dehydroepiandrosterone): DHEA is produced by both the adrenal glands and the ovaries.

Option C. Estrogen: Estrogen is produced by both the adrenal glands and the ovaries (primarily by the ovarian follicles, corpus luteum, and placenta).

Solution for Question 2:

Correct Option B.

- Leptin is a hormone produced by adipose (fat) tissue. It plays a role in regulating energy balance and body weight. Leptin levels are influenced by the amount of body fat, and it communicates with the hypothalamus to regulate appetite and metabolism. Leptin also has an impact on the onset of puberty. Adequate leptin levels are required for the normal initiation and progression of puberty, particularly in females. It acts as a permissive hormone, providing signals to the hypothalamus that the body has sufficient energy reserves for puberty to proceed.

Incorrect options:

Option A. GnRH (Gonadotropin-Releasing Hormone): GnRH is a hormone produced and released by the hypothalamus. It plays a crucial role in the initiation and regulation of the reproductive system. It stimulates the release of luteinizing hormone (LH) and follicle-stimulating hormone (FSH) from the anterior pituitary gland, which in turn stimulates the production of sex hormones (estrogen and testosterone) by the ovaries and testes. While GnRH is necessary for the activation of the reproductive system, it does not directly play a permissive role in the attainment of puberty.

Option C. Insulin: Insulin is a hormone produced by the pancreas, and its primary role is to regulate blood glucose levels. While insulin is not directly involved in the attainment of puberty, it plays a role in overall growth and development. Insulin deficiency or resistance can disrupt normal growth and delay the onset of puberty.

Option D. GH (Growth Hormone): Growth hormone, also known as somatotropin, is produced by the anterior pituitary gland. It plays a significant role in growth, particularly during childhood and adolescence. While GH is not directly involved in the initiation of puberty, it is necessary for the proper development of the skeletal system and overall growth. Growth hormone deficiency can lead to growth retardation and delayed onset of puberty.

Solution for Question 3:

Correct Option A:

- Secondary hyperparathyroidism refers to the excessive secretion of parathyroid hormone (PTH) by the parathyroid glands in response to conditions outside the parathyroid glands themselves. It is important to note that primary hyperparathyroidism, which involves the overproduction of PTH due to a parathyroid adenoma, is not considered secondary hyperparathyroidism. Therefore, parathyroid adenoma is not a cause of secondary hyperparathyroidism.

Incorrect Options:

Option B: Lithium intake: Chronic use of lithium, a medication commonly prescribed for certain psychiatric disorders, can lead to the development of secondary hyperparathyroidism. Lithium interferes with the regulation of PTH secretion and can result in elevated PTH levels.

Option C: Vitamin D deficiency: Vitamin D plays a crucial role in calcium homeostasis, and its deficiency can lead to secondary hyperparathyroidism. Inadequate vitamin D levels impair the absorption of calcium in the intestine, causing a decrease in serum calcium levels. This triggers the release of PTH, which acts to increase calcium levels through various mechanisms.

Option D: Chronic kidney disease: Chronic kidney disease (CKD) is a common cause of secondary hyperparathyroidism. In CKD, there is a decline in kidney function, leading to impaired phosphate excretion and decreased production of active vitamin D. These abnormalities disrupt calcium and phosphate homeostasis, resulting in elevated PTH levels and secondary hyperparathyroidism.

Solution for Question 4:

Correct Answer Option A

- Insulin is a hormone produced by the pancreas that plays a critical role in regulating glucose metabolism. Insulin acts on the cell membrane of target cells, such as muscle, liver, and adipose tissue, to facilitate glucose uptake and utilization.
- Insulin acts on glucose metabolism by increasing the number of glucose transporters, such as GLUT4, on the cell surface of target cells. This leads to an increased uptake of glucose

into the cells by facilitated diffusion. Therefore, option a is the correct choice.

Incorrect Choices

- Option b. ↓permeability of glucose across the cell membrane: Insulin does not decrease the permeability of glucose across the cell membrane. Instead, it increases the number of glucose transporters on the cell surface, which leads to an increased uptake of glucose into the cells.
- Option c. ↑reabsorption of glucose from renal tubules: Insulin does not increase the reabsorption of glucose from renal tubules. Instead, it acts on target cells, such as muscle, liver, and adipose tissue, to facilitate glucose uptake and utilization.
- Option d. ↑glucose reabsorption from intestine: Insulin does not increase glucose reabsorption from the intestine. Instead, it acts on target cells to facilitate glucose uptake and utilization.

Solution for Question 5:

Correct Option. B

- Sertoli cells: Sertoli cells are located in the seminiferous tubules of the testes and play a crucial role in spermatogenesis, the process of sperm cell development. Sertoli cells not only support and nourish developing sperm cells but also secrete inhibin. Inhibin is a hormone that helps regulate the production of follicle-stimulating hormone (FSH) from the anterior pituitary gland. In males, inhibin acts as a negative feedback mechanism, inhibiting the secretion of FSH and thereby regulating spermatogenesis.

Incorrect options

Option A) Leydig cells: Leydig cells, also known as interstitial cells, are found in the testes. They are responsible for producing and secreting testosterone, which is important for the development and maintenance of male reproductive organs and secondary sexual characteristics. Leydig cells do not secrete inhibin.

Option C) Interstitial cells: Interstitial cells, or Leydig cells, as mentioned earlier, are found in the testes and are responsible for producing testosterone. They do not secrete inhibin.

Option D) Peg cells: Peg cells, also known as secretory cells or nonciliated cells, are found in the female reproductive tract, particularly in the fallopian tubes (oviducts). They secrete various substances, including nutrients and protective factors, to support the fertilization and early

development of the embryo. Peg cells do not secrete inhibin.

Solution for Question 6:

Correct Option. C

- Epinephrine: Yes, the adrenal medulla is responsible for the secretion of epinephrine, also known as adrenaline. Epinephrine is a hormone and neurotransmitter that plays a crucial role in the body's response to stress. It is released in response to various stimuli, such as physical or emotional stress, and it acts on various target tissues to increase heart rate, blood pressure, and metabolic rate.

Incorrect options

Option A) Sex hormones: The adrenal medulla does not secrete sex hormones. Sex hormones, such as estrogen and testosterone, are primarily produced by the gonads (ovaries in females and testes in males) under the influence of pituitary hormones.

Option B) Thyroid stimulating hormone: The adrenal medulla does not secrete thyroid-stimulating hormone (TSH). TSH is produced by the anterior pituitary gland and regulates the function of the thyroid gland.

Option D) Glucocorticoids: Glucocorticoids, such as cortisol, are not secreted by the adrenal medulla. They are primarily produced by the outer layer of the adrenal gland called the adrenal cortex. Glucocorticoids are involved in regulating metabolism, immune response, and stress response.

Solution for Question 7:

Correct option B - Thyroxine:

Thyroxine (T4) is a thyroid hormone that crosses the plasma membrane to exert its action. Once inside the target cells, thyroxine is converted to its active form, triiodothyronine (T3), which then binds to nuclear receptors and affects gene transcription, thus influencing various metabolic

processes.

Incorrect options:

Option A - Epinephrine: Epinephrine (adrenaline) is a hormone that acts on cell surface receptors. It binds to specific receptors on the cell surface, triggering a cascade of intracellular signaling events to initiate its physiological effects. It does not cross the plasma membrane.

Option C - Glucagon: Glucagon is a hormone that acts on cell surface receptors. It binds to specific receptors on target cells, primarily in the liver, promoting glycogen breakdown and the release of glucose into the bloodstream. Glucagon does not cross the plasma membrane.

Option D - Insulin: Insulin is a hormone that acts on cell surface receptors. It binds to specific receptors on target cells, such as muscle and adipose tissue, promoting glucose uptake and regulating glucose metabolism. Insulin does not cross the plasma membrane.

Solution for Question 8:

Correct Option A:

Intracellular receptors, also known as nuclear receptors, are located inside the cell and primarily reside within the cytoplasm or nucleus. These receptors can directly bind to specific hormones that are able to pass through the cell membrane and enter the cell. Once inside, these hormones bind to their corresponding intracellular receptors, forming a hormone-receptor complex. This complex then enters the nucleus and influences gene expression by binding to specific DNA sequences.

Thyroxine (T4) is a hormone produced by the thyroid gland. It is lipid-soluble, allowing it to diffuse across the cell membrane and bind to intracellular receptors. Inside the cell, thyroxine binds to specific nuclear receptors known as thyroid hormone receptors (TRs). The hormone-receptor complex then interacts with specific DNA sequences called thyroid hormone response elements (TREs) to regulate gene expression. This process influences various physiological processes, including metabolism, growth, and development.

Incorrect Options:

Option B (Epinephrine): Epinephrine, also known as adrenaline, is a hormone and neurotransmitter involved in the fight-or-flight response. However, it acts primarily through cell surface receptors known as adrenergic receptors, which are located on the cell membrane. Epinephrine does not have intracellular receptors.

Option C (Glucagon): Glucagon is a hormone produced by the pancreas that regulates glucose metabolism. It acts primarily through cell surface receptors known as glucagon receptors, which are located on the cell membrane. Glucagon does not have intracellular receptors.

Option D (Parathormone): Parathormone, also known as parathyroid hormone (PTH), is a hormone produced by the parathyroid glands. It acts primarily through cell surface receptors known as parathyroid hormone receptors, which are located on the cell membrane. Parathormone does not have intracellular receptors.

Solution for Question 9:

Correct Option D. All of the above

Explanation:

- Option A. 11 β -hydroxysteroid dehydrogenase inactivation of glucocorticoids: This enzyme, found in certain tissues such as the liver and adipose tissue, converts active glucocorticoids (such as cortisol) into inactive forms (such as cortisone), regulating local concentrations of these hormones.
- Option B. Thyroid hormone deiodinases inactivating T3: Deiodinase enzymes, specifically type 3 deiodinase (D3), play a crucial role in inactivating thyroid hormones, particularly triiodothyronine (T3), by converting it into inactive forms like 3,3'-diiodothyronine (reverse T3). This process helps regulate local thyroid hormone concentrations and is essential for thyroid hormone homeostasis.
- Option C. Degradation of retinoic acid by Cyp26b1 in primordial germ cell of males: Cyp26b1 is a cytochrome P450 enzyme that degrades retinoic acid, a derivative of vitamin A and an important signaling molecule. Its degradation in specific cells, such as primordial germ cells, helps regulate the local concentration of retinoic acid, which is essential for various developmental processes.

Option A. 11 β -hydroxysteroid dehydrogenase inactivation of glucocorticoids: This enzyme, found in certain tissues such as the liver and adipose tissue, converts active glucocorticoids (such as cortisol) into inactive forms (such as cortisone), regulating local concentrations of these hormones.

Option B. Thyroid hormone deiodinases inactivating T3: Deiodinase enzymes, specifically type 3 deiodinase (D3), play a crucial role in inactivating thyroid hormones, particularly triiodothyronine (T3), by converting it into inactive forms like 3,3'-diiodothyronine (reverse T3). This process

helps regulate local thyroid hormone concentrations and is essential for thyroid hormone homeostasis.

Option C. Degradation of retinoic acid by Cyp26b1 in primordial germ cell of males: Cyp26b1 is a cytochrome P450 enzyme that degrades retinoic acid, a derivative of vitamin A and an important signaling molecule. Its degradation in specific cells, such as primordial germ cells, helps regulate the local concentration of retinoic acid, which is essential for various developmental processes.

Therefore, the degradation of hormones by these mechanisms helps regulate local concentrations, making option D. All of the above the correct answer.

Solution for Question 10:

Correct option: A

- Enterokinase, also known as enteropeptidase, is an enzyme produced in the duodenum of the small intestine. Its primary role is to activate trypsinogen, which is an inactive precursor of trypsin. Trypsinogen is synthesized and released by the pancreas.
- Enterokinase acts on trypsinogen, cleaving it at a specific site to generate active trypsin. Once trypsin is formed, it can then further activate other pancreatic zymogens, such as chymotrypsinogen and procarboxypeptidase, initiating the digestive process.

Incorrect options:

Option B. Trypsin: Trypsin is not activated by enterokinase. Instead, enterokinase is responsible for the activation of trypsinogen to form trypsin. Once trypsin is formed, it can activate other pancreatic zymogens and participate in the digestion of proteins.

Option C. Chymotrypsin: Enterokinase does not directly activate chymotrypsin. Chymotrypsinogen, the inactive precursor of chymotrypsin, is activated by trypsin. Trypsin cleaves chymotrypsinogen to generate active chymotrypsin. Therefore, the activation of chymotrypsinogen is dependent on the presence of trypsin, not enterokinase.

Option D. Antitrypsin: Antitrypsin, also known as alpha-1 antitrypsin, is a protein that functions as an inhibitor of proteases, including trypsin. It is not activated by enterokinase. Instead, antitrypsin acts to regulate and inhibit the activity of trypsin and other proteases to prevent excessive tissue damage.

Solution for Question 11:

Correct option. c

- During puberty, cells are transformed into sperms with a head, midsection, and tail. The sperms use their long tails to push themselves out of the tubes into the epididymis. For the next 2 to 4 weeks the sperm swims through the epididymis as it develops further.

Other options are incorrect.

Solution for Question 12:

Correct option. A

- Glucogenic hormones are hormones that promote the production or release of glucose in the body. They stimulate processes such as glycogenolysis (breakdown of glycogen), gluconeogenesis (synthesis of glucose from non-carbohydrate sources), and the release of glucose into the bloodstream.

Incorrect options:

Option B. Glucagon: Glucagon is a hormone produced by the alpha cells of the pancreas. It acts in opposition to insulin and promotes the breakdown of glycogen into glucose, increases gluconeogenesis, and enhances the release of glucose into the bloodstream.

Option C. Thyroxine (thyroid hormone): Thyroxine is a hormone produced by the thyroid gland. While it primarily affects metabolism and energy utilization, it can indirectly influence gluconeogenesis and the production of glucose.

Option D. Glucocorticoids: Glucocorticoids are a class of hormones, including cortisol, produced by the adrenal glands. They are involved in various metabolic processes, including the regulation of glucose metabolism. Glucocorticoids can stimulate gluconeogenesis and glycogenolysis, leading to an increase in blood glucose levels.

Solution for Question 13:

Correct option. A

- Calcitriol, also known as 1,25-dihydroxyvitamin D, is the active form of Vitamin D3. It is formed in the kidneys from Calcidiol (the intermediate form) through hydroxylation reactions. Calcitriol plays a crucial role in regulating calcium and phosphate levels in the body, promoting their absorption from the intestine and reabsorption from the kidneys.

Incorrect options:

Option B. Calciferol: Calciferol is a term that can refer to various forms of Vitamin D, including both Vitamin D2 (ergocalciferol) and Vitamin D3 (cholecalciferol). However, specifically referring to Calciferol as the active form of Vitamin D3 is not accurate.

Option C. Calcidiol: Calcidiol, also known as 25-hydroxyvitamin D, is an intermediate form of Vitamin D that is produced in the liver from Vitamin D2 or Vitamin D3. It is not the active form of Vitamin D3 itself.

Option D. Ergocalciferol: Ergocalciferol is a form of Vitamin D2 that is derived from plant sources. It undergoes conversion in the body to form the active form of Vitamin D, but specifically referring to Ergocalciferol as the active form of Vitamin D3 is incorrect.

Solution for Question 14:

Correct option. A

- The hormone that has 4 subunits and 2 units for tyrosine kinase receptor binding is Insulin.
- Insulin is a peptide hormone that regulates glucose metabolism in the body. It is composed of two peptide chains, namely the A chain and B chain, which are held together by disulfide bonds. Insulin receptors are tyrosine kinase receptors located on the surface of target cells.
- The insulin receptor is a tetramer, consisting of two α -subunits and two β -subunits. The α -subunits are located extracellularly and contain the insulin-binding sites, while the β -subunits span the cell membrane and possess the tyrosine kinase catalytic activity. When insulin binds to the α -subunits of the receptor, it triggers a series of signaling events leading

to the activation of the tyrosine kinase domain in the β -subunits. This activation initiates downstream signaling pathways involved in glucose uptake, glycogen synthesis, and other cellular responses to insulin.

Incorrect option:

Option B. Glucagon: Glucagon is a peptide hormone that has its own receptor, but it does not bind to tyrosine kinase receptors. Glucagon receptors belong to the G protein-coupled receptor family and activate different signaling pathways compared to tyrosine kinase receptors.

Option C. T3: T3 (triiodothyronine) is a thyroid hormone that acts primarily through nuclear receptors rather than tyrosine kinase receptors. The binding of T3 to its receptors in the nucleus regulates gene expression and metabolic processes.

Option D. ADH (Antidiuretic Hormone, also known as vasopressin): ADH acts through G protein-coupled receptors, specifically V1 and V2 receptors, and does not involve tyrosine kinase receptor signaling.

Solution for Question 15:

Correct Option. B

- Parathormone (Parathyroid Hormone or PTH): Parathormone is a hormone secreted by the parathyroid glands. It plays a crucial role in the regulation of calcium and phosphate levels in the blood. Parathormone acts on the kidneys and stimulates the production of the active form of vitamin D3, 1,25-OH-cholecalciferol. Therefore, parathormone positively regulates blood levels of 1,25-OH-cholecalciferol.

Incorrect options

Option A) Thyroxine: Thyroxine, also known as T4, is a hormone secreted by the thyroid gland. While thyroid hormones play a role in calcium metabolism, they do not directly regulate the blood levels of 1,25-OH-cholecalciferol (active form of vitamin D3).

Option C) Insulin: Insulin is a hormone secreted by the pancreas and plays a key role in glucose metabolism and regulation of blood sugar levels. While insulin can indirectly influence calcium metabolism, it does not directly regulate blood levels of 1,25-OH-cholecalciferol.

Option D) Calcitonin: Calcitonin is a hormone secreted by the thyroid gland and plays a role in calcium homeostasis. Calcitonin helps to decrease blood calcium levels by inhibiting bone

resorption and enhancing calcium excretion by the kidneys. However, it does not directly regulate the blood levels of 1,25-OH-cholecalciferol.

Solution for Question 16:

Correct Option B - Acromegaly: Enlargement of lips, nose, and tongue, prognathism (growth of lower jaw), skull growth, and prominent supraorbital ridges with large frontal sinuses are features suggestive of acromegaly. It occurs due to excess levels of growth hormone after puberty.

Incorrect Options:

Option A- Gigantism: It occurs due to an increase in GH before puberty. Since the epiphyseal plates have not fused, children with this disorder grow significantly taller than their peers.

Option C - Down's syndrome: It is a genetic disorder characterized by Chromosome 21 trisomy. Patients with this disorder present with developmental and intellectual delays.

Option D- Dwarfism: Height for age below the third percentile on standard pediatric growth charts is defined as dwarfism. It occurs due to a deficiency of growth hormone.

Solution for Question 17:

Correct Option C- Adipose tissue:

The question is asking about which body organ or tissue demonstrates insulin dependent uptake.

- In adipose tissue, it is GLUT 4, and GLUT 4 is insulin dependent.

Incorrect Options:

Option A- RBC: In RBC there is GLUT 1 mechanism, so it is incorrect.

Option B- Pancreas: In Pancreas there is GLUT 2 mechanism, so it is incorrect.

Option D- Brain: In RBC there is GLUT 1 and 2 mechanism, so it is incorrect.

Solution for Question 18:

Correct Option D- All are True:

- Increased renal reabsorption of Calcium - PTH increases renal calcium reabsorption (distal tubule) and decreases phosphate reabsorption (proximal tubule)
- Stimulation of osteoclast activity - PTH indirectly increases the activity of osteoclasts. This releases calcium and phosphate from the bone turnover.
- Stimulates Vitamin D synthesis -Increases absorption of calcium and phosphate in the small bowel by activating Vitamin D.

Other Options:

Option A- Increased renal reabsorption of Calcium: This statement is true.

Option B- Stimulation of osteoclast activity: This statement is true.

Option C- Stimulates Vitamin D synthesis: This statement is true.

Solution for Question 19:

Correct Option C - Nuclear:

- This is the correct answer. Thyroid hormones, such as thyroxine (T4) and triiodothyronine (T3), act via nuclear receptors known as thyroid hormone receptors (TRs). These receptors are located within the cell's nucleus. When thyroid hormones bind to these receptors, they influence gene expression, metabolism, and various physiological processes.

Incorrect Options:

Option A - Cytoplasmic: This is not the correct answer. Thyroid hormones do not act via cytoplasmic receptors. Instead, they act primarily through nuclear receptors.

Option B - Tyrosine kinase: This is not the correct answer either. Tyrosine kinase receptors are involved in cell signaling pathways, but they are not the receptors through which thyroid

hormones exert their effects.

Option D - GPCR (G-Protein Coupled Receptor): This is not the correct answer. GPCRs are a different class of receptors that are involved in transmitting signals from the extracellular environment to the inside of the cell through G proteins. Thyroid hormones do not primarily utilize GPCRs for their signaling mechanism.

Solution for Question 20:

Correct Option A - Conn's syndrome (Primary Hyperaldosteronism):

- This condition involves excessive production of aldosterone by the adrenal glands, usually due to a benign tumor in one of the adrenal glands (aldosterone-producing adenoma). Elevated aldosterone levels lead to increased sodium retention and potassium excretion, resulting in hypertension, hypokalemia (low potassium levels), and metabolic alkalosis. The renin-angiotensin-aldosterone system (RAAS) feedback loop is disrupted, leading to suppressed renin levels.

Incorrect Options:

Option B - Cushing's disease: Cushing's disease is caused by excessive production of cortisol, usually due to a pituitary adenoma producing excess adrenocorticotrophic hormone (ACTH). Symptoms include weight gain, central obesity, moon face, buffalo hump, muscle weakness, and hypertension. Hypokalemia and tachycardia are not characteristic features of Cushing's disease.

Option C - Pheochromocytoma: Pheochromocytoma is a tumor of the adrenal medulla that produces excessive amounts of catecholamines (such as epinephrine and norepinephrine). This leads to episodic hypertension, palpitations, tachycardia, headaches, and sweating. Hypokalemia is not a typical feature of pheochromocytoma. Additionally, renin levels would not be suppressed in pheochromocytoma.

Option D - Diabetes Mellitus: Diabetes mellitus is a metabolic disorder characterized by high blood sugar levels due to insufficient insulin production or ineffective use of insulin. While diabetes can lead to secondary hypertension, it does not typically cause hypokalemia or suppressed renin levels.

Solution for Question 21:

Correct Option A - Epinephrine:

- Epinephrine can have both stimulatory and inhibitory effects on insulin secretion depending on the physiological state and the body's immediate needs. During stress, exercise, and hypoglycemia, epinephrine primarily inhibits insulin secretion to raise blood sugar levels and provide energy for the body's demands. In contrast, during periods of rest and digestion, epinephrine levels are low, allowing insulin to facilitate glucose uptake and storage.

Incorrect Options:

Option B - Gastric Inhibitory polypeptide: Gastric Inhibitory Polypeptide, also known as Glucose-Dependent Insulinotropic Peptide, is a hormone released from the gut in response to the ingestion of food, particularly carbohydrates. GIP plays a role in stimulating insulin secretion. It is an incretin hormone, meaning it promotes the release of insulin from pancreatic beta cells in response to elevated blood glucose levels. So, GIP has a stimulatory effect on insulin secretion by helping the body manage post-meal glucose levels.

Option C - Acetylcholine: Acetylcholine is a neurotransmitter with multiple functions, and its impact on insulin secretion is generally stimulatory. In the context of the original question, Acetylcholine would not be considered as having an inhibitory effect on insulin secretion. Instead, it can stimulate insulin release from pancreatic beta cells when activated.

Option D - None of the above

Solution for Question 22:

Correct Option D - a, b c & d are correct:

Hormones of adrenal gland		
Layers (weight)	Hormone production	
Cortex (72%)	Z. Glomerulosa (15%)	Mineralocorticoids: Aldosterone (main) Deoxycorticosterone

Z. Fasciculata (50%)	Glucocorticoids (main) Cortisol corticosterone Sex steroids	
Z. Reticularis (7%)	Sex steroid (main) Dehydroepiandrosterone (DHEA) Androstenedione Glucocorticoids	
Medulla (28%)	Epinephrine (90%) Nor-epinephrine (10%)	

Solution for Question 23:

Correct Option A - 1. SST, 2. GHRH, 3. GH, 4. IGF-1:

- The image represents the feedback regulation of GH secretion. The hypothalamus releases GHRH to stimulate growth hormone (GH) secretion and somatostatin to inhibit it. GH prompts the production of insulin-like growth factor I (IGF-1), which then negatively influences GH secretion. IGF-1 acts at both pituitary and hypothalamus levels.

Incorrect Options:

Options B, C & D are incorrect

Solution for Question 24:

Correct Option A - Growth Hormone (GH):

- GH is known to increase significantly during deep sleep (particularly in non-rapid eye movement or NREM sleep) and also plays a role in stress and starvation responses. It is crucial for post-natal growth and has its peak production during adolescence. It is stimulated by exercise and certain sleep stages.

Incorrect Options:

Option B - Glucocorticoids: These are hormones that are indeed elevated during stress and starvation but are not specifically associated with increased levels during deep sleep. They primarily include cortisol and are involved in the body's response to stress.

Option C - Cortisol: This hormone increases during periods of stress and starvation but does not have a significant increase during deep sleep. It is a type of glucocorticoid and is more associated with the stress response rather than growth or sleep-related processes.

Option D - Prolactin: This hormone primarily increases during pregnancy and breastfeeding. It is not notably increased by stress, starvation, or deep sleep. In fact, prolactin levels can be affected by REM sleep, rather than being elevated during deep sleep.

Solution for Question 25:

Correct Option A - Elevated TSH, low T4 levels:

- In primary hypothyroidism, the thyroid gland fails to produce adequate amounts of thyroid hormones (T3 and T4). Consequently, the pituitary gland releases more TSH (Thyroid-Stimulating Hormone) in an attempt to stimulate the thyroid gland. As a result, TSH levels are elevated while T4 (and often T3) levels are low.

Incorrect Options:

Option B - Elevated TSH, low T3 levels: While T3 levels can also be low in primary hypothyroidism, the hallmark laboratory finding for primary hypothyroidism is elevated TSH with low T4 levels. T3 levels are not the primary focus in the diagnosis of primary hypothyroidism.

Option C - Elevated TSH, elevated T4 levels: Elevated TSH with elevated T4 levels is not consistent with primary hypothyroidism. Elevated TSH typically accompanies low T4 levels in primary hypothyroidism. Elevated T4 levels would suggest hyperthyroidism rather than hypothyroidism.

Option D - Low TSH, low T4 levels: In primary hypothyroidism, TSH levels are high due to the pituitary's attempt to stimulate the underactive thyroid gland. Low TSH with low T4 levels would be more indicative of secondary or tertiary hypothyroidism, where the issue lies with the pituitary or hypothalamus, respectively.

Solution for Question 26:

Correct Option D - GLUT 4:

- This glucose transporter is found predominantly in muscle and adipose tissue. GLUT 4 is regulated by insulin and is crucial for the insulin-mediated uptake of glucose into these tissues. Regular exercise increases the number of GLUT 4 receptors on the cell membrane, improving glucose uptake and insulin sensitivity in individuals with Type 2 Diabetes Mellitus.

Option A - GLUT 1: This glucose transporter is found in various tissues, including the brain and erythrocytes. It helps in the uptake of glucose across cell membranes but is not primarily responsible for the insulin-mediated glucose uptake in muscle cells.

Option B - GLUT 2: This glucose transporter is located primarily in the liver, pancreas, and kidneys. It plays a role in glucose sensing and regulation of blood glucose levels but does not directly mediate the improvement of insulin resistance through physical activity.

Option C - GLUT 3: This glucose transporter is found mainly in neurons and some other tissues. It is responsible for the uptake of glucose into the brain but does not significantly impact insulin resistance in skeletal muscles.

Solution for Question 27:

Correct Option A - Na/I symporter:

- The Na/I symporter (NIS) is responsible for the active transport of iodine into the thyroid follicular cells. It uses the sodium gradient to co-transport iodine into the cell from the bloodstream.

Incorrect Options:

Option B - Pendrin: Pendrin is an iodide/chloride exchanger located on the apical membrane of thyroid follicular cells. It is involved in the movement of iodide into the follicle lumen but not in the transport of iodine into the thyroid follicular cells.

Option C - Cl/I symporter: This option is incorrect as there is no known Cl/I symporter in the thyroid. Iodide transport into the thyroid follicle is not mediated by a chloride/iodide symporter.

Solution for Question 28:

Correct Answer: A) Conn's Syndrome

Explanation:

Conn's syndrome (Primary Hyperaldosteronism) is the most likely diagnosis in a hypertensive patient with:

- Hypernatremia ($\uparrow \text{Na}^+$) $\rightarrow$ due to excessive sodium retention.
- Hypokalemia ($\downarrow \text{K}^+$) $\rightarrow$ due to renal potassium wasting.
- Metabolic alkalosis $\rightarrow$ due to increased H^+ excretion in the kidneys.

Pathophysiology:

- Aldosterone overproduction (usually due to an adrenal adenoma or bilateral adrenal hyperplasia) leads to: $\uparrow \text{Na}^+$ reabsorption $\rightarrow$ Hypertension & Hypernatremia $\uparrow \text{K}^+$ excretion $\rightarrow$ Hypokalemia $\uparrow \text{H}^+$ excretion $\rightarrow$ Metabolic Alkalosis
- $\uparrow \text{Na}^+$ reabsorption $\rightarrow$ Hypertension & Hypernatremia
- $\uparrow \text{K}^+$ excretion $\rightarrow$ Hypokalemia
- $\uparrow \text{H}^+$ excretion $\rightarrow$ Metabolic Alkalosis
- $\uparrow \text{Na}^+$ reabsorption $\rightarrow$ Hypertension & Hypernatremia
- $\uparrow \text{K}^+$ excretion $\rightarrow$ Hypokalemia
- $\uparrow \text{H}^+$ excretion $\rightarrow$ Metabolic Alkalosis

Addison's Disease (Option B) is incorrect because Addison's disease (primary adrenal insufficiency) presents with hypotension, hyponatremia, hyperkalemia, and metabolic acidosis, opposite to this case.

Diabetes Insipidus (Option C) is incorrect because DI causes hypernatremia, but it is due to free water loss (polyuria, polydipsia) rather than aldosterone excess. It is not associated with hypokalemia or metabolic alkalosis.

Cushing's Disease (Option D) is incorrect because while Cushing's disease (excess ACTH and cortisol) can cause hypertension, it does not primarily cause hypokalemia and metabolic alkalosis. However, severe cases with ectopic ACTH production (e.g., small-cell lung cancer) can cause pseudo-Conn's syndrome with a similar presentation.

Solution for Question 29:

Correct Option A:

- Beta-2 adrenergic receptors are G-protein coupled receptors found on the surface of various cells, including pancreatic beta cells. When these receptors are activated by agonists (e.g., epinephrine), they stimulate the production of cyclic adenosine monophosphate (cAMP) through the action of the enzyme adenylyl cyclase.
- The increase in cAMP levels leads to a cascade of intracellular events, including the activation of protein kinase A (PKA). PKA then phosphorylates multiple target proteins, resulting in various physiological responses, including the regulation of glucose metabolism.
- In the context of hyperglycemia, the activation of beta-2 adrenergic receptors and subsequent increase in cAMP levels can lead to the inhibition of insulin release from pancreatic beta cells. Insulin is responsible for promoting glucose uptake and utilization, so its inhibition can result in elevated blood glucose levels.
- In contrast, hypoglycemia is not directly mediated by the beta-2 adrenergic receptor. Instead, hypoglycemia may occur as an indirect effect of beta-2 receptor activation due to increased glycogenolysis (breakdown of glycogen into glucose) and gluconeogenesis (production of glucose from non-carbohydrate sources) mediated by the activation of other signaling pathways, such as the alpha-adrenergic receptors.

Incorrect Options:

Option B- Guanylyl cyclase: Guanylyl cyclase is an enzyme responsible for the production of cyclic guanosine monophosphate (cGMP) from guanosine triphosphate (GTP). It is not directly involved in the signaling pathway associated with beta-2 adrenergic receptors and glucose metabolism. The activation of beta-2 receptors primarily leads to the stimulation of adenylyl cyclase, which increases the production of cyclic adenosine monophosphate (cAMP), not cGMP.

Option C - Phospholipase C: Phospholipase C is an enzyme that cleaves phosphatidylinositol 4,5-bisphosphate (PIP₂) into inositol triphosphate (IP₃) and diacylglycerol (DAG). This leads to the release of calcium ions from intracellular stores and the activation of protein kinase C (PKC). Although phospholipase C plays a role in certain intracellular signaling pathways, it is not directly involved in the signaling pathway associated with beta-2 adrenergic receptors and glucose metabolism.

Option D - Phospholipase A₂: Phospholipase A₂ is an enzyme involved in the hydrolysis of membrane phospholipids, specifically releasing arachidonic acid. Arachidonic acid serves as a

precursor for the synthesis of various inflammatory mediators, such as prostaglandins and leukotrienes. While phospholipase A2 has important roles in inflammation and lipid metabolism, it is not directly involved in the signaling pathway associated with beta-2 adrenergic receptors and glucose metabolism.

Solution for Question 30:

- VIPoma is a rare type of pancreatic neuroendocrine tumor that produces vasoactive intestinal peptide (VIP), a hormone that regulates various physiological processes in the body, including intestinal motility and secretion. VIPoma is also known as Verner-Morrison syndrome or WDHA syndrome (Watery Diarrhea, Hypokalemia, and Achlorhydria).
- The excessive production of VIP by the tumor leads to recurrent episodes of secretory diarrhea, which can be severe and debilitating. Diarrhea is typically watery and voluminous, and it may be associated with abdominal cramping, nausea, and vomiting.

Incorrect Choices:

- Option b. Glucagonoma: Glucagonoma is a type of pancreatic neuroendocrine tumor that produces glucagon, a hormone that regulates glucose metabolism. Glucagonoma is associated with a characteristic rash called necrolytic migratory erythema, as well as diabetes mellitus, weight loss, and anemia. Glucagonoma does not typically cause secretory diarrhea.
 - Option c. Insulinoma: Insulinoma is a type of pancreatic neuroendocrine tumor that produces insulin, a hormone that regulates glucose metabolism. Insulinoma is associated with hypoglycemia, which can cause symptoms such as confusion, weakness, and sweating. Insulinoma does not typically cause secretory diarrhea.
 - Option d. Zollinger Ellison disease: Zollinger Ellison disease is a condition caused by a gastrin-secreting tumor, typically located in the pancreas or duodenum. The excessive production of gastrin leads to hypersecretion of gastric acid, which can cause peptic ulcers and other gastrointestinal symptoms. Zollinger Ellison disease does not typically cause secretory diarrhea.
-

Solution for Question 31:

- T4 (thyroxine) is the main hormone secreted by the thyroid gland, but it is relatively inactive. T3 (triiodothyronine) is the active form of thyroid hormone that exerts most of the biological effects in the body. The conversion of T4 to T3 occurs primarily in the peripheral tissues, such as the liver, kidney, and skeletal muscle.
- The enzyme responsible for the conversion of T4 to T3 is iodothyronine deiodinase I (D1). D1 is expressed in various tissues, including the liver, kidney, and thyroid gland. D1 catalyzes the removal of an iodine atom from the outer ring of T4, producing T3.

Incorrect Choices:

- Option b. Iodothyronine deiodinase III: Iodothyronine deiodinase III (D3) is an enzyme that can inactivate thyroid hormones by removing iodine atoms from the inner ring of T4 and T3. D3 is not involved in the conversion of T4 to T3.
- Option c. Iodothyronine deiodinase II: Iodothyronine deiodinase II (D2) is another enzyme that can convert T4 to T3, but it is primarily expressed in the thyroid gland and pituitary gland. D2 plays a role in the regulation of thyroid hormone levels in these tissues, but it is not as important for the peripheral conversion of T4 to T3.
- Option d. Iodotyrosine deiodinase: Iodotyrosine deiodinase is not a known enzyme involved in the conversion of T4 to T3, it helps the thyroid iodide salvage by catalyzing the deiodination of mono- and diiodotyrosine, the halogenated waste products of the manufacture of thyroid hormone.

Solution for Question 32:

- Puberty is a period of development during which the body undergoes significant changes, including the development of secondary sexual characteristics. Estrogen is a hormone that plays a key role in many of these changes, especially in females.
- Hair growth is not mediated by estrogen during puberty. Androgen hormones, such as testosterone, are responsible for the development of pubic, facial and axillary hair in both males and females.
- Therefore, option a is the correct choice.

Incorrect Choices:

- Option B) Cervical mucus secretion: Cervical mucus secretion is mediated by estrogen during puberty. Estrogen stimulates the production of cervical mucus, which helps to facilitate the transport of sperm through the female reproductive tract.
 - Option C) Vaginal cornification: Vaginal cornification is mediated by estrogen during puberty. Estrogen stimulates the growth and development of the vaginal epithelium, leading to the thickening and cornification of the cells.
 - Option D) Menstruation: Menstruation is mediated by estrogen during puberty. Estrogen promotes the thickening of the endometrial lining of the uterus, which is shed during menstruation.
-

Solution for Question 33:

- Oxytocin is a hormone that is produced by the hypothalamus and released by another gland, the posterior pituitary gland. It plays a role in a variety of physiological processes, including labor and lactation, social bonding, and stress response.
- One of the effects of oxytocin is its ability to stimulate milk ejection (galactokinesis) during lactation. Oxytocin acts on myoepithelial cells in the mammary glands, causing them to contract and expel milk from the alveoli into the ducts.

Incorrect Choices:

- Option b. It can be given by oral route: This statement is incorrect. Oxytocin is not effective when given by oral route because it is rapidly degraded by enzymes in the gastrointestinal tract.
 - Option c. It is secreted by anterior pituitary: This statement is incorrect. Oxytocin is secreted by the posterior pituitary gland, not the anterior pituitary gland.
 - Option d. Oxytocin is known to have antidiuretic properties. Therefore, in higher dose it will result in extreme water intoxication. Hence, option d is incorrect.
-

Solution for Question 34:

- Oxytocin is the hormone responsible for the let-down reflex during breastfeeding. It is released from the posterior pituitary gland in response to nipple stimulation, and it causes the contraction of the myoepithelial cells surrounding the alveoli in the breast. This contraction leads to the ejection of milk from the alveoli into the ducts, where it can be accessed by the infant.
- Therefore, the correct answer is oxytocin.

Incorrect Choices:

- Option a. Progesterone: Progesterone is involved in the development of the mammary glands during pregnancy, but it is not responsible for the cramps that the mother is experiencing during breastfeeding.
 - Option b. Estrogen: Estrogen is involved in the development of the mammary glands during pregnancy, but it is not responsible for the cramps that the mother is experiencing during breastfeeding.
 - Option c. Prolactin: Prolactin is involved in milk production in the mammary glands, but it is not responsible for the cramps that the mother is experiencing during breastfeeding.
-

Solution for Question 35:

- Cervical dilation and delivery during labor are controlled by the contraction of the uterine muscles. These contractions are stimulated by the hormone oxytocin, which is released by the pituitary gland in response to the stretching of the cervix and uterus. Oxytocin causes the uterine muscles to contract, which in turn leads to cervical dilation and delivery.
- The mechanism of cervical dilation and delivery on uterine contraction is similar to milk ejection by oxytocin. During lactation, oxytocin is also released in response to suckling by the infant, causing the contraction of the myoepithelial cells in the mammary glands and the ejection of milk.
- Therefore, the correct choice is option a, milk ejection by oxytocin.

Incorrect Choices:

- Option b. Water retention by ADH is not related to cervical dilation and delivery during labor. ADH is involved in the regulation of water balance in the body by promoting water reabsorption in the kidneys.
- Option c. FSH secretion by the anterior pituitary is involved in the regulation of the menstrual cycle and the growth and development of ovarian follicles, but it is not related to

cervical dilation and delivery during labor.

- Option d. None of the above is an incorrect option as cervical dilation and delivery during labor is similar to milk ejection by oxytocin.

Solution for Question 36:

- Puberty is a stage of development when sexual maturity is reached, and it is driven by a complex interplay of hormones. Estrogen is one of the major hormones involved in female puberty, and it is responsible for several changes in the body.
- During puberty, estrogen stimulates the growth of pubic and underarm hair, but it does not directly mediate hair growth on other parts of the body, such as the legs and arms. Therefore, the hair growth during puberty is not mediated by estrogen, which make option a, the correct choice.

Incorrect Choices:

- Option b. Cervical mucus secretion: Cervical mucus secretion, that is the production and release of fluid from the cervix is regulated by estrogen levels, which increase during puberty. Thus, option b is incorrect choice.
- Option c. Vaginal cornification: Vaginal cornification is regulated by estrogen levels, which increase during puberty.
- Option d. Menstruation: Menstruation is triggered by estrogen, which stimulates the growth of the endometrium in preparation for pregnancy. Thus, option d, is an incorrect choice.

Solution for Question 37:

Correct Option:

Option B. Pubic and axillary hair growth: The growth of pubic and axillary (underarm) hair is influenced by androgens, particularly testosterone. Androgens play a crucial role in the development of secondary sexual characteristics, including body hair. While estrogen does have some influence on body hair growth, the primary driver for the development of pubic and axillary

hair is androgens, particularly testosterone. Estrogen contributes to other pubertal changes, but hair growth is not exclusively caused by estrogen.

Incorrect Options:

Option A. Vaginal cornification: Vaginal cornification, also known as the appearance of cornified epithelial cells in vaginal smears, is influenced by estrogen. This is one of the effects of estrogen on the genital tract during puberty.

Option C. Cervical mucus production: Estrogen is responsible for cervical mucus production during puberty. Increasing levels of estrogen lead to changes in the cervical mucus that are favorable for sperm penetration and fertility.

Option D. Menstruation: Menstruation, or the onset of the menstrual cycle (menarche), is primarily triggered by increased levels of estrogen and its interactions with other hormones. It is considered a key milestone in female puberty and is indeed influenced by estrogen.

So, the correct answer is Option (B) "Pubic and axillary hair growth."

Solution for Question 38:

Correct Option D - Leptin:

Decrease Feeding (Anorexigenic)	Increase Feeding (Orexigenic)
α -Melanocyte-stimulating hormone	Neuropeptide Y
Leptin	Agouti-related protein
Serotonin	Melanin-concentrating hormone
Norepinephrine	Orexins A and B
Corticotropin-releasing hormone	Endorphins
Insulin	Galanin
Cholecystokinin	Amino acids (glutamate and γ -aminobutyric acid)
Glucagon-like peptide	Cortisol
Cocaine and amphetamine-regulated transcript	Ghrelin
Peptide YY	Endocannabinoids

Incorrect Options:

Options A, B, and C are incorrect. They are orexigenic in nature.

Solution for Question 39:

Correct Option C - Skeletal muscle and adipose tissue:

- This is the correct answer. Skeletal muscle and adipose tissue rely on insulin for glucose uptake. Insulin promotes the translocation of glucose transporters (GLUT4) to the cell surface in both of these tissues. GLUT4 allows glucose to enter these cells, where it can be used for energy (in muscle cells) or stored as fat (in adipose tissue).

Incorrect Options:

Option A - Liver: The liver does not solely rely on insulin for glucose uptake. It plays a role in both insulin-dependent and insulin-independent glucose uptake. Insulin suppresses glucose production by the liver (gluconeogenesis) and promotes glycogen synthesis in the liver, which leads to glucose storage.

Option B - Brain: The brain does not rely on insulin for glucose uptake. It is a unique tissue that can take up glucose independently of insulin. Brain cells have glucose transporters (GLUT1) that facilitate glucose transport across the blood-brain barrier.

Option D - Kidneys: The kidneys do not significantly rely on insulin for glucose uptake. The kidneys have glucose transporters (GLUT2) that facilitate glucose reabsorption from the glomerular filtrate back into the bloodstream. This process is not primarily regulated by insulin.

Solution for Question 40:

Correct Answer: C

Incorrect choices:

Option A. Follistatin is a protein that binds to and regulates the activity of certain growth factors, particularly members of the transforming growth factor-beta (TGF-beta) superfamily. It is not

involved in binding thyroxine (T4) or functioning as a thyroxine-binding protein.

Option B. Transthyretin, also known as prealbumin, is a transport protein primarily involved in the transport of thyroid hormones (including thyroxine) and retinol-binding protein (RBP) in the blood. While transthyretin does contribute to the transport of thyroxine, it is not the major thyroxine-binding protein. globulin (TBG): Thyroxine-binding globulin is the major protein responsible for binding and transporting the thyroid hormone thyroxine (T4) in the bloodstream. It is synthesized in the liver and plays a crucial role in maintaining appropriate levels of free thyroxine. TBG binds to thyroxine with high affinity and releases it as needed to meet the metabolic demands of the body. TBG levels can be influenced by various factors, such as pregnancy, estrogen levels, and certain medications.

Option D. Transferrin is a protein involved in the transport of iron in the blood. It binds to iron ions and carries them to various tissues in the body. Transferrin is not directly involved in the binding or transport of thyroxine.

In summary, the major thyroxine-binding protein in the blood is thyroxine-binding globulin (TBG), which specifically binds and transports thyroxine.

Solution for Question 41:

Correct Option A:

- SGLT 1 is primarily found in the small intestine, particularly in the brush border membrane of the enterocytes, which are the absorptive cells lining the intestinal wall. It is responsible for the active transport of glucose and sodium from the intestinal lumen into the enterocytes.
- When oral rehydration solution (ORS) is given to a child with loose stools and signs of dehydration, it contains glucose and electrolytes, including sodium. The glucose in the ORS is absorbed in the small intestine through SGLT 1. The active transport of glucose via SGLT 1 co-transporter is coupled with the absorption of sodium ions. This mechanism helps to enhance the absorption of both glucose and sodium, promoting rehydration.

Incorrect Options

Option B - SGLT 2 (Sodium-Glucose Transporter 2) is primarily found in the kidneys and plays a role in renal glucose reabsorption. It is not directly involved in glucose absorption in the intestines.

Option C - GLUT 4 (Glucose Transporter 4) is an insulin-regulated glucose transporter primarily found in muscle and adipose tissue. It facilitates glucose uptake into these tissues in response to insulin stimulation. GLUT 4 is not directly involved in glucose absorption in the intestines.

Option D - Na-Ca transporter refers to sodium-calcium exchangers, which are responsible for the exchange of sodium and calcium ions across the cell membrane. They are not directly involved in glucose absorption in the intestines.

Solution for Question 42:

Correct option:

Option A.

- GLUT stands for glucose transporter, and GLUT proteins facilitate the transport of glucose across cell membranes. Among the options given, GLUT-4 is the glucose transporter that is primarily found in skeletal muscle and adipose tissue.
- GLUT-4 is responsible for insulin-regulated glucose uptake into skeletal muscle cells. It plays a crucial role in maintaining blood glucose homeostasis. When insulin is released in response to high blood glucose levels, it triggers the translocation of GLUT-4 from intracellular vesicles to the cell membrane of skeletal muscle cells. This allows GLUT-4 to transport glucose from the bloodstream into the muscle cells, where it can be utilized for energy production or stored as glycogen.

Incorrect options:

Option B. GLUT-5: GLUT-5 is primarily found in the small intestine and is responsible for the uptake of fructose, not glucose. It is less relevant to skeletal muscle glucose uptake.

Option C. GLUT-7: GLUT-7 is mainly found in the liver and is involved in glucose uptake and release in hepatocytes. It is not present in skeletal muscle.

Option D. GLUT-2: GLUT-2 is found in various tissues, including the liver, pancreas, and small intestine. It facilitates glucose uptake in these tissues and is involved in glucose sensing and transport across the cell membrane. However, GLUT-2 is not primarily present in skeletal muscle.

Solution for Question 43:

Correct Option B: 24 hours after delivery

- Prolactin levels are highest after delivery, specifically around 24 hours after giving birth. This surge in prolactin secretion stimulates milk production and lactation in breastfeeding mothers. The high levels of prolactin help initiate and maintain breastfeeding and support the production of breast milk.

Incorrect Options:

Option A: Prolactin levels are not highest after 24 hours after ovulation.

Option C: Prolactin levels are not highest after REM sleep.

Option D: Prolactin levels are not highest after After running four 1 hours.

Solution for Question 44:

Correct Answer: c

- During ejaculation, the ischiocavernosus contracts, followed by the relaxation of the external urethral sphincter and then the bulbospongiosus relaxation.

Incorrect Choices:

Option A: This is the incorrect sequence.

Option B: This is the incorrect sequence.

Option D: This is the incorrect sequence.

Solution for Question 45:

Correct Answer: B

- Milk ejection or galactokinase is by oxytocin.

Incorrect Choices:

Option A: The hormone responsible for milk production or galactopoiesis prolactin.

Option C: Oxytocin is not responsible for milk storage.

Option D: ADH is responsible for osmoregulation.

Solution for Question 46:

Correct Option A: Liver

- The liver is the predominant organ producing insulin-like growth factor I (IGF-I).
- IGF-I is synthesized primarily in the liver in response to growth hormone (GH) stimulation.
- It plays a crucial role in growth and development, as well as in the regulation of various metabolic processes.

Incorrect Options:

Option B: Pituitary gland- The pituitary gland does not produce significant amounts of insulin-like growth factor I (IGF-I). Instead, the pituitary gland produces and releases growth hormone (GH), which stimulates the liver to produce IGF-I.

Option C: Pancreas- While the pancreas is responsible for producing and secreting insulin, which plays a role in regulating blood sugar levels, it is not the primary organ producing insulin-like growth factor I (IGF-I). IGF-I is mainly synthesized in the liver.

Option D: Adrenal glands- The adrenal glands are involved in the production and secretion of various hormones, including cortisol and adrenaline, but they do not play a significant role in the production of insulin-like growth factor I (IGF-I).

Solution for Question 47:

Correct Answer: D

- Oxytocin is the hormone in charge of breast cramps when breastfeeding. The milk ducts constrict as a result, releasing milk. The milk ducts contracting can occasionally bring Cramping in the breasts. When a child suckles on the breast, the hormone oxytocin is produced. A naturally occurring hormone called oxytocin controls important parts of both the male and female reproductive systems, such as breastfeeding and labor, as well as some aspects of behavior. Oxytocin is created by the nerve center, which the posterior pituitary organ then stores and distributes into the course. Oxytocin has two primary effects on the body: it causes uterine contractions during work and labor, and childbirth causes breast tissue compressions to help in breastfeeding after labor. Additionally, oxytocin is a chemical messenger in the brain that plays a significant role in numerous human behaviors and social interactions, such as: Sexual arousal Recognition Trust Romantic attachment Parent-infant bonding

- Sexual arousal
- Recognition
- Trust
- Romantic attachment
- Parent-infant bonding
- Sexual arousal
- Recognition
- Trust
- Romantic attachment
- Parent-infant bonding

Incorrect Choices:

Option A. Progesterone: The reproductive system depends heavily on the hormone progesterone. Chemical messengers called hormones instruct the body on how to function. Progesterone promotes menstruation in women and keeps the first trimester of pregnancy stable.

Option B. Estrogen: The development of female sexual traits is brought on by the steroid hormone estrogen, which is connected to the female reproductive system. Estriol, estradiol, and estrone are common names for estrogen.

Option C. Prolactin: Particularly in females, prolactin plays a key role in the physiology of the breast. The ability to produce milk normally depends on the prolactin hormone level. Prolactin level imbalances may impair this capacity. A mother won't be able to make milk if the levels are too low, while non-breastfeeding females or men may experience galactorrhea if the levels are improperly high.

Solution for Question 48:

Correct option is A: Leydig cells.

- Testosterone production is primarily contributed by Leydig cells, also known as interstitial cells, which are found in the testes. Leydig cells are located in the connective tissue surrounding the seminiferous tubules in the testes.
- Leydig cells are responsible for the synthesis and secretion of testosterone, which is the primary male sex hormone. Testosterone plays a crucial role in the development and maintenance of male reproductive organs, secondary sexual characteristics, and overall reproductive function.

Incorrect options:

Option B: Sertoli cells: Sertoli cells are located within the seminiferous tubules of the testes. They have essential functions in supporting and nourishing developing sperm cells, including providing nutrients and regulating their growth. However, Sertoli cells are not primarily responsible for testosterone production.

Option C: Seminiferous tubules: Seminiferous tubules are the structures within the testes where sperm production takes place. While the seminiferous tubules play a critical role in spermatogenesis, they are not directly involved in testosterone production.

Option D: Epididymis: The epididymis is a long, coiled tube located on the back of each testicle. It serves as a storage and maturation site for sperm cells. The epididymis does not have a primary role in testosterone production.

Solution for Question 49:

Correct Answer: D

Option A. Calcitonin: The thyroid gland is responsible for the release of calcitonin. Its primary function is to decrease the levels of serum calcium by increasing the movement of calcium from the blood into the bones. It inhibits bone breakdown (resorption) by osteoclasts. Upon the

binding of the calcitonin to the receptors, a signal is sent to the osteoclasts to slow down their activity. Thus more calcium is deposited in the bones lowering the amount of calcium in the blood.

Option B. Parathormone: Parathyroid hormone (PTH) is a hormone produced by the parathyroid glands which are present in the neck. It works exactly opposite to calcitonin, i.e., it helps to raise serum calcium levels by increasing calcium absorption from the intestines and bones and the reabsorption of calcium by the kidneys.

When PTH binds to these receptors, these cells increase their activity. It promotes bone resorption by activating the osteoclasts. PTH also decreases the excretion of calcium from the kidneys into the urine. It stimulates the activation of vitamin D in the kidneys, further enhancing calcium absorption from the intestines.

Option C. Vitamin D: While not a hormone in the traditional sense, vitamin D plays a crucial role in calcium regulation. Vitamin D is a nutrient that is essential for calcium absorption. When the skin is exposed to sunlight, vitamin D is produced in the body. It is the third component involved in the calcium homeostasis. It works similarly to the PTH.

- Vitamin D helps to increase the absorption of calcium from the intestines by binding to receptors on cells in the small intestine. Upon binding, it signals these cells to increase their activity. This allows more calcium to be absorbed from the intestines, which raises the amount of calcium in the blood. It also promotes calcium reabsorption in the kidneys, reducing its excretion in urine. By enhancing intestinal absorption and renal reabsorption of calcium, vitamin D helps to raise serum calcium levels.

Solution for Question 50:

Correct Option D. Pituitary gland:

The pituitary gland does not have the presence of the sodium iodide symporter. Unlike the thyroid gland and certain other tissues, the pituitary gland does not actively concentrate iodine, and the expression of NIS is not found in this gland.

In summary, while NIS is present in the thyroid gland, placenta, and salivary glands, it is not found in the pituitary gland.

Incorrect Options:

Option A. Thyroid gland: The sodium iodide symporter (NIS) is present in the thyroid gland. NIS is a membrane glycoprotein located in the basolateral membrane of thyroid follicular cells. It plays a crucial role in iodine uptake, which is essential for the synthesis of thyroid hormones (thyroxine/T4 and triiodothyronine/T3).

Option B. Placenta: NIS is also present in the placenta. It is responsible for concentrating iodine in the placental tissue, which is necessary for the development of the fetal thyroid gland and the production of thyroid hormones during pregnancy.

Option C. Salivary gland: NIS is expressed in salivary glands. It is involved in the uptake of iodine by these glands, which is then secreted into saliva.

Solution for Question 51:

- Beta hCG (human chorionic gonadotropin) is structurally similar to LH (luteinizing hormone). Human chorionic gonadotropin (hCG) is a hormone produced during pregnancy. The placenta primarily secretes it and has various functions, including supporting the development of the corpus luteum and maintaining progesterone production.
- Structurally, hCG comprises two subunits, an alpha subunit, and a beta subunit. The beta subunit of hCG shares a significant structural similarity with the beta subunit of luteinizing hormone (LH). In fact, LH and hCG have similar amino acid sequences in their beta subunits, which allows for cross-reactivity in laboratory assays.
- While hCG and LH have similarities in their beta subunits, it's important to note that they have different physiological functions. LH primarily regulates the menstrual cycle and ovulation, while hCG plays a crucial role in supporting pregnancy.

Incorrect Choices:

- Option b. FSH (follicle-stimulating hormone): FSH is also produced by the anterior pituitary gland. It works in conjunction with LH to regulate the menstrual cycle in females and sperm production in males. FSH stimulates the growth and maturation of ovarian follicles in females and the production of sperm cells in males. Unlike beta hCG, FSH is not structurally similar to beta hCG.
- Option c. ACTH (adrenocorticotrophic hormone): ACTH is a hormone secreted by the anterior pituitary gland that stimulates the adrenal glands' production and release of cortisol. It plays a role in the body's response to stress and helps regulate the adrenal cortex. ACTH is not structurally similar to beta hCG.

- Option d. Oxytocin: Oxytocin is a hormone produced by the hypothalamus and released by the posterior pituitary gland. It plays a role in various reproductive and social behaviors. In females, oxytocin is involved in uterine contractions during childbirth and milk letdown during breastfeeding. It also has a role in bonding and social interactions. Oxytocin is not structurally similar to beta hCG.
-

Solution for Question 52:

Correct Option D:

Oxytocin is the hormone in charge of breast cramps when breastfeeding. The milk ducts constrict as a result, releasing milk. The milk ducts contracting can occasionally bring Cramping in the breasts. When a child suckles on the breast, the hormone oxytocin is produced. Additionally, oxytocin is a chemical messenger in the brain that plays a significant role in numerous human behaviors and social interactions, such as:

- Sexual arousal
- Recognition
- Trust
- Romantic attachment
- Parent-infant bonding

Incorrect Options:

Option A. Progesterone: The reproductive system depends heavily on the hormone progesterone. Chemical messengers called hormones instruct the body on how to function. Progesterone promotes menstruation in women and keeps the first trimester of pregnancy stable. Progesterone's major job is to get the endometrium (uterine lining) ready for a fertilized egg to develop and implant. The endometrium loses during menstruation if there is no pregnancy. Progesterone rises if fertilization occurs to support the pregnancy. This is irrelevant to nursing.

Option B. Estrogen: The development of female sexual traits is brought on by the steroid hormone estrogen, which is connected to the female reproductive system. Estriol, estradiol, and estrone are common names for estrogen. Estradiol is the most widely used estrogen hormone for hormone replacement therapy (HRT) in the management of menopausal symptoms among the aforementioned types of estrogen. During the follicular phase of the menstrual cycle, estrogen aids in the proliferation of endometrial cells in the uterus, thickening the uterine lining in

preparation for pregnancy. This is irrelevant to nursing.

Option C. Prolactin: Particularly in females, prolactin plays a key role in the physiology of the breast. Clinically severe pathologic disorders emerge from insufficient or excessive prolactin production. The ability to produce milk normally depends on the prolactin hormone level. Prolactin level imbalances may impair this capacity. A mother won't be able to make milk if the levels are too low, while non-breastfeeding females or men may experience galactorrhea if the levels are improperly high. Prolactin's primary involvement is in milk production; it plays no part in breast contractions during nursing.

Solution for Question 53:

Correct Option A: Insulin

- Among the hormones listed, the one that uses receptor tyrosine kinase signaling is Insulin.
- Insulin is a peptide hormone produced by the beta cells of the pancreas. It acts on target cells by binding to insulin receptors, which are receptor tyrosine kinases. When insulin binds to its receptor, it activates the intrinsic tyrosine kinase activity of the receptor, leading to a cascade of signaling events inside the cell.

Incorrect options:

Option B, C & D: TSH and TRH both belong to the family of G protein-coupled receptors (GPCRs) and signal through G-protein-mediated pathways. TSH acts on the thyroid gland to stimulate the production and release of thyroid hormones, while TRH regulates TSH release from the pituitary gland.

MSH is a peptide hormone that regulates pigmentation and is primarily involved in the activation of melanocortin receptors, which are also GPCRs.

Solution for Question 54:

Correct Option D: Cushing's disease

- The condition that would have the highest levels of ACTH (adrenocorticotrophic hormone) and cortisol is Cushing's disease.
- Cushing's disease is a condition characterized by the overproduction of ACTH by a pituitary tumor (adenoma). This excessive ACTH production leads to increased stimulation of the adrenal glands, causing them to produce and release excessive amounts of cortisol, a stress hormone. Therefore, both ACTH and cortisol levels are elevated in Cushing's disease.

Incorrect options:

Option A: Early morning: In normal individuals, ACTH and cortisol levels are typically highest in the early morning. This is known as the diurnal variation of cortisol secretion, where cortisol levels naturally peak shortly after waking up. However, ACTH levels do not remain consistently elevated throughout the day.

Option B: Normal individuals with dexamethasone administration: Dexamethasone is a synthetic glucocorticoid that is commonly used to suppress cortisol production. In normal individuals, the administration of dexamethasone would lead to a decrease in both ACTH and cortisol levels due to the negative feedback on the hypothalamic-pituitary-adrenal (HPA) axis.

Option C: Addison's disease: Addison's disease is characterized by a deficiency of cortisol production due to adrenal insufficiency. In this condition, both ACTH and cortisol levels are decreased. ACTH levels are elevated as a compensatory mechanism to stimulate the adrenal glands, but the adrenal glands are unable to produce sufficient cortisol.

Solution for Question 55:

Correct option:

Option A.

Ig M (Immunoglobulin M) is an antibody class that is typically produced in response to a recent infection. It is the first antibody produced during the primary immune response and is often associated with an acute or recent infection. Ig M antibodies are generally larger in size and have multiple binding sites, allowing them to efficiently bind to and neutralize pathogens.

Incorrect options:

Option B. IgG (Immunoglobulin G): IgG antibodies are the most abundant and long-lasting antibody class in the bloodstream. They are typically produced during the secondary immune response and persist for an extended period, providing long-term protection against infections.

While IgG antibodies can indicate a previous or ongoing infection, they are not specific to recent infections.

Option C. Ig E (Immunoglobulin E): Ig E antibodies are primarily involved in allergic and hypersensitivity reactions. They are responsible for triggering the release of histamine and other inflammatory substances in response to allergens. IgE antibodies are not directly associated with recent infections.

Option D. Ig A (Immunoglobulin A): Ig A antibodies are predominantly found in mucosal areas such as the respiratory and gastrointestinal tracts. They play a crucial role in protecting mucosal surfaces from pathogens. While IgA antibodies can be involved in fighting infections, they are not specifically indicative of recent infections.

Solution for Question 56:

Correct Option A - Enterochromaffin cell:

- Histamine is primarily secreted by enterochromaffin cells, which are a type of neuroendocrine cells found in the gastrointestinal tract. These cells release histamine in response to gastrin and acetylcholine.
- Histamine has diverse physiological functions and plays a significant role in various processes in the body. In the gastrointestinal tract, histamine acts as a paracrine and neurotransmitter molecule. It can stimulate gastric acid secretion, increase blood flow to the gastric mucosa, and regulate smooth muscle contraction in the gut.

Incorrect Options:

Option B - Parietal cell: Parietal cells are responsible for producing and secreting gastric acid (hydrochloric acid) in the stomach.

Option C - Oxyntic cell: Oxyntic cells, or parietal cells, are another name for the cells that produce gastric acid.

Option D - Chief cell: Chief cells are responsible for secreting pepsinogen, the inactive precursor of the enzyme pepsin, which is involved in protein digestion.

Solution for Question 57:

Correct Option B - Prolactin and growth hormone:

- Inhibitory control by the hypothalamus is exerted on prolactin and growth hormone: The hypothalamus releases inhibitory hormones (dopamine and somatostatin) that act on the anterior pituitary to suppress the secretion of both prolactin and growth hormone.
- The hypothalamus releases inhibitory hormones (dopamine and somatostatin) that act on the anterior pituitary to suppress the secretion of both prolactin and growth hormone.
- The hypothalamus releases inhibitory hormones (dopamine and somatostatin) that act on the anterior pituitary to suppress the secretion of both prolactin and growth hormone.

Incorrect Options:

- Options A, C, and D are incorrect.
-

Solution for Question 58:

Correct Option B - Diabetes Insipidus:

- This is the correct answer. Diabetes insipidus is characterized by the inability of the kidneys to concentrate urine properly, leading to excessive dilute urine output (polyuria) and increased thirst (polydipsia). In this case, the decreased urine osmolarity is a key clue.
- Diabetes insipidus can be central (due to a lack of antidiuretic hormone, ADH) or nephrogenic (due to kidney insensitivity to ADH).

Incorrect Options:

Option A - DM (Diabetes Mellitus): This option is unlikely because the HbA1C level of 5.5% and the blood glucose levels are within the normal range (HbA1C < 6.5% and blood glucose < 126 mg/dL for diagnosis of diabetes).

Option C - Conn's Disease (Primary Hyperaldosteronism): This condition is characterized by excessive production of aldosterone by the adrenal glands, leading to high blood pressure and low blood potassium levels.

Option D - Addison's Disease (Adrenal Insufficiency): This condition involves the adrenal glands not producing enough cortisol and, in some cases, aldosterone. Symptoms often include fatigue,

weight loss, and low blood pressure.

Solution for Question 59:

Correct Option B - cGMP:

- ANP (atrial natriuretic peptide) is a hormone that is produced by the heart and plays a role in regulating blood pressure and fluid balance in the body. ANP acts on the kidneys to increase the excretion of sodium and water, which helps to reduce blood volume and lower blood pressure.
- The mechanism of action of ANP involves the activation of cGMP (cyclic guanosine monophosphate). ANP binds to its receptor on the surface of target cells, which activates the enzyme guanylyl cyclase. Guanylyl cyclase then converts GTP (guanosine triphosphate) to cGMP, which acts as a second messenger to mediate the effects of ANP.

Incorrect Options:

Option A - cAMP: Hormones acting through this pathway are ACTH, TSH, FSH, LH, catecholamines, somatostatin, glucagon, and secretin.

Option C - IP3-DAG: Hormones acting through this pathway are TRH, GnRH, GHRH, Oxytocin, and ADH (V1 receptor).

Option D - JAK/STAT: Hormones acting through this pathway are growth hormone and prolactin hormone.

Solution for Question 60:

Correct Option A - Downregulation of aquaporin 2 production:

- In SIADH, excessive ADH secretion causes water retention. Initially, ADH upregulates aquaporin 2 channels, increasing water reabsorption. Over time, aquaporin 2 production decreases, leading to "vasopressin escape" and allowing water excretion despite continued ADH release.

Incorrect Options:

- Options B, C, and D are incorrect. Refer to Option A for an explanation.
-

Solution for Question 61:

Correct Option A - Supranuclear:

- In serous acinar cells, the Golgi apparatus is typically located above the nucleus, towards the apical (top) region of the cell. This positioning allows for the packaging and modification of proteins produced by the rough endoplasmic reticulum (RER) before they are transported to the apical surface for secretion.

Incorrect Options:

- Option B, C, and D are incorrect.
-

Reproductive Physiology

1. A patient is being evaluated for male infertility. The fertility specialist explains the changes sperm undergo in the epididymis. Which of the following statements correctly describes a change that sperm undergo in the epididymis?

- A. Ability to bind to zona pellucida develops after ejaculation.
 - B. Cytoplasm and cell volume decrease in the testes.
 - C. Sperm gain motility and chemotaxis ability.
 - D. Capacitation for fertilization happens in the epididymis.
-

2. A 15-year-old male is undergoing puberty and starts experiencing physical and hormonal changes associated with this developmental stage. Which hormonal changes and effects are typical during puberty in males?

- A. Decreased testosterone reduces muscle mass.
 - B. Increased LH boosts testosterone production.
 - C. Elevated FSH enlarges penis and scrotum.
 - D. Higher inhibin increases FSH secretion.
-

3. Which of the following statements is true?

- A. Sertoli cells and Leydig cells together form the blood-testis barrier
 - B. Leydig cells secrete testosterone and androgen-binding protein
 - C. Leydig cells can synthesize cholesterol de novo within them
 - D. Sertoli cells secrete Inhibin B to suppress the release of LH
-

4. A 14-year-old male presents with delayed puberty. Physical examination reveals small testicular volume and a lack of secondary sexual characteristics. Hormonal tests indicate low testosterone levels but elevated FSH levels. Which of the following functions of Sertoli cells is most likely compromised in this patient?

- A. Secretion of testosterone in response to LH stimulation
 - B. Production of inhibin, leading to increased FSH levels
 - C. Conversion of testosterone to dihydrotestosterone by 5α -reductase
 - D. Support of spermatogenesis through FSH-mediated stimulation
-

5. Which of the following components is most accurately matched with its contribution to the total volume of semen?

- A. Seminal vesicles contribute 20% of the total volume.
 - B. Prostate contributes 60% of the total volume.
 - C. Bulbourethral glands contribute the majority of the semen volume.
 - D. Seminal vesicles contribute 60% of the total volume.
-

6. At which phase of the menstrual cycle does the primary oocyte complete meiosis I to become a secondary oocyte?

- A. Follicular phase
 - B. Ovulation
 - C. Luteal phase
 - D. Menstrual phase
-

7. A 27-year-old woman with a regular 30-day cycle is in the luteal phase. Which of the following statements accurately reflects the hormonal environment and physiological changes during this phase?

- A. Corpus luteum breaks down, leading to decreased estrogen and progesterone.
 - B. Progesterone levels increase significantly due to the activity of the corpus luteum
 - C. Granulosa and theca interna cells secrete estrogen
 - D. Initiated by increased follicle-stimulating hormone
-

8. According to the Two-Cell-Two-Gonadotropin Hypothesis, which of the following hormones primarily stimulates granulosa cells in the ovaries to promote the production of estrogen?

- A. LH
 - B. FSH
 - C. Progesterone
 - D. Prolactin
-

9. A 45-year-old woman presents with hot flashes, night sweats, and irregular menstrual cycles, suggesting a decrease in estrogen levels. Which of the following is a primary source of estrogen in premenopausal women, and which of the following is an effect of estrogen on the body?

- A. Adrenal cortex; Maintenance of bone density
 - B. Hepatocytes; enlargement of breast
 - C. Granulosa cells, proliferation of endometrium
 - D. Corpus luteum, decreases HDL cholesterol
-

10. Which of the following best describes the feedback regulation mechanism of ovarian function involving estrogen and the hypothalamus-pituitary axis?

- A. High levels of estrogen stimulate the release of LH and FSH from the pituitary gland.
 - B. Low levels of estrogen inhibit the release of GnRH from the hypothalamus
 - C. High levels of estrogen inhibit the release of GnRH from the hypothalamus
 - D. Low levels of estrogen stimulate the release of progesterone from the ovaries
-

Correct Answers

Question	Correct Answer
Question 1	3
Question 2	2
Question 3	3
Question 4	4
Question 5	4
Question 6	2
Question 7	2
Question 8	2
Question 9	3
Question 10	3

Solution for Question 1:

Answer: C) Sperm gain motility and chemotaxis ability.

Explanation:

Sperm acquire motility and the ability to sense chemical signals during their passage through the epididymis.

Change	Description
Motility Acquisition	Sperm leaving the testes are not fully mobile. They gain progressive motility in the epididymis. Motility is suppressed by inhibitory proteins until ejaculation.
Cation Channel Activation	Activation of CatSper family cation channels occurs. Alkaline-sensitive Ca^{2+} channels become active as sperm move to more alkaline cervical mucus.

Chemotaxis Development	Develop the ability to sense chemical signals from the ovaries, helping guide them toward the ovary.
Concentration	Become concentrated due to fluid reabsorption.
Epididymal Fluid Contribution	Epididymis and Sertoli cells secrete nutrient-rich fluid containing hormones, enzymes, and nutrients essential for maturation.
Sperm Storage	Epididymis serves as the primary storage site. Sperm can maintain fertility for at least a month.
Timeframe for Motility	Sperm gain the capability for motility after 18 to 24 hours in the epididymis. Full maturation and hypermotility occur in the female genital tract.
Acrosome Maturation	The acrosome, located at the sperm head and containing enzymes for fertilization, matures while sperm are stored in the epididymis.
Decreased Cytoplasm and Cell Volume	Sperm lose excess cytoplasm and decrease in overall cell volume as they mature in the epididymis.
Zona Pellucida Binding Ability	Sperm acquire receptors during epididymal transit that enable them to bind to the zona pellucida, facilitating penetration and fertilization of the egg.
Capacitation Post-Epididymis	Full fertilization capacity (capacitation) occurs after sperm leave the epididymis. This involves removal of inhibitory factors and changes to the sperm membrane.

Solution for Question 2:

Answer: B) Increased LH boosts testosterone production.

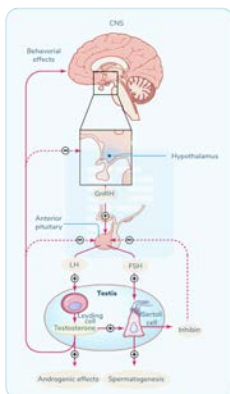
Explanation:

During puberty, increased LH stimulates Leydig cells in the testes to produce more testosterone, leading to the development of secondary sexual characteristics.

Puberty and Male Reproductive System Changes	
Increased Testosterone Production	Surge in testosterone by Leydig cells in testes. Stimulated by LH from anterior pituitary. (Option B)
Primary Sexual Characteristics	Growth of penis, scrotum, and testes. (Option C)

Secondary Sexual Characteristics	Body Hair: Increased growth on pubic area, face, chest, etc. Voice Changes: Deeper voice due to larynx hypertrophy. Skin Changes: Thicker skin, increased sebaceous secretion, acne. Muscle Development: Increased muscle mass (50% greater than females). (Option A) Bone Growth: Enhanced thickness, calcium retention. Metabolic Effects: Increased basal metabolic rate (5-10%), potential increase in red blood cells.
Spermatogenesis	Sperm production begins and continues throughout life within seminiferous tubules of testes.
Role of Hormones in Spermatogenesis	FSH: Stimulates Sertoli cells for sperm support. Testosterone: Essential for germ cell growth. Inhibin: Produced by Sertoli cells to regulate FSH and spermatogenesis rate. (Option D)
Sperm Maturation and Storage	Immature sperm mature in the epididymis, gaining motility (suppressed until ejaculation), stored for at least a month.

Hypothalamic-Pituitary-Testicular Axis



Solution for Question 3:

Answer: C) Leydig cells can synthesize cholesterol de novo within them

Explanation:

Leydig cells meet their cholesterol needs through both de novo synthesis and uptake of extracellular cholesterol from blood for the synthesis of testosterone.

Function of Leydig cells:

- Testosterone secretion is facilitated by Leydig cells.
- Cholesterol serves as the primary precursor for testosterone synthesis. Leydig cells can synthesize cholesterol internally and also absorb it from the blood via LDL and HDL receptors.
- Converts DHEA into testosterone.
- Testosterone moves into the seminiferous tubules (adluminal compartment).

Function of Sertoli cells:

- Only Sertoli cells form the blood-testis barrier (Option A)
- Nourishment and protection to developing sperm cells.
- Sertoli cells contain receptors for FSH (follicle-stimulating hormone) and androgens, which indirectly support spermatogenesis.
- Sertoli cells produce androgen binding protein (ABP). (Option B)
- Sertoli cells express the aromatase enzyme, responsible for converting testosterone to estrogen.
- They produce extracellular matrix components and anti-Müllerian hormone (AMH/MIS), which induces regression of Müllerian ducts during male fetal development.
- Sertoli cells secrete Inhibin B, a crucial factor in the testes that negatively regulates FSH secretion (Option D) through feedback mechanisms.

Solution for Question 4:

Answer: D) Support of spermatogenesis through FSH-mediated stimulation

Explanation:

The most plausible explanation for the observed hormonal imbalances and clinical features is a compromised ability of Sertoli cells to respond to FSH stimulation, leading to both impaired spermatogenesis and, consequently, low testosterone levels. The reduced negative feedback from inhibin due to impaired Sertoli cell function further contributes to the elevated FSH levels.

Hormones of Spermatogenesis:

Follicular stimulating hormone:

- FSH is essential for the initial development of spermatogonia cells.
- Main targets of FSH action include: Proliferation of Sertoli cell Proliferation of Spermatogonia up to primary spermatocyte stage Support the process of meiosis.
- Proliferation of Sertoli cell
- Proliferation of Spermatogonia up to primary spermatocyte stage
- Support the process of meiosis.
- Proliferation of Sertoli cell
- Proliferation of Spermatogonia up to primary spermatocyte stage
- Support the process of meiosis.

Testosterone:

- Testosterone is particularly required for the terminal stages of spermatogenesis
- Main targets of testosterone action include: Spermiogenesis (spermatid to mature sperm) Spermiation (release of sperm) Meiosis (suppression of testosterone leads to block in meiosis)
- Spermiogenesis (spermatid to mature sperm)
- Spermiation (release of sperm)
- Meiosis (suppression of testosterone leads to block in meiosis)
- Spermiogenesis (spermatid to mature sperm)
- Spermiation (release of sperm)
- Meiosis (suppression of testosterone leads to block in meiosis)

Estrogen:

- Main targets of Estrogen include: Fluid resorption and concentration of sperm Acrosome biogenesis of sperm
- Fluid resorption and concentration of sperm
- Acrosome biogenesis of sperm
- Fluid resorption and concentration of sperm
- Acrosome biogenesis of sperm

Growth hormone:

- Growth hormone specifically promotes early division of the spermatogonia

Option A) Secretion of testosterone in response to LH stimulation is a function of Leydig cell

Option B) Production of inhibin leading to increased FSH levels simply does not explain the low testosterone levels seen in this patient

Option C) While testosterone does play a role in spermatogenesis, its production is mainly stimulated by LH acting on Leydig cells.

Solution for Question 5:

Answer: D) Seminal vesicles contribute 60% of the total volume.

Explanation:

The seminal vesicles are responsible for producing approximately 60% of the total volume of semen. The fluid from the seminal vesicles is rich in fructose and other substances that provide energy and aid in the motility of sperm.

Sperm Maturation:

- Spermatogenesis: Begins during adolescence; takes ~74 days.
- Initial Mobility: Sperm are not fully mobile upon leaving the testes.
- Maturation: Occurs in the epididymis where sperm gain motility.
- Motility Mechanism: Involves activation of cation channel proteins (CatSper family) in the sperm tail.
- Alkaline Sensitivity: CatSpers form an alkaline-sensitive Ca^{2+} channel, activated as sperm transition from acidic vagina to alkaline cervical mucus.

Sperm Capacitation

- Location: Occurs in the isthmus of the uterine tubes.
- Purpose: Prepares sperm for the acrosome reaction and increases motility.

Contents of Semen

- Composition: Sperm and fluids from seminal vesicles, prostate, Cowper glands, and potentially urethral glands.
- Volume: 2.5–3.5 mL per ejaculation after several days of abstinence.
- Sperm Count: About 100 million sperm per mL.
- Fluid Source: Seminal vesicles: 60% (Option A) (Option D) Prostate: 20% (Option B) Bulbourethral glands: Remaining portion (Option C)
- Seminal vesicles: 60% (Option A) (Option D)
- Prostate: 20% (Option B)

- Bulbourethral glands: Remaining portion (Option C)
- Seminal vesicles: 60% (Option A) (Option D)
- Prostate: 20% (Option B)
- Bulbourethral glands: Remaining portion (Option C)

Surface Proteins: Sperm contain proteins like fertilin for egg cell membrane fusion

Solution for Question 6:

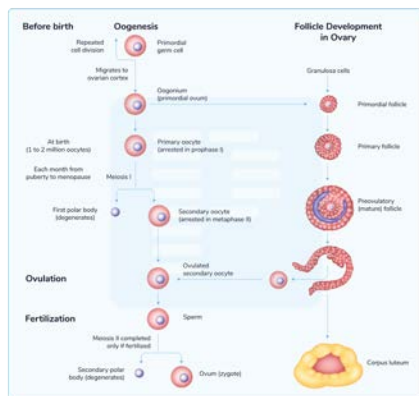
Answer: B) Ovulation

Explanation:

During ovulation, the surge in luteinizing hormone (LH) triggers the completion of meiosis I in the primary oocyte, resulting in the formation of a secondary oocyte and a first polar body. The secondary oocyte then arrests in metaphase II and is released from the follicle.

Table rendering failed

- Menstrual phase: The menstrual phase involves the shedding of the uterine lining if fertilization does not occur, and does not involve the completion of meiosis I in the oocyte. (Option D)



Solution for Question 7:

Answer: B) Progesterone levels increase significantly due to the activity of the corpus luteum

Explanation:

Phase	Description	Hormones	Duration & Events
Menstrual Phase	Occurs when the ovum is not fertilized. (Option A) Corpus luteum breaks down, leading to decreased estrogen and progesterone. Endometrium regresses and sheds.	Decreased Estrogen Decreased Progesterone	Duration: 4-7 days Blood loss: ~40 mL Serous fluid loss: ~35 mL
Follicular Phase (Pre-ovulatory)	Initiated by increased follicle-stimulating hormone (FSH) from the anterior pituitary. (Option D) Stimulates growth of 6-12 primordial follicles. Granulosa and theca interna cells secrete estrogen. (Option C) Estrogen promotes FSH receptor formation and accelerates follicular growth. One follicle becomes dominant (preovulatory follicle).	Follicle-Stimulating Hormone (FSH) Estrogen	Duration: Variable Follicle growth and maturation occur
Luteal Phase (Post-ovulatory)	Begins after ovulation with corpus luteum formation. Corpus luteum secretes progesterone and some estrogen. Prepares endometrium for implantation. If no implantation, corpus luteum involutes, decreasing hormone levels and triggering menstruation.	Luteinizing Hormone (LH) Progesterone Estrogen	Duration: About 12 days Decline in hormones leads to menstruation

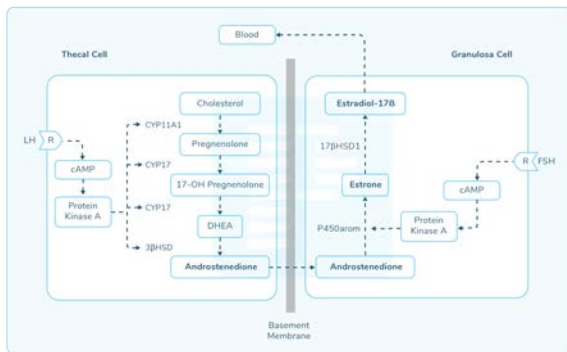
Solution for Question 8:

Answer: B) FSH

Explanation: Granulosa cells have many FSH receptors, and FSH stimulates the secretion of estradiol by increasing the aromatase activity.

Two-Cell-Two-Gonadotropin Hypothesis:

- Naturally occurring estrogens are 17 β -estradiol, estrone, and estrinol.
- Many LH receptors are present on theca interna cells and LH increases conversion of cholesterol to androstenedione.
- The theca interna cells supply androstenedione to the granulosa cells.
- Granulosa cells have many FSH receptors, and FSH (Option B) increases secretion of estradiol by increasing aromatase activity.
- Thus, both LH and FSH are required for the production of estrogen.



LH (Option A) increases progesterone secretion.

Progesterone (Option C) and prolactin (Option D) are not directly involved in production of estrogen in granulosa cells.

Solution for Question 9:

Answer: C) Granulosa cells, proliferation of endometrium

Explanation: Estrogen is primarily secreted by ovaries in granulosa cells and one of its effects is proliferation of endometrium.

	Estrogen	Progesterone
--	----------	--------------

Source	Granulosa cells (Option C), corpus luteum, and placenta	Corpus luteum, placenta and ovarian follicles
Effects on female genitalia	Growth of ovarian follicle Proliferation of endometrium (Option C) Increase the motility of the uterine tubes Increase blood flow in the uterus Increase the amount of uterine muscle and contractility.	Antiestrogenic effect on the myometrium, decreasing their excitability Secretory effect on the endometrium
Female Secondary Sex Characteristics	Enlargement of breasts Female distribution of fat Wide carrying angle	Stimulates the development of lobules and alveoli
Effects on the Central Nervous System	Increase libido Increase the proliferation of dendrites on neurons and the number of synaptic knobs	Thermogenic and responsible for the rise in basal body temperature during ovulation
Effects on Endocrine Organs	Increased secretion of angiotensinogen and thyroid-binding globulin	Large doses of progesterone inhibit LH secretion and potentiate the inhibitory effect of estrogens, preventing ovulation.
Other effects	Retention of salt and water and gain weight just before menstruation Plasma cholesterol-lowering action Vasodilation by increasing the local production of NO. Maintains bone mineral density	Stimulates respiration, and the alveolar PCO ₂ Increases plasma LDL and decreases HDL (Option D)

(Option A) The amount of progestin, androgen, and estrogen produced by the adrenal cortex is a minor fraction of the total amount produced in the body. Hence, not the primary source of estrogen.

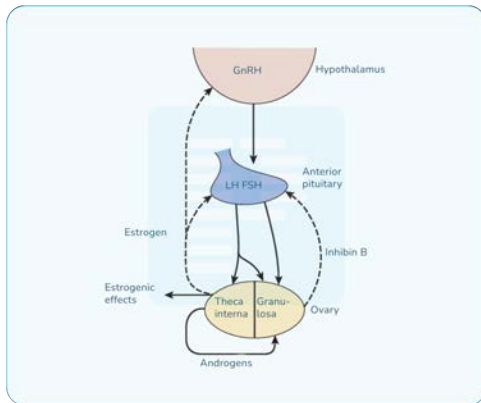
(Option B) Hepatocytes are not involved in estrogen production. Breast enlargement is an effect of estrogen.

(Option D) Corpus luteum secretes more progesterone than estrogen.

Solution for Question 10:

Answer: C) High levels of estrogen inhibit the release of GnRH from the hypothalamus

Explanation: Higher estrogen levels inhibit the release of GnRH from the hypothalamus by negative feedback mechanism.



- Thecal cells are regulated by LH.
- Granulosa cells are regulated by both LH and FSH.
- Circulating estrogens inhibit the secretion of GnRH, LH, and FSH.
- Inhibin from the granulosa cells inhibits FSH secretion.
- A moderate, constant level of circulating estrogen exerts a negative feedback effect on LH secretion.
- At 36–48 h before ovulation, the estrogen feedback effect becomes positive, and this initiates the burst of LH secretion (LH surge) that produces ovulation.